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- (54) **ANTI-NME ANTIBODY AND METHOD OF TREATING CANCER OR CANCER METASTASIS**
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CPC **C07K 16/40** (2013.01); **C07K 2317/31** (2013.01); **C07K 2317/33** (2013.01); **C07K 2317/565** (2013.01); **C07K 2317/622** (2013.01); **C07K 2317/73** (2013.01); **C07K 2317/76** (2013.01)
- (58) **Field of Classification Search**
None
See application file for complete search history.

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(57) **ABSTRACT**

The present application discloses anti-NMR antibodies and their use in treating or preventing diseases.

9 Claims, 59 Drawing Sheets

Specification includes a Sequence Listing.

(56)

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ELISA shows NME7 Dimerizes MUC1*
MUC1* extra cellular domain peptide immobilized on plate was bound by NME7 to saturation; a second MUC1* peptide with a C-terminal His-tag or Biotin tag was added and visualized by HRP labeled antibody to either His-tag or HRP labeled streptavidin

Fig. 1A

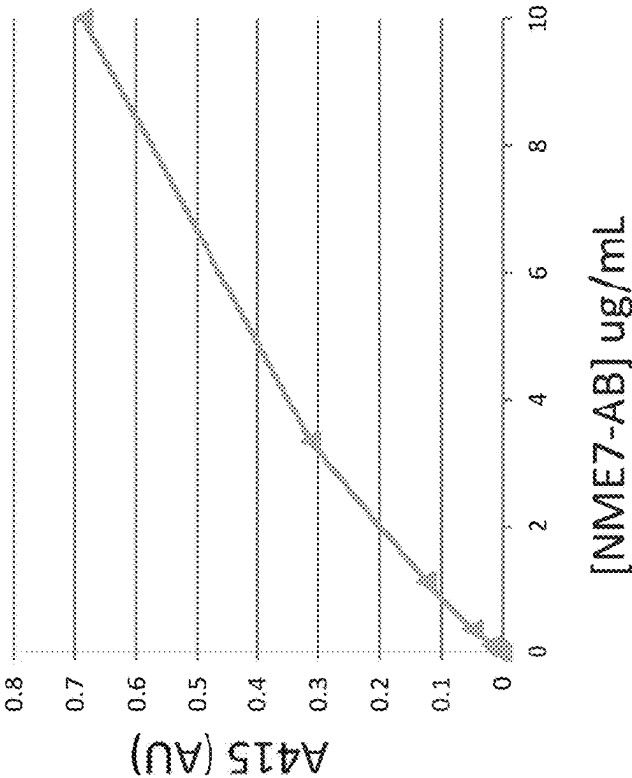
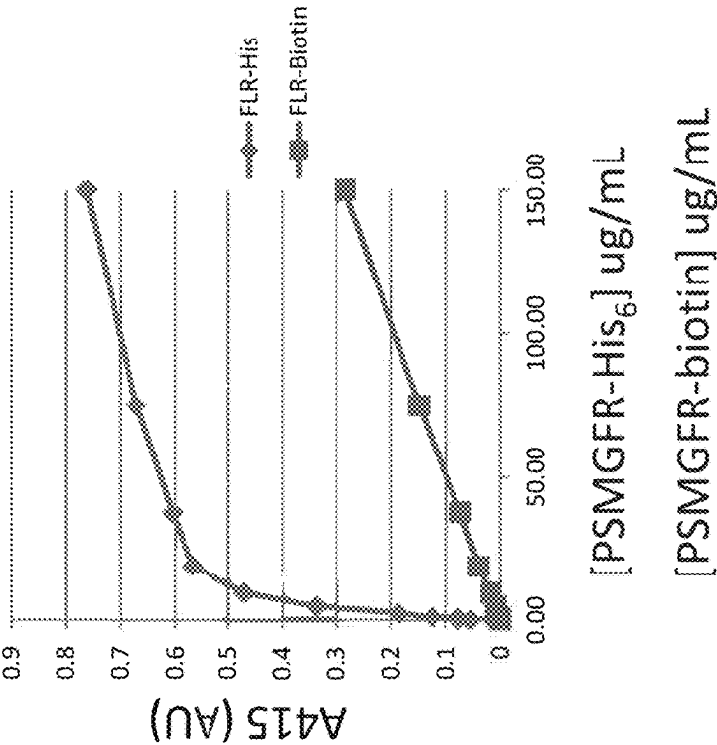


Fig. 1B



Figures 1A - 1B

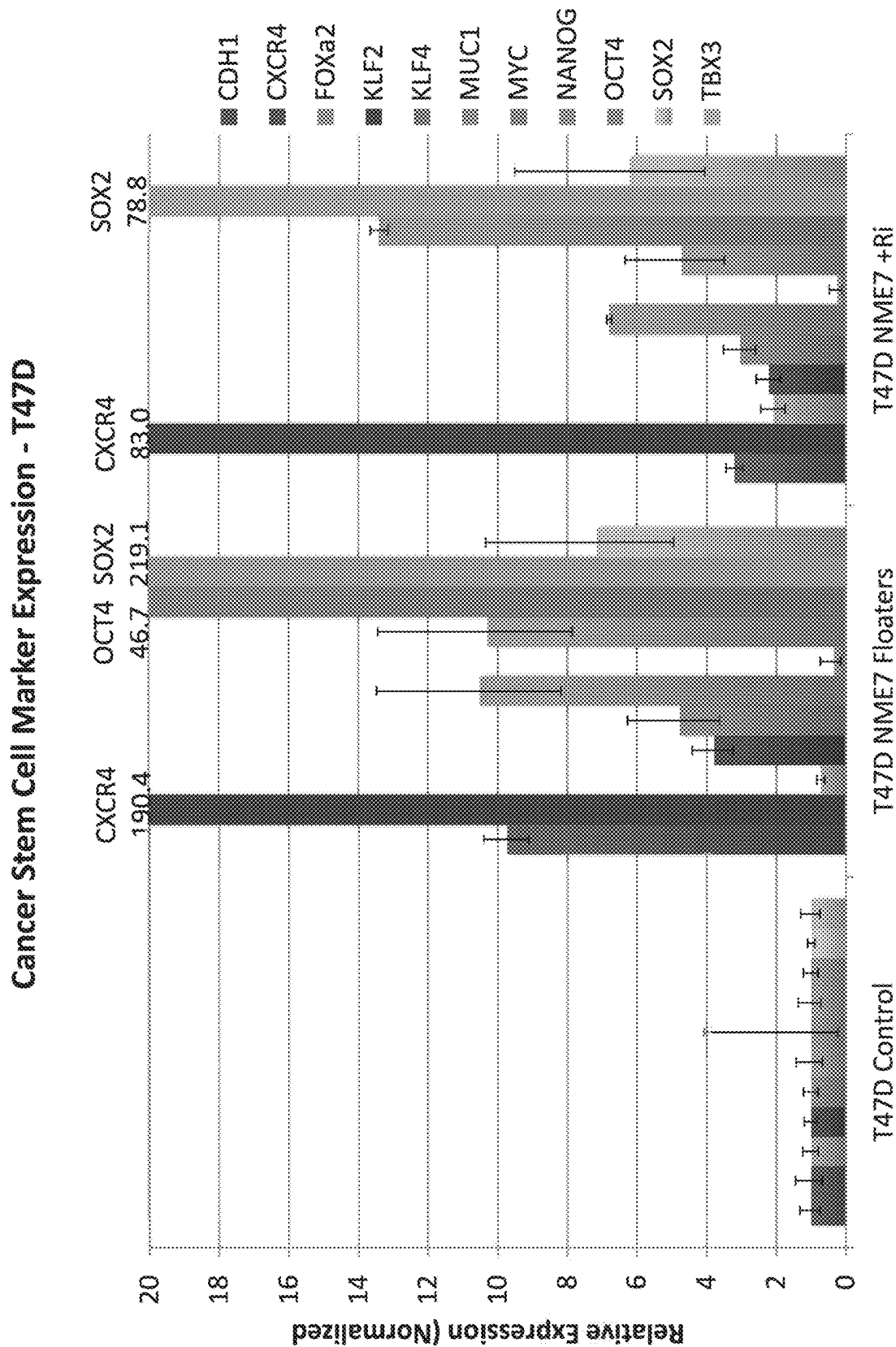


Figure 2

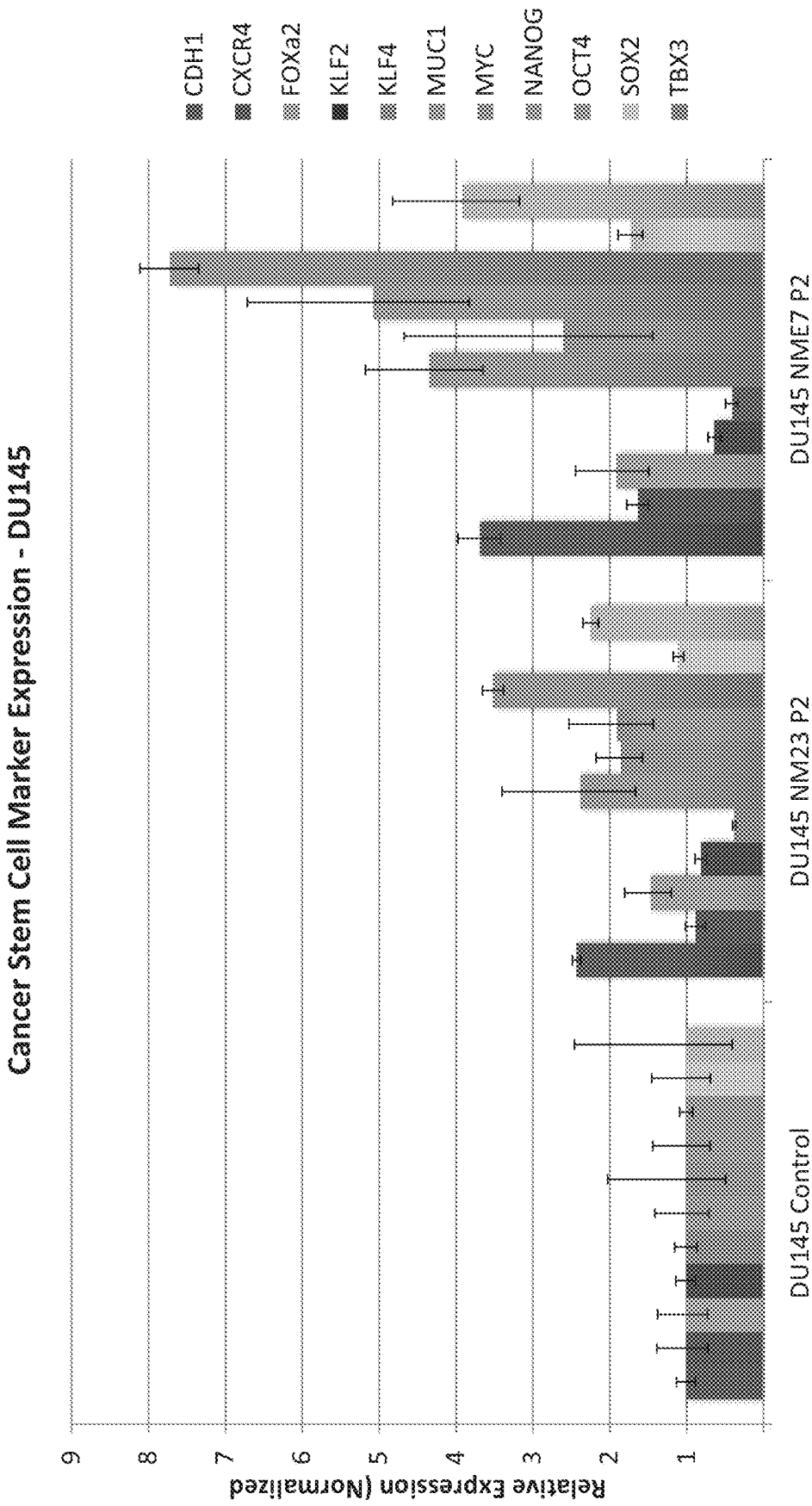


Figure 3

Breast cancer cells are progressed to a more metastatic state by culturing in the presence of 2iL inhibitors or in NME7-AB

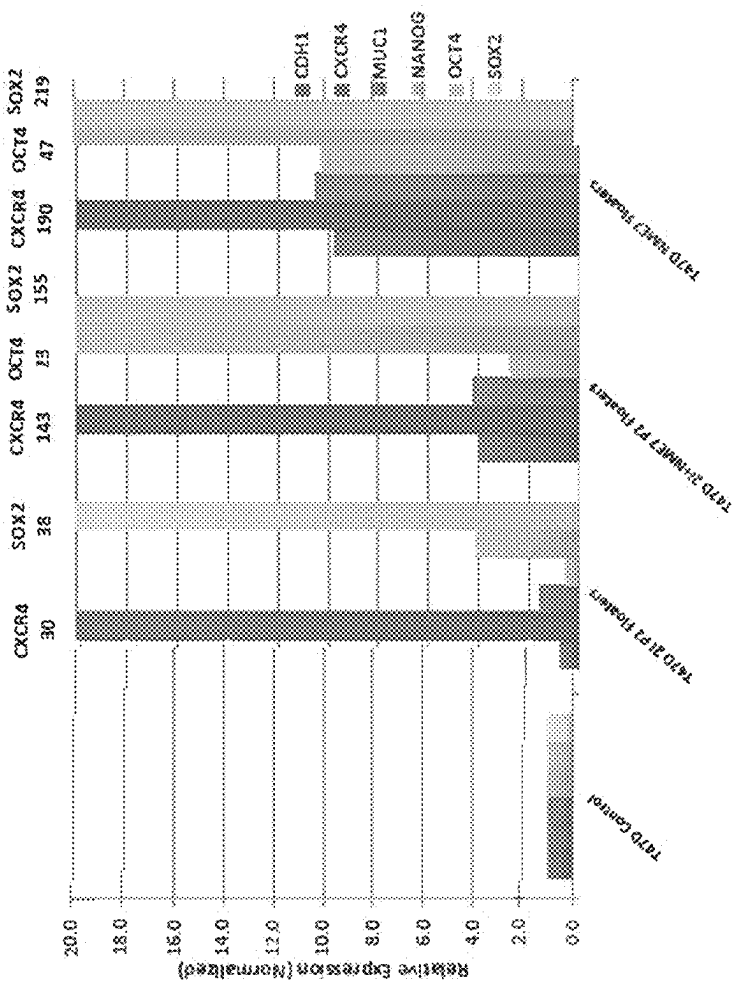


Figure 4

Figure 5

Sequence alignment of human NME1 to human NME7-A and -B domains

```
NME1      MANCERTFIAIKPDGVQRGLVGEIIKRFEQKGFRLVGLKFMQASEDLLNEHYVDLKDRPF 60
NME7A     -----EKTLLAIKPDATSK--AGEIIIEIINRAGFTITKLKMMMLSRKBALDFHVQHQSREFF 54
           *: *:  ****. :. :  .****: :. :  ** :.  ** :*  *..  :. : ** :.***

NME1      FAGLVKYMHSQFVVAMVWEGLVNVVKTGRVMLGETNP---ADSKPGTIRGDFCIQVGRNII 117
NME7A     FNELIQFITTGPPIAMEILRDDAICEWKRLLGPSNSGVARTDASESIRALFGTDGIRNAA 114
           *  *::: : *::**  :. :  : ** :*.  . . : ** . *  :  **

NME1      HGSDSVBSAEKEIGLWFHPPEELVDYTSQAQNWIIYE 152
NME7A     HGPDSFASAAPEMELFF----- 131
           **,**,  ** :*: *:

NME1      MANCERTFIAIKPDGVQRGLVGEIIKRFEQKGFRLVGLKFMQASEDLLKEHYVDLK-DRP 59
NME7B     ----NCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFMMDRVNVVEEFYEVYKGVVT 56
           : *  : **..*..**:*:: :. :  **.:  :::: :.  : :*. *  *  .

NME1      FFAGLVKYMHSQFVVAMVWEGLVNVVKTGRVMLGETNPADS---KPGTIRGDFCIQVGRNII 116
NME7B     EYHDMVTEMYSGPCVAMEIQQNNAKTFRFECGFADPEIARHLRFGTLRAIFGKTKIQNA 116
           : .:*. *:*** **  :  *.** *  : * :*:  :  :***:. *  :*

NME1      IHGSDSVBSAEKEIGLWFHPPEELVDYTSQAQNWIIYE 152
NME7B     VHCTDLPEDGLLEVQYFFKILDN----- 139
           :* :*  *..  * : :*: :
```

Figure 6

NME7 specific peptides for generating antibodies to inhibit NME7 for the treatment or prevention of cancers.

The following peptide sequences are identified as being immunogenic peptides giving rise to antibodies that target human NME7 but not human NME1. The sequences were chosen for their lack of sequence homology to human NME1.

1. LALIKPDA
2. MMMLSRKEALDFHVDHQS
3. ALDFHVDHQS
4. EILRDDAICEWKRL
5. FNELIQFITTGP
6. RDDAICEW
7. SGVARTDASESIRALFGTDGIRNAA
8. ELFFPSSGG
9. KFTNCTCCIVKPHAVSEGLLGKILMA
10. LMAIRDAGFEISAMQMFNMDRVNVEEFYEVYKGVVT
11. EFYEVYKGVVTEYHD
12. EIQQNNATKTFREFCGPADPEIARHLRPGTLRAIFGKTKIQNA
13. YSGPCVAM
14. FREFCGP
15. VHCTDLPEDGLLEVQYFFKILDN
16. IQNAVHCTD
17. TDLPEDGLLEVQYFFKILDN
18. PEDGLLEVQYFFK
19. EIINKAGFTITK
20. MLSRKEALDFHVDHQS
21. NELIQFITT
22. EILRDDAICEWKRL
23. SGVARTDASESIRALFGTDGI
24. SGVARTDASES

Figure 6 (Continued)

25. ALFGTDGI

26. NCTCCIVKPHAVSE

27. LGKILMAIRDA

28. EISAMQMFNMDRVNVE

29. EVYKGVVT

30. EYHDMVTE

31. EFCGPADPEIARHLR

Figure 7

NME7 specific peptides for generating antibodies to inhibit NME7 for the treatment or prevention of cancers.

The following are preferred as they are likely areas that are important for structural integrity or for binding to the MUC1* peptide. Bivalent antibodies wherein each variable region would bind to each one of a pair are preferred.

35. ICEWKRL

36. LGKILMAIRDA

37. HAVSEGLLGK

38. VTEMYSGP

39. NATKTFREF

40. AIRDAGFEI

41. AICEWKRLLG PAN

42. DHQSRPFF

43. AICEWKRLLG PAN

44. VDHQSRPF

45. PDSFAS

46. KAGEIIEIINKAGFTITK

Figure 8

The following peptide sequences are from human NME1 and were selected for their high homology to human NME7 as well as for their homology to other bacterial NME proteins that are able to mimic its function.

- 47. MANCERTFIAIKPDGVQRGLVGEIIRFE
- 48. VDLKDRPF
- 49. HGSDSVESAEKEIGLWF

Especially preferred for their high homology to human NME7-A or -B and also to HSP

593 are:

- 50. ERTFIAIKPDGVQRGLVGEIIRFE
- 51. VDLKDRPFFAGLVKYMHS GPVVAMVWEGLN
- 52. NIIHGSDSVESAEKEIGLWFHPEELV
- 53. KPDGVQRGLVGEII

Figure 9

NME7-AB specific peptides preferred for generating antibodies for the treatment or prevention of cancer.

NME7A peptide 1

MLSRKEALDFHVDHQ5

NME7A peptide 2

SGVARTDASES

NME7B peptide 1

DAGFEISAMQMFNMDRVNVE

NME7B peptide 2 EVYKGVVTEYHDMVTE**NME7B peptide 3**

AIFGKTKIQNAVHCTDLPEDGLLEVQYFF

NME7B peptide 3 Cys to Ser mutation

AIFGKTKIQNAVHSTDLPEDGLLEVQYFF

Anti-NME7 antibodies generated with peptides A1, A2, B1, B2 & B3 bind to NME7 (left) but not to NME1 (right); C20 is an anti-NME1 antibody

Fig. 10A

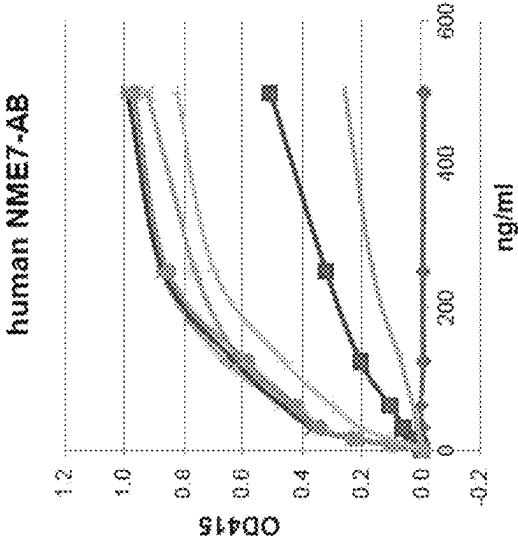
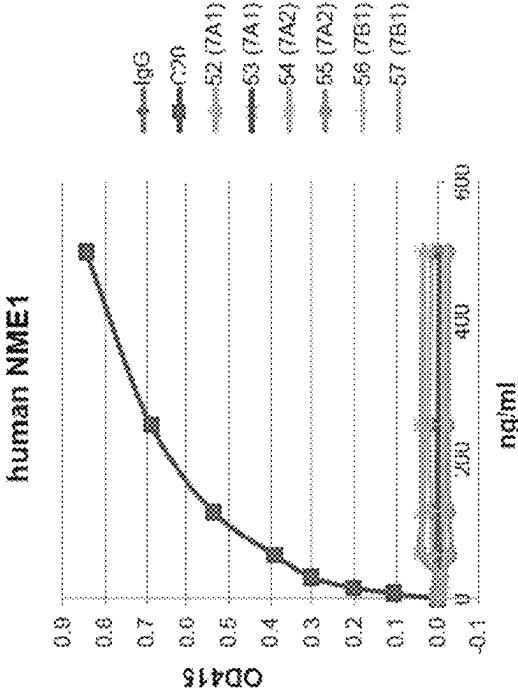


Fig. 10B



Figures 10A-10B

ELISA assay tests NME7 antibodies for ability to block binding of NME7-AB to MUC1* peptide on surface, but not human NME1 dimers

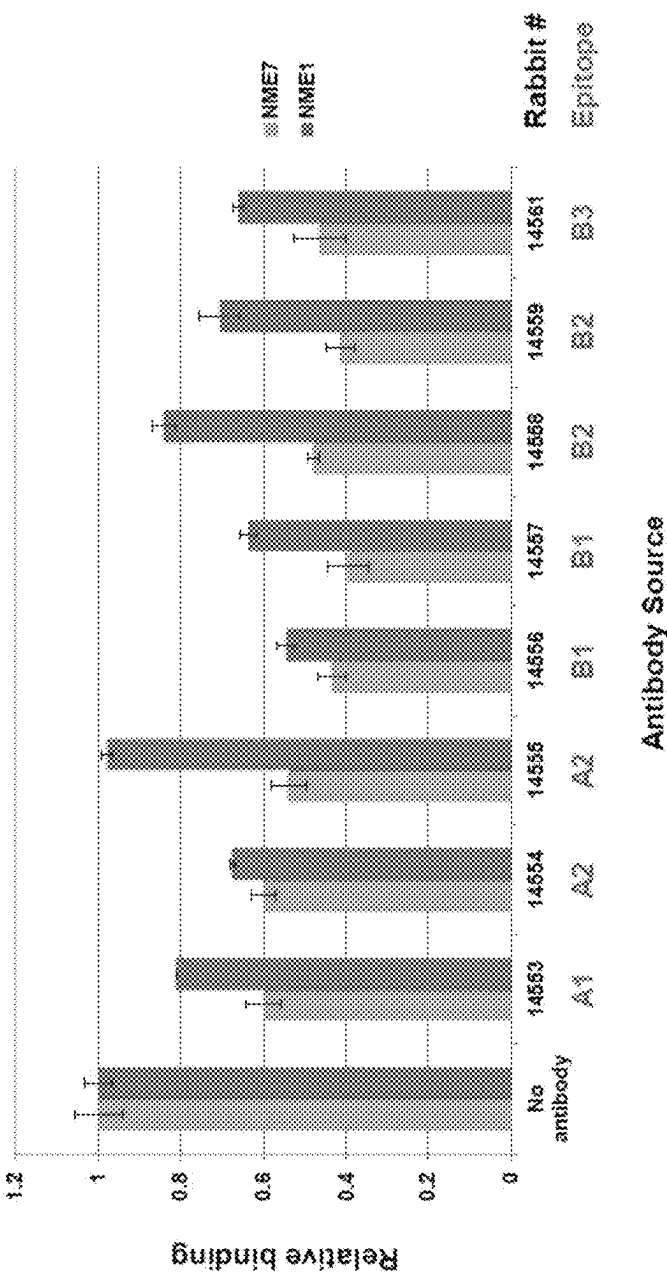


Figure 11

Anti-NME7 antibodies generated with peptides A1, A2, B1, B2 & B3 as well as the immunizing peptides themselves inhibit cancer cell growth

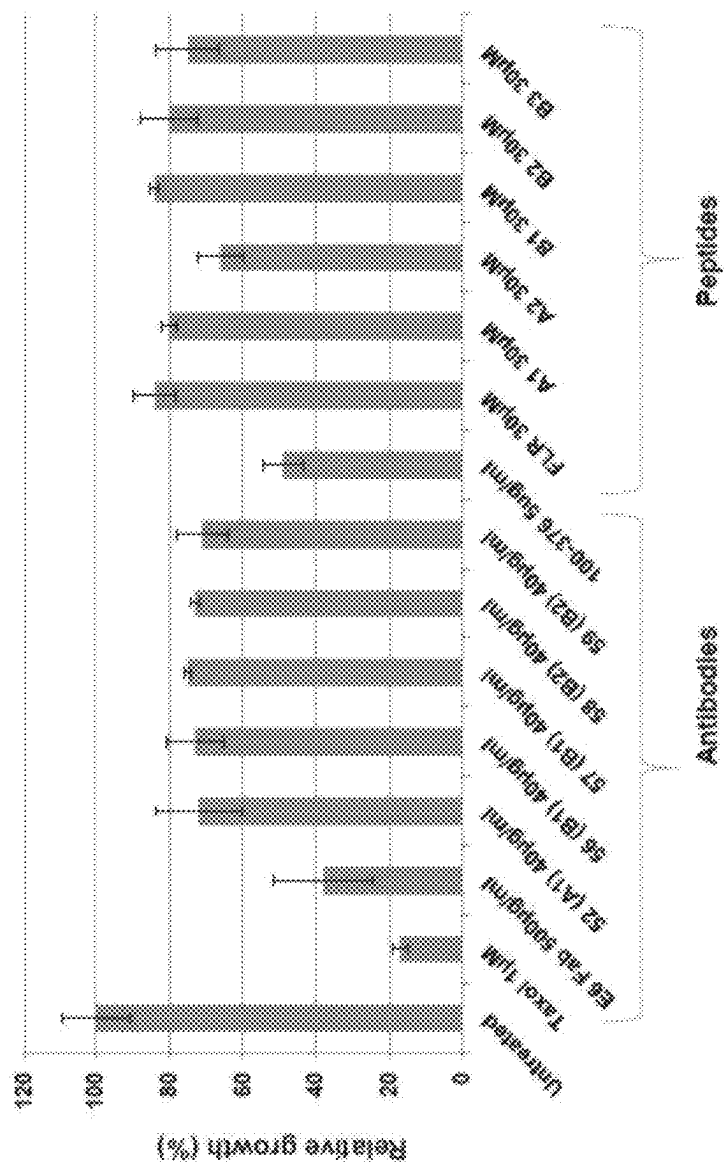
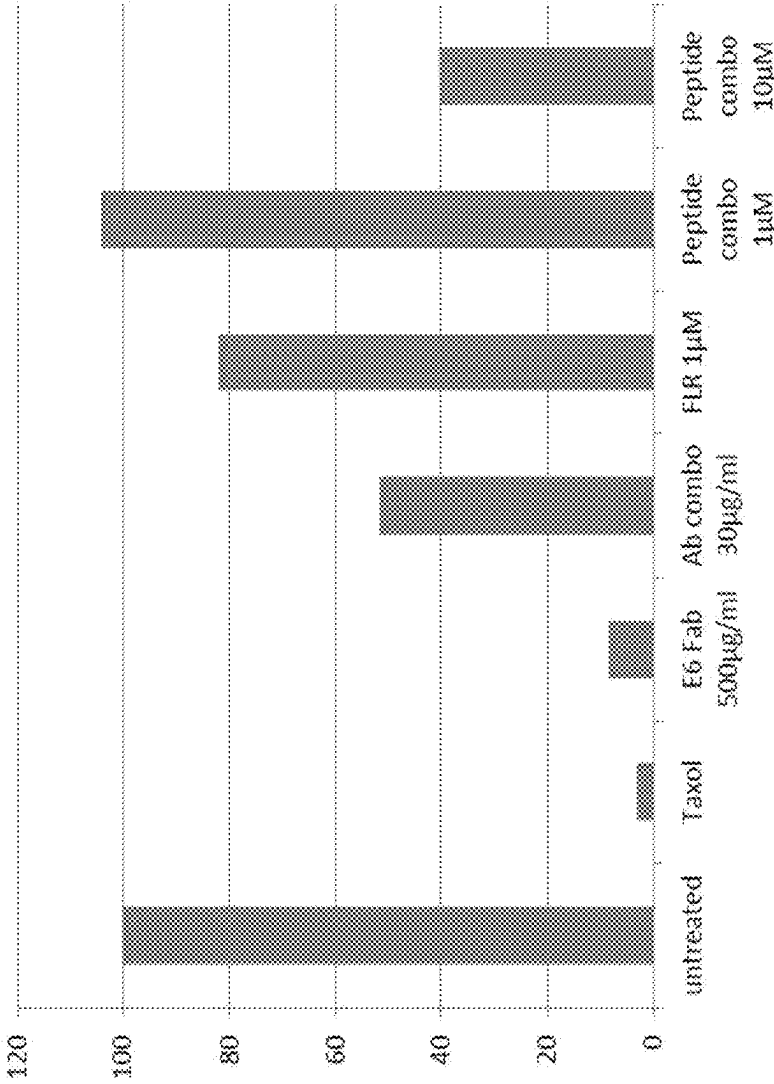


Figure 12

Antibodies generated from NME7-specific peptides inhibit cancer cell growth; peptides themselves also inhibit cancer cell growth



T47D cancer cells. antibody combo 52 (A1), 55 (A2), 56 (B1), 57 (B1), 58 (B2), 59 (B2) at 5ug/ml each 30 ug total. All five peptides were combined and added at 1uM each or 10uM each

Figure 13

Treating cancer cells with anti-NME7 antibodies inhibits transition to “floater” cells, which PCR shows have greatly increased expression of metastatic markers such as CXCR4; xenograft experiments show that the floater cells form a tumor at extremely low copy number – 50 – and thus fulfill the requirement for being classified cancer stem cells or metastatic cancer cells.

Fig. 14A

Antibodies	Floater observation
Control IgG	100%
53,55,57 (A1,A2,B1)	70%
53,57 (A1,B1)	50%
61 (B3)	5%

JR observations

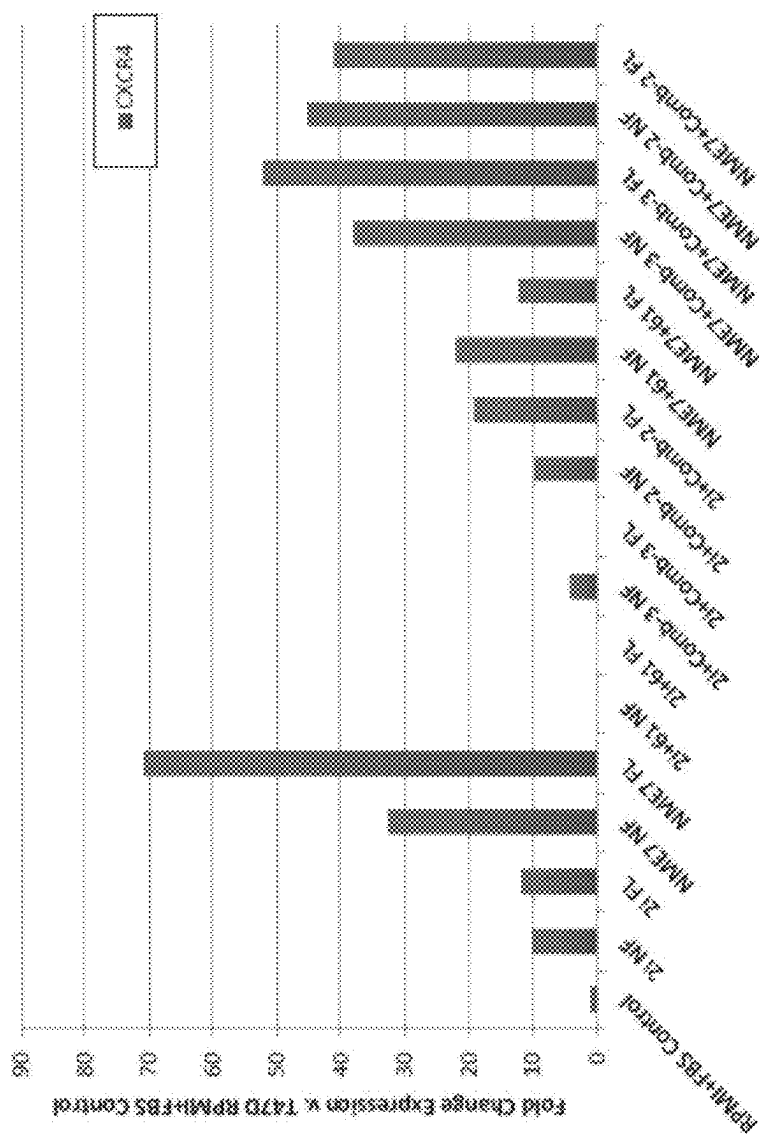
The number of “Floater” cells, which are the ones that have higher expression of metastatic markers and that form tumors in animals at extremely low copy number is typically 20% of the amount of plated cells by Day 7. Here, we define 100% as the number of floater cells that results when a control antibody is added. Other percentages of floater cells is relative to the control in which a control IgG antibody was added.

Antibodies	Floater observation
Control IgG	100%
53,55,57 (A1,A2,B1)	65%
53,57 (A1,B1)	40%
61 (B3)	5%

VH observations

Figures 14A-14B

Treating cancer cells with anti-NME7 antibodies inhibits transition to metastatic cancer cells; CXCR4 is metastasis marker for cancers



T47D breast cancer cells. 2i are MEK and GSK3-beta inhibitors. NF=non-floater; FL=floater. 61= rabbit #61 immunized with B3 peptide derived from NME7. Comb-2 = combination of antibodies from rabbits immunized with A1 and A2. Comb-3 = combination of antibodies from rabbits immunized with peptides A1, A2 and B1.

Figure 15A

Treating cancer cells with NME7-AB peptides inhibits transition to metastatic cancer cells; CXCR4 and SOX2 are metastasis marker for cancers

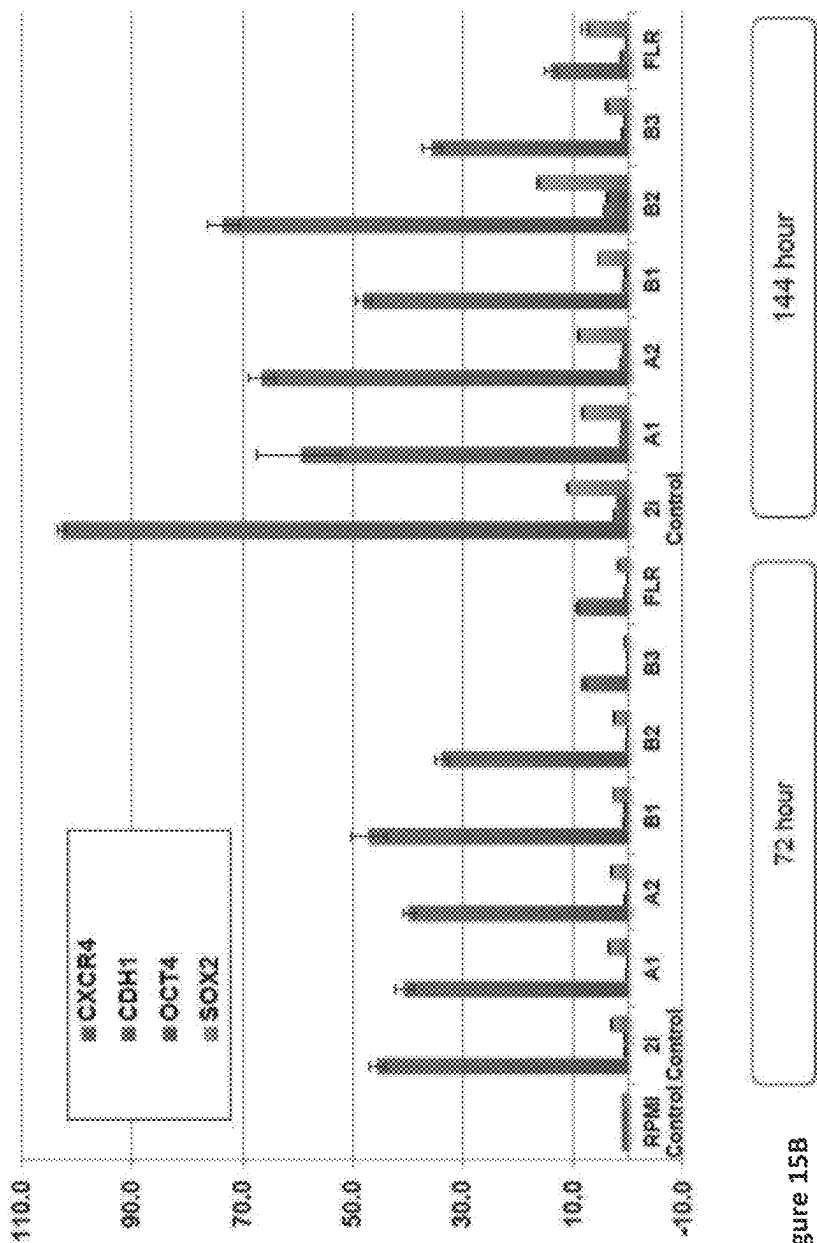


Figure 15B

Treating cancer cells with NME7-AB peptides inhibits transition to metastatic cancer cells; CXCR4 and SOX2 are metastasis marker for cancers

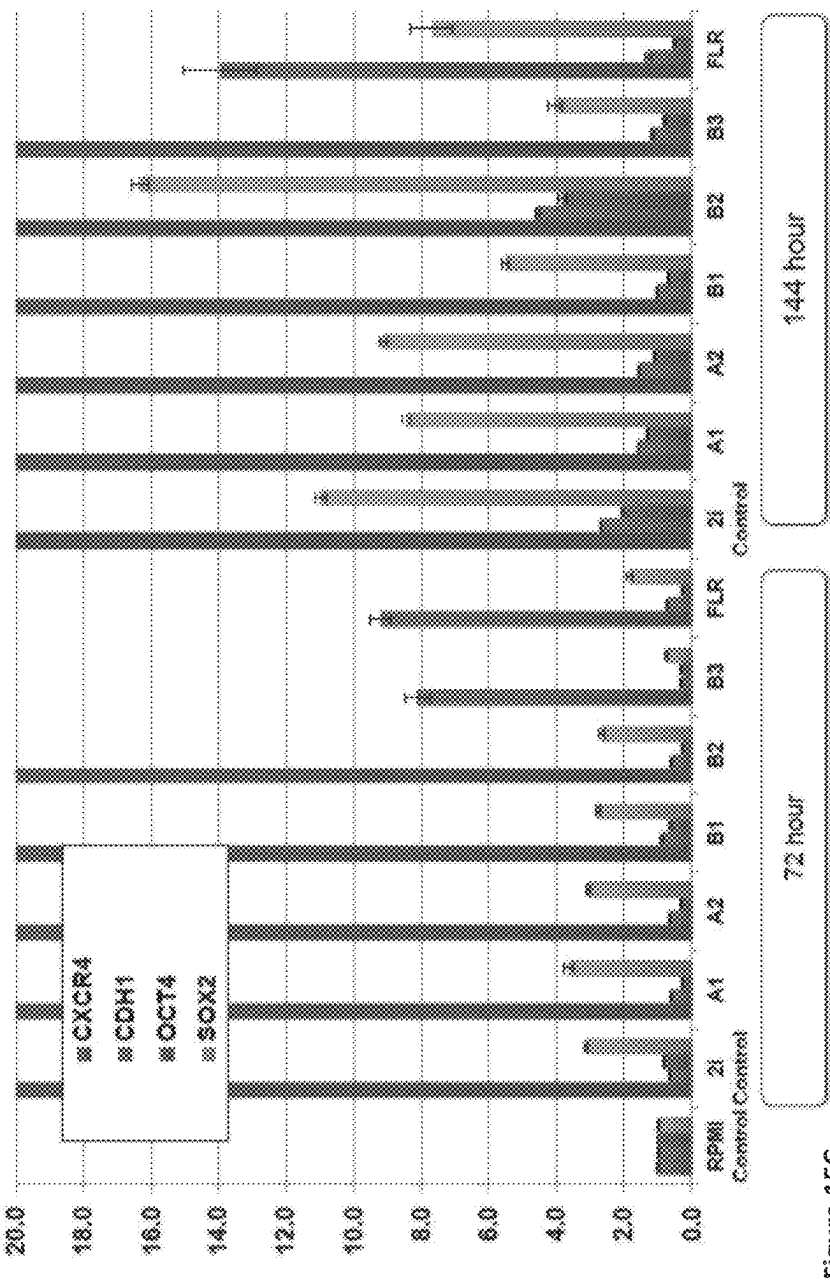


Figure 15C

Cell Line T47D	Medium	Total RNA ng/ μ L	Yield (μ g)	EEF1A1 Threshold Cycle
Control	RPML + 10% FBS	395.8	47.49	15.2
2i attached	MM + 2i	387.8	11.64	14.5
2i Floaters	MM + 2i	234.8	7.04	14.9
NME7 attached	MM+NME7 4nM	334.8	10.04	14.7
NME7 Floaters	MM+NME7 4nM	259.3	7.78	16.0
2i+ Antibody B3 rabbit 61 attached	MM + 2i	2.7	0.08	35.9
2i+ Antibody B3 rabbit 61 Floaters	MM + 2i	3.6	0.11	25.0
2i+ Antibody Combination 3 attached	MM + 2i	44.7	1.34	17.0
2i+ Antibody Combination 3 Floaters	MM + 2i	39.0	1.17	15.9
2i+ Antibody Combination 2 attached	MM + 2i	46.0	1.38	15.6
2i+ Antibody Combination 2 Floaters	MM + 2i	77.8	2.33	16.3
NME7+ Antibody B3 rabbit 61 attached	MM+NME7 4nM	65.7	1.97	17.0
NME7+ Antibody B3 rabbit 61 Floaters	MM+NME7 4nM	15.8	0.47	19.9
NME7+ Antibody Combination 3 attached	MM+NME7 4nM	32.1	0.96	17.0
NME7+ Antibody Combination 3 Floaters	MM+NME7 4nM	109.3	3.28	16.1
NME7+ Antibody Combination 2 attached	MM+NME7 4nM	134.5	4.03	16.1
NME7+ Antibody Combination 2 Floaters	MM+NME7 4nM	139.5	4.19	18.6

Figure 16

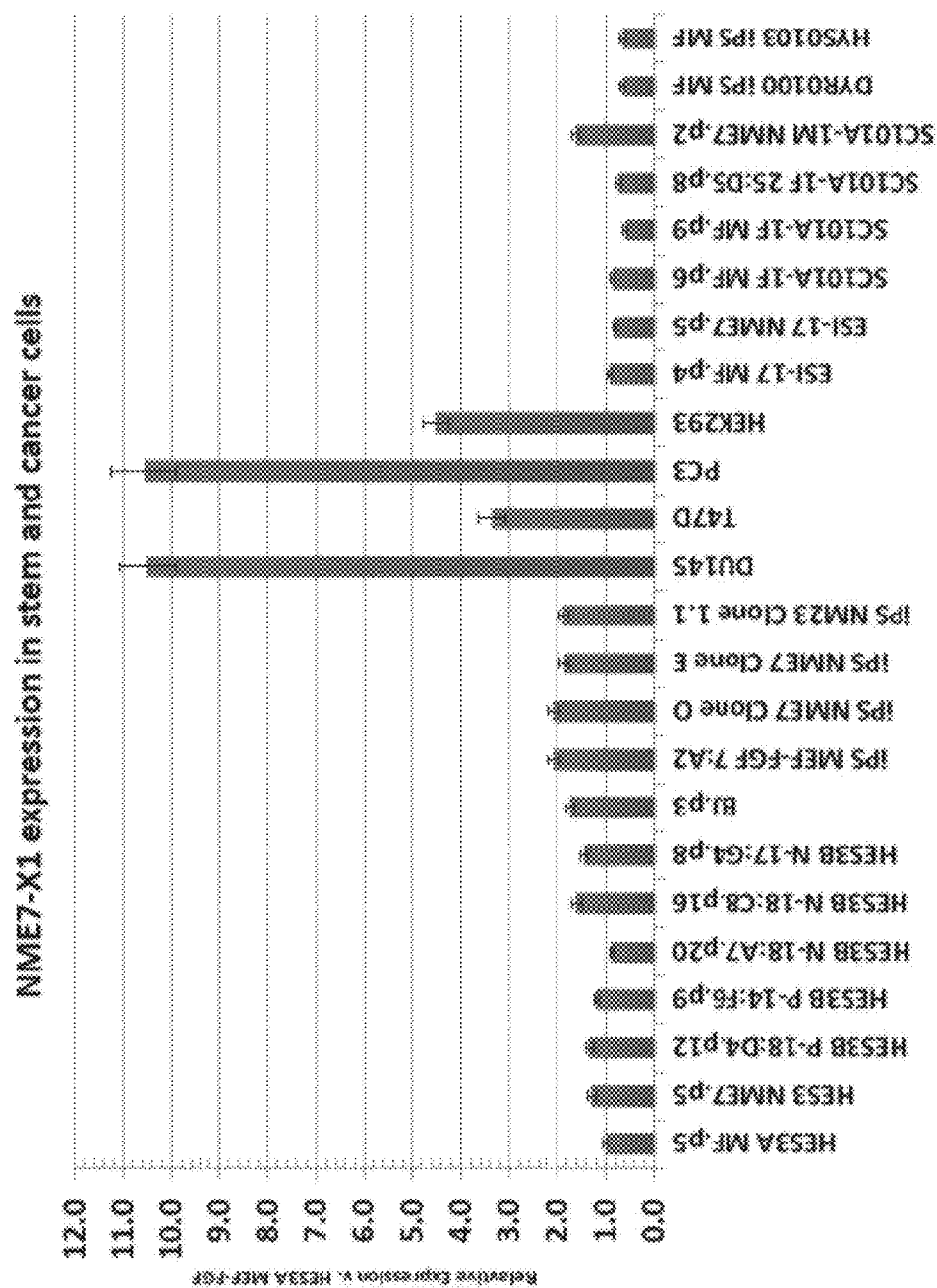


Figure 17

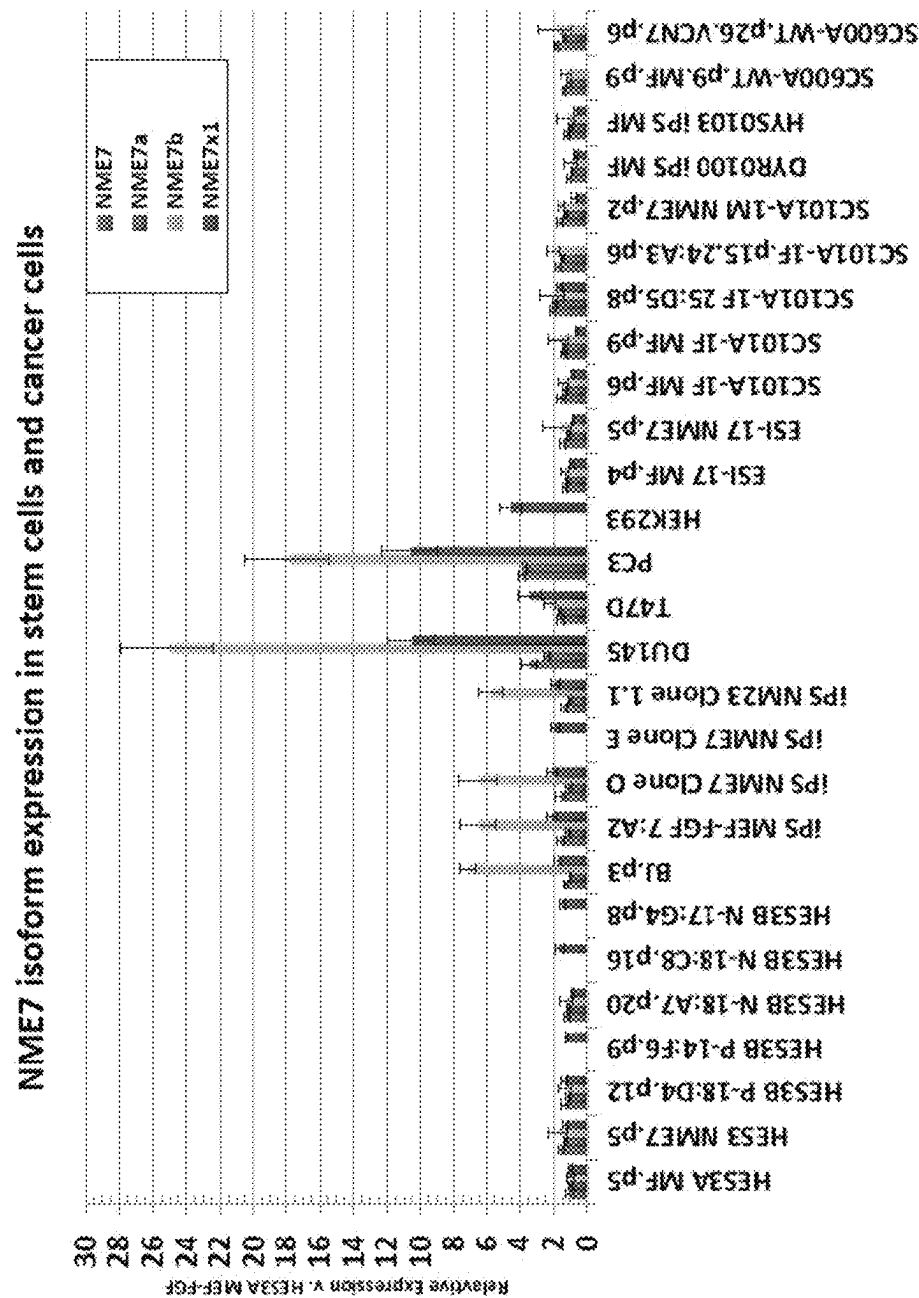
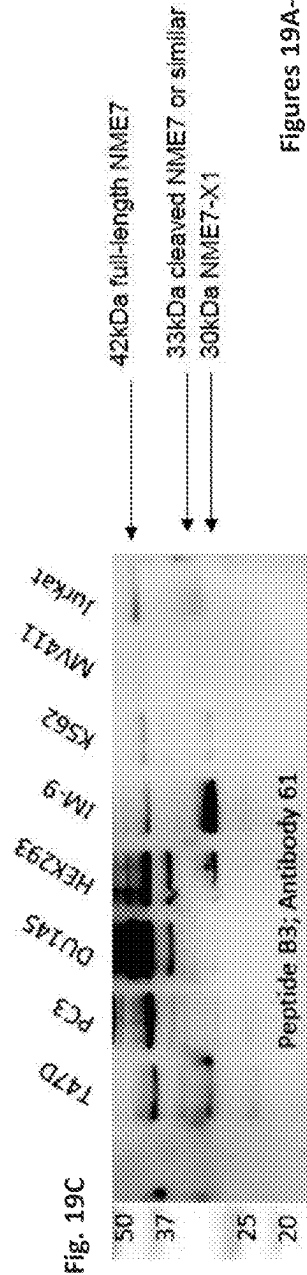
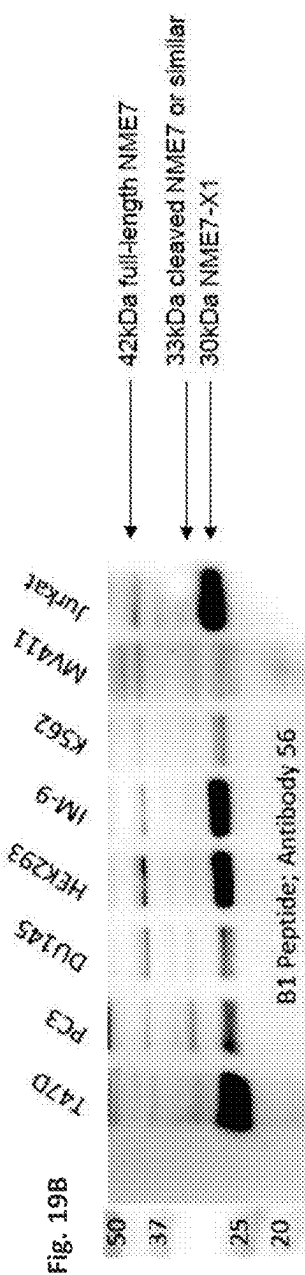
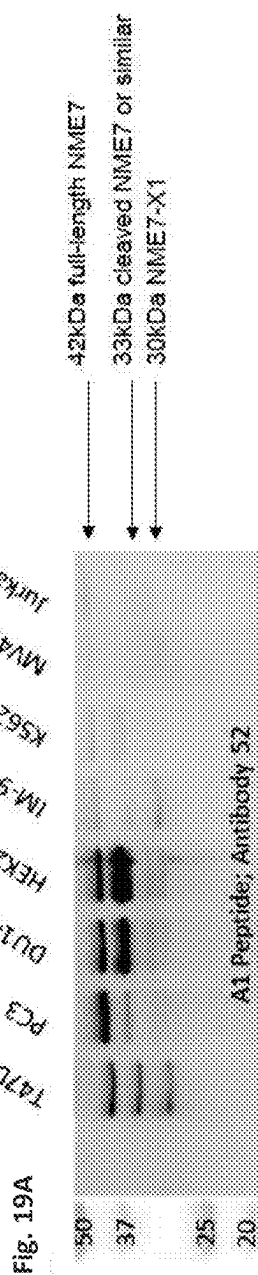


Figure 18



Figures 19A-19C

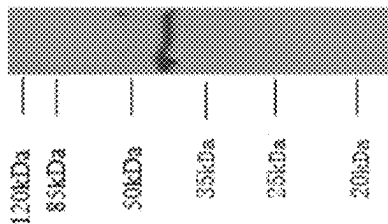


Fig. 19E

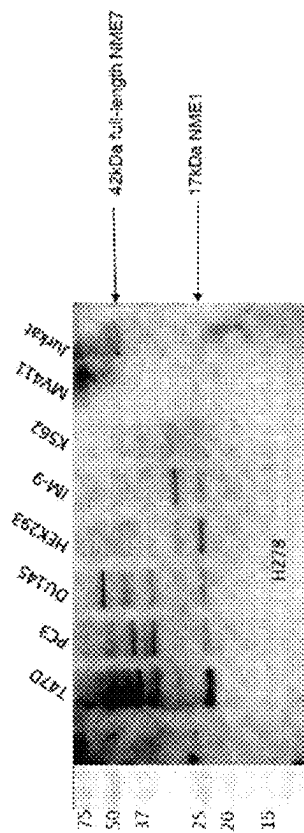


Fig. 19D

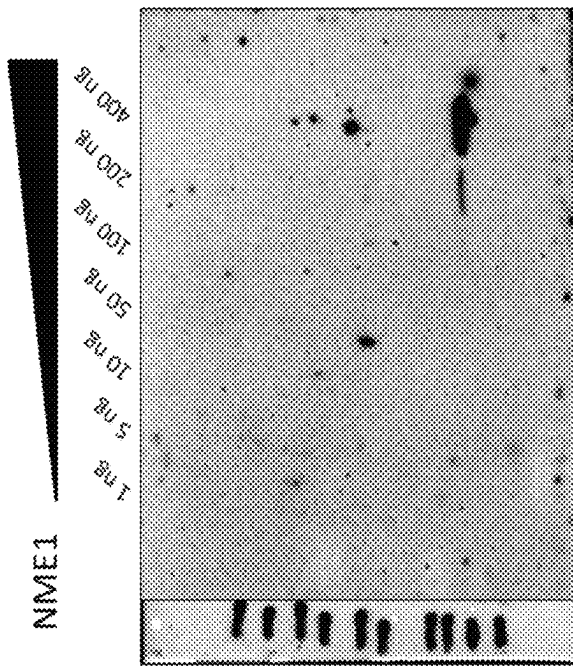


Fig. 19F

Figures 19D-19F

Fig. 20A

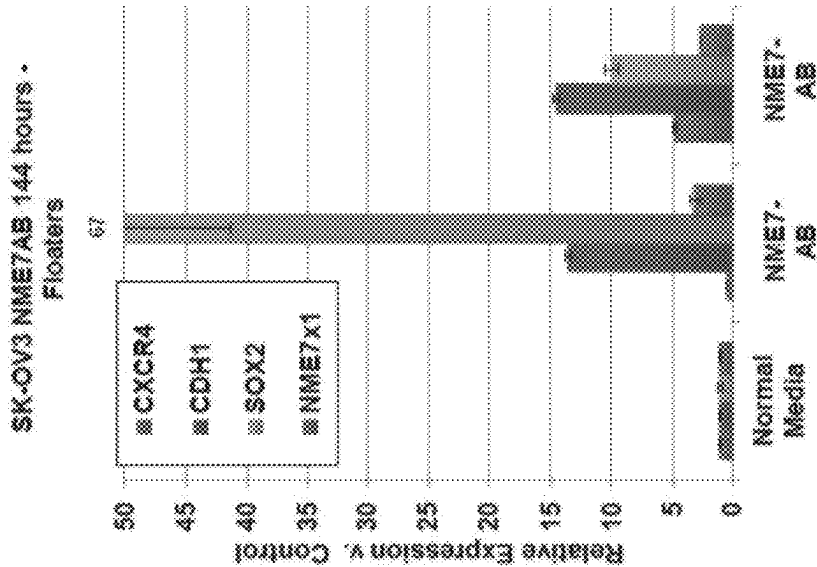
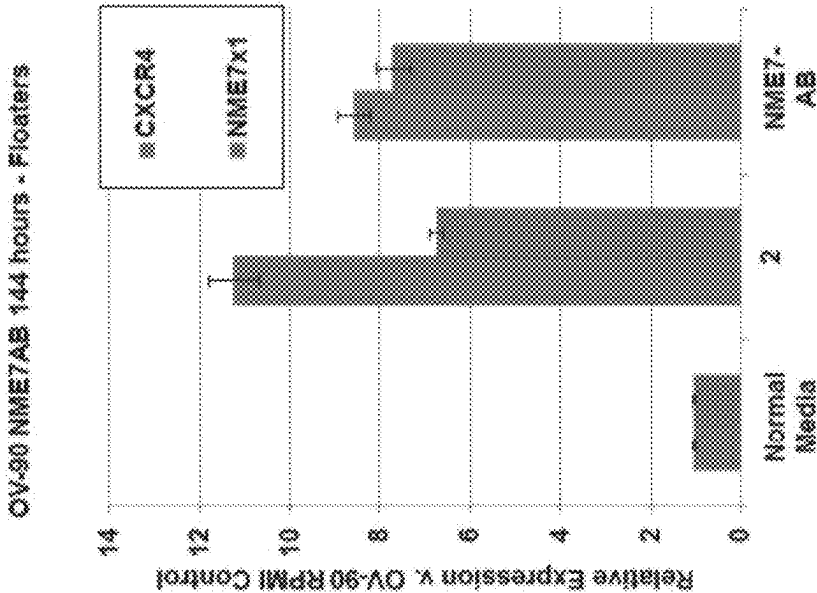


Fig. 20B



Figures 20A-20B

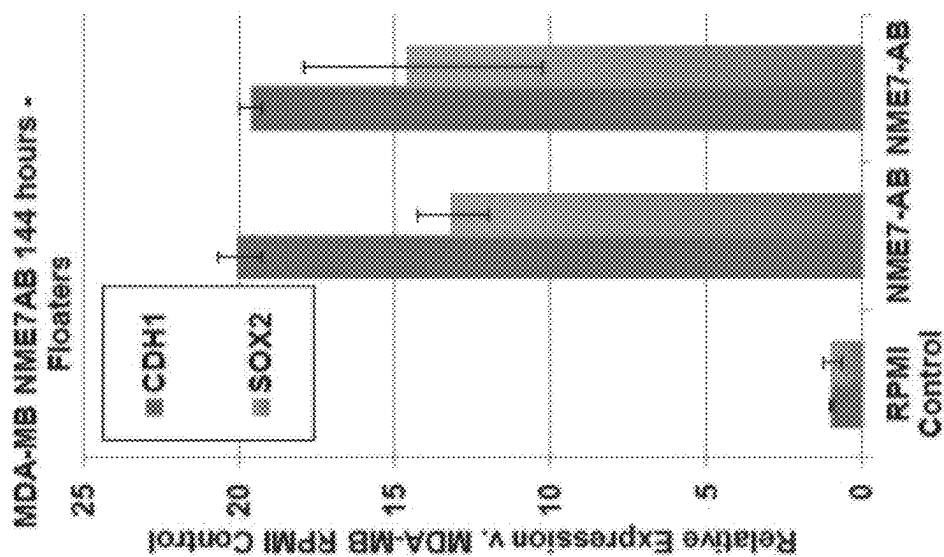


Figure 20C

Various cancer cell lines express both full-length MUC1 and MUC1*

Fig. 21A

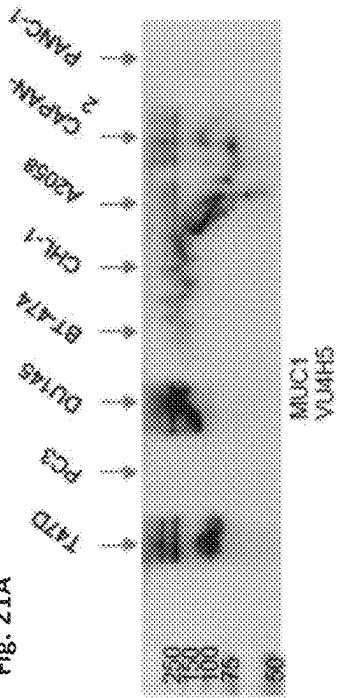
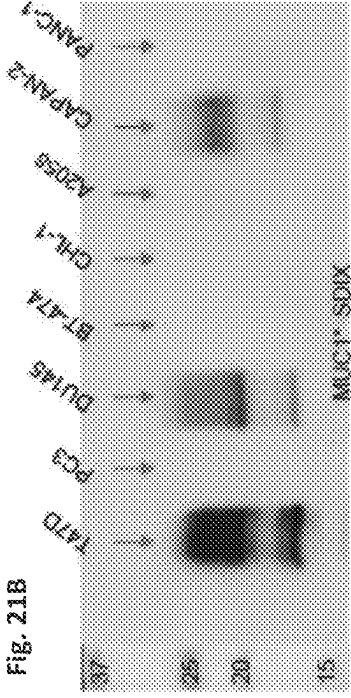


Fig. 21C

T47D – MUC1 positive breast cancer
PC3 – MUC1 negative prostate cancer
DU145 – MUC1 positive prostate cancer
BT-474 – HER2 positive breast cancer
CHL-1 – melanoma
A2058 – melanoma
CAPAN-2 – MUC1 positive pancreatic cancer
PANC-1 – MUC1 negative pancreatic cancer

Fig. 21B



Figures 21A-21C

Fig. 21D

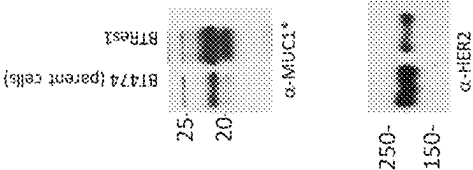


Fig. 21E

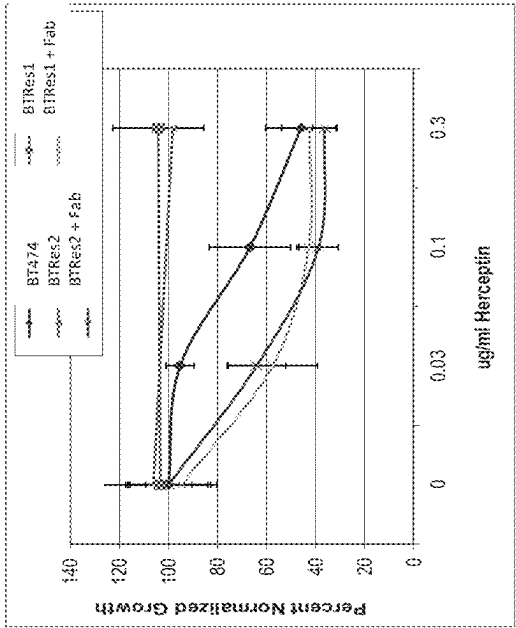
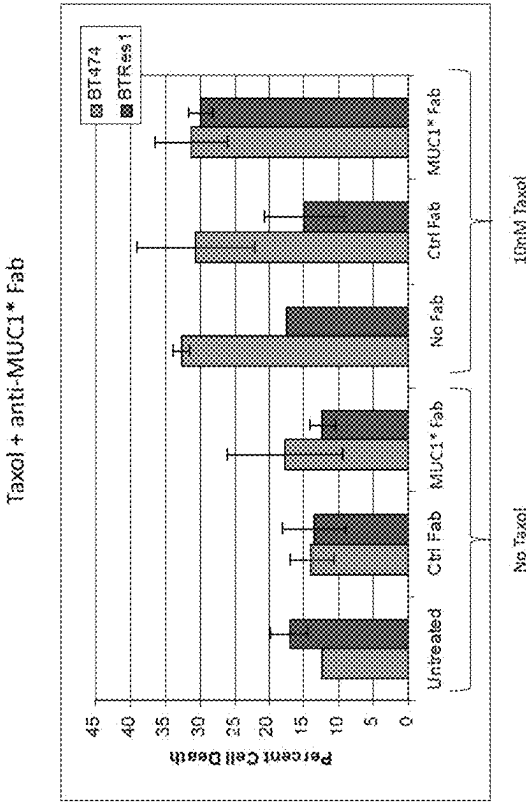
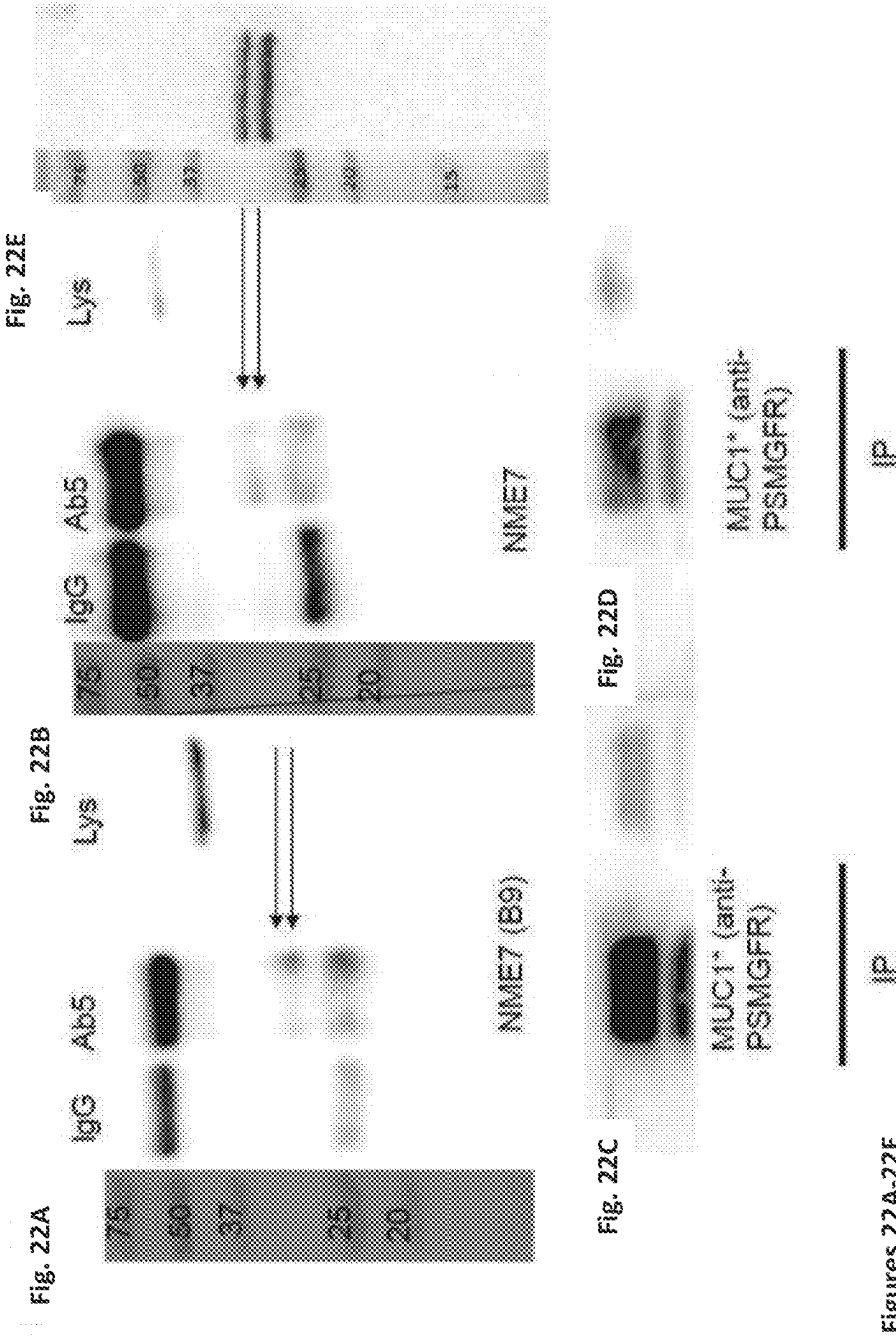


Fig. 21F

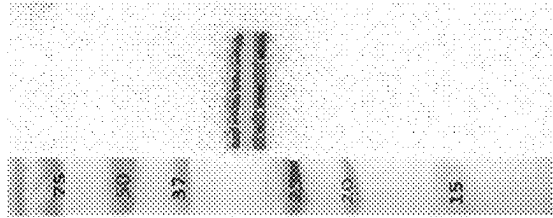
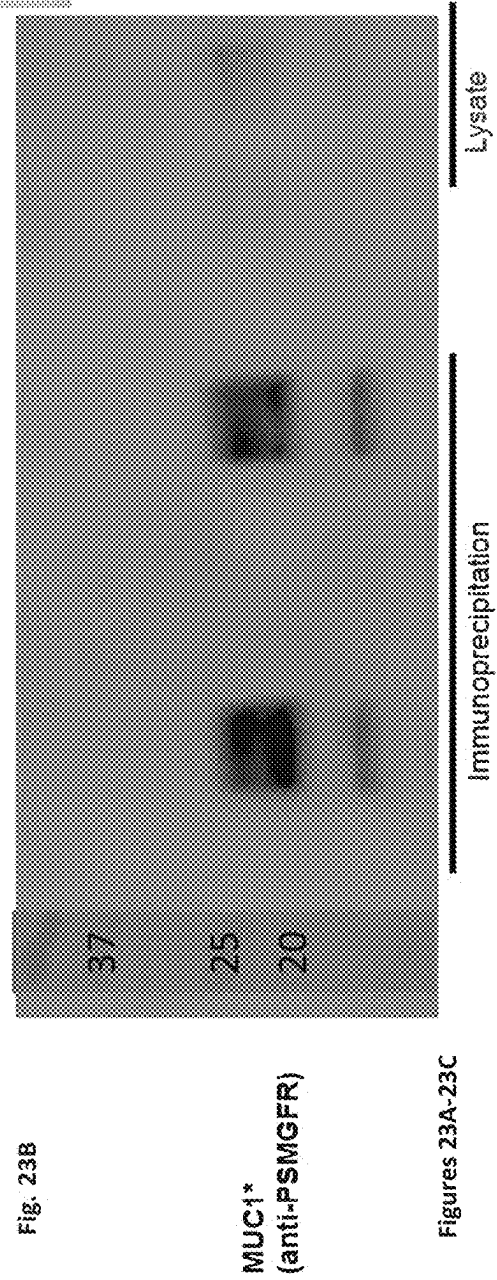
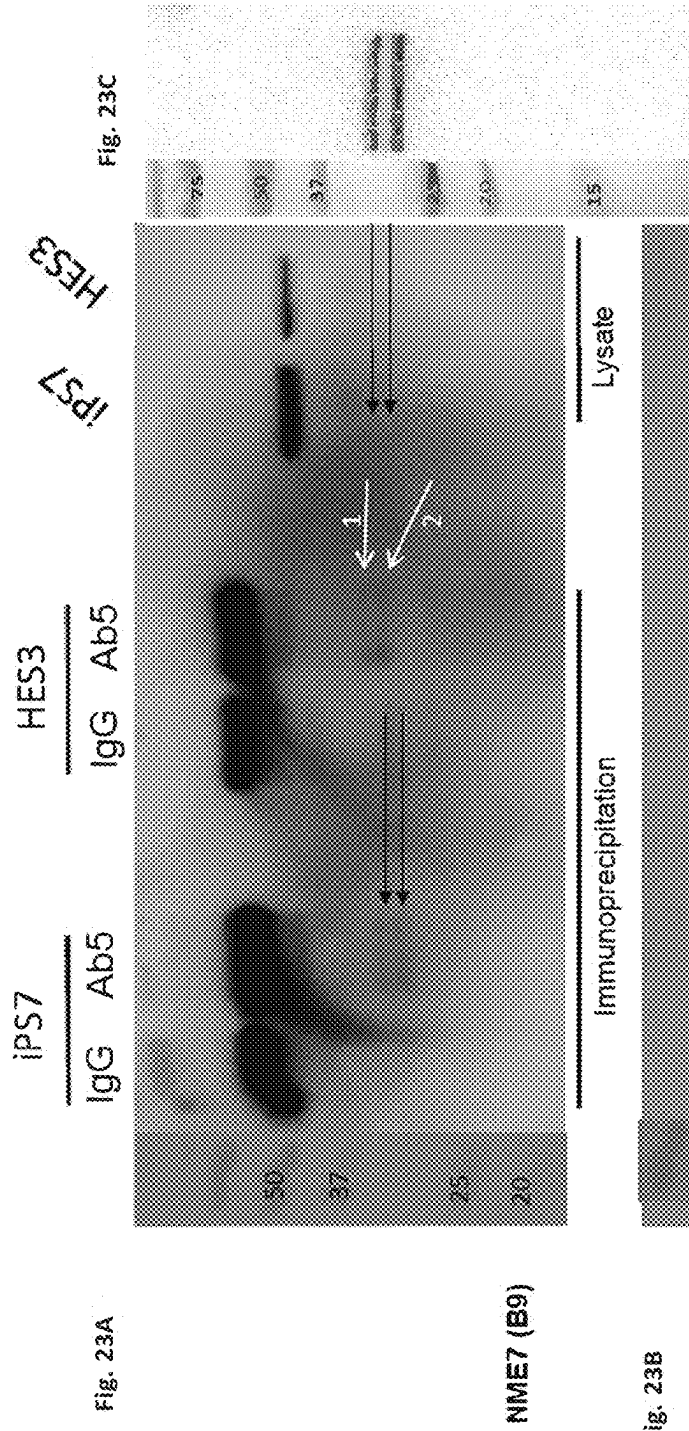


Figures 21D-21F

Co-Immunoprecipitation of MUC1* and NME7 in serum grown T47D breast cancer cells



Co-immunoprecipitation of MUC1* and NME7 from human embryonic stem cells & IPS cells



ELISA: Binding of anti-NME7 B3 antibodies to NME7-AB

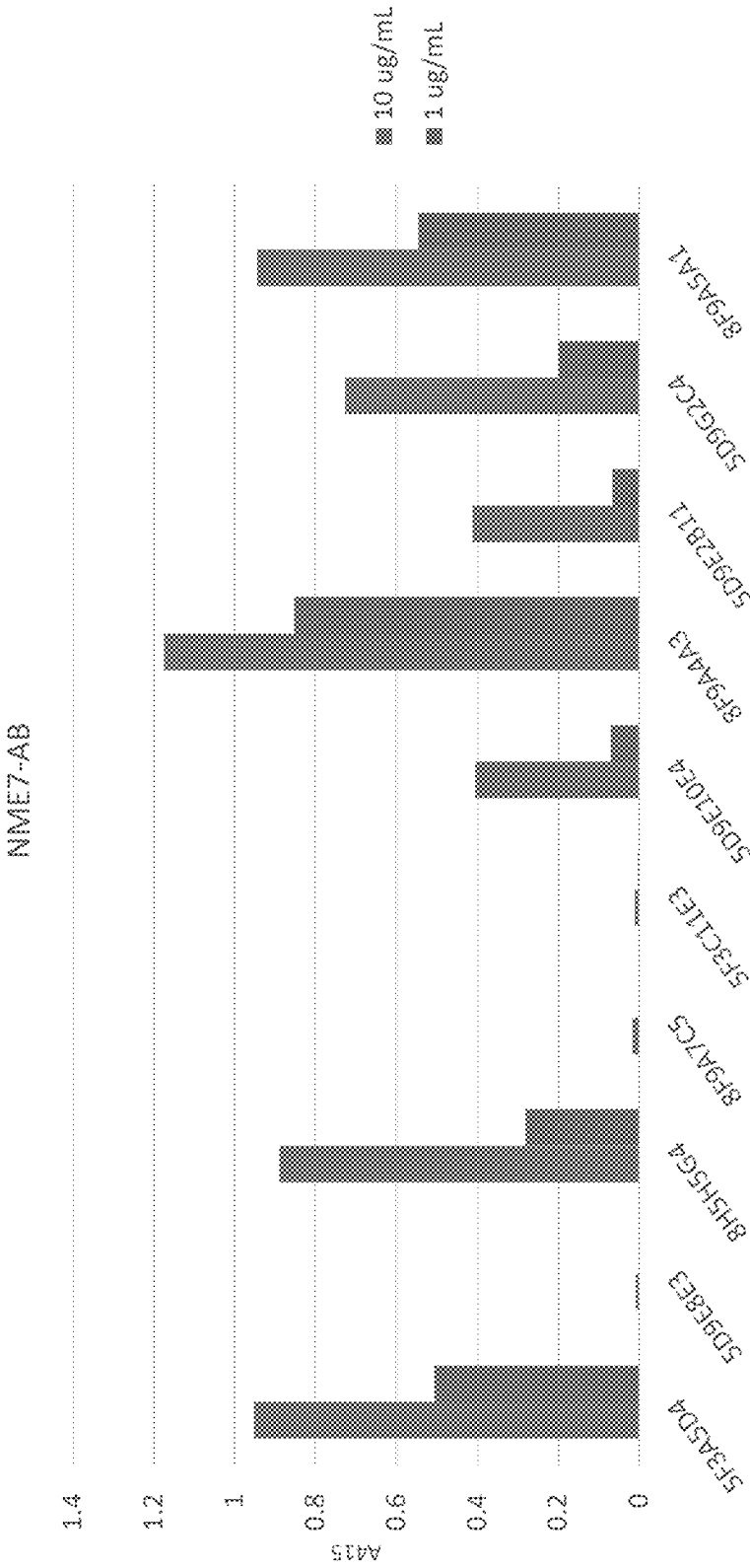


Figure 24

ELISA: Binding of anti-NME7 B3 antibodies to NME7-AB

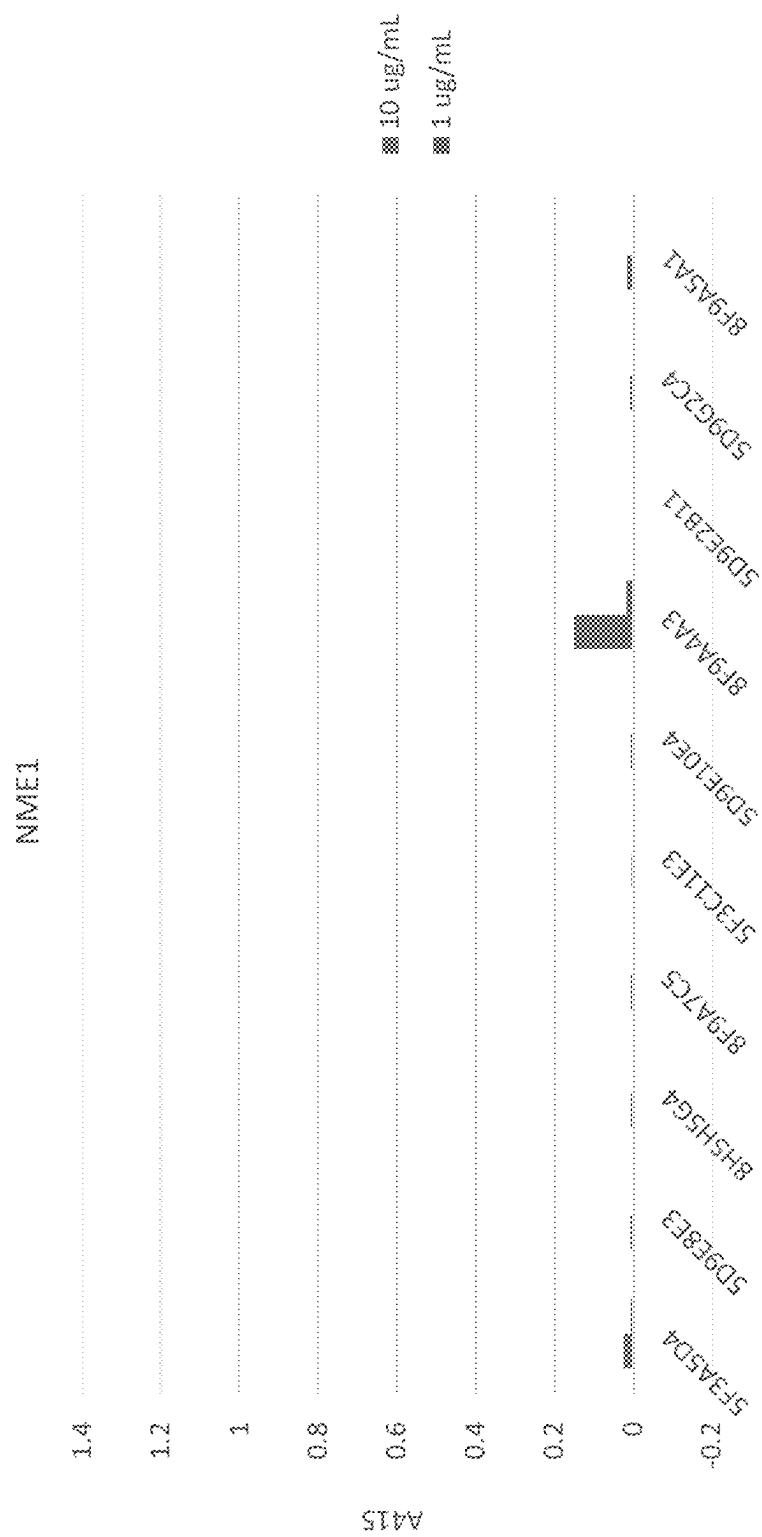


Figure 25

Displacement assay: Anti-NME7 B3 antibodies disrupt binding of NME7-AB to the MUC1* extra cellular domain peptide.
The NME7-AB/anti-NME7 antibody complex was made before adding to plate coated with MUC1* extra cellular domain peptide.

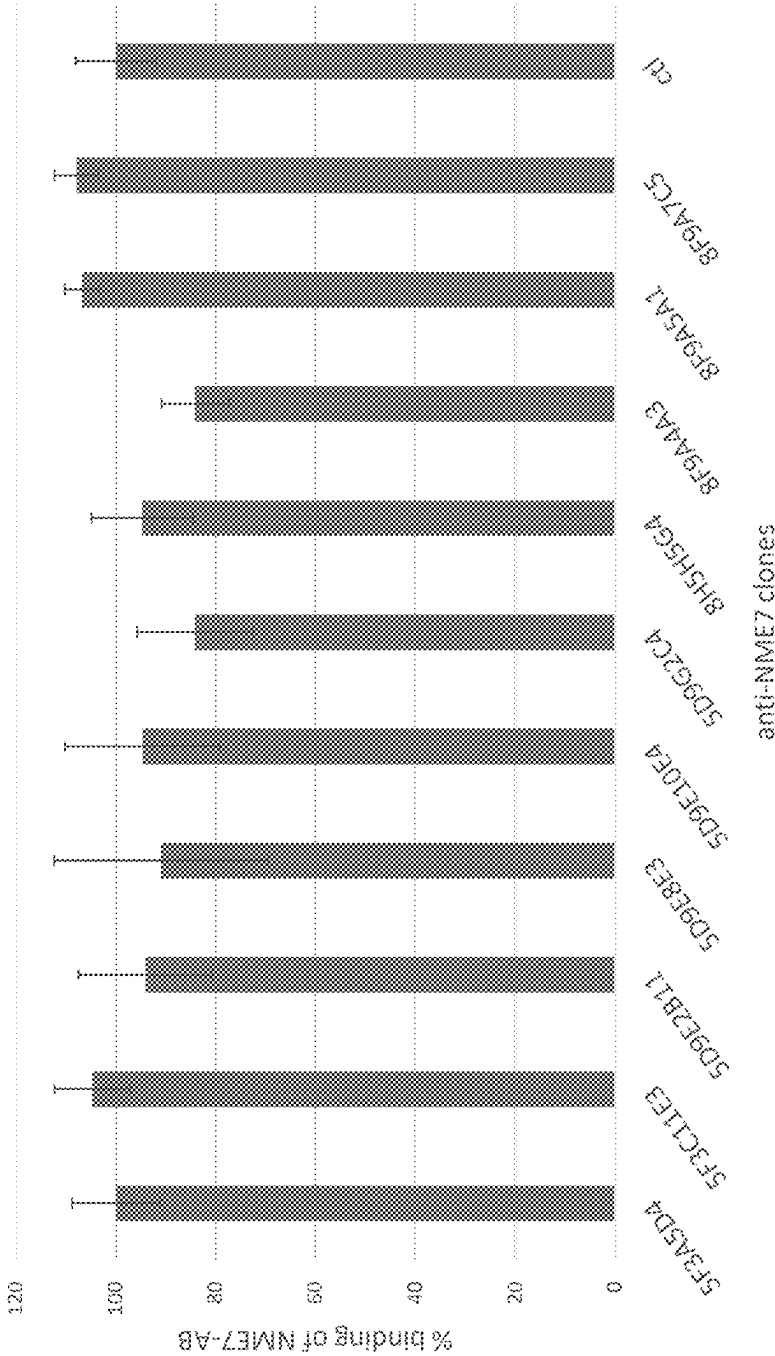


Figure 26

Displacement assay: Anti-NME7 B3 antibodies disrupt binding of NME7-AB to the MUC1* extra cellular domain peptide.
The NME7-AB was first bound to MUC1* extra cellular domain peptide on the plate, then disrupted by the addition of anti-NME7 antibodies.

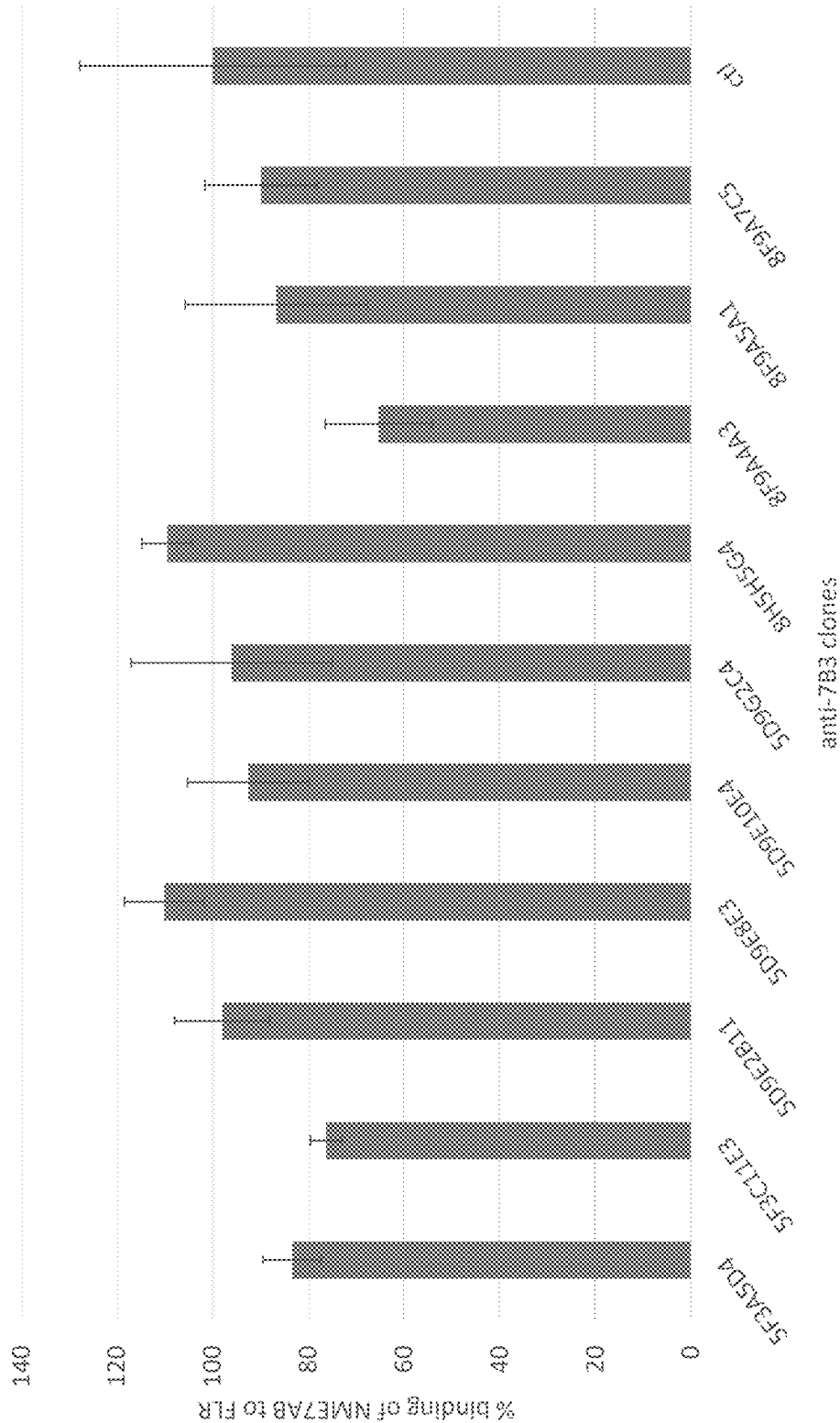


Figure 27

Displacement assay: Anti-NME7 B3 antibodies disrupt binding of NME7-AB to the MUC1* N-10 extra cellular domain peptide. The NME7-AB was first bound to MUC1* N-10 peptide on the plate, then disrupted by the addition of anti-NME7 antibodies.

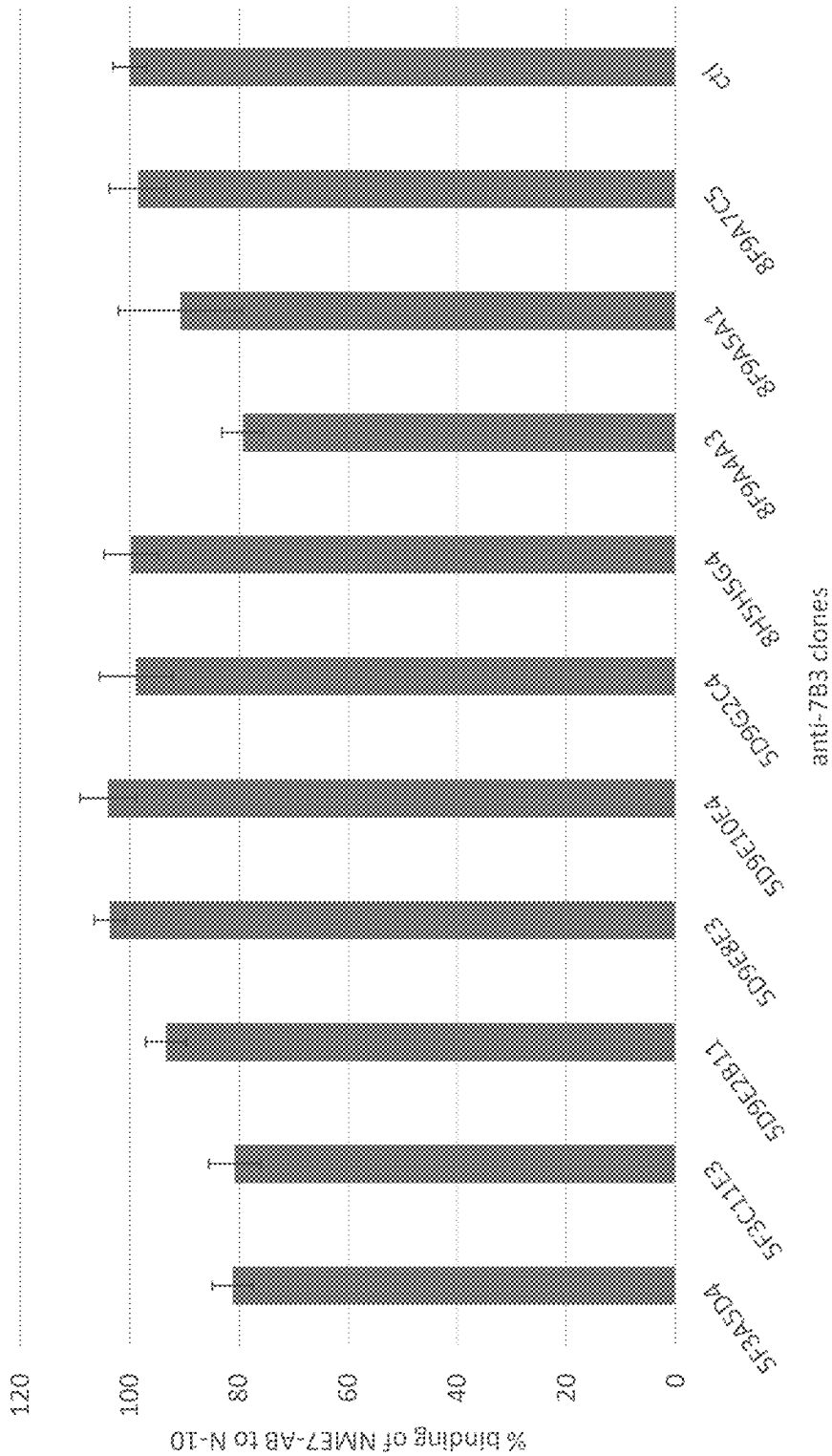


Figure 28

RNA yield, which is an indicator of cell number, from T47D breast cancer cells grown in NME7-AB media in the presence or absence of anti-NME7 B3 antibodies

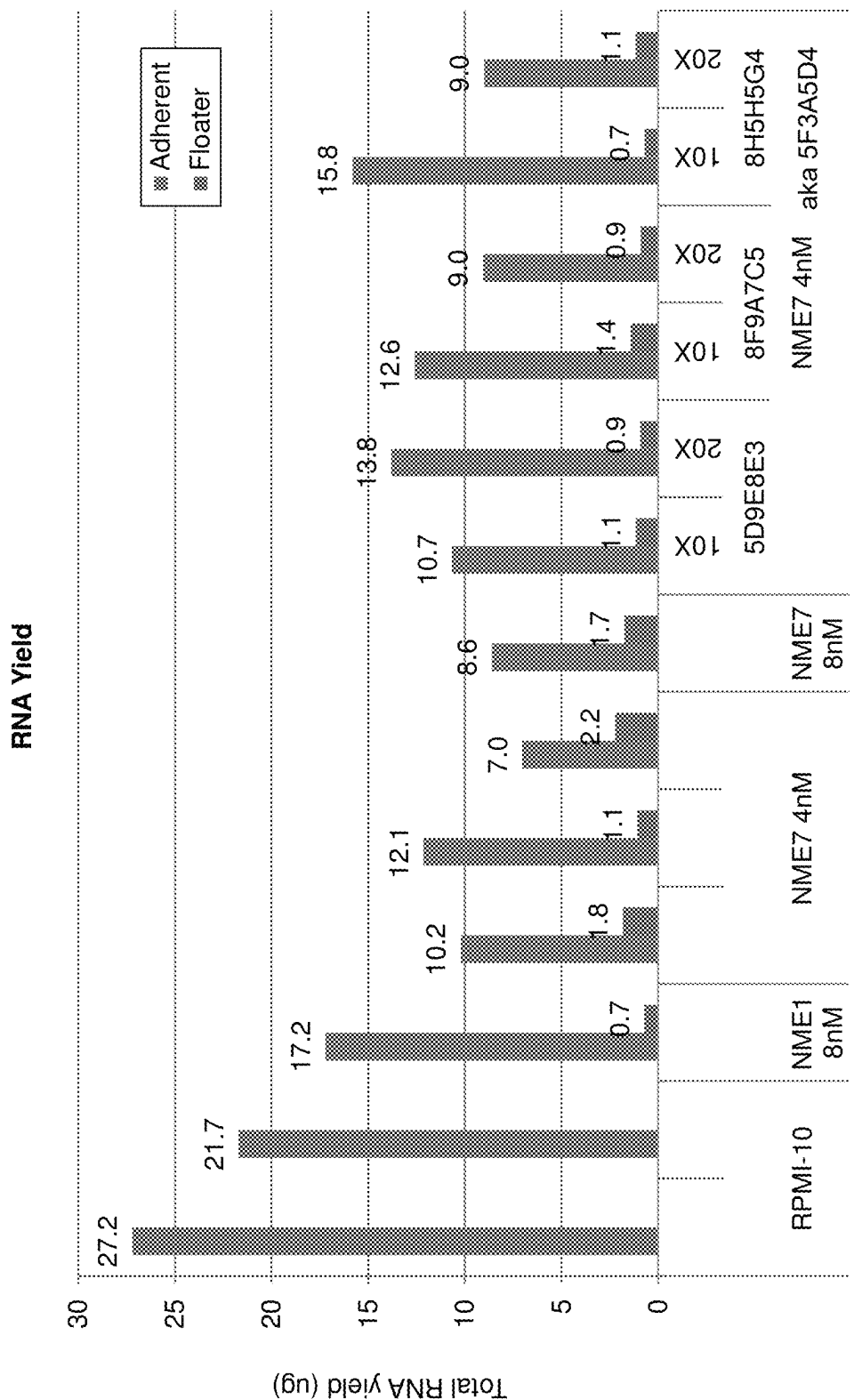


Figure 29

RT-PCR analysis of breast cancer metastasis marker CXCR4 in T47D breast cancer cells grown in NME7-AB media in the presence or absence of anti-NME7 B3 antibodies

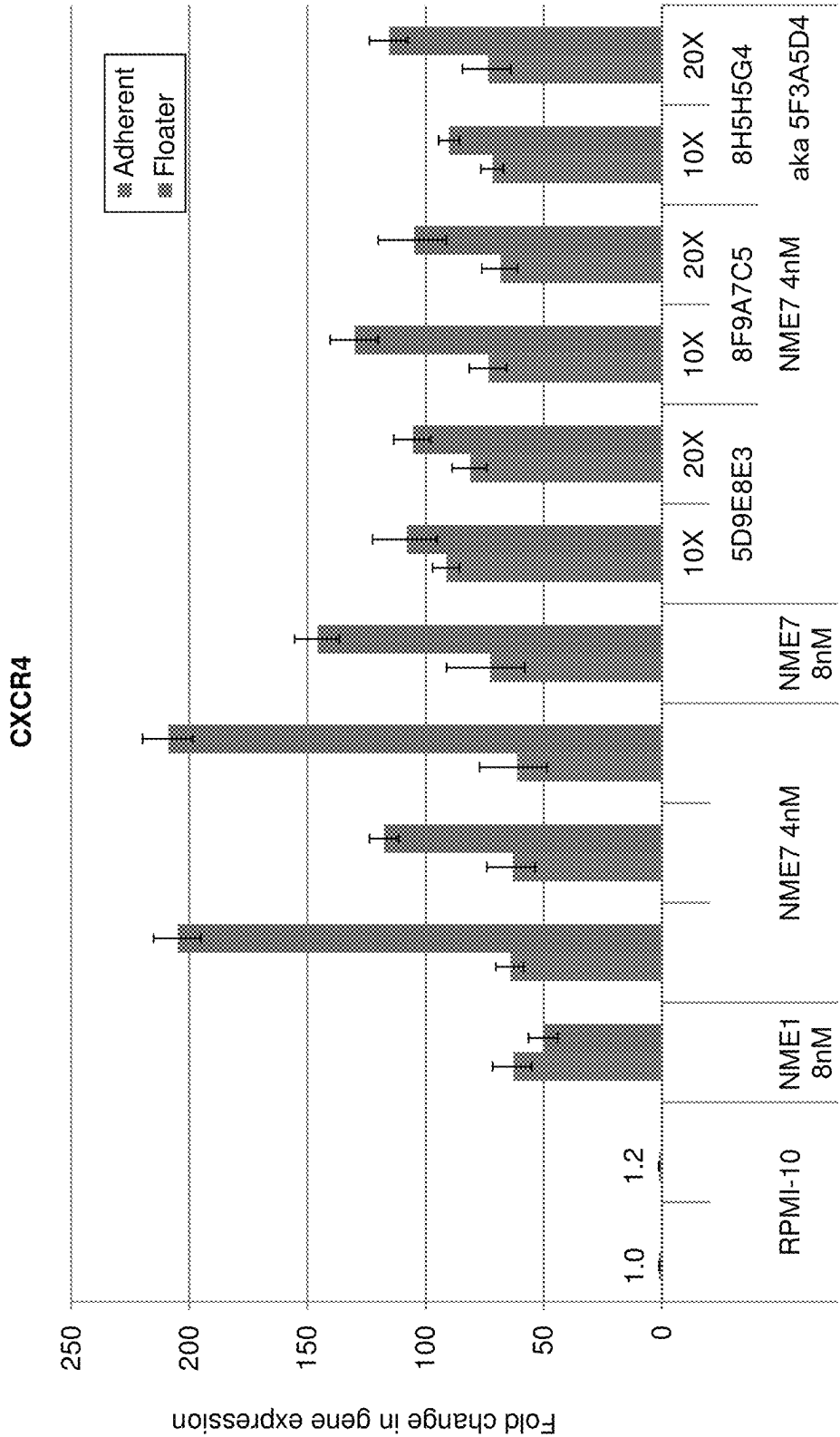


Figure 30

RT-PCR analysis of stem cell and metastasis marker SOX2 in T47D breast cancer cells grown in NME7-AB media in the presence or absence of anti-NME7 B3 antibodies

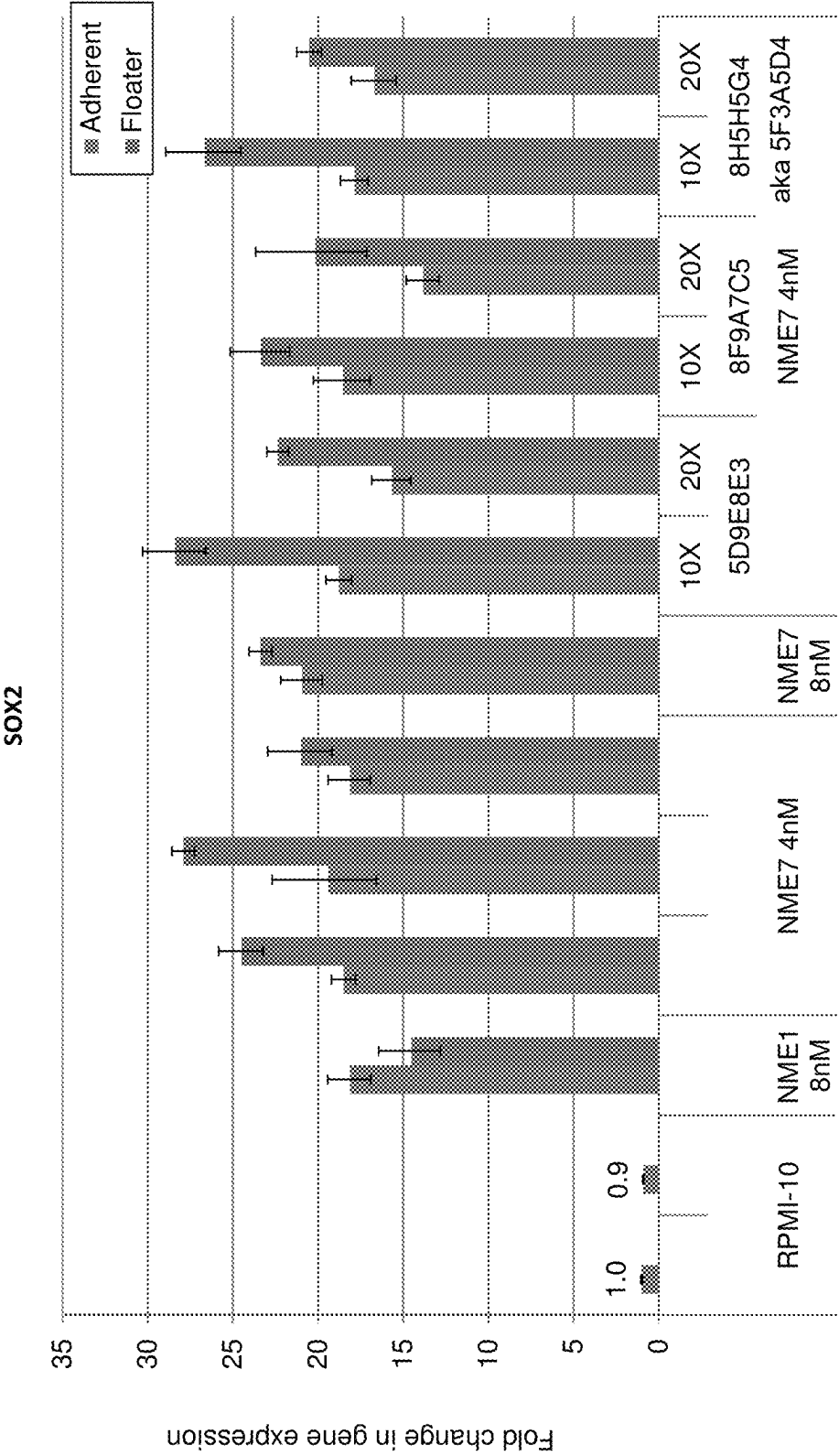


Figure 31

RT-PCR analysis of the NME7-AB cognate receptor MUC1* in T47D breast cancer cells grown in NME7-AB media in the presence or absence of anti-NME7 B3 antibodies

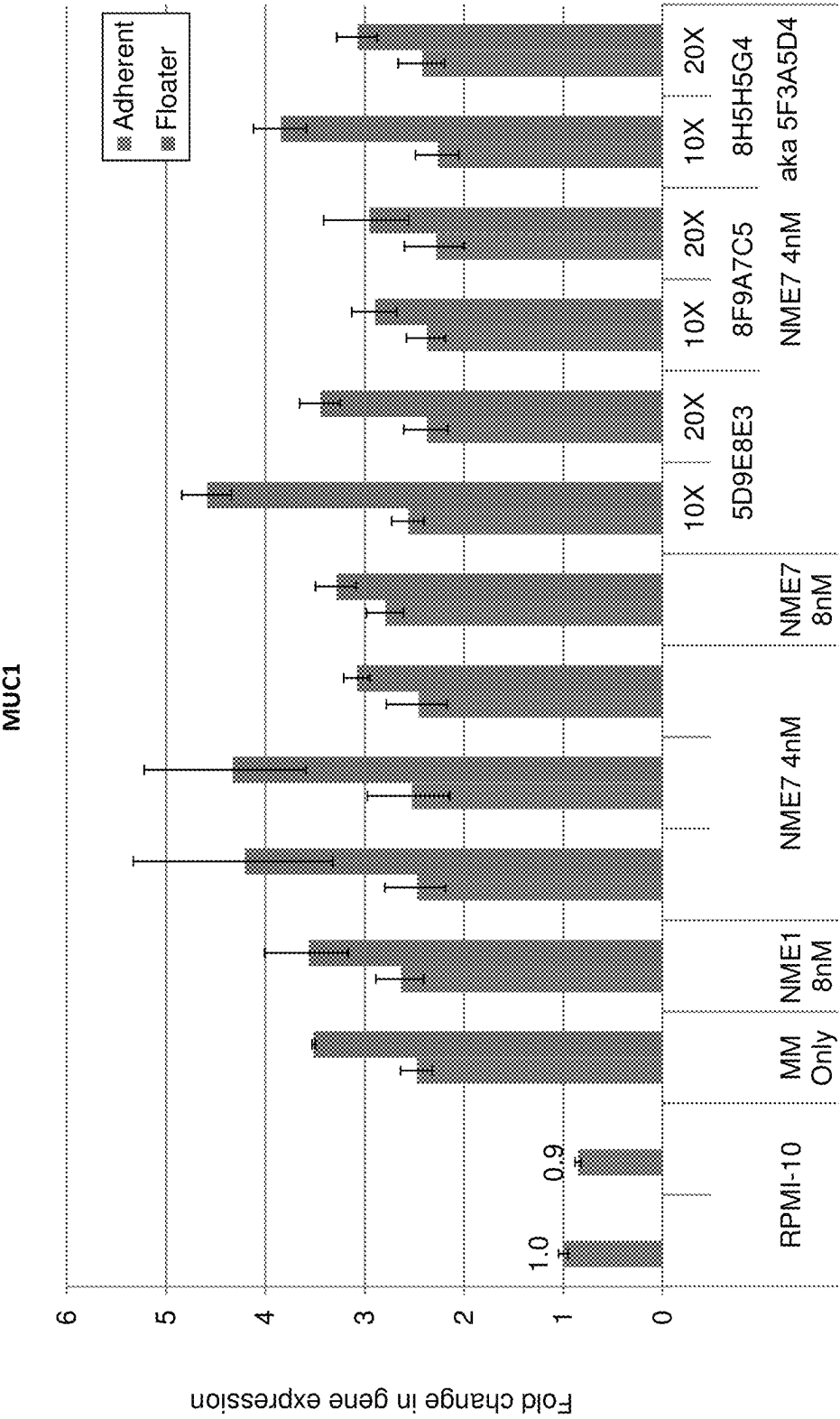


Figure 32

Onco-embryonic growth factor NME7 transforms cancer cells into metastatic cancer stem cells

Day 6

Fig. 33A

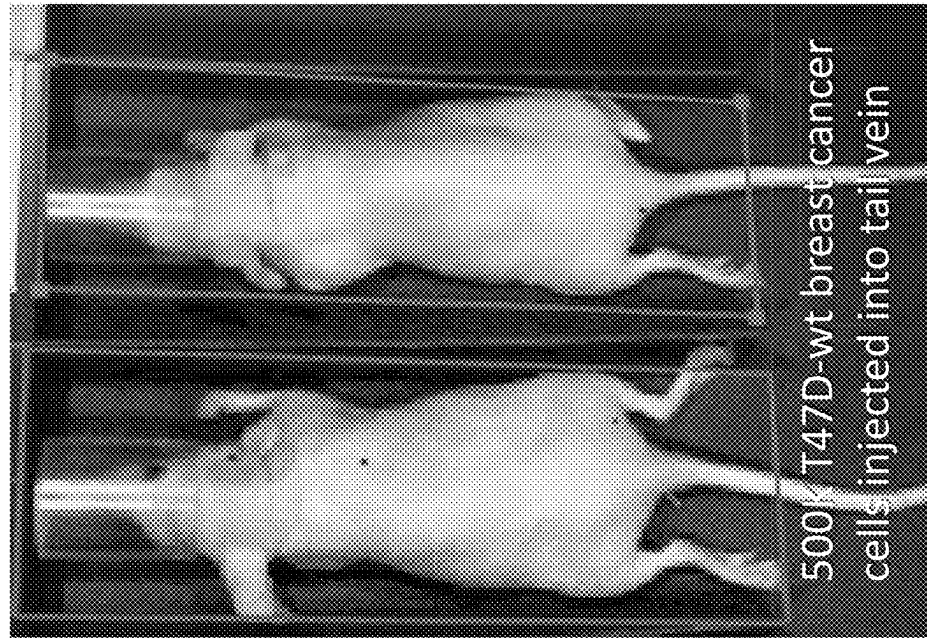
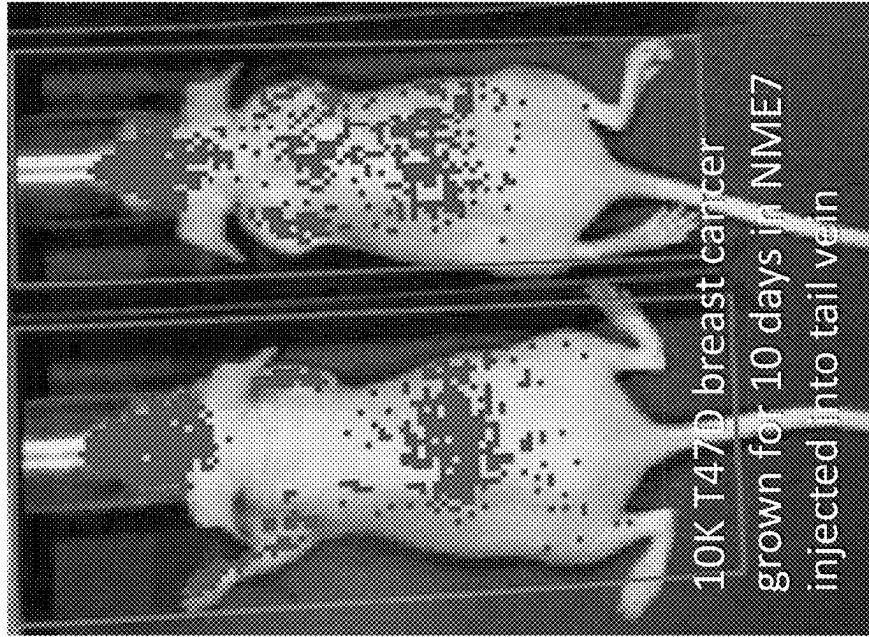


Fig. 33B



Figures 33A-33B

Day 10 T47D-CSC (cancer stem cells) @ 10K vs wt @ 500K +/- anti-NME7 treatment; Day 7 was 1st antibody injection but was done just after injection of NME7

Fig. 34A

T47D-wt
500K i.v.
Day 10
No Antibody

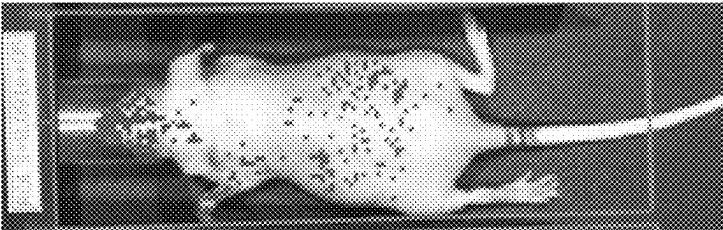


Fig. 34B

T47D-CSC
10K i.v.
Day 10
+ Antibody

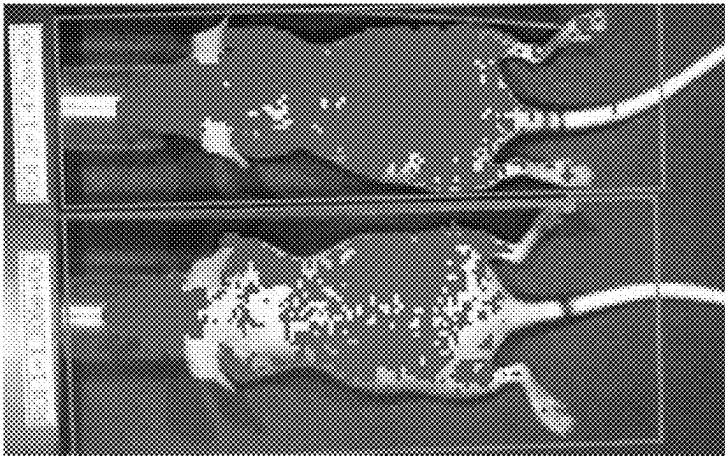


Fig. 34C

Fig. 34D

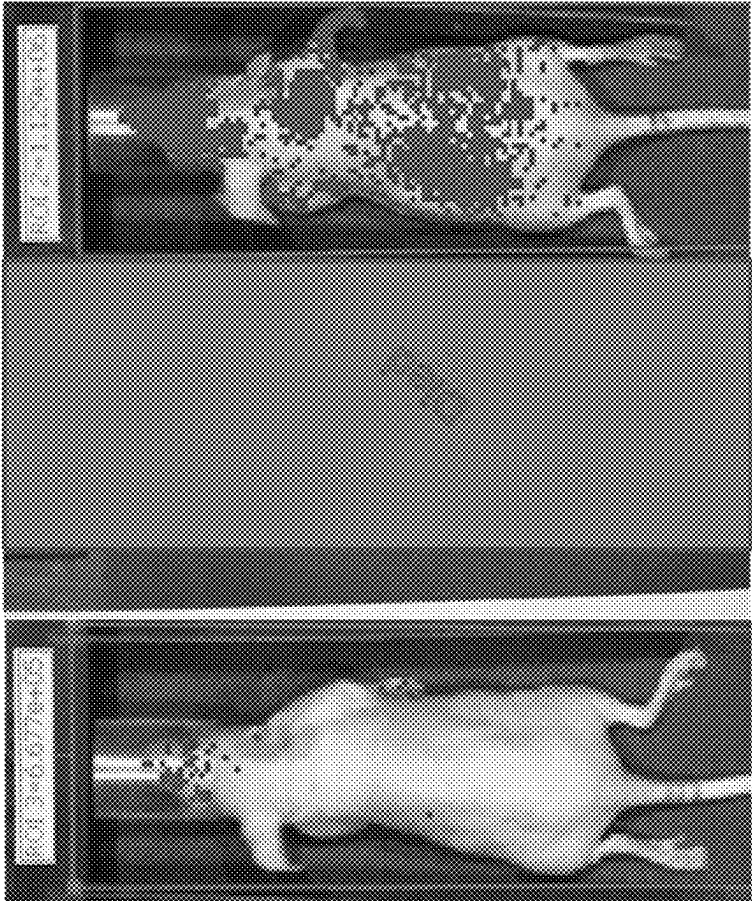
Experiment: Pilot Study: anti-NME7 abs

Cage #	Radiance (E)
13.5 min study	YES → 556
T47D-CSC IV - ab1	0.1306
T47D-CSC IV + ab 2	0.1473
T47D-wt IV 3	0.08091

Figures 34A-34D

Day 12 T47D-CSC @ 10K vs wt @ 500K; CSC mouse was + anti-NME7 treatment;

Fig. 35A	Fig. 35B	Fig. 35C
T47D-wt 500K i.v. Day 12 No Antibody	T47D-CSC 10K i.v. Day 12 No Antibody	T47D-CSC 10K i.v. Day 12 + Antibody



Figures 35A-35C

Day 14 T47D-CSC @ 10K vs wt @ 500K; CSC mouse had 3 anti-NME7 injections, although 1st injection was at same time as NME7 injection

Fig. 36A

T47D-wt
500K i.v.
Day 14
No Antibody

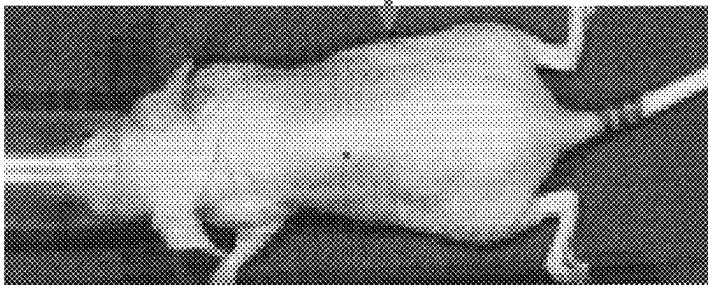
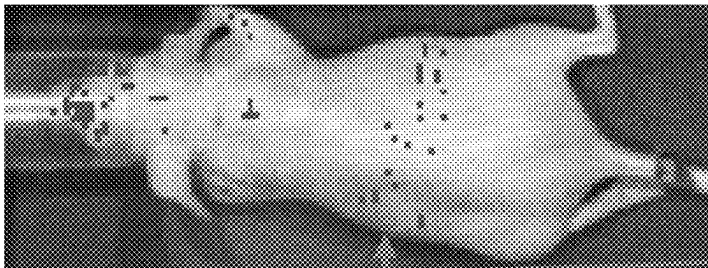
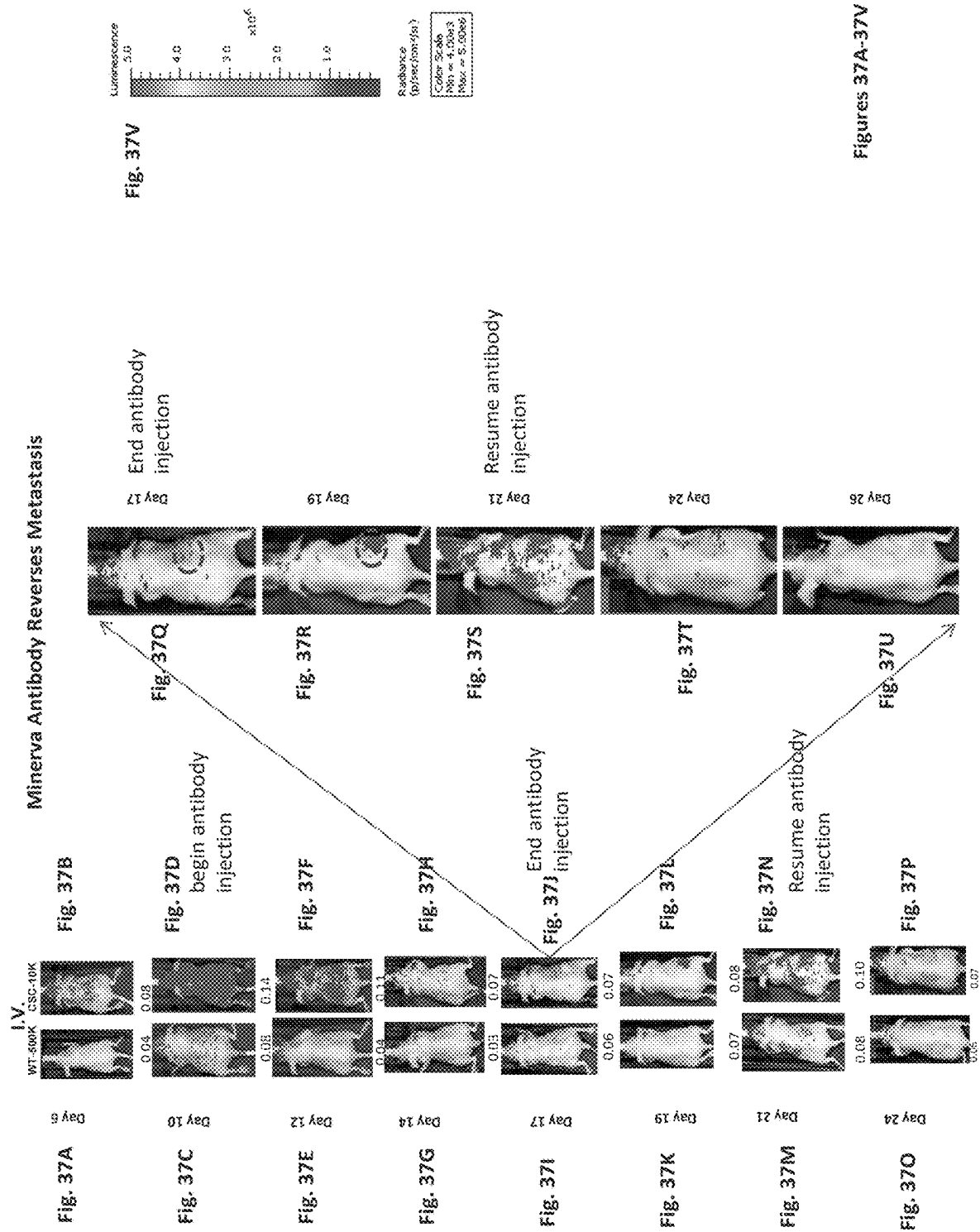


Fig. 36B

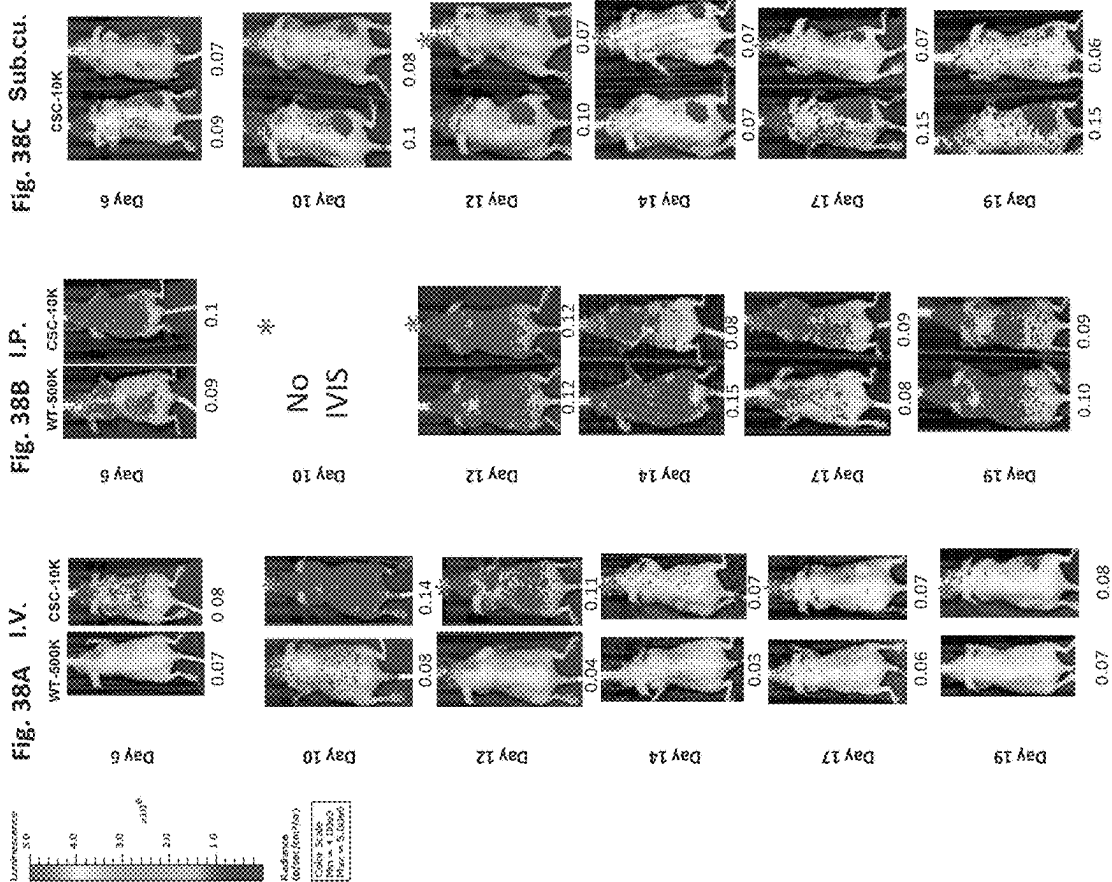


comparison

Figures 36A-36B



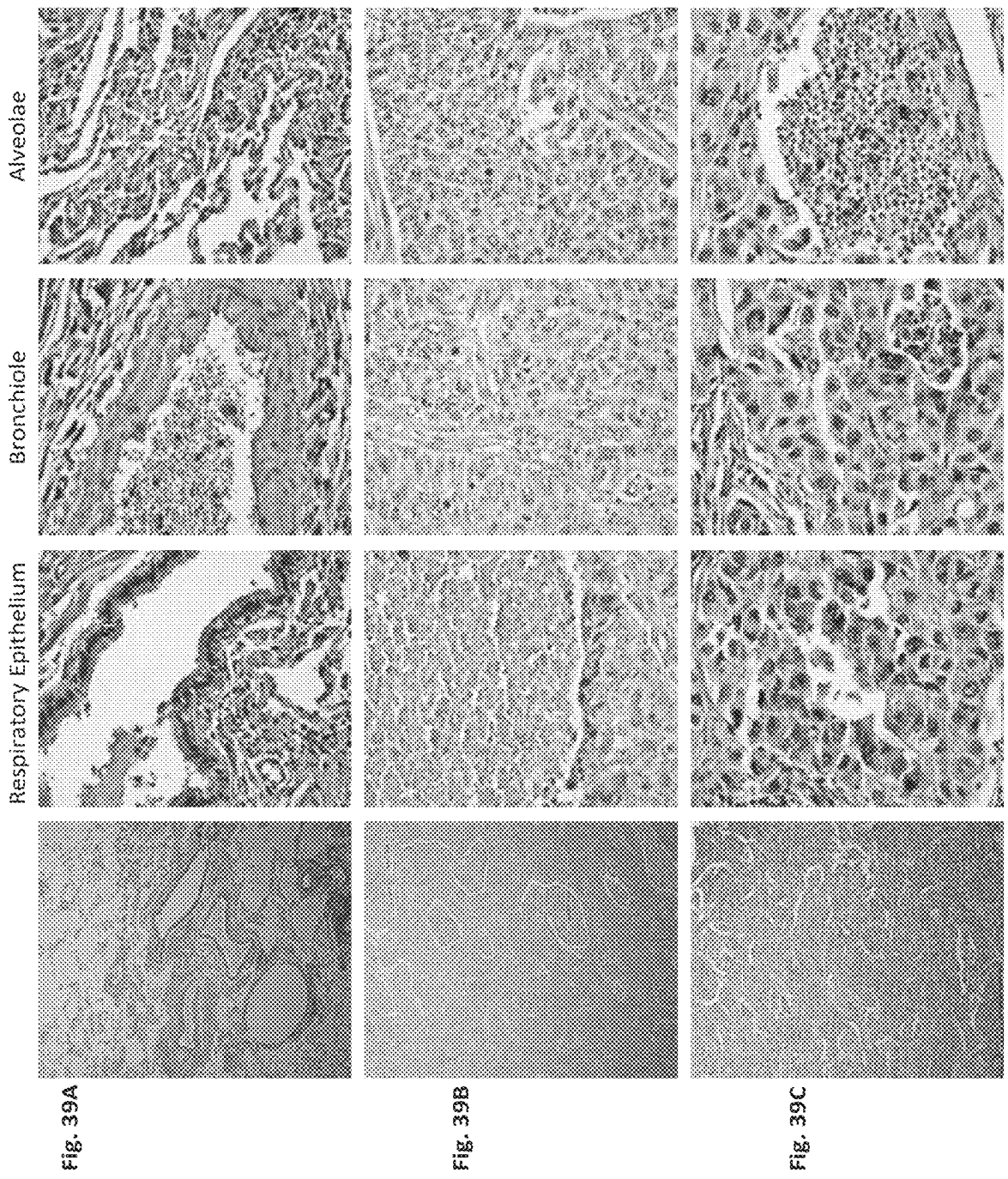
Cancer cells metastasize if grown 10 days in NME7; mets reversed by Minerva's anti-NME7 monoclonal antibodies -- Day 6 -- Day 19



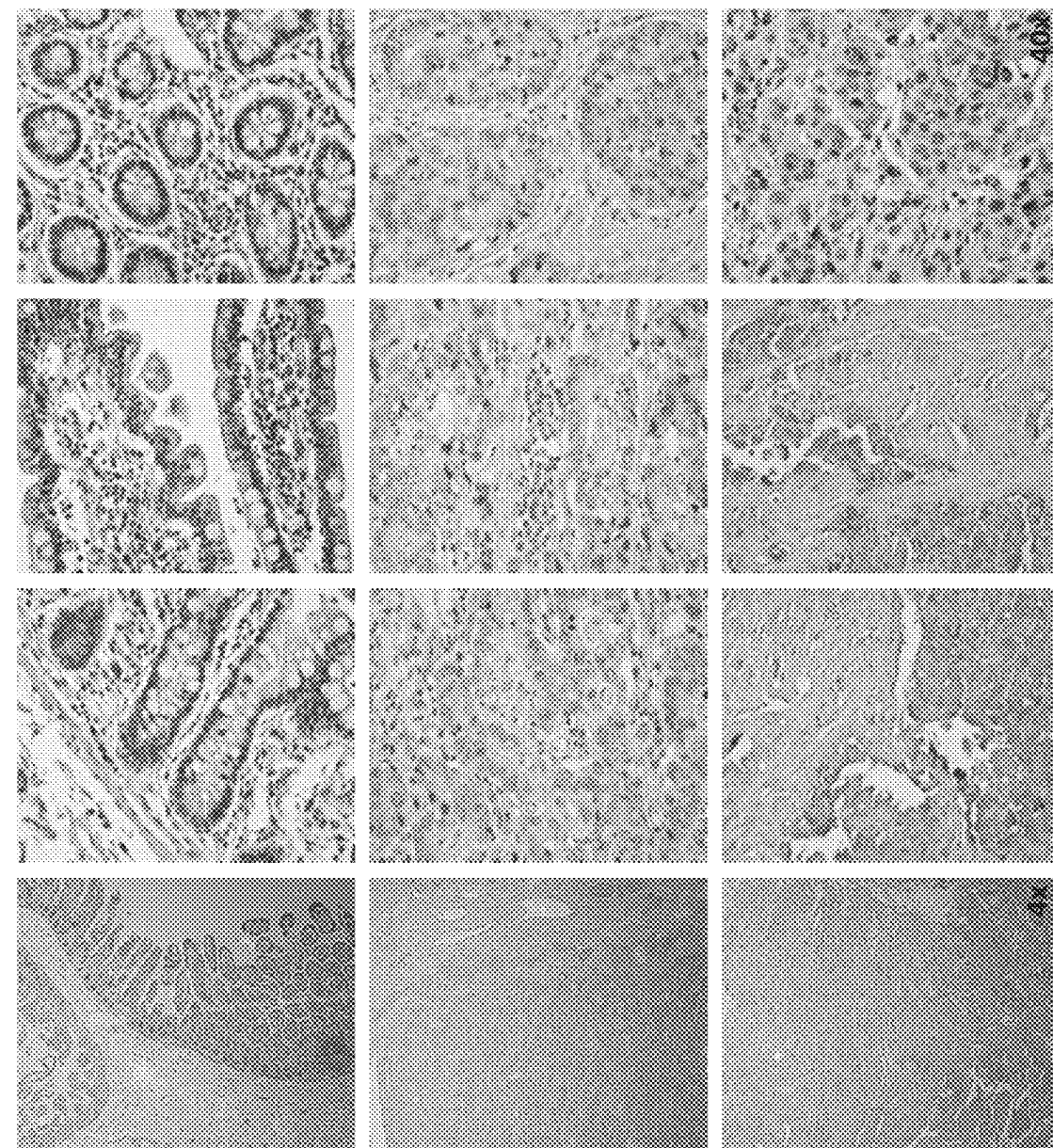
* Indicates
treated with anti-
NME7 antibody

Day 0 – engraftment w 32nM NME7
Day 2, 4, 6 NME7 injections ip 32nM
Day 7, 10, 12, 14 iv, ip or sc anti-NME7
mAb injections

Figures 38A-38C



Lung			
Normal	61	10 ug/ml	61 = Minerva's anti-NME7 antibody
Tumor Grade 2	61	10 ug/ml	
Tumor Grade 3 (T2N0M0) Clinical IB	61	10 ug/ml	
Figures 39A-39C			



Small Intestine

Normal Fig. 40A

61

10 ug/ml

61 = Minerva's anti-NME7 antibody

Tumor Grade 2

Fig. 40B

61

10 ug/ml

Tumor Grade 3
(T3N0M0) Clinical II

Fig. 40C

61

10 ug/ml

Figures 40A-40C

Colon

Normal

61

10 ug/ml

61 = Minerva's anti-NME7 antibody

Tumor Grade 1
(T4N1M1, Clinical IV)

61

10 ug/ml

Tumor Grade 2
(T4N0M0, Clinical IB)

61

10 ug/ml

Tumor Grade 3
(T3N1M1, Clinical IV)

61

10 ug/ml

Figures 41A-41D

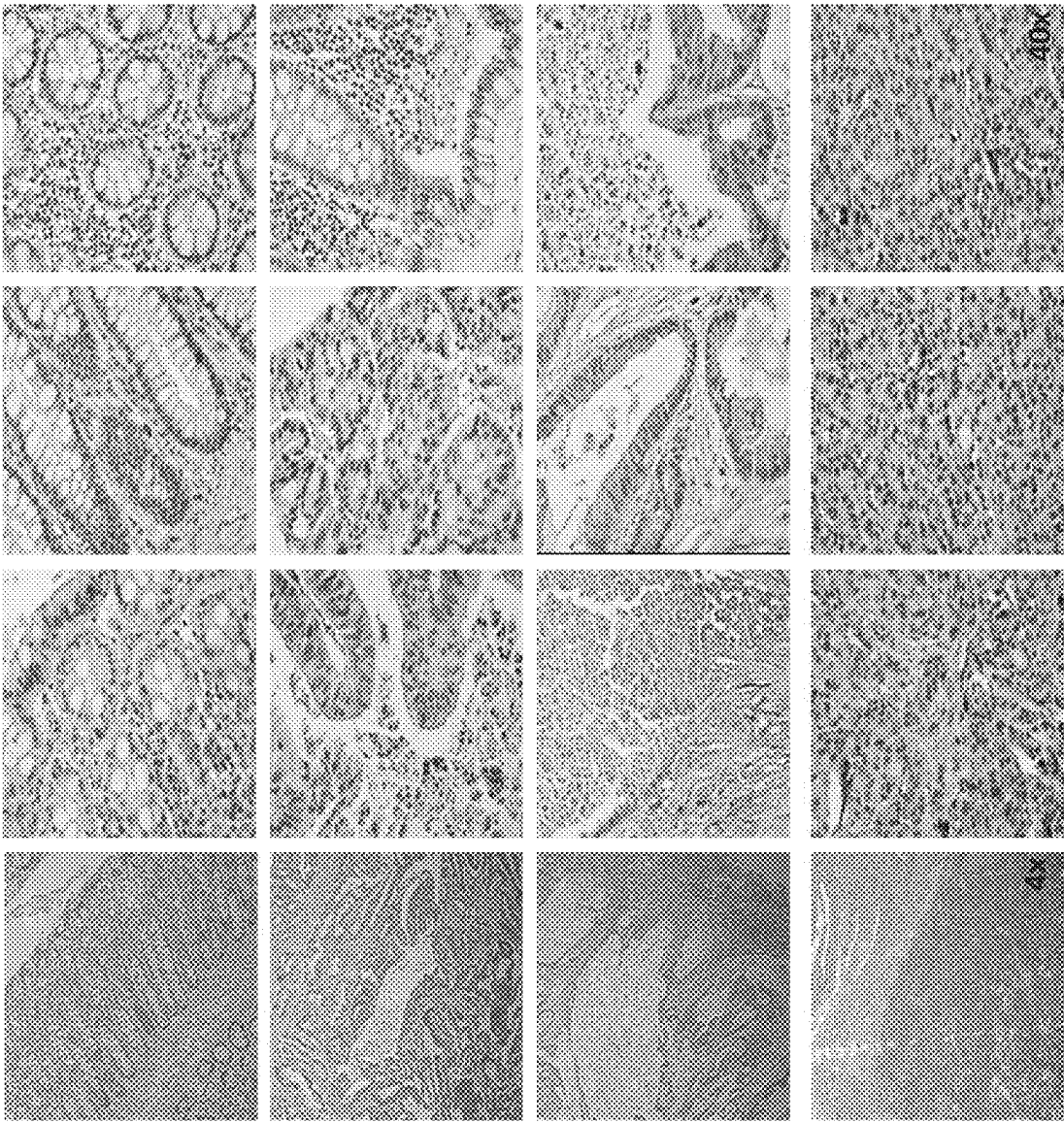


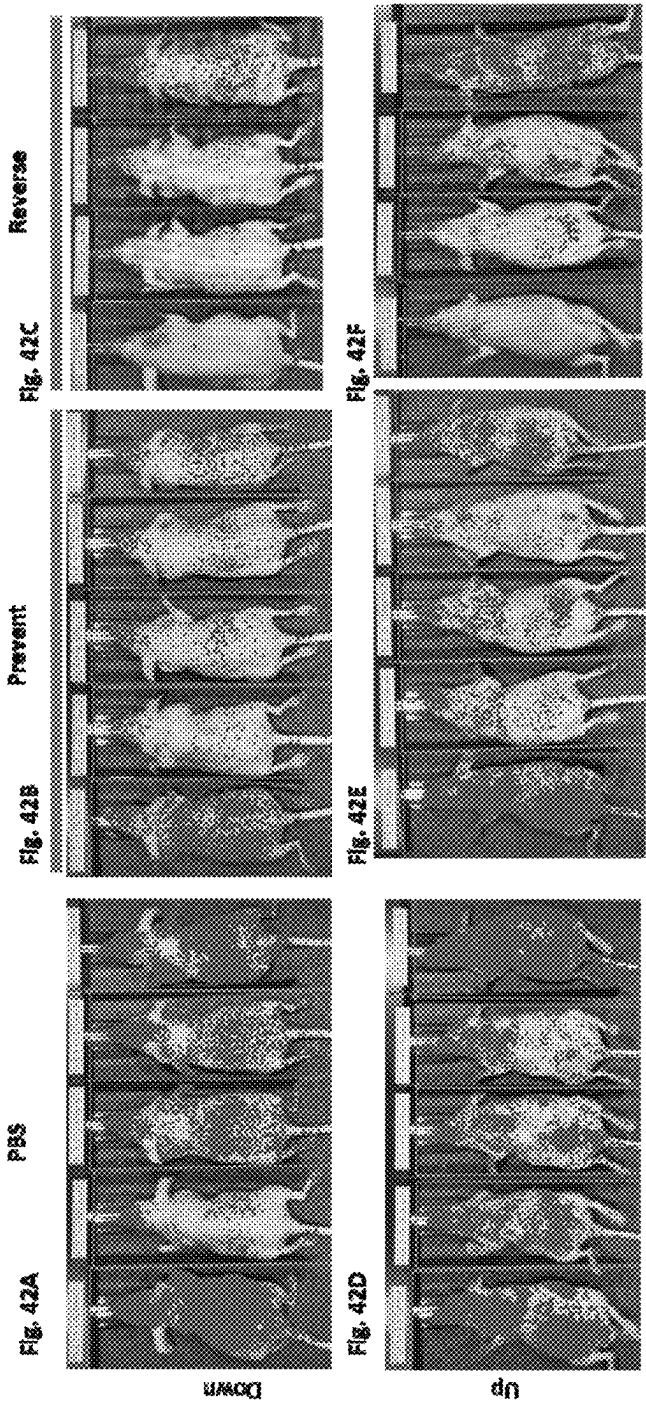
Fig. 41A

Fig. 41B

Fig. 41C

Fig. 41D

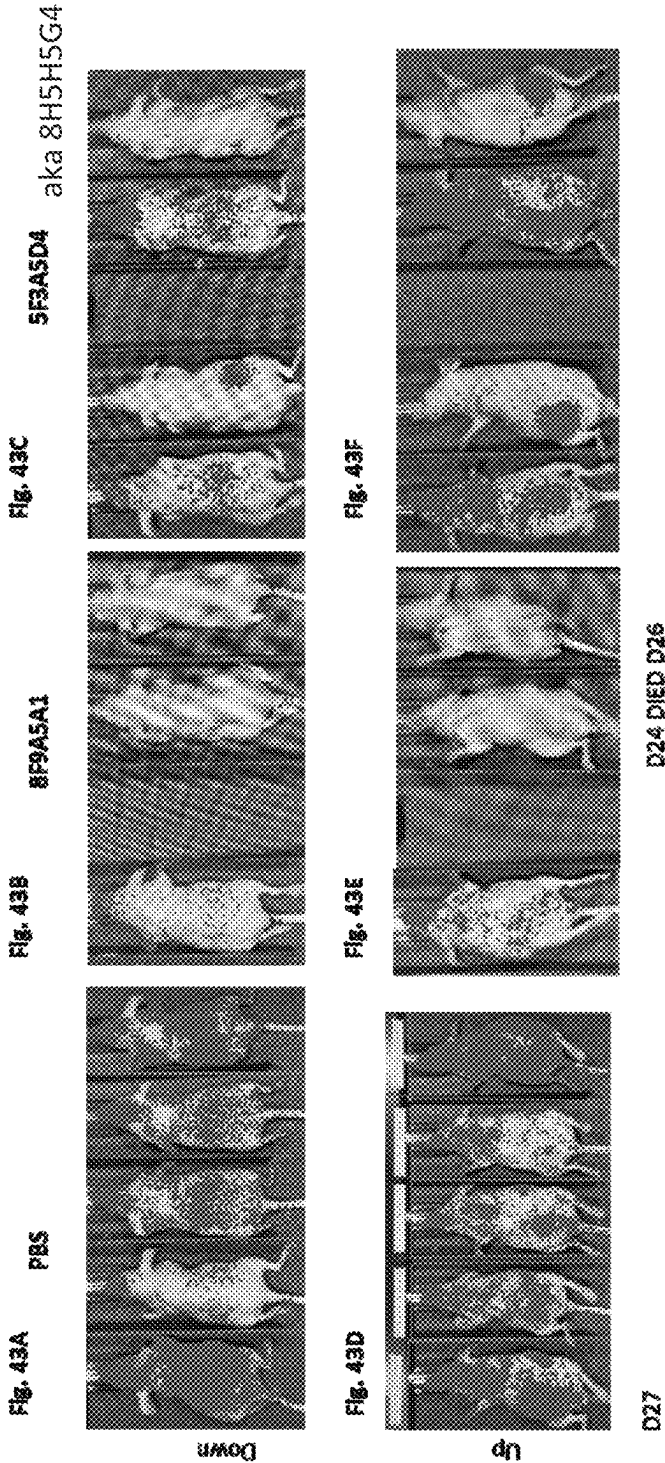
Anti-NME7_{AB} 8F9A4A3 Prevents and Reverses Established Metastasis



Anti-NME7_{AB} antibody 4A3 injected every other day into tail vein at dose equal to 15 mg/kg

Figures 42A-42F

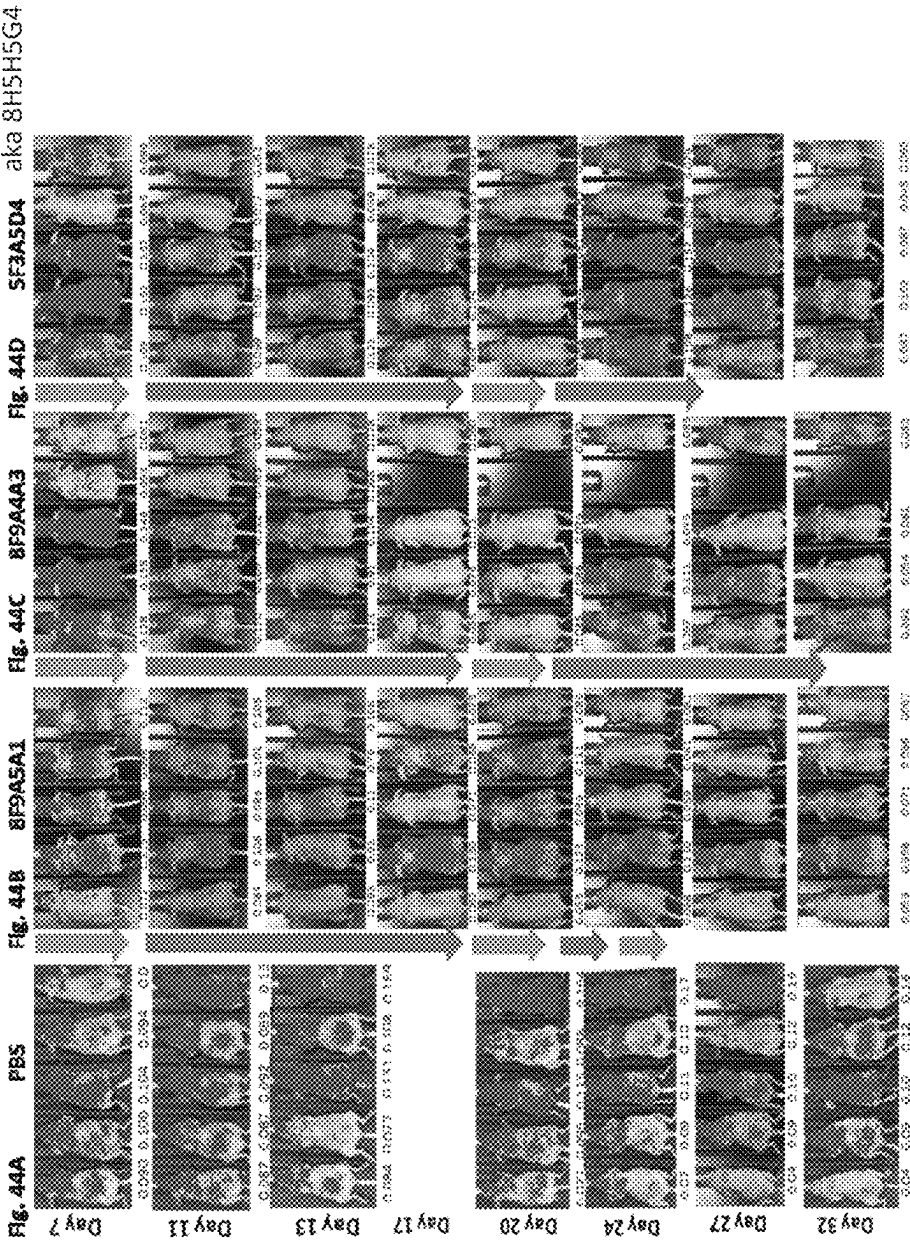
Anti-NME7_{AB} mabs 8F9A5A1 & 5F3A5D4 Inhibit Metastasis in prevention model



Anti-NME7_{AB} antibody injected every other day
into tail vein at dose equal to 15 mg/kg

Figures 43A-43F

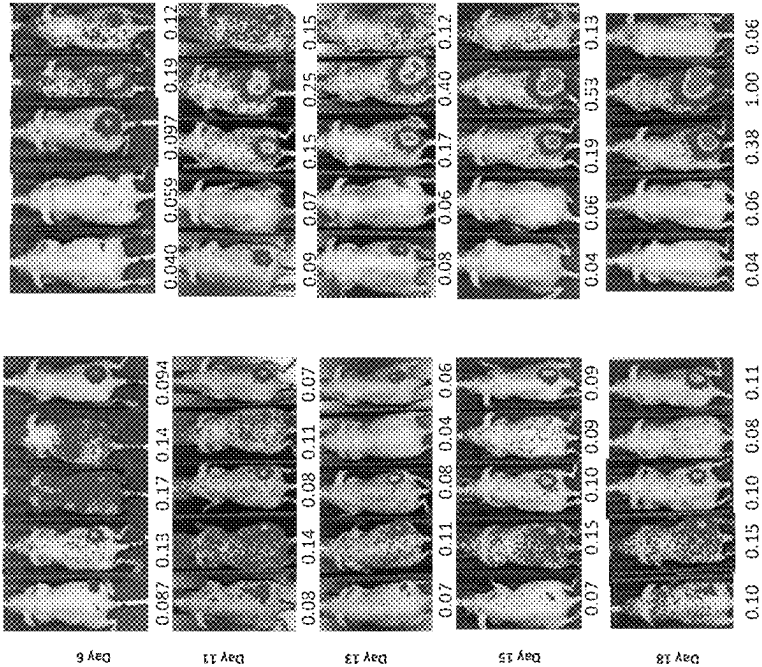
Metastasis reversion model; concentration dependence of anti-NME7 antibodies' ability to reverse established metastases



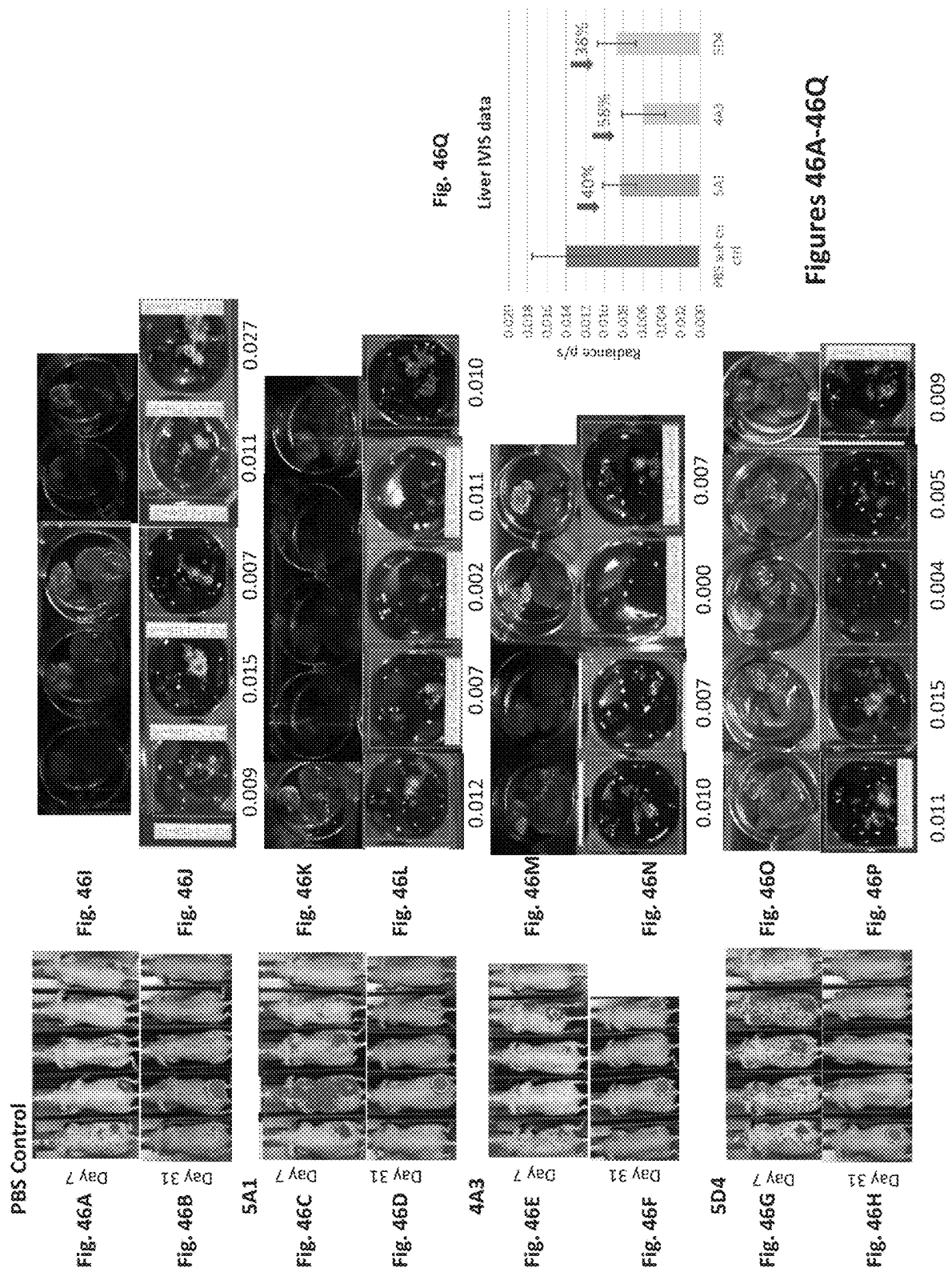
Figures 44A-44D

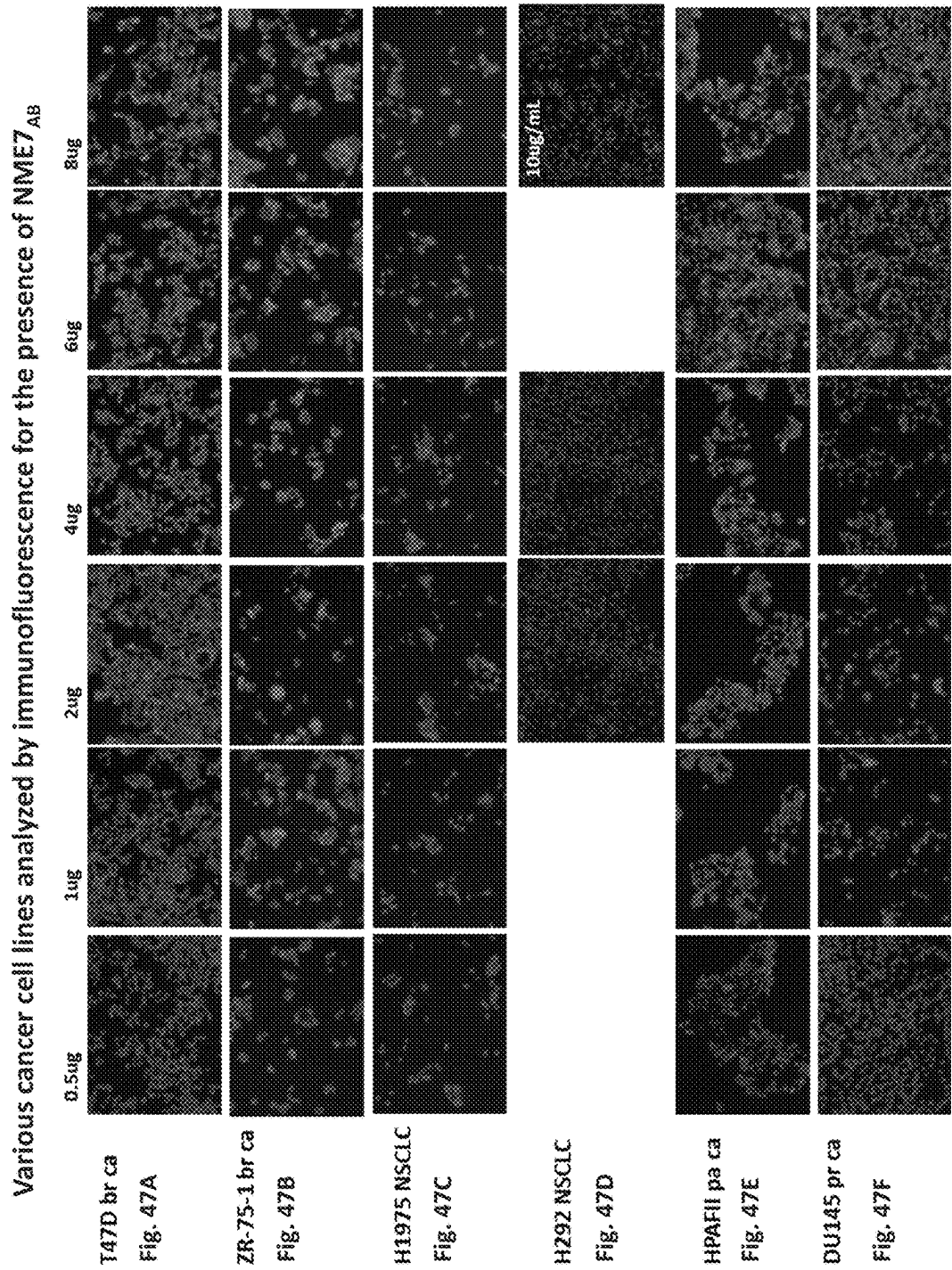
Sub-cu implantation of primary tumor comprised of T47D CSCs; conclusion: antibody cocktail inhibited metastases but did not inhibit growth of the primary

Fig. 45A Control Fig. 45B Anti-NME7_{AB} Antibody

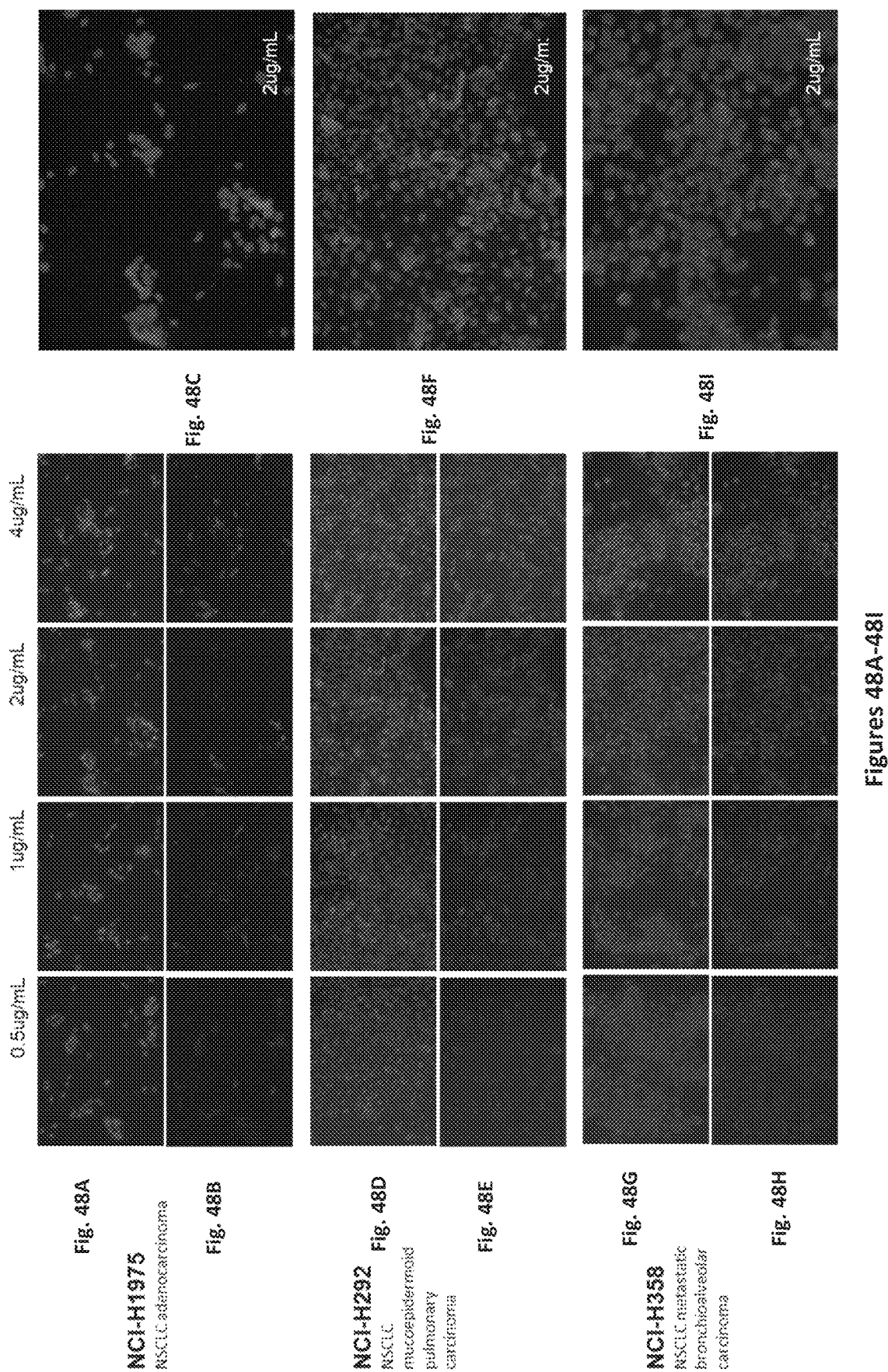


Figures 45A-45B



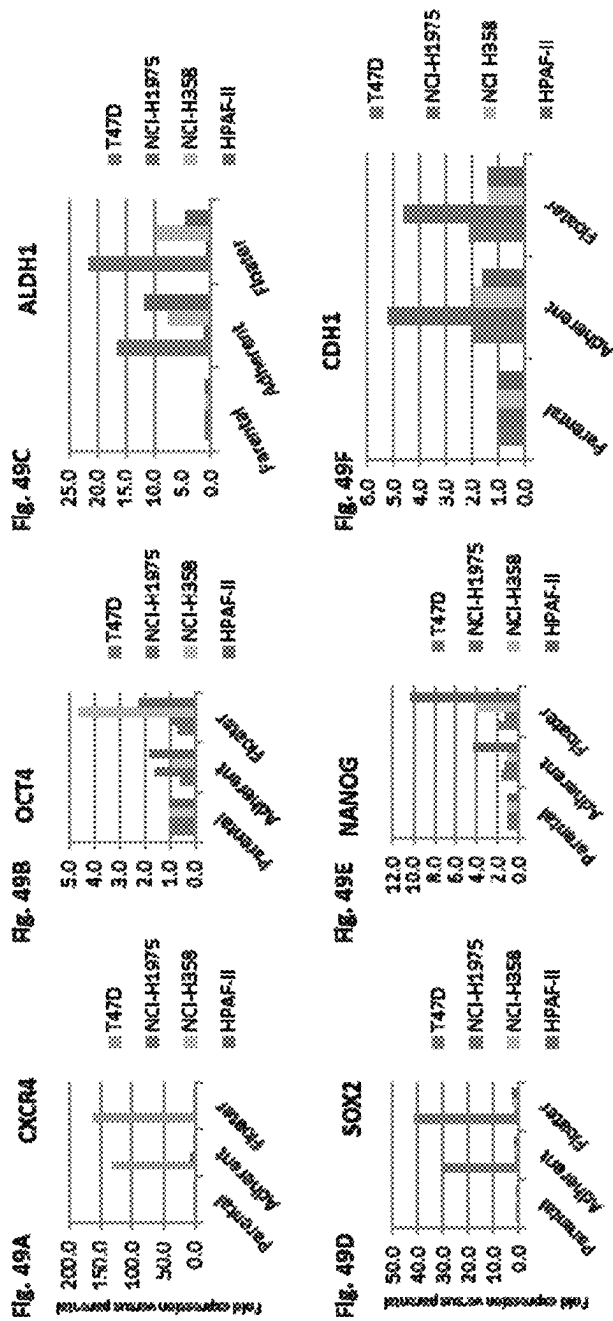


Figures 47A-47F



Figures 48A-48I

PCR measurement of various cancer cell lines before and after culture in NME7_{AB}



Figures 49A-49F

PCR measurement of various cancer cell lines before and after culture in NME7_{A8}

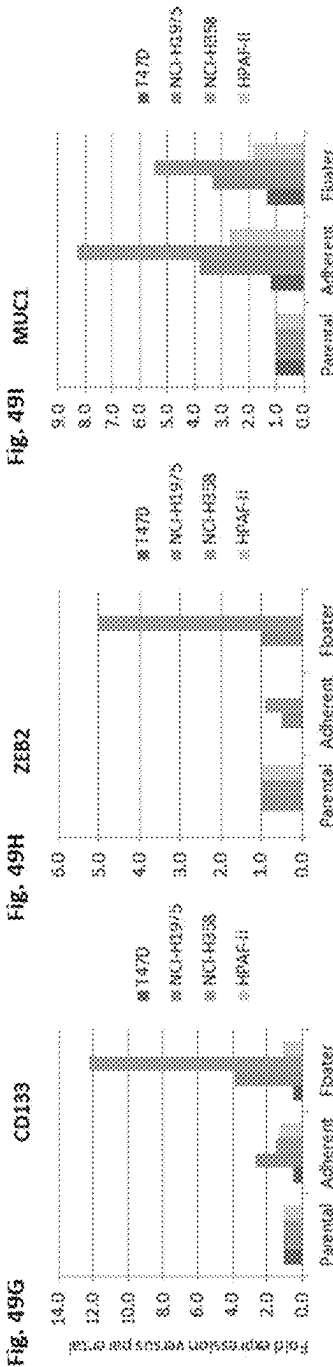
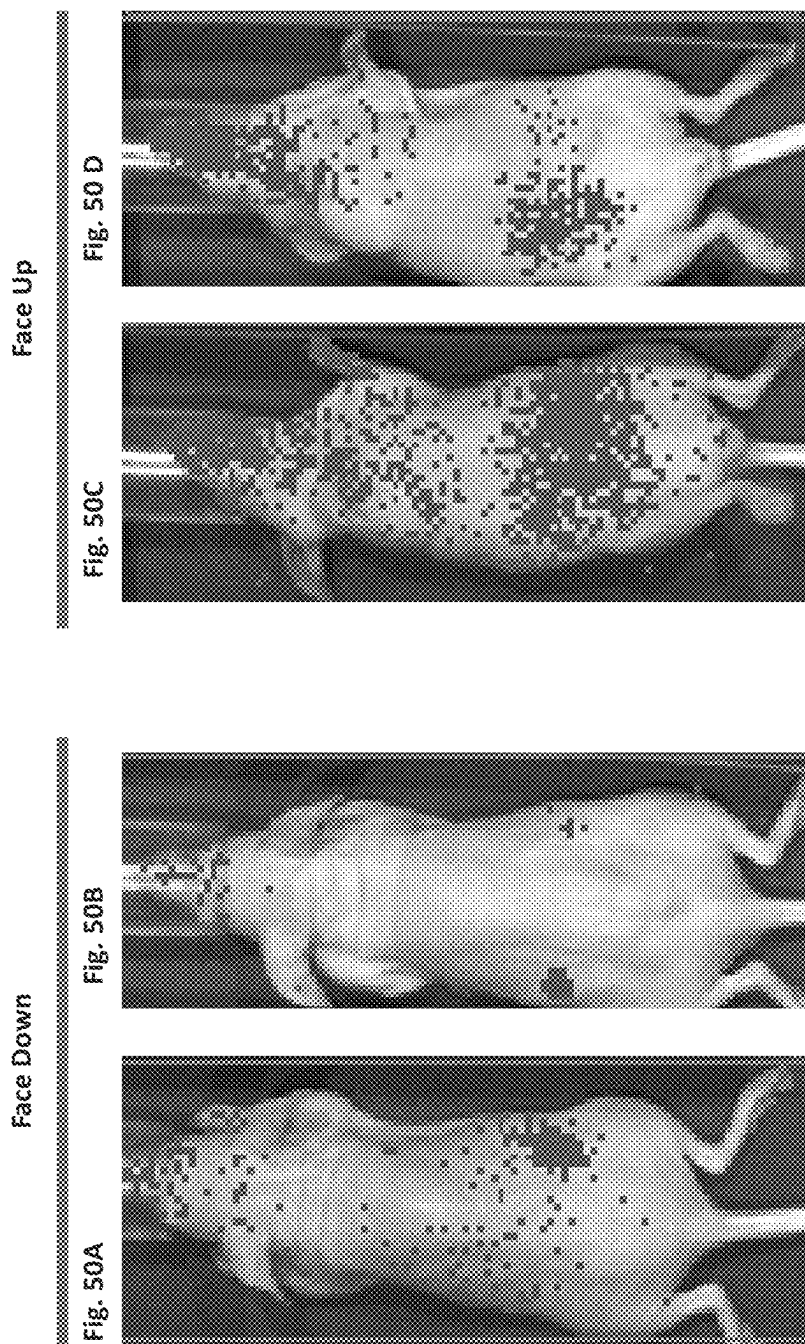


Figure 49G-49I

Day 6 IVIS photographs of NSG mouse injected into tail vein with NCI-H358 lung cancer cells that were cultured for 10 days in NME7_{AB}



Figures 50A-D

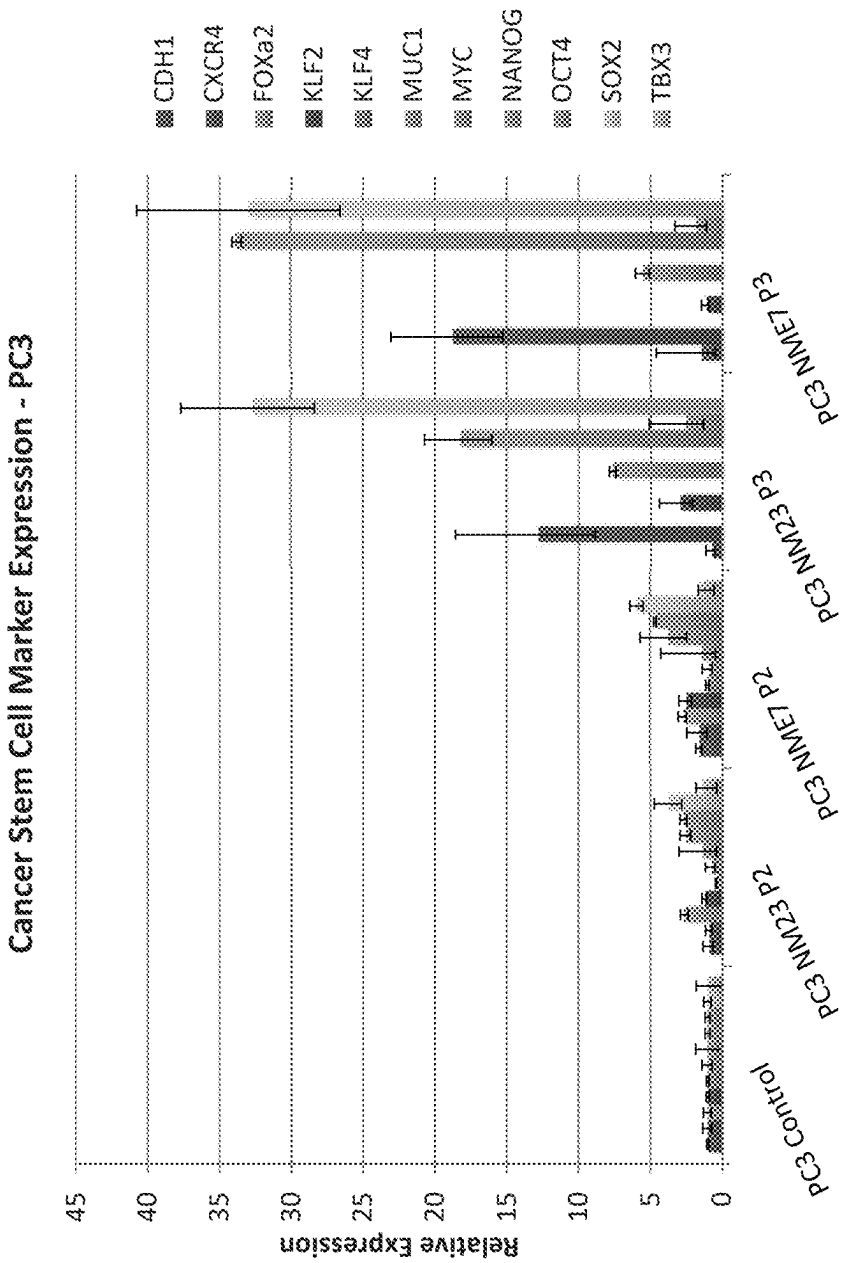


Figure 51

Fig. 52A

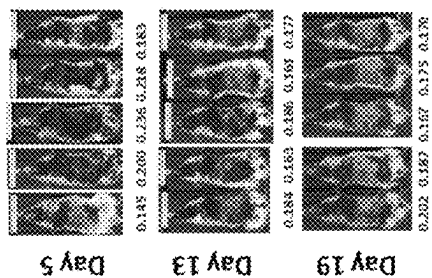


Fig. 52B

hu 4A3 4.8

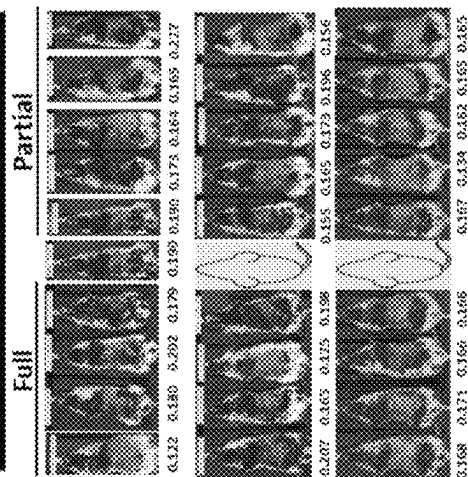


Fig. 52C

hu 4A3 12.3

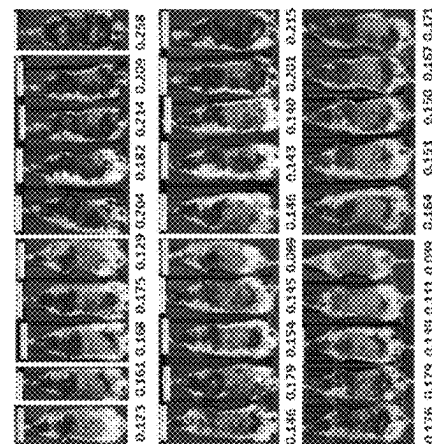


Fig. 52D

mu 4A3

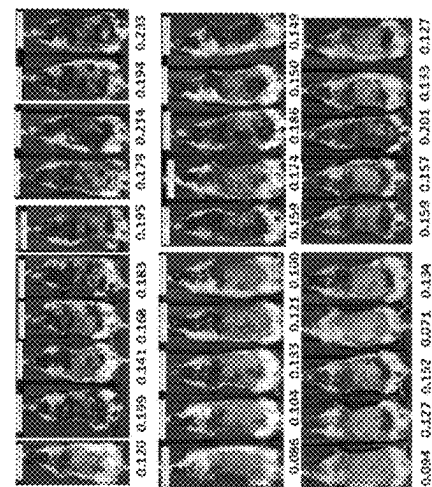


Fig. 52E

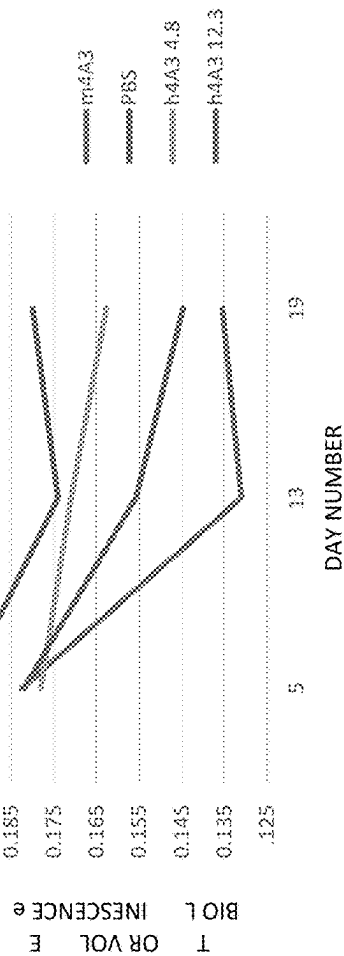
Anti-NME7 antibody 4A3
mouse vs humanized

Figure 52A – 52E

ANTI-NME ANTIBODY AND METHOD OF TREATING CANCER OR CANCER METASTASIS

CROSS REFERENCE

This application is a divisional of U.S. patent application Ser. No. 18/002,832 filed Dec. 21, 2022, which is a National Stage Entry of International Patent Application No. PCT/US2021/039291, filed Jun. 27, 2021, which claims priority to U.S. Patent Application No. 63/046,852, filed Jul. 1, 2020; claims priority to U.S. Patent Application No. 63/044,670, filed Jun. 26, 2020; and is a continuation of International Patent Application No. PCT/US2021/036500, filed Jun. 8, 2021, each of which is incorporated herein by reference in its entirety.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on Dec. 10, 2024, is named 56699-744_401SL.xml and is 998,519 bytes in size.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present application relates to NME proteins, peptides derived from NME proteins, and antibodies generated from the peptides thereof or antibody or antibody fragments selected by virtue of their ability to bind to said peptides. The present application also relates to treating or preventing diseases associated with the expression of NME in a patient.

2. General Background and State of the Art

NDPK (nucleoside diphosphate protein kinase) proteins are a family of proteins grouped together because they all contain an NDPK domain. The first NME proteins discovered, previously called NM23 proteins, were NM23-H1 and NM23-H2. For decades it was unclear whether they induced differentiation or prevented differentiation of hematopoietic cells. The inventors previously discovered that NM23-H1 prevents differentiation when it is a dimer, which binds to the MUC1* growth factor receptor, but at higher concentrations NM23-H1 becomes a hexamer, which does not bind to MUC1*, and it induces differentiation. NM23 used to be called a metastasis suppressor when it was found that it was under-expressed in some very aggressive cancers. The present inventors previously disclosed that NM23-H1 dimers bind to and dimerize the extracellular domain of the MUC1* growth factor receptor that is over expressed on the vast majority of cancers and such binding promotes the growth of cancer cells. Conversely, at higher concentrations, NM23 forms tetramers and hexamers that do not bind to MUC1* and do not promote tumorigenesis. Very recently more NME family proteins (NME 1-10) have been discovered although until now, their functions have not been elucidated. NME7 is a newly discovered NME family protein, but its NDPK domain has no enzymatic activity, unlike other NME family members, NME7 is either not expressed at all in adult tissues or is expressed at extremely low levels.

SUMMARY OF THE INVENTION

The present application is directed to a method of treating or preventing cancer in a subject, comprising administering

to the subject an antibody made against a member of the NME family. The NME family may be NME7 family. The antibody may bind to NME7. The antibody may bind to NME7AB or NME7AB-like protein. The antibody may bind to NME7-X1. The antibody may inhibit binding between NME7 and its cognate binding partner. The cognate binding partner may be MUC1*. The cognate binding partner may be PSMGFR portion of the MUC1* extracellular domain. In one aspect, the antibody may be generated or selected for its ability to bind to a peptide selected from those listed in FIGS. 6-9 (SEQ ID NOS: 88 to 145). Preferably, the peptide may be selected from those listed in FIG. 9 (SEQ ID NOS: 141 to 145).

The peptide may be highly homologous to, or to which is added or subtracted up to 7, up to 6, up to 5, up to 4, up to 3, up to 2, or up to 1 amino acid residues at the N-terminus or C-terminus, of the peptides listed in FIG. 6-9 (SEQ ID NOS: 88 to 145). In one aspect, the antibody may be selected for its ability to bind to NME7AB or NME7-X1 but not to NME1. The antibody may be polyclonal, monoclonal, bivalent, monovalent, bispecific, an antibody fragment containing the variable region, or an antibody mimic. The antibody may be human or humanized. The antibody may be a single chain scFv.

In another aspect, the invention is directed to a method of treating or preventing cancer in a subject, comprising administering to the subject a peptide that is highly homologous or identical to regions of NME7_{AB}. The peptide may be at least 80% homologous to one or more of the peptides listed in FIG. 6. The peptide may be at least 80% homologous to one or more of the peptides listed in FIG. 7. The peptide may be at least 80% homologous to one or more of the peptides listed in FIG. 8. The peptide may be at least 80% homologous to one or more of the peptides listed in FIG. 9. The peptide may be selected from peptides listed in FIG. 6-9 (SEQ ID NOS: 88 to 145). The peptide may be selected from those listed in FIG. 9 (SEQ ID NOS: 141 to 145). Or, the peptide may be highly homologous to, or to which is added or subtracted up to 7, up to 6, up to 5, up to 4, up to 3, up to 2, or up to 1 amino acid residues at the N-terminus or C-terminus, of the peptides listed in FIG. 6-9 (SEQ ID NOS: 88 to 145). The peptide may be connected to another peptide via a spacer or linker.

In another aspect, the invention is directed to a chimeric antigen receptor (CAR), for the treatment or prevention of cancer wherein the targeting extracellular portion of the CAR comprises at least a peptide fragment of a member of the NME family, NME family may be NME7 family. The member of the NME7 family may be NME7. Or, the member of the NME7 family may be NME7AB or NME7AB-like protein. The member of the NME7 family may be also NME7-X1. The targeting extracellular portion of the CAR may include a peptide of the peptides listed in FIG. 6-9 (SEQ ID NOS: 88 to 145). The peptide may be selected from those listed in FIG. 9 (SEQ ID NOS: 141 to 145). The peptide may include a peptide, which is highly homologous to, or to which is added or subtracted up to 7, up to 6, up to 5, up to 4, up to 3, up to 2, or up to 1 amino acid residues at the N-terminus or C-terminus, of the peptides listed in FIG. 6-9 (SEQ ID NOS: 88 to 145). The peptide may be connected to another peptide via a spacer or linker.

In yet another aspect, the invention is directed to a method of treating or preventing cancer or cancer metastasis, comprising engineering the chimeric antigen receptor according to claim 3, into an immune system cell and administering the cell to a subject in need thereof.

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In another aspect, the invention is directed to a chimeric antigen receptor (CAR), for the treatment or prevention of cancer, wherein the targeting extracellular portion of the chimeric antigen receptor comprises a portion of an antibody that binds to NME7_{AB}, NME7_{AB}-like protein or NME7-X1. The portion of the antibody may be a single chain scFv or may be human or humanized.

In yet another aspect, the invention is directed to a method of vaccinating a person against cancer or metastatic cancer comprising immunizing the person with a peptide fragment of a member of the NME family. The NME family may be NME7 family. The member of the NME7 family may be NME7 or NME7b. The member of the NME7 family may be NME7_{AB} or NME7_{AB}-like protein. The NME7 family may be NME7-X1. The immunizing peptide may be a peptide from the peptides listed in FIG. 6-9 (SEQ ID NOS: 88 to 145). Preferably, the peptide may be selected from those listed in FIG. 9 (SEQ ID NOS: 141 to 145). The immunizing peptide may include a peptide, which is highly homologous to, or to which is added or subtracted up to 7, up to 6, up to 5, up to 4, up to 3, up to 2, or up to 1 amino acid residues at the N-terminus or C-terminus, of the peptides listed in FIG. 6-9 (SEQ ID NOS: 88 to 145). The immunizing peptide may be connected to another peptide via a spacer or linker.

In yet another aspect, the invention is directed to a method of treating or preventing cancer in a subject, comprising administering to the subject a nucleic acid that inhibits the expression of NME7, NME7b, NME7_{AB}-like protein or NME7-X1. The nucleic acid may be an anti-sense nucleic acid that suppresses expression of NME7, NME7_{AB}-like protein or NME7-X1. The nucleic acid may be an inhibitory RNA, siRNA, RNAi, or shRNA that inhibits expression of NME7, NME7_{AB}-like protein or NME7-X1.

In another aspect, the invention is directed to a method of treating or preventing cancer in a subject, comprising administering to the subject genetically edited nucleic acids that inhibit expression of NME7, NME7b, NME7_{AB}-like protein or NME7-X1. The genetically edited nucleic acids that inhibit expression of NME7, NME7b, NME7_{AB}-like protein or NME7-X1 may be inserted into cells that may be then administered to the patient. The genetically edited nucleic acids that inhibit expression of NME7, NME7b, NME7_{AB}-like protein or NME7-X1 may be inserted into cells using a viral vector. The viral vector may be a lentiviral system.

In another aspect, the invention is directed to a method of growing cancer cells comprising contacting the cells with NME7_{AB}, NME7b, NME7_{AB}-like protein or NME7-X1, 2i or 5i. The method may include culturing the cells in a medium that contains NME7_{AB}, NME7b, NME7_{AB}-like protein or NME7-X1, 2i or 5i, or growing cells in an animal that expresses human NME7_{AB}, NME7b, NME7_{AB}-like protein or NME7-X1, or to which NME7_{AB}, NME7b, NME7_{AB}-like protein or NME7-X1 is administered. The cancer cells may be breast, prostate, ovarian, colorectal, pancreatic, liver, melanoma or brain cancer cells. Drug candidates may be tested on the cells. The efficacy of the drugs may be assessed by comparing cancer growth to a no drug control or comparing expression levels of metastatic markers or stem cell markers to a no drug control or comparing the ability of the resultant cells to form tumors in animals from low cell copy number compared to a no drug control and determining the efficacy of a candidate drug for the treatment of cancer or metastasis. The cells may be obtained from a patient being assessed for treatment for cancer and drugs that would be effective for that patient are selected based on results using methods described above. The cells may not be obtained

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from a patient being assessed for treatment for cancer but drugs that would be effective for that patient are selected based on results using the methods described above.

In another aspect, the invention is directed to a method of generating antibodies or antibody-like molecules from peptides or peptide mimics having a sequence derived from the sequence of NME. The NME may be NME7. The peptide may be used as an immunogen to generate antibodies or antibody-like molecules. The peptide may be administered to an animal to generate anti-NME7 antibodies. The peptide may be administered to a human to generate anti-NME7 antibodies. The peptide may have a sequence listed in FIG. 6-9 (SEQ ID NOS: 88 to 145). Preferably, the peptide may be selected from those listed in FIG. 9 (SEQ ID NOS: 141 to 145). The peptide may include a peptide, which is highly homologous to, or to which is added or subtracted up to 7, up to 6, up to 5, up to 4, up to 3, up to 2, or up to 1 amino acid residues at the N-terminus or C-terminus, of the peptides listed in FIG. 6-9 (SEQ ID NOS: 88 to 145).

In another aspect, the invention is directed to a method of detecting presence of cancer or the progression of cancer, comprising the steps of:

- 1) obtaining a sample from a patient having cancer or at risk of developing a cancer;
- 2) subjecting that sample to an assay capable of detecting or measuring levels of a member of the NME7 family, or levels of nucleic acids encoding a member of the NME7 family;
- 3) comparing levels of the measured member of the NME7 family or the member of the NME7 family-encoding nucleic acids in the test sample to levels in control patients or control cells;
- 4) determining that the levels of the member of the NME7 family or nucleic acids encoding the member of the NME7 family are elevated compared to the controls; and
- 5) concluding that the donor of the test sample has cancer or has had a progression of cancer if the control to which the test was compared came from a donor previously diagnosed with a cancer. In this method, the detection of the member of the NME7 family in circulation or in a tissue may be an indicator of cancer in a patient. The member of the NME7 family may be NME7, NME7b, NME7-X1, or NME7_{AB}-like protein.

In yet another aspect, the invention is directed to a method comprising:

- detecting presence of a member of the NME7 family or MUC1* in a patient; and
- administering anti-NME7 or anti-MUC1* antibody or antibodies to the patient exhibiting the member of the NME7 family or MUC1* expression. The member of the NME7 family may be NME7, NME7b, NME7-X1, or NME7_{AB}-like protein.

In yet another aspect, the invention is directed to a method for treating or preventing cancer comprising:

- 1) obtaining a sample from a patient suspected of having a cancer or at risk of developing a cancer or at risk of developing a metastatic cancer;
- 2) measuring an amount of the member of an NME7 family or a member of the NME7 family encoding nucleic acid, wherein the measured levels are significantly above those measured in a control sample;
- 3) determining that the patient has a cancer or has developed a more aggressive or a metastatic cancer;
- 4) administering to the patient an effective amount of a therapeutic agent that suppresses expression of the member of the NME7 family, inhibits cleavage of

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NME7 or inhibits NME7 binding to its targets. The target of the member of the NME7 family may be MUC1*. The target of the member of the NME7 family may be PSMGFR portion of the MUC1* extracellular domain. The member of the NME7 family may be NME7, NME7b, NME7-X1, or NME7_{AB}-like protein.

In any of the methods above regarding cancer, cancer may include breast, prostate, ovarian, colorectal, pancreatic, liver, melanoma or brain cancer.

In one aspect, the invention is directed to an NME7 specific antibody or fragment thereof that binds to the NME7 B3 peptide of SEQ ID NO:145 or SEQ ID NO:169. The antibody may be monoclonal antibody or bivalent, monovalent, an Fab, or a single chain variable fragment antibody (scFv). The antibody may be linked to an antibody drug conjugate. The drug may be linked to a toxin or pro-toxin.

The invention is also directed to an isolated nucleic acid encoding the antibody.

The invention is also directed to an isolated hybridoma expressing the monoclonal antibody discussed above. The antibody may specifically bind to NME7_{AB} or NME7-X1, but not to NME1. The antibody may disrupt interaction between NME7_{AB} and MUC1* extra cellular domain or between NME7-X1 and MUC1* extra cellular domain. Or, the antibody may disrupt binding between NME7_{AB} and PSMGFR or between NME7-X1 and PSMGFR. Further, the antibody may disrupt binding between NME7_{AB} and N-10 or between NME7-X1 and N-10.

In another aspect, the antibody may not disrupt interaction between NME7_{AB} and MUC1* extra cellular domain or between NME7-X1 and MUC1* extra cellular domain. NME7_{AB} or NME7-X1 may bind to the N-10 peptide (SEQ ID NO:170) but not to a C-10 peptide (SEQ ID NO: 171). In particular, the antibody may be 5A1, 4A3, 5D4, or 4P3.

The antibody may comprise an amino acid sequence in the heavy chain variable region comprising the following:
in the CDR1 region YFTFTNYGMN (SEQ ID NO:439);
in the CDR2 region WINTYTGEPTYVDDFKG (SEQ ID NO:440); and
in the CDR3 region LRGIRPGPLAY (SEQ ID NO:441);
and

an amino acid sequence in the light chain variable region comprising the following:

in the CDR1 region SASSSVSYMN (SEQ ID NO:444);
in the CDR2 region GISNLAS (SEQ ID NO:445); and
in the CDR3 region QQRSSYPPT (SEQ ID NO:446).

An example of such an antibody above is 5A1.

In another aspect, the antibody may comprise an amino acid sequence in the heavy chain variable region comprising the following:

in the CDR1 region NTFTEYTMH (SEQ ID NO:429);
in the CDR2 region GFNPNGVTNYNQKFKG (SEQ ID NO:430); and
in the CDR3 region RYYHSTYVFYFDS (SEQ ID NO:431); and

an amino acid sequence in the light chain variable region comprising the following:

in the CDR1 region SASQGISNYLN (SEQ ID NO:434);
in the CDR2 region YTSSLHS (SEQ ID NO:435); and
in the CDR3 region QQYSKLPYT (SEQ ID NO:436).

An example of such an antibody above is 5D4.

In another aspect, the antibody may comprise an amino acid sequence in the heavy chain variable region comprising the following:

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in the CDR1 region NTFTEYTMH (SEQ ID NO:388);
in the CDR2 region GFNPNGVTNYNQKFKG (SEQ ID NO:389); and
in the CDR3 region RYYHSLYVFYFDY (SEQ ID NO:390); and

an amino acid sequence in the light chain variable region comprising the following:

in the CDR1 region SASQGISNYLN (SEQ ID NO:434);
in the CDR2 region YTSSLHS (SEQ ID NO:435); and
in the CDR3 region QQYSKLPYT (SEQ ID NO:436).

An example of such an antibody above is 4A3.

In another aspect, the antibody may comprise an amino acid sequence in the heavy chain variable region comprising the following:

in the CDR1 region NTFTEYTMH (SEQ ID NO:388);
in the CDR2 region GFNPNGVTNYNQKFKG (SEQ ID NO:389); and
in the CDR3 region RYYHSLYVFYFDY (SEQ ID NO:390); and

an amino acid sequence in the light chain variable region comprising the following:

in the CDR1 region ITSTDIDDDMN (SEQ ID NO: 1144);
in the CDR2 region EGNTLRP (SEQ ID NO: 1145); and
in the CDR3 region LQSDNLPLT (SEQ ID NO: 1146).

An example of such an antibody above is 4P3.

The antibody may be human, humanized or an engineered antibody mimic.

The antibody may be non-human, such as murine or camelid.

The invention is also directed to a method of administering to a patient for prevention or treatment of cancer comprising administering to the patient a composition comprising the antibody described above.

The invention is also directed to a method for preventing or treating cancer metastasis in a patient, comprising administering to the patient a composition comprising the antibody described above.

The invention is also directed to a method for diagnosing cancer or cancer metastasis comprising contacting a patient specimen and normal specimen with the antibody above, and comparing the results from both specimen, wherein presence of positive binding to the antibody in the patient specimen indicates the presence of cancer or cancer metastasis in the patient. The antibody may be linked to an imaging agent. The patient specimen may be blood, bodily fluid, tissue, circulating cells, in vitro, in vivo, including intra-operative.

The invention is also directed to a cell that is engineered to express an anti-NME7_{AB} antibody or fragment thereof. The cell may be an immune cell, such as T cell or NK cell, or a stem or progenitor cell, preferably stem or progenitor cell that is then differentiated to become a T cell.

The cell may comprise a chimeric antigen receptor (CAR) that recognizes tumor associated antigen. Expression of the anti-NME7 antibody may be inducible. Nucleic acid encoding an anti-NME7_{AB} antibody may be inserted into the Foxp3 enhancer or promoter. The anti-NME7_{AB} antibody may be in an NFAT-inducible system. NFATc1 response element may be inserted upstream of the antibody sequence that is inserted into Foxp3 enhancer or promoter region.

The anti-NME7_{AB} antibody or fragment thereof may bind to the NME7 B3 peptide, or disrupt binding of NME7_{AB} or NME7-X1 to the PSMGFR peptide of the MUC1* extra cellular domain.

The CAR may recognize a tumor associated antigen and an anti-NME7 antibody. The tumor associated antigen may be MUC1*.

The invention is also directed to an anti-cancer vaccine comprising a composition comprising one or more peptides derived from NME7_{AB} listed in FIG. 6-FIG. 9 or a peptide having at least 80%, 85%, 90%, 95%, 97% sequence identity thereof as the immunogenicity eliciting portion. The peptide may be a peptide of SEQ ID NOS: 141-145 or a peptide having at least 80%, 85%, 90%, 95%, 97% sequence identity thereof. The peptide may be a peptide of SEQ ID NO: 145 or a peptide having at least 80%, 85%, 90%, 95%, 97% sequence identity thereof.

In another aspect, the invention is directed to a BiTE comprising the above-described antibody.

In yet another aspect, the invention is directed to a method of generating anti-NME7_{AB} antibodies wherein Cysteine residue in the NME7 B3 peptide is mutated to avoid disulfide bonding.

In yet another aspect, the invention is directed to a method of generating cells with enhanced metastatic potential comprising culturing the cells with NME7_{AB} or NME7-X1.

The invention is also directed to a cell that is engineered to express NME7_{AB} or NME7-X1, a transgenic animal that expresses NME7_{AB} or NME7-X1, wherein the NME7_{AB} or NME7-X1 may be human, and also wherein expression of NME7_{AB} or NME7-X1 may be inducible.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention will become more fully understood from the detailed description given herein below, and the accompanying drawings which are given by way of illustration only, and thus are not limitative of the present invention, and wherein:

FIGS. 1A-1B show a graph of HRP signal from ELISA sandwich assay showing NME7_{AB} dimerizes MUC1* extra cellular domain peptide.

FIG. 2 is a graph of RT-PCR measurements of gene expression for stem cell markers and cancer stem cell markers for T47D cancer cells after being cultured in traditional media or a media containing NME7, wherein cells that became non-adherent (floaters) were analyzed separate from those that remained adherent.

FIG. 3 is a graph of RT-PCR measurements of gene expression for a variety of stem and putative cancer stem cell markers for DU145 prostate cancer cells. Cells were cultured either in traditional media or a media containing NME1 dimers ("NME23") or NME7 (NME7_{AB}). Rho kinase inhibitor was not used because by passage 2, cells remained adherent.

FIG. 4 is a graph of RT-PCR measurement of the metastatic markers and pluripotent stem cell markers showing that the 2i inhibitors (GSK3-beta and MEK inhibitors), which were previously shown to revert stem cells to a more naïve state, also induce cancer cells to a more metastatic state although not as well as NME7_{AB}.

FIG. 5 is a sequence alignment between human NME1 and human NME7-A or -B domain. Figure discloses SEQ ID NOS 1137, 1140, 1137 and 1143, respectively, in order of appearance.

FIG. 6 lists immunogenic peptides from human NME7 with low sequence identity to NME1 and selected for their ability to generate therapeutic anti-NME7 antibodies for the treatment or prevention of cancers. Figure discloses SEQ ID NOS 88-118, respectively, in order of appearance.

FIG. 7 lists immunogenic peptides from human NME7 that may be important for structural integrity or for binding to MUC1* selected for their ability to generate therapeutic anti-NME7 antibodies for the treatment or prevention of

cancers. Figure discloses SEQ ID NOS 122-133, respectively, in order of appearance.

FIG. 8 lists immunogenic peptides from human NME1 that may be important for structural integrity or for binding to MUC1* and selected for their ability to generate therapeutic anti-NME7 antibodies for the treatment or prevention of cancers. Figure discloses SEQ ID NOS 134-140, respectively, in order of appearance.

FIG. 9 lists immunogenic peptides from human NME7 selected for their low sequence identity to NME1 and for their homology to bacterial NME1 proteins that have been implicated in cancers. These peptides are preferred for their ability to generate therapeutic anti-NME7 antibodies for the treatment or prevention of cancers. The peptides shown in this Figure include and added Cysteine covalently bound at the C-terminal end. Figure discloses SEQ ID NOS 141-145, and 169, respectively, in order of appearance.

FIGS. 10A-10B show graphs of ELISA assays in which either NME7_{AB} (FIG. 10A) or NME1 (FIG. 10B) is adsorbed to the plate and anti-NME7 antibodies generated by NME7 peptides A1, A2, B1, B2 and B3 are tested for their ability to bind to NME7 but not to NME1. C20 is an anti-NME1 antibody.

FIG. 11 shows graphs of ELISA assays in which anti-NME7 antibodies generated are tested for their ability to inhibit binding of NME7_{AB} to a surface immobilized MUC1* peptide but not inhibit binding of NME1.

FIG. 12 shows a graph of a cancer cell growth experiment in which breast cancer cells were grown in the presence or absence of NME7 antibodies or short peptides derived from NME7, which were used to generate or select the antibodies. In addition, an antibody generated by immunization with nearly the entire NME7_{AB} peptide, amino acids 100-376, was shown to inhibit cancer cell growth.

FIG. 13 shows a graph of a cancer cell growth experiment in which breast cancer cells were grown in the presence or absence of combinations of NME7 antibodies or combinations of the short peptides derived from NME7, which were used to generate or select the antibodies. Both antibodies as well as their immunizing NME7_{AB} peptides inhibited growth of cancer cells.

FIGS. 14A-14B show tables of scientist observations when cancer cells were grown in either NME7_{AB} or 2i inhibitors, which both are able to transform cancer cells to a more metastatic state, and in the presence or absence of NME7 derived peptides A1, A2, B1, B2 and B3. The NME7_{AB} peptides inhibited the transition of adherent cancer cells to the floater cells, which RT-PCR measurements show have increased expression of metastatic markers, especially CXCR4.

FIGS. 15A-15C show graphs of RT-PCR measurements of expression of CXCR4 and other metastatic markers in T47D breast cancer cells that were grown in either NME7_{AB} or 2i inhibitors, each of which transform cancer cells to a more metastatic state, and the inhibitory effect of anti-NME7 antibodies on the metastatic transformation. FIG. 15A shows PCR graph of CXCR4 expression of T47D cancer cells grown in NME7_{AB} or 2i in the presence or absence of anti-NME7 antibodies. FIG. 15B shows a graph of RT-PCR measurements of CXCR4, CHD1 and SOX2 expression in T47D breast cancer cells that were grown in 2i inhibitors for 72 hours or 144 hours, in the presence of NME7_{AB} immunizing peptides and shows the peptides are themselves inhibitory to the metastatic transformation. Peptides A1, A2 and B1 which were used in the inhibitory Combo 2 and 3 in FIG. 15A are also inhibitory as peptides. Peptide B3 is the most inhibitory and is the immunizing peptide for antibody

61 which was the most inhibitory antibody tested in FIG. 15A. FIG. 15C shows the graph of FIG. 15B with the scale of the Y-axis reduced.

FIG. 16 shows a table of recorded RNA levels in samples that were used for RT-PCR measurement of CXCR4 in FIG. 31 as well as the threshold cycle number for CXCR4 expression as well as for the control housekeeping gene.

FIG. 17 shows a graph of RT-PCR measurement of the expression of NME7-X1 in a panel of human stem cells and cancer cells.

FIG. 18 shows a graph of RT-PCR measurement of the expression of NME7. NME7a, NME7b and NME7-X1 in a panel of human stem cells and cancer cells, NME7a is full-length NME7, NME7b is missing a small portion of the DM10 domain, NME7-X1 is missing all of the DM10 domain and a small portion of the N-terminus of the first NDPK A domain. The bar labeled NME7 means that primers were used that detected both NME7a and NME7b.

FIGS. 19A-19F show photographs of Western blots in which various cancer cell lines are probed for expression of NME7 species using antibodies generated by immunization with NME7 derived peptides. FIG. 19A shows Western blot wherein antibody 52 that binds to the A1 peptide is used to probe a panel of cells for the presence of full-length NME7, NME7_{AB} or NME7-X1. FIG. 19B shows Western blot wherein antibody 56 that binds to the B1 peptide is used to probe a panel of cells for the presence of full-length NME7, NME7_{AB} or NME7-X1. FIG. 19C shows Western blot wherein antibody 61 that binds to the B3 peptide is used to probe a panel of cells for the presence of full-length NME7, NME7_{AB} or NME7-X1. FIG. 19D shows Western blot wherein commercially available polyclonal antibody H278, raised against both the NME7 A and B domain, is used to probe a panel of cells for the presence of NME7. As the figure shows, antibody H278 also recognizes NME1. FIG. 19E shows a gel published on website for commercially available anti-NME7 antibody B9, showing it binds to a species with an apparent molecular weight of full-length NME7. FIG. 19F shows a Western blot in which we used anti-NME7 antibody B9 to probe a gel that was loaded only with NME1. As can be seen in the figure, antibody B9 recognizes NME1 as well as full-length NME7. This is not surprising because like antibody H278, B9 was raised against both A and B domains of NME7 where the A domain of NME1 is highly homologous to the A domain of NME7_{AB}.

FIGS. 20A-20C show graphs of RT-PCR measurements of metastatic markers in cancer cells after being cultured in a serum-free media containing NME7_{AB} compared to the standard media. FIG. 20A shows SK-OV3, a MUC1-positive ovarian cancer cell line increased expression of metastatic markers CXCR4. CDH1 aka E-cadherin, SOX2 and NME7-X1: FIG. 20B shows OV-90 a MUC1-negative ovarian cancer cell line increased expression of metastatic markers CXCR4 and NME7-X1: FIG. 20C shows MDA-MB 55 a breast cancer cell line that expresses minimal levels of MUC1 increased expression of metastatic markers CDH1 aka E-cadherin and SOX2.

FIGS. 21A-21F show photographs of Western blots and description of cancer cell lines analyzed. For Western blots in FIGS. 21A and 21B, all cancer samples were normalized such that they were loaded onto gel at a concentration of 40 ug/mL. In FIG. 21A, various cancer cell lines are probed for the expression of full-length MUC1 using an anti-tandem repeat monoclonal antibody VU4H5. In FIG. 21B, various cancer cell lines are probed for the expression of cleaved form MUC1* using a polyclonal anti-PSMGFR antibody.

FIG. 21C is a description of the cancer cell lines analyzed. FIG. 21D shows that HER2 positive BT474 breast cancer cells, marked "BT474 (parent cells)" express little to no MUC1 or MUC1* until they acquire resistance to Herceptin and other chemotherapy drugs, marked "BTRes1" in figure. Parent cells were made resistant to Herceptin. Taxol. Doxorubicin and cyclophosphamide by culturing the cells in sub-lethal levels of Herceptin. FIG. 21D shows that the expression level of HER2 has not changed but expression of MUC1* has dramatically increased as the cells acquired resistance to Herceptin. FIG. 21E shows a graph of the growth of the parent BT474 cells compared to the drug resistant metastatic cells in response to treatment with Herceptin in the presence or absence of an anti-MUC1* Fab. As can be seen in the figure, the BT474 parent cells show a Herceptin concentration dependent decrease in cell growth, whereas the two Herceptin resistant cell lines. BTRes 1 and BTRes2, show no decrease in cancer cell growth in response to treatment with Herceptin. However, when treated with an anti-MUC1* Fab, the resistant cell lines show a Herceptin concentration dependent decrease in cancer cell growth. FIG. 21F shows a graph of the percent cell death of the parent BT474 cells compared to the drug resistant BTRes1 cells, in response to treatment with Taxol in the presence or absence of an anti-MUC1* Fab.

FIGS. 22A-22E show photographs of Western blots of a co-immunoprecipitation experiment. T47D breast cancer cell extracts were incubated with an antibody against the MUC1 cytoplasmic tail, Ab-5, or a control antibody, IgG, and co-immunoprecipitated. The gels were blotted with two different commercially available anti-NME7 antibodies B9 (FIG. 22A) and CF7 (FIG. 22B). Both gels show unique NME7 bands at ~33 kDa and ~30 kDa. The gels were stripped and re-probed with an antibody against the extracellular domain of MUC1*, anti-PSMGFR (FIG. 22C) and (FIG. 22D), which shows that the NME7 species and MUC1* interact. A recombinant NME7_{AB} and a recombinant NME7-X1 were mixed together and run on a gel, then probed with an anti-NME7 antibody, showing that the two unique NME7 species that are naturally occurring in breast cancer cells and that interact with MUC1* are an NME7_{AB}-like species and NME7-X1 (FIG. 22E).

FIGS. 23A-23C show photographs of Western blots of a co-immunoprecipitation experiment. Human induced pluripotent stem, iPS7, or embryonic stem, HES3, cell extracts were incubated with an antibody against the MUC1 cytoplasmic tail, Ab-5, or a control antibody, IgG, and co-immunoprecipitated. The gel was blotted with a commercially available anti-NME7 antibody B9 (FIG. 23A). Both cell types show unique NME7 bands at ~33 kDa and ~30 kDa. The gel was stripped and re-probed with an antibody against the extracellular domain of MUC1*, anti-PSMGFR (FIG. 23B), which shows that the NME7 species and MUC1* interact. A recombinant NME7_{AB} and a recombinant NME7-X1 were mixed together and run on a gel, then probed with an anti-NME7 antibody, showing that the two unique NME7 species that are naturally occurring in breast cancer cells and that interact with MUC1* are an NME7_{AB}-like species and NME7-X1 (FIG. 23C).

FIG. 24 shows a graph of an ELISA experiment assaying new anti-NME7 antibodies for their ability to bind to NME7_{AB}. NME7_{AB} is known to bind to the extra cellular domain of MUC1*. The surface of the multi-well plate was coated with a recombinant NME7_{AB}. Anti-NME7_{AB} antibodies were separately added to wells. Standard washes were performed and visualized by adding an HRP-conjugated

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secondary antibody. As can be seen, 7 of the 10 new anti-NME7 antibodies bound strongly to NME7_{AB}.

FIG. 25 shows a graph of an ELISA experiment assaying new anti-NME7 antibodies for their ability, or preferably inability, to bind to NME1. The surface of the multi-well plate was coated with a recombinant NME1-S120G dimers, which are also known to bind to the MUC1* extra cellular domain. Anti-NME7_{AB} antibodies were separately added to wells. Standard washes were performed and visualized by adding an HRP-conjugated secondary antibody. As can be seen only one antibody showed just minimal binding to NME1.

FIG. 26 shows a graph of an ELISA competitive inhibition assay, NME7_{AB}/anti-NME7 antibody complexes were made before adding to a multi-well plate coated with MUC1* extra cellular domain peptide. PSMGFR. Recall that NME7_{AB} has two pseudo-identical domains A and B that are each able to bind to MUC1* extra cellular domain. Antibodies that bind to the NME7 B3 peptide, which is in the B domain, do not bind to the NME7 A domain. Therefore, only partial inhibition of the NME7_{AB}/MUC1* interaction is expected.

FIG. 27 shows a graph of an ELISA displacement assay, NME7_{AB} was first bound to surface-immobilized MUC1* extra cellular domain peptide on the plate, then disrupted by the addition of anti-NME7 antibodies.

FIG. 28 shows a graph of an ELISA displacement assay. In this case, the multi-well plate was coated with a truncated MUC1* peptide, N-10, which has the 10 N-terminal amino acids missing of the PSMGFR sequence, NME7_{AB} is known to bind to the N-10 peptide. NME7_{AB} was bound to surface-immobilized N-10 peptide on the plate, then disrupted by the addition of anti-NME7 antibodies.

FIG. 29 shows a graph of the amount of RNA present in samples of T47D breast cancer cells were cultured in either their normal recommended media. RPMI, serum-free media containing only NME7_{AB} as the growth factor at 4 nM, which is optimal, or 8 nM, or serum-free media containing only NME1 S120G dimers as the growth factor at 8 nM: because NME1 is a homodimer and NME7_{AB} is a monomer comprised of two pseudo-identical domains, 8 nM NME1 is the molar equivalent of 4 nM NME7_{AB}. The cancer cells were cultured in the presence or absence of anti-NME7 B3 antibodies. In this experiment, floating cells were separated from the adherent cells and analyzed separately. Significant data argues that the floater cells are the cancer stem cells. An increase or decrease in the amount of RNA in a sample argues that an agent increased or decreased, respectively, the number of cells in a given population that were generated.

FIG. 30 shows a graph of a PCR measurement of metastatic marker CXCR4 in T47D breast cancer cells that were cultured in either their normal recommended media, RPMI, serum-free media containing only NME7_{AB} as the growth factor at 4 nM, which is optimal, or 8 nM, or serum-free media containing only NME1 S120G dimers as the growth factor at 8 nM; because NME1 is a homodimer and NME7_{AB} is a monomer comprised of two pseudo-identical domains, 8 nM NME1 is the molar equivalent of 4 nM NME7_{AB}. The cancer cells were cultured in the presence or absence of anti-NME7 B3 antibodies. In this experiment, floating cells were separated from the adherent cells and analyzed separately. Significant data argues that the floater cells are the cancer stem cells. As can be seen in the figure, growth in NME7_{AB} media increases CXCR4 in the floater population of cells and anti-NME7 B3 antibodies decreased its expression, arguing that anti-NME7 antibodies decreased generation of cancer stem cells.

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FIG. 31 shows a graph of a PCR measurement of stem cell marker and metastatic marker SOX2 in T47D breast cancer cells that were cultured in either their normal recommended media. RPMI, serum-free media containing only NME7_{AB} as the growth factor at 4 nM, which is optimal, or 8 nM, or serum-free media containing only NME1 S120G dimers as the growth factor at 8 nM: because NME1 is a homodimer and NME7_{AB} is a monomer comprised of two pseudo-identical domains, 8 nM NME1 is the molar equivalent of 4 nM NME7_{AB}. The cancer cells were cultured in the presence or absence of anti-NME7 B3 antibodies. In this experiment, floating cells were separated from the adherent cells and analyzed separately. Significant data argues that the floater cells are the cancer stem cells. As can be seen in the figure, growth in NME7_{AB} media increases SOX2 expression in the floater population of cells and anti-NME7 B3 antibodies decreased its expression, arguing that anti-NME7 antibodies decreased generation of cancer stem cells.

FIG. 32 shows a graph of a PCR measurement of stem cell marker and metastatic growth factor receptor MUC1 in T47D breast cancer cells that were cultured in either their normal recommended media. RPMI, serum-free media containing only NME7_{AB} as the growth factor at 4 nM, which is optimal, or 8 nM, or serum-free media containing only NME1 S120G dimers as the growth factor at 8 nM: because NME1 is a homodimer and NME7_{AB} is a monomer comprised of two pseudo-identical domains, 8 nM NME1 is the molar equivalent of 4 nM NME7_{AB}. The cancer cells were cultured in the presence or absence of anti-NME7 B3 antibodies. In this experiment, floating cells were separated from the adherent cells and analyzed separately. Significant data argues that the floater cells are the cancer stem cells. As can be seen in the figure, growth in NME7_{AB} media increases MUC1 expression in the floater population of cells and anti-NME7 B3 antibodies decreased its expression, arguing that anti-NME7 antibodies decreased generation of cancer stem cells.

FIGS. 33A-33B show IVIS photographs of immune compromised nu/nu mice Day 6 post tail vein injection of cancer cells. FIG. 33A shows IVIS photographs of mice injected with 500,000 T47D-wt breast cancer cells. FIG. 33B shows IVIS photographs of mice injected with 10,000 T47D breast cancer cells that were grown for 10 days in NME7_{AB} in a minimal media. The floating cells were collected. These floating cells are referred to herein as cancer stem cells, CSCs. As can be seen in the figure, the mice injected with wild type cancer cells show no signs of metastasis. However, the mice injected with 50-times less cells, but cancer stem cells, show that the injected cancer cells are clearly metastasizing.

FIGS. 34A-34D show IVIS photographs of immune compromised nu/nu mice Day 10 post tail vein injection of cancer cells. FIG. 34A shows IVIS photographs of mouse injected with 500,000 T47D-wt breast cancer cells. FIG. 34B shows IVIS photographs of mouse injected with 10,000 T47D-CSC (cancer stem cells). FIG. 34C shows IVIS photographs of mouse injected with 10,000 T47D-CSC (cancer stem cells) and injected on Day 7 with anti-NME7 antibody. FIG. 34D shows the hand recording of the IVIS measure of emitted photons. As can be seen in the figure, the mouse chosen for treatment is more metastatic than the comparable T47D-CSC mouse. The efficacy of the first antibody injection may have been blocked by the Day 6 injection of free NME7_{AB}. Control mouse injected with 500,000 T47D-wt cells shows some weak emission of photons that may be background or surviving cancer cells.

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FIG. 35A-35C shows IVIS photographs of immune compromised nu/nu mice Day 12 post tail vein injection of cancer cells. FIG. 35A shows IVIS photographs of mouse injected with 500,000 T47D-wt breast cancer cells. FIG. 35B shows that mouse injected with 10,000 T47D-CSC (cancer stem cells) that was not treated with anti-NME7 antibody died from excess tumor burden before IVIS photograph could be taken. FIG. 35C shows IVIS photographs of mouse injected with 10,000 T47D-CSC (cancer stem cells) and injected on Day 7 and Day 10 with anti-NME7 antibody. As can be seen in the figure, the mouse treated with anti-NME7 antibody is clearing away the cancer metastases. Control mouse injected with 500,000 T47D-wt cells shows less photon emissions indicating fewer surviving cancer cells or may be background.

FIGS. 36A-36B shows IVIS photographs of immune compromised nu/nu mice Day 14 post tail vein injection of cancer cells. FIG. 36A shows IVIS photographs of mouse injected with 500,000 T47D-wt breast cancer cells. FIG. 36B shows IVIS photographs of mouse injected with 10,000 T47D-CSC (cancer stem cells) and injected on Day 7, Day 10, and Day 12 with anti-NME7 antibody. As can be seen in the figure, the mouse treated with anti-NME7 antibody nearly completely free of cancer cell metastases. Control mouse injected with 500,000 T47D-wt cells shows no photon emissions.

FIGS. 37A-37V shows time course of IVIS photographs of immune compromised nu/nu mice from Day 6 to Day 26 post cancer cell tail vein injection. FIGS. 37A, 37C, 37E, 37G, 37I, 37K, 37M and 37O show IVIS photographs of mouse that had been injected Day 0 into the tail vein with 500,000 T47D-wt cells. FIGS. 37B, 37D, 37F, 37H, 37J, 37L, 37N and 37P show IVIS photographs of mouse that had been injected Day 0 into the tail vein with 10,000 T47D cancer stem cells, to which anti-NME7 antibody was administered from Day 7 to Day 17, whereupon treatment was suspended, then resumed on Day 21. FIGS. 37Q, 37R, 37S, 37T, and 37U show enlarged IVIS photographs of the treated mouse between Day 17, when anti-NME7 antibody treatment was suspended, through Day 21, when antibody treatment was resumed to Day 26. FIG. 37V shows the scale bar of the IVIS measurements. As can be seen in this time course, cancer cells that had been grown in NME7 readily metastasize and such metastasis can be effectively treated, prevented or reversed by treatment with an antibody that binds to NME7.

FIG. 38A-38C shows time course of IVIS photographs of immune compromised nu/nu mice from Day 6 to Day 19 post injection with either 500,000 T47D wild type breast cancer cells or 10,000 T47D cancer stem cells. FIG. 38A shows mice that were injected into the tail vein (i.v.). FIG. 38B shows mice that were injected intra-peritoneally (i.p.). FIG. 38C shows mice that were injected sub-cutaneously (s.c.).

FIGS. 39A-39C shows human lung tissue specimens stained with an anti-NME7 antibody that binds to the B3 peptide. The figure shows lack of NME7 expression on normal tissues, increasing expression of NME7 as tumor grade and metastasis increase.

FIGS. 40A-40C shows human small intestine tissue specimens stained with an anti-NME7 antibody that binds to the B3 peptide. The figure shows lack of NME7 expression on normal tissues, increasing expression of NME7 as tumor grade and metastasis increase.

FIGS. 41A-41D show human colon tissue specimens stained with an anti-NME7 antibody that binds to the B3 peptide. The figure shows lack of NME7 expression on

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normal tissues, increasing expression of NME7 as tumor grade and metastasis increase.

FIGS. 42A-42F shows photographs of female nu/nu mice weighing approximately 20 g each, which were injected into the tail vein with 10,000 Luciferase positive T47D metastatic breast cancer stem cells and treated with the anti-NME7_{AB} antibody 4A3 also known as 8F9A4A3. To image cancer cells, the Luciferase substrate, Luciferin, is intraperitoneally injected 10 minutes before being photographed in IVIS instrument. FIGS. 42A-42C show IVIS photographs with animals face down. FIG. 42D-42F show IVIS photographs with animals face up. FIGS. 42A and 42D show control animals injected with phosphate buffered saline solution. FIGS. 42B and 42E show a prevention model in which animals were injected with anti-NME7_{AB} antibody 4A3 24 hrs before injection of the metastatic cancer cells, then approximately every other day for a total of 12 antibody injections over 22 days. FIGS. 42C and 42F show a reversal model in which animals were injected with anti-NME7_{AB} antibody 4A3 24 hrs after injection of the metastatic cancer cells, then approximately every other day for a total of 11 antibody injections over 20 days.

FIGS. 43A-43F shows photographs of female nu/nu mice weighing approximately 20 g each, which were injected into the tail vein with 10,000 Luciferase positive T47D metastatic breast cancer stem cells and treated with the anti-NME7_{AB} antibodies 5A1, also known as 8F9A5A1, or 5D4, also known as 5F3A5D4. To image cancer cells, the Luciferase substrate, Luciferin, is intraperitoneally injected 10 minutes before being photographed in IVIS instrument. FIGS. 43A-43C show IVIS photographs with animals face down. FIGS. 43D-43F show IVIS photographs with animals face up. FIGS. 43A and 43D show control animals injected with phosphate buffered saline solution. FIGS. 43B, 43E, 43C and 43F show a prevention model in which animals were injected with anti-NME7_{AB} antibodies 24 hours before injection of the metastatic cancer cells, then approximately every other day for a total of 12 antibody injections over 22 days. Images were taken on Day 27.

FIGS. 44A-44D shows photographs of female nu/nu mice that on Day (were injected into the tail vein with 10,000 Luciferase positive T47D metastatic breast cancer stem cells mixed with NME7_{AB} to a final concentration of 32 nM. On Day 1 and Day 2 animals were injected into the tail vein with more 32 nM NME7_{AB}, which we have shown increases metastases. This is a system to demonstrate reversion of established metastases. On Day 7 animals were treated with individual anti-NME7_{AB} antibodies 8F9A5A1, 8F9A4A3, or 5F3A5D4. FIG. 44A shows control animals injected with phosphate buffered saline solution. FIG. 44B shows animals treated with anti-NME7_{AB} monoclonal antibody 8F9A5A1, also known as 5A1. FIG. 44C shows animals treated with anti-NME7_{AB} monoclonal antibody 8F9A4A3, also known as 4A3. FIG. 44D shows animals treated with anti-NME7_{AB} monoclonal antibody 5F3A5D4, also known as 5D4. Green arrows indicate low antibody dosage (5-7 mg/kg) over the indicated period and Red arrows indicate high dosage (15 mg/kg). As can be seen in the figure, animals treated with anti-NME7_{AB} antibodies have less metastases than the control animals even though many of the animals in the groups to be treated with antibody have more metastasis before any treatment. Higher concentrations of anti-NME7_{AB} antibody are more effective than low concentrations. For example between Day 11 and Day 17, animals were treated with high dose and most of the treated animals have cleared metastases

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by about Day 17. However, 1 low dose of antibody resulted in metastasis recurrence. Animals again respond to high dose treatment by Day 32.

FIGS. 45A-45B shows photographs of female nu/nu mice that on Day 0 were injected sub-cutaneously into the right flank with 10,000 Luciferase positive T47D metastatic breast cancer stem cells, mixed with NME7_{AB} to a final concentration of 32 nM, then mixed in a 1:1 vol:vol with Matrigel. Tumor engraftment was allowed to progress Day 0-Day 6. Animals were then treated i.v. by tail vein injection with anti-NME7_{AB} antibodies. Control animals were injected with PBS. FIG. 45A shows IVIS photographs of control animals. FIG. 45B shows IVIS photographs of animals injected into tail vein with a cocktail of anti-NME7_{AB} antibodies 5A1, 4A3 and 5D4 to a total concentration of 15 mg/kg. Antibodies or PBS were administered 4 times between Day 7 and Day 18. As can be seen in the figure, the anti-NME7_{AB} antibody treated animals show less metastases than the control group. In the treated group, 2 of the 5 animals have primary tumors that are larger than those in the control group. This could be because the anti-NME7_{AB} antibodies prevented the spread of the cancer cells, so they remained concentrated in the primary tumor. In this experiment, PCR analysis showed that after 11 days in culture with NME7_{AB}, the T47D breast cancer cells had upregulated CXCR4 by 109-fold, OCT4 by 2-fold, NANOG by 3.5-fold and MUC1 by 2.7-fold.

FIGS. 46A-46Q shows photographs of female nu/nu mice that on Day 0 were injected sub-cutaneously into the right flank with 10,000 Luciferase positive T47D metastatic breast cancer stem cells, mixed with NME7_{AB} to a final concentration of 32 nM, then mixed in a 1:1 vol:vol with Matrigel. Tumor engraftment was allowed to progress Day 0-Day 6. Animals were then treated i.v. by tail vein injection, with anti-NME7_{AB} antibodies. Control animals were injected with PBS. On Day 38 animals were sacrificed and livers harvested then analyzed by IVIS to detect cancer cells that had metastasized to the liver. FIGS. 46A-46B show whole body IVIS photographs of control animals that were injected with only PBS. FIGS. 46C-46D show whole body IVIS photographs of control animals that were injected with the anti-NME7_{AB} antibody 5A1. FIGS. 46E-46F show whole body IVIS photographs of control animals that were injected with the anti-NME7_{AB} antibody 4A3. FIGS. 46G-46H show whole body IVIS photographs of control animals that were injected with the anti-NME7_{AB} antibody 5D4. FIGS. 46A, 46C, 46E, and 46G are IVIS photographs taken at Day 7 before any treatment. FIGS. 46B, 46D, 46F, and 46H are IVIS photographs taken at Day 31 after anti-NME7_{AB} antibody treatment or mock treatment. As can be seen in the figure, animals in the PBS control group show metastasis (blue dots) in the whole body IVIS photographs, while animals treated with anti-NME7_{AB} antibodies do not. FIGS. 46I-46P show photographs and IVIS photographs of livers and lung harvested from animals after sacrifice. FIGS. 46I, 46K, 46M, and 46O are regular photographs. FIGS. 46J, 46L, 46N, and 46P are IVIS photographs, illuminating the cancer cells that have metastasized there. As can be seen in the figure, the anti-NME7_{AB} antibodies greatly inhibited metastasis to the liver, which is a primary site for breast cancer metastasis. FIG. 46Q is a bar graph of the measured photons emitted and enumerated by IVIS instrument for livers harvested from control animals versus the treated animals.

FIGS. 47A-47F shows photographs of immunofluorescent experiments in which various cancer cell lines are stained for the presence of NME7_{AB}. FIG. 47A shows T47D

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breast cancer cells stained with varying concentrations of anti-NME7_{AB} antibody 5D4. FIG. 47B shows ZR-75-1 breast cancer cells, also known as 1500s, stained with varying concentrations of anti-NME7_{AB} antibody 5D4. FIG. 47C shows H1975 non-small cell lung cancer cells stained with varying concentrations of anti-NME7_{AB} antibody 5D4. FIG. 47D shows H292 non-small cell lung cancer cells stained with varying concentrations of anti-NME7_{AB} antibody 5D4. FIG. 47E shows HPAFII pancreatic cancer cells stained with varying concentrations of anti-NME7_{AB} antibody 5D4. FIG. 47F shows DU145 prostate cancer cells stained with varying concentrations of anti-NME7_{AB} antibody 5D4. As can be seen in the figure, all the cancer cell lines we tested show strong and membranous staining for NME7_{AB}. The monoclonal antibody used in these experiments was 5D4. In parallel, NME7_{AB} antibodies 5A1 and 4A3 were used to stain the same cell lines and produced the same results.

FIGS. 48A-48I shows photographs of immunofluorescent experiments in which various lung cancer cell lines are stained for the presence of NME7_{AB}. FIGS. 48A-48C shows H1975 non-small cell lung cancer cells, which are an adenocarcinoma, stained with varying concentrations of anti-NME7_{AB} antibody 5D4. FIG. 48A is an overlay of DAPI and anti-NME7_{AB} staining. FIG. 48B shows anti-NME7_{AB} staining alone. FIG. 48C is a magnified view of the overlay of DAPI and anti-NME7_{AB} staining. FIGS. 48D-48F shows H292 non-small cell lung cancer cells, which are a mucoepidermoid pulmonary carcinoma, stained with varying concentrations of anti-NME7_{AB} antibody 5D4. FIG. 48D is an overlay of DAPI and anti-NME7_{AB} staining. FIG. 48E shows anti-NME7_{AB} staining alone. FIG. 48F is a magnified view of the overlay of DAPI and anti-NME7_{AB} staining. FIGS. 48G-48I shows H358 non-small cell lung cancer cells, which are a metastatic bronchioalveolar carcinoma, stained with varying concentrations of anti-NME7_{AB} antibody 5D4. FIG. 48G is an overlay of DAPI and anti-NME7_{AB} staining. FIG. 48H shows anti-NME7_{AB} staining alone. FIG. 48I is a magnified view of the overlay of DAPI and anti-NME7_{AB} staining.

FIG. 49A-49I shows PCR graphs of cancer cell lines, breast T47D, Lung H1975, lung H358 and pancreatic HPAFII before and after culture in NME7_{AB}. FIG. 49A measured breast metastatic marker CXCR4. FIG. 49B measured stem cell marker OCT4. FIG. 49C measured metastatic marker ALDH1. FIG. 49D measured stem cell marker SOX2. FIG. 49E measured stem cell marker NANOG. FIG. 49F measured marker CDH1, also known as E-cadherin. FIG. 49G measured metastatic marker CD133. FIG. 49H measured stem cell marker ZEB2. FIG. 49I measured stem, cancer and metastatic marker MUC1. The floater cells, also known as tumor spheres become able to grow anchorage independently and show markers of metastasis that are more elevated than the adherent cells. Animals injected with cancer stem cells are those injected with the NME7_{AB} grown floater cells. As can be seen in the figure markers of metastasis, stem cell markers, or markers of epithelial to mesenchymal transition (EMT) are elevated after culture in NME7_{AB}, indicating a transition to a more metastatic state.

FIG. 50A-50D shows IVIS photographs of NSG mice injected into the tail vein with 10,000 cancer cells that were either NCI-H358 parent cells or NCI-H358 cells after 10 days in culture with NME7_{AB}. FIGS. 50A and 50C show IVIS photographs of the mouse that was injected with the NCI-H358 lung cancer cells that had been grown in NME7_{AB} for 10 days. FIGS. 50B and 50D show IVIS photographs of the mouse that was injected with the parental

NCI-H358 cells. FIGS. 50A and 50B show the IVIS photographs where mice are imaged face down. FIGS. 50C and 50D show the IVIS photographs where mice are imaged face up. As can be seen in the figure, the NME7_{AB} grown cells have greatly increased metastatic potential.

FIG. 51 shows PCR graph of a MUC1 negative prostate cancer line PC3 before and after 2 or 3 passages in culture in either dimeric NM23-H1, also known as NME1, or NME7_{AB}. The graph shows the fold difference in markers of stem cells, cancer cells as well as metastatic markers. As can be seen in the figure, repeated culture in NME1 or NME7_{AB} induces upregulation of stem, cancer and metastatic markers but also upregulates expression of MUC1 by 5-8 times.

FIGS. 52A-52E shows photographs of female nu/nu mice that on Day (were injected into the tail vein with 10,000 Luciferase positive T47D metastatic breast cancer stem cells mixed with NME7_{AB} to a final concentration of 32 nM. On Day 1 and Day 2 animals were injected into the tail vein with more 32 nM NME7_{AB}, which we have shown increases metastases. This is a system to demonstrate reversion of established metastases. On Day 5 animals were injected intravenously with 15 mg/kg of 8F9A4A3, aka 4A3, wherein the antibody was either the murine sequence or humanized 4A3 variant 4.8, or humanized 4A3 variant 12.3. FIG. 52A shows control animals injected with phosphate buffered saline solution. FIG. 52B shows animals treated with anti-NME7_{AB} humanized 4A3 4.8. FIG. 52C shows animals treated with anti-NME7_{AB} humanized 4A3 12.3. FIG. 52D shows animals treated with anti-NME7_{AB} monoclonal antibody murine 4A3. FIG. 52E shows a graph of the quantification of bioluminescence measurement of tumor volume, taken in an IVIS instrument. Animals were injected with either PBS or an anti-NME7 antibody at 15 mg/kg once every 3 days. Antibody hu 4A3 4.8 denotes that the sequences derived from human heavy chain antibody sequence 1.46 (SEQ ID NO: 1102) and light chain sequence 1.6 (SEQ ID NO: 1104) were used. Antibody hu 4A3 12.3 denotes that the sequences derived from human heavy chain antibody sequence 4.4 (SEQ ID NO: 1106) and light chain sequence 4.1 (SEQ ID NO: 1108) were used. Note that half the animals treated with hu 4A3 4.8 were given a partial half dose.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Definitions

In the present application, “a” and “an” are used to refer to both single and a plurality of objects.

As used herein, “about” or “substantially” generally provides a leeway from being limited to an exact number. For example, as used in the context of the length of a polypeptide sequence, “about” or “substantially” indicates that the polypeptide is not to be limited to the recited number of amino acids. A few amino acids add to or subtracted from the N-terminus or C-terminus may be included so long as the functional activity such as its binding activity is present.

As used herein, administration “in combination with” one or more further therapeutic agents includes simultaneous (concurrent) and consecutive administration in any order.

As used herein, “amino acid” and “amino acids” refer to all naturally occurring L- α -amino acids. This definition is meant to include norleucine, ornithine, and homocysteine.

As used herein, in general, the term “amino acid sequence variant” refers to molecules with some differences in their amino acid sequences as compared to a reference (e.g. native

sequence) polypeptide. The amino acid alterations may be substitutions, insertions, deletions or any desired combinations of such changes in a native amino acid sequence.

Substitutional variants are those that have at least one amino acid residue in a native sequence removed and a different amino acid inserted in its place at the same position. The substitutions may be single, where only one amino acid in the molecule has been substituted, or they may be multiple, where two or more amino acids have been substituted in the same molecule.

Substitutes for an amino acid within the sequence may be selected from other members of the class to which the amino acid belongs. For example, the nonpolar (hydrophobic) amino acids include alanine, leucine, isoleucine, valine, proline, phenylalanine, tryptophan and methionine. The polar neutral amino acids include glycine, serine, threonine, cysteine, tyrosine, asparagine and glutamine. The positively charged (basic) amino acids include arginine, lysine and histidine. The negatively charged (acidic) amino acids include aspartic acid and glutamic acid. Also included within the scope of the invention are proteins or fragments or derivatives thereof which exhibit the same or similar biological activity and derivatives which are differentially modified during or after translation, e.g., by glycosylation, proteolytic cleavage, linkage to an antibody molecule or other cellular ligand, and so on.

Insertional variants are those with one or more amino acids inserted immediately adjacent to an amino acid at a particular position in a native amino acid sequence. Immediately adjacent to an amino acid means connected to either the α -carboxy or α -amino functional group of the amino acid.

Deletional variants are those with one or more amino acids in the native amino acid sequence removed. Ordinarily, deletional variants will have one or two amino acids deleted in a particular region of the molecule.

As used herein, “fragments” or “functional derivatives” refers to biologically active amino acid sequence variants and fragments of the polypeptide of the present invention, as well as covalent modifications, including derivatives obtained by reaction with organic derivatizing agents, post-translational modifications, derivatives with nonproteinaceous polymers, and immunoadhesins.

As used herein, “carriers” include pharmaceutically acceptable carriers, excipients, or stabilizers which are non-toxic to the cell or mammal being exposed thereto at the dosages and concentrations employed. Often the pharmaceutically acceptable carrier is an aqueous pH buffered solution. Examples of pharmaceutically acceptable carriers include without limitation buffers such as phosphate, citrate, and other organic acids; antioxidants including ascorbic acid; low molecular weight (less than about 10 residues) polypeptide; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, arginine or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; sugar alcohols such as mannitol or sorbitol; salt-forming counterions such as sodium; and/or nonionic surfactants such as TWEEN®, polyethylene glycol (PEG), and PLURONICS®.

As used herein “pharmaceutically acceptable carrier and/or diluent” includes any and all solvents, dispersion media, coatings antibacterial and antifungal agents, isotonic and absorption delaying agents and the like. The use of such media and agents for pharmaceutical active substances is well known in the art. Except insofar as any conventional

media or agent is incompatible with the active ingredient, use thereof in the therapeutic compositions is contemplated. Supplementary active ingredients can also be incorporated into the compositions.

It is especially advantageous to formulate parenteral compositions in dosage unit form for ease of administration and uniformity of dosage. Dosage unit form as used herein refers to physically discrete units suited as unitary dosages for the mammalian subjects to be treated; each unit containing a predetermined quantity of active material calculated to produce the desired therapeutic effect in association with the required pharmaceutical carrier. The specification for the dosage unit forms of the invention are dictated by and directly dependent on (a) the unique characteristics of the active material and the particular therapeutic effect to be achieved, and (b) the limitations inherent in the art of compounding such an active material for the treatment of disease in living subjects having a diseased condition in which bodily health is impaired.

The principal active ingredient is compounded for convenient and effective administration in effective amounts with a suitable pharmaceutically acceptable carrier in dosage unit form. A unit dosage form can, for example, contain the principal active compound in amounts ranging from 0.5 µg to about 2000 mg. Expressed in proportions, the active compound is generally present in from about 0.5 µg/ml of carrier. In the case of compositions containing supplementary active ingredients, the dosages are determined by reference to the usual dose and manner of administration of the said ingredients.

As used herein, "vector", "polynucleotide vector", "construct" and "polynucleotide construct" are used interchangeably herein. A polynucleotide vector of this invention may be in any of several forms, including, but not limited to, RNA, DNA, RNA encapsulated in a retroviral coat, DNA encapsulated in an adenovirus coat, DNA packaged in another viral or viral-like form (such as herpes simplex, and adeno-structures, such as polyamides).

As used herein, "host cell" includes an individual cell or cell culture which can be or has been a recipient of a vector of this invention. Host cells include progeny of a single host cell, and the progeny may not necessarily be completely identical (in morphology or in total DNA complement) to the original parent cell due to natural, accidental, or deliberate mutation and/or change.

As used herein, "subject" is a vertebrate, preferably a mammal, more preferably a human.

As used herein, "mammal" for purposes of treatment refers to any animal classified as a mammal, including humans, domestic and farm animals, and 200, sports, or pet animals, such as dogs, cats, cattle, horses, sheep, pigs, and so on. Preferably, the mammal is human.

As used herein, "treatment" is an approach for obtaining beneficial or desired clinical results. For purposes of this invention, beneficial or desired clinical results include, but are not limited to, alleviation of symptoms, diminishment of extent of disease, stabilized (i.e., not worsening) state of disease, delay or slowing of disease progression, amelioration or palliation of the disease state, and remission (whether partial or total), whether detectable or undetectable, "Treatment" can also mean prolonging survival as compared to expected survival if not receiving treatment, "Treatment" refers to both therapeutic treatment and prophylactic or preventative measures. Those in need of treatment include those already with the disorder as well as those in which the disorder is to be prevented, "Palliating" a disease means that the extent and/or undesirable clinical manifestations of a

disease state are lessened and/or the time course of the progression is slowed or lengthened, as compared to a situation without treatment.

As used herein, "A1" peptide, "A2" peptide, "B1" peptide, "B2" peptide and "B3" peptide refer to peptides derived from NME7 that are used to generate or select antibodies that bind to human NME7_{AB}, but not (or significantly less) to human NME1. The peptides used to generate these antibodies are common to both NME7_{AB} and NME7-X1, and are set forth as below:

A1 is NME7A peptide 1 (A domain):
MLSRKEALDFHVDHQS (SEQ ID NO: 141)

A2 is NME7A peptide 2 (A domain): SGVARTDASES
(SEQ ID NO:142)

B1 is NME7B peptide 1 (B domain): DAGFEI-SAMQMFNMDRVNVE (SEQ ID NO: 143)

B2 is NME7B peptide 2 (B domain):
EVYKGVVTEYHDMVTE (SEQ ID NO: 144)

B3 is NME7B peptide 3 (B domain): AIFGKTKIQ-
NAVHCTDLPEDGLLEVQYFF (SEQ ID NO:145)

Further, for the sake of clarity, NME7A (with capital letter "A") refers to the subunit A portion of NME7, NME7a (with small letter "a") refers to the full-length NME7 that is described elsewhere in this application. And, NME7B (with capital letter "B") refers to the subunit B portion of NME7, NME7b (with small letter "b") refers to a species of NME7 that is partially devoid of the DM10 region, which is described elsewhere in this application.

As used herein, the term "antibody-like" means a molecule that may be engineered such that it contains portions of antibodies but is not an antibody that would naturally occur in nature. Examples include but are not limited to CAR (chimeric antigen receptor) T cell technology and the Ylanthia® technology. The CAR technology uses an antibody epitope fused to a portion of a T cell so that the body's immune system is directed to attack a specific target protein or cell. The Ylanthia® technology consists of an "antibody-like" library that is a collection of synthetic human fabs that are then screened for binding to peptide epitopes from target proteins. The selected Fab regions can then be engineered into a scaffold or framework so that they resemble antibodies.

As used herein, an "effective amount of an agent to inhibit an NME family member protein" refers to the effective amount of the agent in hindering the activating interaction between the NME family member protein and its cognate receptor such as

As used herein, "NME derived fragment" refers to a peptide sequence that is either a fragment of the NME or is highly homologous to the peptide sequence that is a fragment of the NME.

As used herein, the "MUC1*" extra cellular domain is defined primarily by the PSMGFR sequence (GTINVHD-VETQFNQYKTEAAS-

RYNLITISDVSVSDVPFPFSAQSGA (SEQ ID NO:6)). Because the exact site of MUC1 cleavage depends on the enzyme that clips it, and that the cleavage enzyme varies depending on cell type, tissue type or the time in the evolution of the cell, the exact sequence of the MUC1* extra cellular domain may vary at the N-terminus.

As used herein, the term "PSMGFR" is an acronym for Primary Sequence of MUC1 Growth Factor Receptor as set forth as GTINVHDVETQFNQYKTEAAS-RYNLTISDVSVSDVPFPFSAQSGA (SEQ ID NO:6). In this regard, the "N-number" as in "N-10 PSMGFR" or simply "N-10", "N-15 PSMGFR" or simply "N-15", or "N-20 PSMGFR" or simply "N-20" refers to the number of

amino acid residues that have been deleted at the N-terminal end of PSMGFR. Likewise “C-number” as in “C-10 PSMGFR” or simply “C-10”, “C-15 PSMGFR” or simply “C-15”, or “C-20 PSMGFR” or simply “C-20” refers to the number of amino acid residues that have been deleted at the C-terminal end of PSMGFR. A mixture of deletions and additions is also possible. For instance, N+20/C-27 refers to a peptide fragment of wild-type MUC1 in which 20 amino acids are added to the PSMGFR at the N-terminus and 27 amino acids are deleted from the C-terminus.

As used herein, the “extracellular domain of MUC1*” refers to the extracellular portion of a MUC1 protein that is devoid of the tandem repeat domain. In most cases, MUC1* is a cleavage product wherein the MUC1* portion consists of a short extracellular domain devoid of tandem repeats, a transmembrane domain and a cytoplasmic tail. The precise location of cleavage of MUC1 is not known perhaps because it appears that it can be cleaved by more than one enzyme. The extracellular domain of MUC1* will include most of the PSMGFR sequence but may have an additional 10-20 N-terminal amino acids.

As used herein, “high homology” is considered to be at least 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 97% identity in a designated overlapping region between any two polypeptides.

As used herein, “NME family proteins” or “NME family member proteins”, numbered 1-10, are proteins grouped together because they all have at least one NDPK (nucleotide diphosphate kinase) domain. In some cases, the NDPK domain is not functional in terms of being able to catalyze the conversion of ATP to ADP, NME proteins were formerly known as NM23 proteins, numbered H1 and H2. Recently, as many as ten (10) NME family members have been identified. Herein, the terms NM23 and NME are interchangeable. Herein, terms NME1, NME2, NME5, NME6, NME7, NME8 and NME9 are used to refer to the native protein as well as NME variants. In some cases these variants are more soluble, express better in *E. coli* or are more soluble than the native sequence protein. For example, NME7 as used in the specification can mean the native protein or a variant, such as NME7_{AB} that has superior commercial applicability because variations allow high yield expression of the soluble, properly folded protein in *E. coli*, NME7_{AB} consists primarily of the NME7 A and B domains but is devoid of most of the DM10 domain (SEQ ID NO:39), which is at the N-terminus of the native protein, “NME1” as referred to herein is interchangeable with “NM23-H1”. It is also intended that the invention not be limited by the exact sequence of the NME proteins. The mutant NME1-S120G, also called NM23-S120G, are used interchangeably throughout the application. The S120G mutants and the P96S mutant are preferred because of their preference for dimer formation, but may be referred to herein as NM23 dimers, NME1 dimers, or dimeric NME1, or dimeric NM23.

NME7 as referred to herein is intended to mean native NME7 having a molecular weight of about 42 kDa.

A “family of NME7” refers to full length NME7 as well as naturally occurring or artificially created cleaved form having a molecular weight about 30 kDa, 33 kDa, or a cleaved form having a molecular weight of about 25 kDa, a variant devoid or partially devoid of the DM10 leader sequence (SEQ ID NO:162), which is NME7 amino acids 1-91 of NME7 represented by SEQ ID NO:82 or 147, such as NME7b, NME7-X1, NME7_{AB} or a recombinant NME7 protein, or variants thereof whose sequence may be altered to allow for efficient expression or that increase yield,

solubility or other characteristics that make the NME7 more effective or commercially more viable. The “family of NME7” may also include “NME7_{AB}-like” protein, which is a protein in the range of 30 to 33 kDa that is expressed in cancer cells.

As used herein, an “an agent that maintains stem cells in the naïve state or reverts primed stem cells to the naïve state” refers to a protein, small molecule or nucleic acid that alone or in combination maintains stem cells in the naïve state, resembling cells of the inner cell mass of an embryo. Examples include but are not limited to human NME1 dimers, bacterial, fungal, yeast, viral or parasitic NME proteins that have high sequence identity to human NME proteins, especially NME1, NME7, NME7-X1, NME7_{AB}, NME6, 2i (Silva J et al, 2008; Hanna et al, 2010), 5i (Theunissen T W et al, 2014), nucleic acids such as siRNA that suppress expression of MBD3, CHD4 (Rais Y1 et al, 2013), BRD4, or JMJD6 (Liu W et al 2013).

As used herein, the terms “NME7_{AB}”, “NME7AB” and “NME-AB” are used interchangeably.

As used herein, an “an agent that promotes pluripotency” or “reverts somatic cells to a stem-like or cancer-like state” refers to a protein, small molecule or nucleic acid that alone or in combination induces expression of or suppresses expression of certain genes such that the genetic signature shifts to one that more closely resembles stem cells or cancer cells. Examples include but are not limited to NME1 dimers, NME7, NME7-X1, NME7_{AB}, 2i, 5i, nucleic acids such as siRNA that suppress expression of MBD3, CHD4, BRD4, or JMJD6, microbial NME proteins that have high sequence homology to human NME1, NME2, NME5, NME6, NME7, NME8, or NME9, preferably with the regions that house NDPK domains.

As used herein, in reference to an agent being referred to as a “small molecule”, it may be a synthetic chemical or chemically based molecule having a molecular weight between 50 Da and 2000 Da, more preferably between 150 Da and 1000 Da, still more preferably between 200 Da and 750 Da.

As used herein, in reference to an agent being referred to as a “natural product”, it may be chemical molecule or a biological molecule, so long as the molecule exists in nature.

As used herein, FGF, FGF-2 or bFGF refer to fibroblast growth factor (Xu R H et al, 2005; Xu C et al, 2005).

As used herein, “Rho associated kinase inhibitors” may be small molecules, peptides or proteins (Rath N, et al, 2012). Rho kinase inhibitors are abbreviated here and elsewhere as ROCi or ROCKi, or Ri. The use of specific rho kinase inhibitors are meant to be exemplary and can be substituted for any other rho kinase inhibitor.

As used herein, the term “cancer stem cells” or “tumor initiating cells” refers to cancer cells that express levels of genes that have been linked to a more metastatic state or more aggressive cancers. The terms “cancer stem cells” or “tumor initiating cells” can also refer to cancer cells for which far fewer cells are required to give rise to a tumor when transplanted into an animal. Cancer stem cells and tumor initiating cells are often resistant to chemotherapy drugs.

As used herein, the terms “stem/cancer”, “cancer-like”, “stem-like” refers to a state in which cells acquire characteristics of stem cells or cancer cells, share important elements of the gene expression profile of stem cells, cancer cells or cancer stem cells. Stem-like cells may be somatic cells undergoing induction to a less mature state, such as increasing expression of pluripotency genes. Stem-like cells also refers to cells that have undergone some dedifferentia-

tion or are in a meta-stable state from which they can alter their terminal differentiation. Cancer like cells may be cancer cells that have not yet been fully characterized but display morphology and characteristics of cancer cells, such as being able to grow anchorage-independently or being able to give rise to a tumor in an animal.

As used herein, "spacers" or "linkers" of different lengths can be incorporated anywhere in the peptide. Spacer attachment is usually through an amide linkage but other functionalities are possible.

NME, NME7 and Protein Family of NME7

The present inventors discovered that NME7 and NME7-X1 are highly expressed in early human stem cells and also in most cancer cells (FIG. 17, FIG. 18, FIG. 19A-FIG. 19F, FIG. 22, FIG. 23, FIG. 39, FIG. 40, FIG. 41, FIG. 47, FIG. 48, FIG. 17 shows a graph of RT-PCR measurement of the expression of NME7-X1 in a panel of human stem cells and cancer cells. FIG. 18 shows a graph of RT-PCR measurement of the expression of NME7, NME7a, NME7b and NME7-X1 in a panel of human stem cells and cancer cells, NME7a is full-length NME7, NME7b is missing a small portion of the DM10 domain, NME7-X1 is missing all of the DM10 domain and a small portion of the N-terminus of the first NDPK A domain. The bar labeled NME7 means that primers were used that detected both NME7a and NME7b. FIGS. 19A-19F show photographs of Western blots in which various cancer cell lines are probed for expression of NME7 species using antibodies generated by immunization with NME7 derived short peptides. FIG. 19A shows Western probed with antibody of the invention #52 which binds to NME7 derived peptide A1. FIG. 19B shows Western probed with antibody of the invention #56 which binds to NME7 derived peptide B1. FIG. 19C shows Western probed with antibody of the invention #61 which binds to NME7 derived peptide B3. FIGS. 22A-22E show photographs of Western blots of a co-immunoprecipitation experiment. T47D breast cancer cell extracts were incubated with an antibody against the MUC1 cytoplasmic tail, Ab-5, or a control antibody, IgG, and co-immunoprecipitated. The gels were blotted with two different commercially available anti-NME7 antibodies B9 (FIG. 22A) and CF7 (FIG. 22B). Both gels show unique NME7 bands at ~33 kDa and ~30 kDa. The gels were stripped and re-probed with an antibody against the extracellular domain of MUC1*, anti-PSMGFR (FIG. 22C) and (FIG. 22D), which shows that the NME7 species and MUC1* interact. A recombinant NME7_{AB} and a recombinant NME7-X1 were mixed together and run on a gel, then probed with an anti-NME7 antibody, showing that the two unique NME7 species that are naturally occurring in breast cancer cells and that interact with MUC1* are an NME7_{AB}-like species and NME7-X1 (FIG. 22E). FIGS. 23A-23C show photographs of Western blots of a co-immunoprecipitation experiment. Human induced pluripotent stem, iPS7, or embryonic stem, HES3, cell extracts were incubated with an antibody against the MUC1 cytoplasmic tail, Ab-5, or a control antibody, IgG, and co-immunoprecipitated. The gel was blotted with a commercially available anti-NME7 antibody B9 (FIG. 23A). Both cell types show unique NME7 bands at ~33 kDa and ~30 kDa. The gel was stripped and re-probed with an antibody against the extracellular domain of MUC1*, anti-PSMGFR (FIG. 23B), which shows that the NME7 species and MUC1* interact. A recombinant NME7_{AB} and a recombinant NME7-X1 were mixed together and run on a gel, then probed with an anti-NME7 antibody, showing that the two unique NME7 species that are naturally occurring in breast cancer cells and that interact with MUC1* are an NME7_{AB}-like species and NME7-X1 (FIG.

23C). FIGS. 39A-39C shows human lung tissue specimens stained with an anti-NME7 antibody that binds to the B3 peptide. The figure shows lack of NME7 expression on normal tissues, increasing expression of NME7 as tumor grade and metastasis increase. FIGS. 40A-40C shows human small intestine tissue specimens stained with an anti-NME7 antibody that binds to the B3 peptide. The figure shows lack of NME7 expression on normal tissues, increasing expression of NME7 as tumor grade and metastasis increase. FIGS. 41A-41D show human colon tissue specimens stained with an anti-NME7 antibody that binds to the B3 peptide. The figure shows lack of NME7 expression on normal tissues, increasing expression of NME7 as tumor grade and metastasis increase. FIG. 47 and FIG. 48 show immunofluorescent photographs showing that NME7 is secreted by and binds to an extra cellular receptor of a wide variety of cancer cell lines.

Further, we demonstrated that like NM23-H1, NME7 binds to and dimerizes the MUC1* growth factor receptor on both stem cells and cancer cells (FIG. 1). FIG. 5 shows a sequence alignment of NME1 and NME7 A and B domains.

The inventors recently discovered that NME7 is a primitive form of NME1 (NM23-H1) that is expressed in very early embryonic stem cells, NME7 is either not expressed at all, or is expressed at extremely low levels, in adult tissues. However, the inventors discovered that NME7 is expressed at high levels in cancerous cells and tissues and at even higher levels in metastatic cancer cells and tissues. A cleaved form of NME7 may be a secreted form allowing it to bind to and activate extracellular receptors. We detect full-length NME7, MW 42 kDa, as well as NME7 species that are approximately 33 kDa and 30 kDa. The 33 kDa and 30 kDa species are secreted from cancer cells. Western blots detect full-length NME7 in cell lysates, but smaller 30-33 kDa NME7 species in their conditioned media. Western blots probed with either an antibody that recognizes NME7 or an antibody that only recognizes the DM10 domain show that the lower molecular weight NME7 species that are secreted into the conditioned media are devoid of the DM10 domain. These data are consistent with the idea that naturally occurring NME7 species are comparable to the recombinant NME7_{AB} we generated as they have nearly the same molecular weight, both are secreted and are both devoid of the 91 amino acids of the DM10 domain which may keep the protein retained within the cell.

We discovered a new NME7 isoform, NME7-X1, and also discovered that it is over-expressed in stem cells and cancer cells and is particularly over-expressed in prostate cancers (FIG. 17, FIG. 18, FIG. 19, and FIG. 22), NME7-X1, molecular weight ~30 kDa, comprises NME7 amino acids 125-376, whereas the recombinant NME7_{AB}, molecular weight ~33 kDa, that we generated spans amino acids 92-376, so includes 33 more N-terminal amino acids, NME7b spans amino acids 37-376 and is devoid of only 37 amino acids of the DM10 domain is also overexpressed in prostate cancers (FIG. 18). We generated a human recombinant NME7-X1 and show that it is the secreted 30 kDa NME7 species in cancer cells that runs just lower than a naturally occurring ~33 kDa NME7 species that appears to be a naturally occurring "NME7_{AB}-like" protein that is a cleavage product or alternative isoform.

We tested a panel of cancer cell lines and found that they express high levels of NME7 and lower molecular weight species that may be truncations similar to NME7_{AB}, such as NME7_{AB}-like protein, or alternate isoforms such as NME7-X1.

Whereas NM23-H1 (aka NME1) has to be a dimer, NME7 is a monomer with two binding sites for MUC1* extracellular domain. We generated a recombinant human NME7 that is devoid of the DM10 domain, which we call NME7_{AB}. A sandwich ELISA binding assay that shows that a recombinant NME7_{AB} simultaneously binds to two PSMGFR peptides wherein the extracellular domain of MUC1* is comprised of most or all of the PSMGFR sequence (FIG. 1). In a nanoparticle binding assay, NME7 was also shown to be able to bind to and dimerize the PSMGFR portion of the MUC1* extracellular domain.

Agents that disable NME7, block its interaction with its binding partners or suppress its expression are potent anti-cancer therapeutics. Such agents may be antibodies, small molecules or nucleic acids. They may act on NME7 directly, on molecules that regulate NME7 expression, or on enzymes that cleave NME7 to cancer-promoting forms.

We discovered that like NM23-H1 dimers, a recombinant NME7_{AB} monomer was fully able to support pluripotent human stem cell growth in the absence of any other growth factor, cytokine or serum. Competitively inhibiting the interaction between NME7 and MUC1* extracellular domain, comprised essentially of the PSMGFR sequence, induced differentiation of stem cells, showing that it is the interaction of NME7 and MUC1* that promotes stem cell growth and inhibits differentiation.

Next, we showed that NME7_{AB} alone is also able to fully support human cancer cell growth, NME7_{AB}, when added to regular cancer cell growth media, stimulated cancer cell growth and in particular the growth of MUC1-positive and MUC1*-positive cancer cells. Inhibiting the interaction of NME7 with MUC1* inhibited cancer cell growth. Blocking the MUC1* growth factor receptor with an anti-MUC1* Fab potentially inhibited cancer cell growth. Similarly, antibodies that bind to NME7 inhibit cancer cell growth. In one example of inhibition of cancer growth by anti-NME7 antibody, the polyclonal antibody was generated from immunizing an animal with the portion of NME7 that spans amino acids 100-376 (FIG. 12 and FIG. 13). However, we found that antibodies generated from immunizing with shorter peptides from NME7_{AB} or from NME7-X1 also inhibit cancer growth. In particular, they inhibit the growth of MUC1 and MUC1*-positive cancers. Anti-NME7 antibodies of the invention inhibited the formation of the non-adherent "floater" cells that are able to form tumor spheres and which can travel from primary tumor and metastasize (FIG. 14, FIG. 16, FIG. 29). Anti-NME7 antibodies of the invention inhibited the upregulation of metastatic and stem cell markers, now believed to also be characteristic of metastasis (FIG. 15, FIG. 30, FIG. 31, FIG. 32). NME7 Causes Cancer Metastasis

The inventors further discovered that culturing cancer cells in a minimal media containing NME7_{AB} induced a wide variety of cancer cells to become transformed to a more metastatic state. Evidence of this induced metastatic state include a change from adherent cell growth to non-adherent cell growth, aka, "floater" cells and accompanying up-regulation of specific metastatic markers that were especially upregulated in the floating cells. These metastatic markers that are upregulated after culture in NME7_{AB} include but are not limited to CXCR4, CHD1 aka E-cadherin, MUC1, ALDH1, CD44, and pluripotent stem cell markers such as OCT4, SOX2, NANOG, KLF2/4, FOXa2, TBX3, ZEB2 and c-Myc (FIG. 2, FIG. 3, FIG. 20, FIG. 49, FIG. 51). Cancer cells cultured in NME7_{AB} had dramatically higher engraftment rates when xenografted into test animals, which were over 90%. In addition, very low numbers of

implanted cancer cells formed tumors in the test animals, which is evidence that NME7_{AB} had transformed them into cancer stem cells also known as metastatic cancer cells. Cancer cells cultured in NME7_{AB} and injected into the tail vein of NOD/SCID/GAMMA mice bearing estrogen release pellets metastasized in animals from low numbers of cells compared to the parent cells, grown in regular media (FIG. 33-FIG. 38). Because cancer cells make either an NME7 cleavage product or alternative isoform that is essentially equivalent to NME7_{AB}, the methods described here are not limited to using NME7_{AB}; other NME7 species could work as well. For example, we discovered another NME7 isoform, NME7-X1, is expressed by cancer cells. It is identical to our recombinant NME7_{AB} with the exception that the X1 isoform is missing 33 amino acids from the N-terminus, NME7-X1 is expected to function like NME7_{AB}. "NME7_{AB}-like" protein has also been detected in cancer cells as being about 33 Da species.

We note that the inventors' previous work showed that NME7_{AB} alone is able to revert human stem cells to an earlier naïve state. We discovered that culturing cancer cells in the presence of other reagents that make stem cells revert to a more naïve state, makes the cancer cells transform to a more metastatic state. We demonstrated that culturing cancer cells NME7_{AB} (FIG. 2), or in dimeric NME1 (FIG. 3), or "2i" inhibitors (FIG. 4), are each able to transform regular cancer cells into metastatic cancer cells, which are also called cancer stem cells "CSCs" or tumor initiating cells "TICs". However, NME7_{AB} induced cancer cells to enter a more metastatic state better than NME1, also known as NM23-H1, which was better than 2i.

2i is the name given to two biochemical inhibitors that researchers found made human stem cells revert to a more naïve state. 2i are MEK and GSK3-beta inhibitors PD0325901 and CHIR99021, which are added to culture medium to final concentrations of about 1 mM and 3 mM, respectively, NME7_{AB} and NME7-X1 are at a final concentration of about 4 nM when added to separate batches of minimal medium to make cancer cells transform to metastatic cells, although lower and higher concentrations also work well in the range of about 1 nM to 16 nM. Human or bacterial NME1 dimers are used at a final concentration of 4 nM to 32 nM, with 16 nM typically used in these experiments, wherein the human NME bears the S120G mutation. Lower concentrations may be required if using wild type. It is not intended that these exact concentrations are important. It is important that the NME1 proteins are dimers and the range of concentrations over which this happens is in the low nanomolar range although certain mutations allow higher concentrations to remain as dimers. Similarly, the concentrations of NME7 proteins can vary, NME7_{AB} and NME7-X1 are monomers and concentrations used to transform cancer cells to metastatic cells should allow the proteins to remain as monomers.

In addition to NME7, NME7_{AB}, NME7-X1, and the 2i inhibitors MEKi and GSK3i, other reagents and inhibitors have been shown by others to cause stem cells to revert to a more naïve state. These inhibitors, "i's" include JNKi, p38i, PKCi, ROCKi, BMPi, BRAFi, SRCi as well as growth factors activating and LIF (Gafni et al 2013, Chan et al 2013, Valamehr et al 2014, Ware et al 2014, Theunissen et al 2014). These reagents can also be used to make cancer cells progress to a more metastatic state. Cells that have been induced to transform to a more metastatic state using any single factor or combination of the inhibitors or growth factors, that make stem cells revert to a more naïve state, can

then be used as discovery tools to identify or test drugs to treat or prevent cancer metastasis.

Various molecular markers have been proposed as being indicators of metastatic cancer cells. Different cancer types may have different molecules that are up-regulated. For example, the receptor CXCR4 is up-regulated in metastatic breast cancers while E-cadherin, also known as CHD1, is up-regulated more in metastatic prostate cancers. In addition to these specific metastasis markers, typical markers of pluripotency such as OCT4, SOX2, NANOG, and KLF4 are up-regulated as cancers become metastatic. The starting cancer cells and the later metastatic cancer cells are assayed by PCR to measure expression levels of these genes. We demonstrated that these cancer cells, cultured in agents such as NME7_{AB} that cause them to be transformed to a more metastatic state, as evidenced by increased expression of metastatic markers and pluripotent stem cell markers, function as metastatic cancer cells.

A functional test of whether or not a population of cancer cells is metastatic is to implant very low numbers. e.g. 200, of the cells in immuno-compromised mice and see if they develop into a tumor. Typically 5-6 million cancer cells are required to form a tumor in an immuno-compromised mouse. We showed that as few as 50 of the NME-induced metastatic cancer cells formed tumors in mice. In addition, mice that were injected throughout the test period with human NME7_{AB}, NME1, or NME7-X1 developed remote metastases.

In one particular experiment. T47D human breast cancer cells were cultured in standard RPMI media for 14 days with media changes every 48 hours and passed by trypsinization when approximately 75% confluent. The cells were then plated into 6-well plates and cultured in minimal stem cell media (see Example 1) that was supplemented with 4 nM NME7_{AB}. Media was changed every 48 hours. By about Day 4, some cells become detached from the surface and float. Media is carefully changed so as to retain the "floaters" as these are the cells that have the highest metastatic potential as evidence by RT-PCR measurement of metastatic markers. On Day 7 or 8, the floaters are harvested and counted. Samples are retained for RT-PCR measurement. The key marker measured is CXCR4, which is up-regulated by 40-200-times after being briefly cultured in NME7_{AB}.

The freshly harvested floater metastatic cells were xenografted into the flank of female nu/nu athymic mice that have been implanted with 90-day slow release estrogen pellets. Floater cells were xenografted with 10,000, 1,000, 100 or 50 cells each. Half of the mice in each group of 6 were also injected daily with 32 nM NME7_{AB} near the original implantation site. The parent T47D cells that were cultured in RPMI media without NME7_{AB} were also implanted into mice at 6 million, 10,000 or 100 as controls. Mice implanted with the NME7-induced floater cells developed tumors even when as few as 50 cells were implanted. Mice that were implanted with the floater cells and that received daily injections of NME7_{AB} also developed remote tumors or remote metastases in various organs. 11 out of the 12 mice, or 92%, that were injected with human NME7_{AB} after implantation of the NME7_{AB} cultured cancer cells developed tumors at the injection site. Only 7 out of the 12 mice, or 58%, that were not injected with human NME7_{AB} after implantation developed tumors. 9 out of the 11 mice, or 82%, that exhibited tumors and were injected with human NME7_{AB} developed multiple tumors remote from the injection site. None of the mice that were not injected with NME7_{AB} developed multiple, visible tumors.

After sacrifice. RT-PCR and Western blots showed that the remote bumps on the mice injected with NME7_{AB} were indeed human breast tumors. Similar analysis of their organs showed that in addition to remote bumps, mice had randomly metastasized to the liver and lung with human breast cancer characteristic of the human breast cancer cells that were implanted. As expected, only the mice implanted with 6 million cells grew tumors.

We have demonstrated that human recombinant NME7_{AB} is comparable in size and sequence to NME7-X1 and to a 30-33 kDa NME7 cleavage product. We have shown that NME7_{AB} promotes cancerous growth and causes cancer cells to accelerate to the highly metastatic cancer stem cell (CSC) state also called tumor initiating cells (TIC). Therefore, we conclude that NME7-X1 and an NME7 cleavage product that removes the DM10 domain also promote cancerous growth and causes cancer cells to accelerate to the highly metastatic cancer stem cell (CSC) state also called tumor initiating cells (TIC). In one example, NME7_{AB} was added to cancer cells in a serum-free media and in the absence of any other growth factors or cytokines. Within 7-10 days, the cancer cells had reverted to the highly metastatic CSCs/TICs as evidenced by more than 100-fold increase in the expression of molecular markers such as CXCR4, which are indicators of metastatic cancer cells. In one example, T47D breast cancer cells were cultured in either standard RPMI media or in a Minimal Stem Cell Media (Example 1) to which was added recombinant NME7_{AB} to a final concentration of 16 nM. After 10 days cells were collected and analyzed by RT-PCR for expression of molecular markers of CSCs which were elevated by 10-200-times (FIG. 2). This is a specific, detailed example of how we transformed one cancer cell type to a more metastatic state. It is not intended that the invention be limited by these details as there are a range of cancer cells that are transformed in this way, a range of reagents that revert stem cells to a more naïve state that also progress cancer cells to a more metastatic state and a range of concentrations over which the added reagents transform the cancer cells. Other types of cancer cells have required longer periods of culture in NME7_{AB} for dramatic upregulation of metastatic markers and ability to form tumors from very low numbers of cancer cells implanted. For example, prostate cancer cells cultured in NME7_{AB}, 2i, human NME1 or bacterial NME1 that has high homology to human NME1 or human NME7 showed dramatic increase in metastatic markers after 2-3 passages.

Metastasis marker CXCR4 is particularly elevated in metastatic breast cancer cells, while CHD1 is particularly elevated in metastatic prostate cancer. Here we show that pluripotent stem cell markers such as OCT4, SOX2, NANOG, KLF2/4 and TBX3 are also up-regulated when cancer cells transform to more metastatic cells.

DU145 prostate cancer cells were cultured similarly and those cells cultured in NME7_{AB} also showed dramatic increases in expression of CSC markers (FIG. 3). In prostate cancer cells, CHD1 (aka E-cadherin) and CXCR4 were up-regulated compared to the control cancer cells, which were not grown in NME7_{AB}, along with other pluripotent stem cell markers. FIG. 20A-20C shows that ovarian cancer cell lines SK-OV3, OV-90 and breast cancer cell line MDA-MB all transitioned from adherent to non-adherent floater cells and increased expression of metastatic markers after 72 or 144 hours in culture with NME7_{AB}. Ovarian, prostate, pancreatic cancer cells and melanoma cells were also cultured in NME7_{AB} and were transformed to a more metastatic state after as few as 3 days in culture. FIG. 21 shows that

breast, ovarian, prostate, pancreatic cancer cells and melanoma cells express MUC1 and MUC1*.

Here we have shown that NME7_{AB} transforms a wide range of cancer cells to a more metastatic state. We have also shown that cancer cells express a naturally occurring species that is approximately the same molecular weight as recombinant NME7_{AB} 33 kDa (FIG. 17, FIG. 18, FIG. 19, and FIG. 22 and is also devoid of the DM10 domain like NME7_{AB} and also express an alternative isoform NME7-X1 30 kDa which is the same sequence as NME7_{AB} except is missing 33 amino acids from the N-terminus. A co-immunoprecipitation experiment was performed on T47D breast cancer cells, wherein the cell extracts were incubated with an antibody against the MUC1 cytoplasmic tail. Ab-5, or a control antibody. IgG, and co-immunoprecipitated. The immunoprecipitated species were separated by gel electrophoresis. The gels were blotted with two different commercially available anti-NME7 antibodies. Both gels show unique NME7 bands at ~33 kDa and ~30 kDa (FIG. 22A-22B). The gels were stripped and re-probed with an antibody against the extracellular domain of MUC1*, anti-PSMGFR (FIG. 22C-22D), which shows that the NME7 species and MUC1* interact. A recombinant NME7_{AB} and a recombinant NME7-X1 that we made were mixed together and run on a gel, then probed with an anti-NME7 antibody, showing that the two unique NME7 species that are naturally occurring in breast cancer cells and that interact with MUC1* are an NME7_{AB}-like species and NME7-X1 (FIG. 22E). A similar experiment was carried out in human stem cells. FIG. 23A-23C show photographs of Western blots of a co-immunoprecipitation experiment. Human induced pluripotent stem, iPS7, or embryonic stem. HES3, cell extracts were incubated with an antibody against the MUC1 cytoplasmic tail. Ab-5, or a control antibody, IgG, and co-immunoprecipitated. The gel was blotted with a commercially available anti-NME7 antibody B9 (FIG. 23A). Both cell types show unique NME7 bands at ~33 kDa and ~30 kDa. The gel was stripped and re-probed with an antibody against the extracellular domain of MUC1*, anti-PSMGFR (FIG. 23B), which shows that the NME7 species and MUC1* interact. A recombinant NME7_{AB} and a recombinant NME7-X1 that we made were mixed together and run on a gel, then probed with an anti-NME7 antibody, showing that the two unique NME7 species that are naturally occurring in breast cancer cells and that interact with MUC1* are an NME7_{AB}-like species and NME7-X1 (FIG. 23C). Because NME7_{AB} is a recombinant protein, we do not know if the naturally occurring species may contain an extra 1-15 additional amino acids or devoid of 1-15 additional amino acids than the recombinant NME7_{AB}, yet run with the same apparent molecular weight. By "NME7_{AB}-like", we mean an NME7 species that runs with an apparent molecular weight of approximately 33 kDa that is able to function the way the recombinant NME7_{AB} does, in that it is able to stimulate cancer cell growth, induce transition of cancer cells to a more metastatic state and is able to fully support pluripotent growth of human stem cells.

We conclude that cancer cell lines and cancer cell populations that express NME7 and lower molecular weight NME7 species contain some cancer cells that are CSCs or metastatic cancer cells. These cancers can be made more metastatic or increase the population of cells that are metastatic by culturing the cells in NME7_{AB}, NME7-X1 or lower molecular weight NME7 species. FIG. 19 shows a Western blot of a panel of cancer cells all expressing NME7 as well as lower molecular weight species NME7_{AB}-like at 33 kDa and NME7-X1 at 30 kDa. FIG. 21 shows that cancer cell lines T47D breast cancer. PC3 and DU145 prostate cancer.

BT-474 breast cancer, CHL-1 and A2058 both melanoma cell lines and CAPAN-2 and PANC-1 both pancreatic cell lines all express MUC1 and MUC1*. In FIG. 21A. BT474 cells appear not to express MUC1 or MUC1* however, we previously showed (Fessler et al 2009) that when these HER2 positive breast cancer cells become resistant to chemotherapy drugs. i.e. metastatic, they do so by increasing expression of MUC1* (FIG. 21D). Blocking the MUC1* receptor with an anti-MUC1* Fab reversed their resistance to Herceptin (FIG. 21E). Taxol (FIG. 21F) as well as other chemo agents. These cancer types and other cancer types that express NME7 and lower molecular weight NME7 species such as 33 kDa. 30 kDa can be made more metastatic or increase the population of cells that are metastatic by culturing the cells in NME7_{AB}, NME7-X1 or lower molecular weight NME7 species.

Conversely, the metastatic potential of these and other cancer types that express NME7 and lower molecular weight NME7 species such as 33 kDa or 30 kDa can be reversed by treating the cells with anti-NME7 antibodies. Anti-NME7 antibodies or antibodies that bind to NME7_{AB} or NME7-X1 are administered to a patient for the treatment or prevention of cancers including breast, prostate, ovarian, pancreatic and liver cancers. Because we have shown that NME7_{AB} exerts its tumorigenic effects by binding to and activating the MUC1* growth factor receptor, anti-NME7 antibodies will be effective against any MUC1*-positive cancers, which include but are not limited to breast, lung, liver, pancreatic, gastric colorectal, prostate, brain, melanoma, kidney and others. Anti-NME7, anti-NME7_{AB} or anti-NME7-X1 antibodies are administered to patients for the treatment or prevention of cancers that are NME7_{AB}, NME7_{AB}-like, or NME7-X1 positive or a MUC1* positive.

Testing Patient Cancer Cells for Effective Therapies

NME7_{AB}, NME7-X1 as well as 2i and other reagents that revert stem cells to a more naïve state also induce cancer cells to transform to a more metastatic state. After treatment with any one or combination of these reagents, cancer cells have a higher engraftment rate and require up to 100,000-times less cells to cause a tumor to form in a test animal. Therefore, methods described in this disclosure can be used to enable xenografting of a patient's primary tumor cells into a test animal.

Candidate therapeutic agents can then be tested on the recipient animal. Effective therapeutic agents identified in this way can be used to treat the donor patient or other patients with similar cancers. In one embodiment, a method of identifying effective therapeutics for a particular patient or a particular type of cancer comprises the steps of: 1) cancer cells are obtained from a cell line, a patient or a patient to whom the therapeutic being tested will be administered; 2) cancer cells are cultured in NME7_{AB}, NME7-X1, human NME1, bacterial NME1 that has high homology to human NME1 or NME7, 2i, or other reagents shown to revert stem cells to a more naïve state; 3) resultant cancer cells are implanted into a test animal to which human NME7_{AB}, NME7-X1, human NME1, bacterial NME1 that has high homology to human NME1 or NME7, 2i, or other reagents shown to revert stem cells to a more naïve state may also be administered or animal is transgenic for human NME7_{AB} or NME7-X1; 4) candidate anti-cancer therapeutic agents are administered to the animal; 5) efficacy of the therapeutic agents are assessed; and 6) effective therapeutic agent is administered to the donor patient or to another patient with similar cancer.

Anti-NME7 Antibodies

Anti-NME7 antibodies are potent anti-cancer agents. NME7 is a growth factor that promotes the growth of cancer cells and also promotes their progression to a more metastatic state or a more aggressive state. NME7 and a truncated form of NME7 that is ~ 33 kDa or 30 kDa have been shown to fully support cancer growth even in serum-free media devoid of any other growth factors or cytokines. In pull-down assays, ELISAs and nanoparticle binding experiments, we have shown that the growth factor receptor MUC1* is a binding partner of NME7 and NME7_{AB}. Promotion of this interaction by eliminating all other growth factors or cytokines increased expression of cancer stem cell markers. Blocking this interaction even in the presence of serum, using a polyclonal antibody that specifically binds to NME7 actively killed the cancer cells. Thus, anti-NME7 or anti-NME7_{AB} antibodies are potent anti-cancer agents that can be administered to a patient for the treatment or prevention of cancers. More than 75% of all cancers are MUC1* positive. MUC1* is the transmembrane cleavage product of MUC1 wherein most of the extracellular domain has been shed, leaving a portion of the extracellular domain that contains most of the PSMGFR sequence and may contain 9-20 additional amino acids N-terminal to the boundary of the of the PSMGFR sequence.

One aspect of the invention is a method of treating or preventing cancer in a subject, comprising administering to the subject an effective amount of an anti-NME7 antibody. In one instance, the anti-NME7 antibody is able to bind to NME7_{AB}. In another instance, the anti-NME7 antibody is able to bind to NME7-X1. In yet another instance, the anti-NME7 antibody that is administered to a patient inhibits or prevents its binding to its target in the promotion of cancers. In one case, the target is the extracellular domain of a cleaved MUC1. More specifically, the NME7 target that promotes cancer is the PSMGFR region of the MUC1* extracellular domain. In one aspect, an effective therapeutic agent is one that disrupts or prevents the interaction between an NME7 species and MUC1* extracellular domain, consisting primarily of the PSMGFR portion of MUC1* or the PSMGFR peptide. Agents for the treatment or prevention of cancers are those agents that directly or indirectly inhibit the expression or function of NME7, an NME7_{AB}-like cleavage product or alternative isoform, including NME7-X1. In one case an effective anti-cancer therapeutic agent is one that binds to the NME7 species or disables its tumorigenic activity. An effective therapeutic agent for the treatment or prevention of cancers is an agent that binds to or disables NME7, an NME7_{AB}-like cleavage product or alternative isoform, or NME7-X1. In one aspect, the therapeutic agents that binds to the NME7 species is an antibody. The antibody may be polyclonal, monoclonal, bispecific, bivalent, monovalent, single chain, scFv, or an antibody mimic that may be animal in origin, human-animal chimera, humanized or human. The antibody can be generated by inoculation or immunization with an NME7 species or fragment thereof or selected, for example from a library or a pool of antibodies, for their ability to bind to an NME7 species, including NME7, an NME7_{AB}-like cleavage product or alternative isoform, including NME7-X1.

Generation of Anti-NME7 Antibodies

Anti-NME7 antibodies can be generated outside of the patient such as in a host animal or in a patient. Antibodies can be generated by immunization of NME7 or NME7 fragments or selected from a library or pool of antibodies that may be natural, synthetic, whole or antibody fragments based on their ability to bind to desired NME7 species such

as NME7_{AB} or NME7-X1. In one aspect, the antibody is generated from immunization with, or selected for its ability to bind to, a peptide selected from those listed in FIG. 6-9. In another aspect, the antibody is generated from peptides whose sequences are not identical to those of human NME1 or the antibodies are selected for their ability to bind to NME7 species and their inability to bind to human NME1.

One method used to identify NME7 or NME7-X1 derived peptides that give rise to antibodies that inhibit cancer growth and inhibit transition to metastasis or peptides that are themselves inhibitory is as follows: 1) protein sequences of human NME1, human NME7, human NME7-X1 and several bacterial or fungal NME proteins that have high sequence homology to either human NME1 or human NME7 are aligned; 2) regions of high sequence homology among all the NMEs are identified; 3) peptide sequences that are unique to NME7 or NME7-X1 but are flanking the regions of high sequence homology are identified. The peptides are then synthesized and used to generate antibodies in a human or host animal. The resultant antibodies are selected for therapeutic use if: 1) they bind to NME7_{AB} or NME7-X1, but not to NME1; 2) have the ability to inhibit cancer growth; 3) have the ability to inhibit the transition of cancer cells to a more metastatic state; or 4) inhibit metastasis in vivo. In some cases, antibodies for therapeutic use are selected for their ability to disrupt binding of NME7_{AB} or NME7-X1 to the MUC1* extra cellular domain, to the PSMGFR peptide or to the N-10 peptide.

Use of Anti-NME7 Antibody for Treatment of Cancer

Those antibodies that inhibit cancer growth or transition to a more metastatic state are selected for use as anti-cancer therapeutics and may be administered to a patient for the treatment or prevention of cancers. Selected antibodies may be further optimized for example by engineering or making human chimera antibodies or fully human antibodies. To demonstrate the efficacy of this approach, we selected NME7 peptides from regions of NME7 suspected to be critical to its cancerous function. We then generated antibodies using these peptides and then tested both the resultant antibodies as well as the immunizing peptides for their ability to: a) inhibit cancerous growth; and b) inhibit the induced transition from cancer cells to metastatic cancer cells. NME7 peptides were selected as immunizing agents for antibody production and as inhibitory agents themselves (FIG. 9. Example 7). Peptides A1 (SEQ ID NO:141), A2 (SEQ ID NO:142), B1 (SEQ ID NO:143), B2 (SEQ ID NO:144) and B3 (SEQ ID NO:145), wherein A refers to the domain from which the peptide is derived. i.e. the NDPK A domain and the B denotes that the peptide is derived from the NDPK B domain (FIG. 5). Each peptide was used as an immunogen and injected into 2 rabbits each for production of polyclonal antibodies. The antibodies that were harvested from the blood of the immunized rabbits were purified over a column derivatized with the immunizing peptide. The purified antibodies were then tested for their ability to bind to human NME7. All of the resultant antibodies bound to human NME7 but not human NME1 as desired (FIG. 10A-10B. Example 8). These results show that by choosing peptides whose sequence is found in NME7 but not exactly identical in NME1, antibodies are generated that specifically bind to NME7 but not NME1. Because NME1 has healthy function, it is in most cases desirable to generate antibodies that do not interfere with NME1. The antibodies were also tested for their ability to inhibit the binding of NME7 to a MUC1* extracellular domain peptide. The ELISA experiment shown in FIG. 11 shows that the antibodies inhibited the binding of NME7_{AB} to a MUC1* extracellular domain

peptide much more than they inhibited binding of NME1. Recall that each of the NME7 A domain and B domain can bind to a PSMGFR peptide. Therefore, complete inhibition of NME7_{AB} binding to a PSMGFR peptide cannot be accomplished with a single antibody or peptide that is derived from just one domain. These antibodies and their respective immunizing peptides also inhibited cancer cell growth (FIG. 12-13). These antibodies also inhibited the formation of non-adherent "floater" cells that result from growing cancer cells in NME7_{AB} (FIG. 14). As can be seen in the figure, the polyclonal antibody generated by immunization with the B3 peptide reduced the number of metastatic floater cells by 95%, indicating that anti-NME7 antibodies that bind to the B3 peptide are most effective at inhibiting cancer metastasis. Similarly, the antibodies inhibited the expression of metastatic marker CXCR4 (FIG. 15A). Again, the B3 antibodies were most efficient at inhibiting expression of CXCR4: bar labeled NME7 FL (NME7 floater cells) shows 70-fold increase in CXCR4 that B3 antibody 61 decreased to 20-fold (bar labeled NME7+61 FL). In addition, the immunizing peptides themselves inhibited the upregulation of CXCR4 and other metastatic markers when T47D cancer cells were grown in NME7_{AB} or 2i.

This is but one example of selecting peptides that generate antibodies that inhibit the cancerous function of NME7 and NME7 species. Sequence alignment among human NME1, human NME7, human NME7-X1 and bacterial NME proteins that had high sequence homology to human NME1 or NME7 identified five regions of homology. The fact that peptides A1, A2, B1, B2 and B3 all generated antibodies that inhibited cancer growth or their transition to a metastatic state means that the five regions from which these peptides were derived are regions of NME7 that are important for its function in the promotion of cancer. Other peptides from these regions will also give rise to anti-NME7 antibodies that will inhibit cancer growth and metastasis and are therefore potent anti-cancer therapeutics. Antibodies generated from peptides A1, A2, B1, B2 and B3 were shown to inhibit cancer growth and inhibited the transition to a more metastatic state. Monoclonal antibodies generated by immunization with the same or similar peptides and subsequent testing of the monoclonals will identify antibodies that, after humanizing or other engineering known to those skilled in the art, would be administered to a patient for the treatment or prevention of cancers.

In a particular experiment, the antibodies generated by immunization with peptides A1, A2, B1, B2 and B3, as well as the immunizing peptides themselves, were added to cancer cells in culture to see if the addition of the antibodies or the immunizing peptides would inhibit cancer cell growth. At low concentrations and added separately, the antibodies as well as the immunizing peptides inhibited cancer cells growth (FIG. 12 for one example). However, when added at higher concentrations or combined, the antibodies as well as the immunizing peptides robustly inhibited cancer cell growth (FIG. 13). The corresponding human NME7 amino acid numbers of immunizing peptides A1, A2, B1, B2 and B3 are 127-142, 181-191, 263-282, 287-301, 343-371, respectively, from human full-length NME7 having SEQ ID NO:82 or 147.

To clarify, when residue numbers of NME7 are discussed, they refer to the residue numbers of NME7 as set forth in SEQ ID NO:82 or 147.

The antibody used in cancer growth inhibition experiments and one of the antibodies shown in FIG. 12 was generated by immunizing with NME7 peptide corresponding to amino acids 100-376 of NME7 (SEQ ID NO:82 or

147). To generate higher affinity and specific anti-NME7 antibodies, the following steps are followed: immunize animal with a peptide containing human NME7 amino acids 100-376, then: 1) de-select those antibodies that bind to human NME1; 2) select those antibodies that inhibit NME7_{AB}, 2i, or other NME induced transition of cancer cells to a more metastatic state; 3) select those antibodies that inhibit the growth of cancer cells; 4) select those antibodies that inhibit the growth of MUC1* positive cancer cells; 5) select those antibodies that inhibit binding of NME7_{AB} or NME7-X1 to MUC1* extracellular domain, essentially inhibit binding to the PSMGFR peptide; and/or 6) select those antibodies that bind to one or more of the peptides listed in FIG. 9—A1, A2, B1, B2 or B3 peptides.

Higher affinity monoclonal antibodies or monoclonal antibodies generated from longer peptides may be more effective antibody therapeutics. Alternatively, combinations of anti-NME7, anti-NME7_{AB} or anti-NME7-X1 antibodies are administered to a patient to increase efficacy.

Anti-NME7 Antibodies Inhibit the Transition of Cancer Cells to Metastatic Cancer Cells.

Anti-NME7 antibodies inhibit transition of cancer cells to metastatic cancer cells also called cancer stem cells (CSCs) or tumor initiating cells (TICs). Recall that we have demonstrated that culturing a wide variety of cancer cells in the presence of NME7_{AB} causes them to transition from regular cancer cells to the metastatic CSCs or TICs. Thus, antibodies that bind to NME7, NME7_{AB} or NME7-X1 will inhibit the progression of cancer cells to a more metastatic state.

Cancer cells transform to a more metastatic state when cultured in the presence of agents that revert stem cells to a more naïve state. We have demonstrated that culturing cancer cells in NME7_{AB}, human NME1 dimers, bacterial NME1 dimers or MEK and GSK3-beta inhibitors, called "2i", causes the cells to become more metastatic. As the cells transition to a more metastatic state, they become non-adherent or less adherent and float off of the culture plate. These floating cells, "floaters" were collected separately from those that were adherent and were shown to: a) express much higher levels of metastatic genes; and b) generated tumors when xenografted into mice at very low copy number. RT-PCR measurement of specific metastatic markers such as CXCR4 for breast cancers, CHD1 for prostate cancer, and other pluripotent stem cell markers such as OCT4, SOX2, NANOG, KLF4 and others were dramatically over-expressed in cancer cells that were cultured in NME7_{AB} and most over-expressed in the cells that became non-adherent, called "floaters" here and in figures.

In one example, NME7_{AB} specific antibodies, generated by immunization with NME7-derived peptides A1, A2, B1, B2 and B3, as well as the immunizing peptides themselves, were added into the media along with either NME7_{AB} or 2i to determine if they inhibited the transformation of regular cancer cells to metastatic cancer stem cells. Antibodies and peptides were separately added along with the agent that causes metastatic transformation; in this case NME7_{AB} or the 2i inhibitors PD0325901 and CHIR99021, NME7_{AB} and 2i were separately used to induce the cancer cells to be transformed to a more aggressive metastatic state. 2i was used so that it could not be argued that the antibodies that were added to the media simply sopped up all of the NME7_{AB} so that the causative agent effectively was not there (Example 10).

Visual observation was independently recorded by two scientists as the experiment progressed (FIG. 14). The most striking observation was that the antibodies and the peptides dramatically reduced the number of floater cells, which was

the first indication that the antibodies and peptides inhibit the transformation to metastatic cancer cells. In particular, cells to which the antibody generated from immunization with the B3 peptide barely generated any floater cells, mRNA was extracted from both the floater cells, the adherent cells and the control cancer cells. The amount of mRNA, which indicates cell viability and growth, was measured. Cells that were treated with antibody had much less mRNA, indicating less live dividing cells (FIG. 16), which confirms that anti-NME7_{AB} antibodies inhibit cancer cell growth as well as their transition to a more metastatic state. RT-PCR was used to measure expression levels of metastatic markers, including CXCR4. Treatment with the anti-NME7 antibodies greatly reduced the amount of metastatic markers, such as CXCR4, indicating that the anti-NME7 antibodies or peptides inhibit the transition to metastatic cancer (FIG. 15A-15C). These results show that antibodies that bind to NME7_{AB} can be administered to a patient for the treatment or prevention of metastatic cancers.

Peptides Derived from NME7_{AB} or NME7-X1 Competitively Inhibit the Binding of Intact NME7_{AB} and NME7-X1 and are Anti-Cancer Agents.

In another aspect of the invention, therapeutic agents for the treatment or prevention of cancers are peptides derived from the NME7 sequence, which are administered to a patient for the treatment or prevention of cancers. In one aspect, the NME7-derived peptides are administered to a patient so that the peptides, which should be shorter than the entire NME7 and unable to confer the oncogenic activity of NME7, bind to the targets of NME7 and competitively inhibit the interaction of intact NME7 with its targets, wherein such interactions promote cancer. Since NME7_{AB} is fully able to confer oncogenic activity, the sequence of NME7_{AB} is preferred as the source for the shorter peptide(s), wherein it must be confirmed that the peptides themselves are not able to promote cancerous growth or other tumorigenic or oncogenic activity. In a preferred embodiment, one or more peptides having the sequence of a portion of NME7_{AB} and being preferably about 12-56 amino acids in length are administered to a patient. To increase half-life, the peptides may be peptide mimics, such as peptides with unnatural backbone or D-form amino acids for L. In yet another case, the anti-cancer therapeutic agent is a peptide or peptide mimic wherein the peptide has a sequence highly homologous to at least a portion of NME7, NME7_{AB}, or NME7-X1 or its target the MUC1* extracellular domain, comprising the PSMGFR peptide, also called "FLR" in some cases herein.

FIG. 6-FIG. 9 provide a listing of preferred amino acid sequences that are predicted to inhibit NME7 binding to its cognate target. In a still more preferred embodiment, the peptides that are chosen for administration to a patient suffering from cancer or at risk of developing cancer are chosen because they bind to an NME7 binding partner and they do not themselves confer tumorigenic activity. In a yet more preferred embodiment, the NME7 binding partner is the extracellular domain of MUC1*. In a still more preferred embodiment, the NME7 binding partner is the PSMGFR peptide.

By the term "conferring tumorigenic activity or oncogenic activity", it is meant that the peptides themselves cannot support or promote the growth of cancers. Another way of testing whether or not a peptide or peptides derived from NME7 can promote tumorigenesis is to test whether or not the peptides can support pluripotent growth of human stem cells, NME proteins and peptides that support pluripotent human stem cell growth also support cancer growth. In yet

another method, peptides are de-selected if they can cause somatic cells to revert to a less mature state.

Fragments of NME7_{AB} inhibit cancer cell growth and the transition of cancer cells to a more metastatic state. As a demonstration, NME7 peptides A1, A2, B1, B2 and B3 added separately (FIG. 12) or in combinations (FIG. 13) inhibit the growth of cancer cells. In addition, NME7 peptides A1, A2, B1, B2 and B3 inhibited the transition of cancer cell to a more metastatic state (FIG. 15).

Thus, antibodies generated by immunizing with peptides specific to NME7, and specific to NME7_{AB} or NME7-X1 will block the cancerous action of NME7 species and will be potent anti-cancer agents. Similarly, these results show that the peptides specific to NME7, and specific to NME7_{AB} or NME7-X1 will block the cancerous action of NME7 species. In one aspect of the invention, the peptides are chosen from the list shown in FIG. 6. In one aspect of the invention the peptides are chosen from the list shown in FIG. 7. In one aspect of the invention the peptides are chosen from the list shown in FIG. 8. In yet another aspect of the invention the peptides are chosen from the list shown in FIG. 9. These antibodies may be generated by immunizing or may be generated or selected by other means, then selected for their ability to bind to NME7, NME7_{AB}, NME7-X1 or NME7 derived peptides, including but not limited to NME7 derived peptides A1 (SEQ ID NO: 141), A2 (SEQ ID NO: 142), B1 (SEQ ID NO: 143), B2 (SEQ ID NO: 144) or B3 (SEQ ID NO: 145). Such antibodies may be polyclonal, monoclonal, bispecific, bivalent, monovalent, single chain, scFv, human or humanized or may be an antibody mimic such as protein scaffolds that present recognition regions that bind to a specific target.

Anti-NME7 antibodies for use in the treatment or prevention of cancers can be generated by standard methods known to those skilled in the art wherein those methods are used to generate antibodies or antibody-like molecules that recognize NME7, NME7_{AB} or a shorter form of NME7_{AB} wherein an additional 10-25 amino acids form the N-terminus are not present. Such antibodies may be human or humanized. Such antibodies may be polyclonal, monoclonal, bispecific, bivalent, monovalent, single chain, scFv, human or humanized or may be an antibody mimic such as protein scaffolds that present recognition regions that bind to a specific target.

Anti-NME7 antibodies that are generated by immunization with the NME7 derived peptides A1 (SEQ ID NO: 141), A2 (SEQ ID NO: 142), B1 (SEQ ID NO: 143), B2 (SEQ ID NO: 144) or B3 (SEQ ID NO: 145) or antibodies that bind to the A1, A2, B1, B2 or B3 peptides are antibodies that bind to NME7_{AB} and NME7-X1, but resist binding to NME1 which may be required for the function of some healthy cells. Such antibodies inhibit the binding of NME7_{AB} or NME7-X1 to their target receptor, MUC1*. Antibodies that bind to A1, A2, B1, B2 or B3 peptides are antibodies can be administered to a patient diagnosed with or at risk of developing a cancer or metastasis. Such antibodies may be human or humanized. Such antibodies may be polyclonal, monoclonal, bispecific, bivalent, monovalent, single chain, scFv, or may be an antibody mimic such as protein scaffolds that present recognition regions that bind to a specific target.

Anti-NME7 antibodies that are generated by immunization with the B3 peptide or antibodies that bind to the B3 peptide are especially specific for the recognition of NME7_{AB} and NME7-X1. Such antibodies are also very efficient at inhibiting the binding of NME7_{AB} or NME7-X1 to their target receptor, MUC1*. Antibodies that bind to the B3 peptide are also exceptionally efficient at preventing,

eration of anti-NME7 antibodies. We mutated Cysteine 14 to Serine to make AIFGKTKIQNAVHSTDLPEDG-LLEVQYFF (SEQ ID NO:169) and immunized animals to generate anti-NME7 monoclonal antibodies. The resultant antibodies bind to the native B3 sequence as well as the B3Cys14Ser peptide. Seven (7) high affinity and specific monoclonal antibodies were generated: 8F9A5A1, 8F9A4A3, 5F3A5D4, 5D9E2B11, 5D9E10E4, 5D9G2C4, and 8H5H5G4. However, various sequence alignments showed that there are only three (3) unique sequence antibodies: 8F9A5A1, 8F9A4A3, 8F9A4P3 and 5F3A5D4 as seen below. Bolded and underlined regions indicate CDR sequences.

Note that the polyclonal antibody #61, which was generated in rabbits immunized with the B3 peptide, inhibited the transformation of cancer cells to cancer stem cells as evidenced by antibody #61 blocking upregulation of metastatic marker CXCR4 (FIG. 15).

The B3 peptide (SEQ ID NO: 145) derived from NME7 has a Cysteine at position 14, which complicates the gen-

HEAVY CHAIN ALIGNMENT		
8F9A5A1H	IQLVQSGPELKKPGETVKISKASGYTFTTNYGMNVVKQAPGKGLKWMGWINTYTGPTYV	60
8F9A4A3H	VQLQQSGPELVKPGASVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN	60
5D9E2B11H	VQLQQSGPDLVKPGTSVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN	60
5D9E10E4H	VQLQQSGPDLVKPGTSVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN	60
5D9G2C4H	VQLQQSGPDLVKPGTSVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN	60
5F3A5D4H	VQLQQSGPDLVKPGTSVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN	60
8H5H5G4H	VQLQQSGPDLVKPGTSVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN : ** : ** : * : * : * : * : * : * : * : * : * : * : * : * : *	60
8F9A5A1H	DDFKGRFAFSLETSATCAYLQINNKNEDTSTYFCARLR--GIRPGPLAYWGQGLTVTS	118
8F9A4A3H	QKFKGKATLTVDKSSSIAYMELRSLTSEDSAVYYCARRYHSLYVFYFDYWGQGTTLTVS	120
5D9E2B11H	QKFKGKATLTVDKSSSIAYMELRSLTSEDSAVYYCARRYHSTYVFYFDSWGQGT-----	116
5D9E10E4H	QKFKGKATLTVDKSSSIAYMELRSLTSEDSAVYYCARRYHSTYVFYFDSWGQGTTLTVS	120
5D9G2C4H	QKFKGKATLTVDKSSSIAYMELRSLTSEDSAVYYCARRYHSTYVFYFDSWGQGTTLTVS	120
5F3A5D4H	QKFKGKATLTVDKSSSIAYMELRSLTSEDSAVYYCARRYHSTYVFYFDSWGQGTTLTVS	120
8H5H5G4H	QKFKGKATLTVDKSSSIAYMELRSLTSEDSAVYYCARRYHSTYVFYFDSWGQGTTLTVS : . ** : : : : : * : : ** : : : . . * : : * : * : : : : : : : * : * : *	120
8F9A5A1H	A	119 (SEQ ID NO: 172)
8F9A4A3H	S	121 (SEQ ID NO: 1205)
5D9E2B11H	-	116 (SEQ ID NO: 174)
5D9E10E4H	S	121 (SEQ ID NO: 175)
5D9G2C4H	S	121 (SEQ ID NO: 176)
5F3A5D4H	S	121 (SEQ ID NO: 177)
8H5H5G4H	S	121 (SEQ ID NO: 178)
8F9A4P3H	VQLQQSGPELVKPGASVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN	60
8F9A4A3H	VQLQQSGPELVKPGASVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN	60
5D9E2B11H	VQLQQSGPDLVKPGTSVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN	60
5D9E10E4H	VQLQQSGPDLVKPGTSVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN	60
5D9G2C4H	VQLQQSGPDLVKPGTSVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN	60
5F3A5D4H	VQLQQSGPDLVKPGTSVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN	60
8H5H5G4H	VQLQQSGPDLVKPGTSVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIGGFNPNNNGVTNYN ***** : ***** : ***** : ***** : ***** : ***** : ***** : *****	60
8F9A4P3H	QKFKGKATLTVDKSSSIAYMELRSLTSEDSAVYYCARRYHSLYVFYFDYWGQGTTLTVS	120
8F9A4A3H	QKFKGKATLTVDKSSSIAYMELRSLTSEDSAVYYCARRYHSLYVFYFDYWGQGTTLTVS	120

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5D9E2B11H	<u>QKFKG</u> KATLTVDKSSSTAYMELRSLTSEDSAVYYCARRYHSTYVFYFDSWGQGT-----	116
5D9E10E4H	<u>QKFKG</u> KATLTVDKSSSTAYMELRSLTSEDSAVYYCARRYHSTYVFYFDSWGQGTTLTVS	120
5D9G2C4H	<u>QKFKG</u> KATLTVDKSSSTAYMELRSLTSEDSAVYYCARRYHSTYVFYFDSWGQGTTLTVS	120
5F3A5D4H	<u>QKFKG</u> KATLTVDKSSSTAYMELRSLTSEDSAVYYCARRYHSTYVFYFDSWGQGTTLTVS	120
8H5H5G4H	<u>QKFKG</u> KATLTVDKSSSTAYMELRSLTSEDSAVYYCARRYHSTYVFYFDSWGQGTTLTVS *****	120
8F9A4P3H	S	121 (SEQ ID NO: 173)
8F9A4A3H	S	121 (SEQ ID NO: 1205)
5D9E2B11H	-	116 (SEQ ID NO: 174)
5D9E10E4H	S	121 (SEQ ID NO: 175)
5D9G2C4H	S	121 (SEQ ID NO: 176)
5F3A5D4H	S	121 (SEQ ID NO: 177)
8H5H5G4H	S	121 (SEQ ID NO: 178)
LIGHT CHAIN ALIGNMENT		
8F9A4P3L	ETTVTQSPASLSMAIGKVTIRCI <u>ITSTDIDDDMN</u> WYQKPGEPKLLISEGNTLRPGVPS	60
8F9A4A3L	DIQMTQTSSLSASLGDRVTLS <u>CSASQGISNYLN</u> WYQKPDGTVELLIFYTSSLHSGVPS	60
5D9E2B11L	DIQMTQTSSLSASLGDRVTIS <u>CSASQGISNYLN</u> WFQKPDGTIKLLIYYTSSLHSGVPS	60
5D9E10E4L	DIQMTQTSSLSASLGDRVTIS <u>CSASQGISNYLN</u> WFQKPDGTIKLLIYYTSSLHSGVPS	60
5D9G2C4L	DIQMTQTSSLSASLGDRVTIS <u>CSASQGISNYLN</u> WFQKPDGTIKLLIYYTSSLHSGVPS	60
5F3A5D4L	DIQMTQTSSLSASLGDRVTIS <u>CSASQGISNYLN</u> WFQKPDGTIKLLIYYTSSLHSGVPS	60
8H5H5G4L	DIQMTQTSSLSASLGDRVTIS <u>CSASQGISNYLN</u> WFQKPDGTIKLLIYYTSSLHSGVPS	60
8F9A5A1L	EILLTQSPAIIAASPGKVTITCS <u>SASSV-SYMN</u> WYQKPGSSPKIWEYGISNLASGVPA : **: : : : *:*** *: * : . : **:****. *: * ..* ***:	60
8F9A4P3L	RFSSSGYGTDFVTIENMLSEDVADYYC <u>LQSDNLPLT</u> FGSGTKLEIKR	108 (SEQ ID NO: 185)
8F9A4A3L	RFSGSGSGTDYSLTISNLEPEDIATYYC <u>QQYSKLPYT</u> FGGGTKLEIK	108 (SEQ ID NO: 1109)
5D9E2B11L	RFSGSGSGTDYSLTISNVEPEDIATYYC <u>QQYSKLPYT</u> FGGGTKLEIKR	108 (SEQ ID NO: 186)
5D9E10E4L	RFSGSGSGTDYSLTISNVEPEDIATYYC <u>QQYSKLPYT</u> FGGGTKLEIKR	108 (SEQ ID NO: 187)
5D9G2C4L	RFSGSGSGTDYSLTISNVEPEDIATYYC <u>QQYSKLPYT</u> FGGGTKLEIKR	108 (SEQ ID NO: 188)
5F3A5D4L	RFSGSGSGTDYSLTISNVEPEDIATYYC <u>QQYSKLPYT</u> FGGGTKLEIKR	108 (SEQ ID NO: 189)
8H5H5G4L	RFSGSGSGTDYSLTISNVEPEDIATYYC <u>QQYSKLPYT</u> FGGGTKLEIKR	108 (SEQ ID NO: 190)
8F9A5A1L	RFSGSGSGTFSFTINSMEAEDVATYYC <u>QQRSSYPPT</u> FGGGTKLEIKR ***. ** **: : **:..: **: * ** * .. * **..*****	108 (SEQ ID NO: 191)
5D9E2B11L	DIQMTQTSSLSASLGDRVTIS <u>CSASQGISNYLN</u> WFQKPDGTIKLLIYYTSSLHSGVPS	60
5D9E10E4L	DIQMTQTSSLSASLGDRVTIS <u>CSASQGISNYLN</u> WFQKPDGTIKLLIYYTSSLHSGVPS	60
5D9G2C4L	DIQMTQTSSLSASLGDRVTIS <u>CSASQGISNYLN</u> WFQKPDGTIKLLIYYTSSLHSGVPS	60
5F3A5D4L	DIQMTQTSSLSASLGDRVTIS <u>CSASQGISNYLN</u> WFQKPDGTIKLLIYYTSSLHSGVPS	60
8H5H5G4L	DIQMTQTSSLSASLGDRVTIS <u>CSASQGISNYLN</u> WFQKPDGTIKLLIYYTSSLHSGVPS *****	60
5D9E2B11L	RFSGSGSGTDYSLTISNVEPEDIATYYC <u>QQYSKLPYT</u> FGGGTKLEIKR	108 (SEQ ID NO: 192)
5D9E10E4L	RFSGSGSGTDYSLTISNVEPEDIATYYC <u>QQYSKLPYT</u> FGGGTKLEIKR	108 (SEQ ID NO: 193)
5D9G2C4L	RFSGSGSGTDYSLTISNVEPEDIATYYC <u>QQYSKLPYT</u> FGGGTKLEIKR	108 (SEQ ID NO: 194)
5F3A5D4L	RFSGSGSGTDYSLTISNVEPEDIATYYC <u>QQYSKLPYT</u> FGGGTKLEIKR	108 (SEQ ID NO: 195)
8H5H5G4L	RFSGSGSGTDYSLTISNVEPEDIATYYC <u>QQYSKLPYT</u> FGGGTKLEIKR *****	108 (SEQ ID NO: 196)

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8F9A4P3L  ETTVTQSPASLSMAIGKVTIRCITSTDIDDDMNWYQQKPGPEPKLLISEGNTLRPGVPS    60
8F9A5A1L  EILLTQSPAIIAASPGKVTITCSASSSV-SYMNWYQQKPGSSPKIWIYGISNLAGVPA    60
      * :***** : : :***** * :*::: :*****. ** : * ..* : ***:
8F9A4P3L  RFSSSGYGTDVFTIENMLSEDVADYYCLQSDNLPITFGSGTKLEIKR    108 (SEQ ID NO: 197)
8F9A5A1L  RFSGSGSGTSFSFTINSMEAEDVATYYCQQRSSYPPTFGGSKLEIKR    108 (SEQ ID NO: 198)
      ***.* **.* **.* :* :***** **.* ..* **.******

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Monoclonal antibodies 5D9E2B11, 5D9E10E4, 5D9G2C4, and 8H5H5G4 all have the same sequence as 5F3A5D4, also known as 5D4. Herein, when we refer to antibody 5F3A5D4, aka 5D4, it is understood that it also applies to 5D9E2B11, 5D9E10E4, 5D9G2C4, and 8H5H5G4. As can be seen in FIG. 24 and FIG. 25 anti-NME7 antibodies 8F9A5A1, 8F9A4A3, and 5F3A5D4 all bind to NME7_{AB} but not to NME1. This is important because the A domain of NME7 has high homology to NME1, which is required for normal cell function. For an anti-cancer therapeutic or an anti-metastasis therapeutic it will be imperative to inhibit NME7_{AB} but not NME1.

FIG. 26, FIG. 27 and FIG. 28 show that these anti-NME7 antibodies are also able to disrupt the binding of NME7_{AB} to the MUC1* PSMGFR peptide and the N-10 PSMGFR peptide. As can be seen, there is not a total displacement of NME7_{AB} from the MUC1* peptides. However, recall that NME7_{AB} is comprised of an A domain and a B domain, each of which are capable of binding to MUC1*. These antibodies were designed to disrupt binding of the B domain to MUC1*; the A domain of NME7_{AB} would still be able to bind to the MUC1* peptide on the plate surface. For a useful therapeutic, the antibody would only need to disrupt the binding of one domain to MUC1* and in so doing ligand-induced dimerization and activation of MUC1* growth factor receptor would be blocked. Antibodies or antibody mimics that bind to the NME7 B3 peptide or the B3 Cys14Ser peptide (SEQ ID NO: 169) are antibodies can be administered to a patient diagnosed with or at risk of developing a cancer or metastasis.

It is well known in the field that it is difficult to make cancer cells metastasize in an animal model. It is estimated that in a human tumor only about 1 in 100,000 or even 1 in 1,000,000 cancer cells is able to break away from the tumor and implant elsewhere to initiate a metastasis [Al-Hajj et al., 2003]. Some researchers report that T47D breast cancer cells injected into an immune compromised mouse will metastasize after about 12 weeks [Harrell et al 2006]. Other researchers report that AsPC-1 pancreatic cancer cells will metastasize after about 4 weeks [Suzuki et al, 2013].

Here, we show that T47D breast cancer cells grown for 10 days in a serum-free media containing recombinant NME7_{AB} as the only growth factor. It was observed that when grown in NME7_{AB}, about 25% of the cancer cells began floating, stopped dividing but were still viable. PCR measurement showed that these "floating" cells greatly upregulated expression of the breast cancer metastatic factor CXCR4.

In some of the figures presented herein, these floater cells are referred to as cancer stem cells (CSCs). Immune compromised female nu/nu mice were implanted with 90-day release estrogen pellets. Either 500,000 T47D-wt cells or 10,000 T47D-CSCs (cancer stem cells) were injected into the tail vein (i.v.), sub-cutaneously (s.c.), or into the intra-peritoneal space (i.p.) of the nu/nu mice. These cancer cells were engineered to express Luciferase. To visualize the

tumors or cancer cells, animals are injected with Luciferin, then visualized on an IVIS instrument 10 minutes later. As can be seen in the IVIS measurements of FIG. 33A-FIG. 33B, by Day 6 the 500,000 T47D-wt cells injected into the tail vein show no signs of live cancer cells or cancer cell engraftment.

In stark contrast, the 10,000 T47D-CSC injected into the tail vein have metastasized. Before the Day 6 IVIS measurement, the T47D-CSC mice were injected with 32 nM recombinant NME7_{AB}. The next day, one of the two CSC mice was injected with a cocktail of anti-NME7 monoclonal antibodies 8F9A5A1, 8F9A4A3, and 5F3A5D4 in a volume of 200 uL at a concentration that corresponds to 15 mgs/kg. The nearly coincident injection of NME7_{AB} and anti-NME7 antibody likely nullified the effect of the antibody. FIG. 34 shows that by Day 10, the treated mouse is almost entirely metastatic. As can be seen in the figure, the mouse chosen for treatment is more metastatic than the comparable T47D-CSC mouse.

That animal was again injected with the anti-NME7 antibodies on Day 10. The IVIS measurement of Day 12 (FIG. 35) shows that the antibody treated mouse is beginning to clear the metastases. By Day 14 (FIG. 36) the untreated mouse has died from rampant metastases and the treated mouse has cleared the metastases. FIG. 37 shows the time course of IVIS measurements for the mouse injected with 500,000 T47D-wt cells and the mouse injected with T47D-CSCs that received anti-NME7 treatment until Day 17 when antibody treatment was suspended. As can be seen, on Day 17 there remained a small cluster of cancer cells, which by Day 19 had grown larger. By Day 21 the metastases had spread and antibody treatment was resumed. As is shown in the figure, after resumption of anti-NME7 antibody treatment, the animal was cleared of all metastases and shows no signs of ill health.

FIG. 38 shows the IVIS time course for animals that were injected sub-cutaneously or intra-peritoneally. Antibody injections for animals injected with CSCs sub-cutaneously or intra-peritoneally were also injected with anti-NME7 antibodies s.c. or i.p. . . . In these animals, antibody injections stopped at Day 17 and did not resume. FIG. 39-FIG. 40 show that a polyclonal anti-NME7 antibody generated by immunization with the B3 peptide stains advanced cancers and metastatic cancers but not normal tissues or low-grade cancers, where only 1 in 100,000 or 1 in 1,000,000 cancer cells would be a metastatic cancer cells. Taken together, these data show that anti-NME7 antibodies 8F9A5A1, 8F9A4A3, and 5F3A5D4 or 8F9A5A1, or 8F9A4A3, or 5F3A5D4 administered to a patient diagnosed with or at risk of developing a cancer would prevent, inhibit the formation of, or reverse cancer metastases.

In addition to treating metastatic animals with a cocktail of anti-NME7_{AB} antibodies, we also administered monoclonal anti-NME7_{AB} antibodies individually and showed they were capable of preventing as well as reversing cancer metastases. In one demonstration, female nu/nu mice weigh-

ing approximately 20 g each, were implanted with 90-day estrogen release pellets between 8-10 weeks of age. Cancer cells were made metastatic by culturing for 10-15 days in a serum-free media supplemented with growth factor NME7_{AB}. Both adherent and floating cells show upregulation of metastatic markers and in animals are able to metastasize within 4-7 days. In this case, the floating cells were harvested on Day 11 of in vitro culture and injected into the tail vein of the test animals. To test a prevention model, one group of animals was injected into the tail vein 24 hours before injection of the metastatic cancer cells, with anti-NME7_{AB} antibody 8F9A4A3 at 15 mg/kg and injected thereafter with the same dosage approximately every 48 hours. FIG. 42A-FIG. 42F shows photographs of female nu/nu mice, which were injected into the tail vein with 10,000 Luciferase positive T47D metastatic breast cancer stem cells and treated with the anti-NME7_{AB} antibody 4A3 also known as 8F9A4A3. To image cancer cells, the Luciferase substrate, Luciferin, is intraperitoneally injected 10 minutes before being photographed in IVIS instrument. FIG. 42A-42C show IVIS photographs with animals face down. FIG. 42D-42F show IVIS photographs with animals face up. FIGS. 42A and 42D show control animals injected with phosphate buffered saline solution. FIGS. 42B and 42E show a prevention model in which animals were injected with anti-NME7_{AB} antibody 4A3 24 hrs before injection of the metastatic cancer cells, then approximately every other day for a total of 12 antibody injections over 22 days. FIGS. 42C and 42F show a reversal model in which animals were injected with anti-NME7_{AB} antibody 4A3 24 hrs after injection of the metastatic cancer cells, then approximately every other day for a total of 11 antibody injections over 20 days. As can be seen in the figure, anti-NME7_{AB} antibody 8F9A4A3 can prevent, as well as reverse an established metastasis.

Anti-NME7_{AB} antibodies 5A1 and 5D4 were also tested in a metastasis prevention model and shown to greatly inhibit cancer metastasis. FIG. 43A-43F shows photographs of female nu/nu mice weighing approximately 20 g each, which were injected into the tail vein with 10,000 Luciferase positive T47D metastatic breast cancer stem cells and treated with the anti-NME7_{AB} antibodies 5A1, also known as 8F9A5A1, and 5D4, also known as 5F3A5D4. To image cancer cells, the Luciferase substrate, Luciferin, is intraperitoneally injected 10 minutes before being photographed in IVIS instrument. FIG. 43A-43C show IVIS photographs with animals face down. FIG. 43D-43F show IVIS photographs with animals face up. FIGS. 43A and 43D show control animals injected with phosphate buffered saline solution. FIGS. 43B, 43E, 43C and 43F show a prevention model in which animals were injected with anti-NME7_{AB} antibodies, at 15 mg/kg 24 hours before injection of the metastatic cancer cells, then approximately every other day for a total of 12 antibody injections over 22 days. Photographs were taken either at Day 24 or at Day 27. Specifically, mouse #1 in the group treated with antibody 5A1 was photographed at Day 27 while mouse #2 and #3 were photographed on Day 24 because animals died on Day 26.

Anti-NME7_{AB} antibodies 5A1 and 5D4 were also tested in a metastasis reversal model and shown to greatly inhibit established cancer metastases. In this experiment, animals were injected on Day (into the tail vein with 10,000 T47D metastatic cancer cells mixed with NME7_{AB} at a final concentration of 32 nM. Further, animals were injected twice. Day 3 and Day 4, with more NME7_{AB} which our experiments have shown make the metastasis more difficult to reverse. The first antibody injection was on Day 7.

Because the degree of metastasis in each test animal is somewhat variable, we wanted to make certain that the apparent clearance of metastatic cancer cells was due to the anti-NME7_{AB} treatment. We therefore treated the animals with alternating high dose and low doses. As can clearly be seen in FIG. 44, high dose anti-NME7_{AB} results in clearance of the metastasis, which if not completely eradicated comes back and even increases with lower dose. This experiment shows that all three anti-NME7_{AB} antibodies tested. 5A1, 4A3 and 5D4, which are able to bind to the NME7-B3 peptide, inhibit cancer metastasis in a concentration dependent manner. FIG. 44A-44D show photographs of female nu/nu mice that were injected into the tail vein with 10,000 Luciferase positive T47D metastatic breast cancer stem cells mixed with NME7_{AB} at a final concentration of 32 nM. Animals were then injected into the tail vein with 32 nM NME7_{AB} before being treated with individual anti-NME7_{AB} antibodies. FIG. 44A shows control animals injected with phosphate buffered saline solution. FIG. 44B shows animals treated with anti-NME7_{AB} monoclonal antibody 8F9A5A1. FIG. 44C shows animals treated with anti-NME7_{AB} monoclonal antibody 8F9A4A3. FIG. 44D shows animals treated with anti-NME7_{AB} monoclonal antibody 5F3A5D4. Green arrows indicate low antibody dosage (5-7 mg/kg) over the indicated period and Red arrows indicate high dosage (15 mg/kg). As can be seen in the figure, the metastasis clears considerably when antibody is administered at 15 mg/kg.

In addition to demonstrating that the anti-NME7_{AB} antibodies of the invention can inhibit metastasis, we tested their effect on metastasis from a primary tumor, which would more closely mimic the physiology of cancer metastasis. We generated T47D metastatic breast cancer cells, also known as cancer stem cells (CSCs) by culturing the cancer cells in a minimal serum-free media containing NME7_{AB} for 10-15 days. These T47D CSCs were then implanted sub-cutaneously into the right flank of NSG mice into which had been implanted a 90-day estrogen release pellet. The implanted cancer cells were Luciferase positive so that after injection of the Luciferase substrate, Luciferin, the cancer cells emit photons and can be photographed in an IVIS instrument to measure and locate the implanted cancer cells. FIG. 45A-45B shows photographs of female nu/nu mice that on Day 0 were injected sub-cutaneously into the right flank with 10,000 Luciferase positive T47D metastatic breast cancer stem cells, mixed with NME7_{AB} to a final concentration of 32 nM, then mixed in a 1:1 vol:vol with Matrigel. Tumor engraftment was allowed to progress Day 0-Day 6. Animals were then treated i.v. by tail vein injection with anti-NME7_{AB} antibodies. Control animals were injected with PBS. FIG. 45A shows IVIS photographs of control animals. FIG. 45B shows IVIS photographs of animals injected into tail vein with a cocktail of anti-NME7_{AB} antibodies 5A1, 4A3 and 5D4 to a total concentration of 15 mg/kg. Antibodies or PBS were administered 4 times between Day 7 and Day 18. As can be seen in the figure, the anti-NME7_{AB} antibody treated animals show less metastases (blue dots in whole body) than the control group. 2 of the 5 animals have primary tumors that are larger than those in the control group. This could be because the anti-NME7_{AB} antibodies prevented the spread of the cancer cells, so they remained concentrated in the primary tumor. In this experiment, PCR analysis, performed prior to injection of the cancer cells, showed that after 11 days in culture with NME7_{AB}, the T47D breast cancer cells had upregulated CXCR4 by 109-fold, OCT4 by 2-fold, NANOG by 3.5-fold and MUC1 by 2.7-fold.

In another experiment, we tested the effect of anti-NME7_{AB} antibodies of the invention on metastasis from a primary tumor to organs that breast cancers typically metastasize to. Breast cancers commonly metastasize to liver, lung, bone and brain, in that order. We generated T47D metastatic breast cancer cells by culturing in a minimal serum-free media containing NME7_{AB} for 11 days. These T47D CSCs were then implanted sub-cutaneously into the right flank of NSG mice into which had been implanted a 90-day estrogen release pellet. FIG. 46A-46P shows photographs of female nu/nu mice that on Day 0 were injected sub-cutaneously into the right flank with 10,000 Luciferase positive T47D metastatic breast cancer stem cells, mixed with NME7_{AB} to a final concentration of 32 nM, then mixed in a 1:1 vol:vol with Matrigel. Tumor engraftment was allowed to progress Day 0-Day 6. Animals were then treated i.v., by tail vein injection, with anti-NME7_{AB} antibodies. Control animals were injected with PBS. On Day 38 animals were sacrificed and livers harvested then analyzed by IVIS to detect cancer cells that had metastasized to the liver. FIG. 46A-46B show whole body IVIS photographs of control animals that were injected with only PBS. FIG. 46C-46D show whole body IVIS photographs of control animals that were injected with the anti-NME7_{AB} antibody 5A1. FIG. 46E-46F show whole body IVIS photographs of control animals that were injected with the anti-NME7_{AB} antibody 4A3. FIG. 46G-46H show whole body IVIS photographs of control animals that were injected with the anti-NME7_{AB} antibody 5D4. FIGS. 46A, 46C, 46E, and 46G are IVIS photographs taken at Day 7 before any treatment. FIGS. 46B, 46D, 46F, and 46H are IVIS photographs taken at Day 31 after anti-NME7_{AB} antibody treatment or mock treatment. As can be seen in the figure, animals in the PBS control group show metastasis (blue dots) in the whole body IVIS photographs, while animals treated with anti-NME7_{AB} antibodies do not. FIG. 46I-46P show photographs and IVIS photographs of livers and lung harvested from animals after sacrifice. FIGS. 46I, 46K, 46M, and 46O are regular photographs. FIGS. 46J, 46L, 46N, and 46P are IVIS photographs, illuminating the cancer cells that have metastasized there. As can be seen in the figure, the anti-NME7_{AB} antibodies greatly inhibited metastasis to the liver, which is a primary site for breast cancer metastasis. FIG. 46Q is a bar graph of the measured photons emitted and enumerated by IVIS instrument for livers harvested from control animals versus the treated animals. As can be seen in the inserted graph of IVIS measurements, the inhibition of metastasis to the liver follows the rank order of inhibition of metastasis when cells were injected into the tail vein, which also matches the rank order of potency in being able to disrupt the NME7_{AB}-MUC1* interaction.

FIGS. 52A-52E shows photographs of female nu/nu mice that on Day 0 were injected into the tail vein with 10,000 Luciferase positive T47D metastatic breast cancer stem cells mixed with NME7_{AB} to a final concentration of 32 nM. On Day 1 and Day 2 animals were injected into the tail vein with more 32 nM NME7_{AB}, which we have shown increases metastases. This is a system to demonstrate reversion of established metastases. On Day 5 animals were injected intravenously with 15 mg/kg of 8F9A4A3, aka 4A3, wherein the antibody was either the murine sequence or humanized 4A3 variant 4.8, or humanized 4A3 variant 12.3. FIG. 52A shows control animals injected with phosphate buffered saline solution. FIG. 52B shows animals treated with anti-NME7_{AB} humanized 4A3 4.8. FIG. 52C shows animals treated with anti-NME7_{AB} humanized 4A3 12.3. FIG. 52D shows animals treated with anti-NME7_{AB} mono-

clonal antibody murine 4A3. FIG. 52E shows a graph of the quantification of bioluminescence measurement of tumor volume, taken in an IVIS instrument. Animals were injected with either PBS or an anti-NME7 antibody at 15 mg/kg once every 3 days. Antibody hu 4A3 4.8 denotes that the sequences derived from human heavy chain antibody sequence 1.46 (SEQ ID NO: 1102) and light chain sequence 1.6 (SEQ ID NO: 1104) were used. Antibody hu 4A3 12.3 denotes that the sequences derived from human heavy chain antibody sequence 4.4 (SEQ ID NO: 1106) and light chain sequence 4.1 (SEQ ID NO: 1108) were used. Note that half the animals treated with hu 4A3 4.8 were given a partial half dose.

Humanized 4A3 variant 4.8 was generated from human heavy chain IGHV1-46*01 (IMGT accession number X92343) and from human light chain IGKV1-6*01 (IMGT accession number M64858). Humanized 4A3 variant 4.8 was also called 4A3 146-16.

Humanized 4A3 variant 12.3 was generated from human heavy chain IGHV4-4*01 (IMGT accession number X05713) and from human light chain IGKV4-1*01 (IMGT accession number Z00023). Humanized 4A3 variant 12.3 was also called 4A3 44-41.

The accession numbers are from the IMGT data base (the international Immunogenetics information system).

IGHVx-x or IGKVx-x correspond to the IMGT gene name. *x corresponds to the IMGT allele name

We performed immunofluorescent imaging of many cancer cell lines to determine if cultured cancer cell lines express NME7AB. As FIG. 47A-47F and FIG. 48A-48I clearly show, each MUC1 positive cancer cell line we tested is positive for NME7AB and its binding is membranous, consistent with NME7_{AB} being secreted from cancer cells whereupon it binds to the extra cellular domain of MUC1*. FIG. 47A-47F shows photographs of immunofluorescent experiments in which various cancer cell lines are stained for the presence of NME7AB. FIG. 47A shows T47D breast cancer cells stained with varying concentrations of anti-NME7AB antibody 5D4. FIG. 47B shows ZR-75-1 breast cancer cells, also known as 1500s, stained with varying concentrations of anti-NME7AB antibody 5D4. FIG. 47C shows H1975 non-small cell lung cancer cells stained with varying concentrations of anti-NME7AB antibody 5D4. FIG. 47D shows H292 non-small cell lung cancer cells stained with varying concentrations of anti-NME7AB antibody 5D4. FIG. 47E shows HPAFII pancreatic cancer cells stained with varying concentrations of anti-NME7AB antibody 5D4. FIG. 47F shows DU145 prostate cancer cells stained with varying concentrations of anti-NME7AB antibody 5D4. As can be seen in the figure, all the cancer cell lines we tested show strong and membranous staining for NME7_{AB}. The monoclonal antibody used in these experiments was 5D4. In parallel, NME7_{AB} antibodies 5A1 and 4A3 were used to stain the same cell lines and produced the same results.

FIG. 48A-48I shows photographs of immunofluorescent experiments in which various lung cancer cell lines are stained for the presence of NME7AB. FIG. 48A-48C shows H1975 non-small cell lung cancer cells, which are an adenocarcinoma, stained with varying concentrations of anti-NME7_{AB} antibody 5D4. FIG. 48A is an overlay of DAPI and anti-NME7_{AB} staining. FIG. 48B shows anti-NME7_{AB} staining alone. FIG. 48C is a magnified view of the overlay of DAPI and anti-NME7_{AB} staining. FIG. 48D-48F shows H292 non-small cell lung cancer cells, which are a mucoepidermoid pulmonary carcinoma, stained with varying concentrations of anti-NME7_{AB} antibody 5D4. FIG. 48D

is an overlay of DAPI and anti-NME7_{AB} staining. FIG. 48E shows anti-NME7_{AB} staining alone. FIG. 48F is a magnified view of the overlay of DAPI and anti-NME7_{AB} staining. FIG. 48G-48I shows H358 non-small cell lung cancer cells, which are a metastatic bronchioalveolar carcinoma, stained with varying concentrations of anti-NME7_{AB} antibody 5D4. FIG. 48G is an overlay of DAPI and anti-NME7_{AB} staining. FIG. 48H shows anti-NME7_{AB} staining alone. FIG. 48I is a magnified view of the overlay of DAPI and anti-NME7_{AB} staining.

In addition, culturing these cell lines in a serum-free media containing NME7_{AB} even further increased their expression of stem cell and metastatic markers. In particular, the cells that became non-adherent, referred to here as floaters, have even higher expression of stem cell and metastatic markers than their adherent counterparts. FIG. 49A-49I shows PCR graphs of cancer cell lines, breast T47D, lung H1975, lung H358 and pancreatic HPAFII before and after culture in NME7_{AB}. FIG. 49A measured breast metastatic marker CXCR4. FIG. 49B measured stem cell marker OCT4. FIG. 49C measured metastatic marker ALDH1. FIG. 49D measured stem cell marker SOX2. FIG. 49E measured stem cell marker NANOG. FIG. 49F measured marker CDH1, also known as E-cadherin. FIG. 49G measured metastatic marker CD133. FIG. 49H measured stem cell marker ZEB2. FIG. 49I measured stem, cancer and metastatic marker MUC1. The floater cells, also known as tumor spheres become able to grow anchorage independently and show markers of metastasis that are more elevated than the adherent cells. Animals injected with cancer stem cells are those injected with the NME7_{AB} grown floater cells. As can be seen in the figure markers of metastasis, stem cell markers, or markers of epithelial to mesenchymal transition (EMT) are elevated after culture in NME7_{AB}, indicating a transition to a more metastatic state. FIG. 50 shows Day 6 IVIS photographs of NSG mice injected into the tail vein with either 10,000 H358 lung cancer parent cells or H358 cells after 10-12 days in culture with NME7_{AB}. As can be seen in the figure, the NCI-H358 lung cancer cells grown in NME7_{AB} have greatly increased metastatic potential compared to the parent cells, which are themselves reportedly metastatic cells. The functional increase in metastasis in 6 days from the NCI-H358 NME7_{AB} metastatic cancer stem cells from just 10,000 cells is consistent with FIG. 49, showing that H358 cells greatly increased expression of metastatic markers after culture in NME7_{AB}.

FIG. 51 shows PCR graph of a MUC1 negative prostate cancer line PC3 before and after 2 or 3 passages in culture in either dimeric NM23-H1, also known as NME1, or NME7_{AB}. The graph shows the fold difference in markers of stem cells, cancer cells as well as metastatic markers. As can be seen in the figure, repeated culture in NME1 or NME7_{AB} induces upregulation of stem, cancer and metastatic markers but also upregulates expression of MUC1 by 5-8 times.

Collectively, these data have demonstrated that an NME7 that is devoid of the DM10 domain is secreted by cancer cells and binds to the extra cellular domain of a MUC1 that is devoid of tandem repeat domain, whereupon the NME7 dimerizes the MUC1* extra cellular domain which results in increased cancer cell growth and an increase in the cancer cells metastatic potential. It stands to reason that antibodies that disrupt the interaction between NME7_{AB} and MUC1* extra cellular domain would inhibit cancer cell growth and would inhibit cancer metastasis. Here, we have shown that anti-NME7_{AB} antibodies that inhibit interaction between NME7_{AB} and MUC1* extra cellular domain do in fact

inhibit cancer cell growth and cancer metastasis. Therefore, it follows that anti-NME7_{AB} antibodies can be administered to a patient, diagnosed with or at risk of developing a cancer or metastasis, for the treatment or prevention of cancers.

Because NME1 is expressed in the cytoplasm of all cells and can be lethal if knocked out, and importantly the NME1 A domain has high sequence homology to the NME7 A domain, it is critical that anti-NME7_{AB} antibodies for therapeutic use bind to NME7_{AB} or NME7-X1, but not to NME1. In one aspect of the invention antibodies that would be optimal for therapeutic use were selected for their ability to bind to peptides that were unique to NME7_{AB} or NME7-X1 and were not present in the NME1 sequence. FIG. 6-FIG. 9 lists NME7_{AB} unique peptides.

In a preferred embodiment, antibodies suitable for administration to a patient for the treatment or prevention of cancer or cancer metastasis are selected from the group of antibodies that bind to the NME7 B3 peptide. In yet a more preferred embodiment, antibodies suitable for administration to a patient for the treatment or prevention of cancer or cancer metastasis are selected from the group of antibodies that bind to the NME7 B3 peptide, bind to NME7_{AB} but do not bind to NME1. Examples of antibodies suitable for therapeutic use for the treatment or prevention of cancers or cancer metastasis, which have demonstrated such anti-cancer activity and anti-metastatic activity in vitro and in vivo here, include anti-NME7 antibodies 5A1, 4A3 and 5D4. These are but examples and other antibodies generated as described here and selected as described here will have the same anti-cancer and anti-metastatic activity. Such antibodies may be full antibodies or fragment thereof, including scFvs or antibody mimics wherein the variable domains of the antibody are incorporated into a protein scaffold that mimic an antibody. The antibodies may be of human or non-human species, including murine, camelid, llama, human or humanized and may be monoclonal, polyclonal, scFvs or fragments thereof.

Anti-NME7 antibodies for treatment or prevention of cancers or metastases can be used in many different therapeutic formats. For example, any of the antibodies described herein, or a fragment thereof, can be administered to a patient as a stand-alone antibody or antibody fragment, or attached to a toxin such as an antibody drug conjugate (ADC), or incorporated into a bi-specific antibody or incorporated into a BiTE (bispecific T cell engager), or incorporated into a chimeric antigen receptor (CAR) or engineered to be expressed by a cell that also expresses a CAR. The cell may be an immune cell, a T cell, an NK cell or a stem or progenitor cell, which may then be differentiated into a T cell or an NK cell.

Any of the antibodies described herein, or a fragment thereof, can be used as a diagnostic reagent to probe a bodily fluid, cell, tissue or bodily specimen for the presence of NME7_{AB} or NME7-X1, which would be an indicator of cancer or susceptibility to cancers. Antibodies for diagnostic uses may be connected to an imaging agent, a nucleic acid tag, may be of any species including camelid, and can be used in whole body applications or on a bodily fluid, such as blood, cell, or tissue, in vitro, in vivo or intra-operatively.

The selection criteria, for therapeutically useful or diagnostically useful anti-NME7 antibodies, depends on the format or modality of the therapeutic or diagnostic into which the antibody will be incorporated. If the antibody or antibody fragment is to be administered to a patient as a stand-alone agent for the treatment or prevention of cancers or cancer metastases, then the antibody is selected for its ability to: i) bind to NME7_{AB} or NME7-X1, but not to

NME1; ii) bind to the PSMGFR peptide; iii) bind to the N-10 peptide and iv) disrupt the interaction between NME7_{AB} or NME7-X1 and the MUC1* extra cellular domain or the interaction between NME7_{AB} or NME7-X1 and the N-10 peptide. The antibody may also be selected for its ability to bind to the NME7 B3 peptide. This therapeutic format also encompasses a cell that has been engineered to express a CAR and a secreted anti-NME7 antibody.

Other modalities require other selection criteria for anti-NME7 antibodies. If the anti-NME7 antibody is to be incorporated into an ADC, the ADC must be internalized by the target cell to trigger killing of the target cell. Recall that NME7_{AB} or NME7-X1 will be bound to the extra cellular domain of MUC1*. If the antibody disrupts binding of the NME to MUC1* extra cellular domain, then the toxin-conjugated antibody will not be internalized and the cell will not be killed. Similarly, if the anti-NME7 antibody is to be incorporated into a CAR or a BiTE, the interaction between NME7_{AB} or NME7-X1 cannot be disrupted or the immune cell will no longer be able to direct its killing agents to the cancer cell. If the anti-NME7 antibody is to be used as a diagnostic reagent, the interaction between NME7_{AB} or NME7-X1 cannot be disrupted or antibody and associated label will be washed away. Therefore, for ADC, CAR T, or CAR-NK, BiTEs or diagnostic applications, the anti-NME7 antibody is selected for its ability to: i) bind to NME7_{AB} or NME7-X1, but not to NME1; ii) bind to the PSMGFR peptide; iii) bind to the N-10 peptide and iv) bind to NME7_{AB} or NME7-X1 without disrupting the interaction with the MUC1* extra cellular domain or the interaction between NME7_{AB} or NME7-X1 and the N-10 peptide. The antibody may also be selected for its ability to bind to the NME7 B3 peptide.

In one aspect of the invention, a cell is engineered to express an anti-NME7_{AB} antibody of the invention or fragment thereof. The cell may be an immune cell, such as a T cell or NK cell or it may be a stem or progenitor cell, which may be differentiated into a more mature immune cell such as a T cell or NK cell. In a preferred embodiment, the cell that is engineered to express an anti-NME7_{AB} antibody is also engineered to express a chimeric antigen receptor (CAR). In a preferred embodiment, the CAR recognizes a tumor associated antigen. In a preferred embodiment, the CAR targets MUC1*. In a more preferred embodiment, the CAR is directed to the tumor by anti-MUC1* antibody MNC2. In another aspect of the invention, cell that is engineered to express a CAR is also engineered to inducibly express an anti-NME7 antibody. In one example, the nucleic acid encoding an anti-NME7_{AB} antibody is inserted into the Foxp3 enhancer or promoter. In another example, the anti-NME7_{AB} antibody is in an NFAT-inducible system. In one aspect, the NFAT-inducible system incorporates NFATc1 response elements inserted upstream of an anti-NME7_{AB} antibody sequence. They may be inserted into an IL-2 promoter, a Foxp3 enhancer or promoter or other suitable promoter or enhancer.

In another aspect of the invention, peptides that are unique to NME7_{AB} or NME7-X1 are incorporated into an entity used to immunize or vaccinate people against cancers or cancer metastases. In a preferred embodiment, the peptide

comprises all or part of the NME7 B3 peptide, which may be the NME7 B3 peptide with Cys-14-Ser mutation.

Another aspect of the invention involves a method of generating anti-NME7_{AB} antibodies in a host animal, where the animal is immunized with the NME7 B3 peptide. In a preferred embodiment, the NME7 B3 peptide has Cysteine 14 mutated to Serine (SEQ ID NO: 169) to avoid disulfide bond formation which inhibits NME7 specific antibody generation.

Another aspect of the invention involves a method of generating cells with enhanced metastatic potential involving culturing the cells with NME7_{AB} or NME7-X1. These cells can then be used in many aspects of drug discovery.

Another aspect of the invention involves a cell that is engineered to express NME7_{AB} or NME7-X1. The NME7_{AB} or NME7-X1 may be of human sequence. Their expression may be inducible. In one aspect the cell is an egg which is then developed into an animal that may be a transgenic animal able to express human NME7_{AB} or NME7-X1.

NME7 binds to and dimerizes the extra cellular domain of the MUC1* growth factor receptor. Tissue studies show that MUC1* increases as tumor grade and metastasis increase. Here we show that NME7 expression increases as tumor grade and metastasis increases (FIG. 39-FIG. 41). Here, we have shown that antibodies that inhibit the interaction of NME7 and MUC1* inhibit tumor growth and metastases.

Other NME family members may bind to and dimerize the extra cellular domain of the MUC1* growth factor receptor. For example, we have shown that NME1, NME2 and NME6 can exist as dimers and that they bind to and dimerize the MUC1* extra cellular domain. NME7_{AB} and NME7-X1 have two domains that can bind to the MUC1* extra cellular domain so as monomers they dimerize and activate the MUC1* growth factor receptor. We have now shown that anti-NME7 antibodies inhibit cancer and cancer metastases. Similarly, antibodies or antibody mimics that bind to these other NME proteins may be anti-cancer or anti-metastasis therapeutics that can be administered to a patient diagnosed with or at risk of developing a cancer or a metastasis. In one aspect of the invention, antibodies that can be used therapeutically for the treatment of cancers or metastases are antibodies that bind to NME1, NME2, NME3, NME4, NME5, NME6, NME7, NME8, NME9 or NME10. In one aspect of the invention, the therapeutic antibody or antibody mimic inhibits the binding of the NME protein and its cognate growth factor receptor. In one aspect of the invention, the therapeutic antibody or antibody mimic inhibits the interaction of the NME protein with the extra cellular domain of MUC1*. In another aspect of the invention, the therapeutic antibody or antibody mimic binds to a peptide, derived from NME1, NME2, NME3, NME4, NME5, NME6, NME7, NME8, NME9 or NME10, wherein the peptide is homologous to the NME7 A1, A2, B1, B2 or B3 peptide.

Below is a sequence alignment that shows a homology and identity alignment between NME7 and other NME family members. The underlined or underlined and bolded sequences correspond to NME7 peptides A1 (SEQ ID NO: 141), A2 (SEQ ID NO: 142), B1 (SEQ ID NO: 143), B2 (SEQ ID NO: 144) and B3 (SEQ ID NO: 145).

nucleoside diphosphate kinase 7 isoform a [*Homo sapiens*] (Hu_7)
 MNHSERFVFVIAEWYDFNASLLRRYELLFYPGDSVEMHDVKNHRTFLKRTKYDNLHLEDLFIGNKVNVFSRQLVL
 IDYGDQYTARQLCSRKEKTLALIKPDAISKAGEIIIEIINKAGFTITKLMMMLSRKEALDFHVDHQSRPFFNELI

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7B  N-C-----TCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFMN
      :          :          :          :          :          :
3   MICLVLTIFANLFPSAYSGVNERTFLAVKPDGVQRRLVGEIVRRFERKGFKLVALKLQVQA
      10          20          30          40          50          60

      40          50          60          70          80          90
7B  DRVNVFEFYEYKGVVTEYHDMVTEMYSGPCVAMEIQNNATKTFREFCGPADPEIARHL
      :          :          :          :          :          :
3   SEELLREHY-VELRERPFYSRLVKYMGSGPVVAMVWQGLDVVRASRALIGATDPGDAT--
      70          80          90          100         110

      100         110         120         130
7B  RPGTLRAIFGKTKIQNAVHCTDLPEDGLLEVQYFF----- (SEQ ID NO: 205)
      :          :          :          :          :          :
3   -PGTIRGDFCQVEVGKNVIHGSDSVESAQREIALWFREDELLCWEDSAGHWLYE (SEQ ID NO: 206)
      120         130         140         150         160

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NME4 Theoretical pI/Mw: 10.30/20658.59

MGGLFWRSALRGLRCPAPGPSLLVRHSGSGPSWTRERTLVAVKPDGVQRRLVGDVIQRFERRGFTLVGMKMLQ

APESVLAEHYQDLRRKPFYPALIRYMSSGPVVAMVWEGYNVVRASRAMIGHTDSAEAAPGTIRGDFSVHISRNV

HASDSVEGAQREIQLWFQSSSELVSWADGGQHSSIHQA (SEQ ID NO: 1110)

29.3% identity (68.4% similar) in 133 aa overlap (1-131:56-185)

```

      10          20          30          40          50
7A  EKTALAIKPDASK--AGEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQSRPFFNEL
      :          :          :          :          :          :
4   ERTLAVKPDGVQRRLVGDVIQRFERRGFTLVGMKMLQAPESVLAEHYQDLRRKPFYPAL
      60          70          80          90          100         110

      60          70          80          90          100         110
7A  IQFITGPIIAMEILRDDAICEWKRLGPNSGVARTDASESIRALFGTDGIRNAAHGPD
      :          :          :          :          :          :
4   IRYMSSGPVVAMVWEGYNVVRASRAMIGHTDSAEAA---APGTIRGDFSVHISRNVIHASD
      120         130         140         150         160         170

      120         130
7A  SPASAAAREMELFF (SEQ ID NO: 207)
      :          :          :
4   SVEGAQREIQLWF (SEQ ID NO: 208)
      180

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28.8% identity (56.8% similar) in 132 aa overlap (3-134:40-167)

```

      10          20          30          40          50          60
7B  TCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFMNDRVNVVEEFYEYKGVVTEYHDMV
      :          :          :          :          :          :
4   TLVAVKPDGVQRRLVGDVIQRFERRGFTLVGMKMLQAPESVLAEHYQDLRRK-PFYPALI
      40          50          60          70          80          90

      70          80          90          100         110         120
7B  TEMYSGPCVAMEIQNNATKTFREFCGPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDL
      :          :          :          :          :          :
4   RYMSSGPVVAMVWEGYNVVRASRAMIGHTDSAEAA---PGTIRGDFSVHISRNVIHASDS
      100         110         120         130         140         150

      130
7B  PEDGLLEVQYFF (SEQ ID NO: 209)
      :          :          :
4   VEGAQREIQLWF (SEQ ID NO: 210)
      160

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NME5 Theoretical pI/Mw: 6.08/29296.23

MEISMPPQIYVEKTLAIKPDIVDKKEEIQDIILRSGFTIVQRRKLRLSPEQCSNFYVEKYGKMFFPNLTAYMS

SGPLVAMILARHKAISYWLELLGPNNLSLVAKETHPDSLRAIYGTDDLRLNALHGSNDFAAAEREIRFMFPEVIVEP

IPIGQAADYLNHIMPTLLEGLTELCKQKPADPLFWYMCCRREHWTILRSILLVCMGSRMSLPHCADYCSFVEG

FEIWLADWLLKNNPNKPKLCHHPIVEEPPY (SEQ ID NO: 1111)

44.3% identity (74.8% similar) in 131 aa overlap (1-131:13-143)

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      10          20          30          40          50          60
7A  EKTALAIKPDASKAGEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQSRPFFNELIQ
      :          :          :          :          :          :
5   EKTALAIKPDIVDKKEEIQDIILRSGFTIVQRRKLRLSPEQCSNFYVEKYGKMFFPNLTA
      20          30          40          50          60          70

      70          80          90          100         110         120
7A  FITGPIIAMEILRDDAICEWKRLGPNSGVARTDASESIRALFGTDGIRNAAHGPDSEF
      :          :          :          :          :          :

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NME8 Theoretical pI/Mw: 4.90/67269.94
 MASKKKREVQLQTVINNQSLEWDEMLQNKGLTVIDVTVQAWCGPCRAMQPLFRKKLNELNEDEILHFVAVEADNIVTL
 QPFRDKCEPVFLFSVNGKILEKIQGANAPLVNKKVINLIDEERKIAAGEMARPPQYPEIPLVDSSEVSEESPCES
 VQELYSIAIIKPDAVISKKVLEIKRKITKAGFIEAEHKTVLTEEQVNVFYSRIADQCDFEEFVFSFMTSLGSLYIL
 VVSQGSKHNPPSEETEPQTDTEPNERSEDQPEVEAQVTPGMMNKQDLSQELYERQHLAQLCDIEEDAANVAKFM

-continued

Waterman-Eggert score: 30; 15.1 bits; E(1) < 0.73
 28.9% identity (57.8% similar) in 45 aa overlap (66-108:200-244)

70	80	90	100
7A	PIIAMEILRDDAIC-EWKRLG	PANSGVARTDASES-IRALFGTD	(SEQ ID NO: 241)
10	PIPTTEELKAVRVSTEANRSIVPISRGQRQKSSDESCLVVLFAGD		(SEQ ID NO: 242)
200	210	220	230 240

Waterman-Eggert score: 33; 14.4 bits; E(1) < 0.87
 14.7% identity (52.0% similar) in 75 aa overlap (7-80:35-109)

10	20	30	40	50	60
7A	IKPDAISKAGEIIEIINKAGFTITKLKMM	<u>MMLSRKEALDFHVDHQSR</u>	PPFFNELIQ-FITTG		
10	VDPKDYMFSGLKDETVGRPLPGTVAGQQFLIQDCENCNIYIFDHSATVTIDDC	TNCIIIFLG			
	40	50	60	70	80 90

70	80
7A	PIIAMEILRDDAICE (SEQ ID NO: 243)
10	PVKGSVFFRNCRDCK (SEQ ID NO: 244)
100	

Waterman-Eggert score: 45; 17.5 bits; E(1) < 0.22
 21.6% identity (58.8% similar) in 51 aa overlap (4-50:130-180)

10	20	30	40	50
7B	CCIVKP--HAVSEGLLGKILMAIRDAGFEI--	SAMQMFNMDRVNVEEFYEV	(SEQ ID NO: 245)	
10	CCATQPIIIESSNIKFGCFQWYYPPELAFQPKDAGLSIFNNTWSNIHDFTPV		(SEQ ID NO: 246)	
130	140	150	160	170 180

As an example, antibodies or antibody mimics that bind to the NME7 homologous peptides ("homologous peptides") in particular homologous to A1, A2, B1, B2 or B3 peptides, may be administered to a patient diagnosed with or at risk of developing a cancer or cancer metastasis.

Homologous Peptides to A1, A2, B1, B2 or B3 Peptides

Homologous peptides to A1, A2, B1, B2 or B3 peptides may include without limitation the following:

NME2A1	(amino acids)	
RASEEHLKQHYIDLKD	(SEQ ID NO: 247)	40
NME2A2	(amino acids)	
PADSKPGT	(SEQ ID NO: 248)	45
NME2B1	(amino acids)	
QKGFRVLVAMKFLRASEEHLK	(SEQ ID NO: 249)	50
NME2B2	(amino acids)	
IDLKDRPFPGLVKY	(SEQ ID NO: 250)	55
NME2B3	(amino acids)	
GDFCIQVGRNIIHGSDSVKSAEKEISLWF	(SEQ ID NO: 251)	60
NME3A1	(amino acids)	
QASEELLREHYVELRE	(SEQ ID NO: 252)	65
NME3A1	(amino acids)	
PGDATPGT	(SEQ ID NO: 253)	

-continued

NME3B1	(amino acids)	
RKGFKLVALKLVQASEELLR	(SEQ ID NO: 254)	
NME3B2	(amino acids)	
VELRERPFYSRLVKY	(SEQ ID NO: 255)	
NME3B3	(amino acids)	
GDFCIVEVGKNVIHGSDSVESAQREIALWF	(SEQ ID NO: 256)	
NME4A1	(amino acids)	
QAPESVLAEHYQDLRR	(SEQ ID NO: 257)	
NME4A2	(amino acids)	
SAEAAPGT	(SEQ ID NO: 258)	
NME4B1	(amino acids)	
RRGFTLVGMKMLQAPESVLA	(SEQ ID NO: 259)	
NME4B2	(amino acids)	
QDLRRKPFYPALIRY	(SEQ ID NO: 260)	
NME4B3	(amino acids)	
GDFSVHISRNVIHASDSVEGAQREIQLWF	(SEQ ID NO: 261)	
NME5A1	(amino acids)	
RLSPEQCSNFYVEKYG	(SEQ ID NO: 262)	

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-continued

NME5A2 (amino acids)	(SEQ ID NO: 263)	
SLVAKETHPDS		5
NME5B1 (amino acids)	(SEQ ID NO: 264)	
RSGFTIVQRRKLRSLPEQCS		10
NME5B2 (amino acids)	(SEQ ID NO: 265)	
VEKYGKMFFPNLTAY		15
NME5B3 (amino acids)	(SEQ ID NO: 266)	
AIYGTDDLRLNALHGSNDFAAAREIRFMF		20
NME6A1 (amino acids)	(SEQ ID NO: 267)	
LWRKEDCQRFYREHEG		25
NME6A2 (amino acids)	(SEQ ID NO: 268)	
VFRARHVAPDS		30
NME6B1 (amino acids)	(SEQ ID NO: 269)	
SNKFLIVRMRELLWRKEDCQ		35
NME6B2 (amino acids)	(SEQ ID NO: 270)	
REHEGRFFYQRLVEF		40
NME6B3 (amino acids)	(SEQ ID NO: 271)	
GSFGLTDRNTTHGSDSVVSASREIAAFF		45
NME8A1 (amino acids)	(SEQ ID NO: 272)	
VLSEKEAQALCKEYEN		50
NME8A2 (amino acids)	(SEQ ID NO: 273)	
VEEAIEYFPES		55
NME8A3 (amino acids)	(SEQ ID NO: 274)	
FLTPEQIEKIYPKVTG		60
NME8A4 (amino acids)	(SEQ ID NO: 275)	
PEEAKLLSPDS		65
NME8A5 (amino acids)	(SEQ ID NO: 276)	
VLTEEQVNVFYSRIAD		
NME8B1 (amino acids)	(SEQ ID NO: 277)	
EAGFDLTQVKMFLTPEQIE		
NME8B2 (amino acids)	(SEQ ID NO: 278)	
PKVTGKDFYKDLLEM		
NME8B3		

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-continued

(amino acids)	(SEQ ID NO: 279)	
AQFGISKLNIVH		
NME8B4 (amino acids)	(SEQ ID NO: 280)	
DEDFKILEQRQVVLSEKEAQ		
NME8B5 (amino acids)	(SEQ ID NO: 281)	
KEYENEDYFNKLIEN		
NME8B6 (amino acids)	(SEQ ID NO: 282)	
AQFAMDSLFPVNQLYGSDSLTAEREIQHFF		
NME8B7 (amino acids)	(SEQ ID NO: 283)	
KAGFIIEAEHKTVLTTEEQVV		
NME8B8 (amino acids)	(SEQ ID NO: 284)	
SRIADQCDFEEFVSF		
NME9A1 (amino acids)	(SEQ ID NO: 285)	
TMTEAEVRLFY		
NME9B1 (amino acids)	(SEQ ID NO: 286)	
EAGFEILTNEERTMTAEVR		
NME10A1 (amino acids)	(SEQ ID NO: 287)	
SMKAEDAQRVFREK		
NME10A2 (amino acids)	(SEQ ID NO: 288)	
GQRQKSSDES		
NME10A3 (amino acids)	(SEQ ID NO: 289)	
IQDCENCNIYIPDHSA		
NME10B1	(SEQ ID NO: 290)	
ELAFQFKDAGLSIFNNTWSNIH		

In some cases, peptides derived from other NME proteins can be made more homologous to NME7 A1, A2, B1, B2 or B3 peptides by shifting the frame or extending the NME7 peptides such that the extended peptides are more homologous to the NME7 peptides that gave rise to antibodies that inhibit cancer or cancer metastases. As another example, antibodies or antibody mimics that bind to the NME7 homologous extended peptides ("extended peptides") may be administered to a patient diagnosed with or at risk of developing a cancer or cancer metastasis.

Homologous Extended Peptides to A1, A2, B1, B2 or B3 Peptides

Homologous peptides to A1, A2, B1, B2 or B3 peptides that are extended peptides may include without limitation the following:

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NME2A1
(amino acids) (SEQ ID NO: 291)
RASEEHLKQHYIDLKDRPFFPGL 5

NME2A2
(amino acids) (SEQ ID NO: 292)
LGETNPADSKPGTIRGDF

NME2B1
(amino acids) (SEQ ID NO: 293) 10
GLVGEIIKRFEQKGFRVLVAMKFLRASEEHLKQHY

NME2B2
(amino acids) (SEQ ID NO: 294) 15
YIDLKDRPFFPGLVKYMNSGPPVAM

NME2B3
(amino acids) (SEQ ID NO: 295) 20
PGTIRGDFCIQVGRNIIHGSDSVKSAEKEISLWF

NME3A1
(amino acids) (SEQ ID NO: 296) 25
LKL VQASEELLREHYVELRERPFYSRL

NME3A1
(amino acids) (SEQ ID NO: 297)
LIGATDPGDATPGTIRGDF

NME3B1
(amino acids) (SEQ ID NO: 298) 30
LVGEIVRRFERKGFKLVALKL VQASEELLRE

NME3B2
(amino acids) (SEQ ID NO: 299) 35
EHY-VELRERPFYSRLVKYMGSGPPVAM

NME3B3
(amino acids) (SEQ ID NO: 300) 40
PGTIRGDFCVEVGKNVIHGSDSVESAQREIALWF

NME4A1
(amino acids) (SEQ ID NO: 301) 45
GFTLVGMKMLQAPESVLAHYQDLRRKPF

NME4A2
(amino acids) (SEQ ID NO: 302)
GHTDSAEAAPGTIRGDF

NME4B1
(amino acids) (SEQ ID NO: 303) 50
LVGDVIQRFERRGFTLVGMKMLQAPESVLAHY

NME4B2
(amino acids) (SEQ ID NO: 304) 55
EHYQDLRRKPFYPALIRYMSSGPPVAM

NME4B3
(amino acids) (SEQ ID NO: 305) 60
PGTIRGDFSVHISRNVIIHASDS VEGAQREIQLWF

NME5A1
(amino acids) (SEQ ID NO: 306) 65
GFTIVQRRKLRLSPEQCSNFYVEKYGKMFF

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-continued

NME5A2
(amino acids) (SEQ ID NO: 307)
LLGPNNSLVAKETHPDSLRAIYGTD

NME5B1
(amino acids) (SEQ ID NO: 308)
IQDIILRSGFTIVQRRKLRLSPEQCSNFY

NME5B2
(amino acids) (SEQ ID NO: 309)
FYVEKYGKMFFPNLTAYMSSGPPVAM

NME5B3
(amino acids) (SEQ ID NO: 310)
PDSLRAIYGTDDLRLNALHGSNDFAAAEREIRFMF

NME6A1
(amino acids) (SEQ ID NO: 11)
FLIVMRRELLWRKEDCQRFYREHEGRFFYQRL

NME6A2
(amino acids) (SEQ ID NO: 12)
LMGPTRVFRARHVAPDSIRGSFG

NME6B1
(amino acids) (SEQ ID NO: 13)
ILSNKFLIVMRRELLWRKEDCQRFY

NME6B2
(amino acids) (SEQ ID NO: 314)
FYREHEGRFFYQRLVEFMASGPIRA

NME6B3
(amino acids) (SEQ ID NO: 15)
ARHVAPDSIRGSFGLTDTRNTTHGSDSVVSASREIAAFF

NME8A1
(amino acids) (SEQ ID NO: 16)
FKILEQRQVVLSEKEAQAALCKEYENEDYFNKLI

NME8A2
(amino acids) (SEQ ID NO: 17)
WKQLLGPRTVEEAIEYFPESLCAQFAMD

NME8A3
(amino acids) (SEQ ID NO: 18)
AGFDLTQVKKMFLTPEQIEKIYPKVTGKDFYKDL

NME8A4
(amino acids) (SEQ ID NO: 19)
EWRRLMGPTDPEEAKLLSPDSIRAQFG

NME8A5
(amino acids) (SEQ ID NO: 320)
KAGFIIEAEHKTVLTEBQVVNFYSRIADQCFEE

NME8B1
(amino acids) (SEQ ID NO: 321)
ILKIVKEAGFDLTQVKKMFLTPEQIEKIY

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-continued

NME8B2
(amino acids) (SEQ ID NO: 322)
YPKVTGKDFYKDLLEMLSVGP

NME8B3
(amino acids) (SEQ ID NO: 323)
DPPEAKLLSPDSIRAQFGISKLNIVH

NME8B4
(amino acids) (SEQ ID NO: 324)
LRIKDEDFKILEQRQVVLSEKEAQ

NME8B5
(amino acids) (SEQ ID NO: 325)
KEYENE-DYFNKLIENMTSGPSLA

NME8B6
(amino acids) (SEQ ID NO: 326)
PESLCAQFAMDLPVNLVYSGDSLETAEREIQHFF

NME8B7
(amino acids) (SEQ ID NO: 327)
IKRKITKAGFIIIEAEHKTVLTEEQVVNFY

NME8B8
(amino acids) (SEQ ID NO: 328)
FYSRIADQCDFEEFVSFMTSG

NME9A1
(amino acids) (SEQ ID NO: 329)
AGFEILTNEERTMTAEVRLFY

NME9B1
(amino acids) (SEQ ID NO: 330)
IIMKIQEAGFEILTNEERTMTAEVRLFY

NME10A1
(amino acids) (SEQ ID NO: 331)
GFFLVQTKEVSMKAEDAQRVPREKAP

NME10A2
(amino acids) (SEQ ID NO: 332)
EANSRIVPISRGRQKSSDESCLVVLFGD

NME10A3
(amino acids) (SEQ ID NO: 333)
IQDCENCNIYIFDHSA

NME10B1
(amino acids) (SEQ ID NO: 334)
ELAFQPKDAGLSIFNNTWSNIHDFTPDVCT

Some NME proteins exert a function that is necessary for normal cell growth or development. For example, NME1 is thought to be required for normal cell function. Other NME proteins have catalytic domains whose function is required in normal cells or tissues. In these cases, therapeutic antibodies can be selected based on their ability to bind to the targeted, cancer associated NME, but not to a non-targeted NME. For example, the anti-NME7 antibodies presented here, 8F9A5A1, 8F9A4A3, and 5F3A5D4, were selected for their ability to bind to NME7_{AB} but not to NME1; they were further selected based on their ability to inhibit cancer and cancer metastases.

In another aspect of the invention, anti-NME7 antibodies, antibody fragments, for example scFvs, or fragments of antibody mimics are incorporated into chimeric antigen

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receptors (CARs) which are engineered to be expressed in immune cells. The immune cell can be engineered to express an anti-NME7 CAR, an anti-MUC1* CAR, or both. One of the CARs may be expressed off of an inducible promoter. Alternatively, an immune cell may be engineered to express a CAR such as an anti-MUC1* CAR and an inducible anti-NME7 antibody or antibody fragment. In some instances the inducible promoter may contain NFAT response elements. In one aspect, these engineered species are expressed in T cells, NK cells or dendritic cells. The immune cells may be obtained from the patient or from a donor. In some cases, immune molecules such as MHCs, checkpoint inhibitors or receptors for checkpoint inhibitors are mutated or cut out, for example using CrisPR or CrisPR-like technology. In another aspect, ITAM molecules, Fos, or Jun are mutated or genetically excised via Talens. Sleeping Beauty, CrisPR or CrisPR-like technologies in patient or donor derived immune cells.

In one aspect of the invention, the anti-NME7 antibodies or antibody mimics for use in CAR T format are chosen from among the group of antibodies or antibody mimics that are specific for NME7 but do not disrupt the binding of NME7 to the extra cellular domain of MUC1*. In this way, the anti-NME7 antibody or antibody mimic that targets the CAR T to the tumor will not simply pluck the ligand from the receptor, whereupon the T cell would be unable to inject the target cancer cell with Granzyme B. Such antibodies or antibody mimics are generated by immunizing an animal with an NME7 peptide, such as NME7 peptides A1, A2, B1, B2 or B3 or selected by virtue of their ability to bind to NME7 peptides A1, A2, B1, B2 or B3. Antibodies or antibody mimics can be screened for their ability to specifically bind to NME7, but not to NME1 or NME2, and also for their inability to disrupt binding between NME7 and MUC1* extra cellular domain. For example, in an ELISA setup, the PSMGFR peptide is immobilized to the surface. A labeled NME7_{AB} is allowed to bind to the surface immobilized MUC1* extra cellular domain, and detection of the NME7_{AB} label is measured in the presence or absence of the test antibody or antibody mimic. In one aspect of the invention, an antibody that does not diminish binding between NME7_{AB} and surface-immobilized MUC1* extra cellular domain peptide is selected as an antibody that is incorporated into a CAR and engineered to be expressed in an immune cell and then administered to a patient for the treatment or prevention of cancer or cancer metastases.

In one aspect of the invention, an anti-NME7 antibody or fragment thereof is administered to a patient diagnosed with or at risk of developing a cancer or cancer metastasis. In one aspect, the anti-NME7 antibody or antibody fragment binds to an NME peptide discussed above in particular under sections "Homologous peptides to A1, A2, B1, B2 or B3 peptides" and the "Homologous extended peptides to A1, A2, B1, B2 or B3 peptides".

In another aspect, the antibody, antibody fragment or antibody mimic binds to an NME7 derived peptide chosen from among A1, A2, B1, B2 or B3 (SEQ ID NOS: 141-145). In yet another aspect, the antibody, antibody fragment or antibody mimic binds to an NME7 peptide comprising most or all of the B3 peptide. In one aspect of the invention, the anti-NME7 antibody, antibody fragment or antibody mimic comprises sequences derived from the variable domains of anti-NME7 antibodies 8F9A4A3 ("4A3") (SEQ ID NOS: 1001-1015), 8F9A5A1 ("5A1") (SEQ ID NOS: 1016-1030), or 8H5H5G4 ("5G4") (SEQ ID NOS: 1031-1045) shown below:

Anti-NME7 B3 peptide monoclonal antibodies
Monoclonal antibody 8F9A4A3 "4A3"
Heavy chain variable region sequence

(SEQ ID NO: 1138)

Gaggtcacagctgcaacagctctggacctgaactgggaagcctggggccttcagtgaagatatcctgcaagacttctgg
aaacacattcactgaatacaccatgcactgggtgaagcagagccatggaaagagccttgagtgattggaggttttaacctaacaatg
gtgttactaactacaaccagaagttcaagggcaaggccacattgactgtagacaagtcctccagcacagcctacatggagctccgcag
cctgacatctgaggattctgcagctctattactgtgcaagacggtactaccatagctctctacgtgttttactttgactactggggccaaggca
ccactctcacagtctcctca

Translated protein, wherein the underlined sequence is the complementarity
determining region (CDR):

(SEQ ID NO: 1139)

EVQLQQSGPELVKPGASVKISCKTSGNTFTEYTMHWVKQSHGKSLEWIG

GFNPNNGVITNYNQKFKGKATLTVDKSSSTAYMELRSLTSEDSAVYYCARRYHSLY

VFYFDYWGQTTTLTVSS

Mouse 8F9A4A3 heavy chain variable domain framework 1 (FR1) sequence

(SEQ ID NO: 1113)

gaggtcacagctgcaacagctctggacctgaactgggaagcctggggccttcagtgaagatatcctgcaagacttctgga

(SEQ ID NO: 1114)

EVQLQQSGPELVKPGASVKISCKTSG

Heavy chain variable region CDR1:

(SEQ ID NO: 1115)

Aacacattcactgaatacaccatgcac

(SEQ ID NO: 388)

NTFTEYTMH

Mouse 8F9A4A3 heavy chain variable domain framework 2 (FR2) sequence

(SEQ ID NO: 1116)

Tgggtgaagcagagccatggaaagagccttgagtgattgga

(SEQ ID NO: 1117)

WVKQSHGKSLEWIG

Mouse 8F9A4A3 Heavy chain variable region CDR2:

(SEQ ID NO: 1118)

Ggttttaacctaacaatgggtgttactaactacaaccagaagttcaagggc

(SEQ ID NO: 389)

GFNPNNGVITNYNQKFKG

Mouse 4A3 heavy chain variable domain framework 3 (FR3) sequence

(SEQ ID NO: 1119)

Aaggccacattgactgtagacaagtcctccagcacagcctacatggagctccgcagcctgacatctgaggattctgc

agtctattactgtgcaaga

(SEQ ID NO: 1120)

KATLTVDKSSSTAYMELRSLTSEDSAVYYCAR

Mouse 8F9A4A3 Heavy chain variable region CDR3:

(SEQ ID NO: 1121)

cgggtactaccatagctctctacgtgttttactttgactac

(SEQ ID NO: 390)

RYYHSLYVFYFDY

Mouse 8F9A4A3 Light chain variable region sequence

(SEQ ID NO: 1136)

gatatccagatgacacagactacatcctccctgtctgcctctctggggagacagagtcaccctcagttgcagtgcaagtc

agggcattagcaattatttaaactggatcagcagaaaccagatggaactgttgaaactcctgatctttacacatcaagtttactcagga

gtcccatcaaggttcagtggtcagtggtctgggacagattattctctcaccatcagcaacctggaacctgaagatattgccacttactatt

gtcagcagtatagtaagcttccttacacgttcggaggggggaccaagctggaaataaaa

-continued

Translated protein, wherein the underlined sequence is the complementarity determining region (CDR):

DIQMTQTSSLSASLGDRVTLSCSASQGISNYLNWYQQKPDGTVELLIFYT

(SEQ ID NO: 1109)

SSLHSGVPSRFGSGSGTDYSLTISNLEPEDATYYCQYSKLPYTFGGGKLEIK

Mouse 8F9A4A3 light chain variable domain framework 1 (FR1) sequence

gatatccagatgacacagactacatcctccctgtctgcctctctgggagacagagtcaccctcagttgc

(SEQ ID NO: 1122)

DIQMTQTSSLSASLGDRVTLSC

(SEQ ID NO: 1123)

Mouse 8F9A4A3 Light chain variable region CDR 1:

Agtgcaagtcagggcattagcaattatttaaac

(SEQ ID NO: 1124)

SASQGISNYLN

(SEQ ID NO: 393)

Mouse 4A3 light chain variable domain framework 2 (FR2) sequence

Tggtatcagcagaaaccagatggaactgttgaactcctgatcttt

(SEQ ID NO: 1125)

WYQQKPDGTVELLIF

(SEQ ID NO: 1126)

Mouse 8F9A4A3 Light chain variable region CDR2:

Tacacatcaagtttacactca

(SEQ ID NO: 1127)

YTSSLHS

(SEQ ID NO: 394)

Mouse 8F9A4A3 light chain variable domain framework 3 (FR3) sequence

Ggagtcacatcaagggtcagtgaggcgtgggtctgggacagattattctctcaccatcagcaacctggaacctgaagat

(SEQ ID NO: 1128)

attgccacttactattgt

(SEQ ID NO: 1129)

GVPSRFGSGSGTDYSLTISNLEPEDATYYC

Mouse 8F9A4A3 Light chain variable region CDR3:

Cagcagtatagtaagcttccttacag

(SEQ ID NO: 1130)

QQYSKLPYT

(SEQ ID NO: 395)

Humanized 8F9A4A3 H-ori heavy chain variable domain sequence

Cagggttcagctgggtcagtcgtggtgcagaagtgaagaaacctggcgctctctgtgaaggtgtcctgcaaggtgtccgg

(SEQ ID NO: 1131)

aaataccttcaccgagtagaccatgcactgggtccgacaggccccctggcaaggacttgaatggatggcggttcaacccaacaa

cggcgtgaccaactacaaccagaaattcaagggccgctgaccatgaccgaggacacaagcacagacaccgctacatggaactg

agcagcctgagaagcgaggacaccgctgtactactgcgccagaaggtactaccacagcctgtacgtgttctacttcgactactggg

gccagggcaccttggtcacagttttctct

(SEQ ID NO: 1206)

QVQLVQSGAEVKKPGASVKVCKVSGNTFTFETMHWVRQAPKGLEW

MGGFNPNNGVNTYNQKFGRVTMTEDTSTDYAMELSSLRSEDYAVYYCARRYH

SLYVFYFDYWGGTLTVSS

Humanized 8F9A4A3 H-1.46 heavy chain variable domain sequence

caagtgcagctgggtgcagagcggcgccgaggtgaagaaacctggcgccagcgtgaaagtgctcctgcaaggccag

(SEQ ID NO: 1101)

cggcaatacattcaccgagtagacaatgcactgggtcagacaggcccccgccaggcctggaatggatcggcggtttaaacccea

acaacggcgtgacaaactacaaccagaagttcaagggaaggtgaccatcacaagagacaccagcagcagcaccgtgtacatgga

actgtcttctctgcggagcgaggataccgctgtactattgtgccagacggtactaccacagcctgtacgtgttctacttcgactactgg

ggacagggcaccttggtaccgtgtcctct

QVQLVQSGAEVKKPGASVKVSCKASGNTFTFETMHWVRQAPGQGLEWI
(SEQ ID NO: 1102)
GGFNPNGVTNYNQKFKGKVTITRDTSSSTVYMELSSLRSEDTAVYYCARRYHSL
YVFYFDYWGGTLVTVSS
Humanized 8F9A4A3 H-3.15 heavy chain variable domain sequence
(SEQ ID NO: 1132)
Gaggtgcagctgggtgaaagcggcgccgctggtaagcctggcggatctctgagactgagctgtgccgttctg
gcaataccttcaccgagtagacacatgcactgggtgcggcagggccctggaaaagcctggaatggatcgccggtttaaccccaac
aacggcgtgacaaattacaaccagaaattcaagggaagttcaccatcacagagataagagcaagaacaccctgtacctgcaaatg
aacagcctgaagtccgaggacaccgcccgtgtactactgcgccagacggtagtaccacagcctctatgtgttctacttcgactactggg
ccagggcacactggtcaccgtgtccagc
(SEQ ID NO: 1133)
EVQLVESGGGLVKPGGSLRLSCAASGNTFTFETMHWVRQAPGKGLEWIG
GPNPNNGVTNYNQKFKGFTITRDKSKNTLYLQMNSLKSEDTAVYYCARRYHSLY
VFYFDYWGGTLVTVSS
Humanized 8F9A4A3 H-4.4 heavy chain variable domain sequence
(SEQ ID NO: 1105)
caagtgcagctgcaggagagcggacctggcctggtaagcctggaggcaccctgtctctgacatgtgctgtgtctg
caatacctttaccgagtagacacatgcactgggtgcggcagcctccaggcaaggcctggaatggatcgccggttcaaccccaaca
cggcgtgacaaattacaaccagaaattcaagggaagtgaccatcaccgtggataagtccaagaacaccttcagcctcaagctgag
cagcgtgacagccgcgcacccggtgtactactgcgccagaagatactatcacagcctgtacgtgttctacttcgactactggggc
cagggcacactggtcaccgtgtccagc
(SEQ ID NO: 1106)
QVQLQESGPGLVKPGGTLSTCAVSGNTFTFETMHWVRQPPGKGLEWIG
GPNPNNGVTNYNQKFKGKVTITVDKSKNTFSLKLSSVTAADTAVYYCARRYHSLY
VFYFDYWGGTLVTVSS
Humanized 8F9A4A3 L-1.6 light chain variable domain sequence
(SEQ ID NO: 1103)
gatataccagatgacacagagccctagctccctgagcgcagcgtggcgacgggtcaccattacatgcagcgtt
ctcagggcatctccaactacctgaactggtagcagcagaaacccggcaaggccctaaagtgtgctgatcttctacaccagctctctgca
cagcggcgtgccatctagattcagcggatctggcagcggcaccgactacacctgacctcagctccctccagcctgaggacttcgc
cacctactactgtcagcaatacagaagctgccttataccttggcggcggaacaaaggtggaaatcaag
(SEQ ID NO: 1104)
DIQMTQSPSSLSASVGRVTITCSASQGISNYLNWYQQKPKAPKLLIFYT
SSLHSGVPSRFSGSGSDYTLTISLQPEDFATYYCQYQSKLPYTFGGTKVEIK
Humanized 8F9A4A3 L-3.15 light chain variable domain sequence
(SEQ ID NO: 1134)
Gagatcgtgatgaccagagcccagctacacttagtgtgagtcagggtgaacgggctaccctgtcctgcagcgcca
gccagggcatcagcaactacctgaactggtagcagcagaaacctggcaggccctagactgctgatcttctacaccagcagcctgc
acagcggcatccccccagattcagcggcagcggctctggaacagactacacctgacaatctctagcctgcagctctgaagatttgc
cgtctactactgtcagcaatacagaagctgccttataccttggcggcggaaccaaggtggaaattaag
(SEQ ID NO: 1135)
EIVMTQSPATLSVSPGERATLSCSASQGISNYLNWYQQKPGQAPRLLIFYT
SSLHSGIPARFSGSGSDYTLTISLQSEDFAVYYCQYQSKLPYTFGGTKVEIK
Humanized 8F9A4A3 L-4.1 light chain variable domain sequence
(SEQ ID NO: 1107)
gatatacgtgatgaccagagcccagacagcctggcagtgagtcgggtgagcgtgctacaataactgcagcgcca
gccagggcatctccaactacctgaattggtagcagcagaaacctggcaggctcctaagctgctgatcttctacaccagcagcctgcac

-continued

agcggcgctgccagatagattcagcggcagcggatctggcaccgactacacactgaccatttcttctctccaggccgaggacgtggcc
gtctactactgtcagcaatacagcaagctgccttacaccttggcggaggcacaaggtggaaatcaag

(SEQ ID NO: 1108)

DIVMTQSPDSLAVSLGERATINCSASQGISNYLNWYQQKPGQAPKLLIFYT

SSLHSGVPDRFSGSGSGTDYTLTISSLQAEDVAVYYCQQYSKLPYTFGGGTKVEIK

Monoclonal antibody 5F3A5D4
Heavy chain variable region sequence
H-2,3,4,13,15

(SEQ ID NO: 427)

gtccagctgcaacagctctggacctgatctggtgaagcctgggacttcagtgaagatcctgtaagacttctggaaaca
cattcactgaatacaccatgcactgggtgaagcagagccatggaagagccttgagtggattggaggttttaatcctaacaatggtgta
ctaactacaaccagaagttcaagggaaggccacattgactgtagacaagtcctccagcacagcctacatggagctccgcagcctga
catctgaggattctgcagctctattactgtgcaagacgttactaccatagctacgtgttctactttgactcctggggccaaggcaccact
ctcacagtctcctca

Translated protein, wherein the underlined sequence is the complementarity
determining region (CDR):

(SEQ ID NO: 428)

VOLQQSGPDLVKPGTSVKISCKTSGNTFTEYTMHWVKQSHGKSLIEWIGGF

PNPNNGVNTYNQKFKGKATLTVDKSSSTAYMELRSLTSEDSAVYYCARRYHSTYVFYFDSWGQGTTLTVSS

Heavy chain variable region CDR1:

(SEQ ID NO: 429)

NTFTEYTMH

Heavy chain variable region CDR2:

(SEQ ID NO: 430)

GFNPNNGVNTYNQKFKG

Heavy chain variable region CDR3:

(SEQ ID NO: 431)

RYYHSTYVFYFDS

Light chain variable region sequence
K-1,2,3,4,9

(SEQ ID NO: 432)

gatatccagatgacacagactacatcctccctgtctgcctctctgggagacagagtcaccatcagttgcagtgcaagtc
agggcattagcaattatttaaactgggttcagcagaaaccagatggaactattaagctcctgatctattacacatcaagtttacattcagga
gtcccatcaagggttcagtggcagtggtctctgggacagattattctctcaccatcagtaagtggaaacctgaagatattgccacttactattg
tcagcagtatagtaagcttccttacacgttcggaggggggaccaagctggagataaaaacgg

Translated protein, wherein the underlined sequence is the complementarity
determining region (CDR):

(SEQ ID NO: 433)

DIQMTQTSSLSASLGDRVTISCSASQGISNYLNWYQQKPDGTIKLLIYYS

SLHSGVPSRFSGSGSGTDYSLTISNVEPEDIATYYCQQYSKLPYTFGGGTKLEIKR

Light chain variable region CDR 1:

(SEQ ID NO: 434)

SASQGISNYLN

Light chain variable region CDR2:

(SEQ ID NO: 435)

YTSSLHS

Light chain variable region CDR3:

(SEQ ID NO: 436)

QQYSKLPYT

Mouse 5F3A5D4-V2 heavy chain variable domain sequence:
(DNA)

(SEQ ID NO: 1155)

Gaggtccagctgcaacagctctggacctgatctggtgaagcctgggacttcagtgaagatcctgtaagacttctgga
aacacattcactgaatacaccatgcactgggtgaagcagagccatggaagagccttgagtggattggaggttttaatcctaacaatgg

-continued

tgttactaactacaaccagaagttcaagggcaaggccacattgactgtagacaagtcctccagcacagcctacatggagctccgcagc
ctgacatctgaggattctgcagtcctattactgtgcaagacgttactaccatagtagctacgtgttctactttgactcctggggccaaggcac
cactctcacagtcctctca

(amino acids)

(SEQ ID NO: 1172)

EVQLQQSGPDLVKPGTSVKISCKTSGNTFTEYTMHVKQSHGKSLEWIG

GFNPNNGVITNYNQKFKGKATLTVDKSSSTAYMELRSLTSEDSAVYYCARRYHSTY

VFYFDSWGQGTTLTVSS

Mouse 5F3A5D4-V2 heavy chain variable domain framework 1 (FR1)

sequence

(DNA)

(SEQ ID NO: 1173)

Gaggtccagctgcaacagtcctggacctgatctggtgaagcctgggacttcagtgaagatatcctgtaagacttctgga

(amino acids)

(SEQ ID NO: 1174)

EVQLQQSGPDLVKPGTSVKISCKTSG

Mouse 5F3A5D4-V2 heavy chain variable domain Complementarity-
determining region 1 (CDR1) sequence

(DNA)

(SEQ ID NO: 1187)

Aacacattcactgaatacaccatgcac

(amino acids)

(SEQ ID NO: 429)

NTFTEYTMH

Mouse 5F3A5D4-V2 heavy chain variable domain framework 2 (FR2)
sequence

(DNA)

(SEQ ID NO: 1188)

Tgggtgaagcagagccatggaaagagccttgagtggttgga

(amino acids)

(SEQ ID NO: 1189)

WVKQSHGKSLEWIG

Mouse 5F3A5D4-V2 heavy chain variable domain Complementarity-
determining region 2 (CDR2) sequence

(DNA)

(SEQ ID NO: 1190)

Ggttttaatcctaacaatgggtgttactaactacaaccagaagttcaagggc

(amino acids)

(SEQ ID NO: 430)

GFNPNNGVITNYNQKFKG

Mouse 5F3A5D4-V2 heavy chain variable domain framework 3 (FR3)
sequence

(DNA)

(SEQ ID NO: 1191)

Aaggccacattgactgtagacaagtcctccagcacagcctacatggagctccgcagcctgacatctgaggattctgc

agtcctattactgtgcaaga

(amino acids)

(SEQ ID NO: 1175)

KATLTVDKSSSTAYMELRSLTSEDSAVYYCAR

Mouse 5F3A5D4-V2 heavy chain variable domain Complementarity-
determining region 3 (CDR3) sequence

(DNA)

(SEQ ID NO: 1176)

Cgttactaccatagtagctacgtgttctactttgactcc

(amino acids)

(SEQ ID NO: 431)

RYYHSTYVFYFDS

-continued

Mouse 5F3A5D4-V2 heavy chain variable domain framework 4 (FR4)
sequence
(DNA)

Tggggccaaggcaccactctcacagtctcctca

(SEQ ID NO: 1177)

(amino acids)

WGQGTTLTVSS

(SEQ ID NO: 1178)

Mouse 5F3A5D4-2 light chain variable domain sequence
(DNA)

gatatccagatgacacagactacatcctccctgtctgcctctctgggagacagagtcaccatcagttgcagtgcaagtc

(SEQ ID NO: 1179)

agggcattagcaattatttaaactgggttcagcagaaaccagatggaactattaagctcctgatctattacacatcaagtttacattcagga

gtcccatcaagggttcagtggtcagtggtctctgggacagattattctctcaccatcagtaattgtggaacctgaagatattgccacttactattg

tcagcagtatagtaagcttccttacacgttcggaggggggaccaagctggagataaaa

(amino acids)

(SEQ ID NO: 1180)

DIQMTQTSSLSASLGDRVTISCSASQGISNYLWFPQKPDGTIKLLIYYS

SLHSGVPSRFSGSGTDYSLTISNVEPEDIAITYYCQYSKLPYTFGGGTKLEIK

Mouse 5F3A5D4-2 light chain variable domain framework 1 (FR1) sequence
(DNA)

gatatccagatgacacagactacatcctccctgtctgcctctctgggagacagagtcaccatcagttgc

(SEQ ID NO: 1171)

(amino acids)

(SEQ ID NO: 1147)

DIQMTQTSSLSASLGDRVTISC

Mouse 5F3A5D4-2 light chain variable domain Complementarity-determining
region 1 (CDR1) sequence
(DNA)

agtgcaagtcagggcattagcaattatttaaac

(SEQ ID NO: 1193)

(amino acids)

(SEQ ID NO: 1208)

SASQGISNYLN

Mouse 5F3A5D4-2 light chain variable domain framework 2 (FR2) sequence
(DNA)

tgggttcagcagaaaccagatggaactattaagctcctgatctat

(SEQ ID NO: 1148)

(amino acids)

(SEQ ID NO: 1149)

WFQQKPDGTIKLLIY

Mouse 5F3A5D4-2 light chain variable domain Complementarity-determining
region 2 (CDR2) sequence
(DNA)

tacacatcaagtttacattca

(SEQ ID NO: 1150)

(amino acids)

(SEQ ID NO: 1210)

YTSSLHS

Mouse 5F3A5D4-2 light chain variable domain framework 3 (FR3) sequence
(DNA)

ggagtgccatcaagggttcagtggtcagtggtctctgggacagattattctctcaccatcagtaattgtggaacctgaagata

(SEQ ID NO: 1151)

ttgccacttactattgt

(amino acids)

(SEQ ID NO: 1152)

GVPSRFSGSGSGTDYSLTISNVEPEDIAITYYC

-continued

Mouse 5F3A5D4-2 light chain variable domain Complementarity-determining
region 3 (CDR3) sequence
(DNA)

cagcagtatagtaagcttccttacacg

(SEQ ID NO: 1200)

(amino acids)

(SEQ ID NO: 1212)

QQYSKLPYT

Mouse 5F3A5D4-2 light chain variable domain framework 4 (FR4) sequence
(DNA)

ttcggaggggggaccaagctggagataaaa

(SEQ ID NO: 1153)

(amino acids)

(SEQ ID NO: 1154)

FGGGTKLEIK

Mouse 5F3A5D4-3 light chain variable domain sequence
(DNA)

gatatccagatgacacagactacatcctccctgtctgcctctctgggagacagagtcaccctcagttgcagtgcaagtc

(SEQ ID NO: 1204)

agggcattagcaattattttaactgggtatcagcagaaaccagatggaactgttgaaactcctgatctttacacatcaagtttacactcagga

gtcccatcaagggttcagtggtggcagtggtctctgggacagattattctctcaccatcagcaacctggaacctgaagatattgccacttactatt

gtcagcagtatagtaagcttccttacacgttcggaggggggaccaagctggaaataaaa

(amino acids)

(SEQ ID NO: 1186)

DIQMTQTSSLSASLGDRVTLSCSASQGISNYLNWYQQKPDGTVELLIFYT

SSLHSGVPSRFGSGSGTDYSLTISNLEPEDIATYYCQQYSKLPYTFGGGTKLEIK

Mouse 5F3A5D4-3 light chain variable domain framework 1 (FR1) sequence
(DNA)

gatatccagatgacacagactacatcctccctgtctgcctctctgggagacagagtcaccctcagttgc

(SEQ ID NO: 1156)

(amino acids)

(SEQ ID NO: 1192)

DIQMTQTSSLSASLGDRVTLSC

Mouse 5F3A5D4-3 light chain variable domain Complementarity-determining
region 1 (CDR1) sequence
(DNA)

agtgcaagtcagggcattagcaattatttaaac

(SEQ ID NO: 1194)

(amino acids)

(SEQ ID NO: 1209)

SASQGISNYLN

Mouse 5F3A5D4-3 light chain variable domain framework 2 (FR2) sequence
(DNA)

tggtatcagcagaaaccagatggaactgttgaaactcctgatcttt

(SEQ ID NO: 1195)

(amino acids)

(SEQ ID NO: 1196)

WYQQKPDGTVELLIF

Mouse 5F3A5D4-3 light chain variable domain Complementarity-determining
region 2 (CDR2) sequence
(DNA)

tacacatcaagtttacactca

(SEQ ID NO: 1197)

(amino acids)

(SEQ ID NO: 1211)

YTSSLHS

Mouse 5F3A5D4-3 light chain variable domain framework 3 (FR3) sequence
(DNA)

ggagtgcccatcaagggttcagtggtggcagtggtctctgggacagattattctctcaccatcagcaacctggaacctgaagat

(SEQ ID NO: 1198)

attgccacttactattgt

-continued

(amino acids)

GVPSRFSGSGSGTDYSLTISNLEPEDATYYC

(SEQ ID NO: 1199)

Mouse 5F3A5D4-3 light chain variable domain Complementarity-determining
region 3 (CDR3) sequence
(DNA)

cagcagtatagtaagcttccttacacg

(SEQ ID NO: 1201)

(amino acids)

QQYSKLPYT

(SEQ ID NO: 1213)

Mouse 5F3A5D4-3 light chain variable domain framework 4 (FR4) sequence
(DNA)

ttcggaggggggaccaagctggaaataaaa

(SEQ ID NO: 1157)

(amino acids)

FGGGTKLEIK

(SEQ ID NO: 1207)

Humanized 5F3A5D4-V2 H-1.27 heavy chain variable domain sequence
(DNA)

caagtgcagctgggtccagagcggcgccgaggtgaaaaagcctggcgccagcgtgaaggtgtcctgcaaggtgtct

(SEQ ID NO: 1158)

ggcaataccttcaccgagtagaccatgcactgggtgcgccagggccctggaaaaggcctggaatggatcggcggatttaaccccaa

caacggcgtgacaaactacaaccagaagttcaagggaaggttacactgaccgtggacaccagctcttctaccgctacatggaact

gagcagcctgagaagcgaggatacagcgtgtactattgtgccagaagatactaccacagcacctacgtgttctacttcgacagctgg

ggccagggcacactgggtgacagtgtccagc

(amino acids)

QVQLVQSGAEVKKPGASVKVCKVSGNTFTFETMHWVRQAPGKGLEWI

(SEQ ID NO: 1181)

GGFNPNGVTNYNQKFKGKVTITVDTSSSTAYMELSSLRSEDTAVYYCARRYHST

YVFYFDSWGQGLTVTVSS

Humanized 5F3A5D4-V2 H-1.46 heavy chain variable domain sequence
(DNA)

caagtgcagctgggtgcagagcggagccgaggtgaaaaagccggcgcttctgtgaaggtgtcttgaagccagc

(SEQ ID NO: 1159)

ggcaacaccttcaccgagtagaccatgcactgggtgcgccagggccctggccagggcctggaatggatcggcggctttaatcctaac

aacggcgtgacaaactacaaccagaagttcaagggaaggttacaatcaccagagataccagcagctctaccgtgtacatggaactg

agcagcctgagaagcgaggacaccgcccgtgtattactgcgcagacgggtactaccacagcacctacgtgttctacttcgacagctgg

ggccaggggaacactgggtcacagtgtcctcc

(amino acids)

QVQLVQSGAEVKKPGASVKVCKASGNTFTFETMHWVRQAPGQGLEWI

(SEQ ID NO: 1160)

GGFNPNGVTNYNQKFKGKVTITRDTSSSTVIMELSSLRSEDTAVYYCARRYHST

YVFYFDSWGQGLTVTVSS

Humanized 5F3A5D4-V2 H-3.15 heavy chain variable domain sequence
(DNA)

gaggtgcagctgggtggaaagcggaggaggcctggtaagcctggaggcagcctgagactgagctgtgccgttct

(SEQ ID NO: 1161)

ggcaataccttcaccgagtagaccatgcactgggtgcgccagggccctggcaaaggcctggaatggatcggcggcttcaaccccaa

caacggcgtgacaaactacaaccagaaattcaagggaaggttacaatcaccagagataagtcgaagaacacactctatctgcaaatga

-continued

acagcctgaagtcgaggacaccgccgtgtactactgcccagacgggtactaccacagcacatacgtgttctacttcgacagctggg
gccagggcacccctggtcaccgtgtccagc

(amino acids)

(SEQ ID NO: 1162)

EVQLVESGGGLVKPGGSLRLSCAASGNTFTFETMHWVRQAPGKGLEWIG
GFNPNNGVITNYNQKFKGKFTITRDKSKNTLYLQMSLKSEDTAVYYCARRYHSTY
VFYFDSWGQGLVTVSS

Humanized 5F3A5D4-V2 H-4.4 heavy chain variable domain sequence
(DNA)

(SEQ ID NO: 1163)

caagtgcagctgcaggagagcggacctggcctggcaagcctggcggcacccctgagcctcacctgtgctgtttctg
gcaataaccttcaccgagtagacacatgcactgggtgcggcagcctccaggcaaaggcctggaatggatcgggcgttttaacccaaca
acggcgtgacaaattacaaccagaaattcaagggaagggtgaccatcacagtggaataagccaagaacaccttcagcctgaagctgt
ctagcgtgacagccgccgacaccgccgtgtactactgcccagaagatactatcacagcacctacgtgttctacttcgacagctgggg
acagggcacactggtgacagtgtccagc

(amino acids)

(SEQ ID NO: 1164)

QVQLQESGPGLVKPGGTLSTCAVSGNTFTFETMHWVRQPPGKGLEWIG
GFNPNNGVITNYNQKFKGKVTITVDKSKNTFSLKLSVTAADTAVYYCARRYHSTY
VFYFDSWGQGLVTVSS

Humanized 5F3A5D4-2 L-1.6 light chain variable domain sequence
(DNA)

(SEQ ID NO: 1165)

gatatccagatgacacagagccctagctccctgagcgcagcgtggcgaccgggtcaccattacatgcagcgctt
ctcagggcatctccaactacctgaactgggttcagcagaaacccggcaaggccctaaagctgctgatctattacaccagctctctgcac
agcggcgtgccatctagattcagcggatctggcagcggcaccgactacacctgaccatcagctccctccagcctgaggacttcgcc
acctactactgtcagcaatacagcaagctgccttatacctttggcggcggaacaaagggtggaaatcaag

(amino acids)

(SEQ ID NO: 1166)

DIQMTQSPSSLSASVGDRVTITCSASQGISNYLWVQQKPKAPKLLIYYT
SSLHSGVPSRFGSGSGTDYTLTISSLQPEDFATYYCQQYSKLPYTFGGGTKVEIK

Humanized 5F3A5D4-2 L-3.15 light chain variable domain sequence
(DNA)

(SEQ ID NO: 1167)

gagatcgtgatgaccagagcccagctacacttagtgtagtcagggtgaacgggtaccctgtcctgcagcgcca
gccagggcatcagcaactacctgaactgggttcagcagaaacctggcaggccctagactgctgatctattacaccagcagcctgca
cagcggcatccccgccagattcagcggcagcggctctggaacagactacacctgacaatctctagcctgcagctctgaagattttgc
gtctactactgtcagcaatacagcaagctgccttataccttcggggcggaaccaagggtggaaattaag

(amino acids)

(SEQ ID NO: 1168)

EIVMTQSPATLSVSPGERATLSCSASQGISNYLWVQQKPGQAPRLLIYYT
SSLHSGIPARFGSGSGTDYTLTISSLQSEDFAVYYCQQYSKLPYTFGGGTKVEIK

Humanized 5F3A5D4-2 L-4.1 light chain variable domain sequence
(DNA)

(SEQ ID NO: 1169)

gatatcgtgatgaccagagcccagacagcctggcagtgagtcgggtgagcgtgctacaataactgcagcgcca
gccagggcatctccaactacctgaattgggttcagcagaaacctggcaggctcctaagctgctgatctattacaccagcagcctgcac

-continued

agcggcgtgccagatagattcagcggcagcggatctggcaccgactacacactgaccattttcttctctccaggccgaggacgtggcc

gtctactactgtcagcaatacagcaagctgccttacacctttggcggaggcacaaggtggaaatcaag

(amino acids)

(SEQ ID NO: 1170)

DIVMTQSPDSLAVSLGERATINCSASQGISNYLNWFQQKPGQAPKLLIYYT

SSLHSGVPRFSGSGSDTYTLTISSLQAEDVAVYYCQQYSKLPYTFGGGTKVEIK

Humanized 5F3A5D4-3 L-1.6 light chain variable domain sequence
(DNA)

(SEQ ID NO: 1182)

gatatccagatgacacagagccctagctccctgagcgcagcgtggcgacccgggtcaccattacatgcagcgctt

ctcaggcgcatctccaactacctgaactggtaccagcagaaacccggcaaggccctaaagctgctgatcttctacaccagctctctgca

cagcggcgtgccatctagattcagcggatctggcagcggcaccgactacacccctgaccatcagctccctccagcctgaggactctgc

cacctactactgtcagcaatacagcaagctgccttatacctttggcggcggaacaaaggtggaaatcaag

(amino acids)

(SEQ ID NO: 1183)

DIQMTQSPSSLSASVGDRTVITCSASQGISNYLNWYQQKPGKAPKLLIFYT

SSLHSGVPSRFSGSGSDTYTLTISSLQPEDFATYYCQQYSKLPYTFGGGTKVEIK

Humanized 5F3A5D4-3 L-3.15 light chain variable domain sequence
(DNA)

(SEQ ID NO: 1202)

gagatcgtgatgaccagagcccagctacacttagtgtgagtcagggtgaacgggctaccctgtcctgcagcgcca

gccaggcgcatcagcaactacctgaactggtaccagcagaaacctggccaggccctagactgctgatcttctacaccagcagcctgc

acagcggcatccccgccagattcagcggcagcggctctggaacagactacacccctgacaatctctagcctgcagctctgaagattttgc

cgtctactactgtcagcaatacagcaagctgccttataccttcggcggcggaaccaaggtggaaattaag

(amino acids)

(SEQ ID NO: 1203)

EIVMTQSPATLSVSPGERATLSCSASQGISNYLNWYQQKPGQAPRLIFYT

SSLHSGIPARFSGSGSDTYTLTISSLQSEDFAVYYCQQYSKLPYTFGGGTKVEIK

Humanized 5F3A5D4-3 L-4.1 light chain variable domain sequence
(DNA)

(SEQ ID NO: 1184)

gatatcgtgatgaccagagcccagacagcctggcagtgagtcgggtgagcgtgctacaatcaactgcagcgcca

gccaggcgcatctccaactacctgaattggtatcagcagaaacctggccaggctcctaagctgctgatcttctacaccagcagcctgac

agcggcgtgccagatagattcagcggcagcggatctggcaccgactacacactgaccattttcttctctccaggccgaggacgtggcc

gtctactactgtcagcaatacagcaagctgccttacacctttggcggaggcacaaggtggaaatcaag

(amino acids)

(SEQ ID NO: 1185)

DIVMTQSPDSLAVSLGERATINCSASQGISNYLNWYQQKPGQAPKLLIFYT

SSLHSGVPRFSGSGSDTYTLTISSLQAEDVAVYYCQQYSKLPYTFGGGTKVEIK

Monoclonal antibody 8F9A5A1
Heavy chain variable region sequence
H-3, 4, 6, 10, 11

(SEQ ID NO: 437)

atccagttgggtcagtcctggacctgagctgaagaagcctggagagacagtcgaagatctcctgcaaggcttctgggtat

accttcacaaactatggaatgaactgggtgaagcaggctccaggaaaggggttaaaagtggatgggctggataaacacctacactgga

gagccaacatatgttgatgacttcaagggacgggttgcttctctttggaaacctctgccaccactgcctatttgcagatcaacaacctca

aaaatgaggacacgtctacatatctctgtgcaagattgagggggatacagccgggtcccttggttactggggccaagggactctggtc

actgtctctgca

-continued

Translated protein, wherein the underlined sequence is the complementarity determining region (CDR):

IQLVQSGPELKKPGETVKISKASGYTFTNYGMNVVKQAPGKGLKWMG

(SEQ ID NO: 438)

WINTYTGEPTYVDDFKGRFAFSLETSATTAYLQINNKNEDTSTYFCARLRGIRPGPL

AYWGQGLTVTSA

Heavy chain variable region CDR1:

(SEQ ID NO: 439)

YTFTNYGMN

Heavy chain variable region CDR2:

(SEQ ID NO: 440)

WINTYTGEPTYVDDFKG

Heavy chain variable region CDR3:

(SEQ ID NO: 441)

LRGIRPGPLAY

Light chain variable region sequence
K-1,2,3,4,5

(SEQ ID NO: 442)

gaaattttgctcaccagctctccagcaatcatagctgcattctcctggggagaaggtcaccatcacctgcagtgccagct

caagtgttaagttacatgaactgggtaccagcagaaaccaggatcctccccaaaatatggatttatggtatatccaacctggcttctggag

ttcctgctcgcttcagtgaggctgggtctgggacatctttctcttcacaatcaacagcatggaggctgaagatggtgccacttattactgtc

agcaaaggagtagttaccacccacgttcggaggggggaccaagctggaaataaaacgg

Translated protein, wherein the underlined sequence is the complementarity determining region (CDR):

(SEQ ID NO: 443)

EILLTQSPALIAASPGKVTITCSASSSVSYMNWYQKPGSSPKIWIYGISNL

ASGVPARFSGSGTSFSFTINSMEAEDVATYYCQQRSSYPPTFGGGTKLEIKR

Light chain variable region CDR 1:

(SEQ ID NO: 444)

SASSSVSYMN

Light chain variable region CDR2:

(SEQ ID NO: 445)

GISNLAS

Light chain variable region CDR3:

(SEQ ID NO: 446)

QQRSSYPPT

8F9A4P3 Heavy chain variable region sequence mouse

(SEQ ID NO: 335)

Gtccagctgcaacagctctggacctaagctggaagcctgggcttcagtgagatctcctgcaagacttctggaaa

cacattcactgaatacacccatgcactgggtgaagcagagccatggaaagagccttgagtgagggttttaatacaaatggtg

ttactaactacaaccagaagttcaagggaaggccacattgactgtagacaagtcctccagcacagcctacatggagctccgcagcct

gacatctgaggattctgcagctctattactgtgcaagacggtactaccatagctctctacgtgttttactttgactactggggccaaggcacca

ctctcacagctctcctca

(SEQ ID NO: 1001)

VQLQSGPELVKPGASVKISCKTSGNTFTETMHWVKQSHGKSLIEWIGGF

NPNNGVNTYNQKFKGKATLTVDKSSSTAYMELRSLTSEDSAVYYCARRYHSLYVF

YFDYWGGTTLTVSS

IGHV1-24*01 V-REGION sequence human (closest match hu antibody sequence)

(SEQ ID NO: 336)

Cagggtccagctgggtacagctctgggctgaggtgaagaagcctggggcctcagtgagggtctcctgcaaggtttccg

gatacacccctcactgaattatccatgcactgggtgcgacaggtcctggaaaagggttgagtgagggttttgatcctgaagat

ggtgaacaatctacgcacagaagttccagggcagagtcacatgaccgaggacacatctacagacacagcctacatggagctgag

cagcctgagatctgaggacacggcgtgtattactgtgcaaca

-continued

QVQLVQSGAEVKKPGASVKVSCKVSGYTLTELSMHWVRQAPGKGLEW

(SEQ ID NO: 1002)

MGGFDPEDGETIYAQKFQGRVTMTEDTSTDYAMELSSLRSEDTAVYYCAT

human (closest match hu antibody sequence)
IGHJ4*01 J-REGION sequence

tactttgactactggggccaaggaacctgtggtcacggtctctctca

(SEQ ID NO: 337)

YFDYWGQGLTVTVSS

(SEQ ID NO: 1003)

humanized heavy chain variable seq (SEQ ID NO: 1001 + SEQ ID NO: 1002 + SEQ ID NO: 1003)
humanized 8F9A4P3 Heavy chain variable region sequence
(DNA)

(SEQ ID NO: 338)

cagggtccagctggttacagtcctggggctgaggtgaagaagcctggggcctcagtgagggtctcctgcaaggtttccg

gaaacacattcactgaatacaccatgcactgggtgacagggctcctggaaaagggttgagtgagggttttaatacctaacaat

ggtgttactaactacaaccagaagtcaagggcagagtcaccatgaccgaggacacatctacagacacagcctacatggagctgagc

agcctgagatctgaggacacggccgtgtattactgtgcaagacggtactaccatagtcctctacgtgttttactttgactactggggccaag

gaacctggtcacggtctctctca

(SEQ ID NO: 1004)

QVQLVQSGAEVKKPGASVKVSCKVSGNTFTEYTMHWVRQAPGKGLEW

MGGFNPNGVTNYNQKFQGRVTMTEDTSTDYAMELSSLRSEDTAVYYCARRYH

SLYVFYFDYWGQGLTVTVSS

humanized heavy chain variable seq (codon optimized version of 1004)
humanized 8F9A4P3 Heavy chain variable region sequence (codon optimized)
(DNA)

(SEQ ID NO: 339)

cagggttcagctggttcagtcctggcgccgaagtgaagaaacctggcgctctgtgaaggtgtcctgcaaggtgtccgg

aaataccttcaccgagtagaccatgcactgggtccgacaggccctggcaaggacttgatggatggggcggttcaacccaacaa

cggcgtgaccaactacaaccagaaattcaagggccgctgaccatgaccgaggacacaagcacagacaccgcctacatggaactg

agcagcctgagaagcgaggacaccgccgtgtactactgcccagaaggtactaccacagcctgtacgtgttctacttcgactactggg

gccagggcacctggtcacagtttcttct

(SEQ ID NO: 1005)

QVQLVQSGAEVKKPGASVKVSCKVSGNTFTEYTMHWVRQAPGKGLEW

MGGFNPNGVTNYNQKFQGRVTMTEDTSTDYAMELSSLRSEDTAVYYCARRYH

SLYVFYFDYWGQGLTVTVSS

humanized heavy chain variable seq ("modified" SEQ ID NO: 1005 sequence,
where modified means certain amino acids that are thought to be critical for binding or
structure have been reverted to the mouse sequence).
Modified humanized 8F9A4P3 Heavy chain variable region sequence

(SEQ ID NO: 340)

cagggtccagctggttacagtcctggggctgaggtgaagaagcctggggcctcagtgagggtctcctgcaaggtttccg

gaaacacattcactgaatacaccatgcactgggtgacagggctcctggaaaagggttgagtgagggttttaatacctaacaat

ggtgttactaactacaaccagaagtcaagggcaagtcaccctgaccgtggacacatctagcagcacagcctacatggagctgagc

agcctgagatctgaggacacggccgtgtattactgtgcaagacggtactaccatagtcctctacgtgttttactttgactactggggccaag

gaacctggtcacggtctctctca

(SEQ ID NO: 1006)

QVQLVQSGAEVKKPGASVKVSCKVSGNTFTEYTMHWVRQAPGKGLEWI

GGFNPNGVTNYNQKFQKVTLTVDTSSTAYMELSSLRSEDTAVYYCARRYHSL

YVFYFDYWGQGLTVTVSS

-continued

humanized heavy chain variable seq (SEQ ID NO: 1006 codon optimized)
 Modified humanized 8F9A4P3 Heavy chain variable region sequence (codon optimized)

(SEQ ID NO: 341)

cagggttcagctcgttcagctctggcgccgaagtgaagaaacctggcgccctctgtgaaggtgtcctgcaaggtgtccgg
 aaataaccttcaccgagtagaccatgcactgggtccgacaggcccttggaaggactggaatggatcggcggcttcaacccaaca
 acggcgtagaccaactacaaccagaaattcaagggaagtgaacctgacctggacaccagcagcagcacgcctacatggaact
 gagcagcctgagaagcgaggacaccgccgtgtactactgcgcagaaggtactaccacagcctgtacgtgttctacttcgactactgg
 ggccaggggcacccctgggtcacagtttcttct

(SEQ ID NO: 1007)

QVQLVQSGAEVKKPGASVKVSKVSGNTFTETMHWVRQAPGKGLEWI
 GGFNPNNGVTNYNQKFKGKVTTLTVDTSSTAYMELSSLRSEDTAVYYCARRYHSL
 YVFYFDYWQGTLLTVSS

8F9A4P3 Light chain variable region sequence mouse

(SEQ ID NO: 342)

gaaacaactgtgaccagctctccagcatccctgtccatggctataggagaaaaagtcaccatcagatgcataaccag
 cactgatattgatgatgatatgaactggtagcagcagaagccagggaacctcctaagctccttatttcagaaggcaatactcttcgtcct
 ggagtcacctcccgattctccagcagtggtatggtacagattttgtttttacaattgaaaacatgctctcagaagatgttcgagattactac
 tgtttgcaaagtataacttgccctctcacgttcggctcggggacaaagtggaaataaaacgg

(SEQ ID NO: 1008)

ETTVTQSPASLSMAIGKVTIRCITSTDIDDDMNWYQQKPGEPKLLISEGN
 TLRPGVPSRFSSSGYGTDFVFTIENMLSEDVADYYCLQSDNLPFTFGSGTKLEIKR

human (closest match hu antibody sequence)
 IGKV5-2*01 V-REGION sequence

(SEQ ID NO: 343)

gaaacgacactcacgcagctctccagcattcatgtcagcgactccaggagacaaagtcaacatctcctgcaaagccag
 ccaagacattgatgatgatatgaactggtagcagcagaagcagcagagaagctgctattttcattattcaagaagctactactctcgttcct
 ggaatccacctcgattcagtggcagcgggtatggaacagattttaccctcacaattaataacatagaatctgaggatgctgcattact
 tctgt

(SEQ ID NO: 1009)

ETTLTQSPAFMSATPGDKVNISCKASQDIDDDMNWYQQKPGEAIFIIQEA
 TTLVPGIPPRFSGSGYGTDFTLTINNIESEDAAYYFC

human (closest match hu antibody sequence)
 IGKJ4*02 J-REGION sequence

(SEQ ID NO: 344)

ctcacgttcggcggagggaaggtggagatcaaa

(SEQ ID NO: 1010)

LTFGGGTVKEIK

humanized light chain variable seq (SEQ ID NO: 1008 + SEQ ID NO: 1009 + SEQ ID NO: 1)
 humanized 8F9A4P3 Light chain variable region sequence

(SEQ ID NO: 345)

gaaacgacactcacgcagctctccagcattcatgtcagcgactccaggagacaaagtcaacatctcctgcataaccag
 cactgatattgatgatgatatgaactggtagcagcagaagcagcagagaagctgctattttcattattcaagaaggcaatactcttcgtcctg
 gaatccacctcgattcagtggcagcgggtatggaacagattttaccctcacaattaataacatagaatctgaggatgctgcattactt
 ctgtttgcaaagtataacttgccctctcacgttcggcggagggaaggtggagatcaaacgg

(SEQ ID NO: 1011)

ETTLTQSPAFMSATPGDKVNISCITSTDIDDDMNWYQQKPGEAIFIIQEGN
 TLRPGIPPRFSGSGYGTDFTLTINNIESEDAAYYFCLQSDNLPFTFGGTVKEIKR

-continued

humanized light chain variable seq (codon optimized version of SEQ ID NO: 1011)
humanized 8F9A4P3 Light chain variable region sequence (codon optimized)

(SEQ ID NO: 346)

Gagacaacccctgacacagagccctgccttcatgtctgccacacctggcgacaaagtgaacatcagctgcatcacca
gcaccgacatcgacgacgacatgaactggatcagcagaagcctggcgaggccgccatcttcatcatccaagagggcaacacactg
cggcctggcatccctcctagattttctggcagcggtacggcaccgacttcaccctgaccatcaacaacatcgagagcgaggacgcc
gcctactacttctgcctgcaaagcgacaacctgcctctgacctttggcggaggcaccaaggtggaatcaagcgg

(SEQ ID NO: 1012)

ETTLTQSPAFMSATPGDKVNISCITSTDIDDDMNWYQQKPGEAIFIIQEGN

TLRPGIPPRFSGSGYGTDFTLTINNIESEDAAYYFCLQSDNLP LTFGGG TKVEIKR

humanized light chain variable seq ("modified" SEQ ID NO: 1012 sequence,
where modified means certain amino acids that are thought to be critical for binding or
structure have been reverted to the mouse sequence).
Modified humanized 8F9A4P3 Light chain variable region sequence

(SEQ ID NO: 347)

gaaacgacagtgacgagctctccagcattcatgtcagcgactccaggagacaaagtaccatctcctgcataaccag
cactgatattgatgatgatatgaactggtaccaacagaaaccaggagaagctgctattctgctgattagcgaaggcaataactcttcgtcct
ggaatcccacctcgattcagtagcagcgggtatggaacagattttaccctcacaattaataacatagaatctgaggatgctgcatattact
tctgtttgcaaagtataacttgcctctcacgttcggcggagggaaccaaggtggagatcaaacgg

(SEQ ID NO: 1013)

ETTVTQSPAFMSATPGDKVTISCITSTDIDDDMNWYQQKPGEAAILLISEG

NTRLRPGIPPRFSSSGYGTDFTLTINNIESEDAAYYFCLQSDNLP LTFGGG TKVEIKR

humanized light chain variable seq
Modified humanized 8F9A4P3 Light chain variable region sequence (codon
optimized)

(SEQ ID NO: 348)

gagacaacccgtgacacagagccctgccttcatgtctgccacacctggcgacaaagtgaccatcagctgcatcacca
gcaccgacatcgacgacgacatgaactggatcagcagaagcctggcgaggccgccatcctgcttatctctgagggaaacacactgc
ggcctggcatccctcctagattttccagcagcggctacggcaccgacttcaccctgaccatcaacaacatcgagagcgaggacgccg
cctactacttctgcctgcaaagcgacaacctgcctctgacctttggcggaggcaccaaggtggaatcaagcgg

(SEQ ID NO: 1014)

ETTVTQSPAFMSATPGDKVTISCITSTDIDDDMNWYQQKPGEAAILLISEG

NTRLRPGIPPRFSSSGYGTDFTLTINNIESEDAAYYFCLQSDNLP LTFGGG TKVEIKR

humanized heavy and light chains joined via a flexible linker.
Modified humanized 8F9A4P3 sequence (codon optimized)

(SEQ ID NO: 349)

Caggttcagctgggtcagctctggcgccgaagtgaagaaacctggcgccctctgtgaaggtgtcctgcaaggtgtccgg
aaataccttcaccgagtagaccatgcactgggtccgacaggccccctggcaaaggactggaatggatcggcggttcaaccccaaca
acggcgtgaccaactacaaccagaaattcaagggaagtgacctgacctggacaccagcagcagcacagcctacatggaact
gagcagcctgagaagcgaggacaccgccgtgtactactgcgcagaaggtactaccacagcctgtacgtgttctacttcgactactgg
ggccaggggccccctgggtcacagtttcttctggcggtggcggaagcggaggcgggtggctccgggtggcgaggcagcgaaacgaca
gtgacgcagctctccagcattcatgtcagcgactccaggagacaaagtcaccatctcctgcataaccagcactgatattgatgatgatg
aactggtaccaacagaaaccaggagaagctgctattctgctgattagcgaaggcaataactcttcgctcctggaatcccacctcgattcag
agcagcgggtatggaacagattttaccctcacaattaataacatagaatctgaggatgctgcatattacttctgtttgcaaagtataactt
gcctctcagcttcggcggagggaaccaaggtggagatcaaacgg

(SEQ ID NO: 1015)

QVQLVQSGAEVKKPGASVKVSKVSGNTFTFETMHWVRQAPGKGLEWI

GGFNPNGVTNYNQKFKGKVTLTVDTSSTAYMELSSLRSEDTAVYYCARRYHSL

-continued

YVFYFDYWGQGLVTVSSGGGGSGGGSGGGSETTVTQSPAFMSATPGDKVTISCI

TSTDIDDDMNWYQKPGEAAILLISEGNTLRPGIPPRFSSSGYGTDFLTINNIESEDA

AYYFCLQSDNLPLTFGGGTKVEIKR

8F9A5A1 Heavy chain variable region sequence

(SEQ ID NO: 350)

atccagttgggtgcagtcctggacctgagctgaagaagcctggagagacagtcagatctcctgcaaggcttctgggtat

accttcacaaactatggaatgaactgggtgaagcaggctccaggaaaggtttaaagtggatgggctggataaacacctacactgga

gagccaacatatgttgatgacttcaagggaagggttggcctctctttggaacctctgccaccactgcctatttgcatgaacaacacctca

aaaatgaggacacgtctacatatctctgtgcaagattgagggggatacgaccgggtcccttggttactggggccaagggaactctggtc

actgtctctgca

(SEQ ID NO: 1016)

IQLVQSGPELKKPGETVKISCKASGYTFTNYGMNWVKQAPGKGLKWMG

WINTYTGTEPTYVDDFKGRFAPSLETSATTAYLQINNKNEDTSTYFCARLRGIRPGPL

AYWGQGLTVTVSA

IGHV7-81*01 V-REGION sequence

(SEQ ID NO: 351)

caggtgcagctgggtgcagtcctggccatgaggtgaagcagcctggggcctcagtgaaagtctcctgcaaggcttctg

gttacagtttcaccacctatggtatgaattgggtgccacaggccctggacaagggttgagtggatggggtggttcaaacctacact

gggaaccaacatatgccagggttcacaggacgggttgtcttctccatggacacctctgccagcacagcatacctgcagatcagca

gcctaaaggctgaggacatggccatgtattactgtgcgaga

(SEQ ID NO: 1017)

QVQLVQSGHEVKQPGASVKVCKASGYSFTTYGMNWVPQAPGQGLEW

MGWFNTYTGNTPTYAQGFTGRFVFSMDTSASTAYLQISSLKAEDMAMYCAR

IGHJ4*03 J-REGION sequence

(SEQ ID NO: 352)

tactttgactactggggccaagggaacctgggtcacctctcctca

(SEQ ID NO: 1018)

YFDYWGQGLVTVSS

humanized 8F9A5A1 Heavy chain variable region sequence

(SEQ ID NO: 353)

Caggtgcagctgggtgcagtcctggccatgaggtgaagcagcctggggcctcagtgaaagtctcctgcaaggcttctg

gggtataccttcacaaactatggaatgaactgggtgccacaggccctggacaagggttgagtggatggggtgataaacacctaca

ctggagagccaacatatgttgatgacttcaagggaagggttgtcttctccatggacacctctgccagcacagcatacctgcagatcagc

agcctaaaggctgaggacatggccatgtattactgtgcaagattgagggggatacgaccgggtcccttggttactggggccaaggg

accctgggtcacctctcctca

(SEQ ID NO: 1019)

QVQLVQSGHEVKQPGASVKVCKASGYTFTNYGMNWVPQAPGQGLEW

MGWINTYTGTEPTYVDDFKGRFVFSMDTSASTAYLQISSLKAEDMAMYCARLRGIR

PGPLAYWGQGLVTVSS

humanized 8F9A5A1 Heavy chain variable region sequence (codon optimized)

(SEQ ID NO: 354)

caggttcagctgggtgcagtcctggccacgaagtgaacagcctggcgctctgtgaaggtgtcctgtaaaagccagcg

gtacacctttaccaactacggcatgaactgggtgccccaggctcctggacaaggcttggaatggatgggctggatcaaacctacac

cggcgagcctacctacgtggacgacttcaagggcagattcgtgttcagcatggacaccagcgccagcacagcctacctgcagatca

gctctctgaagccaggatgatggccatgtactactgcgccagactgagagcatcagacctggacctctggcctattggggacagg

gcacactgggtcacagtgtcctct

-continued

QVQLVQSGHEVKQPGASVKVSKASGYTFTNYGMNWPQAPGQGLEW

(SEQ ID NO: 1020)

MGWINTYTGEPTYVDDFKGRFVFSMDTSASTAYLQISSLKAEDMAMYCARLRGIR

PGPLAYWGQGLTVTVSS

Modified humanized 8F9A5A1 Heavy chain variable region sequence

(SEQ ID NO: 355)

cagatccagctggtgcagctctggccccgaggtgaagcagcctggggcctcagtgaaagtctcctgcaaggcttctg

ggtataccttcacaaactatggaatgaactgggtgaagcagggccctggacaagggttgagtggtggatggataaacacctaca

ctggagagccaacatatgttgatgacttcaaggacgggtttgccttctccatggacacctctgccagcacagcatacctgcagatcagc

agcctaaaggctgaggacaccgccacctattactgtgcaagattgagggggatagcaccgggtcccttggttactggggccaagg

accctggtcaccgtctcctca

(SEQ ID NO: 1021)

QIQLVQSGPEVKQPGASVKVSKASGYTFTNYGMNWKQAPGQGLEWM

GWINTYTGEPTYVDDFKGRFAFSMDTSASTAYLQISSLKAEDTATYYCARLRGIRPG

PLAYWGQGLTVTVSS

Modified humanized 8F9A5A1 Heavy chain variable region sequence (codon optimized)

(SEQ ID NO: 356)

cagattcagctggtgcagctctggccccgaagtgaacaacctggcgctctgtgaaggtgtcctgcaaggccagcg

gtcacacctttaccaactacggcatgaactgggtcaagcagggccctggacaaggcctggaatggatgggtggatcaaacacctaca

ccggcgagcctacctacgtggacgacttcaagggcagattcgcttcagcatggacaccagcgccagcacagcctacctgcagatc

agctctctgaaggccgaggacaccgccacctactactgtgccagactgagaggcatcagaccggacctctggcctattggggaca

gggaacactggtcacggtgtcctct

(SEQ ID NO: 1022)

QIQLVQSGPEVKQPGASVKVSKASGYTFTNYGMNWKQAPGQGLEWM

GWINTYTGEPTYVDDFKGRFAFSMDTSASTAYLQISSLKAEDTATYYCARLRGIRPG

PLAYWGQGLTVTVSS

8F9A5A1 Light chain variable region sequence

(SEQ ID NO: 357)

gaaattttgctcaccagctctccagcaatcatagctgcactctcctggggagaaggtcaccatcacctgcagtgccagct

caagtgtaatgtacatgaactgggtaccagcagaaaccaggatcctccccaaaatatggatttatggtatatccaacctggcttctggag

ttcctgctcgcttcagtgggcagtgggctctgggacatctttctcttcacaatcaacagcatggaggctgaagatggtgccacttattactgtc

agcaaaggagtagttaccacccacgttcggaggggggaccaagctggaaataaaacgg

(SEQ ID NO: 1023)

EILLTQSPANIAASPGKVTITCSASSSVSYMNWYQKPGSSPKIWIYGISNL

ASGVPARFSGSGSGTSFSFTINSMEADVATYYCQQRSSYPPTFGGGTKLEIKR

IGKV3D-15*02 V-REGION sequence

(SEQ ID NO: 358)

gaaatagtgatgatgcagctctccagccaccctgtctgtgtctccaggggaaagagccaccctctcctgcaggggccag

tcagagtgtagcagcaacttagcctgtaccagcagaaacctggccaggctccaggctcctcatctatgggtgcattccaccagggcc

actggcatcccagccaggttcagtggcagtggtctgggacagagttcactctcaccatcagcagcctgcagctctgaagattttgcagt

ttattactgtcagcagataataac

(SEQ ID NO: 1024)

EIVMMQSPATLSVSPGERATLSCRASQSVSSNLAWYQQKPGQAPRLLIYG

ASTRATGIPARFSGSGSTEFTLTISLQSEDFAVYYCQQYNN

IGKJ4*02 J-REGION sequence

(SEQ ID NO: 359)

ctcacggttcggcggagggaaggtggagatcaaa

(SEQ ID NO: 1025)

LTFGGGTVKVEIK

-continued

humanized 8F9A5A1 Light chain variable region sequence

(SEQ ID NO: 360)

gaaatagtgtgatgcagtcctccagccaccctgtctgtgtctccaggggaaagagccaccctctcctgcagtgccagc

tcaagtgttaagttacatgaactggtagcagcagaacacctggccaggctcccaggtcctcatctatggtatatccaacctggctctggc

atcccagccaggttcagtggtcagtggtctgggacagagttcactctcaccatcagcagcctgcagtcgaagattttgcagtttattact

gtcagcaaaggagtagttacccaccacgttcggcggagggaaccaaggtggagatcaaacgg

(SEQ ID NO: 1026)

EIVMMQSPATLSVSPGERATLSCSASSSVSYMNWYQKPGQAPRLLIYGIS

NLASGIPARFSGSGSTEFTLTISLQSEDFAVYYCQQRSSYPPTFGGGTKVEIKR

humanized 8F9A5A1 Light chain variable region sequence (codon optimized)

(SEQ ID NO: 361)

gagatcgtgtgatgcagagccccgccacactgagtggtctccaggcgaaagagccacactgtcctgtagcgcca

gcagcagcgtgtcctacatgaactggtagcagcagaagcccgacaggccctagactgctgtagcggcatcagcaatctggcca

gcggcatcctgccagattttctggctctggctccggcaccgagttcaccctgacaatctctagcctgcagagcgaggacttcgccgtg

tactactgccagcagagaagcagctaccctcctacctttggcggaggcaccaggtggaaatcaagcgg

(SEQ ID NO: 1027)

EIVMMQSPATLSVSPGERATLSCSASSSVSYMNWYQKPGQAPRLLIYGIS

NLASGIPARFSGSGSTEFTLTISLQSEDFAVYYCQQRSSYPPTFGGGTKVEIKR

Modified humanized 8F9A5A1 Light chain variable region sequence

(SEQ ID NO: 362)

gaaatagtgtgacccagtcctccagccaccctgtctgtgtctccaggggaaagagccaccctctcctgcagtgccag

ctcaagtgttaagttacatgaactggtagcagcagaacacctggccaggctcccaggtctggatctatggtatatccaacctggcttctgg

catcccagccaggttcagtggtcagtggtctgggacaagcttcagcctcaccatcagcagcctgcagtcgaagattttgcagtttatta

ctgtcagcaaaggagtagttacccaccacgttcggcggagggaaccaaggtggagatcaaacgg

(SEQ ID NO: 1028)

EIVLTQSPATLSVSPGERATLSCSASSSVSYMNWYQKPGQAPRLWIYGIS

NLASGIPARFSGSGSTSFSLTISLQSEDFAVYYCQQRSSYPPTFGGGTKVEIKR

Modified humanized 8F9A5A1 Light chain variable region sequence (codon optimized)

(SEQ ID NO: 363)

gagatcgtgtgacacagtcctccgccacactgagtggtctccaggcgaaagagccacactgtcctgtagcgccag

cagcagcgtgtcctacatgaactggtagcagcagaagcccgacaggccctagactgtggatctacggcatcagcaatctggccag

cggcatcctgccagattttctggctctggctccggcaccagcttcagcctgacaatcagcagcctgcagagcgaggacttcgccgtg

tactactgccagcagagaagcagctaccctcctacctttggcggaggcaccaggtggaaatcaagcgg

(amino acids)

(SEQ ID NO: 1029)

EIVLTQSPATLSVSPGERATLSCSASSSVSYMNWYQKPGQAPRLWIYGIS

NLASGIPARFSGSGSTSFSLTISLQSEDFAVYYCQQRSSYPPTFGGGTKVEIKR

Modified humanized 8F9A5A1 scFV sequence (codon optimized)

(SEQ ID NO: 364)

Cagattcagctgggtgcagtcctggccccgaagtgaacaacacctggcgctctgtgaaggtgtcctgcaaggccagcg

gctacacctttaccaactacggcatgaactgggtcaagcaggccctggacaaggcctggaatggatgggtggatcaaacacctaca

ccggcgagcctacctacgtggacgacttcaagggcagattcgcttcagcatggacaccagcggcagcagcctacctgcagatc

agctctctgaaggccgaggacaccgccactactactgtgccagactgagaggcatcagaccggacctctggcctattggggaca

gggaacactgggtcacctgtcctctggcggtggcggaagcggaggcggtggctccggtggcgaggcagcgagatcgtgctgac

acagtcctcccgccacactgagtggtgtctccaggcgaaagagccacactgtcctgtagcggcagcagcgtgtcctacatgaactg

gtatcagcagaagcccgacaggcccttagactgtggatctacggcatcagcaatctggccagcggcatccctgccagattttctggc

tctggctccggcaccagcttcagcctgacaatcagcagcctgcagagcgaggacttcgccgtgtactactgccagcagagaagcag

ctaccctcctacctttggcggaggcaccaggtggaaatcaagcgg

-continued

(SEQ ID NO: 1030)

QIQLVQSGPEVKQPGASVKVSKKASGYTFITNYGMNWVKQAPGQGLEWM
GWINTYTGEPTYVDDFKGRFAFSDTSASTAYLQISSSLKAEDTATYYCARLRGIRPG
PLAYWGQGLTVTVSSGGGGSGGGSGGGGSEIVLTQSPATLSVSPGERATLSCSASS
SVSYMNWYQQKPGQAPRLWIYGISNLASGIPARFSGSGSGTSFSLTISSLQSEDAVY
YCQQRSSYPPTFGGGTKVEIKR

8H5H5G4 Heavy chain variable region sequence

(SEQ ID NO: 365)

gtccagctgcaacagctctggacctgatctggatgaagcctgggacttcagtgaagatcctgtaagacttctggaaca
cattcactgaatacaccatgcactgggtgaagcagagccatggaagagccttgagtggttgagggttttaactcctaacaatgggtgta
ctaactacaaccagaagttcaagggaaggccacattgactgtagacaagtcctccagcacagcctacatggagctccgcagcctga
catctgaggattctgcagctctattactgtgcaagacgttactaccatagctacgtgttctactttgactcctggggccaaggcaccact
ctcacagctctcctca

(SEQ ID NO: 1031)

VQLQSGPDLVKPGTSVKISCKTSGNTFTEYTMHWVKQSHGKSLWIGGF
NPNNGVTNYNQKFKGKATLTVDKSSSTAYMELRSLTSEDAVYYCARRYHSTYVF
YFDSWGQGLTVTVSS

IGHV1-24*01 V-REGION sequence
(DNA)

(SEQ ID NO: 366)

caggctccagctggtagctctgggctgaggtgaagaagcctggggcctcagtgaaggtctcctgcaaggtttccg
gatacacctcactgaattatccatgcactgggtgcgacaggtcctggaaaagggttgagtggtggaggttttgatcctgaagat
ggtgaacaatctacgcacagaagttccagggcagagtcaccatgaccgaggacacatctacagacacagcctacatggagctgag
cagcctgagatctgaggacacggcgtgtattactgtgcaaca

(amino acids)

(SEQ ID NO: 1032)

QVQLVQSGAEVKKPGASVKVSKVSGYTLTELSMHWVRQAPGKGLEW
MGGFPDPEDGETIYAQKQGRVTMTEDTSTDYAMELSSLRSEDAVYYCAT

IGHJ4*03 J-REGION sequence
(DNA)

(SEQ ID NO: 367)

tactttgactactggggccaagggaacctgggtcacctctcctca

(amino acids)

(SEQ ID NO: 1033)

YFDYWQGLTVTVSS

Humanized 8H5H5G4 Heavy chain variable region sequence
(DNA)

(SEQ ID NO: 368)

caggctccagctggtagctctgggctgaggtgaagaagcctggggcctcagtgaaggtctcctgcaaggtttccg
gaaacacattcactgaatacaccatgcacTgggtgcgacaggtcctggaaaagggttgagtggtggaggttttaactcctaaca
tggtgttactaactacaaccagaagttcaagggcAgagtcaccatgaccgaggacacatctacagacacagcctacatggagctgag
cagcctgagatctgaggacacggcgtgtattactgtGcaagacgttactaccatagctacgtgttctactttgactcctggggcca
agggaacctgggtcacctctcctca

(amino acids)

(SEQ ID NO: 1034)

QVQLVQSGAEVKKPGASVKVSKVSGNTFTEYTMHWVRQAPGKGLEW
MGGFNPNGVTNYNQKFKGRVTMTEDTSTDYAMELSSLRSEDAVYYCARRYH
STYVFYFDSWGQGLTVTVSS

-continued

Humanized 8H5H5G4 Heavy chain variable region sequence (codon optimized)
(DNA)

caggttcagctgggtcagctctggcgccgaagtgaagaaacctggcgccctctgtgaaggtgtcctgcaaggtgtccgg (SEQ ID NO: 369)

aaataccttcaccgagtagacaccatgcactgggtccgacaggccccctggcaaaggacttgaatggatggcggttcaacccaacaa
cggcgtgaccaactacaaccagaaattcaagggcgcgtgaccatgaccgaggacacaagcacagacaccgcctacatggaactg
agcagcctgagaagcgaggacaccgccgtgtactactgcccagaaggtactaccacagcacctacgtgttctacttcgacagctgg
ggccagggcacactggtcacagtttcttct

(amino acids)

(SEQ ID NO: 1035)

QVQLVQSGAEVKKPGASVKVCKVSGNTFTEYTMHWVRQAPGKGLEW

MGGFNPNGVTNYNQKFKGRVTMTEDTSTDYAMELSSLRSEDTAVYYCARRYH

STYVFYFDSWGQGLVTVSS

Modified Humanized 8H5H5G4 Heavy chain variable region sequence
(DNA)

caggttcagctgggtacagctctggggctgaggtgaagaagcctggggcctcagtgaaaggtctcctgcaaggtttccg (SEQ ID NO: 370)

gaaacacattcactgaatacaccatgcactgggtgcgacaggctcctggaaaagggttgtagtggatcgaggttttaactcctaacaat
ggtgttactaactacaaccagaagttcaagggcaaggtcacctgaccgtggacacatctagcagcacagcctacatggagctgagc
agcctgagatctgaggacacggccgtgtattactgtgcaagcgttactaccatagtagcctacgtgttctactttgactcctggggccaa
gggacctgggtcacctgtctctca

(amino acids)

(SEQ ID NO: 1036)

QVQLVQSGAEVKKPGASVKVCKVSGNTFTEYTMHWVRQAPGKGLEWI

GGFNPNGVTNYNQKFKGKVTLTVDTSSTAYMELSSLRSEDTAVYYCARRYHST

YVFYFDSWGQGLVTVSS

Modified Humanized 8H5H5G4 Heavy chain variable region sequence (codon optimized)
(DNA)

caggttcagctgggtcagctctggcgccgaagtgaagaaacctggcgccctctgtgaaggtgtcctgcaaggtgtccgg (SEQ ID NO: 371)

aaataccttcaccgagtagacaccatgcactgggtccgacaggccccctggcaaaggactggaatggatcgcggttcaacccaacaa
acggcgtgaccaactacaaccagaaattcaagggcaaagtgacctgaccgtggacaccagcagcagcacagcctacatggaact
gagcagcctgagaagcgaggacaccgccgtgtactactgcccagaaggtactaccacagcacctacgtgttctacttcgacagctg
gggccagggcacactggtcacagtttcttct

(amino acids)

(SEQ ID NO: 1037)

QVQLVQSGAEVKKPGASVKVCKVSGNTFTEYTMHWVRQAPGKGLEWI

GGFNPNGVTNYNQKFKGKVTLTVDTSSTAYMELSSLRSEDTAVYYCARRYHST

YVFYFDSWGQGLVTVSS

8H5H5G4 Light chain variable region sequence
(DNA)

gatatccagatgacacagactacatcctccctgtctgcctctctgggagacagagtcaccatcagttgcagtgcaagtc (SEQ ID NO: 372)

agggcattagcaattatttaaactgggtttcagcagaaaccagatggaactattaagctcctgatctattacacatcaagtttacattcagga
gtcccatcaaggttcagtggtgggtctgggacagattattctctcaccatcagtaattgtggaacctgaagatattgccacttactattg
tcagcagtatagtagcttccttacacgttcggaggggggaccaagctggagataaaaacgg

(amino acids)

(SEQ ID NO: 1038)

DIQMTQTSSLSASLGDRVTISCSASQGISNYLWFPQQKPDGTIKLLIYYS

SLHSGVPSRFSGSGSDYSLTISNVEPEDIATYYCQQYSKLPYTFGGGTKLEIKR

-continued

IGKV1-27*01 V-REGION sequence
(DNA)

gacatccagatgaccagtcctccatcctccctgtctgcatctgtaggagacagagtcaccatcacttgccgggcgagt
cagggcattagcaattatttagcctggatcagcagaaaccagggaagtctcctaagctcctgatctatgctgcatccactttgcaatcag
gggtcccatctcggttcagtgaggcagtgatctgggacagatttcactctcaccatcagcagcctgcagcctgaagatgttgcaacttatt
actgtcaaaagtataacagtgccct

(amino acids)

DIQMTQSPSSLSASVGDRTITCRASQGISNYLAWYQQKPGKVPKLLIYAA

STLQSGVPSRFRSGSGTDFTLTISLQPEDVATYYCQKYNAP

IGKJ4*02 J-REGION sequence
(DNA)

ctcacgttcggcggaggaccgaagtgagatcaaa

(amino acids)

LTFGGGTKVEIK

humanized 8H5H5G4 Light chain variable region sequence

gacatccagatgaccagtcctccatcctccctgtctgcatctgtaggagacagagtcaccatcacttgcaagtgc
agggcattagcaattatttaaacTggtatcagcagaaaccagggaagtctcctaagctcctgatctattacacatcaagtttacattcagg

gggtcccatctcggttcagtgaggcagtgatctgggacagatttcactctcaccatcagcagcctgcagcctgaagatgttgcaacttatta

ctgtcagcagtatagtaagcttccttacacgttcggcggaggaccgaagtgagatcaaacgg

(amino acids)

(amino acids)

DIQMTQSPSSLSASVGDRTITCSASQGISNYLNWYQQKPGKVPKLLIYYT

SSLHSGVPSRFRSGSGTDFTLTISLQPEDVATYYCQYSKLPYTFGGGTKVEIKR

humanized 8H5H5G4 Light chain variable region sequence (codon optimized)
(DNA)

gacatccagatgacacagagccctagcagcctgtctgccagcgtgggagacagagtgaccatcacatgtagcgcca

gccagggcatcagcaactacctgaactggatcagcagaaaccgggaagtgcccaagctgctgatctactacaccagcagcctg

cacagcggcgtgccagcagattttctggcagcggctctggcaccgacttcaccctgaccatattagcctgcagcctgaggacgtg

gccacctactactgtcagcagtacagcaagctgccctacacctttggcggaggcaccaaggtggaaatcaagcgg

(amino acids)

DIQMTQSPSSLSASVGDRTITCSASQGISNYLNWYQQKPGKVPKLLIYYT

SSLHSGVPSRFRSGSGTDFTLTISLQPEDVATYYCQYSKLPYTFGGGTKVEIKR

Modified humanized 8H5H5G4 Light chain variable region sequence
(DNA)

gacatccagatgaccagtcctccatcctccctgtctgcatctgtaggagacagagtcaccatcacttgcaagtgc

agggcattagcaattatttaactggatcagcagaaaccagggaagtctcctaagctcctgatctattacacatcaagtttacattcagg

gggtcccatctcggttcagtgaggcagtgatctgggacagattacactctcaccatcagcagcctgcagcctgaagatgttgcaacttatta

ctgtcagcagtatagtaagcttccttacacgttcggcggaggaccgaagtgagatcaaacgg

(amino acids)

DIQMTQSPSSLSASVGDRTITCSASQGISNYLNWYQQKPGKVPKLLIYYT

SSLHSGVPSRFRSGSGTDYTLTISLQPEDVATYYCQYSKLPYTFGGGTKVEIKR

-continued

Modified humanized 8H5H5G4 Light chain variable region sequence (codon optimized)
(DNA)

(SEQ ID NO: 378)

gacatccagatgacacagagccctagcagcctgtctgccagcgtgggagacagagtgaccatcacatgtagcgcca
gccagggcatcagcaactacctgaactggatcagcagaaacccggcaaggtgccaagctgctgatctactacaccagcagcctg
cacagcggcgtgccaagcagattttctggcagcggctctggcaccgactacaccctgaccatatctagcctgcagcctgaggacgtg
gccacctactactgtcagcagtagcagaagctgcctacacctttggcggaggcaccaggtggaaatcaagcgg

(amino acids)

(SEQ ID NO: 1044)

DIQMTQSPSSLSASVGDRVTITCSASQGISNYLNWYQQKPGKVPKLLIYYT
SSLHSGVPSRFGSGSGTDYTLTISSLQPEDVATYYCQQYSKLPYTFGGGTKVEIKR

Modified humanized 8H5H5G4 scFV sequence (codon optimized)
(DNA)

(SEQ ID NO: 379)

Caggttcagctgggtcagctctggcgccgaagtgaagaaacctggcgctctgtgaaggtgtcctgcaaggtgtccgg
aaataccttcaccgagtagaccatgcactgggtccgacaggccccctggcgaaggactggaatggatcggcggttcaacccaaca
acggcgtgaccaactacaaccagaaattcaagggcaaagtgacctgacctggacaccagcagcagcacagcctacatggaact
gagcagcctgagaagcaggacaccgccgtgtactactgcgcagaaggtactaccacagcacctacgtgttctacttcgacagctg
ggggcagggcacactgggtcacagtttcttctggcgggtggcggaagcggaggcgggtggctccgggtggcggaggcagcgacatcca
gatgacacagagccctagcagcctgtctgccagcgtgggagacagagtgaccatcacatgtagcgccagggcatcagcaact
acctgaactggatcagcagaaacccggcaaggtgccccagctgtgatctactacaccagcagcctgcacagcggcgtgccaagc
agatcttctggcagcggctctggcaccgactacaccctgaccatatctagcctgcagcctgaggacgtggccacctactactgtcagca
gtacagcaagctgcctacacctttggcggaggcaccaggtggaaatcaagcgg

(amino acids)

(SEQ ID NO: 1045)

QVQLVQSGAEVKKPGASVKVCKVSGNTFTFETMHWVRQAPGKLEWI
GGFNPNGVTNYNQKFKGKVTLTVDTSSTAYMELSSLRSEDAVYYCARRYHST
YVFYFDSWGQGLVTVSSGGGSGGGGSGGGSDIQMTQSPSSLSASVGDRVTITCS
ASQGISNYLNWYQQKPGKVPKLLIYYTSSLHSGVPSRFGSGSGTDYTLTISSLQPED
VATYYCQQYSKLPYTFGGGTKVEIKR

Human IgG1 heavy chain constant region sequence: (for making full antibody -
pair with either kappa or lambda constant region; 2 plasmids, express together)
(DNA)

(SEQ ID NO: 380)

gctagcaccaagggcccatcggtcttccccctggcacccctcctccaagagcacctctgggggcacagcggccctgg
gctgcctggtcaaggactacttccccgaaccggtgacgggtgctgtggaactcaggcgccctgaccagcggcgtgcacaccttcccg
gctgtcctacagtcctcaggactctactccctcagcagcgtggtgacagtgcctccagcagctggggcaccagacctacatctgca
acgtgaatcacaagcccagcaacaccaaggtggacaagaaagttgagccaaatcttgtgacaaaactcacacatgccaccgtgcc
cagcacctgaactcctggggggaccgtcagttcttctcttccccccaaaacccaaggacaccctcatgatctcccgaccctcaggt
cacatgcgtggtggtggacgtgagccacgaagacctgaggtcaagttcaactggtacgtggacggcgtggaggtgcataatgcc
agacaaagccgcgggaggagcagtacaacagcacgtaccgtgtggtcagcgtcctcaccgtcctgcaccaggactggctgaatgg
caaggagtacaagtcaaggtctccaacaaagccctcccagcccccatcgagaaaaccatctccaaagccaaagggcagccccga
gaaccacaggtgtacacctgcccccatccgggaggagatgaccaagaaccaggtcagcctgacctgcctgggtcaaaggctctctat

-continued

cccagcgacatcgccgtggagtgaggagagcaatgggcagccggagagaacaactacaagaccacgcctcccgtgctggactccgac
ggctcctctctctctctacagcaagctcaccgtggacaagagcaggtggcagcaggggaacgtcttctcatgctccgtgatgatgagg
ctctgcacaaccactacacgcagaagagcctctccctgtctccgggtaaatga

(amino acids)

(SEQ ID NO: 1046)

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGV
HTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHT
CPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGV
EVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISK
AKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTT
PVLDSGDSFPLYSKLTVDKSRWQQGNVFCFVMSHEALHNHYTQKSLSLSPGK

Human IgG2 heavy chain constant region sequence: (for making full antibody -
pair with either kappa or lambda constant region; 2 plasmids, express together)
(DNA)

(SEQ ID NO: 381)

gcctccaccaagggcccatcggtcttccccctggcgccctgctccaggagcacctccgagagcacagccgcctg
ggctgctggtcaaggactacttccccgaaccggtagcggtgctggaactcaggcgctctgaccagcgcgtagcaccttccca
gctgtctacagtctcaggactctactccctcagcagcggtgtagcctccagcaactcggcaccagacctacacctgca
acgtagatcacaagcccagcaacaccaaggtggacaagacagtgtagcgcaaatgttgtgtcgagtgcccaccgtgccagcacca
cctgtggcaggaccgtcagttctctcttccccccaaaacccaaggacacctcatgatctccggacccctgaggtcacgtgctggg
tggtggacgtgagccacgaagaccccgaggtccagttcaactggtacgtggacggcgtggaggtgcataatgccaagacaagcc
acgggaggagcagttcaacagcacgttccgtgtggtcagcgtcctcaccgttgtgcaccaggactggctgaacggcaaggagtaca
agtgaaggtctccaacaaaggcctcccagccccatcgagaaaaccatctccaaaacaaaggcgagccccgagaaccacaggt
gtacacctgcccccatcccgaggagatgaccaagaaccaggtcagcctgacctgcctggtcaaaggcttctacccagcgaca
tcgccgtggagtgaggagagcaatgggcagccggagacaactacaagaccacacctccatgctggactccgacggctccttcttc
ctctacagcaagctcaccgtggacaagagcaggtggcagcaggggaacgtcttctcatgctccgtgatgatgaggctctgcacaac
cactacacgcagaagagcctctccctgtctccgggtaaatag

(amino acids)

(SEQ ID NO: 1047)

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVH
TFPAVLQSSGLYSLSSVVTVPSSNFGTQTYTCNVDHKPSNTKVDKTKVERKCCVECPP
CPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVQFNWYVDGVEVHN
AKTKPREEQFNSTFRVSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQP
REPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPMPLDS
DGSFPLYSKLTVDKSRWQQGNVFCFVMSHEALHNHYTQKSLSLSPGK

Human Kappa light chain constant region sequence:
(DNA)

(SEQ ID NO: 382)

aggacgggtggctgcaccatctgtcttcatcttccccccatctgatgagcagttgaaatctggaactgcctctgttgtg
cctgctgaataacttctatcccagagaggccaaagtacagtggaaggtggataacgccctccaatcgggtaactccaggagagtgc
acagagcaggacagcaaggacagcacctacagcctcagcagcacccctgacgctgagcaaagcagactacgagaaacacaaagtc
tacgcctgcgaagtacccatcagggcctgagctcgcccgtcacaaagagcttcaacaggggagagtgtag

(amino acids)

(SEQ ID NO: 1048)

RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSG
NSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC

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Human Lambda light chain constant region sequence:
(DNA)

(SEQ ID NO: 383)

ggtcagcccaaggtgccccctcggtcactctgttcccgccctcctctgaggagcttcaagccaacaaggccacact
gggtgtgtctcataagtgacttctacccgggagccgtgacagtggcctggaaggcagatagcagccccgtcaaggcgggagtgagaga
ccaccacaccctccaacaaaagcaacaacaagtacgcgccagcagctatctgagcctgacgcctgagcagtggaagtccccacag
aagctacagctgccaggtcacgcatgaaggggagcaccgtggagaagacagtggcccctacagaatgttcatag
(amino acids)

(SEQ ID NO: 1049)

GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVANKADSSPVK
AGVETTPSKQSNKKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTVEKTVAPTECS

Human IgG1 Fc region sequence: (to be fused to scFv for homo-dimerizes)
(DNA)

(SEQ ID NO: 384)

gagcccaaatcttgtgacaaaactcacacatgccccacgtgccagcacctgaactcctggggggacgcgtcagtcctt
cctcttccccccaaaacccaaggacaccctcatgatctcccgaccctgaggtcacatgcgtgggtggacgtgagccacgaaga
ccctgaggtcaagttcaactggtacgtggagcggcgtggaggtgcataatgccaaagacaaagccgcgaggagcagtagacaacagc
acgtaccgtgtggtcagcgtcctcacgcctgcaccaggactggctgaatggcaaggagtacaagtgcaggctctccaacaaagcc
ctccagcccccatcgagaaaaccatctccaaagccaaagggcagccccgagaaccacaggtgtacaccctgcccccatcccggtg
aggagatgaccaagaaccaggtcagcctgacctgcctgggtcaaaggcttctatcccagcgacatcgccgtggagtgaggagagcaat
gggcagccgggagaacaactacaagaccacgcctcccgctgctggactccgacggctccttcttctctacagcaagctcacgtggac
aagagcaggtggcagcaggggaacgtcttctcatgctccgtgatgcatgaggctctgcacaaccactacacgcagaagagcctctcc
ctgtctccgggtaaatga

(amino acids)

(SEQ ID NO: 1050)

EPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVS
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKC
KVSNAKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAV
EWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQSVMEALHNH
YTQKSLSLSPGK*

Human IgG2 Fc region sequence:
(DNA)

(SEQ ID NO: 385)

gagcgcaaatgttgtgtcgagtgccccacgtgccagcaccacctgtggcaggaccgtcagtccttctcttccccca
aaacccaaggacaccctcatgatctcccgaccctgaggtcacgtgcgtgggtggacgtgagccacgaagaccccgaggtcca
gttcaactggtacgtggagcggcgtggaggtgcataatgccaaagacaaagccacgggaggagcagttcaacagcacgttccgtgtgg
tcagcgtcctcacgctgtgcaccaggactggctgaacgggaaggagtacaagtgcaggctctccaacaaaggcctccagcccc
atcgagaaaaaccatctccaaaacaaagggcagccccgagaaccacaggtgtacaccctgcccccatcccgaggagatgacca
agaaccaggtcagcctgacctgcctgggtcaaaggcttctacccagcgacatcgccgtggagtgaggagcaatgggcagccgga
gaacaactacaagaccacacctcccatgctggactccgacggctccttcttctctacagcaagctcacgtggacaagagcaggtg
gcagcaggggaacgtcttctcatgctccgtgatgcatgaggctctgcacaaccactacacgcagaagagcctctccctgtctccgggt
aaatag

-continued

(amino acids)

(SEQ ID NO: 1051)

ERKCCVECPPCAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHED
 PEVQFNWYVDGVEVHNAKTKPREEQFNSTFRVSVLTVVHQDWLNGKEYKCKVSN
 KGLPAPIEKTISKTKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWES
 NGQPENNYKTTPMLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQ
 KSLSLSPGK*

In another aspect of the invention, an immune cell engineered to express a CAR is administered to a patient diagnosed with or at risk of developing a cancer or cancer metastasis, wherein the immune cell is also engineered to express an anti-NME7 antibody or antibody fragment, which may be expressed off of an inducible promoter. In one aspect, the CAR is guided by an anti-MUC1* antibody fragment. In one case, the CAR is huMNC2-CAR44. In one aspect, the anti-NME7 antibody or antibody fragment binds to an NME peptide listed under sections "Homologous peptides to A1, A2, B1, B2 or B3 peptides" and the "Homologous extended peptides to A1, A2, B1, B2 or B3 peptides" above. In another aspect, the antibody or antibody fragment binds to an NME7 derived peptide chosen from among A1, A2, B1, B2 or B3 (SEQ ID NOS: 141-145). In yet another aspect, the antibody, antibody fragment or antibody mimic binds to an NME7 peptide comprising the B3 peptide. In one aspect of the invention, the anti-NME7 antibody, antibody fragment or antibody mimic comprises sequences derived from the variable domains of anti-NME7 antibodies 8F9A4A3, 8F9A5A1, or 8H5H5G4.

Such antibodies may be human or humanized. Such antibodies may be polyclonal, monoclonal, bispecific, bivalent, monovalent, single chain, scFv, or may be an antibody mimic such as protein scaffolds that present recognition regions that bind to a specific target. As is appreciated by those skilled in the art, antibodies can be of non-human origin, human or humanized. Methods for humanizing antibodies include fusing all or some of the mouse variable regions to V- and J-regions of a closest match human antibody sequence, for example, as shown in sequences listed as SEQ ID NOS: 1001-1045. Full antibodies, rather than single chain constructs, can also be made. For example, the heavy chain variable mouse sequence is fused to human V- and J-regions then fused to the human heavy chain constant regions of IgG1, IgG2 or IgG3. Similarly, the light chain variable mouse sequences are fused to human V- and J-regions then fused to either the human Kappa or Lambda constant regions of IgG1, IgG2 or IgG3. Plasmids are expressed together and associate to form the full antibody (SEQ ID NOS: 1047-1051).

In another aspect of the invention, small molecules are anti-cancer agents that are selected for their ability to inhibit the tumorigenic effects of NME7, NME7_{AB} or NME7-X1. For example, a high throughput screen identifies small molecules that will treat cancer. In a multi-well plate, small molecules are separately added to wells in which cancer cells are cultured in a medium containing NME7_{AB}. If the small molecule diminishes the amount of cells that become floaters and/or reduces the expression of metastatic markers such as CXCR4, CHD1 or pluripotent stem cell markers, then that small molecule is an anti-cancer drug candidate. Another method of identifying small molecules that are anti-cancer agents is to select those small molecules that bind to NME7, NME7_{AB} or NME7-X1 or suppresses expres-

sion of the NME7 species. Yet another high throughput screen is to select for small molecules that inhibit the binding of NME7_{AB} to the PSMGFR peptide of the MUC1* extracellular domain and those small molecules will be anti-cancer agents.

The sequences of NME7_{AB} and NME7-X1 differ only in that NME7-X1 is missing some of the N-terminal sequence that NME7_{AB} has. Experiments show that there is a naturally occurring NME7 species that is nearly identical to NME7_{AB}, which we call NME7_{AB}-like species. Antibodies that bind to NME7-X1 may also bind to the naturally occurring species that mimics NME7_{AB}, unless there are conformational differences that an antibody can differentiate. Therefore, if it is desired to inhibit NME7-X1 but not NME7_{AB}-like species, or vice versa, siRNA, anti-sense nucleic acids, or genetic editing techniques can be used to inhibit expression of one but not the other.

In one case, the anti-cancer therapeutic agent is a nucleic acid that directly or indirectly suppresses specific expression of NME7, NME7-X1 or NME7_{AB}-like species. Such nucleic acids can be siRNA. RNAi, anti-sense nucleic acids and the like that directly suppress the NME7 species. In another aspect of the invention, the nucleic acid can indirectly suppress the NME7 species for example by altering the expression of a molecule that regulates it. For example, the super enhancer BRD4 suppresses expression of NME7. Therefore, an effective therapeutic for the treatment or prevention of cancer is an agent that increases expression of BRD4. An effective therapeutic may be an agent that increases expression of BRD4's co-factor. JMJD6.

Peptides derived from NME7_{AB} or NME7-X1, or the entire protein, are used to generate anti-NME7 or anti-NME7-X1 antibodies in animals that we have demonstrated inhibit cancer growth and inhibit transition of cancer cells to metastatic cancer cells. Similarly, NME7 derived peptides can be administered to a human such that they generate antibodies that treat or prevent cancer or inhibit transition of cancer cells to metastatic cancer cells, NME7 peptides or proteins are administered to a person as a type of vaccine to stimulate the production of anti-NME7, anti-NME7_{AB} or anti-NME7-X1 antibodies in the recipient. The results shown in FIG. 12 and FIG. 13 indicate that immunizing a person with a collection of peptides derived from NME7, especially in the NME7-X1 or NME7_{AB} sequences may be a more effective vaccine than immunizing with a single peptide. Said peptides or proteins may further be conjugated to a carrier protein or other adjuvant, known to those skilled in the art to aid in the stimulation of an immune response.

NME7 peptides that lie outside of the DM10 domain are preferred to generate antibodies for the treatment or prevention of cancer. Peptides that can be administered to a patient for the prevention of cancer or metastasis contain sequences of the peptides listed in FIG. 6-FIG. 9. A1, A2, B1, B2 and B3 are examples of peptides that generate antibodies that bind to NME7_{AB} and NME7-X1 and are administered to a

patient for the treatment or prevention of cancer. The invention is not limited to peptides of the exact sequence as is naturally occurring in NME7 or NME7-X1. As is known to those skilled in the art, substitution of several amino acids of a peptide sequence can still give rise to antibodies that specifically recognize the natural protein sequence. It is not intended that the invention be limited to the peptides demonstrated herein to inhibit cancer growth or inhibit the transition of regular cancer cells to metastatic cancer cells. The methods used here to identify peptides A1, A2, B1, B2 and B3 can also be used to identify other peptide sequences that could be equally or more effective than the peptides demonstrated here.

Chimeric antigen receptor molecules comprising portions of human NME7_{AB} or NME7-X1 or comprising an antibody fragment that binds to NME7_{AB} or NME7-X1 are anti-cancer therapeutics and are administered to a patient for the treatment or prevention of cancers or cancer metastases.

In one instance, the recognition units or variable regions of anti-NME7 antibodies are fused to molecules of T cells using the technology known as CAR (chimeric antigen receptor) technology or CAR T technology. The salient feature of antibodies or fragments thereof that can be used therapeutically to treat or prevent cancers is the identification of antibody-like variable regions that recognize NME7 and prevent its interaction with targets that promote cancers. In one case, the target is the PSMGFR region of MUC1*.

Antibodies, antibody fragments or single chain antibodies can be engineered into chimeric molecules, including chimeric antigen receptors, also known as CARs, which molecules are then transfected or transduced into an immune system cell, such as a T cell, and administered to a patient. The humanized antibodies or antibody fragments, typically an scFv, comprises much of the extracellular domain of a CAR. The antibody fragment is biochemically fused to immune system signaling molecules, such as CD8 as the transmembrane domain and cytoplasmic signaling motifs such as T cell receptor signaling molecules also called activation domains, or co-stimulatory domains including but not limited to CD3-zeta, CD28, 41bb, OX40. CARs can be transfected into T cells or other cells, preferably immune system cells and administered to a patient. Here we describe CARs in which the extracellular portion contains an anti-NME7, anti-NME7_{AB} or anti-NME7-X1 antibody, antibody fragment or single chain, scFv antibody fragment. In a preferred embodiment, the antibody or antibody fragment is human or humanized.

Effective anti-NME7 or anti-NME7-X1 antibodies or fragments will have the ability to bind to native NME7, NME7_{AB} or NME7-X1. In practice, the parent antibody, from which the extracellular domain of the CAR is engineered, is generated by immunizing an animal with an NME7, NME7_{AB} or NME7-X1 derived peptide. In one aspect of the invention, the immunizing peptide is comprised of NME7 amino acids 1-376. In one aspect of the invention, the immunizing peptide is comprised of NME7 amino acids 92-376. In another aspect of the invention, the immunizing peptide is comprised of NME7 amino acids 125-376. In yet another aspect of the invention, the immunizing peptide is made up of sequences listed in FIG. 6-FIG. 8. In another aspect of the invention, the immunizing peptide is made up of sequences listed in FIG. 9. Alternatively, the parent antibody or the antibody fragment is selected from a library or pool of antibodies, which may be natural, synthetic or fragments of either, wherein they are selected for their ability to bind to NME7, NME7_{AB} or NME7-X1, peptides listed in FIG. 6-FIG. 8, or peptides listed in FIG. 9.

The targeting portion of a CAR need not be an antibody or antibody fragment. Here we describe a CAR wherein the extracellular domain contains an NME7 fragment, NME7-derived peptide(s) are engineered into a different sort of CAR wherein the targeting portion of the extracellular domain is a protein fragment or peptide rather than an antibody or antibody fragment. The peptide CARs are transfected or transduced into an immune system cell, typically a T cell. The NME7 fragments or NME7 derived peptides are selected for their ability to bind to their cognate binding partners but should not be able to function as intact NME7. NME7_{AB} or NME7-X1 and confer tumorigenic activity, NME7 fragments or NME7 derived peptides are biochemically fused to immune system signaling molecules, such as CD8 as the transmembrane domain and cytoplasmic signaling motifs such as T cell receptor signaling molecules also called activation domains, or co-stimulatory domains including but not limited to CD3-zeta, CD28, 41bb, OX40.

In one aspect of the invention, the NME7 fragment is most or all of the NME7 NDPK B domain. In another aspect of the invention, the NME7 fragment is an NME7 peptide that contains one or more of the peptide sequences listed in FIG. 6-FIG. 9. Experiments indicate that, for strategies that use NME7 or fragments of NME7, NME7_{AB}, or NME7-X1 as the targeting portion of a chimeric antigen receptor (CAR) for engineered immune cell therapeutics, fairly large fragments of NME7_{AB} or NME7-X1 would be more effective than shorter peptides, for example peptides less than 15 amino acids in length. Alternatively, a collection of CARs, each bearing a different NME7_{AB} derived peptide can collectively be transfected or transduced into an immune system cell and administered to a patient for the treatment or prevention of cancers. Experiments shown in FIG. 12-FIG. 13 support the validity of this approach.

CARs that contain an NME7 fragment in its extracellular domain are transfected or transduced into an immune system cell, typically a T cell, and administered to a patient for the treatment or prevention of cancers. In one aspect, the cancer is a MUC1*-positive cancer. In another aspect, the cancer is a metastatic cancer.

Agents that inhibit an enzyme that cleaves NME7 can be used to treat or prevent cancers. Some forms of NME7 are sequestered within the cell and therefore are not secreted from the cell whereupon they can act as growth factors to promote cancers. Full-length NME7 is 42 kDa. However, we found that a ~33 kDa NME7 species that is devoid of the DM10 domain and appears to be essentially identical to the recombinant NME7_{AB} that we generated, is secreted from cancer cells and stem cells. This ~33 kDa NME7 species and another ~25 kDa NME7 species may be cleavage products that would be eliminated by an agent that inhibited cleavage of NME7.

The detection of elevated levels of NME7, or an ~33 kDa NME7 species, which we call NME7_{AB}-like species, or NME7-X1 in a patient sample is diagnostic of the presence of cancer or its progression to a more aggressive or metastatic state. The inventors have discovered that both early stage, naïve stem cells and cancer cells, especially MUC1*-positive cancer cells, express high levels of a ~33 kDa NME7 that is devoid of the DM10 domain and NME7-X1.

NME7-X1 was recently listed in a protein database as being a theoretical alternative isoform of NME7, however, it had never been detected in tissues or cells. We designed primers that differentiate NME7-X1 from NME7 by PCR. The expression levels of human NME7, NME7a, NME7b and NME7-X1 were measured by PCR in a panel of cells that included fibroblast cells, human embryonic stem cells,

human iPS cells. T47D human breast cancer cells. DU145 human prostate cancer cells. PC3 human prostate cancer cells. HEK295 human fetal liver cells, and other human stem cell lines, NME7 is expressed at higher levels in cancer cells than in stem cells. Particularly, NME7-X1 is expressed 10-fold higher in prostate cancer cells and 3-fold higher in breast cancer cells, than it is in fibroblast cells or stem cells, NME7-X1 is expressed ~5-fold higher in HEK293 fetal liver cells than it is in fibroblast cells or stem cells and therefore predicts that NME7-X1 is elevated in liver cancers, NME7b is expressed 17-25-times higher in prostate cancer cells than in stem cells.

Detection of elevated levels of NME7 species in a patient sample will be indicators that the patient has a cancer or is at risk of developing a cancer. Levels of NME7 species levels can be measured or assessed by PCR, hybridization schemes, cycling probe technologies. FISH, immunocytochemistry, IHC, Western blot, immunoprecipitation, sandwich assays. ELISA assays and the like. The patient sample may be a fluid sample, a blood sample, milk, urine, cells, liquid biopsy, biopsy and the like. In a patient diagnosed with cancer, elevated levels of NME7 species are indicators of increased metastatic potential. Elevated levels of NME7-X1 are indicators of prostate cancer. Antibodies of the invention are used to detect and distinguish NME7 species and are used as a diagnostic tool.

Because adult cells and tissues do not express significant levels of NME7 or secrete NME7, an effective way to diagnose cancer or to diagnose a more aggressive or metastatic form, or a shift to a more aggressive form, is to measure levels of NME7 in a sample from a patient, from a collection of cells or tissues or from cultured cells, compared to NME7 levels in a healthy sample or compared to levels of NME7 known to exist in healthy adult cells or tissues. Increased levels of NME7 indicate the presence of cancer, the presence of a metastatic cancer or the onset of metastasis. Increased levels of NME7 is also indicative of a MUC1*-positive cancer. The sample assayed for the presence of NME7 may be a collection of cells that may be cultured cell lines or cells from a patient, a bodily fluid, a blood sample, a tissue specimen, or a biopsy specimen. Therefore, a diagnostic assay that will detect the presence of cancer or the progression of cancer, comprises the steps of: 1) obtaining a sample from a patient having cancer or at risk of developing a cancer; 2) subjecting that sample to an assay capable of detecting or measuring levels of NME7, or levels of nucleic acids encoding NME7; 3) comparing levels of the measured NME7 protein or NME7-encoding nucleic acids in the test sample to levels in control patients or control cells; 4) determining that the levels of NME7 or nucleic acids encoding NME7 are elevated compared to the controls; and 5) concluding that the donor of the test sample has cancer or has had a progression of cancer if the control to which the test was compared came from a donor previously diagnosed with a cancer.

In this assay, the control sample to which the test sample is compared can be non-cancerous cells, cultured cells, a sample from a healthy donor, a non-cancerous sample from the donor, or a sample from the donor of the test sample wherein the control sample was taken from the donor at a previous point in time. The source of such samples may be any specimen taken from the patient being tested for the presence or progression of cancer, including bodily fluids, cerebrospinal fluid, bone marrow samples, blood, tissues, cells, biopsy tissues or cells, cultured cells derived from a patient's cells and the like. The source of the sample to which the test sample is compared can be bodily fluids,

cerebrospinal fluid, bone marrow samples, blood, tissues, cells, biopsy tissues or cells, or cultured cells that may be derived from a healthy donor or the test patient wherein the samples were taken at a previous point in time. The measured levels to which the test sample is compared may be from previously recorded data and compiled into lists for comparison to test samples.

Theranostics

Patients diagnosed with elevated levels of NME7 protein or nucleic acids encoding NME7 are then treated with therapeutic agents that suppress expression of NME7, inhibit cleavage of NME7 or inhibit NME7 binding to its targets, wherein such interaction promotes cancers. An important target of NME7 or a cleavage product of NME7, is MUC1*, NME7 binds to and dimerizes the extracellular domain of MUC1*. Therefore, patients diagnosed with elevated levels of NME7 will benefit from treatment with therapeutic agents that inhibit NME7 and/or therapeutic agents that inhibit the dimerization of a cleaved form of MUC1, whose extracellular domain is comprised of some or all of the PSMGFR sequence. Thus assessing suitability of cancer treatments and administration of an effective amount of a therapeutic for the treatment or prevention of cancers would consists of the steps of: 1) obtaining a sample from a patient suspected of having a cancer or at risk of developing a cancer or at risk of developing a metastatic cancer; 2) measuring an amount of NME7 or a cleavage product thereof or an NME7 encoding nucleic acid wherein the measured levels are significantly above those measured in a control sample; 3) determining that the patient has a cancer or has developed a more aggressive or a metastatic cancer; 4) administering to the patient an effective amount of a therapeutic agent that suppresses expression of NME7, inhibits cleavage of NME7 or inhibits NME7 binding to its targets and/or administering to the patient an effective amount of a therapeutic agent that suppresses expression of MUC1, inhibits cleavage of MUC1 to MUC1* or inhibits MUC1* binding to its targets. In a preferred embodiment, the therapeutic agent that inhibits NME7 binding to its targets, inhibits its interaction with MUC1*. In a more preferred embodiment, it inhibits its interaction with the extracellular domain of MUC1* comprised essentially of the PSMGFR sequence. In a preferred embodiment, the therapeutic agent that inhibits MUC1* binding to its targets, inhibits the interaction between MUC1* and NME7. In a more preferred embodiment, the therapeutic agent that inhibits the interaction between MUC1* and NME7 inhibits the binding of MUC1* to the portion of NME7 that is comprised essentially of the sequence of NME7_{AB}.

Chemically Modified Peptides

Polypeptide or antibody therapeutics may suffer from short circulating half-life, and proteolytic degradation and low solubility. To improve the pharmacokinetics and pharmacodynamics properties of the inventive biopharmaceuticals, methods such as manipulation of the amino acid sequence may be made to decrease or increase immunogenicity and decrease proteolytic cleavage: fusion or conjugation of the peptides to immunoglobulins and serum proteins, such as albumin may be made: incorporation into drug delivery vehicles for the biopharmaceuticals such as the inventive peptides and antibodies for protection and slow release may also be made; and conjugating to natural or synthetic polymers are also contemplated. In particular, for synthetic polymer conjugation, pegylation or acylation, such as N-acylation. S-acylation and so forth are also contemplated.

Nucleic Acid Constructs

Also provided is an expression vector comprising a nucleic acid molecule of the invention as described herein, wherein the nucleic acid molecule is operatively linked to an expression control sequence. Also provided is a host-vector system for the production of a polypeptide which comprises the expression vector of the invention which has been introduced into a host cell suitable for expression of the polypeptide. The suitable host cell may be a bacterial cell such as *E. coli*, a yeast cell, such as *Pichia pastoris*, an insect cell, such as *Spodoptera frugiperda*, or a mammalian cell, such as a COS, HEK or CHO cell.

The present invention also provides for methods of producing the polypeptides of the invention by growing cells of the host-vector system described herein, under conditions permitting production of the polypeptide and recovering the polypeptide so produced. The polypeptides useful for practicing the present invention may be prepared by expression in a prokaryotic or eukaryotic expression system.

The recombinant gene may be expressed and the polypeptide purified utilizing any number of methods. The gene may be subcloned into a bacterial expression vector, such as for example, but not by way of limitation, pZER0.

The polypeptides may be purified by any technique which allows for the subsequent formation of a stable, biologically active protein. For example, and not by way of limitation, the factors may be recovered from cells either as soluble proteins or as inclusion bodies, from which they may be extracted quantitatively by 8M guanidinium hydrochloride and dialysis. In order to further purify the factors, any number of purification methods may be used, including but not limited to conventional ion exchange chromatography, affinity chromatography, different sugar chromatography, hydrophobic interaction chromatography, reverse phase chromatography or gel filtration.

When used herein, polypeptide includes functionally equivalent molecules in which amino acid residues are substituted for residues within the sequence resulting in a silent or conservative change. For example, one or more amino acid residues within the sequence can be substituted by another amino acid of a similar polarity, which acts as a functional equivalent, resulting in a silent or conservative alteration. Substitutes for an amino acid within the sequence may be selected from other members of the class to which the amino acid belongs. For example, the nonpolar (hydrophobic) amino acids include alanine, leucine, isoleucine, valine, proline, phenylalanine, tryptophan and methionine. The polar neutral amino acids include glycine, serine, threonine, cysteine, tyrosine, asparagine and glutamine. The positively charged (basic) amino acids include arginine, lysine and histidine. The negatively charged (acidic) amino acids include aspartic acid and glutamic acid. The potential glycosylation amino acids include serine, threonine, and asparagine. Also included within the scope of the invention are proteins or fragments or derivatives thereof which exhibit the same or similar biological activity and derivatives which are differentially modified during or after translation, e.g., by glycosylation, proteolytic cleavage, linkage to an antibody molecule or other cellular ligand, etc.

Any of the methods known to one skilled in the art for the insertion of DNA fragments into a vector may be used to construct expression vectors encoding the polypeptides of the invention using appropriate transcriptional/translational control signals and protein coding sequences. These methods may include in vitro recombinant DNA and synthetic techniques and in vivo recombinations (genetic recombination). Expression of nucleic acid sequence encoding the

polypeptides of the invention may be regulated by a second nucleic acid sequence so that the polypeptide is expressed in a host transformed with the recombinant DNA molecule. For example, expression of the polypeptides described herein may be controlled by any promoter/enhancer element known in the art. Promoters which may be used to control expression of the polypeptide include, but are not limited to the long terminal repeat as described in Squinto et al., (1991, Cell 65:1-20); the SV40 early promoter region (Bernoist and Chambon, 1981, Nature 290:304-310), the CMV promoter, the M-MuLV 5' terminal repeat the promoter contained in the 3' long terminal repeat of Rous sarcoma virus (Yamamoto, et al., 1980, Cell 22:787-797), the herpes thymidine kinase promoter (Wagner et al., 1981, Proc. Natl. Acad. Sci. U.S.A. 78:144-1445), the regulatory sequences of the metallothionein gene (Brinster et al., 1982, Nature 296:39-42); prokaryotic expression vectors such as the β -lactamase promoter (Villa-Komaroff, et al., 1978, Proc. Natl. Acad. Sci. U.S.A. 75:3727-3731), or the tac promoter (DeBoer, et al., 1983, Proc. Natl. Acad. Sci. U.S.A. 80:21-25), see also "Useful proteins from recombinant bacteria" in Scientific American. 1980, 242:74-94; promoter elements from yeast or other fungi such as the Gal 4 promoter, the ADH (alcohol dehydrogenase) promoter, PGK (phosphoglycerol kinase) promoter, alkaline phosphatase promoter, and the following animal transcriptional control regions, which exhibit tissue specificity and have been utilized in transgenic animals: elastase I gene control region which is active in pancreatic acinar cells (Swift et al., 1984, Cell 38:639-646; Ornitz et al., 1986, Cold Spring Harbor Symp. Quant. Biol. 50:399-409; MacDonald, 1987, Hepatology 7:425-515); insulin gene control region which is active in pancreatic beta cells (Hanahan, 1985, Nature 315:115-122), immunoglobulin gene control region which is active in lymphoid cells (Grosschedl et al., 1984, Cell 38:647-658; Adames et al., 1985, Nature 318:533-538; Alexander et al., 1987, Mol. Cell. Biol. 7:1436-1444), mouse mammary tumor virus control region which is active in testicular, breast, lymphoid and mast cells (Leder et al., 1986, Cell 45:485-495), Sendai virus, lenti virus, albumin gene control region which is active in liver (Pinkert et al., 1987, Genes and Devel. 1:268-276), alpha-fetoprotein gene control region which is active in liver (Krumlauf et al., 1985, Mol. Cell. Biol. 5:1639-1648; Hammer et al., 1987, Science 235:53-58); alpha 1-antitrypsin gene control region which is active in the liver (Kelsey et al., 1987, Genes and Devel. 1:161-171), beta-globin gene control region which is active in myeloid cells (Mogam et al., 1985, Nature 315:338-340); Kollias et al., 1986, Cell 46:89-94); myelin basic protein gene control region which is active in oligodendrocyte cells in the brain (Readhead et al., 1987, Cell 48:703-712); myosin light chain-2 gene control region which is active in skeletal muscle (Shani, 1985, Nature 314:283-286), and gonadotropic releasing hormone gene control region which is active in the hypothalamus (Mason et al., 1986, Science 234:1372-1378).

Thus, according to the invention, expression vectors capable of being replicated in a bacterial or eukaryotic host comprising nucleic acids encoding a polypeptide as described herein, are used to transfect the host and thereby direct expression of such nucleic acid to produce polypeptides which may then be recovered in biologically active form. As used herein, a biologically active form includes a form capable of binding to the relevant receptor and causing a differentiated function and/or influencing the phenotype of the cell expressing the receptor.

Expression vectors containing the nucleic acid inserts can be identified by without limitation, at least three general approaches: (a) DNA-DNA hybridization. (b) presence or absence of "marker" gene functions, and (c) expression of inserted sequences. In the first approach, the presence of foreign nucleic acids inserted in an expression vector can be detected by DNA-DNA hybridization using probes comprising sequences that are homologous to an inserted nucleic acid sequences. In the second approach, the recombinant vector/host system can be identified and selected based upon the presence or absence of certain "marker" gene functions (e.g., thymidine kinase activity, resistance to antibiotics, transformation phenotype, occlusion body formation in baculovirus, etc.) caused by the insertion of foreign nucleic acid sequences in the vector. For example, if an *efl* nucleic acid sequence is inserted within the marker gene sequence of the vector, recombinants containing the insert can be identified by the absence of the marker gene function. In the third approach, recombinant expression vectors can be identified by assaying the foreign nucleic acid product expressed by the recombinant constructs. Such assays can be based, for example, on the physical or functional properties of the nucleic acid product of interest, for example, by binding of a ligand to a receptor or portion thereof which may be tagged with, for example, a detectable antibody or portion thereof or binding to antibodies produced against the protein of interest or a portion thereof.

The polypeptide, in particular modified of the present invention, may be expressed in the host cells transiently, constitutively or permanently.

Effective doses useful for treating the diseases or disorders indicated in the present application may be determined using methods known to one skilled in the art (see, for example, Fingl, et al., *The Pharmacological Basis of Therapeutics*, Goodman and Gilman, eds. Macmillan Publishing Co, New York, pp. 1-46 (1975). Pharmaceutical compositions for use according to the invention include the polypeptides described above in a pharmacologically acceptable liquid, solid or semi-solid carrier, linked to a carrier or targeting molecule (e.g., antibody, hormone, growth factor, etc.) and/or incorporated into liposomes, microcapsules, and controlled release preparation prior to administration in vivo. For example, the pharmaceutical composition may comprise a polypeptide in an aqueous solution, such as sterile water, saline, phosphate buffer or dextrose solution. Alternatively, the active agents may be comprised in a solid (e.g. wax) or semi-solid (e.g. gelatinous) formulation that may be implanted into a patient in need of such treatment. The administration route may be any mode of administration known in the art, including but not limited to intravenously, intrathecally, subcutaneously, intrauterinely, by injection into involved tissue, intraarterially, intranasally, orally, or via an implanted device.

Administration may result in the distribution of the active agent of the invention throughout the body or in a localized area. For example, in some conditions, which involve distant regions of the nervous system, intravenous or intrathecal administration of agent may be desirable. In some situations, an implant containing active agent may be placed in or near the lesioned area. Suitable implants include, but are not limited to, gelfoam, wax, spray, or microparticle-based implants.

The present invention also provides for pharmaceutical compositions comprising the polypeptides described herein, in a pharmacologically acceptable vehicle. The compositions may be administered systemically or locally. Any appropriate mode of administration known in the art may be

used, including, but not limited to, intravenous, intrathecal, intraarterial, intranasal, oral, subcutaneous, intraperitoneal, or by local injection or surgical implant. Sustained release formulations are also provided for.

Gene Therapy

Gene therapy refers to therapy performed by the administration to a subject of an expressed or expressible nucleic acid. In this embodiment of the invention, the nucleic acids produce their encoded protein that mediates a therapeutic effect.

Any of the methods for gene therapy available in the art can be used according to the present invention. Exemplary methods are described below.

For general reviews of the methods of gene therapy, see Goldspiel et al., *Clinical Pharmacy* 12:488-505 (1993); Wu and Wu, *Biotherapy* 3:87-95 (1991); Tolstoshev, *Ann. Rev. Pharmacol. Toxicol.* 32:573-596 (1993); Mulligan, *Science* 260:926-932 (1993); and Morgan and Anderson, *Ann. Rev. Biochem.* 62:191-217 (1993); May, *TIBTECH* 11(5):155-215 (1993). Methods commonly known in the art of recombinant DNA technology which can be used are described in Ausubel et al. (eds.), *Current Protocols in Molecular Biology*, John Wiley & Sons, NY (1993); and Kriegler, *Gene Transfer and Expression, A Laboratory Manual*, Stockton Press, NY (1990).

Delivery of the nucleic acids into a patient may be either direct, in which case the patient is directly exposed to the nucleic acid or nucleic acid-carrying vectors, or indirect, in which case, cells are first transformed with the nucleic acids in vitro, then transplanted into the patient. These two approaches are known, respectively, as in vivo or ex vivo gene therapy.

In a specific embodiment, the nucleic acid sequences are directly administered in vivo, where it is expressed to produce the encoded product. This can be accomplished by any of numerous methods known in the art, e.g., by constructing them as part of an appropriate nucleic acid expression vector and administering it so that they become intracellular, e.g., by infection using defective or attenuated retrovirals or other viral vectors, or by direct injection of naked DNA, or coating with lipids or cell-surface receptors or transfecting agents, encapsulation in liposomes, microparticles, or microcapsules, or by administering them in linkage to a peptide which is known to enter the nucleus, by administering it in linkage to a ligand subject to receptor-mediated endocytosis (see, e.g., Wu and Wu, *J. Biol. Chem.* 262:4429-4432 (1987)) (which can be used to target cell types specifically expressing the receptors) and so on. In another embodiment, nucleic acid-ligand complexes can be formed in which the ligand comprises a fusogenic viral peptide to disrupt endosomes, allowing the nucleic acid to avoid lysosomal degradation. In yet another embodiment, the nucleic acid can be targeted in vivo for cell specific uptake and expression, by targeting a specific receptor. Alternatively, the nucleic acid can be introduced intracellularly and incorporated within host cell DNA for expression, by homologous recombination (Koller and Smithies, *Proc. Natl. Acad. Sci. USA* 86:8932-8935 (1989); Zijlstra et al., *Nature* 342:435-438 (1989)).

In a specific embodiment, viral vectors that contain nucleic acid sequences encoding the polypeptide are used. The nucleic acid sequences encoding the polypeptide to be used in gene therapy are cloned into one or more vectors, which facilitates delivery of the gene into a patient. Lentiviral vectors, such as retroviral vectors, and other vectors such as adenoviral vectors and adeno-associated viruses are examples of viral vectors that may be used. Retroviral

vectors contain the components necessary for the correct packaging of the viral genome and integration into the host cell DNA.

Adenoviruses are especially attractive vehicles for delivering genes to respiratory epithelia because they naturally infect respiratory epithelia where they cause a mild disease. Other targets for adenovirus-based delivery systems are liver, the central nervous system, endothelial cells, and muscle. Adenoviruses have the advantage of being capable of infecting non-dividing cells. In addition, adeno-associated virus (AAV) has also been proposed for use in gene therapy.

Another approach to gene therapy involves transferring a gene to cells in tissue culture by such methods as electroporation, lipofection, calcium phosphate mediated transfection, or viral infection. Usually, the method of transfer includes the transfer of a selectable marker to the cells. The cells are then placed under selection to isolate those cells that have taken up and are expressing the transferred gene. Those cells are then delivered to a patient.

In this embodiment, the nucleic acid is introduced into a cell prior to administration in vivo of the resulting recombinant cell. Such introduction can be carried out by any method known in the art, including but not limited to transfection, electroporation, microinjection, infection with a viral or bacteriophage vector containing the nucleic acid sequences, cell fusion, chromosome-mediated gene transfer, microcell-mediated gene transfer, spheroplast fusion and so on. Numerous techniques are known in the art for the introduction of foreign genes into cells and may be used in accordance with the present invention, provided that the necessary developmental and physiological functions of the recipient cells are not disrupted. The technique should provide for the stable transfer of the nucleic acid to the cell, so that the nucleic acid is expressible by the cell and preferably heritable and expressible by its cell progeny.

Cells into which a nucleic acid can be introduced for purposes of gene therapy encompass any desired, available cell type, and include but are not limited to epithelial cells, endothelial cells, keratinocytes, fibroblasts, muscle cells, hepatocytes: blood cells such as T-lymphocytes. B-lymphocytes, monocytes, macrophages, neutrophils, eosinophils, megakaryocytes, granulocytes: various stem or progenitor cells, in particular hematopoietic stem or progenitor cells, e.g., as obtained from bone marrow, umbilical cord blood, peripheral blood, fetal liver, and so on.

In a preferred embodiment, the cell used for gene therapy is autologous to the patient.

In an embodiment in which recombinant cells are used in gene therapy, nucleic acid sequences encoding the polypeptide are introduced into the cells such that they are expressible by the cells or their progeny, and the recombinant cells are then administered in vivo for therapeutic effect. In a specific embodiment, stem or progenitor cells are used. Any stem and/or progenitor cells which can be isolated and maintained in vitro can potentially be used in accordance with this embodiment of the present invention.

In a specific embodiment, the nucleic acid to be introduced for purposes of gene therapy comprises an inducible promoter operably linked to the coding region, such that expression of the nucleic acid is controllable by controlling the presence or absence of the appropriate inducer of transcription.

Therapeutic Composition

The formulation of therapeutic compounds is generally known in the art and reference can conveniently be made to Remington's Pharmaceutical Sciences. 17th ed., Mack Pub-

lishing Co., Easton, Pa., USA. For example, from about 0.05 ng to about 20 mg per kilogram of body weight per day may be administered. Dosage regime may be adjusted to provide the optimum therapeutic response. For example, several divided doses may be administered daily or the dose may be proportionally reduced as indicated by the exigencies of the therapeutic situation. The active compound may be administered in a convenient manner such as by the oral, intravenous (where water soluble), intramuscular, subcutaneous, intra nasal, intra ocular, intradermal or suppository routes or implanting (eg using slow release molecules by the intraperitoneal route or by using cells e.g. monocytes or dendrite cells sensitized in vitro and adoptively transferred to the recipient). Depending on the route of administration, the peptide may be required to be coated in a material to protect it from the action of enzymes, acids and other natural conditions which may inactivate said ingredients.

For example, the low lipophilicity of the peptides will allow them to be destroyed in the gastrointestinal tract by enzymes capable of cleaving peptide bonds and in the stomach by acid hydrolysis. In order to administer peptides by other than parenteral administration, they will be coated by, or administered with, a material to prevent its inactivation. For example, peptides may be administered in an adjuvant, co-administered with enzyme inhibitors or in liposomes. Adjuvants contemplated herein include resorcinols, non-ionic surfactants such as polyoxyethylene oleyl ether and n-hexadecyl polyethylene ether. Enzyme inhibitors include pancreatic trypsin inhibitor, diisopropylfluorophosphate (DEP) and trasylol. Liposomes include water-in-oil-in-water CGF emulsions as well as conventional liposomes.

The active compounds may also be administered parenterally or intraperitoneally. Dispersions can also be prepared in glycerol liquid polyethylene glycols, and mixtures thereof and in oils. Under ordinary conditions of storage and use, these preparations contain a preservative to prevent the growth of microorganisms.

The pharmaceutical forms suitable for injectable use include sterile aqueous solutions (where water soluble) or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersion. In all cases the form must be sterile and must be fluid to the extent that easy syringability exists. It must be stable under the conditions of manufacture and storage and must be preserved against the contaminating action of microorganisms such as bacteria and fungi. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol and liquid polyethylene glycol, and the like), suitable mixtures thereof, and vegetable oils. The proper fluidity can be maintained, for example, by the use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. The prevention of the action of microorganisms can be brought about by various antibacterial and antifungal agents, for example, chlorobutanol, phenol, sorbic acid, thiomersal and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars or sodium chloride. Prolonged absorption of the injectable compositions can be brought about by the use in the composition of agents delaying absorption, for example, aluminium monostearate and gelatin.

Sterile injectable solutions are prepared by incorporating the active compounds in the required amount in the appropriate solvent with various other ingredients enumerated above, as required, followed by filtered sterilization. Generally, dispersions are prepared by incorporating the various

sterile active ingredient into a sterile vehicle which contains the basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, the preferred methods of preparation are vacuum drying and the freeze-drying technique which yield a powder of the active ingredient plus any additional desired ingredient from a previously sterile-filtered solution thereof.

When the peptides are suitably protected as described above, the active compound may be orally administered, for example, with an inert diluent or with an assimilable edible carrier, or it may be enclosed in hard or soft shell gelatin capsule, or it may be compressed into tablets, or it may be incorporated directly with the food of the diet. For oral therapeutic administration, the active compound may be incorporated with excipients and used in the form of ingestible tablets, buccal tablets, troches, capsules, elixirs, suspensions, syrups, wafers, and the like. Such compositions and preparations should contain at least 1% by weight of active compound. The percentage of the compositions and preparations may, of course, be varied and may conveniently be between about 5 to about 80% of the weight of the unit. The amount of active compound in such therapeutically useful compositions is such that a suitable dosage will be obtained. Preferred compositions or preparations according to the present invention are prepared so that an oral dosage unit form contains between about 0.1 μg and 2000 mg of active compound.

The tablets, pills, capsules and the like may also contain the following: A binder such as gum tragacanth, acacia, corn starch or gelatin; excipients such as dicalcium phosphate; a disintegrating agent such as corn starch, potato starch, alginic acid and the like; a lubricant such as magnesium stearate; and a sweetening agent such as sucrose, lactose or saccharin may be added or a flavoring agent such as peppermint, oil of wintergreen, or cherry flavoring. When the dosage unit form is a capsule, it may contain, in addition to materials of the above type, a liquid carrier. Various other materials may be present as coatings or to otherwise modify the physical form of the dosage unit. For instance, tablets, pills, or capsules may be coated with shellac, sugar or both. A syrup or elixir may contain the active compound, sucrose as a sweetening agent, methyl and propylparabens as preservatives, a dye and flavoring such as cherry or orange flavor. Of course, any material used in preparing any dosage unit form should be pharmaceutically pure and substantially non-toxic in the amounts employed. In addition, the active compound may be incorporated into sustained-release preparations and formulations.

Delivery Systems

Various delivery systems are known and can be used to administer a compound of the invention, e.g., encapsulation in liposomes, microparticles, microcapsules, recombinant cells capable of expressing the compound, receptor-mediated endocytosis, construction of a nucleic acid as part of a retroviral or other vector, etc. Methods of introduction include but are not limited to intradermal, intramuscular, intraperitoneal, intravenous, subcutaneous, intranasal, intra ocular, epidural, and oral routes. The compounds or compositions may be administered by any convenient route, for example by infusion or bolus injection, by absorption through epithelial or mucocutaneous linings (e.g., oral mucosa, rectal and intestinal mucosa, etc.) and may be administered together with other biologically active agents. Administration can be systemic or local. In addition, it may be desirable to introduce the pharmaceutical compounds or compositions of the invention into the central nervous

system by any suitable route, including intraventricular and intrathecal injection; intraventricular injection may be facilitated by an intraventricular catheter, for example, attached to a reservoir, such as an Ommaya reservoir. Pulmonary administration can also be employed, e.g., by use of an inhaler or nebulizer, and formulation with an aerosolizing agent.

In a specific embodiment, it may be desirable to administer the pharmaceutical compounds or compositions of the invention locally to the area in need of treatment; this may be achieved by, for example, and not by way of limitation, local infusion during surgery, topical application, e.g., in conjunction with a wound dressing after surgery, by injection, by means of a catheter, by means of a suppository, or by means of an implant, said implant being of a porous, non-porous, or gelatinous material, including membranes, such as sialastic membranes, or fibers. Preferably, when administering a protein, including an antibody or a peptide of the invention, care must be taken to use materials to which the protein does not absorb. In another embodiment, the compound or composition can be delivered in a vesicle, in particular a liposome. In yet another embodiment, the compound or composition can be delivered in a controlled release system. In one embodiment, a pump may be used. In another embodiment, polymeric materials can be used. In yet another embodiment, a controlled release system can be placed in proximity of the therapeutic target, thus requiring only a fraction of the systemic dose.

Sequence Listing Free Text

As regards the use of nucleotide symbols other than a, g, c, t, they follow the convention set forth in WIPO Standard ST.25, Appendix 2, Table 1, wherein k represents t or g; n represents a, c, t or g; m represents a or c; r represents a or g; s represents c or g; w represents a or t and y represents c or t.

[illegible]

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APDTRPAPGS TAPPAHGVTG APDTRPAPGS TAPPAHGVTG
APDTRPAPGS TAPPAHGVTG APDTRPAPGS TAPPAHGVTG
APDTRPAPGS TAPPAHGVTG APDTRPAPGS TAPPAHGVTG
APDTRPAPGS TAPPAHGVTG APDTRPAPGS TAPPAHGVTG
APDTRPAPGS TAPPAHGVTG APDNRPALGS TAPPVHNVTG
ASGSASGSAS TLVHNGTSAR ATTPASKST PFSIPSHHSD
TPTTLASHST KTDASSTHHS SVPLLTSSNH STSPQLSTGV
SFFFLSPHIS NLQFNSSLED PSTDYYQELQ RDISEMFLQI
YKQGGLGLS NIKFRPGSVV VQLTLAFREG TINVHDVETQ
FNQYKTEAAS RYNLTISDVS VSDVPPFSA QSGAGVPGWG
IALLVLVCVL VALAIVYLIA LAVCQCRKN YGQLDIFPAR
DTYHPMEYPT TYHTHGRYVP PSSTDSPYE KVSAGNGGSS

LSYTNPAVAA ASANL
describes full-length MUC1 Receptor (Mucin 1
precursor, Genbank Accession number: P15941).

MTPGTQSPFFLLLLLTVLT (SEQ ID NO: 1)

MTPGTQSPFFLLLLLTVLT VVTA (SEQ ID NO: 2)

MTPGTQSPFFLLLLLTVLT VVTG (SEQ ID NO: 3)

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SEQ ID NOS: 2, 3 and 4 describe N-terminal MUC-1 signaling sequence for directing MUC1 receptor and truncated isoforms to cell membrane surface. Up to 3 amino acid

describes MUC1 cytoplasmic domain nucleotide sequence

tgtcagtgccgccgaaagaactacgggcagctggacatctttccagcccgggatacctaccatcctatgagcgaata

ccccacctaccacacccatgqgcctatgtgccccctagcagtaccgatcgtagcccctatgagaaggatttctgcaggtaacggtagc

aqcagcctctcttacacaaacccagcagtgqgcagccgcttctqccaacttg.

describes MUC1 cytoplasmic domain amino acid sequence

CQCRKNYGQLDIFPARDTYHPMSEYPTYHTHGRYVPPSSSTRSPYEKVS

AGNGGSSLSYTNPAVAAASANL

describes NME7 nucleotide sequence (NME7: GENBANK ACCESSION AB209049)

qagatcctgagacaatqaatcatagtqaaaqattcgttttcattgcagagtggatatgatccaaatgcttcacttcttcgac

ggttatgaagctttttattttaccacaggggatggatctgttgaaatgcatgatgtaaaagaatcatcgacaccttttttaaagcggaccacaaatatgata

acctgcacttggaagattttattttataggcaacaaaagtgaatgtcttttctcgacaactggtattaattgactatggggatcaatatagaagctc

gccagctgggcagtaggaaagaaaaaacgctagccctaattaaaccagatgcaatatcaaaggctggagaaataattgaaataataa

acaaaagctggattttactataaccaaaactcaaaatgatgatgctttcaaggaaaagaagcattggattttcatgtagatcaccagtcgaagacc

ctttttcaatgagctgatccagttttattacaactggtcctattattgccatggagattttaagagatgatgctatatgtgaatggaaaaagactg

ctgggacctgcaaaactctggagtgccacgcacagatgcttctgaaagcattagagccctctttggaacagatggcataagaaatgcag

cgcctggccctgattcttttgccttctgcggccagagaaatggagttgttttttctttcaagtggaggttgtggccggcaaacactgctaa

atttactaattgtacctgttgacattgtttaaaccccatgctgtcagtgaaaggtatgttgaaatacactatattcagtacattttgttaaataggagag

caatggtttattttcttgatgtactttatgtatagaaaataa.

residues may be absent at C-terminal end as indicated by variants in SEO ID NOS: 2, 3 and 4.

GTINVHDTVETOENQYKTEAAS-

RYNLTISDVSVSDVPFPFSAQSGAGVPGW GIAL-
5 VLVCVLVALAIVYLIALA-
VCQCRKKNYGQLDIFPARDTYHPMSEYPTYHTHG
RYVPPSSTDRSPYEKVSAGNGGSSLSYTN-
PAVAAASANL (SEQ ID NO:5) describes a truncated
MUC1 receptor isoform having nat-PSMGFR at its N-ter-
minus and including the transmembrane and cytoplasmic
10 sequences of a full-length MUC1 receptor.

GTINVHDTVETOFNOYKTEAAS-

RYNLTISDVSVDVPFSAQSGA (SEQ ID NO: 6)
describes the extracellular domain of Native Primary
Sequence of the MUC1 Growth Factor Receptor (nat-
PSMGR—an example of “PSMGR”);

TINVHDTVETOFNOYKTEAAS-

RYNLTISDSVSDVPFPFSAQSGA (SEQ ID NO: 7) describes the extracellular domain of Native Primary Sequence of the MUC1 Growth Factor Receptor (nat-PSMGFR—An example of “PSMGFR”), having a single amino acid deletion at the N-terminus of SEQ ID NO:6).

GTINVHDTVETOFNOYK-

25 TEAASPYNLTISDVSVSDVPFPFSAQSGA (SEQ ID NO:
8) describes the extracellular domain of “SPY” functional
variant of the native Primary Sequence of the MUC1
Growth Factor Receptor having enhanced stability (var-
PSMGFR—An example of “PSMGFR”).

30 TINVHDTVETOYNOK-

TEAASPYNLTISQDVSVSDVPFPFSAQSGA (SEQ ID NO: 9) describes the extracellular domain of “SPY” functional variant of the native Primary Sequence of the MUC1 Growth Factor Receptor having enhanced stability (var-PSMGFR—An example of “PSMGFR”), having a single amino acid deletion at the C-terminus of SEQ ID NO: 8).

-continued

describes NME7 amino acid sequence (NME7: GENBANK ACCESSION AB209049)

(SEQ ID NO: 13)

DPETMNHSERFVFIAEWYDPNASLLRRYELLFYPGDGSEMHDKVKNHRT
 FLKRTKYDNLHLEDLFIGNKVNVFSRQLVLIDYGDQYTARQLGSRKEKTLALIKPDAI
 SKAGEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQSRPFFNELIQFITTGPPIAMEIL
 RDDAICEWKRLGPPANSVARTDASESIRALFGTDGIRNAHGPDSFASAAAREMELF
 FPSSGGCGPANTAKFTNCTCCIVKPHAVSEGMLNTLYSVHFVNRRAMFIFLMYFMY
 RK.

describes NM23-H1 nucleotide sequence (NM23-H1: GENBANK
 ACCESSION AF487339)

(SEQ ID NO: 14)

atggtgctactgtctactttagggatcgctctttcaaggcgaggggcctcctatctcaagctgtgatacaggaaccatggc
 caactgtgagcgtacotctcattgcgatcaaaccagatgggggtccagcggggtcttgtgggagagattatcaagcgttttgagcagaaaag
 gattccgccttgttggtctgaaattcatgcaagcttccgaagatcttctcaaggaacactacgttgacctgaaggaccgtccattctttggc
 ggccctggtgaaatacatgcactcagggccggtagttgccatggtctgggaggggctgaatgtggtgaagacgggcccagtcagtc
 ggggagaccaaccctgcagactccaagcctgggaccatccgtggagacttctgcatacaagttggcaggaacattatacatggcagtc
 gattctgtggagagtcagagaaggagatcggtctgtggtttcaccctgaggaactggtagattacacgagctgtgctcagaactggat
 ctatgaatga.

NM23-H1 describes amino acid sequence (NM23-H1:
 GENBANK ACCESSION AF487339)

(SEQ ID NO: 15)

MVLLSTLGIVFQGEPPISCDTGTMANCERTFIAIKPDGVQRGLVGEI IKR
 FEQKGFRVLVGLKFMQASEDLLKEHYVDLKDPPFFAGLVKYMHS GPVVAMVWEG
 NVVKTGRVMLGETNPADSKPGTIRGDFCIQVGRNIIHGSDSVESAEKEIGLWFHPPEEL
 VDYTSQAQNWIIYE.

describes NM23-H1 S120G mutant nucleotide sequence (NM23-
 H1: GENBANK ACCESSION AF487339)

(SEQ ID NO: 16)

atggtgctactgtctactttagggatcgctctttcaaggcgaggggcctcctatctcaagctgtgatacaggaaccatggc
 caactgtgagcgtacotctcattgcgatcaaaccagatgggggtccagcggggtcttgtgggagagattatcaagcgttttgagcagaaaag
 gattccgccttgttggtctgaaattcatgcaagcttccgaagatcttctcaaggaacactacgttgacctgaaggaccgtccattctttggc
 ggccctggtgaaatacatgcactcagggccggtagttgccatggtctgggaggggctgaatgtggtgaagacgggcccagtcagtc
 ggggagaccaaccctgcagactccaagcctgggaccatccgtggagacttctgcatacaagttggcaggaacattatacatggcgggt
 gattctgtggagagtcagagaaggagatcggtctgtggtttcaccctgaggaactggtagattacacgagctgtgctcagaactggat
 ctatgaatga.

describes NM23-H1 S120G mutant amino acid
 sequence (NM23-H1: GENBANK ACCESSION AF487339)

(SEQ ID NO: 17)

MVLLSTLGIVFQGEPPISCDTGTMANCERTFIAIKPDGVQRGLVGEI IKR
 FEQKGFRVLVGLKFMQASEDLLKEHYVDLKDPPFFAGLVKYMHS GPVVAMVWEG
 NVVKTGRVMLGETNPADSKPGTIRGDFCIQVGRNIIHGSDSVESAEKEIGLWFHPPEEL
 VDYTSQAQNWIIYE.

describes NM23-H2 nucleotide sequence
 (NM23-H2: GENBANK ACCESSION AK313448)

(SEQ ID NO: 18)

atggccaacctggagcgcaccttcatcgccatcaagccggacggcgtgcagcgcggcctggtgggcgagatcatc
 aagcgcttcgagcagaaggattccgcctcgtggccatgaagttcctccgggctctgaagaacacctgaagcagcactacattgac
 ctgaaagaccgaccattcttccctgggctggtgaagtacatgaactcagggccggttgtggccatggtctgggaggggctgaacgtg
 gtgaagacagggccgagtgatgcttggggagaccaatccagcagattcaaagccagggcaccattcgtggggacttctgcattcaggtt

-continued

ggcaggaacatcattcatggcagtgattcagtaaaaagtgctgaaaaagaatcagcctatggtttaagcctgaagaactgggttgacta
caagtcttgctcatgactgggtctatgaataa.

describes NM23-H2 amino acid sequence (NM23-H2: GENBANK ACCESSION AK313448)

(SEQ ID NO: 19)

MANLERTFIAIKPDGVQRGLVGEIIRFEQKGFRLVAMKFLRASEEHLKQH

YIDLKDRPFFPGLVKYMNSGPVVMVWEGLNVVKTGRVMLGETNPADSKPGTIRG

DFCIQVGRNI IHGSDSVKSAEKEISLWFKPEELVDYKSCAHDWVYE.

Human NM23-H7-2 sequence optimized for *E. coli* expression:
(DNA)

(SEQ ID NO: 20)

atgcatgacgttaaaaatcacccgtacctttctgaaacgcacgaaatgatgaatctgcatctggaagacctgtttattggc
aacaaagtcaatgtgttctctcgctcagctgggtgctgatcgattatggcgaccagtaaccgcgcgtcaactgggtagtcgcaaagaaa
aacgctggccctgattaaacccggatgcaatctccaaagctggcgaaattatcgaaattatcaacaaagcgggtttcaccatcacgaaac
tgaaaatgatgatgctgagccgtaagaagccctggattttcatgtcgaccaccagctctcgccgtttttcaatgaactgattcaattcatc
accacgggtccgattatcgcaatggaaattctgcgtgatgacgctatctgcgaatggaaacgcctgctggggcccgcaaactcaggtg
ttgcgcgtaccgatgccagtgaaatccattcgctctgtttggcacccatgggtatccgtaatgcagcacatggtcggactcattcgcat
cggcagctcgtgaaatggaaactgtttttcccgagctctggcggttgcgggtccggcaaacaccgccaaatttaccattgtacgtgctgta
ttgtcaaaccgcacgcagtgtcagaaggcctgctgggtaaaattctgatggcaatccgtgatgctggctttgaaatctcgccatgcag
atgttcaacatggacgcgttaacgtcgaagaattctacgaagtttacaaggcgtggttaccgaatatcacgatatggttacggaaatg
tactccggtccgtgctgcgcatggaaattcagcaaaacaatgccacaaaacgtttcgtgaattctgtggtcggcgagatccggaaat
cgcacgtcatctgcgtccgggtacccctgcgcgaatttttggtaaaacgaaatccagaacgctgtgcaactgtaccgatctgccggaa
gacgggtctgctggaagtccaatactttttcaaaattctggataattga

(amino acids)

(SEQ ID NO: 21)

MHDVKNHRTFLKRTKYDNLHLEDLFIGNKVNVFSRQLVLIDYGDQYTAR

QLGSRKEKTLALIKPDAISKAGEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQSRP

FFNELIQFITTGPIIAMEILRDDAICEWKRLGPANSGVARTDASESIRALFGTDGIRNA

AHGPDSFASAAREMELFFPSSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILMAIR

DAGFEISAMQMFNMDRVNVEEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQNN

ATKTFREFCGPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLLEVQYFFKILD

N-

Human NME7-A:
(DNA)

(SEQ ID NO: 22)

atggaaaaaacgctagccctaattaaaccagatgcaatatcaaaggctggagaaataattgaaataataaacaagct
ggatttactataacaaactcaaaatgatgatgctttcaaggaaagaagcattggattttcatgtagatcaccagtcagaccctttttcaat
gagctgatccagtttattacaactgggtcctattattgccatggagattttaagagatgatgctatatgtgaatggaaaagactgctgggacc
tgcaaaactctggagtggcacgcacagatgcttctgaaagcattagagccctctttggaacagatggcataagaaatgcagcgcgatggc
cctgattcttttgccttctgcggccagagaaatggagttgttttttga

(amino acids)

(SEQ ID NO: 23)

MEKTLALIKPDAISKAGEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQSR

RPFFNELIQFITTGPIIAMEILRDDAICEWKRLGPANSGVARTDASESIRALFGTDGIR

NAAHGPDSFASAAREMELFF-

-continued

Human NME7-A1:
(DNA)

(SEQ ID NO: 24)

atggaaaaaacgctagccctaattaaaccagatgcaatatcaaaggctggagaaataattgaaataataaacaagct
ggatttactataaccaaaccaaaatgatgatgctttcaaggaaagaagcattggattttcatgtagatcaccagtcagaccctttttcaat
gagctgatccagtttattacaactggctctattattgccatggagattttaagagatgatgctatatgtgaatggaaaagactgctgggacc
tgcaaacctctggagtggcacgcacagatgcttctgaaagcattagagccctctttggaacagatggcataagaaatgcagcgcatggc
cctgattcttttgccttctgcggccagagaaatggagttgtttttccttcaagtgagggtgtgggcccggcaaacactgctaaatttacttga

(amino acids)

(SEQ ID NO: 25)

MEKTLALIKPDAISKAGEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQS
RPFNFELIQFITTGPIIAMEILRDDAICEWKRLGPPANSVARTDASESIRALFGTDGIR
NAAHGPDSPASAAAREMELFFPSSGGCGPANTAKFT-

Human NME7-A2:
(DNA)

(SEQ ID NO: 26)

atgaatcatagtgaaagattcgttttcattgcagagtggatgatccaaatgcttcacttcttcgacgttatgagcttttatttt
accaggggatggatctgttgaaatgcagatgtaagaatcatcgccctttttaagcggaccaaataatgataacctgcacttggaag
atttatttataggcaacaaagtgaatgtcttttctcgacaactggatttaattgactatggggatcaataacagctcgccagctgggcagta
ggaaagaaaaaacgctagccctaattaaaccagatgcaatatcaaaggctggagaaataattgaaataataaacaagctggatttact
ataaccaaactcaaaatgatgatgctttcaaggaaagaagcattggattttcatgtagatcaccagtcagaccctttttcaatgagctgat
ccagtttattacaactggctctattattgccatggagattttaagagatgatgctatatgtgaatggaaaagactgctgggacctgcaaac
ctggagtggcacgcacagatgcttctgaaagcattagagccctctttggaacagatggcataagaaatgcagcgcacggccctgattct
tttgcttctgcggccagagaaatggagttgttttttga

(amino acids)

(SEQ ID NO: 27)

MNHSERFVPIAEWYDPNASLLRRYELLFYPGDGSVEMHDVKNHRTFLKR
TKYDNLHLEDLFIGNKVNFSRQLVLIDYGDQYTARQLGSRKEKTLALIKPDAISKA
GEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQS RPFNFELIQFITTGPIIAMEILRDD
AICEWKRLGPPANSVARTDASESIRALFGTDGIRNAAHGPDSPASAAAREMELFF-

Human NME7-A3:
(DNA)

(SEQ ID NO: 28)

atgaatcatagtgaaagattcgttttcattgcagagtggatgatccaaatgcttcacttcttcgacgttatgagcttttatttt
accaggggatggatctgttgaaatgcagatgtaagaatcatcgccctttttaagcggaccaaataatgataacctgcacttggaag
atttatttataggcaacaaagtgaatgtcttttctcgacaactggatttaattgactatggggatcaataacagctcgccagctgggcagta
ggaaagaaaaaacgctagccctaattaaaccagatgcaatatcaaaggctggagaaataattgaaataataaacaagctggatttact
ataaccaaactcaaaatgatgatgctttcaaggaaagaagcattggattttcatgtagatcaccagtcagaccctttttcaatgagctgat
ccagtttattacaactggctctattattgccatggagattttaagagatgatgctatatgtgaatggaaaagactgctgggacctgcaaac
ctggagtggcacgcacagatgcttctgaaagcattagagccctctttggaacagatggcataagaaatgcagcgcacggccctgattct
tttgcttctgcggccagagaaatggagttgtttttccttcaagtgagggtgtgggcccggcaaacactgctaaatttacttga

(amino acids)

(SEQ ID NO: 29)

MNHSERFVPIAEWYDPNASLLRRYELLFYPGDGSVEMHDVKNHRTFLKR
TKYDNLHLEDLFIGNKVNFSRQLVLIDYGDQYTARQLGSRKEKTLALIKPDAISKA
GEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQS RPFNFELIQFITTGPIIAMEILRDD
AICEWKRLGPPANSVARTDASESIRALFGTDGIRNAAHGPDSPASAAAREMELFFPSS
GGCGPANTAKFT-

-continued

Human NME7-B:
(DNA)

(SEQ ID NO: 30)

atgaattgtacctgttgcatgtttaaccccatgctgtcagtgaggactgttgggaaagatcctgatggctatccgaga
tgcaggttttgaaatctcagctatgcagatgttcaatatggatcgggttaatgttgaggaaattctatgaagtttataaaggagtagtgaccg
aatatcatgacatggtgacagaaatgtattctggcccttgtagcaatggagattcaacagaataatgctacaaagacatttcgagaattt
tgtggacctgctgacccgaaattgccccgcatttacgccctggaactctcagagcaatctttggtaaaactaagatccagaatgctgttc
actgtactgatctgccagaggatggcctattagagggttcaatacttcttctga

(amino acids)

(SEQ ID NO: 31)

MNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFNMDRVNVEEFY
EVYKGVVTEYHDMVTEMYSGPCVAMEIQNNATKTFREFCGPADPEIARHLRPGTL
RAIFGKTKIQNAVHCTDLPEDGLELVQYFF-

Human NME7-B1:
(DNA)

(SEQ ID NO: 32)

atgaattgtacctgttgcatgtttaaccccatgctgtcagtgaggactgttgggaaagatcctgatggctatccgaga
tgcaggttttgaaatctcagctatgcagatgttcaatatggatcgggttaatgttgaggaaattctatgaagtttataaaggagtagtgaccg
aatatcatgacatggtgacagaaatgtattctggcccttgtagcaatggagattcaacagaataatgctacaaagacatttcgagaattt
tgtggacctgctgacccgaaattgccccgcatttacgccctggaactctcagagcaatctttggtaaaactaagatccagaatgctgttc
actgtactgatctgccagaggatggcctattagagggttcaatacttcttcaagatcttggataattagtgga

(amino acids)

(SEQ ID NO: 33)

MNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFNMDRVNVEEFY
EVYKGVVTEYHDMVTEMYSGPCVAMEIQNNATKTFREFCGPADPEIARHLRPGTL
RAIFGKTKIQNAVHCTDLPEDGLELVQYFFKILDN-

Human NME7-B2:
(DNA)

(SEQ ID NO: 34)

atgccttcaagtgagggttggtggccggcaaacactgctaaatttactaattgtacctgttgcatgtttaaccccatgct
gtcagtgaggactgttgggaaagatcctgatggctatccgagatgcagggtttgaaatctcagctatgcagatgttcaatatggatcgg
gttaatgttgaggaaattctatgaagtttataaaggagtagtgaccgaatatcatgacatggtgacagaaatgtattctggcccttgtagtag
aatggagattcaacagaataatgctacaaagacatttcgagaattttgtggacctgctgacccgaaattgccccgcatttacgccctgga
actctcagagcaatctttggtaaaactaagatccagaatgctgttcaactgtactgatctgccagaggatggcctattagagggttcaatactt
cttctga

(amino acids)

(SEQ ID NO: 35)

MPSSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAM
QMFNMDRVNVEEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQNNATKTFREFC
GPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLELVQYFF-

Human NME7-B3:
(DNA)

(SEQ ID NO: 36)

atgccttcaagtgagggttggtggccggcaaacactgctaaatttactaattgtacctgttgcatgtttaaccccatgct
gtcagtgaggactgttgggaaagatcctgatggctatccgagatgcagggtttgaaatctcagctatgcagatgttcaatatggatcgg
gttaatgttgaggaaattctatgaagtttataaaggagtagtgaccgaatatcatgacatggtgacagaaatgtattctggcccttgtagtag

- continued

aatggagattcaacagaataatgctacaaagacatttcgagaattttgtggacctgctgacacctgaaattgccccgcatttacgccctgga
actctcagagcaatctttggtaaaactaagatccagaatgctgttcactgtactgactgctgccagaggatggcctattagagggtcaatactt
cttcaagatcttggataattagtgga

(amino acids)

(SEQ ID NO: 37)

MPSSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAM

QMFNMDRVNVEEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQQNNATKTFREFC

GPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLEVQYFFKILDN--

Human NME7_{AB}, also known as NME7_{AB}:
(DNA)

(SEQ ID NO: 38)

atggaaaaaacgctagccctaattaaaccagatgcaatatcaaaggctggagaaataattgaaataataaacaagct
ggatttactataaccaaactcaaaatgatgatgctttcaaggaaagaagcattggattttcatgtagatcaccagtcagaccctttttcaat
gagctgacccagtttattacaactggctcctattattgccatggagattttaagagatgatgctatatgtgaatggaaaagactgctgggacc
tgcaaactctggagtgccacgcacagatgcttctgaaagcattagagccctctttggaacagatggcataagaaatgcagcgcatggc
cctgattcttttctgcttctgcgccagagaaatggagttgttttctcctcaagtggaggtgtgggcccggcaaacactgctaaatttactaatt
gtacctgttgcatgtttaaaccccatgctgtcagtgaggactgttgggaaagatcctgatggctatccgagatgcaggttttgaaatctc
agctatgcagatgttcaatatggatcggtttaatgttgaggaaattctatgaagtttataaaggagtagtgaccgaatatcatgacatgggtga
cagaaatgtattctggcccttgtgtagcaatggagattcaacagaataatgctacaaagacatttcgagaattttgtggacctgctgatcct
gaaattgccccgcatattacgccttggaaactctcagagcaatcttggtaaaactaagatccagaatgctgttccactgtactgatctgccag
aggatggcctattagagggttcaatacttcttcaagatcttggataattagtgga

(amino acids)

(SEQ ID NO: 39)

MEKTLALIKPDAISKAGEIIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQ

RPFNFELIQFITTGPIIAMEILRDDAICEWKRLGPNASGVARTDASESIRALFGTDGIR

NAAHGPDSPASAAREMELFFPSSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILM

AIRDAGFEISAMQMFNMDRVNVEEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQQ

NNATKTFREFCGPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLEVQYFFK

ILDN--

Human NME7_{AB1}:
(DNA)

(SEQ ID NO: 40)

atggaaaaaacgctagccctaattaaaccagatgcaatatcaaaggctggagaaataattgaaataataaacaagct
ggatttactataaccaaactcaaaatgatgatgctttcaaggaaagaagcattggattttcatgtagatcaccagtcagaccctttttcaat
gagctgacccagtttattacaactggctcctattattgccatggagattttaagagatgatgctatatgtgaatggaaaagactgctgggacc
tgcaaactctggagtgccacgcacagatgcttctgaaagcattagagccctctttggaacagatggcataagaaatgcagcgcatggc
cctgattcttttctgcttctgcgccagagaaatggagttgttttctcctcaagtggaggtgtgggcccggcaaacactgctaaatttactaatt
gtacctgttgcatgtttaaaccccatgctgtcagtgaggactgttgggaaagatcctgatggctatccgagatgcaggttttgaaatctc
agctatgcagatgttcaatatggatcggtttaatgttgaggaaattctatgaagtttataaaggagtagtgaccgaatatcatgacatgggtga
cagaaatgtattctggcccttgtgtagcaatggagattcaacagaataatgctacaaagacatttcgagaattttgtggacctgctgatcct
gaaattgccccgcatattacgccttggaaactctcagagcaatcttggtaaaactaagatccagaatgctgttccactgtactgatctgccag
aggatggcctattagagggttcaatacttcttctga

(amino acids)

(SEQ ID NO: 41)

MEKTLALIKPDAISKAGEIIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQ

RPFNFELIQFITTGPIIAMEILRDDAICEWKRLGPNASGVARTDASESIRALFGTDGIR

NAAHGPDSPASAAREMELFFPSSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILM

-continued

AIRDAGFEISAMQMFNMDRVNVEEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQQ

NNATKTFREFCGPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLLEVQYFF-

Human NME7-A sequence optimized for *E. coli* expression:
(DNA)

atggaaaaaacgctggccctgattaaaccggatgcaatctccaaagctggcgaaattatcgaaattatcaacaaagcg

(SEQ ID NO: 42)

ggtttccaccatcacgaaactgaaaatgatgatgctgagccgtaaagaagccctggattttcatgtcgaccaccagtctcgccggtttttca

atgaactgattcaattcatcaccacgggtccgattatcgcaatggaaattctgctgatgacgctatctcgcaatggaaacgctgctgg

gcccggcaaaactcaggtgttgccgtaccgatgccagtgaatccattcgcgctctgtttggcaccgatggtatccgtaatgcagcacat

ggtccggactcattcgcatcggcagctcgtgaaatggaactgtttttctga

(amino acids)

(SEQ ID NO: 43)

MEKTLALIKPDAISKAGEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQ

RPFNFELIQFITTGPPIIAMEILRDDAICEWKRLGPNANSGVARTDASESIRALFGTDGIR

NAAHGPDSFASAAREMELFF-

Human NME7-A1 sequence optimized for *E. coli* expression:
(DNA)

atggaaaaaacgctggccctgattaaaccggatgcaatctccaaagctggcgaaattatcgaaattatcaacaaagcg

(SEQ ID NO: 44)

ggtttccaccatcacgaaactgaaaatgatgatgctgagccgtaaagaagccctggattttcatgtcgaccaccagtctcgccggtttttca

atgaactgattcaattcatcaccacgggtccgattatcgcaatggaaattctgctgatgacgctatctcgcaatggaaacgctgctgg

gcccggcaaaactcaggtgttgccgtaccgatgccagtgaatccattcgcgctctgtttggcaccgatggtatccgtaatgcagcacat

ggtccggactcattcgcatcggcagctcgtgaaatggaactgtttttcccgagctctggcggttcgggtccggcaaacaccgccaatt

tacctga

(amino acids)

(SEQ ID NO: 45)

MEKTLALIKPDAISKAGEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQ

RPFNFELIQFITTGPPIIAMEILRDDAICEWKRLGPNANSGVARTDASESIRALFGTDGIR

NAAHGPDSFASAAREMELFFPSSGGCPANTAKFT-

Human NME7-A2 sequence optimized for *E. coli* expression:
(DNA)

atgaatcactccgaacgctttgtttttatcgccgaatggtatgaccgaatgcttcctgctgcccgtacgaactgct

(SEQ ID NO: 46)

gttttatccggcgatggttagcgtggaatgcatgacgttaaaaatcaccttacctttctgaaacgcacgaaatgataatctgcatctg

gaagacctgtttattggcaacaaagtcaatgtgttctctctgctagctgggtgctgatcgattatggcgaccagtagacccgctgcaactg

ggtagtcgcaaagaaaaaacgctggccctgattaaaccggatgcaatctccaaagctggcgaaattatcgaaattatcaacaaagcgg

gtttccaccatcacgaaactgaaaatgatgatgctgagccgtaaagaagccctggattttcatgtcgaccaccagtctcgccggtttttcaa

tgaactgattcaattcatcaccacgggtccgattatcgcaatggaaattctgctgatgacgctatctcgcaatggaaacgctgctggg

cccggcaaaactcaggtgttgccgtaccgatgccagtgaatccattcgcgctctgtttggcaccgatggtatccgtaatgcagcacatg

gtccggactcattcgcatcggcagctcgtgaaatggaactgtttttctga

(amino acids)

(SEQ ID NO: 47)

MNHSEFVFPIAEWYDPNASLLRRYELLFYPGDSVEMHDVKNHRFTFLKR

TKYDNLHLEDLFIGNKVNFSRQLVLIDYGDQYTARQLGSRKEKTLALIKPDAISKA

GEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQSRPFNFELIQFITTGPPIIAMEILRDD

AICEWKRLGPNANSGVARTDASESIRALFGTDGIRNAAHGPDSFASAAREMELFF-

-continued

Human NME7-A3 sequence optimized for *E. coli* expression:
(DNA)

(SEQ ID NO: 48)

atgaatcactccgaacgctttgtttttatcgccgaatggtatgacccgaatgcttcctgctgcgccgctacgaactgct
gtttttatccggcgatggttagcgtggaatgcatgacgttaaaaatcacctacctttctgaaacgcacgaatatgataatctgcatctg
gaagacctgtttattggcaacaaagtcaatgtgttctctcgtagctggtgctgatcgattatggcgaccagtagacacgcgcgtcaactg
ggtagtcgcaaaagaaaaacgctggccctgattaaacggatgcaatctccaaagctggcgaaattatcgaaattatcaacaaagcgg
gtttcaccatcacgaaactgaaaatgatgatgctgagccgtaagaagccctggattttcatgtcgaccaccagtagctgccccgtttttcaa
tgaactgattcaattcatcaccacgggtccgattatcgcaatggaattctgctgatgacgctatctgcaatggaacgcctgctggg
cccgcaaaactcaggtgttgccgtagccgatgccagtgaaatccatcgcgctctgtttggcaccgatggtatccgtaatgcagcacatg
gtccggactcattcgcatcggcagctcgtgaaatggaactgttttcccgagctctggcggttgcggtccggcaaacaccgccaattt
acctga

(amino acids)

(SEQ ID NO: 49)

MNHSERFVFIAEWYDPNASLLRRYELLFYPGDGSVEMHDVKNHRTFLKR
TKYDNLHLEDLFIGNKVNVSRLVLIDYGDQYTRQLGSRKEKTLALIKPDAISKA
GEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQSRPFFNELIQFITTGPIIAMEILRDD
AICEWKRLGPPANSVARTDASESIRALFGTDGIRNAHGPDSFASAAREMELFFPSS
GGCGPANTAKFT-

Human NME7-B sequence optimized for *E. coli* expression:
(DNA)

(SEQ ID NO: 50)

atgaattgtacgtgctgtattgtcaaaccgcacgcagtgtcagaaggcctgctgggtaaaattctgatggcaatccgtg
atgctggctttgaaatctcgccatgcagatgttcaacatggaccgcgttaacgtcgaagaattctacgaagtttacaaaggcgtggtta
ccgaatatcacgatatggttacggaatgtactccgggtccgtgctgcgtaggaaattcagcaaaacaatgccacaaaacgtttcgt
gaattctgtggtccggcagatccggaaatcgacgtcatctgcgtccgggtaccctgcgcgcaatttttggtaaaacgaaaatccagaa
cgctgtgcactgtaccgatctgccgaagacggtctgctggaagtccaatactttttctga

(amino acids)

(SEQ ID NO: 51)

MNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFNMDRVNVEEFY
EYKGVVTEYHDMVTEMYSGPCVAMEIQNNATKTFREFCGPADPEIARHLRPGTL
RAIFGKTKIQNAVHCTDLPEGLLEVQYFF-

Human NME7-B1 sequence optimized for *E. coli* expression:
(DNA)

(SEQ ID NO: 52)

atgaattgtacgtgctgtattgtcaaaccgcacgcagtgtcagaaggcctgctgggtaaaattctgatggcaatccgtg
atgctggctttgaaatctcgccatgcagatgttcaacatggaccgcgttaacgtcgaagaattctacgaagtttacaaaggcgtggtta
ccgaatatcacgatatggttacggaatgtactccgggtccgtgctgcgtaggaaattcagcaaaacaatgccacaaaacgtttcgt
gaattctgtggtccggcagatccggaaatcgacgtcatctgcgtccgggtaccctgcgcgcaatttttggtaaaacgaaaatccagaa
cgctgtgcactgtaccgatctgccgaagacggtctgctggaagtccaatactttttcaaaattctggataattga

(amino acids)

(SEQ ID NO: 53)

MNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFNMDRVNVEEFY
EYKGVVTEYHDMVTEMYSGPCVAMEIQNNATKTFREFCGPADPEIARHLRPGTL
RAIFGKTKIQNAVHCTDLPEGLLEVQYFFKILDN-

Human NME7-B2 sequence optimized for *E. coli* expression :
(DNA)

(SEQ ID NO: 54)

atgccgagctctggcggttgcggtccggcaaacaccgcaaaattaccaattgtacgtgctgtattgtcaaaccgcac
gcagtgtcagaaggcctgctgggtaaaattctgatggcaatccgtgatgctggctttgaaatctcgccatgcagatgttcaacatggac

-continued

cgcgttaacgtcgaagaattctacgaagtttacaaggcgtggttacgaatatcacgatatggttacggaatgtactccggtccgtgc
gtcgcgatggaattcagcaaaacaatgccacaaaacgtttcgtgaattctgtggtccggcagatccggaatcgcacgtcatctgcg
tccgggtaccctgcgcgaatttttggtaaaacgaaatccagaacgctgtgcactgtaccgatctgccggaagacggtctgctggaa
gttcaataactttttctga

(amino acids)

(SEQ ID NO: 55)

MPSSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAM

QMFNMDRVNVEEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQNNATKTFREFC

GPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLEVQYFF-

Human NME7-B3 sequence optimized for *E. coli* expression:
(DNA)

(SEQ ID NO: 56)

atgccgagctctggcgttgcggtccggcaaacaccgcaaatattaccaattgtacgtgctgtattgtcaaaccgcac
gcagtgtcagaaggcctgctgggtaaaattctgatggcaatccgtgatgctggctttgaaatctcgccatgcagatgttcaacatggac
cgcgttaacgtcgaagaattctacgaagtttacaaggcgtggttacgaatatcacgatatggttacggaatgtactccggtccgtgc
gtcgcgatggaattcagcaaaacaatgccacaaaacgtttcgtgaattctgtggtccggcagatccggaatcgcacgtcatctgcg
tccgggtaccctgcgcgaatttttggtaaaacgaaatccagaacgctgtgcactgtaccgatctgccggaagacggtctgctggaa
gttcaataactttttcaaaattctggataattga

(amino acids)

(SEQ ID NO: 57)

MPSSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAM

QMFNMDRVNVEEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQNNATKTFREFC

GPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLEVQYFFKILDN-

Human NME7_{AB}, also known as NME7_{AB} sequence optimized for *E. coli*
expression:
(DNA)

(SEQ ID NO: 58)

atggaaaaaacgctggccctgattaaaccggatgcaatctccaaagctggcgaaattatcgaaattatcaacaaagcg
ggtttccaccatcacgaaactgaaatgatgatgctgagccgtaaagaagccctggattttcaatgtcgaccaccagtctcgccggttttca
atgaactgattcaattcatcaccacgggtccgattatcgcaatggaaattctgcgtgatgacgctatctgcgaatggaaacgctgctgg
gccccgcaaaactcaggtgttgcggtaccgatgccagtgaaatccattcgcgctctgtttggcaccgatggtatccgtaatgcagcacat
ggtccggactcattcgcatcggcagctcgtgaaatggaactgtttttcccgagctctggcgttgcggtccggcaaacaccgcaaat
taccaattgtacgtgctgtattgtcaaacccgcacgagtgtcagaaggcctgctgggtaaaattctgatggcaatccgtgatgctggctt
gaaatctcgccatgcagatgttcaacatggaccggttaacgtcgaagaattctacgaagtttacaaggcgtggttaccgaatatca
cgatatggttacggaatgtactccggtccgtgctgcgatggaaatcagcaaaacaatgccacaaaacgtttcgtgaattctgtgg
tccggcagatccggaatcgcacgtcatctgcgtccgggtaccctgcgcgaatttttggtaaaacgaaatccagaacgctgtgcact
gtaccgatctgccggaagacggtctgctggaagtcaataactttttcaaaattctggataattga

(amino acids)

(SEQ ID NO: 59)

MEKTLALIKPDAISKAGEIIEIINKAGFTITKLMMMLSRKEALDFVHDQS

RPPFNELIQFITTGPPIAMEILRDDAICEWKRLGPNANSGVARTDASESIRALFGTDGIR

NAAHGPDSPASAREMELFFPSSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILM

AIRDAGFEISAMQMFNMDRVNVEEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQQ

NNATKTFREFCGPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLEVQYFFK

ILDN-

Human NME7_{AB1}, also known as NME7_{AB1} sequence optimized for *E. coli* expression:
(DNA)

Atggaaaaaacgctggccctgattaaacccggatgcaatctccaagctggcgaaattatcgaaattatcaacaagc
gggtttcaccatcacgaaactgaaaatgatgatgctgagccgtaagaagccctggattttcatgtcgaccaccagctctcgccggttttc
aatgaactgattcaattcatcaccacgggtccgattatcgcaatggaaattctgcgtgatgacgctatctgcgaatggaaacgcctgctg
ggcccggaactcaggtgttgcgcgtagccgatgccagtgaaatccattcgcgctctgtttggcaccgatggtatccgtaatgcagcaca
tggtccggactcattcgcatcggcagctcgtgaaatggaaactgtttttcccgagctctggcggttgcggtccggcaaacaccgccaaat
ttaccaattgtacgtgctgtattgtcaaaccgcacgcagtgtcagaaggcctgctgggtaaaattctgatggcaatccgtagtctggctt
tgaaatctcggccatgcagatgttcaacatggaccgcgttaacgtcgaagaattctacgaagttacaaaggcgtggttaccgaatatca
cgatatggttacgggaaatgtactccggtccgtgctgcgatggaaattcagcaaaaacatgccacaaaaacgcttctgtgaattctgtgg
tccgcgagatccggaaatcgcacgtcatctgcgtccgggtaccctgcgcgcaatttttggtaaaaacgaaatccagaacgctgtgcact
gtaccgatctgccggaagacggtctgctggaagtcaatactttttctga

(amino acids)

MEKTLALIKPDAISKAGEIEIIEINKAGFTITKLKMMMLSRKEALDPHVDHQS
RPFNFELIQFITTGPIIAMEILRDDAICEWKRLGLPANGSGVARTDASESIRALFGTDGIR
NAAHGPDSFASAAREMELFPSPSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILM
AIRDAGFEISAMQMFNMDRVNVEEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQQ
NNATKTFREFCGPADPEIARHLRPGTLRAIFGKTKIONAVHCTDLPEDGLLEVOYFF-

Mouse NME6
(DNA)

Atgacctccatctctgccaagtccccagctcttcagctcacactagccctgatcaagcctgatgcagttgccaccca
ctgatcctggaggctgttcatcagcagattctgagcaacaagtctctcattgtacgaacgagggaaactgcagtggaagctggaggact
gccggagggtttaccgagagcatgaaggcggtttttctatcagcggtggtggagttcatgacaagtgggccaatccgagcctatatc
cttccccaaaagatgccatccaactttggaggacactgatgggaccaccagagtatttcgagcacgctatatagcccagattcaat
tcgtggaagtttgggcctcactgacacccgaaatactaccatggctcagactccgtggtttccgccagagagagattgcagccttctt
ccctgacttcagtgaaacagcgctggatgaggaggaggaaccccagctgcggtgtggtcctgtgcactacagtcacaggaaggtat
ccactgtgcacgtgaaacaggaagccacaaacaacctaacaaaacctag

(amino acids)

MTSILRSPQALQLTLALIKPDVAHPLILEAVHQQILSNKFLIVRTRELQWK
LEDCCRFFYREHEGRFFYQRLVEFMTSGPIRAYILAHKDAIQLWRTLMGPTRVFRARY
IAPDSIRGSLGLTDTRNTHGSDSVVSASREIAAFFPDFSEQRWYEEEEPQLRCGPVHY
SPEEGIHCAAETGGHKOPNKT-

Human NME6 :
(DNA)

Atgaccagaatctggggagtgagatggcctcaatcttggaagccctcaggctctccagctcactctagccctgat
caagcctgacgcagctcgccatccactgattctggaggctgttcacagcagattctaagcaacaagttctctgattgtacgaatgagag
aactactgtggagaaaggaagattgccagagggtttaccgagagcatgaagggcggtttttctatcagaggctggtggagttcatggcc
agcgggccaatccgagcctacatccttgcccacaaggatgccatccagctctggaggacgctcatgggaccaccagagtgtccga
gcacgccatgtqcccccaqattctatccgtqqgaqgttccgacctactgacaccqcaacaccacccatggttcggactctgtggtttc

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agccagcagagagattgcagccttcttccctgacttcagtgaaacagcgctggtatgaggaggaagagccccagttgcgctgtggccct
gtgtgctatagcccagaggaggtgtccactatgtagctggaacaggaggcctaggaccagcctga

(amino acids)

(SEQ ID NO: 65)

MTQNLGSEMASILRSPQALQLTLALIKPDAVAHPLILEAVHQQILSNKFLIV

RMRELLWRKEDCQRFYREHEGRFFYQRLVEFMASGPIRAYILAHKDAILWRTLGM

PTRVFRARHVAPDSIRGSFGLTDRNTTHGSDSVVSASREIAAFPPDFSEQRWYEEEE

PQLRCGPVCYSPEGGVHYVAGTGGLGPA-

Human NME6 1:

(DNA)

(SEQ ID NO: 66)

Atgaccagaatctggggagtggatggcctcaatcttggaagccctcaggtctccagctcactctagccctgat

caagcctgacgcagtcgccccatccactgattctggaggctgttcacagcagattctcaagcaacaagttcctgattgtacgaatgagag

aactactgtggagaaaggaagattgccagaggttttaccgagagcatgaaggcggtttttctatcagaggctggtggagtccatggcc

agcgggccaatccgagcctacatccttgcccacaaggatgccatccagctctggaggacgctcatgggaccaccagagtgtccga

gcacgccatgtggccccagattctatccgtgggagtttcggcctcactgacaccgcaacaccacccatggttcggactctgtggttc

agccagcagagagattgcagccttcttccctgacttcagtgaaacagcgctggtatgaggaggaagagccccagttgcgctgtggccct

gtgtga

(amino acids)

(SEQ ID NO: 67)

MTQNLGSEMASILRSPQALQLTLALIKPDAVAHPLILEAVHQQILSNKFLIV

RMRELLWRKEDCQRFYREHEGRFFYQRLVEFMASGPIRAYILAHKDAILWRTLGM

PTRVFRARHVAPDSIRGSFGLTDRNTTHGSDSVVSASREIAAFPPDFSEQRWYEEEE

PQLRCGPV-

Human NME6 2:

(DNA)

(SEQ ID NO: 68)

Atgtcactctagccctgatcaagcctgacgcagtcgccccatccactgattctggaggctgttcacagcagattctaa

gcaacaagttcctgattgtacgaatgagagaactactgtggagaaaggaagattgccagaggttttaccgagagcatgaaggcggtttt

ttctatcagaggctggtggagtccatggccagcgggccaatccgagcctacatccttgcccacaaggatgccatccagctctggagga

cgctcatgggaccaccagagtgttcgagcacgccatgtggccccagattctatccgtgggagtttcggcctcactgacaccgcaa

caccacccatggttcggactctgtggtttcagccagcagagagattgcagccttcttccctgacttcagtgaacagcgctggtatgagg

aggaagagccccagttgcgctgtggccctgtgtga

(amino acids)

(SEQ ID NO: 69)

MLTLALIKPDAVAHPLILEAVHQQILSNKFLIVRMRELLWRKEDCQRFYRE

HEGRFFYQRLVEFMASGPIRAYILAHKDAILWRTLGMGPTRVFRARHVAPDSIRGSF

GLTDRNTTHGSDSVVSASREIAAFPPDFSEQRWYEEEPQLRCGPV-

Human NME6 3:

(DNA)

(SEQ ID NO: 70)

Atgtcactctagccctgatcaagcctgacgcagtcgccccatccactgattctggaggctgttcacagcagattctaa

gcaacaagttcctgattgtacgaatgagagaactactgtggagaaaggaagattgccagaggttttaccgagagcatgaaggcggtttt

ttctatcagaggctggtggagtccatggccagcgggccaatccgagcctacatccttgcccacaaggatgccatccagctctggagga

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caccacccatggttcggactctgtggtttcagccagcagagagattgcagccttcttccctgacttcagtgaacagcgctggtatgagg

aggaagagccccagttgcgctgtggccctgtgtgctatagccagaggaggtgtccactatgtagctggaacaggaggcctagga

ccagcctga

-continued

(amino acids)

(SEQ ID NO: 71)

MLTLALIKPDAVAHPLILEAVHQILSNKFLIVRMRELLWRKEDCQRFYRE

HEGRFFYQRLVEFMASGPIRAYILAHKDAIQLWRTLMPTRVFRARHVAPDSIRGSF

GLTDTRNTTHGSDSVVSASREIAAFFPDFSEQRWYEEEEPQLRCGPVCYSPEGGVHY

VAGTGGLGPA-

Human NME6 sequence optimized for *E. coli* expression:
(DNA)

(SEQ ID NO: 72)

Atgacgcaaaatctgggctcggaatggcaagatcctgcgctccccgcaagcactgcaactgacctggctctgat

caaaccggacgctgttgcctcatccgctgattctggaagcggccaccagcaaattctgagcaacaaatttctgatcgtgcgtatgcgcg

aactgctgtggcgtaaagaagattgccagcgtttttatcgcaacatgaaggccgtttcttttatcaacgcctggttgaattcatggcctct

ggctccgattcgcgcataatctggctcacaagatgcgattcagctgtggcgtaacctgatgggtccgacgcgcgtcttctgctgcacgt

catgtggcaccggactcaatccgtggctcgctcggtctgaccgatacgcgcaataaccacgcacggtagcgactctgttgtagtgcgtc

ccgtgaaatcgcgcccttttcccggaacttctccgaacagcgttggtacgaagaagaagaaccgcaactgcgctgtggcccggtctgtt

attctccggaagtggtgtccattatgtggcgggcacgggtggtctgggtccggcatga

(amino acids)

(SEQ ID NO: 73)

MTQNLGSEMASILRSPQALQLTLALIKPDAVAHPLILEAVHQILSNKFLIV

RMRELLWRKEDCQRFYREHEGRFFYQRLVEFMASGPIRAYILAHKDAIQLWRTLMP

PTRVFRARHVAPDSIRGSFGLTDTRNTTHGSDSVVSASREIAAFFPDFSEQRWYEEEE

PQLRCGPVCYSPEGGVHYVAGTGGLGPA-

Human NME6 1 sequence optimized for *E. coli* expression:
(DNA)

(SEQ ID NO: 74)

Atgacgcaaaatctgggctcggaatggcaagatcctgcgctccccgcaagcactgcaactgacctggctctgat

caaaccggacgctgttgcctcatccgctgattctggaagcggccaccagcaaattctgagcaacaaatttctgatcgtgcgtatgcgcg

aactgctgtggcgtaaagaagattgccagcgtttttatcgcaacatgaaggccgtttcttttatcaacgcctggttgaattcatggcctct

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catgtggcaccggactcaatccgtggctcgctcggtctgaccgatacgcgcaataaccacgcacggtagcgactctgttgtagtgcgtc

ccgtgaaatcgcgcccttttcccggaacttctccgaacagcgttggtacgaagaagaagaaccgcaactgcgctgtggcccggtctga

(amino acids)

(SEQ ID NO: 75)

MTQNLGSEMASILRSPQALQLTLALIKPDAVAHPLILEAVHQILSNKFLIV

RMRELLWRKEDCQRFYREHEGRFFYQRLVEFMASGPIRAYILAHKDAIQLWRTLMP

PTRVFRARHVAPDSIRGSFGLTDTRNTTHGSDSVVSASREIAAFFPDFSEQRWYEEEE

PQLRCGPV-

Human NME6 2 sequence optimized for *E. coli* expression:
(DNA)

(SEQ ID NO: 76)

Atgctgacctggctctgatcaaacggacgctgttgcctcatccgctgattctggaagcggccaccagcaaattctg

agcaacaaatttctgatcgtgcgtatgcgcgaactgctgtggcgtaaagaagattgccagcgtttttatcgcaacatgaaggccgtttct

tttatcaacgcctggttgaattcatggcctctggctccgattcgcgcataatcctggctcacaagatgcgattcagctgtggcgtaacctg

atgggtccgacgcgcgtcttctgtgcacgtcatgtggcaccggactcaatccgtggctcgctcggtctgaccgatacgcgcaataccac

gcacggtagcgactctgttgtagtgcgtcccgtaaatcgcgcccttttcccggaacttctccgaacagcgttggtacgaagaagaag

aaccgcaactgcgctgtggcccggtctga

-continued

(amino acids)

(SEQ ID NO: 77)

MLTLALIKPDAVAHPLILEAVHQILSNKFLIVRMRELLWRKEDCQRFYRE

HEGRFFYQRLVEFMASGPIRAYILAHKDAIQLWRTL MGPTRVFRARHVAPDSIRGSF

GLTDTRNTTHGSDSVVSASREIAAFFPDFSEQRWYEEEEPQLRCGPV-

Human NME6 3 sequence optimized for *E. coli* expression:
(DNA)

(SEQ ID NO: 78)

Atgctgacccctggctctgatcaaacccggacgctgttgctcatccgctgattctggaagcggccaccagcaaattctg

agcaacaaattttctgatcgctgctatgcgcgaactgctgtggcgtaaagaagattgccagcgtttttatcgcgaaatgaaggccggtttct

tttatcaacgctggttgaaatcatggcctctggtccgattcgcgcataatctctggctcacaagatgcgattcagctgtggcgtaccctg

atgggtccgacgcgcgtctttcgtgcacgtcatgtggcaccggactcaatccgtggctcgttcggtctgaccgatcgcgcaataaccac

gcacggtagcgactctgttgtagtgctcccgtaaatcgcggcctttttcccgacttctccgaacagcgttggtacgaagaagaag

aaccgcaactgcgctgtggcccggtctgttattctccggaaggtggtgtccattatgtgggggcacgggtggtctgggtccggcatg

a

(amino acids)

(SEQ ID NO: 79)

MLTLALIKPDAVAHPLILEAVHQILSNKFLIVRMRELLWRKEDCQRFYRE

HEGRFFYQRLVEFMASGPIRAYILAHKDAIQLWRTL MGPTRVFRARHVAPDSIRGSF

GLTDTRNTTHGSDSVVSASREIAAFFPDFSEQRWYEEEEPQLRCGPVCYSPEGGVHY

VAGTGGLGPA-

OriGene-NME7-1 full length
(DNA)

(SEQ ID NO: 80)

gacgttgtatacgactcctatagggcggccgggaattcgtcgactggatccggtaccgaggagatctgccgcgcg

atcgccatgaatcatagtgaaagattcgttttcattgcagagtggatgatccaaatgcttcacttcttcgacgttatgagcttttattttacc

aggggatggatctgttgaaatgcgatgataaagaatcatcgaccccttttaaaagcggaacaaatgataacctgcacttggaagattta

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gtttattacaactggctcctattattgccatggagattttaagagatgatgctatatgtgaatggaaaagactgctgggacctgcaaacctctg

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caaatgatatcctggattacaaggatgacgacgataagggttaa

(amino acids)

(SEQ ID NO: 81)

MNHSERFVFIAEWYDPNASLLRRYELLFYPGDSVEMHDVKNHRTFLKR

TKYDNLHLEDLFIGNKVNVSRLVLIDYGDQYARQLGSRKEKTLALIKPDAISKA

GEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQSRPFFNELIQFITTGPIIAMEILRDD

AICEWKRLGPPANSVARTDASESIRALFGTDGIRNAAGPDSFASAAAREMELFFPSS

GGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFNMDRVNV

EEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQNNATKTFREFCGPADPEIARHLR

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PGTLRAIFGKTKIQNAVHCTDLPEDGLLEVQYFFKILDNTRTRRLEQKLISEEDLAAN

DILDYKDDDDKV

Abnova NME7-1 Full length
(amino acids)

(SEQ ID NO: 82)

MNHSEFVFIAEWYDPNASLLRRYELLFYPGDGSVEMHDVKNHRTFLKR

TKYDNLHLEDLFIGNKVNVFSRQLVLIDYGDQYTARQLGSRKEKTLALIKPDAISKA

GEIIEIINKAGFTITKLMMMLSRKEALDFHVDHQSRPFFNELIQFITTGPPIAMEILRDD

AICEWKRLGPPANSVARTDASESIRALFGTDGIRNAAHGPDSPASAREMELFFPSS

GGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFNMDRVNV

EEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQQNNATKTFREFCGPADPEIARHLR

PGTLRAIFGKTKIQNAVHCTDLPEDGLLEVQYFFKILDN

Abnova Partial NME7-B
(amino acids)

(SEQ ID NO: 83)

DRVNVEEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQQNNATKTFREF

CGPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLLEVQYFFKIL

Histidine Tag

(SEQ ID NO: 84)

(ctcgag) caccaccaccaccactga

Strept II Tag

(SEQ ID NO: 85)

(accggt) tggagccatcctcagttcgaaaagtaatga

N-10 peptide:

(SEQ ID NO: 86)

QFNQYKTEAASRYNLTISDVSVSDVPPFSAQSGA

C-10 peptide

(SEQ ID NO: 87)

GTINVHDVETQFNQYKTEAASRYNLTISDVSVSDV

LALIKPDA

(SEQ ID NO: 88)

MMMLSRKEALDFHVDHQS

(SEQ ID NO: 89)

ALDFHVDHQS

(SEQ ID NO: 90)

EILRDDAICEWKRL

(SEQ ID NO: 91)

FNELIQFITTGP

(SEQ ID NO: 92)

RDDAICEW

(SEQ ID NO: 93)

SGVARTDASESIRALFGTDGIRNAA

(SEQ ID NO: 94)

ELFFPSSGG

(SEQ ID NO: 95)

KFTNCTCCIVKPHAVSEGLLGKILMA

(SEQ ID NO: 96)

LMAIRDAGFEISAMQMFNMDRVNVEEFYEVYKGVVT

(SEQ ID NO: 97)

EFYEVYKGVVTEYHD

(SEQ ID NO: 98)

EIQQNNATKTFREFCGPADPEIARHLRPGTLRAIFGKTKIQNA

(SEQ ID NO: 99)

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YSGPCVAM	(SEQ ID NO: 100)
FREFCGP	(SEQ ID NO: 101)
VHCTDLPEDGGLLEVQYFFKILDN	(SEQ ID NO: 102)
IQNAVHCTD	(SEQ ID NO: 103)
TDLPEDGGLLEVQYFFKILDN	(SEQ ID NO: 104)
PEDGGLLEVQYFFK	(SEQ ID NO: 105)
EIINKAGFTITK	(SEQ ID NO: 106)
MLSRKEALDFHVDHQS	(SEQ ID NO: 107)
NELIQFITT	(SEQ ID NO: 108)
EILRDDAICEWKRL	(SEQ ID NO: 109)
SGVARTDASESIRALFGTDGI	(SEQ ID NO: 110)
SGVARTDASES	(SEQ ID NO: 111)
ALFGTDGI	(SEQ ID NO: 112)
NCTCCIVKPHAVSE	(SEQ ID NO: 113)
LGKILMAIRDA	(SEQ ID NO: 114)
EISAMQMFNMDRVNVE	(SEQ ID NO: 115)
EVYKGVVT	(SEQ ID NO: 116)
EYHDMVTE	(SEQ ID NO: 117)
EFCGPADPEIARHLR	(SEQ ID NO: 118)
AIFGKTKIQNAV	(SEQ ID NO: 119)
LPEDGGLLEVQYFFKILDN	(SEQ ID NO: 120)
GPDSFASAAREMELFFP	(SEQ ID NO: 121)
Immunizing peptides derived from human NME7	(SEQ ID NO: 122)
ICEWKRL	(SEQ ID NO: 123)
LGKILMAIRDA	(SEQ ID NO: 124)
HAVSEGLLGK	(SEQ ID NO: 125)
VTEMYSGP	(SEQ ID NO: 126)
NATKTFREF	(SEQ ID NO: 126)

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AIRDAGFEI	(SEQ ID NO: 127)
AICEWKRLLG PAN	(SEQ ID NO: 128)
DHQSRPFF	(SEQ ID NO: 129)
AICEWKRLLG PAN	(SEQ ID NO: 130)
VDHQSRPF	(SEQ ID NO: 131)
PDSFAS	(SEQ ID NO: 132)
KAGEIIEIINKAGFTITK	(SEQ ID NO: 133)
Immunizing peptides derived from human NME1	
MANCERTFIAIKPDGVQRLVGEIIRFE	(SEQ ID NO: 134)
VDLKDRPF	(SEQ ID NO: 135)
HGSDSVESAEKEIGLWF	(SEQ ID NO: 136)
ERTFIAIKPDGVQRLVGEIIRFE	(SEQ ID NO: 137)
VDLKDRPFFAGLVKYMHS GPV VAMVWEGLN	(SEQ ID NO: 138)
NIIHGSDSVESAEKEIGLWFHPEELV	(SEQ ID NO: 139)
KPDGVQRLVGEII	(SEQ ID NO: 140)
Immunizing peptide derived from human NME7, but which does not bind NME1 peptide A1	
MLSRKEALDFHVDHQS	(SEQ ID NO: 141)
peptide A2	
SGVARTDASES	(SEQ ID NO: 142)
peptide B1	
DAGFEISAMQMFNM DRVNVE	(SEQ ID NO: 143)
peptide B2	
EVYKGVVTEYHDMVTE	(SEQ ID NO: 144)
peptide B3	
AIFGKTKIQNAVHCTDLPEDGLLEVQYFF	(SEQ ID NO: 145)
Human NME7 a (DNA)	
atgaatcatagtgaaagattcgttttcattgcagagtggatgatccaaatgcttcacttcttcgacgttatgagcttttat	(SEQ ID NO: 146)
accaggggatggatctgttgaaatgcatgatgtaagaatcatcgccactttttaagcggaccaaataatgataacctgcaacttggag	
atattttataggcaacaaagtgaatgtctttctcgacaactggatttaattgactatggggatcaataacacgctcgccagctgggcagta	
ggaaagaaaaaacgctagccctaattaaaccagatgcaatatcaaaggctggagaaataattgaaataataaacaagctggatttact	
ataaccaaactcaaaatgatgatgctttcaaggaaagaagcattggattttcatgtagatcaccagtcaagacctttttcaatgagctgat	
ccagtttattacaactggctcattattgccatggagattttaagagatgatgctatatgtgaatggaaaagactgctgggacctgcaaaact	
ctggagtggcagcagacagatgcttctgaaagcattagagccctctttggaacagatggcataagaaatgcagcgcacatggccctgattct	

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tttgcttctgcggccagagaaatggagttgttttttcttcaagtggaggtgtgtggccggcaaacactgctaaatttactaattgtacctgtt
gcattgttaaaccctatgctgtcagtgaggactgttgggaaagatcctgatggctatccgagatgcaggttttgaaatctcagctatgca
gatgttcaatatggatcgggttaatgttgaggaattctatgaagttataaaaggagtagtgaccgaatatcatgacatggtgacagaaatgt
attctggcccttgtgtagcaatggagattcaacagaataatgctacaaagacatttcgagaattttgtggacctgctgatcctgaaattggc
cggcatttacgccctggaactctcagagcaatctttggtaaaactaagatccagaatgctgttcactgtactgatctgccagaggatggc
ctattagaggttcaatacttcttcaagatcttggataattag

(amino acids)

(SEQ ID NO: 147)

MNHSERFVFIAEWYDPNASLLRRYELLFYPGDGSVEMHDVKNHRTFLKR
TKYDNLHLEDLFIGNKVNVFSRQLVLIDYGDQYTARQLGSRKEKTLALIKPDAISKA
GEIIEIINKAGFTITKLMMMLSRKEALDFHVDHQSRPFFNELIQFITTGPIIAMEILRDD
AICEWKRLGPGANSGVARTDASESIRALFGTDGIRNAAHGPDSPASAAAREMELFFPSS
GGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFNMDRVNV
EEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQQNNATKTFREFCGPADPEIARHLR
PGLRAIFGKTKIQNAVHCTDLPEDGLLEVQYFFKILDN

Human NME7 b
(DNA)

(SEQ ID NO: 148)

atgcatgatgtaaagaatcatcgacacctttttaagcggaccacaaatagataacctgcacttggaagatttatttataggc
aacaaagtgaatgtcttttctcgacaactgggtattaattgactatggggatcaatatacagctcgccagctgggcagtaggaaagaaaa
acgctagccctaattaaaccagatgcaatatcaaaggctgggaaataaattgaaataataaaacaaagctggatttactataaccaaactc
aaaatgatgatgctttcaaggaaagaagcattggattttcatgtagatcaccagtcaagacctttttcaatgagctgatccagtttattaca
actggctcctattattgccatggagattttaagagatgatgctatatgtgaatggaaaagactgctgggacctgcaaacctctggagtgga
cgcacagatgcttctgaaagcattagagccctctttggaacagatggcataagaaatgcagcgcattggccctgatcttttgcctctgcg
gccagagaaatggagttgttttttcttcaagtggaggtgtgtggccggcaaacactgctaaatttactaattgtacctgttgcatgtttaa
ccccatgctgtcagtgaggactgttgggaaagatcctgatggctatccgagatgcaggttttgaaatctcagctatgcagatgttcaata
tggatcgggttaatgttgaggaattctatgaagttataaaaggagtagtgaccgaatatcatgacatggtgacagaaatgtattctggccct
tgtgtagcaatggagattcaacagaataatgctacaaagacatttcgagaattttgtggacctgctgatcctgaaattgcccgccatttac
gccctggaactctcagagcaatctttggtaaaactaagatccagaatgctgttccactgtactgatctgccagaggatggcctattagagg
ttcaatacttcttcaagatcttggataattag

(amino acids)

(SEQ ID NO: 149)

MHDVKNHRTFLKRTKYDNLHLEDLFIGNKVNVFSRQLVLIDYGDQYTAR
QLGSRKEKTLALIKPDAISKAGEIIEIINKAGFTITKLMMMLSRKEALDFHVDHQSRP
FFNELIQFITTGPIIAMEILRDDAICEWKRLGPGANSGVARTDASESIRALFGTDGIRNA
AHGPDSPASAAAREMELFFPSSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILMAIR
DAGFEISAMQMFNMDRVNVEEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQQNN
ATKTFREFCGPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLLEVQYFFKILD

N

Human NME7_{AB} also known as NME7_{AB}
(DNA)

(SEQ ID NO: 150)

atggaaaaaacgctagccctaattaaaccagatgcaatatcaaaggctggagaaataaattgaaataataaaacaaagct
ggatttactataaccaaactcaaaatgatgatgctttcaaggaaagaagcattggattttcatgtagatcaccagtcaagacctttttcaat
gagctgatccagtttattacaactggctcctattattgccatggagattttaagagatgatgctatatgtgaatggaaaagactgctgggacc

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tgcaaaactctggagtggcacgcacagatgcttctgaaagcattagagccctcttggaaacagatggcataaagaaatgcagcgcgatggc
cctgattcttttctgcttctgcggccagagaaatggagttgttttcttcaagtggaggtgtgggcccggcaaacactgctaatttactaatt
gtacctgttgcatgttaaaccccatgctgtcagtgaggactgttgggaaagatcctgatggctatccgagatgcaggttttgaaatctc
agctatgcagatgttcaataggatcgggttaatgttgaggaattctatgaagtttataaaggagtagtgaccgaatatcatgacatggatga
cagaaatgtattctggcccttgtgtagcaatggagattcaacagaataatgctacaaagacatttcgagaattttgtggacctgctgatcct
gaaattgcccggcatttacgccttgaactctcagagcaatcttggtaaaactaagatccagaatgctgttccactgtactgatctgccag
aggatggcctattagaggttcaatacttcttcaagatcttggataattag

(amino acids)

(SEQ ID NO: 151)

MEKTLALIKPDAISKAGEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQS
RPFFNELIQFITTGPPIAMEILRDDAICEWKRLGPNANSGVARTDASESIRALFGTDGIR
NAAHGPDSPASAAREMELFFPSSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILM
AIRDAGFEISAMQMFNMDRVNVEEFYEVYKGVVTEYHDMVTEMSGPCVAMEIQQ
NNATKTFREFCGPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLLEVQYFFK
ILDN

Human NME7-X1
(DNA)

(SEQ ID NO: 152)

atgatgatgctttcaaggaaagaagcattggattttcatgtagatcaccagtcgaagacccttttcaatgagctgatccag
tttattacaactggctcattattgccatggagattttaagagatgatgctatatgtgaatggaaaagactgctgggacctgcaaaactctgg
agtggcacgcacagatgcttctgaaagcatttagagccctcttggaaacagatggcataaagaaatgcagcgcgatggccctgatcttttgc
ttctgcggccagagaaatggagttgttttcttcaagtggaggtgtgggcccggcaaacactgctaatttactaattgtacctgttgcat
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ctggcccttgtgtagcaatggagattcaacagaataatgctacaaagacatttcgagaattttgtggacctgctgatcctgaaattgcccg
gcatttacgccttgaactctcagagcaatcttggtaaaactaagatccagaatgctgttccactgtactgatctgccagaggatggccta
ttagaggttcaatacttcttcaagatcttggataattag

(amino acids)

(SEQ ID NO: 153)

MMMLSRKEALDFHVDHQSRRPFFNELIQFITTGPPIAMEILRDDAICEWKRL
LGPANSGVARTDASESIRALFGTDGIRNAAHGPDSPASAAREMELFFPSSGGCGPANT
AKFTNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFNMDRVNVEEFYEVYK
GVVTEYHDMVTEMSGPCVAMEIQQNNATKTFREFCGPADPEIARHLRPGTLRAIFG
KTKIQNAVHCTDLPEDGLLEVQYFFKILDN*

Human NME7 a (optimized for *E coli* expression)
(DNA)

(SEQ ID NO: 154)

atgaatcactccgaacgctttgtttttatcgccgaatggtagtaccgcaatgcttcctgctgcgccgctacgaactgct
gttttatccgggcgatggtagcgtggaaatgcagtgacgttaaaaaatcacctacctttctgaaacgcacgaaatagataatctgcatctg
gaagacctgtttattggcaacaagaatgaatgtgttctctcgtcagctggtagctgatcgattatggcgaccagtagaccgcgctcaactg
ggtagtcgcaaaaaaacgctggccctgattaaacggatgcaatctccaaagctggcgaaattatcgaaattatcaacaaagcgg
gtttcaccatcacgaaactgaaatgatgatgctgagccgtaaagaagccctggattttcatgtcgaccaccagtagtgcgccgtttttcaa
tgaactgattcaattcatcaccacgggtccgattatcgcaatggaaattctgcgtgatgacgctatctgcgaatggaaacgcctgctggg
cccggcaaacctcaggtgttgccgctaccgatccagtgaaatccatctcgcgctctgtttggcaccgatggtagcgaatgcagcacatg
gtccggactcattcgcatcggcagctcgtgaaatggaactgtttttcccgagctctggcggttgccggtccggcaaacaccgccaattt
accaattgtacgtgctgtattgtcaaacccgcacgcagtgtagaaggcctgctgggtgaaattctgatggcaatccgtgatgctggcctt

-continued

gaaatctcggccatgcagatgttcaacatggaccgcttaacgtcgaagaattctacgaagtttacaaggcgtggttacccaatatca
cgatatggttacggaaatgtactccggtccgtgcgtcgcgatggaaattcagcaaaacaatgccacaaaacgtttcgtgaattctgtgg
tccggcagatccggaaatcgacgtcatctgcgtccgggtaccctgcgcgaattttggtaaaacgaaaatccagaacgctgtgcact
gtaccgatctgccgaagacggtctgctggaagtcaatactttttcaaaattctggataat

(amino acids)

(SEQ ID NO: 155)

MNHSERFVPIAEWYDPNASLLRRYELLFYPGDGSVEMHDVKNHRTFLKR
TKYDNLHLEDLFIGNKVNVSRLVLIDYGDQYTARQLGSRKEKTLALIKPDAISKA
GEIIEIINKAGFTITKLMMMLSRKEALDFVHDHQSRRPFNFELIQFITTGPIIAMEILRDD
AICEWKRLGPPANSVARTDASESIRALFGTDGIRNAHGPDSFASAAREMELFFPSS
GGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFNMMDRVNV
EEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQNNATKTFREFCGPADPEIARHLR
PGTLRAIFGKTKIQNAVHCTDLPEDGLLEVQYFFKILDNTG

Human NME7 b (optimized for *E coli* expression)
(DNA)

(SEQ ID NO: 156)

atgcatgacgttataaaatcacgctacctttctgaaacgcacgaaatgatgaatctgcatctggaagacctgttatggc
aacaagtcaatgtgtctctcgtcagctggtgctgatcgattatggcgaccagttacaccgcgcgtcaactgggtagtcgcaaagaaa
aacgctggccctgattaaaccggatgcaatctccaaagctggcgaaattatcgaaattatcaacaagcgggtttcaccatcacgaaac
tgaaaatgatgatgctgagccgttaagaagccctggattttcatgtcgaccaccagttctgcccgtttttcaatgaactgattcaattcatc
accacgggtccgattatcgcaatggaaattctgcgtgatgacgctatctgcgaatggaaacgcctgctgggcccgggcaaactcaggtg
ttgcgcgtaccgatgccagtgaaatccattcgcgctctgtttggcaccgatggtatccgtaatgcagcacatggtccggactcattcgcac
cggcagctcgtgaaatggaactgtttttcccgagctctggcggttgcggtccggcaaacaccgcaaatttaccaattgtacgtgctgta
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tactccggtccgtgcgtcgcatggaaatccagcaaaacaatgccacaaaacgtttcgtgaattctgtggtccggcagatccggaaat
cgcacgtcatctgcgtccgggtaccctgcgcgaattttggtaaaacgaaaatccagaacgctgtgcactgtaccgatctgccggaa
gacggtctgctggaagtcaatactttttcaaaattctggataat

(amino acids)

(SEQ ID NO: 157)

MHDVKNHRTFLKRTKYDNLHLEDLFIGNKVNVSRLVLIDYGDQYTAR
QLGSRKEKTLALIKPDAISKAGEIIEIINKAGFTITKLMMMLSRKEALDFVHDHQSRRP
FFNELIQFITTGPIIAMEILRDDAICEWKRLGPPANSVARTDASESIRALFGTDGIRNA
AHGPDSFASAAREMELFFPSSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILMAIR
DAGFEISAMQMFNMMDRVNVEEFYEVYKGVVTEYHDMVTEMYSGPCVAMEIQNN
ATKTFREFCGPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLLEVQYFFKILD
NTG

Human NME7_{AB} also known as NME7_{AB} (optimized for *E coli* expression)
(DNA)

(SEQ ID NO: 158)

atggaaaaaacgctggccctgattaaaccggatgcaatctccaaagctggcgaaattatcgaaattatcaacaaagcg
ggtttcaccatcacgaaactgaaaatgatgatgctgagccgttaaagaagccctggattttcatgtcgaccaccagttctgcccgtttttca
atgaactgattcaattcatcaccacgggtccgattatcgcaatggaaattctgcgtgatgacgctatctgcgaatggaaacgcctgctgg
gcccggcaaacctcaggtgttgcggtaccgatgccagtgaaatccattcgcgctctgtttggcaccgatggtatccgtaatgcagcacat
ggtccggactcattcgcacgcgcagctcgtgaaatggaactgtttttcccgagctctggcggttgcggtccggcaaacaccgcaaat

167

168

-continued

taccaattgtacgtgctgtattgtcaaacgcacgcagtgtcagaaggcctgctgggtaaaattctgatggcaatccgtgatgctggttt
 gaaatctcggccatgcagatgttcaacatggaccgcttaacgtcgaagaattctacgaagtttacaaggcgtggttaccgaatatca
 cgatatggttacggaaatgtactccggtccgtgctgcgatggaaattcagcaaaacaatgccacaaaacgtttcgtgaattctgtgg
 tccggcagatccggaatcgcacgtcatctgcgtccgggtaccctgcgcgaattttggtaaacgaaaatccagaacgctgtgcact
 gtaccgatctgcggaagacgggtctgctggaagtcaatactttttcaaaattctggataat

(amino acids)

(SEQ ID NO: 159)

MEKTLALIKPDAISKAGEIIEIINKAGFTITKLKMMMLSRKEALDFHVDHQ
 RPPFNELIQFITTGPPIAMEILRDDAICEWKRLGPPANSVARTDASESIRALFGTDGIR
 NAAHGPDSFASAAAREMELFFPSSGGCGPANTAKFTNCTCCIVKPHAVSEGLLGKILM
 AIRDAGFEISAMQMFNMDRVNVEEFYEVYKGVVTEYHDMVTEMSGPCVAMEIQ
 NNATKTFREFCGPADPEIARHLRPGTLRAIFGKTKIQNAVHCTDLPEDGLLEVQYFFK
 ILDNTG

Human NME7-X1 (optimized for *E coli* expression)
 (DNA)

(SEQ ID NO: 160)

atgatgatgctgagccgtaaagaagccctggattttcatgtcgaccaccagtcctcgccggttttcaatgaactgattcaa
 ttcacaccacgggtccgattatcgcaatggaaattctgcgtgatgacgctatctcgcaatggaaacgcctgctgggcccggcgaactc
 aggtgttgcgctaccgatgccagtgaaatccattcgcgctctgtttggcaccgatggatccgtaatgcagcacatggccggactcatt
 cgcacccggcagctcgtgaaatggaaactgttttcccgagctctggcggttgcggtccggcaaacaccgcaaatctaccaattgtacgt
 gctgtattgtcaaacgcacgcagtgtcagaaggcctgctgggtaaaattctgatggcaatccgtgatgctggctttgaaatctcggcca
 tgcagatgttcaacatggaccgcttaacgtcgaagaattctacgaagtttacaaggcggtgtaccgaatatcacgatatggttacgg
 aaatgtactccggtccgtgctgcgatggaaattcagcaaaacaatgccacaaaacggttctgtgaattctgtggtccggcagatccg
 gaaatcgcacgtcatctgcgtccgggtaccctgcgcgaattttggtaaacgaaaatccagaacgctgtgcactgtaccgatctgcc
 ggaagacgggtctgctggaagtcaatactttttcaaaattctggataat

(amino acids)

(SEQ ID NO: 161)

MMMLSRKEALDFHVDHQSRPPFNELIQFITTGPPIAMEILRDDAICEWKRL
 LGPPANSVARTDASESIRALFGTDGIRNAAHGPDSFASAAAREMELFFPSSGGCGPANT
 AKFTNCTCCIVKPHAVSEGLLGKILMAIRDAGFEISAMQMFNMDRVNVEEFYEVYK
 GVVTEYHDMVTEMSGPCVAMEIQNNATKTFREFCGPADPEIARHLRPGTLRAIFG
 KTKIQNAVHCTDLPEDGLLEVQYFFKILDNTG

DM10 domain of NME7
 (amino acids)

(SEQ ID NO: 162)

MNHSERFVPIAEWYDPNASLLRRYELLFYPGDGSVEMHDVKNHRTFLKR
 TKYDNLHLEDLFIGNKVNVFSRQLVLIDYGDQYTARQLGSRK

a fragment or variation of PSMGFR peptide

(SEQ ID NO: 163)

SNIKFRPGSVVVQLTLAFREGTINVHVDVETQFNQYKTEAASRY;

a fragment or variation of PSMGFR peptide

(SEQ ID NO: 164)

SVVVQLTLAFREGTINVHVDVETQFNQYKTEAASRY;

a fragment or variation of PSMGFR peptide

(SEQ ID NO: 165)

VQLTLAFREGTINVHVDVETQFNQY;

a fragment or variation of PSMGFR peptide

(SEQ ID NO: 166)

SNIKFRPGSVVVQLTLAFREGTIN;

-continued

a fragment or variation of PSMGFR peptide

(SEQ ID NO: 167)

SNIKFRPGSVVQLTLAFREGTINVHDTVETQFNQYKTE;

a fragment or variation of PSMGFR peptide

(SEQ ID NO: 168)

VQLTLAFREGTINVHDTVETQFNQYKTEAASRYNLTISDVSVSDVP.

Cys at residue 14 is mutated to Ser of NME7B peptide 3 (B domain):

(SEQ ID NO: 169)

AIFGKTKIQNAVHSTDLPEDGLLEVQYFF

N-10 peptide

(SEQ ID NO: 170)

QFNQYKTEAASRYNLTISDVSVSDVPFPFSAQSGA

C-10 peptide

(SEQ ID NO: 171)

GTINVHDTVETQFNQYKTEAASRYNLTISDVSVSDV

EXAMPLES

Example 1—Components of Minimal Serum-Free Base (“MM”) (500 Mls)

400 ml DME/F12/GlutaMAX I (Invitrogen #10565-018)
 100 ml Knockout Serum Replacement (KO-SR, Invitrogen #10828-028)
 5 ml 100×MEM Non-essential Amino Acid Solution (Invitrogen #11140-050)
 0.9 ml (0.1 mM) β-mercaptoethanol (55 mM stock, Invitrogen #21985-023).

Example 2—Generation of Protein Constructs

For generating recombinant NME7, first, constructs were made to make a recombinant NME7 that could be expressed efficiently and in soluble form. The first approach was to make a construct that would encode the native NME7 (a) or an alternative splice variant NME7 (b), which has an N-terminal deletion. In some cases, the constructs carried a histidine tag or a strep tag to aid in purification, NME7-a, full-length NME7 expressed poorly in *E. coli* and NME7-b did not express at all in *E. coli*. However, a novel construct was made in which the DM10 sequence was deleted and the NME7 comprised essentially the NDPK A and B domains having a calculated molecular weight of 33 kDa.

This novel NME7_{AB} expressed very well in *E. coli* and existed as the soluble protein, NME7_{AB} was first purified over an NTA-Ni column and then further purified by size exclusion chromatography (FPLC) over a Sephadex 200 column. Fractions were collected and tested by SDS-PAGE to identify fractions with the highest and purest expression of NME7_{AB}. The FPLC trace for the combined fractions that were the most pure were combined. The purified NME7_{AB} protein was then tested and shown to fully support the growth of human stem cells and further reverts them to the most naïve, pre-X-inactivation state. The purified NME7_{AB} was also shown to accelerate the growth of cancer cells.

Example 3—ELISA Assay Showing NME7_{AB} Simultaneously Binds to Two MUC1* Extra Cellular Domain Peptides

Results are shown in FIG. 1. The PSMGFR peptide bearing a C-terminal Cysteine (PSMGFR-Cys) was covalently coupled to BSA using Imject Maleimide activated BSA kit (Thermo Fisher). PSMGFR-Cys coupled BSA was diluted to 10 ug/mL in 0.1M carbonate/bicarbonate buffer

pH 9.6 and 50 uL was added to each well of a 96 well plate. After overnight incubation at 4° C., the plate was washed twice with PBS-T and a 3% BSA solution was added to block remaining binding site on the well. After 1 h at RT the plate was washed twice with PBS-T and NME7, diluted in PBS-T+1% BSA, was added at different concentrations. After 1h at RT the plate was washed 3× with PBS-T and anti-NM23-H7 (B-9). Santa Cruz. Biotechnology), diluted in PBS-T+1% BSA, was added at 1/500 dilution. After 1 h at RT the plate was washed 3× with PBS-T and goat anti mouse-HRP, diluted in PBS-T+1% BSA, was added at 1/3333 dilution. After 1 h at RT the plate was washed 3× with PBS-T and binding of NME7 was measured at 415 nm using ABTS solution (Pierce).

ELISA MUC1* dimerization: The protocol for NME7 binding was used, and NME7 was used at 11.6 ug/mL.

After 1 h at RT the plate was washed 3× with PBS-T and His-Tagged PSMGFR peptide (PSMGFR-His) or biotinylated PSMGFR peptide (PSMGFR-biotin), diluted in PBS-T+1% BSA, was added at different concentration. After 1 h at RT the plate was washed 3× with PBS-T and anti-Histag-HRP (Abcam) or streptavidin-HRP (Pierce), diluted in PBS-T+1% BSA, was added at a concentration of 1/5000. After 1 h at RT the plate was washed 3× with PBS-T and binding of PSMGFR peptide to NME7 already bound to another PSMGFR peptide (which could not signal by anti-His antibody or by streptavidin) coupled BSA was measured at 415 nm using a ABTS solution (Pierce).

Example 4—Functional Testing of Human Recombinant NME7_{AB}

For testing recombinant NME7_{AB} for ability to maintain pluripotency and inhibit differentiation, a soluble variant of NME7, NME7_{AB}, was generated and purified. Human stem cells (iPS cat #SC101a-1, System Biosciences) were grown per the manufacturer's directions in 4 ng/ml bFGF over a layer of mouse fibroblast feeder cells for four passages. These source stem cells were then plated into 6-well cell culture plates (Vita™, Thermo Fisher) that had been coated with 12.5 ug/well of a monoclonal anti-MUC1* antibody. MN-C3. Cells were plated at a density of 300,000 cells per well. The base media was Minimal Stem Cell Media consisting of: 400 ml DME/F12/GlutaMAX I (Invitrogen #10565-018), 100 ml Knockout Serum Replacement (KO-SR, Invitrogen #10828-028), 5 ml 100×MEM Non-essential Amino Acid Solution (Invitrogen #11140-050) and 0.9 ml (0.1 mM) β-mercaptoethanol (55 mM stock, Invitrogen #21985-023). The base media can be any media. In a

preferred embodiment, the base media is free of other growth factors and cytokines. To the base media was added either 8 nM of NME7_{AB} or 8 nM NM23-H1 refolded and purified as stable dimers. Media was changed every 48 hours and due to accelerated growth, had to be harvested and passaged at Day 3 post-plating. Comparable pluripotent stem cell growth was achieved when stem cells were grown in NM23-H1 dimers or in NME7 monomers.

NME7 and NM23-H1 (NME1) dimers both grew pluripotently and had no differentiation even when 100% confluent. As can be seen in the photos, NME7 cells grew faster than the cells grown in NM23-H1 dimers. Cell counts at the first harvest verified that culture in NME7 produced 1.4-times more cells than culture in NM23-H1 dimers. ICC staining for the typical pluripotent markers confirmed that NME7_{AB} fully supported human stem cell growth, pluripotency, and resisted differentiation.

The NME7 species of ~30-33 kDa may be an alternative splice isoform or a post translational modification such as cleavage, which may enable secretion from the cell.

Example 5—Inducing Transition of Cancer Cells to Metastatic Cancer Cells by Culturing Cells Under Conditions that Revert Stem Cells to a More Naïve State

Cancer cells are normally cultured in a serum-containing media such as RPMI. We discovered that culturing cancer cells in the presence of reagents that make stem cells revert to a more naïve state, makes the cancer cells transform to a more metastatic state.

We demonstrated that NME7_{AB}, human NME1 dimers, bacterial NME1 dimers. NME7-X1 and “2i” inhibitors were each able to transform regular cancer cells into metastatic cancer cells, which are also called cancer stem cells “CSCs” or tumor initiating cells “TICs”. 2i is the name given to two biochemical inhibitors that researchers found made human stem cells revert to a more naïve state. 2i are MEK and GSK3-beta inhibitors PD0325901 and CHIR99021, which are added to culture medium to final concentrations of about 1 mM and 3 mM, respectively.

NME7_{AB} and NME7-X1 are at a final concentration of about 4 nM when added to separate batches of minimal medium to make cancer cells transform to metastatic cells, although lower and higher concentrations also work well in the range of about 1 nM to 16 nM. Human or bacterial NME1 dimers are used at a final concentration of 4 nM to 32 nM, with 16 nM typically used in these experiments, wherein the human NME bears the S120G mutation. Lower concentrations may be required if using wild type. It is not intended that these exact concentrations are important. It is important that the NME1 proteins are dimers and the range of concentrations over which this happens is in the low nanomolar range although certain mutations allow higher concentrations to remain as dimers.

Similarly, the concentrations of NME7 proteins can vary, NME7_{AB} and NME7-X1 are monomers and concentrations used to transform cancer cells to metastatic cells should allow the proteins to remain as monomers. Various molecular markers have been proposed as being indicators of metastatic cancer cells. Different cancer types may have different molecules that are up-regulated. For example, the receptor CXCR4 is up-regulated in metastatic breast cancers while E-cadherin, also known as CHD1, is up-regulated more in metastatic prostate cancers.

In addition to these specific metastasis markers, typical markers of pluripotency such as OCT4, SOX2, NANOG,

and KLF4 are up-regulated as cancers become metastatic. The starting cancer cells and the later metastatic cancer cells can be assayed by PCR to measure expression levels of these genes.

FIG. 2 shows a graph of RT-PCR measurements of T47D breast cancer cells that were cultured in a media that contained NME7_{AB}. A rho I kinase inhibitor, ROCi, ROCKi or Ri, was added to prevent the transformed cells from floating off the plate. Expression levels of various metastatic markers as well as pluripotent stem cell markers were measured for the parent cells and for the NME7_{AB} cultured cells. The results show that the floater cells express higher amounts of metastatic and pluripotency markers compared to the cells that received ROCi. We reasoned it was because those measurements were the average of cells that did not transform and those that did but the ROCi made them remain adherent. This can clearly be seen in figures wherein “—Ri” means adherent cells that did not receive ROCi and so were not mixed with the highly metastatic cells that float.

Prostate cancer cells also transitioned to a more metastatic state when cultured in media containing NM23, aka NME1, or NME7_{AB}. Here we show that for every cell line tested so far, culture in NME7_{AB}, human NME1 dimers, or bacterial NMEs that have high sequence homology to human, induces transition to a more metastatic state.

FIG. 4 shows a graph of RT-PCR measurements of expression levels of metastatic and pluripotency markers for breast cancer cells that are cultured in media containing either 2i inhibitors, NME7_{AB} or both. As can be seen, 2i inhibitors are also able to induce the transition of cancer cells to a more metastatic state. Ovarian cancer cell lines SK-OV3, OV-90, pancreatic cancer cell lines CAPAN-2 and PANC-1, breast cancer cell line MDA-MB all displayed the morphological transition of going from adherent to non-adherent when cultured in NME7_{AB} and or 2i inhibitors.

FIG. 20 shows graphs of RT-PCR measurement of metastatic or pluripotency markers for various cancer cell lines cultured for 72 or 144 hours in NME7_{AB}. FIG. 20A shows that SK-OV3 cells increase expression of metastatic markers CHD1, SOX2 and NME7-X1 when cultured in NME7_{AB}. FIG. 20B shows that OV-90 cells increase expression of metastatic markers CXCR4 and NME7-X1 after culture in NME7_{AB}.

Example 6—Demonstration that Cancer Cells Cultured in NME7 Become Metastatic

A functional test of whether or not a population of cancer cells is metastatic is to implant very low numbers, e.g. 200, of the cells in immuno-compromised mice and see if they develop into a tumor. Typically 5-6 million cancer cells are required to form a tumor in an immuno-compromised mouse. We showed that as few as 50 of the NME-induced metastatic cancer cells formed tumors in mice. In addition, mice that were injected throughout the test period with human NME7_{AB}, NME1, or NME7-X1 developed remote metastases.

T47D human breast cancer cells were cultured in standard RPMI media for 14 days with media changes every 48 hours and passed by trypsinization when approximately 75% confluent. The cells were then plated into 6-well plates and cultured in minimal stem cell media (see Example 1) that was supplemented with 4 nM NME7_{AB}. Media was changed every 48 hours. By about Day 4, some cells become detached from the surface and float. Media is carefully changed so as to retain the “floaters” as these are the cells that have the highest metastatic potential as evidenced by

RT-PCR measurement of metastatic markers. On Day 7 or 8, the floaters are harvested and counted. Samples are retained for RT-PCR measurement. The key marker measured is CXCR4 which is up-regulated by 40-200 times after being briefly cultured in NME7_{AB}.

The freshly harvested floater metastatic cells are xenografted into the flank of female nu/nu athymic mice that have been implanted with 90-day slow release estrogen pellets. Floater cells were xenografted as 10,000, 1,000, 100 or 50 cells each. Half of the mice in each group of 6 were also injected daily with 32 nM NME7_{AB} near the original implantation site. The parent T47D cells that were cultured in RPMI media without NME7_{AB} were also implanted into mice as 6 million, 10,000 or 100 as controls. Mice implanted with the NME7-induced floater cells developed tumors even when as few as 50 cells were implanted. Mice that were implanted with the floater cells and that received daily injections of NME7_{AB} also developed remote tumors or remote metastases in various organs. 11 out of the 12 mice, or 92%, that were injected with human NME7_{AB} after implantation of the NME7_{AB} cultured cancer cells, developed tumors at the injection site. Only 7 out of the 12 mice, or 58%, that were not injected with human NME7_{AB} after implantation developed tumors. 9 out of the 11 mice, or 82%, that got tumors and were injected with human NME7_{AB} developed multiple tumors remote from the injection site. None of the mice that were not injected with NME7_{AB} developed multiple, visible tumors.

After sacrifice, RT-PCR and Western blots showed that the remote bumps on the mice injected with NME7_{AB} were indeed human breast tumors. Similar analysis of their organs showed that in addition to remote bumps, mice had randomly metastasized to the liver and lung with human breast cancer characteristic of the human breast cancer cells that were implanted. As expected, only the mice implanted with 6 million cells grew tumors.

Several experiments like the one described above were performed with essentially the same results. In each experiment, there were either 24 or 52 mice, including all proper controls.

Example 7—Peptides Selected Because their Sequence is Unique to NME7, A1, A2, B1, B2 and B3, Inhibit the Binding of NME7 Species to MUC1* Extracellular Domain Peptide

NME7 peptides were selected as immunizing agents for antibody production. NME7 peptides A1, A2, B1, B2 and B3 (FIG. 9) were chosen using a process of sequence alignment among human NME1, human NME7 and several bacterial NMEs that were homologous to human NME1 or human NME7. Five regions that had high sequence homology among all were identified. However, to prevent selecting peptides that would give rise to antibodies that would inhibit human NME1 as well as human NME7, we chose NME7 sequences that were adjacent to the homologous regions wherein those peptides had sequences that were different from human NME1. We did ELISA assays to see if the peptides on their own could bind to a synthetic MUC1* peptide on the surface and inhibit the binding of human NME7 or human NME1 to the immobilized peptide (FIG. 11). FIG. 11 shows that the peptides inhibited the binding of NME7 and NME1 to the immobilized PSMGFR peptide. Recall that each of the NME7 A domain and B domain can bind to a PSMGFR peptide. Therefore complete inhibition of NME7_{AB} binding to a PSMGFR peptide cannot be accomplished with a single antibody or peptide that is derived from

just one domain. This showed that those regions from which the peptides were derived were the regions that interacted with MUC1* and would give rise to antibodies that would bind to those regions of NME7 and inhibit its binding to MUC1* receptor.

In another experiment, the free peptides A1, A2, B1, B2 and B3 were added to cancer cells in culture that were undergoing transition to a more metastatic state by culturing in either NME7_{AB} or 2i. FIG. 14 shows a table of scientist observations when cancer cells are grown in either NME7_{AB} or 2i inhibitors, and shows that the free peptides inhibited the morphological change from adherent cells to floaters, which for breast cancer cells is directly correlated to increased expression of metastatic markers, especially CXCR4. RT-PCR measurements confirm that the NME7_{AB} peptides inhibited the increase in expression of metastasis marker CXCR4.

FIG. 15 shows a graph of RT-PCR measurements of CXCR4 expression in T47D breast cancer cells that were grown in either NME7_{AB} or 2i inhibitors, each of which transform cancer cells to a more metastatic state, and the inhibitory effect of NME7-derived peptides, A1, A2, B1, B2 and B3, on the metastatic transformation. FIG. 32 shows a table of recorded RNA levels in samples that were used for RT-PCR measurement of CXCR4 in FIG. 15 as well as the threshold cycle number for CXCR4 expression as well as for the control housekeeping gene.

Example 8—Anti-NME7 Antibodies Specifically Bind to Human NME7 but not to Human NME1

A standard ELISA assay was performed to determine whether or not the NME7 antibodies we generated by immunization with NME7_{AB} peptides A1, A2, B1, B2, and B3 would bind specifically to NME7_{AB}, but not to human NME1 as it has healthy functions and it may be detrimental to a human to block it with an antibody. The ELISAs of FIG. 24-25 show that all of the NME7 antibodies that were generated from peptides A1, A2, B1, B2, and B3 bind to human NME7_{AB} (FIG. 24) but not to human NME1 (FIG. 25). The peptides used to generate these antibodies are common to both NME7_{AB} and NME7-X1. This assays show that the antibodies generated from peptides A1, A2, B1, B2, and B3 specifically bind to NME7_{AB} and by extension will bind to NME7-X1.

NME7A peptide 1 (A domain):
MLSRKEALDFHVDHQS (SEQ ID NO:141)
NME7A peptide 2 (A domain): SGVARTDASES (SEQ ID NO:142)
NME7B peptide 1 (B domain): DAGFEI-SAMQMFNMDRVNVE (SEQ ID NO: 143)
NME7B peptide 2 (B domain):
EYVKGVVTEYHDMVTE (SEQ ID NO:144)
NME7B peptide 3 (B domain): AIFGKTKIQ-NAVHCTDLPEDGLLEVQYFF (SEQ ID NO:145)

Example 9—Anti-NME7 Specific Antibodies and the Peptides that Generated them Inhibit Cancer Cell Growth

Rabbits were immunized with NME7 peptides A1, A2, B1, B2, and B3 and antibodies were generated, collected and purified over a column to which the immunizing peptide had been conjugated. T47D breast cancer cells were plated and cultured according to ATCC protocols in RPMI media supplemented with serum. Antibodies generated from immunization with peptides A1, A2, B1, B2, and B3 were

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added at the concentrations indicated in FIG. 12. Immunizing peptides A1, A2, B1, B2, and B3, and the PSMGFR extracellular domain peptide of MUC1*, "FLR" here, were also added separately to growing T47D breast cancer cells. Taxol and the B6 anti-MUC1* Fab were added as controls. The graph of FIG. 12 shows that the antibodies generated, as well as the free peptides, potently inhibited the growth of the cancer cells. Note the comparison to inhibition using Taxol, which is a chemotherapy agent that kills healthy and cancer cells alike. Also, for comparison, a polyclonal antibody generated using a large stretch of NME7 from amino acid 100 to 376 is shown. Although this antibody is a potent inhibitor of cancer growth it could have non-specific effects since it can bind to NME1 as well as to NME7.

In a similar experiment, combinations of the antibodies generated from immunization with peptides A1, A2, B1, B2, and B3 as well as the peptides themselves were added to growing cancer cells at the concentrations indicated. The graphs of cell growth shown in FIG. 13 show that the combinations of antibodies and peptides potently inhibited the growth of cancer cells. In these two experiments, the cells were MUC1* positive breast cancer cells.

Example 10—Anti-NME7 Antibodies Inhibit the Transition of Cancer Cells to Metastatic Cancer Cells

Cancer cells transform to a more metastatic state when cultured in the presence of agents that revert stem cells to a more naïve state. We have demonstrated that culturing cancer cells in NME7_{AB}, human NME1 dimers, bacterial NME1 dimers or MEK and GSK3-beta inhibitors, called "2i", causes the cells to become more metastatic. As the cells transition to a more metastatic state, they become non-adherent and float off of the culture plate. These floating cells, "floaters" were collected separately from those that were adherent and were shown to: a) express much higher levels of metastatic genes; and b) when xenografted into mice, the floater cells were able to generate tumors when implanted at very low numbers. RT-PCR measurement of specific metastatic markers such as CXCR4 in breast cancers, CHD1 in prostate cancer, and other pluripotent stem cell markers such as OCT4, SOX2, NANOG, KLF4, c-Myc and others were dramatically over-expressed in cancer cells that were cultured in NME7_{AB} and most over-expressed in the cells that became non-adherent, called "floaters" here and in figures.

Here we show that the NME7-specific antibodies, generated by immunization with NME7-derived peptides A1, A2, B1, B2 and B3, as well as the peptides themselves, inhibit the transition from cancer cell to metastatic cancer cells. In the first of these experiments, the antibodies generated by immunization with A1, A2, B1, B2 and B3 were tested for their ability to inhibit the metastatic transition induced by culture of T47D breast cancer cells in NME7_{AB} or in 2i inhibitors. The most striking observation was that the antibodies and the peptides dramatically reduced the number of floater cells, which was the first indication that the antibodies and peptides had inhibited the transformation to metastatic cancer cells. In particular, cells to which the antibody generated from immunization with the B3 peptide barely generated any floater cells.

FIG. 14 shows the recorded observations of the percentage of floater cells visible for each antibody relative to the control wells that did not receive any antibody treatment. mRNA was extracted from both the floater cells and the adherent cells. RT-PCR was used to measure expression

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levels of metastatic markers, including CXCR4. Treatment with the anti-NME7 antibodies greatly reduced the amount of metastatic markers, such as CXCR4, indicating the antibodies inhibited the transition to metastatic cancer. (See FIG. 15). Notably, the antibody generated by immunization with peptide B3, aka antibody #61, essentially completely inhibited the transition to a more metastatic state. FIG. 15B shows that breast cancer cells that were treated with the NME7_{AB} peptides, A1, A2, B1, B2 and B3, alone were able to potently inhibit the transition to a more metastatic state induced by culturing the cells in a media containing the 2i inhibitors. Peptide B3 was especially effective as was antibody #61 that it generated. FIG. 15C shows the same graph but with the Y-axis expanded to show the peptide inhibition of metastatic markers. The amount of mRNA, which indicates cell viability and growth, was measured. Cells that were treated with antibody had much less mRNA, indicating that in addition to inhibiting the transition to a more metastatic state, the anti-NME7_{AB} antibodies inhibited the growth of the cancer cells. FIG. 16 shows a table of the amounts of RNA recovered for the inhibition experiment shown in FIG. 15A.

Example 11—Anti-NME7 Antibodies Generated with NME7-Derived Peptides A1, A2, B1, B2 and B3 Identify Novel NME7 Species not Detectable Using any Commercially Available Antibodies

As is known to those skilled in the art, some antibodies recognize a linear portion of the target protein and can be used in Western blot assays while other antibodies recognize a non-linear conformational motif and can be used in pull-down or immunoprecipitation assays. Previous to this application, cleaved NME7 or isoform NME7-X1 was not known to exist. Using antibodies that were commercially available at the time of filing shows that existing antibodies could not specifically detect these important NME7 species. B9 (Santa Cruz Biotechnology) is a monoclonal antibody raised against NME7 amino acids 100-376. FIG. 19D-19F shows that it only detects full-length 42 kDa NME7. Another commercially available antibody, H278, is a rabbit polyclonal raised against NME7 amino acids 100-376, which includes amino acid sequences that are not unique to NME7. FIG. 19D-19F shows that this antibody also stains NME1, which is 17 kDa as well as full-length NME7 and other bands that do not appear to be specific to NME7_{AB}.

NME7 antibodies generated by immunization with NME7_{AB} peptides A1, A2, B1, B2 or B3 identify new NME7 species including the full-length 42 kDa protein, a ~33 kDa NME7 species that may be a cleavage product or alternative isoform, a ~30 kDa NME7 species that may be a cleavage product or alternative isoform, wherein the ~30 kDa species appears to be NME7-X1. FIG. 19A-C shows that antibodies generated by peptides A1, B1 and B3 identify the secreted forms of NME7, NME7_{AB} and NME7-X1 in a wide range of cancer cell lines, including T47D breast cancer cells, PC3 and DU145 prostate cancer cells, HEK293 fetal liver cells, and leukemia cells IM-9, K562, and MV411.

Example 12—Generation of Anti-NME7 Antibodies

A synthetic peptide having the sequence of the B3 region of NME7, AIFGKTKIQNAVHCTDLPEDGLLEVQYFFC (SEQ ID NO: 1142), was used to immunize rabbits. Antibodies that resulted from immunization with NME7 peptide B3 inhibited the growth of MUC1* positive cancer cells and also inhibited the formation of cancer stem cells, which are

characterized by upregulation of metastatic markers, ability to grow anchorage independently, and are able to form tumors in animals from as few as 200 cells, whereas regular cancer cells typically require implantation of about 4 million cells for tumor engraftment.

In some cases, the NME7 B3 peptide was made with a C14A or C14V mutation. This sequence more reproducibly generated anti-NME7 antibodies.

Monoclonal antibodies were generated in mice according to standard methods by immunizing with NME7 B3, B3 with C14A mutation, or B3 with C14V mutation. The antibodies listed were selected because of their ability to bind to NME7, NME7-X1, NME7_{AB}, but importantly did not bind to NME1, which is thought to be required for some normal cellular functions. These antibodies also bind to the NME7 derived peptides B3, B3 with C14A mutation, and B3 with C14V mutation.

Experiments showed that these anti-NME7 antibodies inhibited the binding of NME7 to the MUC1* extra cellular domain, but did not block the binding of NME1 to the MUC1* extra cellular domain peptide. Further, the antibodies inhibited the formation of cancer stem cells.

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- Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the invention specifically described herein. Such equivalents are intended to be encompassed in the scope of the claims.

SEQUENCE LISTING

Sequence total quantity: 1213

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 note = MUC1 Receptor - Mucin 1 precursor, Genbank Accession number: P15941
 source 1..1255
 mol_type = protein
 organism = synthetic construct

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                      organism = synthetic construct

SEQUENCE: 13
DPETMNHSEF FVFIAEWYDP NASLLRRYEL LFYPGDSGVE MHDVKNHRTF LKRTKYDNLH 60
LEDLPIGNKV NVFSRQLVLI DYGDQYTARQ LGSRKETLA LIKPDASKA GEIIEIINKA 120
GFTITKLKMM MLRKEALDF HVDHQSRPFF NELIQFITG PIIAMEILRD DAICEWKRL 180
GPANSGVART DASESIRALF GTDGIRNAAH GPDSFASAA EMELFFPSSG GCGPANTAKF 240
TNCTCCIVKP HAVSEGMLNT LYSVHFVNRR AMFIFLMYFM YRK                                           283

SEQ ID NO: 14         moltype = DNA length = 534
FEATURE               Location/Qualifiers
misc_feature          1..534
                      note = NM23-H1 nucleotide sequence (NM23-H1: GENBANK
                      ACCESSION AF487339)
source                1..534
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 14
atgggtgctac tgtctacttt agggatcgct tttcaaggcg aggggctccc tatctcaagc 60
tgtgatacac gaaccatggc caactgtgag cgtaccttca ttgcgatcaa accagatggg 120
gtccagcggg gtcttctggg agagattatc aagcgttttg agcagaaagg attccgcctt 180
gttggtctga aatcctatga agcttccgaa gatcttctca aggaacacta cgttgacctg 240
aaggaccgtc cattcttttg cggcctgggt aaatacatgc actcaggggc ggtagttgcc 300
atgggtctggg aggggctgaa tgggtggaag acgggcccag tcatgctcgg ggagaccaac 360
cctgcagact ccaagcctgg gaccatccgt ggagacttct gcatacaagt tggcagggaac 420
attatcatg gcagtgattc tgtggagagt gcagagaagg agatcggtgt gtggtttcac 480
cctgaggaaac tggtagatta cagcagctgt gctcagaact ggatctatga atga 534

SEQ ID NO: 15         moltype = AA length = 177
FEATURE               Location/Qualifiers
REGION               1..177
                      note = NM23-H1 describes amino acid sequence (NM23-H1:
                      GENBANK ACCESSION AF487339)
source                1..177
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 15
MVLSTLGLIV FQEGGPIISS CDTGTMANCE RTFIAIKPDG VQRGLVGEII KRFEQKGFRL 60
VGLKFMQASE DLLKEHYVDL KDRPFFAGLV KYMHSGPVVA MVWGLNVVK TGRVMLGETN 120
PADSKPGTIR GDFCIQVGRN IIHGSDSVES AEKEIGLWFH PEELVDYTSQ AQNWIYE 177

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SEQ ID NO: 16      moltype = DNA length = 534
FEATURE           Location/Qualifiers
misc_feature       1..534
                   note = NM23-H1 S120G mutant nucleotide sequence (NM23-H1:
                   GENBANK ACCESSION AF487339)
source            1..534
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 16
atggtgctac tgtctacttt agggatcgctc tttcaaggcg aggggcctcc tatctcaagc 60
tgtgatacac gaaccatggc caactgtgag cgtacottca ttgcgatcaa accagatggg 120
gtccagcggg gtcttgtggg agagattatc aagcgttttg agcagaaagg attccgcctt 180
gttggcttga aattcatgca agcttccgaa gatcttctca aggaacacta cgttgacctg 240
aaggaccgtc cattctttgc cggcctgggt aaatacatgc actcagggcc ggtagttgcc 300
atggtctggg aggggctgaa tgtggtgaag acgggccgag tcatgctcgg ggagaccaac 360
ctgcagact ccaagcctgg gaccatccgt ggagacttct gcatacaagt tggcaggaac 420
attatacatg gcggtgattc tgtggagagt gcagagaagg agatcggctt gtggtttcac 480
cctgaggaac tggtagatta cacgagctgt gctcagaact ggatctatga atga 534

```

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SEQ ID NO: 17      moltype = AA length = 177
FEATURE           Location/Qualifiers
REGION           1..177
                   note = NM23-H1 S120G mutant amino acid sequence (NM23-H1:
                   GENBANK ACCESSION AF487339)
source            1..177
                   mol_type = protein
                   organism = synthetic construct

```

```

SEQUENCE: 17
MVLSTLGLIV PQEGGPPISS CDTGTMANCE RTFIAIKPDG VQRGLVGEII KRFEQKGFRL 60
VGLKFMQASE DLLKEHYVDL KDRPFFAGLV KYMHS GPVVA MVWEGLVVK TGRVMLGETN 120
PADSKPGTIR GDFCIQVGRN IHHGDSVES AEKEIGLWFH PEELVDYTSC AQNWIYE 177

```

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SEQ ID NO: 18      moltype = DNA length = 459
FEATURE           Location/Qualifiers
misc_feature       1..459
                   note = NM23-H2 nucleotide sequence (NM23-H2: GENBANK
                   ACCESSION AK313448)
source            1..459
                   mol_type = other DNA
                   organism = synthetic construct

```

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SEQUENCE: 18
atggccaacc tggagcgcac cttcatcgcc atcaagccgg acggcgtgca gcgcggcctg 60
gtggcgcgaga tcatcaagcg cttcgagcag aagggattcc gcctcgtggc catgaagtgc 120
ctccggggcct ctgaagaaca cctgaagcag cactacattg acctgaaaga ccgaccattc 180
ttccctggggc tgggtgaagta catgaactca gggccgggtg tggccatggt ctgggagggg 240
ctgaacgtgg tgaagacagg ccgagtgatg cttggggaga ccaatccagc agattcaaag 300
ccaggcacca ttctgtggga cttctgcatt cagggttgga ggaacatcat tcatggcagt 360
gattcagtaa aaagtgtcga aaaagaaatc agcctatggt ttaagcctga agaactgggt 420
gactacaagt cttgtgctca tgactgggtc tatgaataa 459

```

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SEQ ID NO: 19      moltype = AA length = 152
FEATURE           Location/Qualifiers
REGION           1..152
                   note = NM23-H2 amino acid sequence (NM23-H2: GENBANK
                   ACCESSION AK313448)
source            1..152
                   mol_type = protein
                   organism = synthetic construct

```

```

SEQUENCE: 19
MANLERTFIA IKPDGVQRGL VGEIIKRFEQ KGFRVLVAMKF LRASEEHLKQ HYIDLKDRPF 60
FPGLVKYMNS GPVVAMVWEG LNVVKTGRVM LGETNPADSK PGTIRGDFCI QVGRNIIHGS 120
DSVKSAAKEI SLWFKPEELV DYKSCAHDWV YE 152

```

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SEQ ID NO: 20      moltype = DNA length = 1023
FEATURE           Location/Qualifiers
misc_feature       1..1023
                   note = Human NM23-H7-2 sequence optimized for E. coli
                   expression
source            1..1023
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 20
atgcatgacg ttaaaaaatca ccgtaccttt ctgaaacgca cgaaatatga taatctgcat 60
ctggaagacc tgtttatttg caacaaagtc aatgtgttct ctcgctcagct ggtgctgac 120
gattatggcg accagtacac cgcgcgtcaa ctgggtatgc gcaaagaaaa aacgctggcc 180
ctgattaaac cggatgcaat ctccaaagct ggcgaaatta tcgaaattat caacaaagcg 240
ggtttcacca tcacgaaatc gaaaatgatg atgctgagcc gtaaagaagc cctggatttt 300
catgtcgacc accagtctcg ccggtttttc aatgaactga ttcaattcat caccacgggt 360
ccgattatcg caatggaaat tctgcgtgat gacgctatct gcgaatggaa acgcctgctg 420

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ggccccggcaa actcaggtgt tgcgcgtacc gatgccagt gattccattcg cgctctgttt 480
ggcacccgatg gtatocgtaa tgcagacat ggtccggact cattcgcatc ggcagctcgt 540
gaaatggaaac tgtttttccc gagctctgtgc ggttgcggtc cggcaaacac cgccaaattt 600
accaattgta cgtgctgtat tgtcaaaccg cagcagagt cagaaggcct gctgggtaaa 660
attctgatgg caatocgtga tgcgtgcttt gaaatctcgg ccatgcagat gttcaacatg 720
gaccgcgtta acgtcgaaga attctacgaa gtttacaagg gcgtgggttac cgaatatcac 780
gatatggta cggaaatgta ctccggtccg tgcgtcgcga tggaaattca gcaaaacaat 840
gccacaaaaa cgtttcgtga attctgtggt cgggcagatc cggaaatcgc acgtcatctg 900
cgteccggta ccctgcgcgc aatttttggg aaacgaaaaa tccagaacgc tgtgactgt 960
accgatctgc cggaagacgg tctgctggaa gttcaatact ttttcaaat tctggataat 1020
tga 1023

SEQ ID NO: 21      moltype = AA length = 340
FEATURE           Location/Qualifiers
REGION           1..340
                  note = Human NM23-H7-2 sequence optimized for E. coli
                  expression
source           1..340
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 21
MHDVKNHRTF LKRTKYDNLH LEDLFIGNKV NVFSRQLVLI DYGDQYTARQ LGSRKEKTLA 60
LIKPDaisKA GEIIEIINKA GFTITKLKMM MLsrKEALDF HVDHQSRPFF NELIQFITTG 120
PIIAMEILRD DAICEWKRLl GPANSGVART DASESIRALF GTDGIRNAAH GPDSFASAAr 180
EMELFFPSSG GCGPANTAKF TNCTCCIVKP HAVSEGLLGK ILMAIRDAGF EISAMQMFNM 240
DRVNVVEFYE VYKGVVTEYH DMVTEMYSGP CVAMEIQQNN ATKTFREFCG PADPEIARHL 300
RPGTLRAIFG KTKIQNAVHC TDLPEdGLLE VQYFFKILDN 340

SEQ ID NO: 22      moltype = DNA length = 399
FEATURE           Location/Qualifiers
misc_feature      1..399
                  note = Human NME7-A
source           1..399
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 22
atggaaaaaa cgctagccct aattaaacca gatgcaatat caaaggctgg agaaataatt 60
gaaataataa acaagctggg atttactata accaaactca aaatgatgat gctttcaagg 120
aaagaagcat tggattttca tgtagatcac cagtcaagac cctttttcaa tgagctgatc 180
cagtttatta caactggtcc tattattgcc atggagattt taagagatga tgctatatgt 240
gaatggaaaa gactgctggg acctgcaaac tctggagtgg cagcacaga tgcttctgaa 300
agcattagag ccctcttggg aacagatggc ataagaaatg cagcgcatgg ccctgattct 360
tttgcttctg cggccagaga aatggagttg tttttttga 399

SEQ ID NO: 23      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION           1..132
                  note = Human NME7-A
source           1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 23
MEKTLALIKP DAISKAGEII EIINKAGFTI TKLKMMLSR KEALDFHVDH QSRPFFNELI 60
QFITTGPIIA MEILRDDAIC EWKRLlGPAN SGVARTDASE SIRALFGTDG IRNAAHGPDs 120
FASAREMEL FF 132

SEQ ID NO: 24      moltype = DNA length = 444
FEATURE           Location/Qualifiers
misc_feature      1..444
                  note = Human NME7-A1
source           1..444
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 24
atggaaaaaa cgctagccct aattaaacca gatgcaatat caaaggctgg agaaataatt 60
gaaataataa acaagctggg atttactata accaaactca aaatgatgat gctttcaagg 120
aaagaagcat tggattttca tgtagatcac cagtcaagac cctttttcaa tgagctgatc 180
cagtttatta caactggtcc tattattgcc atggagattt taagagatga tgctatatgt 240
gaatggaaaa gactgctggg acctgcaaac tctggagtgg cagcacaga tgcttctgaa 300
agcattagag ccctcttggg aacagatggc ataagaaatg cagcgcatgg ccctgattct 360
tttgcttctg cggccagaga aatggagttg ttttttcctt caagtggagg ttgtgggccg 420
gcaaacactg ctaaatttac ttga 444

SEQ ID NO: 25      moltype = AA length = 147
FEATURE           Location/Qualifiers
REGION           1..147
                  note = Human NME7-A1
source           1..147
                  mol_type = protein

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                organism = synthetic construct
SEQUENCE: 25
MEKTLALIKP DAISKAGEII EIINKAGFTI TKLKMMMLSR KEALDFHVDH QSRPFFNELI 60
QFITTGPIIA MEILRDDAIC EWKRLGPN SGVARTDASE SIRALFGTDG IRNAAHGPD 120
FASAAREMEL FFPSSGGCGP ANTAKFT 147

SEQ ID NO: 26      moltype = DNA length = 669
FEATURE           Location/Qualifiers
misc_feature      1..669
                  note = Human NME7-A2
source            1..669
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 26
atgaatcata gtgaagatt cgttttcatt gcagagtgg atgatccaaa tgcttcactt 60
cttcgacgtt atgagctttt attttaccca ggggatggat ctgttgaat gcatgatgta 120
aagaatcatc gcaccttttt aaagcggacc aaatatgata acctgcactt ggaagattta 180
tttataggca acaaagttaa tgccttttct cgacaactgg tattaattga ctatggggat 240
caatatacag ctgcgccagt gggcagtagg aaagaaaaaa cgctagccct aattaaacca 300
gatgcaatat caaaggctgg agaaataatt gaaataataa acaaagctgg atttactata 360
accaaactca aaatgatgat gctttcaagg aaagaagcat tggattttca tgtagatcac 420
cagtcaagac cctttttcaa tgagctgac cagtttatta caactggtcc tattattgcc 480
atggagattt taagagatga tgctatatgt gaatggaaaa gactgctggg acctgcaaac 540
tctggagtgg cagcacaga tgcttctgaa agcattagag ccctctttgg aacagatggc 600
ataagaaatg cagcgcattg ccctgattct tttgcttctg cggccagaga aatggagtgt 660
tttttttga 669

SEQ ID NO: 27      moltype = AA length = 222
FEATURE           Location/Qualifiers
REGION            1..222
                  note = Human NME7-A2
source            1..222
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 27
MNHSERFVFI AEWYDPNASL LRRYELLFYP GDGSVEMHDV KNHRTFLKRT KYDNLHLEDL 60
FIGNKVNVFS RQLVLIDYGD QYTARQLGSR KEKTLALIKP DAISKAGEII EIINKAGFTI 120
TKLKMMMLSR KEALDFHVDH QSRPFFNELI QFITTGPIIA MEILRDDAIC EWKRLGPN 180
SGVARTDASE SIRALFGTDG IRNAAHGPD FASAAREMEL FF 222

SEQ ID NO: 28      moltype = DNA length = 714
FEATURE           Location/Qualifiers
misc_feature      1..714
                  note = Human NME7-A3
source            1..714
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 28
atgaatcata gtgaagatt cgttttcatt gcagagtgg atgatccaaa tgcttcactt 60
cttcgacgtt atgagctttt attttaccca ggggatggat ctgttgaat gcatgatgta 120
aagaatcatc gcaccttttt aaagcggacc aaatatgata acctgcactt ggaagattta 180
tttataggca acaaagttaa tgccttttct cgacaactgg tattaattga ctatggggat 240
caatatacag ctgcgccagt gggcagtagg aaagaaaaaa cgctagccct aattaaacca 300
gatgcaatat caaaggctgg agaaataatt gaaataataa acaaagctgg atttactata 360
accaaactca aaatgatgat gctttcaagg aaagaagcat tggattttca tgtagatcac 420
cagtcaagac cctttttcaa tgagctgac cagtttatta caactggtcc tattattgcc 480
atggagattt taagagatga tgctatatgt gaatggaaaa gactgctggg acctgcaaac 540
tctggagtgg cagcacaga tgcttctgaa agcattagag ccctctttgg aacagatggc 600
ataagaaatg cagcgcattg ccctgattct tttgcttctg cggccagaga aatggagtgt 660
tttttcctt caagtggagg tttgtggccg gcaaacactg ctaaatttac ttga 714

SEQ ID NO: 29      moltype = AA length = 237
FEATURE           Location/Qualifiers
REGION            1..237
                  note = Human NME7-A3
source            1..237
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 29
MNHSERFVFI AEWYDPNASL LRRYELLFYP GDGSVEMHDV KNHRTFLKRT KYDNLHLEDL 60
FIGNKVNVFS RQLVLIDYGD QYTARQLGSR KEKTLALIKP DAISKAGEII EIINKAGFTI 120
TKLKMMMLSR KEALDFHVDH QSRPFFNELI QFITTGPIIA MEILRDDAIC EWKRLGPN 180
SGVARTDASE SIRALFGTDG IRNAAHGPD FASAAREMEL FFPSSGGCGP ANTAKFT 237

SEQ ID NO: 30      moltype = DNA length = 408
FEATURE           Location/Qualifiers
misc_feature      1..408
                  note = Human NME7-B
source            1..408

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 30
atgaattgta cctgttgcat tgttaaacc ccatgctgtca gtgaaggact gttgggaaag 60
atcctgatgg ctatccgaga tgcaggtttt gaaatctcag ctatgcagat gttcaatatg 120
gatcgggtta atgttgagga attctatgaa gtttataaag gagtagtgac cgaatatcat 180
gacatggtga cagaaatgta ttctggccct tgtgtagcaa tggagattca acagaataat 240
gctacaaaga catttcgaga attttgtgga cctgctgac ctgaaattgc ccggcattta 300
cgccctggaa ctctcagagc aatctttgtt aaaactaaga tccagaatgc tgttcactgt 360
actgatctgc cagaggatgg cctattagag gttcaatact tcttctga 408

SEQ ID NO: 31      moltype = AA  length = 135
FEATURE           Location/Qualifiers
REGION            1..135
                  note = Human NME7-B
source            1..135
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 31
MNCTCCIVKP HAVSEGLLGK ILMAIRDAGF EISAMQMFNM DRVNVVEEYF VYKGVVTEYH 60
DMVTEMYSGP CVAMEIQNNN ATKTFREFCG PADPEIARHL RPYTLRAIFG KTKIQNAVHC 120
TDLPEDGLLE VQYFF      135

SEQ ID NO: 32      moltype = DNA  length = 426
FEATURE           Location/Qualifiers
misc_feature       1..426
                  note = Human NME7-B1
source            1..426
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 32
atgaattgta cctgttgcat tgttaaacc ccatgctgtca gtgaaggact gttgggaaag 60
atcctgatgg ctatccgaga tgcaggtttt gaaatctcag ctatgcagat gttcaatatg 120
gatcgggtta atgttgagga attctatgaa gtttataaag gagtagtgac cgaatatcat 180
gacatggtga cagaaatgta ttctggccct tgtgtagcaa tggagattca acagaataat 240
gctacaaaga catttcgaga attttgtgga cctgctgac ctgaaattgc ccggcattta 300
cgccctggaa ctctcagagc aatctttgtt aaaactaaga tccagaatgc tgttcactgt 360
actgatctgc cagaggatgg cctattagag gttcaatact tcttcaagat cttggataat 420
tagtga      426

SEQ ID NO: 33      moltype = AA  length = 140
FEATURE           Location/Qualifiers
REGION            1..140
                  note = Human NME7-B1
source            1..140
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 33
MNCTCCIVKP HAVSEGLLGK ILMAIRDAGF EISAMQMFNM DRVNVVEEYF VYKGVVTEYH 60
DMVTEMYSGP CVAMEIQNNN ATKTFREFCG PADPEIARHL RPYTLRAIFG KTKIQNAVHC 120
TDLPEDGLLE VQYFFKILDN      140

SEQ ID NO: 34      moltype = DNA  length = 453
FEATURE           Location/Qualifiers
misc_feature       1..453
                  note = Human NME7-B2
source            1..453
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 34
atgccttcaa gtggagggtg tgggccggca aacactgcta aatttactaa ttgtacctgt 60
tgcattgtta aaccccatgc tgtcagttaa ggactgttgg gaaagatcct gatggctatc 120
cgagatgcag gttttgaaat ctcagctatg cagatgttca atatggatcg ggttaatgtt 180
gaggaattct atgaagttaa taaaggagta gtgaccgaat atcatgacat ggtgacagaa 240
atgtattctg gcccttgtgt agcaatggag attcaacaga ataatgctac aaagacattt 300
cgagaatttt gtggacctgc tgatcctgaa attgcccggc atttacgcc tggaaactctc 360
agagcaatct ttggtaaaac taagatccag aatgctgttc actgtactga tctgccagag 420
gatggcctat tagaggttca atactcttc tga      453

SEQ ID NO: 35      moltype = AA  length = 150
FEATURE           Location/Qualifiers
REGION            1..150
                  note = Human NME7-B2
source            1..150
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 35
MPSSGGCGPA NTAKFTNCTC CIVKPHAVSE GLLGKILMAI RDAGFEISAM QMFNMDRVNV 60
EEFYEVYKGV VTEYHDMVTE MYSGPCVAME IQQNNATKTF REFCGPADPE IARHLRPGTL 120

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RAIFGKTKIQ NAVHCTDLPE DGLLEVQYFF 150

SEQ ID NO: 36 moltype = DNA length = 471
 FEATURE Location/Qualifiers
 misc_feature 1..471
 note = Human NME7-B3
 source 1..471
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 36

atgccttcaa	gtggaggttg	tgggccggca	aacactgcta	aatttactaa	ttgtacctgt	60
tgcattgtta	aaccccatgc	tgtcagtgaa	ggactgttgg	gaaagatcct	gatggctatc	120
cgagatgcag	gttttgaat	ctcagctatg	cagatgttca	atatggatcg	ggttaatgtt	180
gaggaattct	atgaagttta	taaaggagta	gtgaccgaat	atcatgacat	ggtgacagaa	240
atgtattctg	gcccttgtgt	agcaatggag	attcaacaga	ataatgctac	aaagacattt	300
cgagaatttt	gtggacctgc	tgatcctgaa	attgcccggc	atttacgccc	tggaactctc	360
agagcaatct	ttggtaaaac	taagatccag	aatgctgttc	actgtactga	tctgccagag	420
gatggcctat	tagaggttca	atacttcttc	aagatcttgg	ataattagt	a	471

SEQ ID NO: 37 moltype = AA length = 155
 FEATURE Location/Qualifiers
 REGION 1..155
 note = Human NME7-B3
 source 1..155
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 37

MPSSGGCGPA	NTAKFTNCTC	CIVKPHAVSE	GLLGKILMAI	RDAGFEISAM	QMFMNDRVNV	60
EEFYEVYKGV	VTEYHDMVTE	MYSGPCVAME	IQNNNATKTF	REFCGPADPE	IARHLRPGTL	120
RAIFGKTKIQ	NAVHCTDLPE	DGLLEVQYFF	KILDN			155

SEQ ID NO: 38 moltype = DNA length = 864
 FEATURE Location/Qualifiers
 misc_feature 1..864
 note = Human NME7-AB, also known as NME7AB
 source 1..864
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 38

atggaaaaaa	cgctagccct	aattaaacca	gatgcaatat	caaaggctgg	agaaataatt	60
gaaataataa	acaaagctgg	atttactata	accaaactca	aaatgatgat	gctttcaagg	120
aaagaagcat	tggattttca	tgtagatcac	cagtcaagac	cctttttcaa	tgagctgac	180
cagtttatta	caactgggtcc	tattattgcc	atggagattt	taagagatga	tgctatatgt	240
gaatggaaaa	gactgctggg	acctgcaaac	tctggagtgg	cacgcacaga	tgcttctgaa	300
agcatttagag	ccctctttgg	aacagatggc	ataagaaatg	cagcgcatgg	ccctgattct	360
tttgcttctg	cgccagagaga	aatggagttg	tttttctctt	caagtggagg	ttgtgggccc	420
gcaaacactg	ctaaatttac	taattgtacc	tggttcattg	ttaaacccca	tgctgtcagt	480
gaaggactgt	tggaagagat	cctgatggct	atccgagatg	caggttttga	aatctcagct	540
atgcagatgt	tcaatatgga	tcgggttaat	gttgaggaat	tctatgaagt	ttataaagga	600
gtagtgcacc	aatatcatga	catgggtgaca	gaaatgtatt	ctggcccttg	tgtagcaatg	660
gagattcaac	agaataatgc	tacaaagaca	tttcgagaat	tttgtggacc	tgctgacct	720
gaaattggccc	ggcattttacg	ccttggaact	ctcagagcaa	tccttggtta	aactaagatc	780
cagaatgctg	tttactgtac	tgatctgcc	gaggatggcc	tattagaggt	tcaatacttc	840
ttcaagatct	tggaataatta	gtga				864

SEQ ID NO: 39 moltype = AA length = 286
 FEATURE Location/Qualifiers
 REGION 1..286
 note = Human NME7-AB, also known as NME7AB
 source 1..286
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 39

MEKTLALIKP	DAISKAGEII	EIINKAGFTI	TKLKMMMLSR	KEALDPHVDH	QSRPFFNELI	60
QFITTGPIIA	MEILRDAIC	EWKRLGPN	SGVARTDASE	SIRALFGTDG	IRNAAHGPD	120
FASAREMEL	FFPSSGGCGP	ANTAKFTNCT	CCIVKPHAVS	EGLLGKILMA	IRDAGFEISA	180
MQMFMNDRVN	EEFYEVYKGV	VTEYHDMVT	EMYSGPCVAM	EIQNNNATKT	FREFCGPADP	240
EIARHLRPGT	LRAIFGKTKI	QNAVHCTDLP	EDGLLEVQYF	FKILDN		286

SEQ ID NO: 40 moltype = DNA length = 846
 FEATURE Location/Qualifiers
 misc_feature 1..846
 note = Human NME7-AB1
 source 1..846
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 40

atggaaaaaa	cgctagccct	aattaaacca	gatgcaatat	caaaggctgg	agaaataatt	60
gaaataataa	acaaagctgg	atttactata	accaaactca	aaatgatgat	gctttcaagg	120

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aaagaagcat tggattttca tgtagatcac cagtcaagac cctttttcaa tgagctgate 180
cagtttatta caactgggtcc tattattgcc atggagattt taagagatga tgctatatgt 240
gaatggaaaa gactgctggg acctgcaaac tctggagtgg cagcgcacaga tgcttctgaa 300
agcattagag cctcttttgg aacagatggc ataagaaatg cagcgcattg cctgattct 360
tttgcttctg cggccagaga aatggagttg ttttttcctt caagtggagg ttgtgggccc 420
gcaaacactg ctaaatttac taattgtacc tgttgcatgg ttaaacccca tgctgtcagt 480
gaaggactgt tgggaaagat cctgatggct atccgagatg caggttttga aatctcagct 540
atgcagatgt tcaatatgga tcgggttaat gttgaggaat tctatgaagt ttataaagga 600
gtagtgcacc aatatcatga catggtgaca gaaatgtatt ctggcccttg tgtagcaatg 660
gagattcaac agaataatgc tacaaagaca ttctcgagaat ttgtggacc tgctgatcct 720
gaaattgccc ggcatttacg ccttggaact ctacagagcaa tctttggtaa aactaagatc 780
cagaatgctg ttcactgtac tgatctgcca gaggatggcc tattagaggt tcaatacttc 840
ttctga 846

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SEQ ID NO: 41      moltype = AA length = 281
FEATURE           Location/Qualifiers
REGION            1..281
                  note = Human NME7-AB1
source            1..281
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 41
MEKTLALIKP DAISKAGEII EIINKAGFTI TKLKMMMLSR KEALDFHVDH QSRPFFNELI 60
QFITTGPIIA MEILRDAIC EWKRLGPN SGVARTDASE SIRALFGTDG IRNAAHGPD 120
FASAAREMEL FFPSSGGCGP ANTAKFTNCT CCIVKPHAVS ELLGKILMA IRDAGFEISA 180
MQMFNMDRVN VEEFYEVYKG VVTEYHDMVT EMYSGPCVAM EIQQNNATKT FREFCGPADP 240
ETARHLRPGT LRAIFGKTKI QNAVHCTDLP EDGLLEVQYF F 281

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SEQ ID NO: 42      moltype = DNA length = 399
FEATURE           Location/Qualifiers
misc_feature       1..399
                  note = Human NME7-A sequence optimized for E. coli
                  expression
source            1..399
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 42
atggaaaaaa cgctggccct gattaaaccg gatgcaatct ccaaagctgg cgaaattatc 60
gaaattatca acaagcgagg ttccaccatc acgaaactga aaatgatgat gctgagccgt 120
aaagaagccc tggattttca tgtcgaccac cagtctcgcc cgtttttcaa tgaactgatt 180
caattcatca ccacgggttc gattatcgca atggaaattc tgcgtgatga cgctatctgc 240
gaatggaaac gcctgctggg cccggcaaac tcaggtgttg cgcgtaccga tgccagtga 300
tccattcgcg ctctgttttg caccgatggt atccgtaatg cagcacatgg tccggactca 360
ttcgcatcgg cagctcgtga aatggaactg tttttctga 399

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SEQ ID NO: 43      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = Human NME7-A sequence optimized for E. coli
                  expression
source            1..132
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 43
MEKTLALIKP DAISKAGEII EIINKAGFTI TKLKMMMLSR KEALDFHVDH QSRPFFNELI 60
QFITTGPIIA MEILRDAIC EWKRLGPN SGVARTDASE SIRALFGTDG IRNAAHGPD 120
FASAAREMEL FF 132

```

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SEQ ID NO: 44      moltype = DNA length = 444
FEATURE           Location/Qualifiers
misc_feature       1..444
                  note = Human NME7-A1 sequence optimized for E. coli
                  expression
source            1..444
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 44
atggaaaaaa cgctggccct gattaaaccg gatgcaatct ccaaagctgg cgaaattatc 60
gaaattatca acaagcgagg ttccaccatc acgaaactga aaatgatgat gctgagccgt 120
aaagaagccc tggattttca tgtcgaccac cagtctcgcc cgtttttcaa tgaactgatt 180
caattcatca ccacgggttc gattatcgca atggaaattc tgcgtgatga cgctatctgc 240
gaatggaaac gcctgctggg cccggcaaac tcaggtgttg cgcgtaccga tgccagtga 300
tccattcgcg ctctgttttg caccgatggt atccgtaatg cagcacatgg tccggactca 360
ttcgcatcgg cagctcgtga aatggaactg tttttcccga gctctggcgg ttgcggtccg 420
gcaaacaccg ccaaatttac ctga 444

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SEQ ID NO: 45      moltype = AA length = 147
FEATURE           Location/Qualifiers
REGION            1..147

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note = Human NME7-A1 sequence optimized for E. coli
expression
source      1..147
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 45
MEKTLALIKP DAISKAGEII EIINKAGFTI TKLKMMMLSR KEALDFHVDH QSRPFFNELI 60
QFITTGPIIA MEILRDDAIC EWKRLLG PAN SGVARTDASE SIRALFGTDG IRNAAHGPD 120
FASAAREMEL FFPSSGGCGP ANTAKFT 147

SEQ ID NO: 46      moltype = DNA length = 669
FEATURE            Location/Qualifiers
misc_feature        1..669
                    note = Human NME7-A2 sequence optimized for E. coli
                    expression
source              1..669
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 46
atgaatcact ccgaacgctt tgtttttatc gccgaatggt atgacccgaa tgettccctg 60
ctgcgccgct acgaactgct gttttatccg ggcgatggta gcgtggaaat gcatgacgtt 120
aaaaatcacc gtacctttct gaaacgcacg aaatatgata atctgcatct ggaagacctg 180
tttattggca acaaagtcac tgtgttctct cgtcagctgg tgctgatcga ttatggcgac 240
cagtacaccg cgcgctcaact gggtagtcgc aaagaaaaaa cgctggccct gattaaaccg 300
gatgcaatct ccaaagctgg cgaaattatc gaaattatca acaaagcggg ttccaccatc 360
acgaaactga aaatgatgat gctgagccgt aaagaagccc tggattttca tgcgaccac 420
cagtcctcgcc cgtttttcaa tgaactgatt caattcatca ccacgggtcc gattatcgca 480
atggaaatct tgcgtgatga cgctatctgc gaatggaaac gcctgctggg cccggcaaac 540
tcaggtgttg cgcgtaaccg tgcagtgaa tccattcgcg ctctgtttgg caccgatggt 600
atccgtaatg cagcacatgg tccggactca ttcgcatcgg cagctcgtga aatggaactg 660
tttttctga 669

SEQ ID NO: 47      moltype = AA length = 222
FEATURE            Location/Qualifiers
REGION              1..222
                    note = Human NME7-A2 sequence optimized for E. coli
                    expression
source              1..222
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 47
MNHSERFVFI AEWYDPNASL LRRYELLFYP GDGSVEMHDV KNHRTFLKRT KYDNLHLEDL 60
FIGNKVNVFS RQLVLIDYGD QYTARQLGSR KEKTLALIKP DAISKAGEII EIINKAGFTI 120
TKLKMMMLSR KEALDFHVDH QSRPFFNELI QFITTGPIIA MEILRDDAIC EWKRLLG PAN 180
SGVARTDASE SIRALFGTDG IRNAAHGPD FASAAREMEL FF 222

SEQ ID NO: 48      moltype = DNA length = 714
FEATURE            Location/Qualifiers
misc_feature        1..714
                    note = Human NME7-A3 sequence optimized for E. coli
                    expression
source              1..714
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 48
atgaatcact ccgaacgctt tgtttttatc gccgaatggt atgacccgaa tgettccctg 60
ctgcgccgct acgaactgct gttttatccg ggcgatggta gcgtggaaat gcatgacgtt 120
aaaaatcacc gtacctttct gaaacgcacg aaatatgata atctgcatct ggaagacctg 180
tttattggca acaaagtcac tgtgttctct cgtcagctgg tgctgatcga ttatggcgac 240
cagtacaccg cgcgctcaact gggtagtcgc aaagaaaaaa cgctggccct gattaaaccg 300
gatgcaatct ccaaagctgg cgaaattatc gaaattatca acaaagcggg ttccaccatc 360
acgaaactga aaatgatgat gctgagccgt aaagaagccc tggattttca tgcgaccac 420
cagtcctcgcc cgtttttcaa tgaactgatt caattcatca ccacgggtcc gattatcgca 480
atggaaatct tgcgtgatga cgctatctgc gaatggaaac gcctgctggg cccggcaaac 540
tcaggtgttg cgcgtaaccg tgcagtgaa tccattcgcg ctctgtttgg caccgatggt 600
atccgtaatg cagcacatgg tccggactca ttcgcatcgg cagctcgtga aatggaactg 660
tttttccgga gctctggcgg ttgcggtccg gcaaacaccg ccaaatttac ctga 714

SEQ ID NO: 49      moltype = AA length = 237
FEATURE            Location/Qualifiers
REGION              1..237
                    note = Human NME7-A3 sequence optimized for E. coli
                    expression
source              1..237
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 49
MNHSERFVFI AEWYDPNASL LRRYELLFYP GDGSVEMHDV KNHRTFLKRT KYDNLHLEDL 60
FIGNKVNVFS RQLVLIDYGD QYTARQLGSR KEKTLALIKP DAISKAGEII EIINKAGFTI 120

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TKLKMMLSR KEALDFHVDH QSRPFFNELI QFITTGPPIA MEILRDDAIC EWKRLLG PAN 180
 SGVARTDASE SIALRFGTDG IRNAAHGPD S FASAAREMEL FFPSSGGCGP ANTAKFT 237

SEQ ID NO: 50 moltype = DNA length = 408
 FEATURE Location/Qualifiers
 misc_feature 1..408
 note = Human NME7-B sequence optimized for E. coli
 expression
 source 1..408
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 50
 atgaattgta cgtgctgtat tgtcaaacccg caccgagtg cagaaggcct gctgggtaaa 60
 attctgatgg caatccgtga tgcctggcttt gaaatctcgg ccatgcagat gttcaacatg 120
 gaccgcgtta acgtcgaaga attctacgaa gtttacaagg gcgtgggttac cgaatatcac 180
 gatattggtta cggaaatgta ctccggtccg tgcgtcgcga tggaaattca gcaaaacaat 240
 gccacaaaaa cgtttcgtga attctgtggt ccggcagatc cggaaatcgc acgtcatctg 300
 cgtccgggta ccctgcgcgc aatttttggg aaaacgaaaa tccagaacgc tgtgcactgt 360
 accgatctgc cggaagacgg tctgctggaa gttcaatact ttttctga 408

SEQ ID NO: 51 moltype = AA length = 135
 FEATURE Location/Qualifiers
 REGION 1..135
 note = Human NME7-B sequence optimized for E. coli
 expression
 source 1..135
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 51
 MNCTCCIVKP HAVSEGLLGK ILMAIRDAGF EISAMQMFNM DRNVVEEFYE VYKGVVTEYH 60
 DMVTEMYSGP CVAMEIQNN ATKTFREFCG PADPEIARHL RPTGLRAIFG KTKIQNAVHC 120
 TDLPEDGLLE VQYFF 135

SEQ ID NO: 52 moltype = DNA length = 423
 FEATURE Location/Qualifiers
 misc_feature 1..423
 note = Human NME7-B1 sequence optimized for E. coli
 expression
 source 1..423
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 52
 atgaattgta cgtgctgtat tgtcaaacccg caccgagtg cagaaggcct gctgggtaaa 60
 attctgatgg caatccgtga tgcctggcttt gaaatctcgg ccatgcagat gttcaacatg 120
 gaccgcgtta acgtcgaaga attctacgaa gtttacaagg gcgtgggttac cgaatatcac 180
 gatattggtta cggaaatgta ctccggtccg tgcgtcgcga tggaaattca gcaaaacaat 240
 gccacaaaaa cgtttcgtga attctgtggt ccggcagatc cggaaatcgc acgtcatctg 300
 cgtccgggta ccctgcgcgc aatttttggg aaaacgaaaa tccagaacgc tgtgcactgt 360
 accgatctgc cggaagacgg tctgctggaa gttcaatact ttttcaaat tctggataat 420
 tga 423

SEQ ID NO: 53 moltype = AA length = 140
 FEATURE Location/Qualifiers
 REGION 1..140
 note = Human NME7-B1 sequence optimized for E. coli
 expression
 source 1..140
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 53
 MNCTCCIVKP HAVSEGLLGK ILMAIRDAGF EISAMQMFNM DRNVVEEFYE VYKGVVTEYH 60
 DMVTEMYSGP CVAMEIQNN ATKTFREFCG PADPEIARHL RPTGLRAIFG KTKIQNAVHC 120
 TDLPEDGLLE VQYFFKILDN 140

SEQ ID NO: 54 moltype = DNA length = 453
 FEATURE Location/Qualifiers
 misc_feature 1..453
 note = Human NME7-B2 sequence optimized for E. coli
 expression
 source 1..453
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 54
 atgcgcgagct ctggcggttg cggtcgggca aacaccgcca aatttaccaa ttgtacgtgc 60
 tgtattgtca aaccgcacgc agtgtcagaa ggctgtgctg gtaaaattct gatggcaatc 120
 cgtgatgctg gctttgaaat ctccggccatg cagatgttca acatggaccg cgtaaacgtc 180
 gaagaattct acgaagttaa caaaggcgtg gttaccgaat atcacgatat ggtaacggaa 240
 atgtactcgc gtcctgtcgt cgcgatggaa attcagcaaa acaatgccac caaaacgttt 300
 cgtgaattct gtggtccggc agatccggaa atcgcacgtc atctgcgtcc gggtaacctg 360

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cgcgcaattt ttggtaaaac gaaaatccag aacgctgtgc actgtaccga tctgccggaa 420
gacggtctgc tgggaagtca atactttttc tga 453

SEQ ID NO: 55      moltype = AA  length = 150
FEATURE           Location/Qualifiers
REGION            1..150
                  note = Human NME7-B2 sequence optimized for E. coli
                  expression
source            1..150
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 55
MPSSGGCGPA NTAKFTNCTC CIVKPHAVSE GLLGKILMAI RDAGFEISAM QMFNMDRVNV 60
EEFYEVYKGV VTEYHDMVTE MYSGPCVAME IQQNNATKTF REFCGPADPE IARHLRPGTL 120
RAIFGKTKIQ NAVHCTDLPE DGLLEVQYFF 150

SEQ ID NO: 56      moltype = DNA  length = 468
FEATURE           Location/Qualifiers
misc_feature      1..468
                  note = Human NME7-B3 sequence optimized for E. coli
                  expression
source            1..468
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 56
atgccgagct ctggcggttg cggtcgggca aacaccgccca aatttaccaa ttgtacgtgc 60
tgtattgtca aaccgcacgc agtgtcagaa ggcctgctgg gtaaaattct gatggcaatc 120
cgtgatgctg gctttgaaat ctggccatg cagatgttca acatggaccg cgtaaagctc 180
gaagaattct acgaagtta caaaggcgtg gttaccgaat atcacgatat ggttacggaa 240
atgtactccg gtccgtgcgt cgcgatggaa attcagcaaa acaatgccac caaaacgttt 300
cgtgaattct gtggtccggc agatccggaa atcgcacgtc atctgcgtcc gggtaccctg 360
cgcgcaattt ttggtaaaac gaaaatccag aacgctgtgc actgtaccga tctgccggaa 420
gacggtctgc tgggaagtca atactttttc aaaattctgg ataattga 468

SEQ ID NO: 57      moltype = AA  length = 155
FEATURE           Location/Qualifiers
REGION            1..155
                  note = Human NME7-B3 sequence optimized for E. coli
                  expression
source            1..155
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 57
MPSSGGCGPA NTAKFTNCTC CIVKPHAVSE GLLGKILMAI RDAGFEISAM QMFNMDRVNV 60
EEFYEVYKGV VTEYHDMVTE MYSGPCVAME IQQNNATKTF REFCGPADPE IARHLRPGTL 120
RAIFGKTKIQ NAVHCTDLPE DGLLEVQYFF KILDN 155

SEQ ID NO: 58      moltype = DNA  length = 861
FEATURE           Location/Qualifiers
misc_feature      1..861
                  note = Human NME7-AB, also known as NME7AB sequence
                  optimized for E. coli expression
source            1..861
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 58
atggaaaaaa cgctggccct gattaaaccc gatgcaatct ccaaagctgg cgaaattatc 60
gaaattatca acaaagccggg ttccaccatc acgaaactga aaatgatgat gctgagccgt 120
aaagaagccc tggattttca tgtcgaccac cagtctcgcc cgtttttcaa tgaactgatt 180
caattcatca ccacgggtcc gattatcgca atggaaattc tgcgtgatga cgctatctgc 240
gaatggaaac gcctgctggg cccggcaaac tcaggtgttg cgcgtaccga tgccagtga 300
tccattcgcg ctctgtttgg caccgatggt atccgtaatg cagcacatgg tccggactca 360
ttcgatcggc cagctcgtga aatggaaactg tttttcccga gctctggcgg ttgcggtccg 420
gcaaacaccg ccaaatttac caattgtacg tgctgtattg tcaaacccga cgcagtgtca 480
gaaggcctgc tgggtaaaat tctgatggca atccgtgatg ctggctttga aatctcggcc 540
atgcagatgt tcaacatgga ccgcgttaac gtcgaagaat tctacgaagt ttacaaggc 600
gtggttaccc aatatcacga tatggttacg gaaatgtact ccggtccgtg cgtcgcgatg 660
gaaattcagc aaaacaatgc caccaaaaac tttctgtaat tctgtggtcc ggcagatccg 720
gaaatcgcac gtcactctgc tccgggtacc ctgcgcgcaa ttttggttaa aacgaaaatc 780
cagaacgctg tgcactgtac cgatctgccg gaagacggtc tgctggaagt tcaatacttt 840
tcaaaaatc tggataattg a 861

SEQ ID NO: 59      moltype = AA  length = 286
FEATURE           Location/Qualifiers
REGION            1..286
                  note = Human NME7-AB, also known as NME7AB sequence
                  optimized for E. coli expression
source            1..286
                  mol_type = protein

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organism = synthetic construct

SEQUENCE: 59
MEKTLALIKP DAISKAGEII EIINKAGFTI TKLKMMMLSR KEALDFHVDH QSRPFFNELI 60
QFITTGPIIA MEILRDDAIC EWKRLGPGAN SGVARTDASE SIRALFGTDG IRNAAHGPD5 120
FASAAAREMEL FFPSSGGCGP ANTAKFNTCT CCIVKPHAVS EGLLGKILMA IRDAGFEISA 180
MQMFNMDRVN VEEFYEVYKG VVTEYHDMVT EMYSGPCVAM EIQQNNATKT FREFCGPADP 240
EIAHRLRPGT LRAIFGKTKI QNAVHCTDLP EDGLLEVQYF FKILDN 286

SEQ ID NO: 60      moltype = DNA length = 846
FEATURE           Location/Qualifiers
misc_feature       1..846
                   note = Human NME7-AB1, also known as NME7AB1 sequence
                   optimized for E. coli expression
source            1..846
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 60
atggaaaaaa cgctggccct gattaaaccg gatgcaatct ccaaagctgg cgaaattatc 60
gaaattatca acaaagcggg ttccaccatc acgaaactga aaatgatgat gctgagccgt 120
aaagaagccc tggattttca tgtcgaccac cagtctcgcc cgtttttcaa tgaactgatt 180
caattcatca ccacgggtcc gattatcgca atggaaaattc tgcgtgatga cgctatctgc 240
gaatggaaac gcctgctggg cccggcaaac tcaggtgttg cgcgtaaccg tgccagtga 300
tccttcgcg cctctgtttg caccgatgtg atccgtaatg cagcacatgg tccggactca 360
tccgatcggg cagctcgatg aatggaaactg tttttccgca gctctggcgg ttgcggtccg 420
gcaaacacccg ccaaatttac caattgtacg tctgttattg tcaaaccgca cgcagtgtca 480
gaaggcctgc tgggtaaaat tctgatggca atccgtgatg ctggctttga aatctcggcc 540
atgcagatgt tcaacatgga ccgcgttaac gtcgaagaat tctacgaagt ttacaaggc 600
gtggttacgc aatatcacga tatggttacg gaaatgtact ccggtccgtg cgtcgcgatg 660
gaaattcagc aaaacaatgc caccaaaacg tttcgtgaat tctgtggtcc ggcagatccg 720
gaaatcgcac gtcactctgc tccgggtacc ctgcgcgcaa tttttggtaa aacgaaaatc 780
cagaacgctg tgcactgtac cgatctgccc gaagacggtc tgctggaagt tcaatacttt 840
ttctga 846

SEQ ID NO: 61      moltype = AA length = 281
FEATURE           Location/Qualifiers
REGION            1..281
                   note = Human NME7-AB1, also known as NME7AB1 sequence
                   optimized for E. coli expression
source            1..281
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 61
MEKTLALIKP DAISKAGEII EIINKAGFTI TKLKMMMLSR KEALDFHVDH QSRPFFNELI 60
QFITTGPIIA MEILRDDAIC EWKRLGPGAN SGVARTDASE SIRALFGTDG IRNAAHGPD5 120
FASAAAREMEL FFPSSGGCGP ANTAKFNTCT CCIVKPHAVS EGLLGKILMA IRDAGFEISA 180
MQMFNMDRVN VEEFYEVYKG VVTEYHDMVT EMYSGPCVAM EIQQNNATKT FREFCGPADP 240
EIAHRLRPGT LRAIFGKTKI QNAVHCTDLP EDGLLEVQYF F 281

SEQ ID NO: 62      moltype = DNA length = 570
FEATURE           Location/Qualifiers
misc_feature       1..570
                   note = Mouse NME6
source            1..570
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 62
atgacctcca tcttgccaag tcccccaagct cttcagctca cactagccct gatcaagcct 60
gatgcagttg cccaccactc gatcctggag gctgttcacg agcagattct gagcaacaag 120
ttcctcattg tacgaacgag ggaactgcag tggaaagctgg aggactgccg gaggttttac 180
cgagagcatg aagggcgctt tttctatcag cggctggttg agttcatgac aagtgggcca 240
atccgagcct atatccttgc ccacaaagat gccatccaac tttggaggac actgatggga 300
cccaccagag tatttcgagc acgctatata gcccagatt caattcgttg aagtttgggc 360
ctcactgaca cccgaaatac taccatggc tcagactccg tggtttcgcg cagcagagag 420
attgcagcct tcttcctgga cttcagtgaa cagcgtcgtg atgaggagga ggaacccag 480
ctgcggtgtg gtctctgtga ctacagtcca gaggaaggta tccactgtgc agctgaaaca 540
ggaggccaca aacaacctaa caaacctag 570

SEQ ID NO: 63      moltype = AA length = 189
FEATURE           Location/Qualifiers
REGION            1..189
                   note = Mouse NME6
source            1..189
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 63
MTSILRSPQA LQLTLALIKP DAVAHPLILE AVHQQLSNK FLIVRTRELQ WKLEDCCRFF 60
REHEGRFFYQ RLVEFMTSGP IRAYILAHKD AIQLWRTLGM PTRVFRARYI APDSIRGSLG 120
LTDTRNTTHG SDSVVSASRE IAFFPDSE QRWYEEEPQ LRCGPVHYSP EEGIHCAET 180
GGHKQPNKT 189

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SEQ ID NO: 64      moltype = DNA length = 585
FEATURE           Location/Qualifiers
misc_feature       1..585
                   note = Human NME6
source            1..585
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 64
atgaccaga atctggggag tgagatggcc tcaatcttgc gaagccctca ggctctccag 60
ctcaactctag ccctgatcaa gctgacgca gtcgcccac cactgattct ggaggctgtt 120
catcagcaga ttctaagcaa caagttcctg attgtacgaa tgagagaact actgtggaga 180
aaggaagatt gccagagggt ttaccgagag catgaaggcg gttttttcta tcagaggctg 240
gtggagttca tggccagcgg gccaatccga gcctacatcc ttgccacaaa ggatgccatc 300
cagctctgga ggacgctcat gggacccacc agagtgttcc gagcacgcca tgtggcccca 360
gattctatcc gtgggagttt cggcctcact gacacccgca acaccaccca tggttcggac 420
tctgtggttt cagccagcag agagattgca gccttcttcc ctgacttcag tgaacagcgc 480
tggtatgagg aggaagagcc ccagttgcgc tgtggccctg tgtgctatag cccagaggga 540
ggtgtccact atgtagctgg aacaggaggc ctaggaccag cctga 585

SEQ ID NO: 65      moltype = AA length = 194
FEATURE           Location/Qualifiers
REGION            1..194
                   note = Human NME6
source            1..194
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 65
MTQNLGSEMA SILRSPQALQ LTLALIKPDA VAHPLILEAV HQQILSNKFL IVRMRELLWR 60
KEDCQRFYRE HEGRFFYQRL VEFMASGPIR AYILAHKDAI QLWRTLGMPT RVFRARHVAP 120
DSIRGSFGLT DTRNTTHGSD SVVSASREIA AFFPDFSEQR WYEEEPQLR CGPVCYSPEG 180
GVHYVAGTGG LGPA 194

SEQ ID NO: 66      moltype = DNA length = 525
FEATURE           Location/Qualifiers
misc_feature       1..525
                   note = Human NME6 1
source            1..525
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 66
atgaccaga atctggggag tgagatggcc tcaatcttgc gaagccctca ggctctccag 60
ctcaactctag ccctgatcaa gctgacgca gtcgcccac cactgattct ggaggctgtt 120
catcagcaga ttctaagcaa caagttcctg attgtacgaa tgagagaact actgtggaga 180
aaggaagatt gccagagggt ttaccgagag catgaaggcg gttttttcta tcagaggctg 240
gtggagttca tggccagcgg gccaatccga gcctacatcc ttgccacaaa ggatgccatc 300
cagctctgga ggacgctcat gggacccacc agagtgttcc gagcacgcca tgtggcccca 360
gattctatcc gtgggagttt cggcctcact gacacccgca acaccaccca tggttcggac 420
tctgtggttt cagccagcag agagattgca gccttcttcc ctgacttcag tgaacagcgc 480
tggtatgagg aggaagagcc ccagttgcgc tgtggccctg tgtga 525

SEQ ID NO: 67      moltype = AA length = 174
FEATURE           Location/Qualifiers
REGION            1..174
                   note = Human NME6 1
source            1..174
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 67
MTQNLGSEMA SILRSPQALQ LTLALIKPDA VAHPLILEAV HQQILSNKFL IVRMRELLWR 60
KEDCQRFYRE HEGRFFYQRL VEFMASGPIR AYILAHKDAI QLWRTLGMPT RVFRARHVAP 120
DSIRGSFGLT DTRNTTHGSD SVVSASREIA AFFPDFSEQR WYEEEPQLR CGPV 174

SEQ ID NO: 68      moltype = DNA length = 468
FEATURE           Location/Qualifiers
misc_feature       1..468
                   note = Human NME6 2
source            1..468
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 68
atgctcactc tagccctgat caagcctgac gcagtcgccc atccactgat tctggaggct 60
gttcatcagc agattctaag caacaagttc ctgattgtac gaatgagaga actactgtgg 120
agaaaggaag attgccagag gttttaccga gagcatgaag ggcgtttttt ctatcagagg 180
ctgggtggagt tcatggccag cggggccaatc cgagcctaca tccttgccca caaggatgcc 240
atccagctct ggaggagcct catgggaccc accagagtgt tccgagcacg ccatgtggcc 300
ccagattcta tccgtgggag ttctggcctc actgacaccc gcaacaccac ccatgtgttc 360
gactctgtgg tttcagccag cagagagatt gcagccttct tccctgactt cagtgaacag 420
cgctggtatg aggaggaaga gccccagttg cgctgtggcc ctgtgtga 468

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SEQ ID NO: 69      moltype = AA  length = 155
FEATURE           Location/Qualifiers
REGION            1..155
                  note = Human NME6 2
source            1..155
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 69
MLTLALIKPD AVAHPLILEA VHQQILSNKF LIVRMRELLW RKEDCQRFYR EHEGRFFYQR 60
LVEFMASGPI RAYILAHKDA IQLWRTLMPG TRVFRARHVA PDSIRGSFGL TDRNTTHGS 120
DSVVSASREI AAFPPDFSEQ RWYEEEEPQL RCGPV 155

SEQ ID NO: 70      moltype = DNA  length = 528
FEATURE           Location/Qualifiers
misc_feature      1..528
                  note = Human NME6 3
source            1..528
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 70
atgctcactc tagccctgat caagcctgac gcagtcgccc atccactgat tctggaggct 60
gttcatcagc agattctaag caacaagttc ctgattgtac gaatgagaga actactgtgg 120
agaaaggaag attgccagag gttttaccga gagcatgaag ggcgtttttt ctatcagagg 180
ctggtggagt tcatggccag cgggccaatc cgagcctaca tccttgccca caaggatgcc 240
atccagctct ggaggacgct catggggaccc accagagtgt tccgagcagc ccatgtggcc 300
ccagattcta tccgtgggag ttctcgccctc actgacaccc gcaacaccac ccatggttcg 360
gactctgtgg tttcagccag cacagagatt gcagccttct tccctgactt cagtgaacag 420
cgctgggatg aggaggaaga gccccagttg cgctgtggcc ctgtgtgcta tagcccagag 480
ggaggtgtcc actatgtagc tggaacagga ggcctaggac cagcctga 528

SEQ ID NO: 71      moltype = AA  length = 175
FEATURE           Location/Qualifiers
REGION            1..175
                  note = Human NME6 3
source            1..175
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 71
MLTLALIKPD AVAHPLILEA VHQQILSNKF LIVRMRELLW RKEDCQRFYR EHEGRFFYQR 60
LVEFMASGPI RAYILAHKDA IQLWRTLMPG TRVFRARHVA PDSIRGSFGL TDRNTTHGS 120
DSVVSASREI AAFPPDFSEQ RWYEEEEPQL RCGPVCYSPE GGVHYVAGTG GLGPA 175

SEQ ID NO: 72      moltype = DNA  length = 585
FEATURE           Location/Qualifiers
misc_feature      1..585
                  note = Human NME6 sequence optimized for E. coli expression
source            1..585
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 72
atgacgcaaa atctggggctc ggaaatggca agtatccctgc gctccccgca agcactgcaa 60
ctgaccctgg ctctgatcaa accggacgct gttgctcctc cgctgattct ggaagcggtc 120
caccagcaaa ttctgagcaa caaatttctg atcgtgcgta tgcgcgaact gctgtggcgt 180
aaagaagatt gccagcgttt ttatcgcgaa catgaaggcc gtttctttta tcaacgcctg 240
gttgaattca tggcctctgg tccgattcgc gcataatcc ttggctcaca agatgcgatt 300
cagctgtggc gtaccctgat gggtcgcgac cgcgtcttcc gtgcacgta tgtggcaccg 360
gactcaatcc gtggtcgttt cggctctgacc gatacgcgca ataccacgca cggtagcgac 420
tctgttgta gtgcgtcccg tgaatcgcg gcctttttcc cggacttctc cgaacagcgt 480
tggtacgaag aagaagaacc gcaactgcgc tgtggcccgg tctgttatcc tccggaaggt 540
ggtgtccatt atgtggcggg cacgggtggt ctgggtccgg catga 585

SEQ ID NO: 73      moltype = AA  length = 194
FEATURE           Location/Qualifiers
REGION            1..194
                  note = Human NME6 sequence optimized for E. coli expression
source            1..194
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 73
MTQNLGSEMA SILRSPQALQ LTLALIKPDA VAHPLILEAV HQQILSNKFL IVRMRELLWR 60
KEDCQRFYRE HEGRFFYQRL VEFMASGPIR AYILAHKDAI QLWRTLMPGT RVFRARHVAP 120
DSIRGSFGLT DTRNTTHGSD SVVSASREIA AFPDFSEQR WYEEEEPQLR CGPVCYSPEG 180
GVHYVAGTGG LGPA 194

SEQ ID NO: 74      moltype = DNA  length = 525
FEATURE           Location/Qualifiers
misc_feature      1..525
                  note = Human NME6 1 sequence optimized for E. coli

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expression
source      1..525
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 74
atgacgcaaa atctggggctc ggaaatggca agtatcctgc gctccccgca agcactgcaa 60
ctgacctggy ctctgatcaa accggacgct gttgctcatc cgctgattct ggaagcggtc 120
caccagcaaa ttctgagcaa caaatctctg atcgtgcgta tgcgcgaact gctgtggcgt 180
aaagaagatt gccagcgctt ttatcgcgaa catgaaggcc gtttctttta tcaacgcctg 240
gttgaattca tggcctctgg tccgattcgc gcataatcc tggctcacia agatgcgatt 300
cagctgtggc gtacctgat gggtcggacg cgcgtctttc gtgcacgtca tgtggcaccg 360
gactcaatcc gtggtcgtt cggctcgacc gatacgcgca ataccacgca cggtagcgac 420
tctgttgta gtgcgtcccg tgaatcgcg gcctttttcc cggacttctc cgaacagcgt 480
tggtagcaag aagaagaacc gcaactgcgc tgtggcccg tctga 525

SEQ ID NO: 75      moltype = AA length = 174
FEATURE           Location/Qualifiers
REGION           1..174
                note = Human NME6 1 sequence optimized for E. coli
                expression
source          1..174
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 75
MTQNLGSEMA SILRSPQALQ LTLALIKPDA VAHPLILEAV HQQILSNKFL IVRMRELLWR 60
KEDCQRFYRE HEGRFFYQRL VEFMASGPIR AYILAHKDAI QLWRTL MGPT RVFRARHVAP 120
DSIRGSFGLT DTRNTHGSD SVVSASREIA AFFPDFSEQR WYEEEPQLR CGPV 174

SEQ ID NO: 76      moltype = DNA length = 468
FEATURE           Location/Qualifiers
misc_feature     1..468
                note = Human NME6 2 sequence optimized for E. coli
                expression
source          1..468
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 76
atgctgaccc tggctctgat caaacccggac gctgttgctc atccgctgat tctggaagcg 60
gtccaccacg aaattctgag caacaaattt ctgatcgtgc gtatgcgcga actgctgtgg 120
cgtaaagaag attgccagcg tttttatcgc gaacatgaag gccgtttctt ttatcaacgc 180
ctggttgaat tcatggcctc tgggtccgatt cgcgcataata tccctggctca caaagatgcg 240
attcagctgt ggcgtaccct gatgggtccg acgcgcgtct tctgtgcacg tcatgtggca 300
ccggactcaa tccgtggctc gttcggctcg accgatacgc gcaataccac gcacggtagc 360
gactctgttg ttagtgctgc ccgtgaaatc gcggcctttt tcccggaact ctccgaacag 420
cgttggtagc aagaagaaga accgcaactg cgctgtggcc cggctctga 468

SEQ ID NO: 77      moltype = AA length = 155
FEATURE           Location/Qualifiers
REGION           1..155
                note = Human NME6 2 sequence optimized for E. coli
                expression
source          1..155
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 77
MLTLALIKPD AVAHPILILEA VHQQILSNKF LIVRMRELLW RKEDCQRFYR EHEGRFFYQR 60
LVEFMASGPI RAYILAHKDA IQLWRTL MGPT TRVFRARHVA PDSIRGSFGL TDTRNTHGS 120
DSVVSASREI AAFPDPDFSEQ RWYEEEPQL RCGPV 155

SEQ ID NO: 78      moltype = DNA length = 528
FEATURE           Location/Qualifiers
misc_feature     1..528
                note = Human NME6 3 sequence optimized for E. coli
                expression
source          1..528
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 78
atgctgaccc tggctctgat caaacccggac gctgttgctc atccgctgat tctggaagcg 60
gtccaccacg aaattctgag caacaaattt ctgatcgtgc gtatgcgcga actgctgtgg 120
cgtaaagaag attgccagcg tttttatcgc gaacatgaag gccgtttctt ttatcaacgc 180
ctggttgaat tcatggcctc tgggtccgatt cgcgcataata tccctggctca caaagatgcg 240
attcagctgt ggcgtaccct gatgggtccg acgcgcgtct tctgtgcacg tcatgtggca 300
ccggactcaa tccgtggctc gttcggctcg accgatacgc gcaataccac gcacggtagc 360
gactctgttg ttagtgctgc ccgtgaaatc gcggcctttt tcccggaact ctccgaacag 420
cgttggtagc aagaagaaga accgcaactg cgctgtggcc cggctctgta ttctccgaa 480
ggtggtgtcc attatgtggc gggcacgggt ggtctgggtc cggcatga 528

SEQ ID NO: 79      moltype = AA length = 175

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FEATURE          Location/Qualifiers
REGION           1..175
                 note = Human NME6 3 sequence optimized for E. coli
                 expression
source           1..175
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 79
MTLLALIKPD AVAHPLILEA VHQQILSNKF LIVRMRELLW RKEDCQRFYR EHEGRFFYQR 60
LVEFMASGPI RAYILAHKDA IQLWRTLMGP TRVFRARHVA PDSIRGSFGL TDRNTTHGS 120
DSVVSASREI AAFPPDFSEQ RWYEEEEEPQL RCGPVCYSPE GGVHYVAGTG GLGPA 175

SEQ ID NO: 80    moltype = DNA length = 1306
FEATURE          Location/Qualifiers
misc_feature     1..1306
                 note = OriGene-NME7-1 full length
source           1..1306
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 80
gacgttgat acgactccta tagggcggcc gggaattcgt cgactggatc cggtaaccgag 60
gagatctgcc gccgcgatcg ccatgaatca tagtgaaaga ttcgttttca ttgcagagtg 120
gtatgatcca aatgcttcac ttcttcgacg ttatgagctt ttattttacc caggggatgg 180
atctgttgaa atgcatgatg taaagaatca tcgcacctt ttaaagcggg ccaaatatga 240
taacctgcac ttggaagatt tatttatagg caacaaagtg aatgtcttct ctgcacaact 300
ggtattaat gactatgggg atcaatatac agctcgccag ctgggcagta ggaaagaaaa 360
aacgctagcc ctaattaaac cagatgcaat atcaaaggct ggagaaataa ttgaaataat 420
aaacaaagct ggatttacta taacaaact caaaatgatg atgctttcaa ggaaagaagc 480
attggatttt catgtagatc accagtcaag accctttttc aatgagctga tccagtttat 540
tacaactggt cctattattg ccatggagat tttaagagat gatgctatat gtgaatggaa 600
aagactgctg ggacctgcaa actctggagt ggcacgcaca gatgcttctg aaagcattag 660
agccctcttt ggaacagatg gcataagaaa tgcagcgcat ggcctgatt cttttgcttc 720
tgcggccaga gaaatggagt tgttttttcc ttcaagtgga ggttggtggc cggaacaacac 780
tgttaaatat actaatgtga cctgttgcat tgttaaacc catgctgtca gtgaaggact 840
gttgggaaag atcctgatgg ctatccgaga tgcaggtttt gaaatctcag ctatgcagat 900
gttcaatag gatcggttga atgttgagga attctatgaa gtttataaag gagtagtgac 960
cgaatatcat gacatggtga cagaaatgta ttctggccct tgtgtagcaa tggagattca 1020
acagaataat gctacaaaga catttcgaga attttgtgga cctgctgac ctgaaattgc 1080
ccggcattta cgcctggaa cctcagagc aatctttggt aaaactaaga tccagaatgc 1140
tgttcactgt actgatctgc cagaggatgg cctattagag gttcaatact tcttcaagat 1200
cttgataaat acgcgtacgc ggccgctcga gcagaaactc atctcagaag aggatctggc 1260
agcaaatgat atcctggatt acaaggatga cgacgataag gtttaa 1306

SEQ ID NO: 81    moltype = AA length = 407
FEATURE          Location/Qualifiers
REGION           1..407
                 note = OriGene-NME7-1 full length
source           1..407
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 81
MNHSERFVFI AEWYDPNASL LRRYELLYFP GDGSVEMHDV KNHRTFLKRT KYDNLHLEDL 60
FIGNKVNVFS RQLVLIDYGD QYTARQLGSR KEKTLALIKP DAISKAGEII EIINKAGFTI 120
TKLKMMLLSR KEALDFHVDH QSRPFFNELI QFITTGPIIA MEILRDDAIC EWKRLGSPAN 180
SGVARTDASE SIRALFGTDG IRNAAHGPDS FASAAREMEL FFPSSGGCGP ANTAKFTNCT 240
CCIVKPHAVS EGLLGKILMA IRDAGFEISA MQMFNMDRVN VEEFYEVYKG VVTEYHDMVT 300
EMYSGPCVAM EIQQNNATKT FREFCGPADP EIAHRLRPGT LRAIFGKTKI QNAVHCTDLP 360
EDGLEEVQYF FKILDNTRTR RLEQKLISEE DLAANDILDY KDDDDKV 407

SEQ ID NO: 82    moltype = AA length = 376
FEATURE          Location/Qualifiers
REGION           1..376
                 note = Abnova NME7-1 Full length
source           1..376
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 82
MNHSERFVFI AEWYDPNASL LRRYELLYFP GDGSVEMHDV KNHRTFLKRT KYDNLHLEDL 60
FIGNKVNVFS RQLVLIDYGD QYTARQLGSR KEKTLALIKP DAISKAGEII EIINKAGFTI 120
TKLKMMLLSR KEALDFHVDH QSRPFFNELI QFITTGPIIA MEILRDDAIC EWKRLGSPAN 180
SGVARTDASE SIRALFGTDG IRNAAHGPDS FASAAREMEL FFPSSGGCGP ANTAKFTNCT 240
CCIVKPHAVS EGLLGKILMA IRDAGFEISA MQMFNMDRVN VEEFYEVYKG VVTEYHDMVT 300
EMYSGPCVAM EIQQNNATKT FREFCGPADP EIAHRLRPGT LRAIFGKTKI QNAVHCTDLP 360
EDGLEEVQYF FKILDN 376

SEQ ID NO: 83    moltype = AA length = 98
FEATURE          Location/Qualifiers
REGION           1..98
                 note = Abnova Partial NME7-B

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source          1..98
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 83
DRVNVVEEPEYK VYKGVVTEYH DMVTEMYSGP CVAMEIQQNN ATKTFREFCG PADPEIARHL 60
RPGTLRAIFG KTKIQNAVHC TDLPEDGLE VQYFFKIL 98

SEQ ID NO: 84      moltype = DNA length = 27
FEATURE           Location/Qualifiers
misc_feature       1..27
                   note = Histidine Tag
source            1..27
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 84
ctcgagcacc accaccacca ccactga 27

SEQ ID NO: 85      moltype = DNA length = 36
FEATURE           Location/Qualifiers
misc_feature       1..36
                   note = Strept II Tag
source            1..36
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 85
accgggtgga gccatcctca gttcgaaaag taatga 36

SEQ ID NO: 86      moltype = AA length = 35
FEATURE           Location/Qualifiers
REGION            1..35
                   note = N-10 peptide
source            1..35
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 86
QFNQYKTEAA SRYNLTISDV SVSDVPPFFS AQSGA 35

SEQ ID NO: 87      moltype = AA length = 35
FEATURE           Location/Qualifiers
REGION            1..35
                   note = C-10 peptide
source            1..35
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 87
GTINVHDTVET QFNQYKTEAA SRYNLTISDV SVSDV 35

SEQ ID NO: 88      moltype = AA length = 8
FEATURE           Location/Qualifiers
REGION            1..8
                   note = Peptide of NME7AB
source            1..8
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 88
LALIKPDA 8

SEQ ID NO: 89      moltype = AA length = 18
FEATURE           Location/Qualifiers
REGION            1..18
                   note = Peptide of NME7AB
source            1..18
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 89
MMMLSRKEAL DFHVDHQS 18

SEQ ID NO: 90      moltype = AA length = 10
FEATURE           Location/Qualifiers
REGION            1..10
                   note = Peptide of NME7AB
source            1..10
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 90
ALDFHVDHQS 10

SEQ ID NO: 91      moltype = AA length = 14
FEATURE           Location/Qualifiers

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REGION	1..14	
	note = Peptide of NME7AB	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 91		
EILRDDAICE WKRL		14
SEQ ID NO: 92	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
REGION	1..12	
	note = Peptide of NME7AB	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 92		
FNELIQFITT GP		12
SEQ ID NO: 93	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = Peptide of NME7AB	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 93		
RDDAICEW		8
SEQ ID NO: 94	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
REGION	1..25	
	note = Peptide of NME7AB	
source	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 94		
SGVARTDASE SIRALFGTDG IRNAA		25
SEQ ID NO: 95	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Peptide of NME7AB	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 95		
ELFFPSSGG		9
SEQ ID NO: 96	moltype = AA length = 26	
FEATURE	Location/Qualifiers	
REGION	1..26	
	note = Peptide of NME7AB	
source	1..26	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 96		
KFTNCTCCIV KPHAVSEGLL GKILMA		26
SEQ ID NO: 97	moltype = AA length = 36	
FEATURE	Location/Qualifiers	
REGION	1..36	
	note = Peptide of NME7AB	
source	1..36	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 97		
LMAIRDAGFE ISAMQMFNMD RVNVEEFYEV YKGVVT		36
SEQ ID NO: 98	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Peptide of NME7AB	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 98		
EFYEVYKGVV TEYHD		15
SEQ ID NO: 99	moltype = AA length = 43	

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FEATURE	Location/Qualifiers	
REGION	1..43	
source	note = Peptide of NME7AB 1..43 mol_type = protein organism = synthetic construct	
SEQUENCE: 99		
EIQQNNATKT FREFCGPADP EIAHRLRPGT LRAIFGKTKI QNA		43
SEQ ID NO: 100	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
source	note = Peptide of NME7AB 1..8 mol_type = protein organism = synthetic construct	
SEQUENCE: 100		
YSGPCVAM		8
SEQ ID NO: 101	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
source	note = Peptide of NME7AB 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 101		
FREFCGP		7
SEQ ID NO: 102	moltype = AA length = 23	
FEATURE	Location/Qualifiers	
REGION	1..23	
source	note = Peptide of NME7AB 1..23 mol_type = protein organism = synthetic construct	
SEQUENCE: 102		
VHCTDLPEDG LLEVQYFFKI LDN		23
SEQ ID NO: 103	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
source	note = Peptide of NME7AB 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 103		
IQNAVHCTD		9
SEQ ID NO: 104	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
source	note = Peptide of NME7AB 1..20 mol_type = protein organism = synthetic construct	
SEQUENCE: 104		
TDLPEDGLLE VQYFFKILDN		20
SEQ ID NO: 105	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
source	note = Peptide of NME7AB 1..13 mol_type = protein organism = synthetic construct	
SEQUENCE: 105		
PEDGLLEVQY FFK		13
SEQ ID NO: 106	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
REGION	1..12	
source	note = Peptide of NME7AB 1..12 mol_type = protein organism = synthetic construct	
SEQUENCE: 106		
EIINKAGFTI TK		12

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SEQ ID NO: 107	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Peptide of NME7AB	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 107		
MLSRKEALDF HVDHQS		16
SEQ ID NO: 108	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Peptide of NME7AB	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 108		
NELIQFITT		9
SEQ ID NO: 109	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Peptide of NME7AB	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 109		
EILRDDAICE WKRL		14
SEQ ID NO: 110	moltype = AA length = 21	
FEATURE	Location/Qualifiers	
REGION	1..21	
	note = Peptide of NME7AB	
source	1..21	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 110		
SGVARTDASE SIRALFGTDG I		21
SEQ ID NO: 111	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Peptide of NME7AB	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 111		
SGVARTDASE S		11
SEQ ID NO: 112	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = Peptide of NME7AB	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 112		
ALFGTDGI		8
SEQ ID NO: 113	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Peptide of NME7AB	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 113		
NCTCCIVKPH AVSE		14
SEQ ID NO: 114	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Peptide of NME7AB	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 114		
LGKILMAIRD A		11

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SEQ ID NO: 115	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Peptide of NME7AB	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 115		
EISAMQMFNM DRVNVE		16
SEQ ID NO: 116	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = Peptide of NME7AB	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 116		
EVYKGVVT		8
SEQ ID NO: 117	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = Peptide of NME7AB	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 117		
EYHDMVTE		8
SEQ ID NO: 118	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Peptide of NME7AB	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 118		
EFCGPADPEI ARHLR		15
SEQ ID NO: 119	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
REGION	1..12	
	note = Peptide of NME7AB	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 119		
AIFGKTKIQN AV		12
SEQ ID NO: 120	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
REGION	1..18	
	note = Peptide of NME7AB	
source	1..18	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 120		
LPEDGLLEVQ YFFKILDN		18
SEQ ID NO: 121	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Peptide of NME7AB	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 121		
GPDSFASAAR EMELFFP		17
SEQ ID NO: 122	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Immunizing peptides derived from human NME7	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 122		

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ICEWKRL	7
SEQ ID NO: 123	moltype = AA length = 11
FEATURE	Location/Qualifiers
REGION	1..11
	note = Immunizing peptides derived from human NME7
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 123	
LGKILMAIRD A	11
SEQ ID NO: 124	moltype = AA length = 10
FEATURE	Location/Qualifiers
REGION	1..10
	note = Immunizing peptides derived from human NME7
source	1..10
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 124	
HAVSEGLLGK	10
SEQ ID NO: 125	moltype = AA length = 8
FEATURE	Location/Qualifiers
REGION	1..8
	note = Immunizing peptides derived from human NME7
source	1..8
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 125	
VTEMYSGP	8
SEQ ID NO: 126	moltype = AA length = 9
FEATURE	Location/Qualifiers
REGION	1..9
	note = Immunizing peptides derived from human NME7
source	1..9
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 126	
NATKTFREF	9
SEQ ID NO: 127	moltype = AA length = 9
FEATURE	Location/Qualifiers
REGION	1..9
	note = Immunizing peptides derived from human NME7
source	1..9
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 127	
AIRDAGFEI	9
SEQ ID NO: 128	moltype = AA length = 13
FEATURE	Location/Qualifiers
REGION	1..13
	note = Immunizing peptides derived from human NME7
source	1..13
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 128	
AICEWKRLLG PAN	13
SEQ ID NO: 129	moltype = AA length = 8
FEATURE	Location/Qualifiers
REGION	1..8
	note = Immunizing peptides derived from human NME7
source	1..8
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 129	
DHQSRPFF	8
SEQ ID NO: 130	moltype = AA length = 13
FEATURE	Location/Qualifiers
REGION	1..13
	note = Immunizing peptides derived from human NME7
source	1..13
	mol_type = protein
	organism = synthetic construct

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SEQUENCE: 130 AICEWKRLLG PAN	13
SEQ ID NO: 131 FEATURE REGION source	moltype = AA length = 8 Location/Qualifiers 1..8 note = Immunizing peptides derived from human NME7 1..8 mol_type = protein organism = synthetic construct
SEQUENCE: 131 VDHQSRPF	8
SEQ ID NO: 132 FEATURE REGION source	moltype = AA length = 6 Location/Qualifiers 1..6 note = Immunizing peptides derived from human NME7 1..6 mol_type = protein organism = synthetic construct
SEQUENCE: 132 PDSFAS	6
SEQ ID NO: 133 FEATURE REGION source	moltype = AA length = 18 Location/Qualifiers 1..18 note = Immunizing peptides derived from human NME7 1..18 mol_type = protein organism = synthetic construct
SEQUENCE: 133 KAGEIIEIIN KAGFTITK	18
SEQ ID NO: 134 FEATURE REGION source	moltype = AA length = 29 Location/Qualifiers 1..29 note = Immunizing peptides derived from human NME1 1..29 mol_type = protein organism = synthetic construct
SEQUENCE: 134 MANCERTFIA IKPDGVQRGL VGEIIKRFE	29
SEQ ID NO: 135 FEATURE REGION source	moltype = AA length = 8 Location/Qualifiers 1..8 note = Immunizing peptides derived from human NME1 1..8 mol_type = protein organism = synthetic construct
SEQUENCE: 135 VDLKDRPF	8
SEQ ID NO: 136 FEATURE REGION source	moltype = AA length = 17 Location/Qualifiers 1..17 note = Immunizing peptides derived from human NME1 1..17 mol_type = protein organism = synthetic construct
SEQUENCE: 136 HGSDSVESAE KEIGLWF	17
SEQ ID NO: 137 FEATURE REGION source	moltype = AA length = 25 Location/Qualifiers 1..25 note = Immunizing peptides derived from human NME1 1..25 mol_type = protein organism = synthetic construct
SEQUENCE: 137 ERTFIAIKPD GVQRGLVGEI IKRFE	25
SEQ ID NO: 138 FEATURE REGION source	moltype = AA length = 30 Location/Qualifiers 1..30 note = Immunizing peptides derived from human NME1 1..30 mol_type = protein

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organism = synthetic construct	
SEQUENCE: 138	
VDLKDRPFPA GLVKYMHS GP VVAMVWEGLN	30
SEQ ID NO: 139	moltype = AA length = 26
FEATURE	Location/Qualifiers
REGION	1..26
	note = Immunizing peptides derived from human NME1
source	1..26
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 139	
NIIHGSDSVE SAEKEIGLWF HPPELV	26
SEQ ID NO: 140	moltype = AA length = 14
FEATURE	Location/Qualifiers
REGION	1..14
	note = Immunizing peptides derived from human NME1
source	1..14
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 140	
KPDGVQRGLV GEII	14
SEQ ID NO: 141	moltype = AA length = 16
FEATURE	Location/Qualifiers
REGION	1..16
	note = NME7A peptide 1 (A domain)
source	1..16
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 141	
MLSRKEALDF HVDHQS	16
SEQ ID NO: 142	moltype = AA length = 11
FEATURE	Location/Qualifiers
REGION	1..11
	note = NME7A peptide 2 (A domain)
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 142	
SGVARTDASE S	11
SEQ ID NO: 143	moltype = AA length = 20
FEATURE	Location/Qualifiers
REGION	1..20
	note = NME7B peptide 1 (B domain)
source	1..20
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 143	
DAGFEISAMQ MFNMDRVNVE	20
SEQ ID NO: 144	moltype = AA length = 16
FEATURE	Location/Qualifiers
REGION	1..16
	note = NME7B peptide 2 (B domain)
source	1..16
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 144	
EVYKGVVTEY HDMVTE	16
SEQ ID NO: 145	moltype = AA length = 29
FEATURE	Location/Qualifiers
REGION	1..29
	note = NME7B peptide 3 (B domain)
source	1..29
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 145	
AIFGKTKIQN AVHCTDLPED GLLEVQYFF	29
SEQ ID NO: 146	moltype = DNA length = 1131
FEATURE	Location/Qualifiers
misc_feature	1..1131
	note = Human NME7 a
source	1..1131

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 146
atgaatcata gtgaagatt cgttttcatt gcagagtggg atgatccaaa tgcttcactt 60
cttcgacgtt atgagctttt attttaccca ggggatggat ctgttgaaat gcatgatgta 120
aagaatcaco gcaccttttt aaagcggacc aaatatgata acctgcactt ggaagattta 180
tttataggca acaaagtga tgtcttttct cgacaactgg tattaattga ctatggggat 240
caatatacag ctccgacgtt gggcagtagg aaagaaaaaa cgctagccct aattaaacca 300
gatgcaatat caaaggctgg agaaataatt gaaataataa acaaagctgg atttactata 360
accaaactca aaatgatgat gctttcaagg aaagaagcat tggattttca tgtagatcac 420
cagtcaagac cctttttcaa tgagctgatc cagtttatta caactgggcc tattattgcc 480
atggagattt taagagatga tgcctatagt gaatggaaaa gactgctggg acctgcaaac 540
tctggagtgg cagcgcacaga tgcctctgaa agcattagag ccctctttgg aacagatggc 600
ataagaaatg cagcgcatgg cctcgattct tttgcttctg cggccagaga aatggagtgt 660
ttttttcctt caagtggagg ttgtgggccc gcaaacactg ctaaatttac taattgtacc 720
tgttgcattg ttaaacccca tgcgtcagt gaaggactgt tgggaaagat cctgatggct 780
atccgagatg caggttttga aatctcagct atgcagatgt tcaatatgga tcgggttaat 840
gttgaggaat tctatgaagt ttataaagga gtagtgaccg aatatcatga catggtgaca 900
gaaatgtatt ctggcccttg ttagcaatg gagattcaac agaataatgc tacaagaca 960
tttcgagaat tttgtggacc tgctgacctt gaaattgccc ggcatttacg ccctggaaact 1020
ctcagagcaa tctttggtaa aactaagatc cagaatgctg ttcactgtac tgatctgcc 1080
gaggatggcc tattagaggt tcaatacttc ttcaagatct tggataatta g 1131

SEQ ID NO: 147      moltype = AA length = 376
FEATURE            Location/Qualifiers
REGION             1..376
                   note = Human NME7 a
source             1..376
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 147
MNHSERFVFI AEWYDPNASL LRRYELLFYP GDGSVEMHDV KNHRTFLKRT KYDNLHLEDL 60
FIGNKVNVFS RQLVLIDYGD QYTARQLGSR KEKTLALIKP DAISKAGEII EIINKAGFTI 120
TKLKMMMLSR KEALDFHVDH QSRPFNFELI QFITTGPIIA MEILRDDAIC EWKRLLG PAN 180
SGVARTDASE SIRALFGTDG IRNAAHGPD S FASAAREMEL FFPSSGGCGP ANTAKFTNCT 240
CCIVKPHAVS EGLGKILMA IRDAGFEISA MQMFNMDRV N VEEFYEVYKG VVTEYHDMVT 300
EMYSGPCVAM EIQQNNATKT FRFCGPADP EIARHLRPGT LRAIFGKTKI QNAVHCTDLP 360
EDGLLEVQYF FKILDN                                     376

SEQ ID NO: 148      moltype = DNA length = 1023
FEATURE            Location/Qualifiers
misc_feature       1..1023
                   note = Human NME7 b
source             1..1023
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 148
atgcatgatg taaagaatca tcgcaccttt ttaaagcgga ccaaatatga taacctgcac 60
ttggaagatt tatttatagg caacaaagtg aatgtctttt ctgcacaaact ggtattaatt 120
gactatgggg atcaatatac agctcgccag ctgggcagta ggaaagaaaa aacgctagcc 180
ctaattaaac cagatgcaat atcaaaaggct ggagaaataa ttgaaataat aaacaaagct 240
ggatttacta taaccaaact caaaatgatg atgctttcaa ggaaagaagc attggatttt 300
catgtagatc accagtcaag accctttttc aatgagctga tccagtttat tacaactggc 360
cctattattg ccatggagat tttaagagat gatgctatat gtgaatggaa aagactgctg 420
ggactcgcaa actctggagt ggcacgcaca gatgcttctg aaagcattag agccctcttt 480
ggaacagatg gcataagaaa tgcagcgcat ggcctgattt cttttgcttc tgcggccaga 540
gaaatggagt tgttttttcc ttcaagtggg ggttgtgggc cggcaaacac tgctaaattt 600
actaattgta cctgttgcat tgttaaaccc catgctgtca gtgaaggact gttgggaaag 660
atcctgatgg ctatccgaga tgcaggtttt gaaatctcag ctatgcagat gttcaatatg 720
gatcgggtta atgttgagga attctatgaa gtttataaag gagtagtgac cgaatatcat 780
gacatggtga cagaaatgta ttctggccct tgtgtagcaa tggagattca acagaataat 840
gtacaaaaga catttcgaga attttgtgga cctgctgac ctgaaattgc ccggcattta 900
cgccctggaa ctctcagagc aatccttggg aaaactaaga tccagaatgc tgttcactgt 960
actgatctgc cagaggatgg cctattagag gttcaatact tcttcaagat cttggataat 1020
tag                                                    1023

SEQ ID NO: 149      moltype = AA length = 340
FEATURE            Location/Qualifiers
REGION             1..340
                   note = Human NME7 b
source             1..340
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 149
MHDVKNHRTF LKRTKYDNLH LEDLFIGNKV NVFSRQLVLI DYGDQYTARQ LGSRKEKTLA 60
LIKPDALSKA GEIIEIINKA GFTITKLKMM MLSRKEALDF HVDHQSRPFF NELIQFITTG 120
PIIAMEILRD DAICEWKRLG GPANSGVART DASESIRALF GTDGIRNAAH GPDSFASAA R 180
EMELFFPSSG GCGPANATAK TNCTCCIVKP HAVSEGLLGK ILMAIRDAGF EISAMQMFMN 240
DRVNVVEEYF VYKGVVTEYH DMVTEMYSGP CVAMEIQQNN ATKTFREFCG PADPEIARHL 300

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RPKTLRAIFG KTKIQNAVHC TDLPEDGLLE VQYFFKILDN 340

SEQ ID NO: 150 moltype = DNA length = 861
 FEATURE Location/Qualifiers
 misc_feature 1..861
 note = Human NME7-AB also known as NME7AB
 source 1..861
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 150
 atggaaaaaa cgctagccct aattaaacca gatgcaatat caaaggctgg agaaataatt 60
 gaaataataa acaagctgg atttactata accaaactca aaatgatgat gctttcaagg 120
 aaagaagcat tggattttca tgtagatcac cagtcaagac cctttttcaa tgagctgac 180
 cagtttatta caactgggtcc tattattgcc atggagattt taagagatga tgctatatgt 240
 gaatggaaaa gactgctggg acctgcaaac tctggagtg cagcacaga tgcttctgaa 300
 agcattagag cctcttttgg aacagatggc ataagaaatg cagcgcattg ccctgattct 360
 ttgtcttctg cggccagaga aatggagtg ttttttcctt caagtggagg ttgtgggccc 420
 gcaaacactg ctaaatttac taattgtacc tgttgcattg ttaaacccca tgctgtcagt 480
 gaaggactgt tgggaaagat cctgatggct atccgagatg caggttttga aatctcagct 540
 atgcagatgt tcaatatgga tcgggttaat gttgaggaat tctatgaagt ttataaagga 600
 gtagtgaccg aatatcatga catggtgaca gaaatgtatt ctggcccttg ttagcaaatg 660
 gagattcaac agaataatgc tacaagaca ttctgagaat ttgtggacc tgctgaccc 720
 gaaattgccc ggcatttacg ccttggaact ctccagagcaa tctttggtaa aactaagatc 780
 cagaatgctg ttcactgtac tgatctgcca gaggatggcc tattagaggt tcaatacttc 840
 ttcaagatct tggataatta g 861

SEQ ID NO: 151 moltype = AA length = 286
 FEATURE Location/Qualifiers
 REGION 1..286
 note = Human NME7-AB also known as NME7AB
 source 1..286
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 151
 MEKTLALIKP DAISKAGEII EIINKAGFTI TKLKMMLSR KEALDFHVDH QSRPPFNELI 60
 QPITTGPIIA MEILRDAIC EWKRLGPN SGVARTDASE SIRALFGTDG IRNAAHGPDG 120
 FASAREMEL FFPSSGGCGP ANTAKFTNCT CCIVKPHAVS EGLLGKILMA IRDAGFEISA 180
 MQMFNMDRVN VEEFYEVYKG VVTEYHDMVT EMYSGPCVAM EIQQNNATKT FREFCGPADP 240
 ETARHLRPGT LRAIFGKTKI QNAVHCTDLP EDGLLEVQYF FKILDN 286

SEQ ID NO: 152 moltype = DNA length = 759
 FEATURE Location/Qualifiers
 misc_feature 1..759
 note = Human NME7-X1
 source 1..759
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 152
 atgatgatgc tttcaaggaa agaagcattg gattttcatg tagatcacca gtcaagaccc 60
 tttttcaatg agctgatcca gtttattaca actggtccta ttattgccat ggagatttta 120
 agagatgatg ctatatgtga atggaaaaga ctgctgggac ctgcaaacctc tggagtgcca 180
 cgccacagatg cttctgaaag cattagagcc ctctttggaa cagatggcat aagaaatgca 240
 ggcgatggcc ctgattcttt tgcctctgcg gccagagaaa tggagttggt ttttccttca 300
 agtggaggtt gtgggcccgc aaacactgct aaatttacta attgtacctg ttgcattgtt 360
 aaaccccatg ctgtcagtgga agcactgttg ggaaagatcc tgatggctat ccgagatgca 420
 gggtttgaaa tctcagctat gcagatgttc aatatggatc ggggttaagt tgaggaattc 480
 tatgaagttt ataaaggagt agtgaccgaa tatcatgaca tggtgacaga aatgtattct 540
 tgcccttggt tagcaatgga gattcaacag aataatgcta caaagacatt tcgagaattt 600
 tgtggacctg ctgatcctga aattgcccgg catttacgcc ctggaactct cagagcaatc 660
 tttggtaaaa ctaagatcca gaatgctgtt cactgtactg atctgccaga ggatggccta 720
 ttagaggttc aatactcttt caagatcttg gataattag 759

SEQ ID NO: 153 moltype = AA length = 252
 FEATURE Location/Qualifiers
 REGION 1..252
 note = Human NME7-X1
 source 1..252
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 153
 MMMLSRKEAL DFVHDHQRSP FFNELIQFIT TGPPIAMEIL RDDAICEWKR LLGPANSGVA 60
 RTDASESIRA LFGTDGIRNA AHGPDSPASA AREMELFFPS SGGCGPANTA KFTNCTCCIV 120
 KPHAVSEGLL GKILMAIRDA GFELISAMQM NMDRVNVEEF YEVYKGVVTE YHDMVT EMY 180
 GPCVAMEIQQ NNATKTFREF CGPADPEIAR HLRPGTLRAI FGKTKIQNAV HCTDLPEDGL 240
 LEVQYFFKIL DN 252

SEQ ID NO: 154 moltype = DNA length = 1128
 FEATURE Location/Qualifiers
 misc_feature 1..1128

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note = Human NME7 a (optimized for E coli expression)
source      1..1128
            mol_type = other DNA
            organism = synthetic construct

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SEQUENCE: 154
atgaatcact ccgaacgctt tgtttttatc gccgaatggt atgacccgaa tgcttccctg 60
ctgcgcgcgt acgaactgct gttttatccg ggcgatggta gcgtggaaat gcatgacgtt 120
aaaaatcacc gtacotttct gaaacgcacg aaatatgata atctgcatct ggaagacctg 180
tttattggca acaaaagtcaa tgtgttctct cgtcagctgg tgctgatcga ttatggcgac 240
cagtacaccg cgcgtcaact gggtagtcgc aaagaaaaaa cgctggccct gattaaaccg 300
gatgcaatct ccaaagctgg cgaattatc gaaattatca acaaaagcggg ttccaccatc 360
acgaacttga aaatgatgat gctgagccgt aaagaagccc tggattttca tgcgaccac 420
cagtcctgcc cgtttttcaa tgaactgatt caattcatca ccacgggtcc gattatcgca 480
atggaaattc tgcgtgatga cgtatctgca gaatggaaac gcctgctggg cccggcaaac 540
tcagggtgtg cgcgtaccga tgcagtgtaa tccattcgcg ctctgtttgg caccgatggt 600
atccgtaatg cagcacatgg tccggactca ttcgcatcgg cagctcgtga aatggaaactg 660
ttttcccgga gctctggcgg ttgcggtccg gcaaacaccg ccaaatttac caattgtacg 720
tgctgtattg tcaaacccga cgcagtgtaa gaaggcctcg tgggtaaaat tctgatggca 780
atccgtgatg tctggtttga aatctcggcc atgcagatgt tcaacatgga ccgcgttaac 840
gtcgaagaat tctcgaagt ttacaaagcg gtggttacgg aatatcacga tatggttacg 900
gaaatgtact ccggtccgtg cgtcgcgatg gaaattcagc aaaaacaatg caccaaaacg 960
tttcgtgaat tctgtggtcc ggcagatccg gaaatcgac gtcactgctg tccgggtacc 1020
ctgcgcgcaa tttttggtta aacgaaaatc cagaacgctg tgcactgtac cgatctgccc 1080
gaagacggtc tgctggaagt tcaatacttt ttcaaaatcc tggataat 1128

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SEQ ID NO: 155      moltype = AA length = 378
FEATURE            Location/Qualifiers
REGION            1..378
                  note = Human NME7 a (optimized for E coli expression)
source            1..378
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 155
MNHSERFVFI AEWYDPNASL LRRYELLFYP GDGSVMHVDV KNHRTFLKRT KYDNLHLEDL 60
FIGNKVNVFS RQLVLIDYGD QYTARQLGSR KEKTLALIKP DAISKAGEII EIINKAGFTI 120
TKLKMMMLSR KEALDFHVDH QSRPFNFELI QFITTGPIIA MEILRDDAIC EWKRLGPGAN 180
SGVARTDASE SIRALFGTDG IRNAAHGPDG FASAAREMEL FFPSSGGCGP ANTAKFTNCT 240
CCIVKPHAVS EGLLGKILMA IRDAGFEISA MQMFNMDRVN VEEFYEVYKG VVTEYHDMVT 300
EMYSGPCVAM EIQQNNATKT FREFCGPADP EIARHLRPGT LRAIFGKTKI QNAVHCTDLP 360
EDGLLEVQYF FKILDNTG 378

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SEQ ID NO: 156      moltype = DNA length = 1020
FEATURE            Location/Qualifiers
misc_feature       1..1020
                  note = Human NME7 b (optimized for E coli expression)
source            1..1020
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 156
atgcatgacg ttaaaaaatca ccgtaccttt ctgaaacgca cgaatatga taatctgcat 60
ctggaagacc tgtttattgg caacaaagtc aatgtgttct ctcgtcagct ggtgctgac 120
gattatggcg accagatcac cgcgcgtcaa ctgggtagtc gcaagaaaaa aacgctggcc 180
ctgattaaac cggatgcaat ctccaaagct ggcgaaatta tcgaaattat caacaaagcg 240
ggtttcacca tcacgaaatc gaaaatgatg atgctgagcc gtaaaagaagc cctggatttt 300
catgtcgacc accagtctcg ccggtttttc aatgaactga ttcaattcat caccacgggt 360
ccgattatcg caatggaaat tctgctgatg gacgctatct gcgaatggaa acgcctgctg 420
ggcccgccga actcaggtgt tgcgcgtacc gatgccagtg aatccattcg cgctctgttt 480
ggcaccgatg gtatccgtaa tgcagcacat ggtccggact cattegcate ggcagctcgt 540
gaaatggaa cgtttttccc gagctctggc ggttgcggtc cggcaaacac cgccaaattt 600
accaattgta cgtgctgtat tgtcaaaccc cagcagtggt cagaaggcct gctgggttaa 660
attctgatgg caatccgtga tgctggcttt gaaatctcgg ccatgcagat gttcaacatg 720
gaccgcgtta acgtcgaaga attctacgaa gtttacaaag gcgtgggttac cgaatatcac 780
gatattggtta cggaaatgta ctccggtccg tgcgtcgaga tggaaattca gcaaaacaat 840
gccacccaaa cgtttcgtga attctgtggt ccggcagatc cggaaatcgc acgtcatctg 900
cgtccgggta ccctgcgcgc aatttttggg aaacgaaaaa tccagaacgc tgtgcactgt 960
accgatctgc cgggaagacg tctgctggaa gttcaatact ttttcaaaat tctggataat 1020

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SEQ ID NO: 157      moltype = AA length = 342
FEATURE            Location/Qualifiers
REGION            1..342
                  note = Human NME7 b (optimized for E coli expression)
source            1..342
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 157
MHDVKNHRTF LKRTKYDNLH LEDLFIGNKV NVFSRQLVLI DYGDQYTRQ LGSRKETLA 60
LIKPDAlSKA GEIIEIINKA GFTITKLKMM MLSRKALDF HVDHQSRPFF NELIQFITG 120
PIIAMEILRD DAICEWKRLD GPANSGVART DASESIRALF GTDGIRNAAH GPDSFASAA 180
EMELFFPSSG GCGPANTAKF TNCTCCIVKP HAVSEGLLGK ILMAIRDAGF EISAMQMFNM 240

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DRVNVEEFYE VYKGVVTEYH DMVTEMYSGP CVAMEIQNN ATKTFREFCG PADPEIARHL 300
RPGTLRAIFG KTKIQNAVHC TDLPEGLLE VQYFFKILDN TG 342

SEQ ID NO: 158      moltype = DNA length = 858
FEATURE            Location/Qualifiers
misc_feature        1..858
                    note = Human NME7-AB also known as NME7AB (optimized for E
                    coli expression)
source              1..858
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 158
atggaaaaaa cgctggccct gattaaaccg gatgcaatct ccaaagctgg cgaaattatc 60
gaaattatca acaaagcggg ttccaccatc acgaaactga aaatgatgat gctgagccgt 120
aaagaagccc tggattttca tgcgaccac cagtctcgcc cgtttttcaa tgaactgatt 180
caatcatca ccacgggtcc gattatcgca atggaattc tgcgtgatga cgctatctgc 240
gaatggaaac gcctgctggg cccggcaaac tcaggtgttg cgcgtacgga tgccagtga 300
tccattcgcg ctctgtttgg caccgatggt atccgtaatg cagcacatgg tccggactca 360
ttcgcatcgg cagctcgtga aatggaactg ttttcccgga gctctggcgg ttgcgggtccg 420
gcaaacaccg ccaaatttac caattgtacg tgcgtgtattg tcaaacccga cgcagtgtca 480
gaaggcctgc tgggtaaaat tctgatggca atccgtgatg ctggctttga aatctcgccc 540
atgcagatgt tcaacatgga ccgcgttaac gtcgaagaat tctacgaagt ttacaaaggc 600
gtggttacgg aatatcacga tatggttacg gaaatgtact ccggtccgtg cgtcgcgatg 660
gaaattcagc aaaacaatgc caccaaaacg tttcgtgaat tctgtggtcc gccagatccg 720
gaaatcgcac gtcactctgc tccgggtacc ctgcgcgcaa ttttggttaa aacgaaaatc 780
cagaacgctg tgcaactgtac cgatctgccg gaagacggtc tgcgtggaagt tcaatacttt 840
ttcaaaatc tgataat 858

SEQ ID NO: 159      moltype = AA length = 288
FEATURE            Location/Qualifiers
REGION             1..288
                    note = Human NME7-AB also known as NME7AB (optimized for E
                    coli expression)
source              1..288
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 159
MEKTLALIKP DAISKAGEII EIINKAGFTI TKLKMMMLSR KEALDPHVDH QSRPPFNELI 60
QPIITGPIIA MEILRDDAIC EWKRLGPGAN SGVARTDASE SIRALFGTDG IRNAAHGPD 120
FASAREMEL FFPSSGGCGP ANTAKFTNCT CCIVKPHAVS EGLLGKILMA IRDAGFEISA 180
MQMFNMDRVN VEEFYEVYKG VVTEYHDMVT EMYSGPCVAM EIQQNNATKT FREFCGPADP 240
ETARHLRPGT LRAIFGKTKI QNAVHCTDLP EDGLLEVQYF FKILDNTG 288

SEQ ID NO: 160      moltype = DNA length = 756
FEATURE            Location/Qualifiers
misc_feature        1..756
                    note = Human NME7-X1 (optimized for E coli expression)
source              1..756
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 160
atgatgatgc tgagccgtaa agaagccctg gattttcatg tgcaccacca gtctcgccc 60
tttttcaatg aactgattca attcatcacc acgggtccga ttatcgcaat ggaaattctg 120
cgtgatgacg ctatctcgca atggaaacgc ctgctggggc cggcaaaactc aggtgttgcg 180
cgtaccgatg ccagtgaatc cattcgcgct ctgtttggca ccatggttat ccgtaatgca 240
gcacatggtc cggactcatt cgcacatggc gctcgtgaaa tggaaactgt tttcccgagc 300
tctggcggtt gcggtccggc aaacaccgcc aaatttacca attgtacgtg ctgtattgtc 360
aaaccgcacg cagtgtcaga aggcctgctg ggtaaaatc tcatggcaat ccgtgatgct 420
ggctttgaaa tctcgcccat gcagatgttc aacatggacc gcgttaacct cgaagaatc 480
tacgaagttt acaaaggcgt ggttacggaa tatcacgata tggttacgga aatgtactcc 540
ggtcctgtcg tccgatggga aattcagcaa aacaatgcc aaaaacgtt tctgtaattc 600
tgtggtccgg cagatccgga aatcgcacgt catctgcgtc cgggtaccct gcgcgcaatt 660
tttggtaaaa cgaaaatcca gaacgctgtg cactgtaccg atctgccgga agacggtctg 720
ctggaagtgc aatacttttt caaattctg gataat 756

SEQ ID NO: 161      moltype = AA length = 254
FEATURE            Location/Qualifiers
REGION             1..254
                    note = Human NME7-X1 (optimized for E coli expression)
source              1..254
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 161
MMMLSRKEAL DFHVDHQSRP FFNELIQFIT TGPIIAMEIL RDDAICEWKR LLGPANSGVA 60
RTDASESIRA LFGTDGIRNA AHGPDSPFASA AREMELFFPS SGGCGPANTA KFTNCTCCIV 120
KPHAVSEGLL GKILMAIRDA GFELISAMQM NMDRVNVEEF YEVYKGVVTE YHDMVTEMY 180
GPCVAMEIQ NNATKTFREF CGPADPEIAR HLRPGTLRAI FGKTKIQNAV HCTDLPEDGL 240
LEVQYFFKIL DNTG 254

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SEQ ID NO: 162 moltype = AA length = 91
 FEATURE Location/Qualifiers
 REGION 1..91
 note = DM10 domain of NME7
 source 1..91
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 162
 MNHSERFVFI AEWYDPNASL LRRYELLFYP GDGSVEMHDV KNHRTFLKRT KYDNLHLEDL 60
 FIGNKVNVFS RQLVLIDYGD QYTARQLGSR K 91

SEQ ID NO: 163 moltype = AA length = 43
 FEATURE Location/Qualifiers
 REGION 1..43
 note = Fragment or variation of PSMGFR peptide
 source 1..43
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 163
 SNIKFRPGSV VVQLTLAPRE GTINVHDTV QFNQYKTEAA SRY 43

SEQ ID NO: 164 moltype = AA length = 35
 FEATURE Location/Qualifiers
 REGION 1..35
 note = Fragment or variation of PSMGFR peptide
 source 1..35
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 164
 SVVVLTLAP REGTINVHDV ETQFNQYKTE AASRY 35

SEQ ID NO: 165 moltype = AA length = 24
 FEATURE Location/Qualifiers
 REGION 1..24
 note = Fragment or variation of PSMGFR peptide
 source 1..24
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 165
 VQLTLAPREG TINVHDTVETQ FNQY 24

SEQ ID NO: 166 moltype = AA length = 24
 FEATURE Location/Qualifiers
 REGION 1..24
 note = Fragment or variation of PSMGFR peptide
 source 1..24
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 166
 SNIKFRPGSV VVQLTLAPRE GTIN 24

SEQ ID NO: 167 moltype = AA length = 38
 FEATURE Location/Qualifiers
 REGION 1..38
 note = Fragment or variation of PSMGFR peptide
 source 1..38
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 167
 SNIKFRPGSV VVQLTLAPRE GTINVHDTV QFNQYKTE 38

SEQ ID NO: 168 moltype = AA length = 45
 FEATURE Location/Qualifiers
 REGION 1..45
 note = Fragment or variation of PSMGFR peptide
 source 1..45
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 168
 VQLTLAPREG TINVHDTVETQ FNQYKTEAAS RYNLTISDVS VSDVP 45

SEQ ID NO: 169 moltype = AA length = 29
 FEATURE Location/Qualifiers
 REGION 1..29
 note = Ser of NME7B peptide 3 (B domain)
 source 1..29
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 169

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AIFGKTKIQN AVHSTDLPED GLLEVQYFF	29
SEQ ID NO: 170	moltype = AA length = 35
FEATURE	Location/Qualifiers
REGION	1..35
	note = N-10 peptide
source	1..35
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 170	
QFNQYKTEAA SRYNLTISDV SVSDVPPPPS AQSGA	35
SEQ ID NO: 171	moltype = AA length = 35
FEATURE	Location/Qualifiers
REGION	1..35
	note = C-10 peptide
source	1..35
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 171	
GTINVHDTVET QFNQYKTEAA SRYNLTISDV SVSDV	35
SEQ ID NO: 172	moltype = AA length = 119
FEATURE	Location/Qualifiers
REGION	1..119
	note = 8F9A5A1H heavy chain
source	1..119
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 172	
IQLVQSGPEL KKPGETVKIS CKASGYFTN YGMNWKQAP GKGLKWMGWI NTYTGEPTYV	60
DDFKGRFAPS LETSATTAYL QINNLTNEDT STYFCARLRG IRPGPLAYWG QGTLVTVSA	119
SEQ ID NO: 173	moltype = AA length = 121
FEATURE	Location/Qualifiers
REGION	1..121
	note = 8F9A4P3H heavy chain
source	1..121
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 173	
VQLQQSGPEL VKPGASVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNNGVTNYN	60
QKFKGKATLT VDKSSSTAYM ELRSLTSEDS AVYYCARRY HSLYVFYFDY WQGTTTLTVS	120
S	121
SEQ ID NO: 174	moltype = AA length = 116
FEATURE	Location/Qualifiers
REGION	1..116
	note = 5D9E2B11H
source	1..116
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 174	
VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNNGVTNYN	60
QKFKGKATLT VDKSSSTAYM ELRSLTSEDS AVYYCARRY HSTYVFYFDS WQGTT	116
SEQ ID NO: 175	moltype = AA length = 121
FEATURE	Location/Qualifiers
REGION	1..121
	note = 5D9E10E4H
source	1..121
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 175	
VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNNGVTNYN	60
QKFKGKATLT VDKSSSTAYM ELRSLTSEDS AVYYCARRY HSTYVFYFDS WQGTTTLTVS	120
S	121
SEQ ID NO: 176	moltype = AA length = 121
FEATURE	Location/Qualifiers
REGION	1..121
	note = 5D9G2C4H
source	1..121
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 176	
VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNNGVTNYN	60
QKFKGKATLT VDKSSSTAYM ELRSLTSEDS AVYYCARRY HSTYVFYFDS WQGTTTLTVS	120
S	121

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SEQ ID NO: 177 moltype = AA length = 121
 FEATURE Location/Qualifiers
 REGION 1..121
 note = 5F3A5D4H
 source 1..121
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 177
 VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNGVTNYN 60
 QKFKGKATLT VDKSSSTAYM ELRLTSEDS AVYYCARRY HSTYVFYFDS WGQGTTLTVS 120
 S 121

SEQ ID NO: 178 moltype = AA length = 121
 FEATURE Location/Qualifiers
 REGION 1..121
 note = 8H5H5G4H
 source 1..121
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 178
 VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNGVTNYN 60
 QKFKGKATLT VDKSSSTAYM ELRLTSEDS AVYYCARRY HSTYVFYFDS WGQGTTLTVS 120
 S 121

SEQ ID NO: 179 moltype = AA length = 121
 FEATURE Location/Qualifiers
 REGION 1..121
 note = 8F9A4A3H
 source 1..121
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 179
 VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNGVTNYN 60
 QKFKGKATLT VDKSSSTAYM ELRLTSEDS AVYYCARRY HSLYVFYFDY WGQGTTLTVS 120
 S 121

SEQ ID NO: 180 moltype = AA length = 116
 FEATURE Location/Qualifiers
 REGION 1..116
 note = 5D9E2B11H
 source 1..116
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 180
 VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNGVTNYN 60
 QKFKGKATLT VDKSSSTAYM ELRLTSEDS AVYYCARRY HSTYVFYFDS WGQGT 116

SEQ ID NO: 181 moltype = AA length = 121
 FEATURE Location/Qualifiers
 REGION 1..121
 note = 5D9E10E4H
 source 1..121
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 181
 VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNGVTNYN 60
 QKFKGKATLT VDKSSSTAYM ELRLTSEDS AVYYCARRY HSTYVFYFDS WGQGTTLTVS 120
 S 121

SEQ ID NO: 182 moltype = AA length = 121
 FEATURE Location/Qualifiers
 REGION 1..121
 note = 5D9G2C4H
 source 1..121
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 182
 VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNGVTNYN 60
 QKFKGKATLT VDKSSSTAYM ELRLTSEDS AVYYCARRY HSTYVFYFDS WGQGTTLTVS 120
 S 121

SEQ ID NO: 183 moltype = AA length = 121
 FEATURE Location/Qualifiers
 REGION 1..121
 note = 5F3A5D4H
 source 1..121
 mol_type = protein
 organism = synthetic construct

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SEQUENCE: 183
VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNNGVTNYN 60
QKFKGKATLT VDKSSSTAYM ELRLTSEDS AVYYCARRY HSTYVFYFDS WQGTTTLTVS 120
S 121

SEQ ID NO: 184      moltype = AA length = 121
FEATURE            Location/Qualifiers
REGION              1..121
                    note = 8H5H5G4H
source              1..121
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 184
VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNNGVTNYN 60
QKFKGKATLT VDKSSSTAYM ELRLTSEDS AVYYCARRY HSTYVFYFDS WQGTTTLTVS 120
S 121

SEQ ID NO: 185      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION              1..108
                    note = 8F9A4P3L
source              1..108
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 185
ETTVTQSPAS LSMAIGKVT IRCITSTDID DDMNWyQQKP GEPPKLLISE GNTLRPGVPS 60
RFSSSGYGTD FVFTIENMLS EDVADYYCLQ SDNLPLTFGS GTKLEIKR 108

SEQ ID NO: 186      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION              1..108
                    note = 5D9E2B11L
source              1..108
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 186
DIQMTQTSS LSASLGDRVT ISCSASQGIS NYLWFWQQKP DGTIKLLIYY TSSLHSGVPS 60
RFSGSGSGTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR 108

SEQ ID NO: 187      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION              1..108
                    note = 5D9E10E4L
source              1..108
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 187
DIQMTQTSS LSASLGDRVT ISCSASQGIS NYLWFWQQKP DGTIKLLIYY TSSLHSGVPS 60
RFSGSGSGTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR 108

SEQ ID NO: 188      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION              1..108
                    note = 5D9G2C4L
source              1..108
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 188
DIQMTQTSS LSASLGDRVT ISCSASQGIS NYLWFWQQKP DGTIKLLIYY TSSLHSGVPS 60
RFSGSGSGTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR 108

SEQ ID NO: 189      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION              1..108
                    note = 5F3A5D4L
source              1..108
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 189
DIQMTQTSS LSASLGDRVT ISCSASQGIS NYLWFWQQKP DGTIKLLIYY TSSLHSGVPS 60
RFSGSGSGTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR 108

SEQ ID NO: 190      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION              1..108
                    note = 8H5H5G4L
source              1..108
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 190
DIQMTQTTSS LSASLGDRVT ISCSASQGIS NYLWNFQQKP DGTIKLLIYY TSSLHSGVPS 60
RFGSGSGSTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR 108

SEQ ID NO: 191      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = 8F9A5A1L
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 191
EILLTQSPAI IAASPGKEVT ITCSASSSVS YMNWYQQKPG SSPKIWIYGI SNLASGVPAR 60
FSGSGSGTSF SFTINSMEAE DVATYYCQQR SSYPPTFGGG TKLEIKR 107

SEQ ID NO: 192      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION             1..108
                   note = 5D9E2B11L
source             1..108
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 192
DIQMTQTTSS LSASLGDRVT ISCSASQGIS NYLWNFQQKP DGTIKLLIYY TSSLHSGVPS 60
RFGSGSGSTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR 108

SEQ ID NO: 193      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION             1..108
                   note = 5D9E10E4L
source             1..108
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 193
DIQMTQTTSS LSASLGDRVT ISCSASQGIS NYLWNFQQKP DGTIKLLIYY TSSLHSGVPS 60
RFGSGSGSTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR 108

SEQ ID NO: 194      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION             1..108
                   note = 5D9G2C4L
source             1..108
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 194
DIQMTQTTSS LSASLGDRVT ISCSASQGIS NYLWNFQQKP DGTIKLLIYY TSSLHSGVPS 60
RFGSGSGSTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR 108

SEQ ID NO: 195      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION             1..108
                   note = 5F3A5D4L
source             1..108
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 195
DIQMTQTTSS LSASLGDRVT ISCSASQGIS NYLWNFQQKP DGTIKLLIYY TSSLHSGVPS 60
RFGSGSGSTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR 108

SEQ ID NO: 196      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION             1..108
                   note = 8H5H5G4L
source             1..108
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 196
DIQMTQTTSS LSASLGDRVT ISCSASQGIS NYLWNFQQKP DGTIKLLIYY TSSLHSGVPS 60
RFGSGSGSTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR 108

SEQ ID NO: 197      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION             1..108
                   note = 8F9A4P3L
source             1..108
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 197
ETTVTQSPAS LSMAIGKEVT IRCITSTDID DDMNWWYQQKP GEPPKLLISE GNTLRPGVPS 60

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RFSSSGYGTD FVFTIENMLS EDVADYYCLQ SDNLPLTFGS GTKLEIKR 108

SEQ ID NO: 198 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = 8F9A5A1L
 source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 198
 EILLTQSPAI IAASPGKEKVT ITCSSSSVS YMNWYQQKPG SSPKIWIYGI SNLASGVPAR 60
 FSGSGSGTGF SFTINSMEAE DVATYYCQQR SSYPPTFGGG TKLEIKR 107

SEQ ID NO: 199 moltype = AA length = 131
 FEATURE Location/Qualifiers
 REGION 1..131
 note = NME7A partial sequence
 source 1..131
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 199
 EKTALIKPD AISKAGEIIE IINKAGFTIT KLKMMMLSRK EALDFHVDHQ SRPFFNELIQ 60
 FITTGPIIAM EILRDDAICE WKRLGFPANS GVARTDASES IRALFGTDGI RNAAHGPDSE 120
 ASAAREMELF F 131

SEQ ID NO: 200 moltype = AA length = 152
 FEATURE Location/Qualifiers
 REGION 1..152
 note = NME2 partial sequence
 source 1..152
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 200
 MANLERTFIA IKPDGVQGRGL VGEIIKRFEQ KGFRVLVAMKF LRASEEHLKQ HYIDLKDRPF 60
 FPGLVKYMNS GPVVAMVWEG LNVVKTGRVM LGETNPADSK PGITRGDFCI QVGRNIIHGS 120
 DSVKSAEKEI SLWFKPEELV DYKSCAHDWV YE 152

SEQ ID NO: 201 moltype = AA length = 134
 FEATURE Location/Qualifiers
 REGION 1..134
 note = NME7B partial sequence
 source 1..134
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 201
 NCTCCIVKPH AVSEGLLGKI LMAIRDAGFE ISAMQMFNMD RVNVVEEFYEV YKGVVTEYHD 60
 MVTEMYSGPC VAMEIQQNNNA TKTFREFCGP ADPEIARHLR PGTLRIFAIGK TKIQNAVHCT 120
 DLPEDGLLEV QYFF 134

SEQ ID NO: 202 moltype = AA length = 152
 FEATURE Location/Qualifiers
 REGION 1..152
 note = NME2 partial sequence
 source 1..152
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 202
 MANLERTFIA IKPDGVQGRGL VGEIIKRFEQ KGFRVLVAMKF LRASEEHLKQ HYIDLKDRPF 60
 FPGLVKYMNS GPVVAMVWEG LNVVKTGRVM LGETNPADSK PGITRGDFCI QVGRNIIHGS 120
 DSVKSAEKEI SLWFKPEELV DYKSCAHDWV YE 152

SEQ ID NO: 203 moltype = AA length = 131
 FEATURE Location/Qualifiers
 REGION 1..131
 note = NME7A partial sequence
 source 1..131
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 203
 EKTALIKPD AISKAGEIIE IINKAGFTIT KLKMMMLSRK EALDFHVDHQ SRPFFNELIQ 60
 FITTGPIIAM EILRDDAICE WKRLGFPANS GVARTDASES IRALFGTDGI RNAAHGPDSE 120
 ASAAREMELF F 131

SEQ ID NO: 204 moltype = AA length = 130
 FEATURE Location/Qualifiers
 REGION 1..130
 note = NME3 partial sequence
 source 1..130
 mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 204
ERTFLAVKPD  GVQRRLVGEI  VRRFERKGFK  LVALKLQAS  EELLREHYVE  LRERPFYSRL  60
VKYMGSGPVV  AMVWQGLDVV  RASRALIGAT  DPGDATPGTI  RGDFCPEVGK  NVIHGSDSVE  120
SAQREIALWF                                     130

SEQ ID NO: 205                moltype = AA  length = 134
FEATURE                      Location/Qualifiers
REGION                        1..134
                                note = NME7B partial sequence
source                        1..134
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 205
NCTCCIVKPH  AVSEGLLGKI  LMAIRDAGFE  ISAMQMFNMD  RVNVEEFYEV  YKGVVTEYHD  60
MVTEMYSGPC  VAMEIQNNA  TKTFRFCGP  ADPEIARHLR  PGTLRIFGK  TKIQNAVHCT  120
DLPEDGLLEV  QYFF                                     134

SEQ ID NO: 206                moltype = AA  length = 169
FEATURE                      Location/Qualifiers
REGION                        1..169
                                note = NME3 partial sequence
source                        1..169
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 206
MICLVLTIFA  NLFPSAYSGV  NERTFLAVKP  DGVQRRLVGE  IVRRFERKGF  KLVALKLQVA  60
SEELLREHV  ELRERPFYSR  LVKYMGS GPV  VAMVWQGLDV  VRASRALIGA  TDPGDATPGT  120
IRGDFCPEVG  KNVIHGSDSV  ESAQREIALW  FREDELLCWE  DSAGHWLYE   169

SEQ ID NO: 207                moltype = AA  length = 131
FEATURE                      Location/Qualifiers
REGION                        1..131
                                note = NME7A partial sequence
source                        1..131
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 207
EKTALALIKPD  AISKAGEIIE  IINKAGFTIT  KLKMMMLSRK  EALDFHVDHQ  SRPFFNELIQ  60
FITTGPIIAM  EILRDDAICE  WKRLGFPANS  GVRTDASES  IRALFGTDGI  RNAAHGPD SF  120
ASAAREMELF  F                                     131

SEQ ID NO: 208                moltype = AA  length = 130
FEATURE                      Location/Qualifiers
REGION                        1..130
                                note = NME4 partial sequence
source                        1..130
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 208
ERTLVAVKPD  GVQRRLVGDV  IQRFERRGFT  LVGMKMLQAP  ESVLAEHYQD  LRRKPFYPAL  60
IRYMSSGPVV  AMVWEGYNV  RASRAMIGHT  DSAEAPGTI  RGDFSVHISR  NVIHASDSVE  120
GAQREIQLWF                                     130

SEQ ID NO: 209                moltype = AA  length = 132
FEATURE                      Location/Qualifiers
REGION                        1..132
                                note = NME7B partial sequence
source                        1..132
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 209
TCCIVKPHAV  SEGLLGKILM  AIRDAGFEIS  AMQMFNMDRV  NVEEFYEVYK  GVVTEYHDMV  60
TEMYSGPCVA  MEIQNNATK  TFREFCGPAD  PEIARHLRPG  TLRAIFGKTK  IQNAVHCTDL  120
PEDGLLEVQY  FF                                     132

SEQ ID NO: 210                moltype = AA  length = 128
FEATURE                      Location/Qualifiers
REGION                        1..128
                                note = NME4 partial sequence
source                        1..128
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 210
TLVAVKPDGV  QRRLVGDIVQ  RFERRGFTLV  GMKMLQAPES  VLAEHYQDLR  RKPFPYPALIR  60
YMSSGPVVAM  VWEGYNVRA  SRAMIGHTDS  AEAAPGTIRG  DFSVHISRN  IHASDSVEGA  120
QREIQLWF                                     128

SEQ ID NO: 211                moltype = AA  length = 131

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FEATURE	Location/Qualifiers	
REGION	1..131	
	note = NME7A partial sequence	
source	1..131	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 211		
EKTLALIKPD AISKAGEIIE IINKAGFTIT KLKMMMLSRK EALDFHVDHQ SRPFFNELIQ		60
FITGPIIAM EILRDDAICE WKRLGPNAS GVARTDASES IRALFGTDGI RNAAHGPDSE		120
ASAAREMELF F		131
SEQ ID NO: 212	moltype = AA length = 131	
FEATURE	Location/Qualifiers	
REGION	1..131	
	note = NME5 partial sequence	
source	1..131	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 212		
EKTLAIKPD IVDKEEIQD IILRSGFTIV QRRKLRLSPE QCSNFYVEKY GKMFPPNLTA		60
YMSSGPLVAM ILARHKAISY WLELLGPNNS LVAKETHPDS LRATYGTDDL RNALHGSNDF		120
AAAREIRFM F		131
SEQ ID NO: 213	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = NME7B partial sequence	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 213		
TCCIVKPHAV SEGGLGKILM AIRDAGFEIS AMQMFNMDRV NVEEFYEVYK GVVTEYHDMV		60
TEMYSGPCVA MEIQNNATK TREFCGPAD PEIARHLRPG TLRAIFGKTK IQNAVHCTDL		120
PEDGLEEVQY FF		132
SEQ ID NO: 214	moltype = AA length = 129	
FEATURE	Location/Qualifiers	
REGION	1..129	
	note = NME5 partial sequence	
source	1..129	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 214		
TLAIKPDIV DKEEIQDII LRSFTIVQR RKLRLSPEQC SNFYVEKYGK MFPNLTAYM		60
SSGPLVAMIL ARHKAISYWL ELLGPNNSLV AKETHPDSL AIYGTDDLRLN ALHGSNDFAA		120
AREIRFMF		129
SEQ ID NO: 215	moltype = AA length = 129	
FEATURE	Location/Qualifiers	
REGION	1..129	
	note = NME7A partial sequence	
source	1..129	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 215		
TLALIKPDAI SKAGEIIEII NKAGFTITKL KMMMLSRKEA LDFHVDHQSR PFFNELIQFI		60
TTGPIIAMEI LRDDAICEWK RLLGPANSGV ARTDASESIR ALFGTDGIRN AAHGPDSEFAS		120
AAREMELFF		129
SEQ ID NO: 216	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = NME6 partial sequence	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 216		
TLALIKPDAV AHPLILEAVH QQILSNKFLI VMRELLWRK EDCQRFYREH EGRFFYQRLV		60
EFMASGPIRA YILAHKDAIQ LWRTLMGPTR VFRARHVAPD SIRGSFGLTD TRNTTHGSDS		120
VVSASREIAA FF		132
SEQ ID NO: 217	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = NME7B partial sequence	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 217		

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TCIVKPHAV	SEGLGKILM	AIRDAGFEIS	AMQMFNMDRV	NVEEFYEVYK	GVVTEYHDMV	60
TEMYSGPCVA	MEIQONNATK	TFREFCGPAD	PEIARHLRPG	TLRAIFGKTK	IQNAVHCTDL	120
PEDGLLEVQY	FF					132
SEQ ID NO: 218	moltype = AA length = 132					
FEATURE	Location/Qualifiers					
REGION	1..132					
	note = NME6 partial sequence					
source	1..132					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 218						
TLALIKPDV	AHPLILEAVH	QQILSNKFLI	VRMRELLWRK	EDCQRFYREH	EGRFFYQRLV	60
EFMASGPIRA	YILAHKDAIQ	LWRTLMGPTR	VFRARHVAPD	SIRGSFGLTD	TRNTTHGSDS	120
VVSASREIAA	FF					132
SEQ ID NO: 219	moltype = AA length = 131					
FEATURE	Location/Qualifiers					
REGION	1..131					
	note = NME7A partial sequence					
source	1..131					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 219						
EKTLALIKPD	AISKAGEIIE	IINKAGFTIT	KLKMMMLSRK	EALDFHVDHQ	SRPFFNELIQ	60
FITTGPIIAM	EILRDDAICE	WKRLGPPANS	GVARTDASES	IRALFGTDGI	RNAAHGPDSP	120
ASAAAREMELF	F					131
SEQ ID NO: 220	moltype = AA length = 133					
FEATURE	Location/Qualifiers					
REGION	1..133					
	note = NME8 partial sequence					
source	1..133					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 220						
EKTLALLRPN	LFHERKDDVL	RIIKDEDFKI	LEQRQVVLSE	KEAQALCKEY	ENEDYFNKLI	60
ENMTSGPSLA	LVLRLDNLQ	YWKQLLGPR	VEEAIEYPPE	SLCAQFAMDS	LPVNQLYGSD	120
SLETAEREIQ	HFF					133
SEQ ID NO: 221	moltype = AA length = 127					
FEATURE	Location/Qualifiers					
REGION	1..127					
	note = NME7A partial sequence					
source	1..127					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 221						
EKTLALIKPD	AISKAGEIIE	IINKAGFTIT	KLKMMMLSRK	EALDFHVDHQ	SRPFFNELIQ	60
FITTGPIIAM	EILRDDAICE	WKRLGPPANS	GVARTDASES	IRALFGTDGI	RNAAHGPDSP	120
ASAAAREM						127
SEQ ID NO: 222	moltype = AA length = 127					
FEATURE	Location/Qualifiers					
REGION	1..127					
	note = NME8 partial sequence					
source	1..127					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 222						
QSTLGLIKPH	ATSEQREQIL	KIVKEAGFDL	TQVKMFLTP	EQIEKIYPKV	TGKDFYKDLL	60
EMLSVGSPMV	MILTKWNAVA	EWRLMGPTD	PEEAKLLSPD	SIRAQFGISK	LKNIVHGASN	120
AYEAKEV						127
SEQ ID NO: 223	moltype = AA length = 63					
FEATURE	Location/Qualifiers					
REGION	1..63					
	note = NME7A partial sequence					
source	1..63					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 223						
TLALIKPDAI	SKAGEIIEII	NKAGFTITKL	KMMLSRKEA	LDFHVDHQSR	PPFNELIQFI	60
TTG						63
SEQ ID NO: 224	moltype = AA length = 65					
FEATURE	Location/Qualifiers					
REGION	1..65					
	note = NME8 partial sequence					

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source                1..65
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 224
STAIKPDVAV ISKKVLEIKR KITKAGFIIE AEHKTVLTEE QVVFYSRIA DQCDFEEFVS 60
FMTSG                                           65

SEQ ID NO: 225        moltype = AA length = 116
FEATURE              Location/Qualifiers
REGION              1..116
                      note = NME7B partial sequence
source              1..116
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 225
TCCIVKPHAV SEGLLGKILM AIRDAGFEIS AMQMFNMDRV NVEEFYEVYK GVVTEYHDMV 60
TEMYSGPCVA MEIQNNATK TFREFCGPAD PEIARHLRPG TLRAIFGKTK IQNAVH      116

SEQ ID NO: 226        moltype = AA length = 114
FEATURE              Location/Qualifiers
REGION              1..114
                      note = NME8 partial sequence
source              1..114
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 226
TLGLIKPHAT SEQREQILKI VKEAGFDLTQ VKKMFLTPEQ IEKIYPKVTG KDFYKDLLEM 60
LSVGPSMVM I LTKWNAVAEW RRLMGPTDPE EAKLLSPDSI RAQPGISKLK NIVH      114

SEQ ID NO: 227        moltype = AA length = 115
FEATURE              Location/Qualifiers
REGION              1..115
                      note = NME7B partial sequence
source              1..115
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 227
ILMAIRDAGF EISAMQMFNM DRVNVEEFYE VYKGVVTEYH DMVTEMYSGP CVAMEIQNN 60
ATKTFREFCG PADPEIARHL RPGTLRAIFG KTKIQNAVHC TDLPEDGLLE VQYFF      115

SEQ ID NO: 228        moltype = AA length = 115
FEATURE              Location/Qualifiers
REGION              1..115
                      note = NME8 partial sequence
source              1..115
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 228
VLRIKDEDF KILEQRQVVL SEKEAQALCK EYENEDYFNK LIENMTSGPS LALVLLRDNG 60
LQYWKQLLGP RTVEEAIEYF PESLCAQFAM DSLPVNQLYG SDSLETAERE IQHFF      115

SEQ ID NO: 229        moltype = AA length = 100
FEATURE              Location/Qualifiers
REGION              1..100
                      note = NME7B partial sequence
source              1..100
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 229
IVKPHAVSEG LLGKILMAIR DAGFEISAMQ MFNMDRVNVE EFYEVYKGVV TEYHDMVTEM 60
YSGPCVAMEI QQNNATKTFR EFCGPADPEI ARHLRPGTLR                        100

SEQ ID NO: 230        moltype = AA length = 110
FEATURE              Location/Qualifiers
REGION              1..110
                      note = NME8 partial sequence
source              1..110
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 230
IIKPDVAVISK KVLEIKRKIT KAGFIIEAEH KTVLTTEEQVV NFYSRIADQC DFEEFVSFMT 60
SGLSYILVVS QGSKHNPPE ETEPQTDTEP NERSEDQPEV EAQVTPGMMK            110

SEQ ID NO: 231        moltype = AA length = 174
FEATURE              Location/Qualifiers
REGION              1..174
                      note = NME9
source              1..174
                      mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 231
MLSSKGLTVV DVYQGWCGPC KPVVSLFQKM RIEVGLDLLH FALAEADRLD VLEKYRGKCE 60
PTFLFYAIKD EALSDEDECV SHGKNNGEDE DMVSSERTCT LAIIKPDAVA HGKTDEIIMK 120
IQEAGFEILT NEERTMTEAE VRLFYQHKAG ESPSSVRHRN ALQCRPWKPG QRRC 174

SEQ ID NO: 232      moltype = AA length = 58
FEATURE            Location/Qualifiers
REGION             1..58
                   note = NME7A partial sequence
source             1..58
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 232
TLALIKPDAI SKAGEIIEII NKAGFTITKL KMMMLSRKEA LDFHAMEILR DDAICEWK 58

SEQ ID NO: 233      moltype = AA length = 60
FEATURE            Location/Qualifiers
REGION             1..60
                   note = NME9 partial sequence
source             1..60
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 233
TLAIKPDVA AHGKTDEIIM KIQEAGFEIL TNEERTMTEA EVRLFYTLAI IKPDVAHKG 60

SEQ ID NO: 234      moltype = AA length = 31
FEATURE            Location/Qualifiers
REGION             1..31
                   note = NME7A partial sequence
source             1..31
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 234
ISKAGEIIEI INKAGFTITK LKMMMLSRKE A 31

SEQ ID NO: 235      moltype = AA length = 29
FEATURE            Location/Qualifiers
REGION             1..29
                   note = NME9 partial sequence
source             1..29
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 235
IQEAGFEILT NEERTMTEAE VRLFYQHKA 29

SEQ ID NO: 236      moltype = AA length = 53
FEATURE            Location/Qualifiers
REGION             1..53
                   note = NME7B partial sequence
source             1..53
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 236
NCTCCIVKPH AVSEGLLGKI LMAIRDAGFE ISAMQMFNMD RVNVEEFYEV YKG 53

SEQ ID NO: 237      moltype = AA length = 53
FEATURE            Location/Qualifiers
REGION             1..53
                   note = NME9 partial sequence
source             1..53
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 237
TCTLAIKPD AVAHGKTDEI IMKIQEAGFE ILTNEERTMT EAEVRLFYQH KAG 53

SEQ ID NO: 238      moltype = AA length = 350
FEATURE            Location/Qualifiers
REGION             1..350
                   note = NME10
source             1..350
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 238
MGCFFSKRRK ADKESRPENE EERPKQYSWD QREKVDPKDY MFSGLKDETV GRLPGTVAGQ 60
QFLIQDCENC NIYIFDHSAT VTIDDCNTCI IFLGPVKGSV FFRNCRDCKC TLACQQFRVR 120
DCRKLEVFLC CATQPIIESS SNIKFGCFQW YYPELAFQFK DAGLSIFNNT WSNIHDFTPV 180
SGELNWSLLP EDAVVQDYVP IPTTEELKAV RVSTEANRSI VPISRGQRQK SSDESCLVVL 240
FAGDYTIANA RKLIDEMVGK GFPLVQTKEV SMKAEDAQRV FREKAPDFLP LLNKGPIVAL 300

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EFNGDGAVEV CQLIVNEIFN GTKMFVSESK ETASGDVDSF YNFADIQMG I 350

SEQ ID NO: 239 moltype = AA length = 68
 FEATURE Location/Qualifiers
 REGION 1..68
 note = NME7A partial sequence
 source 1..68
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 239
 AISKAGEIIE IINKAGFTIT KLKMMMLSRK EALDFHVDHQ SRPFFNELIQ FITTGPIIAM 60
 EILRDDAI 68

SEQ ID NO: 240 moltype = AA length = 63
 FEATURE Location/Qualifiers
 REGION 1..63
 note = NME10 partial sequence
 source 1..63
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 240
 TIANARKLID EMVGKGFFLV QTKEVSMKAE DAQRVFREKA PDFLPLLNGK PVIALEFNGD 60
 GAV 63

SEQ ID NO: 241 moltype = AA length = 43
 FEATURE Location/Qualifiers
 REGION 1..43
 note = NME7A partial sequence
 source 1..43
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 241
 PIIAMEILRD DAICEWKRLI GPANSGVART DASESIRALF GTD 43

SEQ ID NO: 242 moltype = AA length = 45
 FEATURE Location/Qualifiers
 REGION 1..45
 note = NME10 partial sequence
 source 1..45
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 242
 PIPTTEELKA VRVSTEANRS IVPISRGQRQ KSSDESCLVV LFAGD 45

SEQ ID NO: 243 moltype = AA length = 74
 FEATURE Location/Qualifiers
 REGION 1..74
 note = NME7A partial sequence
 source 1..74
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 243
 IKPDAISKAG EIIEIINKAG FTITKLKMMM LSRKEALDFH VDHQSRPFFN ELIQFITTGP 60
 IIAMEILRDD AICE 74

SEQ ID NO: 244 moltype = AA length = 75
 FEATURE Location/Qualifiers
 REGION 1..75
 note = NME10 partial sequence
 source 1..75
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 244
 VDPKDYMFSG LKDETVGRLP GTVAGQQFLI QDCENCNIYI FDHSATVTID DCTNCIIFLG 60
 PVKGSVFFRN CRDCK 75

SEQ ID NO: 245 moltype = AA length = 47
 FEATURE Location/Qualifiers
 REGION 1..47
 note = NME7B partial sequence
 source 1..47
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 245
 CCIVKPHAVS ELLGKILMA IRDAGFEISA MQMFNMDRVN VEEFYEV 47

SEQ ID NO: 246 moltype = AA length = 51
 FEATURE Location/Qualifiers
 REGION 1..51

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source	note = NME10 partial sequence 1..51 mol_type = protein organism = synthetic construct	
SEQUENCE: 246		
CCATQPIIES SSNIKFGCFQ WYYPELAFQF KDAGLSIFNN TWSNIHDFTP V		51
SEQ ID NO: 247	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = NME2A1	
source	1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 247		
RASEEHLKQH YIDLKD		16
SEQ ID NO: 248	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = NME2A2	
source	1..8 mol_type = protein organism = synthetic construct	
SEQUENCE: 248		
PADSKPGT		8
SEQ ID NO: 249	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = NME2B1	
source	1..20 mol_type = protein organism = synthetic construct	
SEQUENCE: 249		
QKGFRLVAMK FLRASEEHLK		20
SEQ ID NO: 250	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = NME2B2	
source	1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 250		
IDLKDRPFPG LVKY		14
SEQ ID NO: 251	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
REGION	1..29	
	note = NME2B3	
source	1..29 mol_type = protein organism = synthetic construct	
SEQUENCE: 251		
GDFCIQVGRN IIHGSDSVKS AEKEISLWF		29
SEQ ID NO: 252	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = NME3A1	
source	1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 252		
QASEELLREH YVELRE		16
SEQ ID NO: 253	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = NME3A1	
source	1..8 mol_type = protein organism = synthetic construct	
SEQUENCE: 253		
PGDATPGT		8
SEQ ID NO: 254	moltype = AA length = 20	
FEATURE	Location/Qualifiers	

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REGION	1..20	
	note = NME3B1	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 254		
RKGFKLVALK LVQASELLR		20
SEQ ID NO: 255	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = NME3B2	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 255		
VELRERPFYS RLVKY		15
SEQ ID NO: 256	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
REGION	1..29	
	note = NME3B3	
source	1..29	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 256		
GDFCFEVGKN VIHGSDSVES AQREIALWF		29
SEQ ID NO: 257	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = NME4A1	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 257		
QAPESVLAEH YQDLRR		16
SEQ ID NO: 258	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = NME4A2	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 258		
SAEAAPGT		8
SEQ ID NO: 259	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = NME4B1	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 259		
RRGFTLVGMK MLQAPESVLA		20
SEQ ID NO: 260	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = NME4B2	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 260		
QDLRRKPFYP ALIRY		15
SEQ ID NO: 261	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
REGION	1..29	
	note = NME4B3	
source	1..29	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 261		
GDFSVHISRN VIHASDSVEG AQREIQLWF		29
SEQ ID NO: 262	moltype = AA length = 16	

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FEATURE	Location/Qualifiers	
REGION	1..16	
	note = NME5A1	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 262		
RLSPEQCSNF YVEKYG		16
SEQ ID NO: 263	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = NME5A2	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 263		
SLVAKETHPD S		11
SEQ ID NO: 264	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = NME5B1	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 264		
RSGFTIVQRR KLRLSPEQCS		20
SEQ ID NO: 265	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = NME5B2	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 265		
VEKYGKMFFP NLTAY		15
SEQ ID NO: 266	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
REGION	1..29	
	note = NME5B3	
source	1..29	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 266		
AIYGTDDLRLN ALHGSNDFAA AEREIRFMF		29
SEQ ID NO: 267	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = NME6A1	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 267		
LWRKEDCQRF YREHEG		16
SEQ ID NO: 268	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = NME6A2	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 268		
VFRARHVAPD S		11
SEQ ID NO: 269	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = NME6B1	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 269		
SNKFLIVMR ELLWRKEDCQ		20

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SEQ ID NO: 270	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = NME6B2	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 270		
REHEGRFFYQ RLVEF		15
SEQ ID NO: 271	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
REGION	1..29	
	note = NME6B3	
source	1..29	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 271		
GSFGLTDRN TTHGSDSVVS ASREIAAFF		29
SEQ ID NO: 272	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = NME8A1	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 272		
VLSEKEAQAL CKEYEN		16
SEQ ID NO: 273	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = NME8A2	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 273		
VEEAIEYFPE S		11
SEQ ID NO: 274	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = NME8A3	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 274		
FLTPEQIEKI YPKVTG		16
SEQ ID NO: 275	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = NME8A4	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 275		
PEEAKLLSPD S		11
SEQ ID NO: 276	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = NME8A5	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 276		
VLTEEQVNF YSRIAD		16
SEQ ID NO: 277	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = NME8B1	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 277		
EAGFDLTQVK KMFLTPEQIE		20

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SEQ ID NO: 278	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = NME8B2	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 278		
PKVTGKDFYK DLLEM		15
SEQ ID NO: 279	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = NME8B3	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 279		
AQFGISKLNK IVH		13
SEQ ID NO: 280	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = NME8B4	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 280		
DEDFKILEQR QVVLSEKEAQ		20
SEQ ID NO: 281	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = NME8B5	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 281		
KEYENEDYFN KLIEN		15
SEQ ID NO: 282	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
REGION	1..30	
	note = NME8B6	
source	1..30	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 282		
AQFAMDSLVPV NQLYGSDSLE TAEREIQHFF		30
SEQ ID NO: 283	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = NME8B7	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 283		
KAGFIIAEH KTVLTERQVV		20
SEQ ID NO: 284	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = NME8B8	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 284		
SRIADQCDFE EFVSF		15
SEQ ID NO: 285	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = NME9A1	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 285		

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TMTEAEVRLF Y		11
SEQ ID NO: 286	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = NME9B1	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 286		
EAGFEILTNE ERTMTEAEVR		20
SEQ ID NO: 287	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = NME10A1	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 287		
SMKAEDAQRV FREK		14
SEQ ID NO: 288	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
REGION	1..10	
	note = NME10A2	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 288		
GQRQKSSDES		10
SEQ ID NO: 289	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = NME10A3	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 289		
IQDCENCNIY IFDHSA		16
SEQ ID NO: 290	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
REGION	1..22	
	note = NME10B1	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 290		
ELAFQFKDAG LSIFNNTWSN IH		22
SEQ ID NO: 291	moltype = AA length = 23	
FEATURE	Location/Qualifiers	
REGION	1..23	
	note = NME2A1	
source	1..23	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 291		
RASEEHLKQH YIDLKDRPFF PGL		23
SEQ ID NO: 292	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
REGION	1..18	
	note = NME2A2	
source	1..18	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 292		
LGETNPADSK PGTIRGDF		18
SEQ ID NO: 293	moltype = AA length = 34	
FEATURE	Location/Qualifiers	
REGION	1..34	
	note = NME2B1	
source	1..34	
	mol_type = protein	
	organism = synthetic construct	

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SEQUENCE: 293		
GLVGEIIKRF EQKGFRLVAM KFLRASEEHL KQHY		34
SEQ ID NO: 294	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
REGION	1..25	
source	note = NME2B2	
	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 294		
YIDLKDRPFF PGLVKYMSG PVVAM		25
SEQ ID NO: 295	moltype = AA length = 34	
FEATURE	Location/Qualifiers	
REGION	1..34	
source	note = NME2B3	
	1..34	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 295		
PGTIRGDFCI QVGRNIIHGS DSVKSAAKEI SLWF		34
SEQ ID NO: 296	moltype = AA length = 27	
FEATURE	Location/Qualifiers	
REGION	1..27	
source	note = NME3A1	
	1..27	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 296		
LKLVQASEEL LREHYVELRE RPFYSRL		27
SEQ ID NO: 297	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
source	note = NME3A1	
	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 297		
LIGATDPGDA TPGTIRGDF		19
SEQ ID NO: 298	moltype = AA length = 31	
FEATURE	Location/Qualifiers	
REGION	1..31	
source	note = NME3B1	
	1..31	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 298		
LVGEIVRRFE RKGFKLVALK LVQASEELLR E		31
SEQ ID NO: 299	moltype = AA length = 27	
FEATURE	Location/Qualifiers	
REGION	1..27	
source	note = NME3B2	
	1..27	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 299		
EHYVELRERP FYSRLVKYMG SGPVVAM		27
SEQ ID NO: 300	moltype = AA length = 34	
FEATURE	Location/Qualifiers	
REGION	1..34	
source	note = NME3B3	
	1..34	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 300		
PGTIRGDFCV EVGKNVIHGS DSVESAQREI ALWF		34
SEQ ID NO: 301	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
REGION	1..29	
source	note = NME4A1	
	1..29	
	mol_type = protein	

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SEQUENCE: 301	organism = synthetic construct	
GFTLVGMKML QAPESVLAEH YQDLRRKPF		29
SEQ ID NO: 302	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
source	note = NME4A2	
	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 302		
GHTDSAEAAP GTIRGDF		17
SEQ ID NO: 303	moltype = AA length = 33	
FEATURE	Location/Qualifiers	
REGION	1..33	
source	note = NME4B1	
	1..33	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 303		
LVGDDVIQRPE RRGFTLVGMK MLQAPESVLA EHY		33
SEQ ID NO: 304	moltype = AA length = 27	
FEATURE	Location/Qualifiers	
REGION	1..27	
source	note = NME4B2	
	1..27	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 304		
EHYQDLRRKP FYPALIRYMS SGPVVAM		27
SEQ ID NO: 305	moltype = AA length = 34	
FEATURE	Location/Qualifiers	
REGION	1..34	
source	note = NME4B3	
	1..34	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 305		
PGTIRGDFSV HISRNVIHAS DSVEGAQREI QLWF		34
SEQ ID NO: 306	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
REGION	1..30	
source	note = NME5A1	
	1..30	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 306		
GFTIVQRRKL RLSPEQCSNF YVEKYGKMFF		30
SEQ ID NO: 307	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
REGION	1..25	
source	note = NME5A2	
	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 307		
LLGPNNLSVA KETHPDSLRA IYGTD		25
SEQ ID NO: 308	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
REGION	1..29	
source	note = NME5B1	
	1..29	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 308		
IQDIILRSGF TIVQRRKLRL SPEQCSNFY		29
SEQ ID NO: 309	moltype = AA length = 26	
FEATURE	Location/Qualifiers	
REGION	1..26	
source	note = NME5B2	
	1..26	

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	mol_type = protein organism = synthetic construct	
SEQUENCE: 309 FYVEKYGKMF FPNLTAYMSS GPLVAM		26
SEQ ID NO: 310 FEATURE REGION	moltype = AA length = 34 Location/Qualifiers 1..34 note = NME5B3	
source	1..34 mol_type = protein organism = synthetic construct	
SEQUENCE: 310 PDSLRAIYGT DDLRNALHGS NDFAAAEREI RFMF		34
SEQ ID NO: 311 FEATURE REGION	moltype = AA length = 32 Location/Qualifiers 1..32 note = NME6A1	
source	1..32 mol_type = protein organism = synthetic construct	
SEQUENCE: 311 FLIVRMRELL WRKEDCQRFY REHEGRFFYQ RL		32
SEQ ID NO: 312 FEATURE REGION	moltype = AA length = 23 Location/Qualifiers 1..23 note = NME6A2	
source	1..23 mol_type = protein organism = synthetic construct	
SEQUENCE: 312 LMGPTRVFRA RHVAPDSIRG SFG		23
SEQ ID NO: 313 FEATURE REGION	moltype = AA length = 25 Location/Qualifiers 1..25 note = NME6B1	
source	1..25 mol_type = protein organism = synthetic construct	
SEQUENCE: 313 ILSNKFLIVR MRELLWRKED CQRFY		25
SEQ ID NO: 314 FEATURE REGION	moltype = AA length = 25 Location/Qualifiers 1..25 note = NME6B2	
source	1..25 mol_type = protein organism = synthetic construct	
SEQUENCE: 314 FYREHEGRFF YQRLVEFMAS GPIRA		25
SEQ ID NO: 315 FEATURE REGION	moltype = AA length = 39 Location/Qualifiers 1..39 note = NME6B3	
source	1..39 mol_type = protein organism = synthetic construct	
SEQUENCE: 315 ARHVAPDSIR GSFGLTDTRN TTHGSDSVVS ASREIAAFF		39
SEQ ID NO: 316 FEATURE REGION	moltype = AA length = 33 Location/Qualifiers 1..33 note = NME8A1	
source	1..33 mol_type = protein organism = synthetic construct	
SEQUENCE: 316 FKILEQRQVV LSEKEAQCALC KEYENEDYFN KLI		33
SEQ ID NO: 317 FEATURE REGION	moltype = AA length = 28 Location/Qualifiers 1..28 note = NME8A2	

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source	1..28	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 317		
WKQLLGPRTV EEAIEYFPES LCAQFAMD		28
SEQ ID NO: 318	moltype = AA length = 34	
FEATURE	Location/Qualifiers	
REGION	1..34	
	note = NME8A3	
source	1..34	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 318		
AGFDLTQVKK MFLTPEQIEK IYPKVTGKDF YKDL		34
SEQ ID NO: 319	moltype = AA length = 27	
FEATURE	Location/Qualifiers	
REGION	1..27	
	note = NME8A4	
source	1..27	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 319		
EWRRLMGPTD PEEAKLLSPD SIRAQFG		27
SEQ ID NO: 320	moltype = AA length = 34	
FEATURE	Location/Qualifiers	
REGION	1..34	
	note = NME8A5	
source	1..34	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 320		
KAGFIIIEAH KTVLTEEQVV NFYSRIADQC DFEE		34
SEQ ID NO: 321	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
REGION	1..29	
	note = NME8B1	
source	1..29	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 321		
ILKIVKEAGF DLTQVKKMFL TPEQIEKIY		29
SEQ ID NO: 322	moltype = AA length = 21	
FEATURE	Location/Qualifiers	
REGION	1..21	
	note = NME8B2	
source	1..21	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 322		
YPKVTGKDFY KDLLEMLSVG P		21
SEQ ID NO: 323	moltype = AA length = 27	
FEATURE	Location/Qualifiers	
REGION	1..27	
	note = NME8B3	
source	1..27	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 323		
DPPEAKLLSP DSIRAQFGIS KLKNIVH		27
SEQ ID NO: 324	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
REGION	1..25	
	note = NME8B4	
source	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 324		
LRIIKDEDFK ILEQRQVVLS EKEAQ		25
SEQ ID NO: 325	moltype = AA length = 23	
FEATURE	Location/Qualifiers	
REGION	1..23	

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source	note = NME8B5 1..23 mol_type = protein organism = synthetic construct	
SEQUENCE: 325		
KEYENEDYFN KLIENMTSGP SLA		23
SEQ ID NO: 326	moltype = AA length = 35	
FEATURE	Location/Qualifiers	
REGION	1..35	
source	note = NME8B6 1..35 mol_type = protein organism = synthetic construct	
SEQUENCE: 326		
PESLCAQFAM DSLPVNQLYG SDSLETAERE IQHFF		35
SEQ ID NO: 327	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
REGION	1..29	
source	note = NME8B7 1..29 mol_type = protein organism = synthetic construct	
SEQUENCE: 327		
IKRKITKAGF IIEAEHKTVL TEEQVVNFY		29
SEQ ID NO: 328	moltype = AA length = 21	
FEATURE	Location/Qualifiers	
REGION	1..21	
source	note = NME8B8 1..21 mol_type = protein organism = synthetic construct	
SEQUENCE: 328		
FYSRIADQCD FEEFVSFMTS G		21
SEQ ID NO: 329	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
REGION	1..22	
source	note = NME9A1 1..22 mol_type = protein organism = synthetic construct	
SEQUENCE: 329		
AGFEILTNEE RTMTAEVRL FY		22
SEQ ID NO: 330	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
REGION	1..29	
source	note = NME9B1 1..29 mol_type = protein organism = synthetic construct	
SEQUENCE: 330		
IIMKIQEAGF EILTNEERTM TEAEVRLFY		29
SEQ ID NO: 331	moltype = AA length = 26	
FEATURE	Location/Qualifiers	
REGION	1..26	
source	note = NME10A1 1..26 mol_type = protein organism = synthetic construct	
SEQUENCE: 331		
GFFLVQTKEV SMKAEDAQRV FREKAP		26
SEQ ID NO: 332	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
REGION	1..30	
source	note = NME10A2 1..30 mol_type = protein organism = synthetic construct	
SEQUENCE: 332		
EANRSIVPIS RGQRQKSSDE SCLVVLFAGD		30
SEQ ID NO: 333	moltype = AA length = 16	
FEATURE	Location/Qualifiers	

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REGION 1..16
note = NME10A3
source 1..16
mol_type = protein
organism = synthetic construct

SEQUENCE: 333
IQDCENCNIY IFDHSA 16

SEQ ID NO: 334 moltype = AA length = 30
FEATURE Location/Qualifiers
REGION 1..30
note = NME10B1
source 1..30
mol_type = protein
organism = synthetic construct

SEQUENCE: 334
ELAFQFKDAG LSIFNNTWSN IHDFTPVDCT 30

SEQ ID NO: 335 moltype = DNA length = 363
FEATURE Location/Qualifiers
misc_feature 1..363
note = 8F9A4P3 Heavy chain variable region sequence mouse
source 1..363
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 335
gtccagctgc aacagtctgg acctgaactg gtgaagcctg gggcttcagt gaagatatcc 60
tgcaagactt ctggaaacac attcactgaa tacaccatgc actgggtgaa gcagagccat 120
ggaaagagcc ttgagtgatg tggaggtttt aatcctaaca atggtgttac taactacaac 180
cagaagttca agggcaaggc cacattgact gtagacaagt cctccagcac agcctacatg 240
gagctccgca gcctgacatc tgaggattct gcagtctatt actgtgcaag acggtactac 300
catagtctct acgtgtttta ctttgactac tggggccaag gcaccaactct cacagtctcc 360
tca 363

SEQ ID NO: 336 moltype = DNA length = 294
FEATURE Location/Qualifiers
misc_feature 1..294
note = IGHV1-24*01 V-REGION sequence human (closest match
hu antibody sequence)
source 1..294
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 336
cagggtccagc tggtagacgc tggggctgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg ttccgggata caccctcact gaattatcca tgcactgggt gcgacaggct 120
cctggaaaag ggcttgagtg gatgggaggt ttgatccctg aagatgggtg aacaatctac 180
gcacagaagt tccagggcag agtcaccatg accgaggaca catctacaga cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtgt attactgtgc aaca 294

SEQ ID NO: 337 moltype = DNA length = 45
FEATURE Location/Qualifiers
misc_feature 1..45
note = IGHJ4*01 J-REGION sequence human (closest match hu
antibody sequence)
source 1..45
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 337
tactttgact actggggcca aggaaccctg gtcaccgtct cctca 45

SEQ ID NO: 338 moltype = DNA length = 366
FEATURE Location/Qualifiers
misc_feature 1..366
note = humanized 8F9A4P3 Heavy chain variable region
sequence
source 1..366
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 338
cagggtccagc tggtagacgc tggggctgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg ttccgggaaa cacattcact gaatacacca tgcactgggt gcgacaggct 120
cctggaaaag ggcttgagtg gatgggaggt tttaatccta acaatgggtg tactaactac 180
aaccagaagt tcaagggcag agtcaccatg accgaggaca catctacaga cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtgt attactgtgc aagacgggtac 300
taccatagtc tctacgtggt ttactttgac tactggggcc aaggaaccct ggtaaccgtc 360
tcctca 366

SEQ ID NO: 339 moltype = DNA length = 366
FEATURE Location/Qualifiers

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misc_feature      1..366
                  note = humanized 8F9A4P3 Heavy chain variable region
                  sequence (codon optimized)
source            1..366
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 339
caggttcagc tggttcagtc tggcgccgaa gtgaagaaac ctggcgccctc tgtgaagggtg 60
tcctgcaagg tgtccggaat taccttcacc gactacacca tgcactgggt cgcacaggcc 120
cctggcaaa gacttgaatg gatggcgccg ttcaacccca acaacggcgt gaccaactac 180
aaccagaaat tcaagggccg cgtgaccatg accgaggaca caagcacaga caccgcctac 240
atggaaactg gcagcctgag aagcgaggac accgccgtgt actactgcgc cagaagggtac 300
taccacagcc tgtacgtgtt ctacttcgac tactggggcc agggcaccct ggtcacagtt 360
tcttct      366

SEQ ID NO: 340      moltype = DNA length = 366
FEATURE            Location/Qualifiers
misc_feature        1..366
                  note = Modified humanized 8F9A4P3 Heavy chain variable
                  region sequence
source              1..366
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 340
caggtcacgc tgggtacagtc tggggctgag gtgaagaagc ctggggccctc agtgaagggtc 60
tcctgcaagg tttccggaaa cacattcact gaatacacca tgcactgggt ggcacaggct 120
cctggaaaag ggcttgagtg gattggaggt tttaatccta acaatgggtg tactaactac 180
aaccagaagt tcaagggcaa agtcaccctg accgtggaca catctagcag cacagcctac 240
atggagctga gcagcctgag atctgaggac accggccgtgt attactgtgc aagacgggtac 300
taccatagtc tctacgtgtt ttactttgac tactggggcc aaggaaacct ggtcacccgtc 360
tcctca      366

SEQ ID NO: 341      moltype = DNA length = 366
FEATURE            Location/Qualifiers
misc_feature        1..366
                  note = Modified humanized 8F9A4P3 Heavy chain variable
                  region sequence (codon optimized)
source              1..366
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 341
caggttcagc tggttcagtc tggcgccgaa gtgaagaaac ctggcgccctc tgtgaagggtg 60
tcctgcaagg tgtccggaaa taccttcacc gactacacca tgcactgggt cgcacaggcc 120
cctggcaaa gacttgaatg gatcgccggc ttcaacccca acaacggcgt gaccaactac 180
aaccagaaat tcaagggcaa agtgaccctg accgtggaca ccagcagcag cacagcctac 240
atggaaactg gcagcctgag aagcgaggac accgccgtgt actactgcgc cagaagggtac 300
taccacagcc tgtacgtgtt ctacttcgac tactggggcc agggcaccct ggtcacagtt 360
tcttct      366

SEQ ID NO: 342      moltype = DNA length = 324
FEATURE            Location/Qualifiers
misc_feature        1..324
                  note = 8F9A4P3 Light chain variable region sequence mouse
source              1..324
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 342
gaaacaactg tgacccagtc tccagcatcc ctgtccatgg ctataggaga aaaagtcacc 60
atcagatgca taaccagcac tgatattgat gatgatatga actgggtacca gcagaagcca 120
ggggaacctc ctaagtcctt tatctcagaa ggcaatactc ttcgtcctgg agtcccctcc 180
cgattctcca gcagtggcta tggtagagat ttgtttttta caattgaaaa catgctctca 240
gaagatgttg cagattacta ctgtttgcaa agtgataact tgcctctcac gttcggctcg 300
gggacaaaag tggaaataaa acgg      324

SEQ ID NO: 343      moltype = DNA length = 264
FEATURE            Location/Qualifiers
misc_feature        1..264
                  note = IGKV5-2*01 V-REGION sequence human (closest match hu
                  antibody sequence)
source              1..264
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 343
gaaacgacac tcacgcagtc tccagcattc atgtcagcga ctccaggaga caaagtcac 60
atctcctgca aagccagcca agacattgat gatgatatga actgggtacca acagaaacca 120
ggagaagctg ctattttcat tattcaagaa gctactactc tcgttcctgg aatcccacct 180
cgattcagtg gcagcgggta tggaaacagat ttaccctca caattaataa catagaatct 240
gaggatgctg catattactt ctgt      264

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SEQ ID NO: 344      moltype = DNA length = 36
FEATURE            Location/Qualifiers
misc_feature        1..36
                    note = IGKJ4*02 J-REGION sequence human (closest match hu
                    antibody sequence)
source              1..36
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 344
ctcacgttcg gcgaggggac caaggtggag atcaaa                      36

SEQ ID NO: 345      moltype = DNA length = 324
FEATURE            Location/Qualifiers
misc_feature        1..324
                    note = humanized 8F9A4P3 Light chain variable region
                    sequence
source              1..324
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 345
gaaacgacac tcacgcagtc tccagcattc atgtcagcga ctccaggaga caaagtcaac 60
atctcctgca taaccagcac tgatattgat gatgatatga actggtacca acagaaacca 120
ggagaagctg ctattttcat tattcaagaa ggcaatactc ttcgtcctgg aatcccacct 180
cgattcagtg gcagcgggta tggaaacagat ttaccctca caattaataa catagaatct 240
gaggatgctg catattactt ctgtttgcaa agtgataact tgcctctcac gttcggcgga 300
gggaccaagg tggagatcaa acgg                      324

SEQ ID NO: 346      moltype = DNA length = 324
FEATURE            Location/Qualifiers
misc_feature        1..324
                    note = humanized 8F9A4P3 Light chain variable region
                    sequence (codon optimized)
source              1..324
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 346
gagacaaccc tgacacagag ccttcgcttc atgtctgccca cacctggcga caaagtgaac 60
atcagctgca tcaccagcac cgacatcgac gacgacatga actgggtatca gcagaagcct 120
ggcgaggccg ccattcttcat catccaagag ggcaacacac tgcgggcttg catccctcct 180
agattttctg gcagcgggta cgccaccgac ttcacctga ccatcaacaa catcgagagc 240
gaggacgccg cctactactt ctgcctgcaa agcgacaacc tgcctctgac ctttggcgga 300
ggcaccaagg tggaaatcaa gcgg                      324

SEQ ID NO: 347      moltype = DNA length = 324
FEATURE            Location/Qualifiers
misc_feature        1..324
                    note = Modified humanized 8F9A4P3 Light chain variable
                    region sequence
source              1..324
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 347
gaaacgacag tgacgcagtc tccagcattc atgtcagcga ctccaggaga caaagtcacc 60
atctcctgca taaccagcac tgatattgat gatgatatga actggtacca acagaaacca 120
ggagaagctg ctattctgct gattagcgaa ggcaatactc ttcgtcctgg aatcccacct 180
cgattcagta gcagcgggta tggaaacagat ttaccctca caattaataa catagaatct 240
gaggatgctg catattactt ctgtttgcaa agtgataact tgcctctcac gttcggcgga 300
gggaccaagg tggagatcaa acgg                      324

SEQ ID NO: 348      moltype = DNA length = 324
FEATURE            Location/Qualifiers
misc_feature        1..324
                    note = Modified humanized 8F9A4P3 Light chain variable
                    region sequence (codon optimized)
source              1..324
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 348
gagacaaccc tgacacagag ccttcgcttc atgtctgccca cacctggcga caaagtgaac 60
atcagctgca tcaccagcac cgacatcgac gacgacatga actgggtatca gcagaagcct 120
ggcgaggccg ccattctgct tatctctgag ggaaacacac tgcgggcttg catccctcct 180
agattttcca gcagcgggta cgccaccgac ttcacctga ccatcaacaa catcgagagc 240
gaggacgccg cctactactt ctgcctgcaa agcgacaacc tgcctctgac ctttggcgga 300
ggcaccaagg tggaaatcaa gcgg                      324

SEQ ID NO: 349      moltype = DNA length = 735
FEATURE            Location/Qualifiers
misc_feature        1..735
                    note = Modified humanized 8F9A4P3 sequence (codon

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optimized) - humanized heavy and light chains joined via a flexible linker

source 1..735
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 349

cagggttcagc	tgggttcagtc	tggcgccgaa	gtgaagaaac	ctggcgccctc	tgtgaagggtg	60
tcctgcaagg	tgtccggaaa	taccttcacc	gagtacacca	tgcactgggt	ccgacaggcc	120
cctggcaaa	gactgggaatg	gatcgccggc	ttcaacccca	acaacggcgt	gaccaactac	180
aaccagaaat	tcaagggcaa	agtgaccctg	accgtggaca	ccagcagcag	cacagcctac	240
atggaaactga	gcagcctgag	aagcgaggag	accgcccgtg	actactgcgc	cagaagggtac	300
taccacagcc	tgtacgtgtt	ctacttcgac	tactggggcc	agggcaccct	ggtcacagtt	360
tcttctggcg	gtggcgggaag	cggaggcggg	ggctccgggtg	gcggaggcag	cgaacgcaca	420
gtgacgcagt	ctccagcatt	catgtcagcg	actccaggag	acaaagtcac	catctcctgc	480
ataaccagca	ctgatattga	tgatgatattg	aactgggtacc	aacagaaacc	aggagaagct	540
gctattctgc	tgattagcga	aggcaatact	cttcgtcctg	gaatcccacc	tcgattcagt	600
agcagcgggt	atggaacaga	ttttaccctc	acaattaata	acatagaatc	tgaggatgct	660
gcatattact	tctgtttgca	aagtgtataac	ttgcctctca	cggtcggcgg	agggaccaag	720
gtggagatca	aacgg					735

SEQ ID NO: 350 moltype = DNA length = 357
FEATURE Location/Qualifiers
misc_feature 1..357
note = 8F9A5A1 Heavy chain variable region sequence
source 1..357
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 350

atccagttgg	tgcagtctgg	acctgagctg	aagaagcctg	gagagacagt	caagatctcc	60
tgcaaggctt	ctgggtatatac	cttcacaaaac	tatggaatga	actgggtgaa	gcaggctcca	120
ggaaagggtt	taaaagtggat	gggctggata	aacacctaca	ctggagagcc	aacatatgtt	180
gatgacttca	aggagcgggt	tgccctctct	ttggaaacct	ctgccaccac	tgcttatgtt	240
cagatcaaca	acctcaaaaa	tgaggacacg	tctacatatt	tctgtgcaag	attgaggggg	300
atcacgccgg	gtcccttggc	ttactggggc	caagggactc	tggtcactgt	ctctgca	357

SEQ ID NO: 351 moltype = DNA length = 294
FEATURE Location/Qualifiers
misc_feature 1..294
note = IGHV7-81*01 V-REGION sequence
source 1..294
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 351

cagggtgcagc	tgggtgcagtc	tggccatgag	gtgaagcagc	ctggggccctc	agtgaaggtc	60
tcctgcaagg	cttctgggtta	cagtttcacc	acctatggta	tgaattgggt	gccacaggcc	120
cctggacaag	ggcttgagtg	gatgggatgg	ttcaacacct	acactgggaa	cccaacatat	180
gcccagggtc	tcacaggacg	gtttgtcttc	tccatggaca	cctctgccag	cacagcatac	240
ctgcagatca	gcagccctaaa	ggctgaggac	atggccatgt	attactgtgc	gaga	294

SEQ ID NO: 352 moltype = DNA length = 45
FEATURE Location/Qualifiers
misc_feature 1..45
note = IGHJ4*03 J-REGION sequence
source 1..45
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 352

tactttgact	actggggcca	agggaccctg	gtcaccgtct	cctca	45
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SEQ ID NO: 353 moltype = DNA length = 360
FEATURE Location/Qualifiers
misc_feature 1..360
note = humanized 8F9A5A1 Heavy chain variable region sequence
source 1..360
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 353

cagggtgcagc	tgggtgcagtc	tggccatgag	gtgaagcagc	ctggggccctc	agtgaaggtc	60
tcctgcaagg	cttctgggtta	taccttcaca	aactatggaa	tgaactgggt	gccacaggcc	120
cctggacaag	ggcttgagtg	gatgggatgg	ataaacacct	acactggaga	gccaacatat	180
gttgatgact	tcaagggacg	gtttgtcttc	tccatggaca	cctctgccag	cacagcatac	240
ctgcagatca	gcagccctaaa	ggctgaggac	atggccatgt	attactgtgc	aagattgagg	300
gggatcacgac	cgggtccctt	ggcttactgg	ggccaaggga	ccctggtcac	cgtctcctca	360

SEQ ID NO: 354 moltype = DNA length = 360
FEATURE Location/Qualifiers
misc_feature 1..360
note = humanized 8F9A5A1 Heavy chain variable region

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sequence (codon optimized)
source          1..360
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 354
caggttcagc tgggtgcagtc tggccacgaa gtgaacacagc ctggcgccctc tgtgaagggtg 60
tcctgtaaag ccagcggcta cacctttacc aactacggca tgaactgggt gccccaggct 120
cctggacaag gcttggaatg gatgggctgg atcaacacct acaccggcga gcctacctac 180
gtggacgact tcaagggcag attcgtgttc agcatggaca ccagcgccag cacagcctac 240
ctgcagatca gctctctgaa ggccgaggat atggccatgt actactgcgc cagactgaga 300
ggcatcagac ctggacctct ggctatttgg ggacagggca cactgggtcac agtgtcctct 360

SEQ ID NO: 355      moltype = DNA length = 360
FEATURE            Location/Qualifiers
misc_feature       1..360
                   note = Modified humanized 8F9A5A1 Heavy chain variable
                   region sequence

source            1..360
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 355
cagatccagc tgggtgcagtc tggcccccag gtgaagcagc ctggggccctc agtgaagggtc 60
tcctgcaagg cttctgggta taccttcaca aactatggaa tgaactgggt gaagcaggcc 120
cctggacaag ggcttgagtg gatgggatgg ataaacacct aactggaga gccaacatat 180
gttgatgact tcaagggagc gtttgccttc tccatggaca cctctgccag cacagcatac 240
ctgcagatca gcagcctaaa ggctgaggac accgccacct attactgtgc aagattgagg 300
gggatacgac cgggtccctt ggcttactgg ggccaaggga ccctgggtcac cgtctcctca 360

SEQ ID NO: 356      moltype = DNA length = 360
FEATURE            Location/Qualifiers
misc_feature       1..360
                   note = Modified humanized 8F9A5A1 Heavy chain variable
                   region sequence (codon optimized)

source            1..360
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 356
cagattcagc tgggtgcagtc tggcccccga gtgaacaacac ctggcgccctc tgtgaagggtg 60
tcctgcaagg ccagcggcta cacctttacc aactacggca tgaactgggt caagcaggcc 120
cctggacaag gcctggaaat gatgggctgg atcaacacct acaccggcga gcctacctac 180
gtggacgact tcaagggcag attcgccttc agcatggaca ccagcgccag cacagcctac 240
ctgcagatca gctctctgaa ggccgaggac accgccacct actactgtgc cagactgaga 300
ggcatcagac ccggacctct ggctatttgg ggacagggaa cactgggtcac cgtgtcctct 360

SEQ ID NO: 357      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature       1..321
                   note = 8F9A5A1 Light chain variable region sequence

source            1..321
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 357
gaaattttgc tcaccagtc tccagcaatc atagctgcat ctctgggga gaaggtcacc 60
atcacctgca gtgccagctc aagtgttaag tacatgaact ggtaccagca gaaaccaggga 120
tcctcccca aaatatggat ttatggtata tccaacctgg cttctggagt tcctgctcgc 180
ttcagtggca gtgggtctgg gacatcttct tctttcaca tcaacagcat ggaggctgaa 240
gatgttgcca cttattactg tcagcaaaag agtagttacc caccacggt cggagggggg 300
accaagctgg aaataaaacg g 321

SEQ ID NO: 358      moltype = DNA length = 279
FEATURE            Location/Qualifiers
misc_feature       1..279
                   note = IGKV3D-15*02 V-REGION sequence

source            1..279
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 358
gaaatagtga tgatgcagtc tccagccacc ctgtctgtgt ctccaggga aagagccacc 60
ctctcctgca gggccagtc gagtgtagc agcaacttag cctggtacca gcagaacct 120
ggccaggctc ccaggctcct catctatggt gcattccaca gggccactgg catcccagcc 180
aggttcagtg gcagtggttc tgggacagag ttcactctca ccatacgag cctgcagctc 240
gaagattttg cagtttatta ctgtcagcag tataataac 279

SEQ ID NO: 359      moltype = DNA length = 36
FEATURE            Location/Qualifiers
misc_feature       1..36
                   note = IGKJ4*02 J-REGION sequence

source            1..36
                  mol_type = other DNA

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                organism = synthetic construct
SEQUENCE: 359
ctcacgttcg gcggaggag caaggtggag atcaaa          36

SEQ ID NO: 360      moltype = DNA  length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = humanized 8F9A5A1 Light chain variable region
                    sequence
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 360
gaaatagtga tgatgcagtc tccagccacc ctgtctgtgt ctccaggga aagagccacc  60
ctctcctgca gtgccagctc aagtgttaagt tacatgaact ggtaccagca gaaacctggc 120
caggetccca ggctcctcat ctatggtata tccaacctgg cttctggcat ccagccagg 180
ttcagtgcca gtgggtctgg gacagagttc actctcacca tcagcagcct gcagtctgaa 240
gattttgcag tttattactg tcagcaaagg agtagttacc caccacggt cgcgaggagg 300
accaaggtgg agatcaaagc g          321

SEQ ID NO: 361      moltype = DNA  length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = humanized 8F9A5A1 Light chain variable region
                    sequence (codon optimized)
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 361
gagatcgtga tgatgcagag ccccgccaca ctgagtgtgt ctccaggcga aagagccaca  60
ctgtcctgta gcgccagcag cagcgtgttc tacatgaact ggtaccagca gaagcccgga 120
caggccccca gactgctgat ctacggcatc agcaatctgg ccagcggcat ccctgccaga 180
ttttctggct ctggctccgg caccgagttc accctgacaa tctctagcct gcagagcgag 240
gacttcgccg tgtactactg ccagcagaga agcagctacc ctctacctt tggcgaggagc 300
accaaggtgg aaatcaagcg g          321

SEQ ID NO: 362      moltype = DNA  length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = Modified humanized 8F9A5A1 Light chain variable
                    region sequence
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 362
gaaatagtgc tgaccagctc tccagccacc ctgtctgtgt ctccaggga aagagccacc  60
ctctcctgca gtgccagctc aagtgttaagt tacatgaact ggtaccagca gaaacctggc 120
caggetccca ggctcctgat ctatggtata tccaacctgg cttctggcat ccagccagg 180
ttcagtgcca gtgggtctgg gacaagcttc agcctcacca tcagcagcct gcagtctgaa 240
gattttgcag tttattactg tcagcaaagg agtagttacc caccacggt cgcgaggagg 300
accaaggtgg agatcaaagc g          321

SEQ ID NO: 363      moltype = DNA  length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = Modified humanized 8F9A5A1 Light chain variable
                    region sequence (codon optimized)
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 363
gagatcgtgc tgacacagtc tcccgccaca ctgagtgtgt ctccaggcga aagagccaca  60
ctgtcctgta gcgccagcag cagcgtgttc tacatgaact ggtaccagca gaagcccgga 120
caggccccca gactgtggat ctacggcatc agcaatctgg ccagcggcat ccctgccaga 180
ttttctggct ctggctccgg caccagcttc agcctgacaa tcagcagcct gcagagcgag 240
gacttcgccg tgtactactg ccagcagaga agcagctacc ctctacctt tggcgaggagc 300
accaaggtgg aaatcaagcg g          321

SEQ ID NO: 364      moltype = DNA  length = 726
FEATURE            Location/Qualifiers
misc_feature        1..726
                    note = Modified humanized 8F9A5A1 scFV sequence (codon
                    optimized)
source              1..726
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 364
cagattcagc tgggtcagtc tggccccgaa gtgaaacaac ctggcgccctc tgtgaagggtg  60
tcctgcaagg ccagcggtcta cacctttacc aactacggca tgaactgggt caagcaggcc 120

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cctggacaag gcctggaatg gatgggctgg atcaaacacct acaccggcga gcctacctac 180
gtggacgact tcaagggcag attcgcttc agcatggaca ccagcgccag cacagcctac 240
ctgcagatca gctctctgaa ggccgaggac accgccacct actactgtgc cagactgaga 300
ggcatcagac ccggacctct ggccatttgg ggacaggga cactgggtcac cgtgtcctct 360
ggcggtggcg gaagcggagg cgggtggctcc ggtggcggag gcagcgagat cgtgctgaca 420
cagtctcccg ccacactgag tgtgtctcca ggcgaaagag ccacactgtc ctgtagcgcc 480
agcagcagcg tgtcctacat gaactgggat cagcagaagc ccggacaggc ccctagactg 540
tggatctacg gcacagcaa tctggccagc ggcacccctg ccagattttc tggctctggc 600
tccggcacca gcttcagcct gacaatcagc agcctgcaga gcgaggactt cgcctgttac 660
tactgccagc agagaagcag ctaccctcct acctttggcg gaggcaccaa ggtggaatc 720
aagcgg 726

SEQ ID NO: 365      moltype = DNA length = 363
FEATURE            Location/Qualifiers
misc_feature       1..363
                    note = 8H5H5G4 Heavy chain variable region sequence
source             1..363
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 365
gtccagctgc aacagtctgg acctgatctg gtgaagcctg ggacttcagt gaagatatcc 60
tgtaagactt ctggaaacac attcaactgaa tacaccatgc actgggtgaa gcagagccat 120
ggaaagagcc ttgattggat tggaggtttt aatcctaaca atgggtgttac taactacaac 180
cagaagtcca agggcaaggc cacattgact gtagacaagt cctccagcac agcctacatg 240
gagctccgca gcctgacatc taggagattct gcagtctatt actgtgcaag acgttactac 300
catagtacct acgtgttcta ctttgactcc tggggccaag gcaccactct cacagtctcc 360
tca 363

SEQ ID NO: 366      moltype = DNA length = 294
FEATURE            Location/Qualifiers
misc_feature       1..294
                    note = IGHV1-24*01 V-REGION sequence
source             1..294
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 366
caggctccagc tggtagacatc tggggctgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg tttccggata caccctcact gaattatcca tgcactgggt gcgacaggct 120
cctggaaaag ggcttgagtg gatgggaggt tttgatcctg aagatgggtg aacaatctac 180
gcacagaagt tccagggcag agtcaccatg accgaggaca catctacaga cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtgt attactgtgc aaca 294

SEQ ID NO: 367      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature       1..45
                    note = IGHJ4*03 J-REGION sequence
source             1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 367
tactttgact actggggcca agggaccctg gtcaccgtct cctca 45

SEQ ID NO: 368      moltype = DNA length = 366
FEATURE            Location/Qualifiers
misc_feature       1..366
                    note = Humanized 8H5H5G4 Heavy chain variable region
                    sequence
source             1..366
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 368
caggctccagc tggtagacatc tggggctgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg tttccggaaa cacattcact gaatacacca tgcactgggt gcgacaggct 120
cctggaaaag ggcttgagtg gatgggaggt tttaatccta acaatgggtg tactaactac 180
aaccagaagt tcaagggcag agtcaccatg accgaggaca catctacaga cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtgt attactgtgc aagacgttac 300
taccatagta cctacgtgtt ctactttgac tcctggggcc aagggaccct ggtcaccgtc 360
tcctca 366

SEQ ID NO: 369      moltype = DNA length = 366
FEATURE            Location/Qualifiers
misc_feature       1..366
                    note = Humanized 8H5H5G4 Heavy chain variable region
                    sequence (codon optimized)
source             1..366
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 369
caggttcagc tggttcagtc tggcgccgaa gtgaagaaac ctggcgccctc tgtgaagggtg 60

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tcttgcaagg tgtccggaat taccttcacc gagtacacca tgcactgggt cgcacaggcc 120
cctggcaaaag gacttgaatg gatgggcggc ttcaacccca acaacggcgt gaccaactac 180
aaccagaaat tcaagggcgc cgtgaccatg accgaggaca caagcacaga caccgcctac 240
atggaactga gcagcctgag aagcgaggac accgcccgtg actactgcgc cagaagggtac 300
taccacagca cctacgtgtt ctacttcgac agctggggcc agggcacact ggtcacagtt 360
tcttct 366

SEQ ID NO: 370      moltype = DNA length = 366
FEATURE            Location/Qualifiers
misc_feature        1..366
                    note = Modified Humanized 8H5H5G4 Heavy chain variable
                    region sequence
source              1..366
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 370
caggtccagc tggtagatgc tggggctgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcttgcaagg ttcccgaaa cacattcact gaatacacca tgcactgggt ggcacaggct 120
cctggaaaag ggcttgagtg gatcggaggt tttaatccta acaatgggtg tactaactac 180
aaccagaagt tcaagggcaa ggtcacccct accgtggaca catctagcag cacagcctac 240
atggagctga gcagcctgag atctgaggac accgcccgtg attactgtgc aagacgttac 300
taccatagta cctacgtgtt ctactttgac tcttggggcc aaggggacct ggtcaccgtc 360
tcctca 366

SEQ ID NO: 371      moltype = DNA length = 366
FEATURE            Location/Qualifiers
misc_feature        1..366
                    note = Modified Humanized 8H5H5G4 Heavy chain variable
                    region sequence (codon optimized)
source              1..366
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 371
caggttcagc tgggttcagtc tggcgccgaa gtgaagaaac ctggcgccctc tgtgaagggtg 60
tcttgcaagg tgtccggaat taccttcacc gagtacacca tgcactgggt cgcacaggcc 120
cctggcaaaag gactggaatg gatcggcggc ttcaacccca acaacggcgt gaccaactac 180
aaccagaaat tcaagggcaa agtgaccctg accgtggaca ccagcagcag cacagcctac 240
atggaactga gcagcctgag aagcgaggac accgcccgtg actactgcgc cagaagggtac 300
taccacagca cctacgtgtt ctacttcgac agctggggcc agggcacact ggtcacagtt 360
tcttct 366

SEQ ID NO: 372      moltype = DNA length = 324
FEATURE            Location/Qualifiers
misc_feature        1..324
                    note = 8H5H5G4 Light chain variable region sequence
source              1..324
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 372
gatatccaga tgacacagac tacatcctcc ctgtctgcct ctctgggaga cagagtcacc 60
atcagttgca gtgcaagtca gggcattagc aattatttaa actggtttca gcagaaacca 120
gatggaacta ttaagctcct gatctattac acatcaagtt tacattcagg agtcccatca 180
agggttcagtg gcagtgggtc tgggacagat tattctctca ccatcagtaa tgtggaacct 240
gaagatattg ccacttacta ttgtcagcag tatagtaagc ttccttacac gttcggaggg 300
gggaccaagc tggagataaa acgg 324

SEQ ID NO: 373      moltype = DNA length = 285
FEATURE            Location/Qualifiers
misc_feature        1..285
                    note = IGKV1-27*01 V-REGION sequence
source              1..285
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 373
gacatccaga tgaccacagtc tccatcctcc ctgtctgcat ctgtaggaga cagagtcacc 60
atcacttgcc gggcgagtcg gggcattagc aattatttag cctggatca gcagaaacca 120
gggaaagttc ctaagctcct gatctatgct gcattccatt tgcaatcagg ggtcccatct 180
cggttcagtg gcagtggatc tgggacagat ttcactctca ccatcagcag cctgcagcct 240
gaagatgttg caacttatta ctgtcaaaag tataacagtg ccct 285

SEQ ID NO: 374      moltype = DNA length = 36
FEATURE            Location/Qualifiers
misc_feature        1..36
                    note = IGKJ4*02 J-REGION sequence
source              1..36
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 374
ctcacgttcg gcggaggagc caaggtggag atcaaa 36

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SEQ ID NO: 375      moltype = DNA length = 324
FEATURE            Location/Qualifiers
misc_feature        1..324
                    note = humanized 8H5H5G4 Light chain variable region
                    sequence
source              1..324
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 375
gacatccaga tgaccagtc tccatcctcc ctgtctgcat ctgtaggaga cagagtcacc 60
atcacttgca gtgcaagtca gggcattagc aattatttaa actggtatca gcagaaacca 120
gggaaagttc ctaagctcct gatctattac acatcaagtt tacattcagg ggtcccatct 180
cggttcagtg gcagtggatc tgggacagat ttcactctca ccatcagcag cctgcagcct 240
gaagatgttg caacttatta ctgtcagcag tatagtaagc ttccttacac gttcggcgga 300
gggaccaagg tggagatcaa acgg                                     324

SEQ ID NO: 376      moltype = DNA length = 324
FEATURE            Location/Qualifiers
misc_feature        1..324
                    note = humanized 8H5H5G4 Light chain variable region
                    sequence (codon optimized)
source              1..324
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 376
gacatccaga tgacacagag ccctagcagc ctgtctgcca gcgtgggaga cagagtgacc 60
atcacatgta ggcgcaagcca gggcatcagc aactacctga actggtatca gcagaaaccc 120
ggcaaggtgc ccaagctgct gatctactac accagcagcc tgcacagcgg cgtgccaagc 180
agattttctg gcagcggctc tggcaccgac ttcaccctga ccatatctag cctgcagcct 240
gaggacgtgg ccacctacta ctgtcagcag tacagcaagc tgccttacac ctttggcgga 300
ggcaccaagg tggaaatcaa gcgg                                     324

SEQ ID NO: 377      moltype = DNA length = 324
FEATURE            Location/Qualifiers
misc_feature        1..324
                    note = Modified humanized 8H5H5G4 Light chain variable
                    region sequence
source              1..324
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 377
gacatccaga tgaccagtc tccatcctcc ctgtctgcat ctgtaggaga cagagtcacc 60
atcacttgca gtgcaagtca gggcattagc aattatttaa actggtatca gcagaaacca 120
gggaaagttc ctaagctcct gatctattac acatcaagtt tacattcagg ggtcccatct 180
cggttcagtg gcagtggatc tgggacagat tacactctca ccatcagcag cctgcagcct 240
gaagatgttg caacttatta ctgtcagcag tatagtaagc ttccttacac gttcggcgga 300
gggaccaagg tggagatcaa acgg                                     324

SEQ ID NO: 378      moltype = DNA length = 324
FEATURE            Location/Qualifiers
misc_feature        1..324
                    note = Modified humanized 8H5H5G4 Light chain variable
                    region sequence (codon optimized)
source              1..324
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 378
gacatccaga tgacacagag ccctagcagc ctgtctgcca gcgtgggaga cagagtgacc 60
atcacatgta ggcgcaagcca gggcatcagc aactacctga actggtatca gcagaaaccc 120
ggcaaggtgc ccaagctgct gatctactac accagcagcc tgcacagcgg cgtgccaagc 180
agattttctg gcagcggctc tggcaccgac tacaccctga ccatatctag cctgcagcct 240
gaggacgtgg ccacctacta ctgtcagcag tacagcaagc tgccttacac ctttggcgga 300
ggcaccaagg tggaaatcaa gcgg                                     324

SEQ ID NO: 379      moltype = DNA length = 735
FEATURE            Location/Qualifiers
misc_feature        1..735
                    note = Modified humanized 8H5H5G4 scFV sequence (codon
                    optimized)
source              1..735
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 379
caggttcagc tggttcagtc tggcgccgaa gtgaagaaac ctggcgccctc tgtgaaggtg 60
tcctgcaagg tgtccgaaa taccttcacc gactacacca tgcactgggt cgcagaggcc 120
cctggcaaa gactggaatg gatcgccggc ttcaacccca acaacggcgt gaccaactac 180
aaccagaaat tcaagggcaa agtgaccctg accgtggaca ccagcagcag cacagcctac 240
atggaactga gcagcctgag aagcgaggac accgccgtgt actactgcgc cagaaggtac 300

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taccacagca cctacgtgtt ctacttcgac agctggggcc agggcacact ggtcacagtt 360
tctttctggcg gtggcggaag cggaggcggt ggctcgggtg gcggaggcag cgacatccag 420
atgacacaga gccctagcag cctgtctgcc agcgtgggag acagagtgcac catcacatgt 480
agcgccagcc agggcatcag caactacctg aactgggtatc agcagaaacc cggcaagggtg 540
cccaagctgc tgatctacta caccagcagc ctgcacagcg gcgtgccaaag cagattttct 600
ggcagcggt ctggcaccga ctacaccctg accatatcta gcctgcagcc tggaggacgtg 660
gccacctact actgtcagca gtacagcaag ctgcccctaca ctttggcggt aggcaccaag 720
gtggaaatca agcggg                                     735

SEQ ID NO: 380      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature       1..993
                    note = Human IgG1 heavy chain constant region sequence:
                    (for making full antibody - pair with either kappa or
                    lambda constant region; 2 plasmids, express together)
source             1..993
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 380
gctagcacca agggcccac cgtcttcccc ctggcaccct cctccaagag cacctctggg 60
ggcacagcgg ccttgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgtcg 120
tggaactcag gcgcctgcag cagcggcggtg cacaccttcc cggctgtcct acagtctctca 180
ggactctact ccttcagcag cgtgggtgaca gtgcccctcca gcagcttggg caccagacc 240
tacactctga acgtgaatca caagcccagc aacaccaagg tggacaagaa agttgagccc 300
aaatcttgtg acaaaactca cacatgcccc ccgtgccag cactgaact cctgggggga 360
ccgtcagtc tctcttctcc cccaaaaccc aaggacaccc tcatgatctc ccggaccct 420
gaggtcacat gcgtgggtgt ggacgtgagc cagcaagacc ctgaggtcaa gttcaactgg 480
tacgtggacg gcgtggaggt gcataatgcc aagacaaagc cgcgggagga gcagtacaac 540
agcacgtacc gtgtgggtcag cgtctcacc gtctcgcacc aggaactggct gaatggcaag 600
gagtagaagt gcaagggtct caacaaagcc ctcccagccc ccacgcagaa aaccatctcc 660
aaagccaaag ggcagccccg agaaccacag gtgtacaccc tgcccccatc ccgggaggag 720
atgaccaaga accaggtcag cctgacctgc ctgggtcaaa gcttctatcc cagcgacatc 780
gccgtggagt gggagagcaa tgggcagccg gagaacaact acaagaccac gcctcccgtg 840
ctggactcgt acggctctct ctctctctac agcaagctca ccgtggacaa gagcaggttg 900
cagcagggga acgtctctc atgctccgtg atgcatgagg ctctgcacaa ccactacacg 960
cagaagagcc tctccctgtc tccgggtaaa tga                                     993

SEQ ID NO: 381      moltype = DNA length = 981
FEATURE            Location/Qualifiers
misc_feature       1..981
                    note = Human IgG2 heavy chain constant region sequence:
                    (for making full antibody - pair with either kappa or
                    lambda constant region; 2 plasmids, express together)
source             1..981
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 381
gcctccacca agggcccac cgtcttcccc ctggcgccct gctccaggag cacctccgag 60
agcacagccg ccttgggctg cctgggtcaag gactacttcc ccgaaccggg gacgggtgtcg 120
tggaactcag gcgtctgtac cagcggcggtg cacaccttcc cagctgtcct acagtctctca 180
ggactctact ccttcagcag cgtgggtgacc gtgcccctcca gcaacttcgg caccagacc 240
tacactctga acgtgaatca caagcccagc aacaccaagg tggacaagac agttgagcgc 300
aaatgttgtg tcgagtgcgc accgtgcccc gcaccacctg tggcaggacc gtcagtcttc 360
ctcttcccc caaaacccaa ggacacctc atgatctccc ggacccctga ggtcactgtc 420
gtgggtgttg acgtgagcca cgaagacccc gaggtccagt tcaactggta cgtggacggc 480
gtggaggtgc ataagtcaca gacaaagcca cgggaggagc agttcaacag cacttccgt 540
gtggtcagcg tcttcaccgt tgtgcaccag gactgggtga acggcaagga gtacaagtgc 600
aaggtctcca acaaggcctt cccagcccc atcgagaaaa ccatctccaa aaccaagggt 660
cagccccgag aaccacaggt gtacacctg ccccatccc gggaggagat gaccaagaac 720
caggtcagcc tgacctgcct ggtcaaagcc ttctacccca gcgacatcgc cgtggaggtg 780
gagagcaatg ggcagccgga gaacaactac aagaccacac ctcccattgt ggactccgac 840
ggctccttct tctctctacg caagctcacc gtggacaaga gcaggtggca gcaggggaac 900
gtcttctcat gctccgtgat gcatgaggct ctgcacaacc actacacgca gaagagcctc 960
tccctgtctc cgggtaataa g                                     981

SEQ ID NO: 382      moltype = DNA length = 324
FEATURE            Location/Qualifiers
misc_feature       1..324
                    note = Human Kappa light chain constant region sequence
source             1..324
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 382
aggacgggtg ctgcaccatc tgtcttcac ttcctcgccat ctgatgagca gttgaaatct 60
ggaactgcct ctgttgtgtg cctgtgtaat aactctatc ccagagaggg caaagtacag 120
tggaagggtg ataacgcctt ccaatcgggt aactcccagg agagtgtcac agagcaggac 180
agcaaggaca gcacctacag cctcagcagc accctgacgc tgagcaaagc agactacgag 240
aaacacaaag tctacgcctg cgaagtcacc catcagggcc tgagctcgcc cgtcacaaag 300
agcttcaaca ggggagagtg ttatg                                     324

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mol_type = protein
organism = synthetic construct

SEQUENCE: 393
SASQGISNYL N 11

SEQ ID NO: 394
FEATURE
REGION
note = Monoclonal antibody 8F9A4A3 Light chain variable
region CDR2
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 394
YTSSLHS 7

SEQ ID NO: 395
FEATURE
REGION
note = Monoclonal antibody 8F9A4A3 Light chain variable
region CDR3
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 395
QQYSKLPYT 9

SEQ ID NO: 396
FEATURE
misc_feature
note = Monoclonal antibody 5D9E2B11 - Heavy chain variable
region sequence - H-1,4,7,8,12
source
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 396
gtccagctgc aacagtcttg acctgatctg gtgaagcctg ggacttcagt gaagatatcc 60
tgtaagactt ctggaaacac attcactgaa tacaccatgc actgggtgaa gcagagccat 120
ggaaagagcc ttgagtggat tggaggtttt aatcctaaca atggtgttac taactacaac 180
cagaagtcca agggcaaggc cacattgact gtagacaagt cctccagcac agcctacatg 240
gagctccgca gacctgacatc tgaggattct gcagtctatt actgtgcaag acgttactac 300
catagtacct acgtgttcta ctttgactcc tggggccaag gcaccactct cacagtctcc 360
tca 363

SEQ ID NO: 397
FEATURE
REGION
note = Monoclonal antibody 5D9E2B11 Translated protein,
wherein the underlined sequence is the complementarity
determining region (CDR)
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 397
VQLQQSGPDL VKPGTSVKIS CKTSGNFTFE YTMHWVKQSH GKSLEWIGGF NPNNGVTNYN 60
QKFKGKATLT VDKSSSTAYM ELRLTSEDS AVYYCARRY HSTYVFYFDS WQGGTTLTVS 120
S 121

SEQ ID NO: 398
FEATURE
REGION
note = Monoclonal antibody 5D9E2B11 Heavy chain variable
region CDR1
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 398
NTFTEYTMH 9

SEQ ID NO: 399
FEATURE
REGION
note = Monoclonal antibody 5D9E2B11 Heavy chain variable
region CDR2
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 399
GPNPNNGVTN YNQKPKG 17

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source                1..9
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 406
QQYSKLPYT                                                    9

SEQ ID NO: 407        moltype = DNA length = 363
FEATURE              Location/Qualifiers
misc_feature          1..363
                      note = Monoclonal antibody 5D9E10E4 - Heavy chain variable
                      region sequence - H-2,4,7,10,12

source                1..363
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 407
gtccagctgc aacagtctgg acctgatctg gtgaagcctg ggacttcagt gaagatatcc 60
tgtaagactt ctggaaacac attcactgaa tacaccatgc actgggtgaa gcagagccat 120
ggaaagagcc ttgagtggat tggaggtttt aatcctaaca atggtgttac taactacaac 180
cagaagtcca agggcaaggc cacattgact gtagacaagt cctccagcac agcctacatg 240
gagctccgca gcctgacatc tgaggattct gcagtctatt actgtgcaag acgttactac 300
catagtacct acgtgttcta ctttgactcc tggggccaag gcaccactct cacagtctcc 360
tca                                                         363

SEQ ID NO: 408        moltype = AA length = 121
FEATURE              Location/Qualifiers
REGION              1..121
                      note = Monoclonal antibody 5D9E10E4 Translated protein,
                      wherein the underlined sequence is the complementarity
                      determining region (CDR)

source                1..121
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 408
VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNNGVTNYN 60
QKFKGKATLT VDKSSSTAYM ELRSLTSEDS AVYYCARRY YSTYVFYFDS WQGQTTLTVS 120
S                                                         121

SEQ ID NO: 409        moltype = AA length = 9
FEATURE              Location/Qualifiers
REGION              1..9
                      note = Monoclonal antibody 5D9E10E4 Heavy chain variable
                      region CDR1

source                1..9
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 409
NTFTEYTMH                                                    9

SEQ ID NO: 410        moltype = AA length = 17
FEATURE              Location/Qualifiers
REGION              1..17
                      note = Monoclonal antibody 5D9E10E4 Heavy chain variable
                      region CDR2

source                1..17
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 410
GFNPNNGVIN YNQKFKG                                          17

SEQ ID NO: 411        moltype = AA length = 13
FEATURE              Location/Qualifiers
REGION              1..13
                      note = Monoclonal antibody 5D9E10E4 Heavy chain variable
                      region CDR3

source                1..13
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 411
RYYHSTYVFY FDS                                              13

SEQ ID NO: 412        moltype = DNA length = 324
FEATURE              Location/Qualifiers
misc_feature          1..324
                      note = Monoclonal antibody 5D9E10E4 Light chain variable
                      region sequence - K-2,6,8,14,15

source                1..324
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 412

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SEQUENCE: 418
VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNNGVTNYY 60
QKFKGKATLT VDKSSSTAYM ELRLTSEDS AVYYCARRY HSTYVFYFDS WGQGTTLTVS 120
S 121

SEQ ID NO: 419 moltype = AA length = 9
FEATURE Location/Qualifiers
REGION 1..9
 note = Monoclonal antibody 5D9G2C4 Heavy chain variable
 region CDR1
source 1..9
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 419
NTFTEYTMH 9

SEQ ID NO: 420 moltype = AA length = 17
FEATURE Location/Qualifiers
REGION 1..17
 note = Monoclonal antibody 5D9G2C4 Heavy chain variable
 region CDR2
source 1..17
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 420
GFNPNNGVNTN YNQKFKG 17

SEQ ID NO: 421 moltype = AA length = 13
FEATURE Location/Qualifiers
REGION 1..13
 note = Monoclonal antibody 5D9G2C4 Heavy chain variable
 region CDR3
source 1..13
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 421
RYYHSTYVFY FDS 13

SEQ ID NO: 422 moltype = DNA length = 324
FEATURE Location/Qualifiers
misc_feature 1..324
 note = Monoclonal antibody 5D9G2C4 Light chain variable
 region sequence - K-4,6,7,8,10
source 1..324
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 422
gatatccaga tgacacagac tacatcctcc ctgtctgcct ctctgggaga cagagtcacc 60
atcagttgca gtgcaagtc gggcattagc aattatttaa actgggttca gcagaaacca 120
gatggaacta ttaagctcct gatctattac acatcaagtt tacattcagg agtcccatca 180
agggttcagtg gcagtgggtc tgggacagat tattctctca ccacacagtaa tgtggaacct 240
gaagatatgt ccacttacta ttgtcagcag tataagtaagc ttccttacac gttcggaggg 300
gggaccaagc tggagataaa acgg 324

SEQ ID NO: 423 moltype = AA length = 108
FEATURE Location/Qualifiers
REGION 1..108
 note = Monoclonal antibody 5D9G2C4 Translated protein,
 wherein the underlined sequence is the complementarity
 determining region (CDR)
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 423
DIQMTQTSS LSASLGDRVT ISCSASQGIS NYLWFWQKP DGTIKLLIYY TSSLHSGVPS 60
RFSGSGSGTD YSLTISNVEP EDIATYICQQ YSKLPYTFGG GTKLEIKR 108

SEQ ID NO: 424 moltype = AA length = 11
FEATURE Location/Qualifiers
REGION 1..11
 note = Monoclonal antibody 5D9G2C4 Light chain variable
 region CDR 1
source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 424
SASQGISNYL N 11

SEQ ID NO: 425 moltype = AA length = 7

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FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Monoclonal antibody 5D9G2C4 Light chain variable	
	region CDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 425		
YTSSLHS		7
SEQ ID NO: 426	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Monoclonal antibody 5D9G2C4 Light chain variable	
	region CDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 426		
QQYSKLPYT		9
SEQ ID NO: 427	moltype = DNA length = 363	
FEATURE	Location/Qualifiers	
misc_feature	1..363	
	note = Monoclonal antibody 5F3A5D4 - Heavy chain variable	
	region sequence - H-2,3,4,13,15	
source	1..363	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 427		
gtccagctgc aacagtcttg accatgctg gtgaagcctg ggacttcagt gaagatatcc		60
tgtaagactt ctggaaacac attcactgaa tacaccatgc actgggtgaa gcagagccat		120
ggaaagagcc ttgagtggat tggaggtttt aatcctaaca atggtgttac taactacaac		180
cagaagtcca agggcaaggc cacattgact gtagacaagt cctccagcac agcctacatg		240
gagctccgca gcctgacatc tgaggattct gcagtctatt actgtgcaag acgttactac		300
catagtacct acgtgttcta ctttgactcc tggggccaag gcaccactct cacagtctcc		360
tca		363
SEQ ID NO: 428	moltype = AA length = 121	
FEATURE	Location/Qualifiers	
REGION	1..121	
	note = Monoclonal antibody 5F3A5D4 Translated protein,	
	wherein the underlined sequence is the complementarity	
	determining region (CDR)	
source	1..121	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 428		
VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNNGVTNYN		60
QKFKGKATLT VDKSSSTAYM ELRLTSEDS AVYYCARRY YSTYVFYFDS WQGGTTLTVS		120
S		121
SEQ ID NO: 429	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Monoclonal antibody 5F3A5D4 Heavy chain variable	
	region CDR1	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 429		
NTFTEYTMH		9
SEQ ID NO: 430	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Monoclonal antibody 5F3A5D4 Heavy chain variable	
	region CDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 430		
GFNPNNGVIN YNQKFKG		17
SEQ ID NO: 431	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Monoclonal antibody 5F3A5D4 Heavy chain variable	
	region CDR3	

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source                1..13
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 431
RYYHSTYVIFY FDS                                           13

SEQ ID NO: 432        moltype = DNA  length = 324
FEATURE              Location/Qualifiers
misc_feature          1..324
                      note = Monoclonal antibody 5F3A5D4 Light chain variable
                      region sequence - K-1,2,3,4,9

source                1..324
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 432
gatatccaga tgacacagac tacatcctcc ctgtctgcct ctctgggaga cagagtcacc 60
atcagttgca gtgcaagtca gggcatttagc aattatttaa actgggttca gcagaaacca 120
gatggaacta ttaagctcct gatctattac acatcaagtt tacattcagg agtcccatca 180
aggttcagtg gcagtggggtc tgggacagat tattctctca ccatcagtaa tgtggaacct 240
gaagatatgg ccacttacta ttgtcagcag tatagtaagc ttccttacac gttcggaggg 300
gggaccaagc tggagataaa acgg                                           324

SEQ ID NO: 433        moltype = AA  length = 108
FEATURE              Location/Qualifiers
REGION                1..108
                      note = Monoclonal antibody 5F3A5D4 Translated protein,
                      wherein the underlined sequence is the complementarity
                      determining region (CDR)

source                1..108
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 433
DIQMTQTTS LSASLGDRVT ISCSASQGIS NYLNWFQQKP DGTIKLLIYY TSSLHSGVPS 60
RFGSGSGSTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR           108

SEQ ID NO: 434        moltype = AA  length = 11
FEATURE              Location/Qualifiers
REGION                1..11
                      note = Monoclonal antibody 5F3A5D4 Light chain variable
                      region CDR1

source                1..11
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 434
SASQGISNYL N                                           11

SEQ ID NO: 435        moltype = AA  length = 7
FEATURE              Location/Qualifiers
REGION                1..7
                      note = Monoclonal antibody 5F3A5D4 Light chain variable
                      region CDR2

source                1..7
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 435
YTSSLHS                                           7

SEQ ID NO: 436        moltype = AA  length = 9
FEATURE              Location/Qualifiers
REGION                1..9
                      note = Monoclonal antibody 5F3A5D4 Light chain variable
                      region CDR3

source                1..9
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 436
QQYSKLPYT                                           9

SEQ ID NO: 437        moltype = DNA  length = 357
FEATURE              Location/Qualifiers
misc_feature          1..357
                      note = Monoclonal antibody 8F9A5A1 - Heavy chain variable
                      region sequence - H-3,4,6,10,11

source                1..357
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 437
atccagttgg tgcagctcgg acctgagctg aagaagcctg gagagacagt caagatctcc 60
tgcaaggctt ctgggtatgc cttcacaaac tatggaatga actgggtgaa gcaggctcca 120

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ggaagggtt taaagtggat gggctggata aacacctaca ctggagagcc aacatatgtt 180
gatgacttca agggacgggtt tgcttctct ttggaaacct ctgccaccac tgctatttg 240
cagatcaaca acctcaaaaa tgaggacacg tctacatatt tctgtgcaag attgaggggg 300
atacgaccgg gtcctctggc ttactggggc caagggactc tggctactgt ctctgca 357

SEQ ID NO: 438      moltype = AA length = 119
FEATURE            Location/Qualifiers
REGION            1..119
                    note = Monoclonal antibody 8F9A5A1 Translated protein,
                    wherein the underlined sequence is the complementarity
                    determining region (CDR)
source            1..119
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 438
IQLVQSGPEL KKPGETVKIS CKASGYFTTN YGMNWKQAP GKGLKWMGWI NTYTGEPTYV 60
DDFKGRFAPS LETSATAYL QINNKNEDT STYFCARLRG IRPGPLAYWG QGTLVTVSA 119

SEQ ID NO: 439      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION            1..9
                    note = Monoclonal antibody 8F9A5A1 Heavy chain variable
                    region CDR1
source            1..9
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 439
YTFTNYGMN                                     9

SEQ ID NO: 440      moltype = AA length = 17
FEATURE            Location/Qualifiers
REGION            1..17
                    note = Monoclonal antibody 8F9A5A1 Heavy chain variable
                    region CDR2
source            1..17
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 440
WINTYTGEPT YVDDFKG                                     17

SEQ ID NO: 441      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION            1..11
                    note = Monoclonal antibody 8F9A5A1 Heavy chain variable
                    region CDR3
source            1..11
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 441
LRGIRPGPLA Y                                     11

SEQ ID NO: 442      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature       1..321
                    note = Monoclonal antibody 8F9A5A1 Light chain variable
                    region sequence - K-1,2,3,4,5
source            1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 442
gaaattttgc tcacccagtc tccagcaatc atagctgcat ctctgggga gaaggtcacc 60
atcacctgca gtgccagctc aagtgtaatg tacatgaact ggtaccagca gaaaccagga 120
tcctcccca aaatatggat ttatggata tccaacctgg cttctggagt tcctgctcgc 180
ttcagtgcca gtgggtctgg gacatctttc tctttcaca tcaacagcat ggaggctgaa 240
gatgttgcca cttattactg tcagcaaagg agtagttacc caccacggtt cggagggggg 300
accaagctgg aaataaaacg g                                     321

SEQ ID NO: 443      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION            1..107
                    note = Monoclonal antibody 8F9A5A1 Translated protein,
                    wherein the underlined sequence is the complementarity
                    determining region (CDR)
source            1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 443
EILLTQSPAI IAASPGKVT ITCSASSSVS YMNWYQOKPG SSPKIWIYGI SNLASGVPAR 60
FSGSGSGTSF SFTINSMEAE DVATYYCQQR SSYPPTFGGG TKLEIKR 107

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SEQ ID NO: 444      moltype = AA  length = 10
FEATURE            Location/Qualifiers
REGION             1..10
                   note = Monoclonal antibody 8F9A5A1 Light chain variable
                   region CDR 1
source             1..10
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 444
SASSSVSYMN                               10

SEQ ID NO: 445      moltype = AA  length = 7
FEATURE            Location/Qualifiers
REGION             1..7
                   note = Monoclonal antibody 8F9A5A1 Light chain variable
                   region CDR2
source             1..7
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 445
GISNLAS                               7

SEQ ID NO: 446      moltype = AA  length = 9
FEATURE            Location/Qualifiers
REGION             1..9
                   note = Monoclonal antibody 8F9A5A1 Light chain variable
                   region CDR3
source             1..9
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 446
QQRSSYPPT                               9

SEQ ID NO: 447      moltype = DNA  length = 363
FEATURE            Location/Qualifiers
misc_feature       1..363
                   note = Monoclonal antibody 8H5H5G4 - Heavy chain variable
                   region sequence - H-1,3,5,6,10
source             1..363
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 447
gtccagctgc aacagtctgg acctgatctg gtgaagcctg ggacttcagt gaagatatcc 60
tgtaagactt ctggaaacac attcaactgaa tacaccatgc actgggtgaa gcagagccat 120
ggaaagagcc ttgagtggat tggaggtttt aatcctaaca atggtgttac taactacaac 180
cagaagttca agggcaaggc cacattgact gtagacaagt cctccagcac agcctacatg 240
gagctccgca gcctgacatc tgaggattct gcagtctatt actgtgcaag acgttactac 300
catagtacct acgtgttcta ctttgactcc tggggccaag gcaccactct cacagtctcc 360
tca                                           363

SEQ ID NO: 448      moltype = AA  length = 121
FEATURE            Location/Qualifiers
REGION             1..121
                   note = Monoclonal antibody 8H5H5G4 Translated protein,
                   wherein the underlined sequence is the complementarity
                   determining region (CDR)
source             1..121
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 448
VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNNGVTNYN 60
QKFKGKATLT VDKSSSTAYM ELRSLTSEDS AVYICARRY HSTYVFYFDS WQGQTTTLTVS 120
S                                           121

SEQ ID NO: 449      moltype = AA  length = 9
FEATURE            Location/Qualifiers
REGION             1..9
                   note = Monoclonal antibody 8H5H5G4 Heavy chain variable
                   region CDR1
source             1..9
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 449
NTFTEYTMH                               9

SEQ ID NO: 450      moltype = AA  length = 17
FEATURE            Location/Qualifiers
REGION             1..17

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note = Monoclonal antibody 8H5H5G4 Heavy chain variable
  region CDR2
source      1..17
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 450
GFNPNNNGVTN YNQKPKG                               17

SEQ ID NO: 451      moltype = AA length = 13
FEATURE            Location/Qualifiers
REGION            1..13
note = Monoclonal antibody 8H5H5G4 Heavy chain variable
  region CDR3
source      1..13
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 451
RYYHSTYVFY FDS                                     13

SEQ ID NO: 452      moltype = DNA length = 324
FEATURE            Location/Qualifiers
misc_feature      1..324
note = Monoclonal antibody 8H5H5G4 Light chain variable
  region sequence - K-2,5,8,9,15
source      1..324
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 452
gatatccaga tgacacagac tacatcctcc ctgtctgcct ctctgggaga cagagtcacc 60
atcagttgca gtgcaagtca gggcattagc aattatttaa actgggttca gcagaaacca 120
gatggaacta ttaagctcct gatctattac acatcaagtt tacattcagg agtcccatca 180
aggttcagtg gcagtggggc tgggacagat tattctctca ccatcagtaa tgtggaacct 240
gaagatatcg ccacttacta ttgtcagcag tatagtaagc ttccttacac gttcggaggg 300
gggaccaagc tggagataaa acgg                                     324

SEQ ID NO: 453      moltype = AA length = 108
FEATURE            Location/Qualifiers
REGION            1..108
note = Monoclonal antibody 8H5H5G4 Translated protein,
  wherein the underlined sequence is the complementarity
  determining region (CDR)
source      1..108
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 453
DIQMTQTTS LSASLGDRVT ISCSASQGIS NYLNWFQKP DGTIKLLIY TSSLHSGVPS 60
RPSGSGSGTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR          108

SEQ ID NO: 454      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION            1..11
note = Monoclonal antibody 8H5H5G4 Light chain variable
  region CDR 1
source      1..11
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 454
SASQGISNYL N                                         11

SEQ ID NO: 455      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION            1..7
note = Monoclonal antibody 8H5H5G4 Light chain variable
  region CDR2
source      1..7
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 455
YTSSLHS                                             7

SEQ ID NO: 456      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION            1..9
note = Monoclonal antibody 8H5H5G4 Light chain variable
  region CDR3
source      1..9
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 456

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                    Heavy chain variable region sequence mouse
source              1..121
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1001
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QKFKGKATLT VDKSSSTAYM ELRSLTSEDS AVYYCARRY HSLYVFYFDY WQGTTLTVS 120
S                                                    121

SEQ ID NO: 1002     moltype = AA  length = 98
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source              1..98
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1002
QVQLVQSGAE VKKPGASVKV SCKVSGYTLT ELSMHWVRQA PGKGLEWMGG FDPEDGETIY 60
AQKFQGRVTM TEDTSTDYAY MELSSLRSED TAVYYCAT                               98

SEQ ID NO: 1003     moltype = AA  length = 15
FEATURE             Location/Qualifiers
REGION              1..15
                    note = IGHJ4*01 J-REGION sequence human (closest match hu
                    antibody sequence)
source              1..15
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1003

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YFDYWGQGT L VTVSS 15

SEQ ID NO: 1004 moltype = AA length = 122
 FEATURE Location/Qualifiers
 REGION 1..122
 note = humanized 8F9A4P3 Heavy chain variable region
 sequence
 source 1..122
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1004
 QVQLVQSGAE VKKPGASVKV SCKVSGNTFT EYTMHWVRQA PGKGLEWMGG FNPNGVTNY 60
 NQKFKGRVTM TEDTSTDYAY MELSSLRSED TAVYYCARRY YHSLYVFYFD YWGQGT LVTV 120
 SS 122

SEQ ID NO: 1005 moltype = AA length = 122
 FEATURE Location/Qualifiers
 REGION 1..122
 note = humanized 8F9A4P3 Heavy chain variable region
 sequence (codon optimized)
 source 1..122
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1005
 QVQLVQSGAE VKKPGASVKV SCKVSGNTFT EYTMHWVRQA PGKGLEWMGG FNPNGVTNY 60
 NQKFKGRVTM TEDTSTDYAY MELSSLRSED TAVYYCARRY YHSLYVFYFD YWGQGT LVTV 120
 SS 122

SEQ ID NO: 1006 moltype = AA length = 122
 FEATURE Location/Qualifiers
 REGION 1..122
 note = Modified humanized 8F9A4P3 Heavy chain variable
 region sequence
 source 1..122
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1006
 QVQLVQSGAE VKKPGASVKV SCKVSGNTFT EYTMHWVRQA PGKGLEWIGG FNPNGVTNY 60
 NQKFKGKVT LTVDTSSSTAY MELSSLRSED TAVYYCARRY YHSLYVFYFD YWGQGT LVTV 120
 SS 122

SEQ ID NO: 1007 moltype = AA length = 122
 FEATURE Location/Qualifiers
 REGION 1..122
 note = Modified humanized 8F9A4P3 Heavy chain variable
 region sequence (codon optimized)
 source 1..122
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1007
 QVQLVQSGAE VKKPGASVKV SCKVSGNTFT EYTMHWVRQA PGKGLEWIGG FNPNGVTNY 60
 NQKFKGKVT LTVDTSSSTAY MELSSLRSED TAVYYCARRY YHSLYVFYFD YWGQGT LVTV 120
 SS 122

SEQ ID NO: 1008 moltype = AA length = 108
 FEATURE Location/Qualifiers
 REGION 1..108
 note = 8F9A4P3 Light chain variable region sequence mouse
 source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1008
 ETTVTQSPAS LSMAIGKVT IRCITSTDID DDMNYYQQK GEPPKLLISE GNTLRPGVPS 60
 RFSSSGYGT D FVFTIENMLS EDVADYYCLQ SDNLPLTFGS GTKLEIKR 108

SEQ ID NO: 1009 moltype = AA length = 88
 FEATURE Location/Qualifiers
 REGION 1..88
 note = IGKV5-2*01 V-REGION sequence human (closest match hu
 antibody sequence)
 source 1..88
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1009
 ETTLTQSPAF MSATPGDKVN ISCKASQDID DDMNYYQQK GEAAIFIIQE ATTLVPGIPP 60
 RFSSSGYGT D FTLTINNIES EDAAYYFC 88

SEQ ID NO: 1010 moltype = AA length = 12
 FEATURE Location/Qualifiers

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REGION 1..12
note = IGKJ4*02 J-REGION sequence human (closest match hu
antibody sequence)

source 1..12
mol_type = protein
organism = synthetic construct

SEQUENCE: 1010
LTFGGGTVKE IK 12

SEQ ID NO: 1011 moltype = AA length = 108
FEATURE Location/Qualifiers
REGION 1..108
note = humanized 8F9A4P3 Light chain variable region
sequence

source 1..108
mol_type = protein
organism = synthetic construct

SEQUENCE: 1011
ETTLTQSPAF MSATPGDKVN ISCITSTDID DDMNWWYQKPK GEAAIFIIQE GNTLRPGIPP 60
RFGSGGYGTD FTLTINNIES EDAAAYFCLQ SDNLPLTFGG GTKVEIKR 108

SEQ ID NO: 1012 moltype = AA length = 108
FEATURE Location/Qualifiers
REGION 1..108
note = humanized 8F9A4P3 Light chain variable region
sequence (codon optimized)

source 1..108
mol_type = protein
organism = synthetic construct

SEQUENCE: 1012
ETTLTQSPAF MSATPGDKVN ISCITSTDID DDMNWWYQKPK GEAAIFIIQE GNTLRPGIPP 60
RFGSGGYGTD FTLTINNIES EDAAAYFCLQ SDNLPLTFGG GTKVEIKR 108

SEQ ID NO: 1013 moltype = AA length = 108
FEATURE Location/Qualifiers
REGION 1..108
note = Modified humanized 8F9A4P3 Light chain variable
region sequence

source 1..108
mol_type = protein
organism = synthetic construct

SEQUENCE: 1013
ETTVTQSPAF MSATPGDKVT ISCITSTDID DDMNWWYQKPK GEAAILLISE GNTLRPGIPP 60
RFSSSGYGTD FTLTINNIES EDAAAYFCLQ SDNLPLTFGG GTKVEIKR 108

SEQ ID NO: 1014 moltype = AA length = 108
FEATURE Location/Qualifiers
REGION 1..108
note = Modified humanized 8F9A4P3 Light chain variable
region sequence (codon optimized)

source 1..108
mol_type = protein
organism = synthetic construct

SEQUENCE: 1014
ETTVTQSPAF MSATPGDKVT ISCITSTDID DDMNWWYQKPK GEAAILLISE GNTLRPGIPP 60
RFSSSGYGTD FTLTINNIES EDAAAYFCLQ SDNLPLTFGG GTKVEIKR 108

SEQ ID NO: 1015 moltype = AA length = 245
FEATURE Location/Qualifiers
REGION 1..245
note = Modified humanized 8F9A4P3 sequence (codon
optimized) - humanized heavy and light chains joined via a
flexible linker

source 1..245
mol_type = protein
organism = synthetic construct

SEQUENCE: 1015
QVQLVQSGAE VKKPGASVKV SCKVSGNTFT EYTMHWVRQA PGKGLEWIGG FNPNGVTNY 60
NQKFKGKVTI TVDTSSSTAY MELSSLRSED TAVYYCARRY YHSLYVFYFD YWGQGLVTV 120
SSGGGGSGGG GSGGGGSETT VTQSPAFMSA TPGDKVTISC ITSTDIDDDM NWYQKPKGEA 180
AILLISEGNT LRPGIPPRFS SSGYGTDFTL TINNIESEDA AYYFCLQSDN LPLTFGGGTK 240
VEIKR 245

SEQ ID NO: 1016 moltype = AA length = 119
FEATURE Location/Qualifiers
REGION 1..119
note = 8F9A5A1 Heavy chain variable region sequence

source 1..119
mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 1016
IQLVQSGPEL KKPGETVKIS CKASGYTFTN YGMNWKQAP GKGLKWMGWI NTYTGEPTYV 60
DDFKGRFAPS LETSATTAYL QINNLIKNETD STYFCARLRG IRPGPLAYWG QGTLVTVSA 119

SEQ ID NO: 1017      moltype = AA length = 98
FEATURE             Location/Qualifiers
REGION              1..98
                    note = IGHV7-81*01 V-REGION sequence
source              1..98
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1017
QVQLVQSGHE VKQPGASVKV SCKASGYSFT TYGMNWPQA PGQGLEWMGW FNTYTGNPTY 60
AQGFTGRFVF SMDTSASTAY LQISSLKAED MAMYCAR 98

SEQ ID NO: 1018      moltype = AA length = 15
FEATURE             Location/Qualifiers
REGION              1..15
                    note = IGHJ4*03 J-REGION sequence
source              1..15
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1018
YFDYWGGTL VTVSS 15

SEQ ID NO: 1019      moltype = AA length = 120
FEATURE             Location/Qualifiers
REGION              1..120
                    note = humanized 8F9A5A1 Heavy chain variable region
                    sequence
source              1..120
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1019
QVQLVQSGHE VKQPGASVKV SCKASGYTFT NYGMNWPQA PGQGLEWMGW INTYTGEPTY 60
VDDFKGRFVF SMDTSASTAY LQISSLKAED MAMYCARLR GIRPGPLAYW GQGLVTVSS 120

SEQ ID NO: 1020      moltype = AA length = 120
FEATURE             Location/Qualifiers
REGION              1..120
                    note = humanized 8F9A5A1 Heavy chain variable region
                    sequence (codon optimized)
source              1..120
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1020
QVQLVQSGHE VKQPGASVKV SCKASGYTFT NYGMNWPQA PGQGLEWMGW INTYTGEPTY 60
VDDFKGRFVF SMDTSASTAY LQISSLKAED MAMYCARLR GIRPGPLAYW GQGLVTVSS 120

SEQ ID NO: 1021      moltype = AA length = 120
FEATURE             Location/Qualifiers
REGION              1..120
                    note = Modified humanized 8F9A5A1 Heavy chain variable
                    region sequence
source              1..120
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1021
QVQLVQSGPE VKQPGASVKV SCKASGYTFT NYGMNWKQA PGQGLEWMGW INTYTGEPTY 60
VDDFKGRFAP SMDTSASTAY LQISSLKAED TATYYCARLR GIRPGPLAYW GQGLVTVSS 120

SEQ ID NO: 1022      moltype = AA length = 120
FEATURE             Location/Qualifiers
REGION              1..120
                    note = Modified humanized 8F9A5A1 Heavy chain variable
                    region sequence (codon optimized)
source              1..120
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1022
QVQLVQSGPE VKQPGASVKV SCKASGYTFT NYGMNWKQA PGQGLEWMGW INTYTGEPTY 60
VDDFKGRFAP SMDTSASTAY LQISSLKAED TATYYCARLR GIRPGPLAYW GQGLVTVSS 120

SEQ ID NO: 1023      moltype = AA length = 107
FEATURE             Location/Qualifiers
REGION              1..107
                    note = 8F9A5A1 Light chain variable region sequence
source              1..107

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mol_type = protein
organism = synthetic construct

SEQUENCE: 1023
EILLTQSPAI IAASPGKEKVT ITCSASSSVS YMNWYQQKPG SSPKIWIYGI SNLASGVPAR 60
FSGSGSGTSF SFTINSMEAE DVATYYCQQR SSYPPTFGGG TKLEIKR 107

SEQ ID NO: 1024      moltype = AA length = 93
FEATURE             Location/Qualifiers
REGION              1..93
                    note = IGKV3D-15*02 V-REGION sequence
source              1..93
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1024
EIVMMQSPAT LSVSPGERAT LSCRASQSVS SNLAWYQQKPG QAPRLLIYG ASTRATGIPA 60
RFSGSGSGTE FTLTISSLQS EDFAVYYCQQ YNN 93

SEQ ID NO: 1025      moltype = AA length = 12
FEATURE             Location/Qualifiers
REGION              1..12
                    note = IGKJ4*02 J-REGION sequence
source              1..12
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1025
LTFGGGTVKE IK 12

SEQ ID NO: 1026      moltype = AA length = 107
FEATURE             Location/Qualifiers
REGION              1..107
                    note = humanized 8F9A5A1 Light chain variable region
                    sequence
source              1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1026
EIVMMQSPAT LSVSPGERAT LSCSASSSVS YMNWYQQKPG QAPRLLIYGI SNLASGIPAR 60
FSGSGSGTEF TLTISLQSE DFAVYYCQQR SSYPPTFGGG TKVEIKR 107

SEQ ID NO: 1027      moltype = AA length = 107
FEATURE             Location/Qualifiers
REGION              1..107
                    note = humanized 8F9A5A1 Light chain variable region
                    sequence (codon optimized)
source              1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1027
EIVMMQSPAT LSVSPGERAT LSCSASSSVS YMNWYQQKPG QAPRLLIYGI SNLASGIPAR 60
FSGSGSGTEF TLTISLQSE DFAVYYCQQR SSYPPTFGGG TKVEIKR 107

SEQ ID NO: 1028      moltype = AA length = 107
FEATURE             Location/Qualifiers
REGION              1..107
                    note = Modified humanized 8F9A5A1 Light chain variable
                    region sequence
source              1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1028
EIVLTQSPAT LSVSPGERAT LSCSASSSVS YMNWYQQKPG QAPRLWIYGI SNLASGIPAR 60
FSGSGSGTSF SLTISSLQSE DFAVYYCQQR SSYPPTFGGG TKVEIKR 107

SEQ ID NO: 1029      moltype = AA length = 107
FEATURE             Location/Qualifiers
REGION              1..107
                    note = Modified humanized 8F9A5A1 Light chain variable
                    region sequence (codon optimized)
source              1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1029
EIVLTQSPAT LSVSPGERAT LSCSASSSVS YMNWYQQKPG QAPRLWIYGI SNLASGIPAR 60
FSGSGSGTSF SLTISSLQSE DFAVYYCQQR SSYPPTFGGG TKVEIKR 107

SEQ ID NO: 1030      moltype = AA length = 242
FEATURE             Location/Qualifiers
REGION              1..242
                    note = Modified humanized 8F9A5A1 scFV sequence (codon

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source          optimized)
                1..242
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 1030
QIQLVQSGPE VKQPGASVKV SCKASGYTFT NYGMNWKQA PGQGLEWMGW INTYTGEPTY 60
VDDFKGRFAF SMDTSASTAY LQISLKAED TATYYCARLR GIRPGPLAYW GQGTLLTVSS 120
GGGGSGGGGS GGGGSEIVLT QSPATLSVSP GERATLSCSA SSSVSYMNWY QQKPGQAPRL 180
WIYGISNLAS GIPARFSGSG SGTFSFLTIS SLQSEDAFVY YCQQRSSYPY TFGGGTKVEI 240
KR                                                    242

SEQ ID NO: 1031      moltype = AA length = 121
FEATURE             Location/Qualifiers
REGION              1..121
                    note = 8H5H5G4 Heavy chain variable region sequence
source              1..121
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1031
VQLQQSGPDL VKPGTSVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNNGVTNRY 60
QKFKGKATLT VDKSSSTAYM ELRSLTSEDS AVYYCARRY HSTYVFYFDS WGQGTLLTVS 120
S                                                    121

SEQ ID NO: 1032      moltype = AA length = 98
FEATURE             Location/Qualifiers
REGION              1..98
                    note = IGHV1-24*01 V-REGION sequence
source              1..98
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1032
QVQLVQSGAE VKKPGASVKV SCKVSGYTLT ELSMHWVRQA PGKGLEWMGG FDPEDGETIY 60
AQKFKGRVTM TEDTSTDYAY MELSSLRSED TAVYYCAT 98

SEQ ID NO: 1033      moltype = AA length = 15
FEATURE             Location/Qualifiers
REGION              1..15
                    note = IGHJ4*03 J-REGION sequence
source              1..15
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1033
YFDYWGGQTL VTVSS 15

SEQ ID NO: 1034      moltype = AA length = 122
FEATURE             Location/Qualifiers
REGION              1..122
                    note = Humanized 8H5H5G4 Heavy chain variable region
                    sequence
source              1..122
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1034
QVQLVQSGAE VKKPGASVKV SCKVSGNTFT EYTMHWVRQA PGKGLEWMGG FNPNGVTNY 60
NQKFKGRVTM TEDTSTDYAY MELSSLRSED TAVYYCARRY YHSTYVFYFD SWGQGTLLTV 120
SS                                                    122

SEQ ID NO: 1035      moltype = AA length = 122
FEATURE             Location/Qualifiers
REGION              1..122
                    note = Humanized 8H5H5G4 Heavy chain variable region
                    sequence (codon optimized)
source              1..122
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1035
QVQLVQSGAE VKKPGASVKV SCKVSGNTFT EYTMHWVRQA PGKGLEWMGG FNPNGVTNY 60
NQKFKGRVTM TEDTSTDYAY MELSSLRSED TAVYYCARRY YHSTYVFYFD SWGQGTLLTV 120
SS                                                    122

SEQ ID NO: 1036      moltype = AA length = 122
FEATURE             Location/Qualifiers
REGION              1..122
                    note = Modified Humanized 8H5H5G4 Heavy chain variable
                    region sequence
source              1..122
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1036

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QVQLVQSGAE VKKPGASVKV SCKVSGNTFT EYTMHWVRQA PGKGLEWIGG FNPNGVTNY 60
 NQKFKGKVTL TVDTSSSTAY MELSSLRSED TAVYYCARRY YHSTYVPYFD SWGQGLVTV 120
 SS 122

SEQ ID NO: 1037 moltype = AA length = 122
 FEATURE Location/Qualifiers
 REGION 1..122
 note = Modified Humanized 8H5H5G4 Heavy chain variable
 region sequence (codon optimized)
 source 1..122
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1037
 QVQLVQSGAE VKKPGASVKV SCKVSGNTFT EYTMHWVRQA PGKGLEWIGG FNPNGVTNY 60
 NQKFKGKVTL TVDTSSSTAY MELSSLRSED TAVYYCARRY YHSTYVPYFD SWGQGLVTV 120
 SS 122

SEQ ID NO: 1038 moltype = AA length = 108
 FEATURE Location/Qualifiers
 REGION 1..108
 note = 8H5H5G4 Light chain variable region sequence
 source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1038
 DIQMTQTSS LSASLGDRVT ISCSASQGIS NYLNWFQQKP DGTIKLLIYY TSSLHSGVPS 60
 RFGSGSGTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIKR 108

SEQ ID NO: 1039 moltype = AA length = 95
 FEATURE Location/Qualifiers
 REGION 1..95
 note = IGKV1-27*01 V-REGION sequence
 source 1..95
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1039
 DIQMTQSPSS LSASVGDVRT ITCRASQGIS NYLAWYQQKP GKVPKLLIYA ASTLQSGVPS 60
 RFGSGSGTD FTLTISSLQP EDVATYYCQK YNSAP 95

SEQ ID NO: 1040 moltype = AA length = 12
 FEATURE Location/Qualifiers
 REGION 1..12
 note = IGKJ4*02 J-REGION sequence
 source 1..12
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1040
 LTFGGGTKVE IK 12

SEQ ID NO: 1041 moltype = AA length = 108
 FEATURE Location/Qualifiers
 REGION 1..108
 note = humanized 8H5H5G4 Light chain variable region
 sequence
 source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1041
 DIQMTQSPSS LSASVGDVRT ITCRASQGIS NYLNWYQQKP GKVPKLLIYY TSSLHSGVPS 60
 RFGSGSGTD FTLTISSLQP EDVATYYCQQ YSKLPYTFGG GTKVEIKR 108

SEQ ID NO: 1042 moltype = AA length = 108
 FEATURE Location/Qualifiers
 REGION 1..108
 note = humanized 8H5H5G4 Light chain variable region
 sequence (codon optimized)
 source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1042
 DIQMTQSPSS LSASVGDVRT ITCRASQGIS NYLNWYQQKP GKVPKLLIYY TSSLHSGVPS 60
 RFGSGSGTD FTLTISSLQP EDVATYYCQQ YSKLPYTFGG GTKVEIKR 108

SEQ ID NO: 1043 moltype = AA length = 108
 FEATURE Location/Qualifiers
 REGION 1..108
 note = Modified humanized 8H5H5G4 Light chain variable
 region sequence
 source 1..108

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mol_type = protein
organism = synthetic construct

SEQUENCE: 1043
DIQMTQSPSS LSASVGDVRT ITCSASQGIS NYLWYQQKP GKVPKLLIYY TSSLHSGVPS 60
RFGSGSGSDT YTLTISSLQP EDVATYYCQQ YSKLPYTFGG GTKVEIKR 108

SEQ ID NO: 1044      moltype = AA length = 108
FEATURE             Location/Qualifiers
REGION              1..108
                    note = Modified humanized 8H5H5G4 Light chain variable
                    region sequence (codon optimized)
source              1..108
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1044
DIQMTQSPSS LSASVGDVRT ITCSASQGIS NYLWYQQKP GKVPKLLIYY TSSLHSGVPS 60
RFGSGSGSDT YTLTISSLQP EDVATYYCQQ YSKLPYTFGG GTKVEIKR 108

SEQ ID NO: 1045      moltype = AA length = 245
FEATURE             Location/Qualifiers
REGION              1..245
                    note = Modified humanized 8H5H5G4 scFV sequence (codon
                    optimized)
source              1..245
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1045
QVQLVQSGAE VKKPGASVKV SCKVSGNTFT EYTMHWVRQA PGKGLEWIGG FNPNGVTNY 60
NQKFKGKVTL TVDTSSSTAY MELSSLRSED TAVYYCARRY YHSTYVFYFD SWGQGTLLTV 120
SSGGGGSGGG GSGGGGSDIQ MTQSPSSLSA SVGDRVTITC SASQGISNYL NWYQQKPGKV 180
PKLLIYYTSS LHSVGPSPRF SSGSGTDYTL TISLQPEDV ATYYCQQYSK LPYTFGGGTK 240
VEIKR 245

SEQ ID NO: 1046      moltype = AA length = 330
FEATURE             Location/Qualifiers
REGION              1..330
                    note = Human IgG1 heavy chain constant region sequence:
                    (for making full antibody - pair with either kappa or
                    lambda constant region; 2 plasmids, express together)
source              1..330
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1046
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPELLGG 120
PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 180
STYRVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGGSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

SEQ ID NO: 1047      moltype = AA length = 326
FEATURE             Location/Qualifiers
REGION              1..326
                    note = Human IgG2 heavy chain constant region sequence:
                    (for making full antibody - pair with either kappa or
                    lambda constant region; 2 plasmids, express together)
source              1..326
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1047
ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 60
GLYSLSSVVT VPSSNFGTQT YTCNVDHKPS NTKVDKTVR KCCVECPPCP APPVAGPSVF 120
LPPPKPKDTL MISRTPEVTC VVVDVSHEDP EVQFNWYVDG VEVHNAKTKP REEQFNSTFR 180
VSVLTVVHQ DWLNGKEYKC KVSNGKGLPAP IEKTISKTKG QPREPQVYTL PPSREEMTKN 240
QVSLTCLVKG FYPSDIAVEW ESNQGPENNY KTTTPMLDSD GSFFLYSKLT VDKSRWQQGN 300
VFSCSVMEHA LHNHYTQKSL SLSPGK 326

SEQ ID NO: 1048      moltype = AA length = 107
FEATURE             Location/Qualifiers
REGION              1..107
                    note = Human Kappa light chain constant region sequence
source              1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1048
RTVAAPSVFI FPPSDEQLKS GTASVVCLLN NFYPREAKVQ WKVDNALQSG NSQESVTEQD 60
SKDSTYLSLS TLTLKADYE KHKVYACEVT HQGLSSPVTK SFNRGEC 107

SEQ ID NO: 1049      moltype = AA length = 106

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FEATURE Location/Qualifiers
 REGION 1..106
 note = Human Lambda light chain constant region sequence
 source 1..106
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1049
 GQPKAAPSVT LFPSPSEELQ ANKATLVCLI SDFYPGAVTV AWKADSSPVK AGVETTTPSK 60
 QSNKYAASS YLSLTPEQWK SHRSYSCQVT HEGSTVEKTV APTECS 106

SEQ ID NO: 1050 moltype = AA length = 232
 FEATURE Location/Qualifiers
 REGION 1..232
 note = Human IgG1 Fc region sequence: (to be fused to scFv
 for homo-dimerizes)
 source 1..232
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1050
 EPKSCDKTHT CPPCPAPELL GGPSVFLFPP KPKDTLMISR TPEVTCVVVD VSHEDPEVKF 60
 NWYVDGVEVH NAKTKPREEQ YNSTYRVVSV LTVLHQDWLN GKEYKCKVSN KALPAPIEKT 120
 ISKAKGQPRE PQVYTLPPSR EEMTKNQVSL TCLVKGFYPS DIAVEWESNG QPENNYKTP 180
 PVLDSGSGSF LYSLKTVDKS RWQQGNVFSC SVMHEALHNNH YTQKSLSLSP GK 232

SEQ ID NO: 1051 moltype = AA length = 228
 FEATURE Location/Qualifiers
 REGION 1..228
 note = Human IgG2 Fc region sequence
 source 1..228
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1051
 ERKCCVECP CPAPPVAGPS VFLEPPKPKD TLMISRTPEV TCVVVDVSHE DPEVQFNWYV 60
 DGVEVHNAKT KPREEQFNST FRVSVLTVV HQDWLNGKEY KCKVSNKGLP APIEKTISK 120
 KGQPREPQVY TLPPSREEMT KNQVSLTCLV KGFYPSDIAV EWESNGQPEN NYKTPPMLD 180
 SDGSFPLYSK LTVDKSRWQQ GNVFSCSVMH EALHNHYTQK SLSLSPGK 228

SEQ ID NO: 1052 moltype = length =
 SEQUENCE: 1052
 000

SEQ ID NO: 1053 moltype = length =
 SEQUENCE: 1053
 000

SEQ ID NO: 1054 moltype = length =
 SEQUENCE: 1054
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SEQ ID NO: 1055 moltype = length =
 SEQUENCE: 1055
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SEQ ID NO: 1056 moltype = length =
 SEQUENCE: 1056
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SEQ ID NO: 1057 moltype = length =
 SEQUENCE: 1057
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SEQ ID NO: 1058 moltype = length =
 SEQUENCE: 1058
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SEQ ID NO: 1059 moltype = length =
 SEQUENCE: 1059
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SEQ ID NO: 1060 moltype = length =
 SEQUENCE: 1060
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SEQ ID NO: 1061 moltype = length =
 SEQUENCE: 1061
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SEQ ID NO: 1062 moltype = length =
 SEQUENCE: 1062

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SEQ ID NO: 1063 moltype = length =
SEQUENCE: 1063
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SEQ ID NO: 1064 moltype = length =
SEQUENCE: 1064
000

SEQ ID NO: 1065 moltype = length =
SEQUENCE: 1065
000

SEQ ID NO: 1066 moltype = length =
SEQUENCE: 1066
000

SEQ ID NO: 1067 moltype = length =
SEQUENCE: 1067
000

SEQ ID NO: 1068 moltype = length =
SEQUENCE: 1068
000

SEQ ID NO: 1069 moltype = length =
SEQUENCE: 1069
000

SEQ ID NO: 1070 moltype = length =
SEQUENCE: 1070
000

SEQ ID NO: 1071 moltype = length =
SEQUENCE: 1071
000

SEQ ID NO: 1072 moltype = length =
SEQUENCE: 1072
000

SEQ ID NO: 1073 moltype = length =
SEQUENCE: 1073
000

SEQ ID NO: 1074 moltype = length =
SEQUENCE: 1074
000

SEQ ID NO: 1075 moltype = length =
SEQUENCE: 1075
000

SEQ ID NO: 1076 moltype = length =
SEQUENCE: 1076
000

SEQ ID NO: 1077 moltype = length =
SEQUENCE: 1077
000

SEQ ID NO: 1078 moltype = length =
SEQUENCE: 1078
000

SEQ ID NO: 1079 moltype = length =
SEQUENCE: 1079
000

SEQ ID NO: 1080 moltype = length =
SEQUENCE: 1080
000

SEQ ID NO: 1081 moltype = length =
SEQUENCE: 1081
000

SEQ ID NO: 1082 moltype = length =

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SEQUENCE: 1082
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SEQ ID NO: 1083	moltype =	length =
SEQUENCE: 1083		
000		

SEQ ID NO: 1084	moltype =	length =
SEQUENCE: 1084		
000		

SEQ ID NO: 1085	moltype =	length =
SEQUENCE: 1085		
000		

SEQ ID NO: 1086	moltype =	length =
SEQUENCE: 1086		
000		

SEQ ID NO: 1087	moltype =	length =
SEQUENCE: 1087		
000		

SEQ ID NO: 1088	moltype =	length =
SEQUENCE: 1088		
000		

SEQ ID NO: 1089	moltype =	length =
SEQUENCE: 1089		
000		

SEQ ID NO: 1090	moltype =	length =
SEQUENCE: 1090		
000		

SEQ ID NO: 1091	moltype =	length =
SEQUENCE: 1091		
000		

SEQ ID NO: 1092	moltype =	length =
SEQUENCE: 1092		
000		

SEQ ID NO: 1093	moltype =	length =
SEQUENCE: 1093		
000		

SEQ ID NO: 1094	moltype =	length =
SEQUENCE: 1094		
000		

SEQ ID NO: 1095	moltype =	length =
SEQUENCE: 1095		
000		

SEQ ID NO: 1096	moltype =	length =
SEQUENCE: 1096		
000		

SEQ ID NO: 1097	moltype =	length =
SEQUENCE: 1097		
000		

SEQ ID NO: 1098	moltype =	length =
SEQUENCE: 1098		
000		

SEQ ID NO: 1099	moltype =	length =
SEQUENCE: 1099		
000		

SEQ ID NO: 1100	moltype =	length =
SEQUENCE: 1100		
000		

SEQ ID NO: 1101	moltype = DNA	length = 366
FEATURE	Location/Qualifiers	
misc_feature	1..366	
	note = Humanized 8F9A4A3 H-1.46 heavy chain variable domain	

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```

sequence
source      1..366
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 1101
caagtgcagc tgggtgcagag cggcgccgag gtgaagaaac ctggcgccag cgtgaaagt 60
tcctgcaagg ccagcggcaa tacattcacc gactacacaa tgcactgggt cagacaggcc 120
cccgccagg gcctggaatg gatcgccgga tttaacccca acaacggcgt gacaaactac 180
aaccagaagt tcaagggcaa ggtgaccatc acaagagaca ccagcagcag caccgtgtac 240
atggaactgt cttctctgcg gagcgaggat accgcccgtg actattgtgc cagacggtag 300
taccacagcc tgtacgtgtt ctacttcgac tactggggac agggcacccct ggttaccgtg 360
tcctct      366

SEQ ID NO: 1102      moltype = AA length = 122
FEATURE              Location/Qualifiers
REGION               1..122
                    note = Humanized 8F9A4A3 H-1.46 heavy chain variable domain
                    sequence
source      1..122
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 1102
QVQLVQSGAE VKKPGASVKV SCKASGNTFT EYTMHWVRQA PGQGLEWIGG FNPNGVTNY 60
NQKFKGKVTI TRDTSSSTVY MELSSLRSED TAVYYCARRY YHSLYVPYFD YWQGQTLVTV 120
SS      122

SEQ ID NO: 1103      moltype = DNA length = 321
FEATURE              Location/Qualifiers
misc_feature         1..321
                    note = Humanized 8F9A4A3 L-1.6 light chain variable domain
                    sequence
source      1..321
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 1103
gatatccaga tgacacagag ccctagctcc ctgagcgcca gcgtgggcca ccgggtcacc 60
attacatgca gcgcttctca gggcatctcc aactacctga actggtacca gcagaaaccc 120
ggcaaggccc ctaagctgct gatcttctac accagctctc tgcacagcgg cgtgccatct 180
agattcagcg gatctggcag cggcacccgac tacaccctga ccatcagctc cctccagcct 240
gaggacttcg ccacctaata ctgtcagcaa tacagcaagc tgccttatac ctttggcggc 300
ggaacaaagg tggaaatcaa g      321

SEQ ID NO: 1104      moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION               1..107
                    note = Humanized 8F9A4A3 L-1.6 light chain variable domain
                    sequence
source      1..107
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 1104
DIQMTQSPSS LSASVGDVRT ITCSASQGIS NYLNWYQKP GKAPKLLIFY TSSLHSGVPS 60
RFSGSGSGTD YTLTISSLQP EDFATYYCQQ YSKLPYTFGG GTKVEIK      107

SEQ ID NO: 1105      moltype = DNA length = 366
FEATURE              Location/Qualifiers
misc_feature         1..366
                    note = Humanized 8F9A4A3 H-4.4 heavy chain variable domain
                    sequence
source      1..366
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 1105
caagtgcagc tgcaggagag cggacctggc ctggttaagc ctggaggcac cctgtctctg 60
acatgtgctg tgtctggcaa tacctttacc gactacacca tgcactgggt gcggcagcct 120
ccaggcaagg gcctggaatg gatcgccggc ttcaacccca acaacggcgt gacaaattac 180
aaccagaaat tcaagggaaa agtgaccatc accgtggata agtccaagaa caccttcagc 240
ctcaagctga gcagcgtgac agccgcgcag accgcccgtg actactgcgc cagaagatac 300
tatcacagcc tgtacgtgtt ctacttcgac tactggggcc agggcacact ggtcaccgtg 360
tccagc      366

SEQ ID NO: 1106      moltype = AA length = 122
FEATURE              Location/Qualifiers
REGION               1..122
                    note = Humanized 8F9A4A3 H-4.4 heavy chain variable domain
                    sequence
source      1..122
            mol_type = protein
            organism = synthetic construct

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SEQUENCE: 1106
 QVQLQESGPG LVKPGGTLST TCAVSGNTFT EYTMHWVRQP PGKGLEWIGG FNPNGVTNY 60
 NQKFKGKVTI TVDKSKNTFS LKLSSVTAAD TAVYYCARRY YHSLYVPYFD YWGQGLVTV 120
 SS 122

SEQ ID NO: 1107 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Humanized 8F9A4A3 L-4.1 light chain variable domain
 sequence
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 1107
 gatatacgtga tgacccagag cccagacagc ctggcagtgta gtctgggtga gcgtgctaca 60
 atcaactgca ggcagcagcca gggcatctcc aactacatga attggatatca gcagaaacct 120
 ggccaggctc ctaagctgct gatcttctac accagcagcc tgcacagcgg cgtgccagat 180
 agattcagcg gcagcggatc tggcaccgac tacacactga ccatttcttc tctccaggcc 240
 gaggacgtgg ccgtctacta ctgtcagcaa tacagcaagc tgccttacac ctttggcgga 300
 ggcacaaagg tggaaatcaa g 321

SEQ ID NO: 1108 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Humanized 8F9A4A3 L-4.1 light chain variable domain
 sequence
 source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1108
 DIVMTQSPDS LAVSLGERAT INCSASQGIS NYLNWYQQKP GQAPKLLIFY TSSLHSGVPD 60
 RFGSGSGGTD YTLTISSLQA EDVAVYYCQQ YSKLPYTFGG GTKVEIK 107

SEQ ID NO: 1109 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = 8F9A4A3L
 source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1109
 DIQMTQTTSS LSASLGDRVT LSCSASQGIS NYLNWYQQKP DGTVELLIFY TSSLHSGVPS 60
 RFGSGSGGTD YSLTISNLEP EDIATYYCQQ YSKLPYTFGG GTKLEIK 107

SEQ ID NO: 1110 moltype = AA length = 187
 FEATURE Location/Qualifiers
 REGION 1..187
 note = NME4
 source 1..187
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1110
 MGGLFWRSAL RGLRCGPAP GPSLLVRHGS GGPSWTRERT LVAVKPDGVQ RRLVGDVIQR 60
 FERRGFTLVG MKMLQAPESV LAEHYQDLRR KPFYPALIRY MSSGPVVAMV WEGYNVVRAS 120
 RAMIGHTDSA EAAPGTIRGD FSVHISRNI HASDSVEGAQ REIQLWFQSS ELVSWADGGQ 180
 HSSIHPA 187

SEQ ID NO: 1111 moltype = AA length = 254
 FEATURE Location/Qualifiers
 REGION 1..254
 note = NME5
 source 1..254
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1111
 MEISMPPPIQ YVEKTLAIK PDIVDKEEEI QDIILRSGFT IVQRRKLRLS PEQCSNFYVE 60
 KYGKMFFPNL TAYMSSGPLV AMILARHKAI SYWLELLGPN NSLVAKETHP DSLRAIYGTD 120
 DLRNALHGSN DFAAAEREIR FMFPEVIVEP IPIGQAADY LNLHIMPTLL EGLTELCKQK 180
 PADPLFWYMC CRREHWTLRS ILLVCMGIR MSLPHCADYC SFVEGFELWL ADWLLKNNPN 240
 KPKLCHHPIV EEPY 254

SEQ ID NO: 1112 moltype = AA length = 588
 FEATURE Location/Qualifiers
 REGION 1..588
 note = NME8
 source 1..588
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1112

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MASKKREVQL QTVINNQSLW DEMLQNKGLT VIDVYQAWCG PCRAMQPLFR KLKNELNEDE 60
ILHFAVAEAD NIVTLQPPRD KCEPVFLFSV NGKIIEKIQQ ANAPLVNKKV INLIDEERKI 120
AAGEMARPQY PEIPLVSDS EVSEESPCES VQELYSIAII KPDAVISKV LEIKRKITKA 180
GFIIIEAHEKT VLTEEQVNF YSRIADQCDF EEFVSFMTSG LSYILVVSQG SKHNPSEET 240
EPQTDTEPNE RSEDOPEVEA QVTPGMMKNK QDSLQEYLER QHLAQLCDIE EDAANVAKFM 300
DAFFPDFKKM KSMKLEKTLA LLRPNLFHER KDDVLRRIKD EDFKILEQRQ VVLSEKEAQA 360
LCKEYENEDY FNKLIENMTS GPSLALVLLR DNGLQYWKQL LGPRTVEEAI EYFPESLCAQ 420
FAMDSLVPVQ LYGSDSLETA EREIQHFFPL QSTLGLIKPH ATSEQREQIL KIVKEAGFDL 480
TQVKKMFLTP EQIEKIYPKV TGKDFYKDLL EMLSVGPSMV MILTKWNAVA EWRRLMGPTD 540
PEEAKLLSPD SIRAQFGISK LKNIVHGASN AYEAEVVRN LFEDPEEN 588

```

```

SEQ ID NO: 1113      moltype = DNA length = 78
FEATURE
misc_feature        1..78
                    note = Mouse 8F9A4A3 heavy chain variable domain framework 1
source              1..78
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 1113
gaggtccagc tgcaacagtc tggacctgaa ctggtgaagc ctggggcttc agtgaagata 60
tcctgcaaga cttctgga                                     78

```

```

SEQ ID NO: 1114      moltype = AA length = 26
FEATURE
REGION              1..26
                    note = Mouse 8F9A4A3 heavy chain variable domain framework 1
source              1..26
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 1114
EVQLQQSGPE LVKPGASVKI SCKTSG                                     26

```

```

SEQ ID NO: 1115      moltype = DNA length = 27
FEATURE
misc_feature        1..27
                    note = Heavy chain variable region CDR1
source              1..27
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 1115
aacacattca ctgaatacac catgcac                                     27

```

```

SEQ ID NO: 1116      moltype = DNA length = 42
FEATURE
misc_feature        1..42
                    note = Mouse 8F9A4A3 heavy chain variable domain framework 2
source              1..42
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 1116
tgggtgaagc agagccatgg aaagagcctt gagtggattg ga                                     42

```

```

SEQ ID NO: 1117      moltype = AA length = 14
FEATURE
REGION              1..14
                    note = Mouse 8F9A4A3 heavy chain variable domain framework 2
source              1..14
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 1117
WVKQSHGKSL EWIG                                     14

```

```

SEQ ID NO: 1118      moltype = DNA length = 51
FEATURE
misc_feature        1..51
                    note = Mouse 8F9A4A3 Heavy chain variable region CDR2
source              1..51
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 1118
ggttttaatc ctaacaatgg tgttactaac tacaaccaga agttcaaggg c                                     51

```

```

SEQ ID NO: 1119      moltype = DNA length = 96
FEATURE
misc_feature        1..96
                    note = Mouse 4A3 heavy chain variable domain framework 3
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct

```

-continued

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SEQUENCE: 1119
aaggccacat tgactgtaga caagtcctcc agcacagcct acatggagct ccgcagcctg 60
acatctgagg attctgcagt ctattactgt gcaaga 96

SEQ ID NO: 1120      moltype = AA  length = 32
FEATURE             Location/Qualifiers
REGION              1..32
                    note = Mouse 4A3 heavy chain variable domain framework 3
source              1..32
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1120
KATLTVDKSS STAYMELRSL TSEDSAVYYC AR 32

SEQ ID NO: 1121      moltype = DNA  length = 39
FEATURE             Location/Qualifiers
misc_feature        1..39
                    note = Mouse 8F9A4A3 Heavy chain variable region CDR3
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1121
cgttactacc atagtctcta cgtgttttac ttgactac 39

SEQ ID NO: 1122      moltype = DNA  length = 69
FEATURE             Location/Qualifiers
misc_feature        1..69
                    note = Mouse 8F9A4A3 light chain variable domain framework 1
source              1..69
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1122
gatatccaga tgacacagac tacatcctcc ctgtctgcct ctctgggaga cagagtcacc 60
ctcagttgc 69

SEQ ID NO: 1123      moltype = AA  length = 23
FEATURE             Location/Qualifiers
REGION              1..23
                    note = Mouse 8F9A4A3 light chain variable domain framework 1
source              1..23
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1123
DIQMTQTSS LSASLGDRVT LSC 23

SEQ ID NO: 1124      moltype = DNA  length = 33
FEATURE             Location/Qualifiers
misc_feature        1..33
                    note = Mouse 8F9A4A3 Light chain variable region CDR 1
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1124
agtgaagtc agggcattag caattattta aac 33

SEQ ID NO: 1125      moltype = DNA  length = 45
FEATURE             Location/Qualifiers
misc_feature        1..45
                    note = Mouse 4A3 light chain variable domain framework 2
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1125
tggtatcagc agaaaccaga tggaactgtt gaactcctga tcctt 45

SEQ ID NO: 1126      moltype = AA  length = 15
FEATURE             Location/Qualifiers
REGION              1..15
                    note = Mouse 4A3 light chain variable domain framework 2
source              1..15
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1126
WYQQKPDGTV ELLIF 15

SEQ ID NO: 1127      moltype = DNA  length = 21
FEATURE             Location/Qualifiers
misc_feature        1..21
                    note = Mouse 8F9A4A3 Light chain variable region CDR2

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source                1..21
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 1127
tacacatcaa gtttacactc a                                21

SEQ ID NO: 1128      moltype = DNA length = 96
FEATURE              Location/Qualifiers
misc_feature          1..96
                      note = Mouse 8F9A4A3 light chain variable domain framework 3
source                1..96
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 1128
ggagtcctcat caagggttcag tggcagtggtg tctgggacag attattctct caccatcagc 60
aacctggaac ctgaagatat tgccacttac tattgt                                96

SEQ ID NO: 1129      moltype = AA length = 32
FEATURE              Location/Qualifiers
REGION                1..32
                      note = Mouse 8F9A4A3 light chain variable domain framework 3
source                1..32
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 1129
GVPSRFSGSG SGTDSLITIS NLEPEDIATY YC                                32

SEQ ID NO: 1130      moltype = DNA length = 27
FEATURE              Location/Qualifiers
misc_feature          1..27
                      note = Mouse 8F9A4A3 Light chain variable region CDR3
source                1..27
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 1130
cagcaggtata gtaagcttcc ttacacg                                27

SEQ ID NO: 1131      moltype = DNA length = 366
FEATURE              Location/Qualifiers
misc_feature          1..366
                      note = Humanized 8F9A4A3 H-ori heavy chain variable domain
                      sequence
source                1..366
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 1131
caggttcagc tgggttcagtc tgggtgcagaa gtgaagaaac ctggcgccctc tgtgaaggtg 60
tctgtcaaag tgctcggaaa taccttcacc gaggtaacca tgcactgggt cgcgacggcc 120
cctggcaaaag gacttgaatg gatgggcggc ttcaacccca acaacggcgt gaccaactac 180
aaccagaaat tcaagggcgg cgtgaccatg accgaggaca caagcacaga caccgcctac 240
atggaactga gcagcctgag aagcgaggac accgcccgtgt actactgcgc cagaagggtac 300
taccacagcc tgtacgtgtt ctacttcgac tactggggcc agggcacccct ggtcacagtt 360
tcttct                                366

SEQ ID NO: 1132      moltype = DNA length = 366
FEATURE              Location/Qualifiers
misc_feature          1..366
                      note = Humanized 8F9A4A3 H-3.15 heavy chain variable domain
                      sequence
source                1..366
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 1132
gaggtgcagc tgggtgaaaag cggcggcggc ctgggttaagc ctggcggtac tctgagactg 60
agctgtgccc cttctggcaa taccttcacc gaggtaacca tgcactgggt ggcgacggcc 120
cctggaaaag gcctggaatg gatcggcgga tttaacccca acaacggcgt gacaaattac 180
aaccagaaat tcaagggcaa gtccaccatc acaagagata agagcaagaa caccctgtac 240
ctgcaaatga acagcctgaa gtccgaggac accgcccgtgt actactgcgc cagacgggtac 300
taccacagcc tctatgtgtt ctacttcgac tactggggcc agggcacact ggtcaccgtg 360
tccagc                                366

SEQ ID NO: 1133      moltype = AA length = 122
FEATURE              Location/Qualifiers
REGION                1..122
                      note = Humanized 8F9A4A3 H-3.15 heavy chain variable domain
                      sequence
source                1..122
                      mol_type = protein
                      organism = synthetic construct

```


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```

mol_type = protein
organism = synthetic construct

SEQUENCE: 1139
EVQLQQSGPE LVKPGASVKI SCKTSGNTFT EYTMHWVKQS HGKSLEWIGG FNPNGVTNY 60
NQKFKGKATL TVDKSSSTAY MELRSLTSED SAVYYCARRY YHSLYVFYFD YWGQGTTLTV 120
SS 122

SEQ ID NO: 1140      moltype = AA length = 131
FEATURE             Location/Qualifiers
source              1..131
                    mol_type = protein
                    organism = Homo sapiens

SEQUENCE: 1140
EKTALAIKPD AISKAGEIIE IINKAGFTIT KLKMMMLSRK EALDFHVDHQ SRPFFNELIQ 60
FITTGPIIAM EILRDDAICE WKRLLGPNAS GVARTDASES IRALFGTDGI RNAAHGPDSE 120
ASAAREMELF F 131

SEQ ID NO: 1141      moltype = AA length = 376
FEATURE             Location/Qualifiers
source              1..376
                    mol_type = protein
                    organism = Homo sapiens

SEQUENCE: 1141
MNHSERFVFI AEWYDPNASL LRRYELLFYP GDGSVEMHDV KNRHFLKRT KYDNLHLEDL 60
FIGNKVNVFS RQLVLIDYGD QYTARQLGSR KEKTLAIKPD DAISKAGEII EIINKAGFTI 120
TKLKMMLSRK KEALDFHVDH QSRPFFNELI QFITTGPIIA MEILRDDAIC EWKRLLGPNAS 180
SGVARTDASE SIRALFGTDG IRNAAHGPDSE FASAAREMEL FFPSSGGCGP ANTAKFTNCT 240
CCIVKPHAVS EGLLGKILMA IRDAGFEISA MQMFNMDRVN VEEFYEVYKG VVTEYHDMVT 300
EMYSGPCVAM EIQQNNATKT FREFCGPADP EIAHRLRPQT LRAIFGKTKI QNAVHCTDLP 360
EDGLEEVQYF FKILDN 376

SEQ ID NO: 1142      moltype = AA length = 30
FEATURE             Location/Qualifiers
REGION             1..30
                    note = B3 region of NME7
source             1..30
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1142
AIFGKTKIQN AVHCTDLPED GLLEVQYFFC 30

SEQ ID NO: 1143      moltype = AA length = 139
FEATURE             Location/Qualifiers
source              1..139
                    mol_type = protein
                    organism = Homo sapiens

SEQUENCE: 1143
NCTCCIVKPH AVSEGLLGKI LMAIRDAGFE ISAMQMFNMD RVNVEEFYEV YKGVVTEYHD 60
MVTEMYSGPC VAMEIQNNA TKTFREFCGP ADPEIARHLR PGTLRRAIFGK TKIQNAVHCT 120
DLPEDGLLEV QYFFKILDN 139

SEQ ID NO: 1144      moltype = AA length = 11
FEATURE             Location/Qualifiers
REGION             1..11
                    note = Light Chain CDR1 region
source             1..11
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1144
ITSTDIDDM N 11

SEQ ID NO: 1145      moltype = AA length = 7
FEATURE             Location/Qualifiers
REGION             1..7
                    note = Light Chain CDR2 region
source             1..7
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1145
EGNTLRP 7

SEQ ID NO: 1146      moltype = AA length = 9
FEATURE             Location/Qualifiers
REGION             1..9
                    note = Light Chain CDR3 region
source             1..9
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1146

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-continued

LQSDNLPLT 9

SEQ ID NO: 1147 moltype = AA length = 23
 FEATURE Location/Qualifiers
 REGION 1..23
 note = Mouse 5F3A5D4-2 light chain variable domain
 framework 1
 source 1..23
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1147
 DIQMTQTTSS LSASLGDRVT ISC 23

SEQ ID NO: 1148 moltype = DNA length = 45
 FEATURE Location/Qualifiers
 misc_feature 1..45
 note = Mouse 5F3A5D4-2 light chain variable domain
 framework 2
 source 1..45
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 1148
 tggtttcagc agaaaccaga tgggaactatt aagctcctga tctat 45

SEQ ID NO: 1149 moltype = AA length = 15
 FEATURE Location/Qualifiers
 REGION 1..15
 note = Mouse 5F3A5D4-2 light chain variable domain
 framework 2
 source 1..15
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1149
 WFQQKPDGTI KLLIY 15

SEQ ID NO: 1150 moltype = DNA length = 21
 FEATURE Location/Qualifiers
 misc_feature 1..21
 note = Mouse 5F3A5D4-2 light chain variable domain
 Complementarity -determining region 2
 source 1..21
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 1150
 tacacatcaa gtttacattc a 21

SEQ ID NO: 1151 moltype = DNA length = 96
 FEATURE Location/Qualifiers
 misc_feature 1..96
 note = Mouse 5F3A5D4-2 light chain variable domain
 framework 3
 source 1..96
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 1151
 ggagtcctcat caagggttcag tggcagtggtg tctgggacag attattctct caccatcagt 60
 aatgtggaac ctgaagatat tgccacttac tattgt 96

SEQ ID NO: 1152 moltype = AA length = 32
 FEATURE Location/Qualifiers
 REGION 1..32
 note = Mouse 5F3A5D4-2 light chain variable domain
 framework 3
 source 1..32
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1152
 GVPSRFSGSG SGTDSLITIS NVEPEDIATY YC 32

SEQ ID NO: 1153 moltype = DNA length = 30
 FEATURE Location/Qualifiers
 misc_feature 1..30
 note = Mouse 5F3A5D4-2 light chain variable domain
 framework 4
 source 1..30
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 1153
 ttcggagggg ggaccaagct ggagataaaa 30

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SEQ ID NO: 1154      moltype = AA  length = 10
FEATURE             Location/Qualifiers
REGION              1..10
                    note = Mouse 5F3A5D4-2 light chain variable domain
                    framework 4
source              1..10
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1154
FGGGTKLEIK                                     10

SEQ ID NO: 1155      moltype = DNA  length = 366
FEATURE             Location/Qualifiers
misc_feature         1..366
                    note = Mouse 5F3A5D4-V2 heavy chain variable domain sequence
source              1..366
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1155
gagggtccagc tgcaacagtc tggacctgat ctggtgaagc ctgggacttc agtgaagata 60
tcctgtgaaga cttctggaaa cacattcact gaatacacca tgcactgggt gaagcagagc 120
catggaaaga gccttgagtg gattggaggt tttaatccta acaatgggtg tactaactac 180
aaccagaagt tcaagggcaa ggccacattg actgtagaca agtcctccag cacagcctac 240
atggagctcc gcagcctgac atctgaggat tctgcagtct attactgtgc aagacgttac 300
taccatagta cctacgtgtt ctactttgac tectggggcc aaggcaccac tctcacagtc 360
tcctca                                           366

SEQ ID NO: 1156      moltype = DNA  length = 69
FEATURE             Location/Qualifiers
misc_feature         1..69
                    note = Mouse 5F3A5D4-3 light chain variable domain
                    framework 1
source              1..69
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1156
gatatccaga tgacacagac tacatcctcc ctgtctgcct ctctgggaga cagagtcacc 60
ctcagttgc                                           69

SEQ ID NO: 1157      moltype = DNA  length = 30
FEATURE             Location/Qualifiers
misc_feature         1..30
                    note = Mouse 5F3A5D4-3 light chain variable domain
                    framework 4
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1157
ttcggagggg ggaccaagct ggaataaaaa                                     30

SEQ ID NO: 1158      moltype = DNA  length = 366
FEATURE             Location/Qualifiers
misc_feature         1..366
                    note = Humanized 5F3A5D4-V2 H-1.27 heavy chain variable
                    domain sequence
source              1..366
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1158
caagtgcagc tgggtccagag cggcgccgag gtgaaaaagc ctggcgccag cgtgaagggtg 60
tcctgcaagg tgtctggcaa taccttcacc gaggtaacca tgcactgggt gcggcaggcc 120
cctggaaaag gcctgggaatg gatcgcgga ttttaaccca acaacggcgt gaccaactac 180
aaccagaagt tcaagggcaa ggttacactg accgtggaca ccagctcttc taccgcctac 240
atggaaactg gcagcctgag aagcagagat acagccgtgt actattgtgc cagaagatac 300
taccacagca cctacgtgtt ctacttcgac agctggggcc agggcacact ggtgacagtg 360
tccagc                                           366

SEQ ID NO: 1159      moltype = DNA  length = 366
FEATURE             Location/Qualifiers
misc_feature         1..366
                    note = Humanized 5F3A5D4-V2 H-1.46 heavy chain variable
                    domain sequence
source              1..366
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1159
caagtgcagc tgggtccagag cggagccgag gtgaaaaagc ccggcgcttc tgtgaagggtg 60
tcttgtaaag ccagcggcaa caccttcacc gaggtaacca tgcactgggt gcggcaggcc 120

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cctggccagg gcctggaatg gatcgggcgc tttaatccta acaacggcgt gacaaactac 180
aaccagaagt tcaagggcaa ggttacaatc accagagata ccagcagctc taccgtgtac 240
atggaactga gcagcctgag aagcgaggac accgccgtgt attactgcgc cagacggtac 300
taccacagca cctacgtgtt ctacttcgac agctggggcc agggaaact ggtcacagtg 360
tcctcc 366

SEQ ID NO: 1160      moltype = AA  length = 122
FEATURE             Location/Qualifiers
REGION              1..122
                    note = Humanized 5F3A5D4-V2 H-1.46 heavy chain variable
                    domain sequence
source              1..122
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1160
QVQLVQSGAE VKKPGASVKV SCKASGNTFT EYTMHWVRQA PGQGLEWIGG FNPNGVTNY 60
NQKFKGKVTI TRDTSSTVY MELSSLRSED TAVYYCARRY YHSTYVFYFD SWGQGLTVTV 120
SS 122

SEQ ID NO: 1161      moltype = DNA  length = 366
FEATURE             Location/Qualifiers
misc_feature        1..366
                    note = Humanized 5F3A5D4-V2 H-3.15 heavy chain variable
                    domain sequence
source              1..366
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1161
gaggtgcagc tgggtgaaag cggaggaggc ctggttaagc ctggaggcag cctgagactg 60
agctgtgccg cttctggcaa taccttcacc gactacacca tgcactgggt gcggcaggcc 120
cctggcaaaag gcctggaatg gatcgggcgc ttcaacccca acaacggcgt gacaaattac 180
aaccagaaat tcaagggcaa gtttacaatc accagagata agtctaagaa cacactctat 240
ctgcaaatga acagcctgaa gtccgaggac accgccgtgt actactgcgc cagacggtac 300
taccacagca catacgtgtt ctacttcgac agctggggcc agggcaccct ggtcacctg 360
tccagc 366

SEQ ID NO: 1162      moltype = AA  length = 122
FEATURE             Location/Qualifiers
REGION              1..122
                    note = Humanized 5F3A5D4-V2 H-3.15 heavy chain variable
                    domain sequence
source              1..122
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1162
EVQLVESGGG LVKPGGSLRL SCAASGNTFT EYTMHWVRQA PGKGLEWIGG FNPNGVTNY 60
NQKFKGKFTI TRDKSKNTLY LQMSLKSED TAVYYCARRY YHSTYVFYFD SWGQGLTVTV 120
SS 122

SEQ ID NO: 1163      moltype = DNA  length = 366
FEATURE             Location/Qualifiers
misc_feature        1..366
                    note = Humanized 5F3A5D4-V2 H-4.4 heavy chain variable
                    domain sequence
source              1..366
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1163
caagtgcagc tgcaggagag cggacctggc ctggtaaacg ctggcggcac cctgagcctc 60
acctgtgctg tttctggcaa taccttcacc gactacacca tgcactgggt gcggcagcct 120
ccaggcaaaag gcctggaatg gatcggcgga tttaacccca acaacggcgt gacaaattac 180
aaccagaaat tcaagggcaa ggtgaccatc acagtggata agtccaagaa caccttcagc 240
ctgaagctgt ctacgctgac agccgccgac accgccgtgt actactgcgc cagaagatac 300
tatcacagca cctacgtgtt ctacttcgac agctggggac agggcacact ggtgacagt 360
tccagc 366

SEQ ID NO: 1164      moltype = AA  length = 122
FEATURE             Location/Qualifiers
REGION              1..122
                    note = Humanized 5F3A5D4-V2 H-4.4 heavy chain variable
                    domain sequence
source              1..122
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1164
QVQLQESGPG LVKPGGTLST TCAVSGNTFT EYTMHWVRQP PGKGLEWIGG FNPNGVTNY 60
NQKFKGKVTI TVDKSKNTFS LKLSSVTAAD TAVYYCARRY YHSTYVFYFD SWGQGLTVTV 120
SS 122

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SEQ ID NO: 1165 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Humanized 5F3A5D4-2 L-1.6 light chain variable
 domain sequence
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 1165
 gatatccaga tgacacagag ccctagctcc ctgagcgcca gcgtgggcga ccgggtcacc 60
 attacatgca gcgcttctca gggcatctcc aactacctga actgggttca gcagaaaccc 120
 ggcaaggccc ctaagctgct gatctattac accagctctc tgcacagcgg cgtgccatct 180
 agattcagcg gatctggcag cggcaccgac tacaccctga ccacagctc cctccagcct 240
 gaggacttcg ccacctaacta ctgtcagcaa tacagcaagc tgccttatac ctttggcggc 300
 ggaacaaagg tggaaatcaa g 321

SEQ ID NO: 1166 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Humanized 5F3A5D4-2 L-1.6 light chain variable
 domain sequence
 source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1166
 DIQMTQSPSS LSASVGDVRT ITCSASQGIS NYLWVQQKP GKAPKLLIYY TSSLHSGVPS 60
 RFGSGSGSTD YTLTISSLQP EDFATYYCQQ YSKLPYTFGG GTKVEIK 107

SEQ ID NO: 1167 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Humanized 5F3A5D4-2 L-3.15 light chain variable
 domain sequence
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 1167
 gagatcgtga tgacccagag cccagctaca cttagtgtga gtccagggtga acgggctacc 60
 ctgtcctgca gcgccagcca gggcatcagc aactacctga actgggttca gcagaaacct 120
 ggccaggccc ctagactgct gatctattac accagcagcc tgcacagcgg catccccgcc 180
 agattcagcg gcagcggctc ttggaacagac tacaccctga caatctctag cctgcagtct 240
 gaagattttg ccgtctacta ctgtcagcaa tacagcaagc tgccttatac ctttggcggc 300
 ggaaccaagg tggaaattaa g 321

SEQ ID NO: 1168 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Humanized 5F3A5D4-2 L-3.15 light chain variable
 domain sequence
 source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1168
 EIVMTQSPAT LSVSPGERAT LSCSASQGIS NYLWVQQKP GQAPRLLIYY TSSLHSGIPA 60
 RFGSGSGSTD YTLTISSLQS EDFAVYYCQQ YSKLPYTFGG GTKVEIK 107

SEQ ID NO: 1169 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Humanized 5F3A5D4-2 L-4.1 light chain variable
 domain sequence
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 1169
 gatatcgtga tgacccagag cccagacagc ctggcagtga gtctgggtga gcgtgtctaca 60
 atcaactgca gcgccagcca gggcatctcc aactacctga attgggttca gcagaaacct 120
 ggccaggctc ctaagctgct gatctattac accagcagcc tgcacagcgg cgtgccagat 180
 agattcagcg gcagcggatc ttggcaccgac tacacactga ccattttctc tctccaggcc 240
 gaggacgtgg ccgtctacta ctgtcagcaa tacagcaagc tgccttacac ctttggcgga 300
 ggcacaaagg tggaaatcaa g 321

SEQ ID NO: 1170 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Humanized 5F3A5D4-2 L-4.1 light chain variable
 domain sequence
 source 1..107
 mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 1170
DIVMTQSPDS LAVSLGERAT INCSASQGIS NYLNWFQQKP GQAPKLLIYY TSSLHSGVPD 60
RFGSGSGGTD YTLTISSLQA EDVAVYYCQQ YSKLPYTFGG GTKVEIK 107

SEQ ID NO: 1171      moltype = DNA length = 69
FEATURE             Location/Qualifiers
misc_feature         1..69
                     note = Mouse 5F3A5D4-2 light chain variable domain
                     framework 1
source               1..69
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 1171
gatatccaga tgacacagac tacatcctcc ctgtctgcct ctctgggaga cagagtcacc 60
atcagttgc 69

SEQ ID NO: 1172      moltype = AA length = 122
FEATURE             Location/Qualifiers
REGION              1..122
                     note = Mouse 5F3A5D4-V2 heavy chain variable domain sequence
source              1..122
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1172
EVQLQQSGPD LVKPGTSVKI SCKTSGNTFT EYTMHWVKQS HGKSLEWIGG FNPNGVTNY 60
NQKFKGKATL TVDKSSSTAY MELRSLTSED SAVYYCARRY YHSTYVPYFD SWGQGTTLTV 120
SS 122

SEQ ID NO: 1173      moltype = DNA length = 78
FEATURE             Location/Qualifiers
misc_feature         1..78
                     note = Mouse 5F3A5D4-V2 heavy chain variable domain
                     framework 1
source               1..78
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 1173
gaggtccagc tgcaacagtc tggacctgat ctggtgaagc ctgggacttc agtgaagata 60
tcctgtaaga cttctgga 78

SEQ ID NO: 1174      moltype = AA length = 26
FEATURE             Location/Qualifiers
REGION              1..26
                     note = Mouse 5F3A5D4-V2 heavy chain variable domain
                     framework 1
source               1..26
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1174
EVQLQQSGPD LVKPGTSVKI SCKTSG 26

SEQ ID NO: 1175      moltype = AA length = 32
FEATURE             Location/Qualifiers
REGION              1..32
                     note = Mouse 5F3A5D4-V2 heavy chain variable domain
                     framework 3
source               1..32
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1175
KATLTVDKSS STAYMELRSL TSEDSAVYYC AR 32

SEQ ID NO: 1176      moltype = DNA length = 39
FEATURE             Location/Qualifiers
misc_feature         1..39
                     note = Mouse 5F3A5D4-V2 heavy chain variable domain
                     Complementarity -determining region 3
source               1..39
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 1176
cgttactacc atagtaccta cgtgttctac ttgactcc 39

SEQ ID NO: 1177      moltype = DNA length = 33
FEATURE             Location/Qualifiers
misc_feature         1..33
                     note = Mouse 5F3A5D4-V2 heavy chain variable domain
                     framework 4

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source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 1177
tggggccaaag gcaccactct cacagtctcc tca                               33

SEQ ID NO: 1178        moltype = AA  length = 11
FEATURE               Location/Qualifiers
REGION                1..11
                      note = Mouse 5F3A5D4-V2 heavy chain variable domain
                      framework 4
source                1..11
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 1178
WGQGTTLTVS S                                                    11

SEQ ID NO: 1179        moltype = DNA  length = 321
FEATURE               Location/Qualifiers
misc_feature          1..321
                      note = Mouse 5F3A5D4-2 light chain variable domain sequence
source                1..321
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 1179
gatatccaga tgacacagac tacatcctcc ctgtctgcct ctctgggaga cagagtcacc 60
atcagttgca gtgcaagtca gggcattagc aattatttaa actgggttca gcagaaacca 120
gatggaacta ttaagctcct gatctattac acatcaagtt tacattcagg agtcccatca 180
agggtcagtg gcagtgggtc tgggacagat tattctctca ccatcagtaa tgtggaaacct 240
gaagatatgt ccacttacta ttgtcagcag tatagtaagc ttccttacac gttcggagggt 300
gggaccaaacg tggagataaaa a                                           321

SEQ ID NO: 1180        moltype = AA  length = 107
FEATURE               Location/Qualifiers
REGION                1..107
                      note = Mouse 5F3A5D4-2 light chain variable domain sequence
source                1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 1180
DIQMTQTTS LSASLGDRVT ISCSASQGIS NYLWNFQQKP DGTIKLLIYY TSSLHSGVPS 60
RPSGSGSGTD YSLTISNVEP EDIATYYCQQ YSKLPYTFGG GTKLEIK              107

SEQ ID NO: 1181        moltype = AA  length = 122
FEATURE               Location/Qualifiers
REGION                1..122
                      note = Humanized 5F3A5D4-V2 H-1.27 heavy chain variable
                      domain sequence
source                1..122
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 1181
QVQLVQSGAE VKKPGASVKV SCKVSGNTFT EYTMHWVRQA PGKGLEWIGG FNPNGVTNY 60
NQKFKGKVTL TVDTSSTSTAY MELSSLRSED TAVYYCARRY YHSTYVFYFD SWGQGTLLTV 120
SS                                                    122

SEQ ID NO: 1182        moltype = DNA  length = 321
FEATURE               Location/Qualifiers
misc_feature          1..321
                      note = Humanized 5F3A5D4-3 L-1.6 light chain variable
                      domain sequence
source                1..321
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 1182
gatatccaga tgacacagag ccctagctcc ctgagcgcca gcgtggggcga ccgggtcacc 60
attacatgca gcgcttctca gggcatctcc aactacctga actgggtacca gcagaaaccc 120
ggcaaggccc ctaagctgct gatcttctac accagctctc tgcacagcgg cgtgccatct 180
agattcagcg gatctggcag cggcaccgac tacaccctga ccatcagctc cctccagcct 240
gaggacttcg ccacctacta ctgtcagcaa tacagcaagc tgccttatac ctttggcggc 300
ggaacaaaagg tggaaatcaa g                                           321

SEQ ID NO: 1183        moltype = AA  length = 107
FEATURE               Location/Qualifiers
REGION                1..107
                      note = Humanized 5F3A5D4-3 L-1.6 light chain variable
                      domain sequence
source                1..107
                      mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 1183
DIQMTQSPSS LSASVGRVT ITCASQGIS NYLNWYQQKP GKAPKLLIFY TSSLHSGVPS 60
RFGSGSGTD YLTISSLPQ EDFATYYCQQ YSKLPYTFGG GTKVEIK 107

SEQ ID NO: 1184      moltype = DNA length = 321
FEATURE             Location/Qualifiers
misc_feature         1..321
                     note = Humanized 5F3A5D4-3 L-4.1 light chain variable
                     domain sequence
source               1..321
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 1184
gatatcgtga tgacccagag cccagacagc ctggcagtga gtctgggtga gcgtgctaca 60
atcaactgca ggcagagcca gggcatctcc aactacatga attggatatca gcagaaacct 120
ggccaggctc ctaagctgct gatcttctac accagcagcc tgcacagcgg cgtgccagat 180
agattcagcg gcagcggatc tggcaccgac tacacactga ccatttcttc tctccaggcc 240
gaggacgtgg ccgtctacta ctgtcagcaa tacagcaagc tgccttacac ctttggcgga 300
ggcacaaggg tggaaatcaa g 321

SEQ ID NO: 1185      moltype = AA length = 107
FEATURE             Location/Qualifiers
REGION              1..107
                     note = Humanized 5F3A5D4-3 L-4.1 light chain variable
                     domain sequence
source               1..107
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1185
DIVMTQSPDS LAVSLGERAT INCSASQGIS NYLNWYQQKP GQAPKLLIFY TSSLHSGVPD 60
RFGSGSGTD YLTISSLQA EDVAVYYCQQ YSKLPYTFGG GTKVEIK 107

SEQ ID NO: 1186      moltype = AA length = 107
FEATURE             Location/Qualifiers
REGION              1..107
                     note = Mouse 5F3A5D4-3 light chain variable domain sequence
source               1..107
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1186
DIQMTQTTS LSASLGRVT LSCASQGIS NYLNWYQQKP DGTVELLIFY TSSLHSGVPS 60
RFGSGSGTD YSLTISNLEP EDIATYYCQQ YSKLPYTFGG GTKLEIK 107

SEQ ID NO: 1187      moltype = DNA length = 27
FEATURE             Location/Qualifiers
misc_feature         1..27
                     note = Mouse 5F3A5D4-V2 heavy chain variable domain
                     Complementarity -determining region 1
source               1..27
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 1187
aacacattca ctgaatacac catgcac 27

SEQ ID NO: 1188      moltype = DNA length = 42
FEATURE             Location/Qualifiers
misc_feature         1..42
                     note = Mouse 5F3A5D4-V2 heavy chain variable domain
                     framework 2
source               1..42
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 1188
tgggtgaagc agagccatgg aaagagcctt gagtggattg ga 42

SEQ ID NO: 1189      moltype = AA length = 14
FEATURE             Location/Qualifiers
REGION              1..14
                     note = Mouse 5F3A5D4-V2 heavy chain variable domain
                     framework 2
source               1..14
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1189
WVKQSHGKSL EWIG 14

SEQ ID NO: 1190      moltype = DNA length = 51
FEATURE             Location/Qualifiers

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misc_feature      1..51
                  note = Mouse 5F3A5D4-V2 heavy chain variable domain
                  Complementarity -determining region 2
source            1..51
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 1190
ggttttaatc ctaacaatgg tgttactaac tacaaccaga agttcaaggg c          51

SEQ ID NO: 1191    moltype = DNA length = 96
FEATURE           Location/Qualifiers
misc_feature      1..96
                  note = Mouse 5F3A5D4-V2 heavy chain variable domain
                  framework 3
source            1..96
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 1191
aaggccacat tgactgtaga caagtcctcc agcacagcct acatggagct ccgcagcctg  60
acatctgagg attctgcagt ctattactgt gcaaga                             96

SEQ ID NO: 1192    moltype = AA length = 23
FEATURE           Location/Qualifiers
REGION           1..23
                  note = Mouse 5F3A5D4-3 light chain variable domain
                  framework 1
source            1..23
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 1192
DIQMTQTSS LSASLGDRVT LSC                                          23

SEQ ID NO: 1193    moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = Mouse 5F3A5D4-2 light chain variable domain
                  Complementarity -determining region 1
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 1193
agtgaagtc agggcattag caattattta aac                                33

SEQ ID NO: 1194    moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = Mouse 5F3A5D4-3 light chain variable domain
                  Complementarity -determining region 1
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 1194
agtgaagtc agggcattag caattattta aac                                33

SEQ ID NO: 1195    moltype = DNA length = 45
FEATURE           Location/Qualifiers
misc_feature      1..45
                  note = Mouse 5F3A5D4-3 light chain variable domain
                  framework 2
source            1..45
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 1195
tggatcagc agaaaccaga tggaactgtt gaactcctga tcttt                45

SEQ ID NO: 1196    moltype = AA length = 15
FEATURE           Location/Qualifiers
REGION           1..15
                  note = Mouse 5F3A5D4-3 light chain variable domain
                  framework 2
source            1..15
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 1196
WYQKPDGTV ELLIF                                                  15

SEQ ID NO: 1197    moltype = DNA length = 21
FEATURE           Location/Qualifiers
misc_feature      1..21

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note = Mouse 5F3A5D4-3 light chain variable domain
  Complementarity -determining region 2
source      1..21
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 1197
tacacatcaa gtttacactc a                               21

SEQ ID NO: 1198      moltype = DNA length = 96
FEATURE             Location/Qualifiers
misc_feature         1..96
note = Mouse 5F3A5D4-3 light chain variable domain
framework 3
source              1..96
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1198
ggagtcctcat caagggttcag tggcagtggtg tctgggacag attattctct caccatcagc 60
aacctggaac ctgaagatat tgccacttac tattgt                               96

SEQ ID NO: 1199      moltype = AA length = 32
FEATURE             Location/Qualifiers
REGION              1..32
note = Mouse 5F3A5D4-3 light chain variable domain
framework 3
source              1..32
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1199
GVPSRFGSGG SGTDYSLTIS NLEPEDIATY YC                               32

SEQ ID NO: 1200      moltype = DNA length = 27
FEATURE             Location/Qualifiers
misc_feature         1..27
note = Mouse 5F3A5D4-2 light chain variable domain
  Complementarity -determining region 3
source              1..27
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1200
cagcagtata gtaagcttcc ttacacg                               27

SEQ ID NO: 1201      moltype = DNA length = 27
FEATURE             Location/Qualifiers
misc_feature         1..27
note = Mouse 5F3A5D4-3 light chain variable domain
  Complementarity -determining region 3
source              1..27
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1201
cagcagtata gtaagcttcc ttacacg                               27

SEQ ID NO: 1202      moltype = DNA length = 321
FEATURE             Location/Qualifiers
misc_feature         1..321
note = Humanized 5F3A5D4-3 L-3.15 light chain variable
  domain sequence
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 1202
gagatcgtga tgacccagag cccagctaca cttagtgtga gtccaggtga acgggctacc 60
ctgtcctgca ggcgagcca gggcatcagc aactacctga actggtacca gcagaaacct 120
ggccaggccc ctgactgtct gatcttctac accagcagcc tgcacagcgg catccccgcc 180
agattcagcg gcagcggctc tggaaacagac tacacctga caatctctag cctgcagtct 240
gaagattttg ccgtctacta ctgtcagcaa tacagcaagc tgccttatac ctctggcgcc 300
ggaaccaagg tggaaattaa g                               321

SEQ ID NO: 1203      moltype = AA length = 107
FEATURE             Location/Qualifiers
REGION              1..107
note = Humanized 5F3A5D4-3 L-3.15 light chain variable
  domain sequence
source              1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 1203
EIVMTQSPAT LSVSPGERAT LSCSASQGIS NYLNWYQQKP GQAPRLLIIFY TSSLHSGIPA 60

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RFSGSGSGTD YTLTISSLQS EDFAVYVCQQ YSKLPYTFGG GTKVEIK 107

SEQ ID NO: 1204      moltype = DNA  length = 321
FEATURE             Location/Qualifiers
misc_feature         1..321
                     note = Mouse 5F3A5D4-3 light chain variable domain sequence
source               1..321
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 1204
gatatccaga tgacacagac tacatcctcc ctgtctgcct ctctgggaga cagagtcacc 60
ctcagttgca gtgcaagtca gggcattagc aattatttaa actgggtatca gcagaaacca 120
gatggaactg ttgaactcct gatcttttac acatcaagtt tacactcagg agtcccatca 180
aggttcagtg gcagtgggtc tgggacagat tattctctca ccatacagaa cctgggaacct 240
gaagatattg ccacttacta ttgtcagcag tatagtaagc ttccttacac gttcggaggg 300
gggaccaagc tggaaataaa a 321

SEQ ID NO: 1205      moltype = AA  length = 121
FEATURE             Location/Qualifiers
REGION              1..121
                     note = 8F9A4A3H
source              1..121
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1205
VQLQQSGPEL VKPGASVKIS CKTSGNTFTE YTMHWVKQSH GKSLEWIGGF NPNNGVTNRY 60
QKFKGKATLT VDKSSSTAYM ELRLTSEDS AVYYCARRY HSLYVFYFDY WQGQTLTVS 120
S 121

SEQ ID NO: 1206      moltype = AA  length = 122
FEATURE             Location/Qualifiers
REGION              1..122
                     note = Humanized 8F9A4A3 H-ori heavy chain variable domain
                           sequence
source              1..122
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1206
QVQLVQSGAE VKKPGASVKV SCKVSGNTFT EYTMHWVRQA PGKGLEWMGG FNPNGVTNRY 60
NQKFKGRVTM TEDTSTDYAY MELSSLRSED TAVYYCARRY YHSLYVFYFD YWQGQTLVTV 120
SS 122

SEQ ID NO: 1207      moltype = AA  length = 10
FEATURE             Location/Qualifiers
REGION              1..10
                     note = Mouse 5F3A5D4-3 light chain variable domain
                           framework 4 (FR4) sequence
source              1..10
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1207
FGGGTKLEIK 10

SEQ ID NO: 1208      moltype = AA  length = 11
FEATURE             Location/Qualifiers
REGION              1..11
                     note = Mouse 5F3A5D4-2 light chain variable domain
                           Complementarity -determining region 1 (CDR1) sequence
source              1..11
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1208
SASQGISNYL N 11

SEQ ID NO: 1209      moltype = AA  length = 11
FEATURE             Location/Qualifiers
REGION              1..11
                     note = Mouse 5F3A5D4-3 light chain variable domain
                           Complementarity -determining region 1 (CDR1) sequence
source              1..11
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1209
SASQGISNYL N 11

SEQ ID NO: 1210      moltype = AA  length = 7
FEATURE             Location/Qualifiers
REGION              1..7
                     note = Mouse 5F3A5D4-2 light chain variable domain

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source	Complementarity -determining region 2 (CDR2) sequence 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 1210 YTSSLHS		7
SEQ ID NO: 1211 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = Mouse 5F3A5D4-3 light chain variable domain Complementarity -determining region 2 (CDR2) sequence	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 1211 YTSSLHS		7
SEQ ID NO: 1212 FEATURE REGION	moltype = AA length = 9 Location/Qualifiers 1..9 note = Mouse 5F3A5D4-2 light chain variable domain Complementarity -determining region 3 (CDR3) sequence	
source	1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 1212 QQYSKLPYT		9
SEQ ID NO: 1213 FEATURE REGION	moltype = AA length = 9 Location/Qualifiers 1..9 note = Mouse 5F3A5D4-3 light chain variable domain Complementarity -determining region 3 (CDR3) sequence	
source	1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 1213 QQYSKLPYT		9

The invention claimed is:

1. A nucleic acid encoding a monoclonal antibody, a bispecific antibody, a bispecific T-cell engager, a Fab, or a single chain variable fragment antibody (scFv) comprising:

(a) a heavy chain variable region (HCV) comprising the sequence of any one of SEQ ID NOs: 1160, 1162, 1164, and 1181; and

a light chain variable region (LCV) comprising complementarity determining regions (CDRs) comprising a light chain CDR1 (LC-CDR1) comprising the sequence of SEQ ID NO: 434, an LC-CDR2 comprising the sequence of SEQ ID NO: 435, and an LC-CDR3 comprising the sequence of SEQ ID NO: 436; or

(b) an HCV comprising CDRs comprising a heavy chain CDR1 (HC-CDR1) comprising the sequence of SEQ ID NO: 429, an HC-CDR2 comprising the sequence of SEQ ID NO: 430, and an HC-CDR3 comprising the sequence of SEQ ID NO: 431; and

an LCV comprising the sequence of any one of SEQ ID NOs: 1166, 1168, 1170, 1183, 1203, and 1185; wherein the monoclonal antibody, the bispecific antibody, the bispecific T-cell engager, the Fab, or the scFv binds to NME7.

2. The nucleic acid of claim 1, wherein the HCV comprises the sequence of any one of SEQ ID NOs: 1160, 1162, 1164, and 1181; and the LCV comprises the sequence of any one of SEQ ID NOs: 1166, 1168, 1170, 1183, 1203, and 1185.

3. The nucleic acid of claim 1, wherein the HCV comprises the sequence of SEQ ID NO: 1164, and the LCV comprises the sequence of any one of SEQ ID NOs: 1166, 1168, 1170, 1183, 1203, and 1185.

4. The nucleic acid of claim 1, wherein the HCV comprises the sequence of any one of SEQ ID NOs: 1160, 1162, 1164, and 1181; and the LCV comprises the sequence of SEQ ID NO: 1185.

5. The nucleic acid of claim 1, wherein the nucleic acid encodes the monoclonal antibody.

6. The nucleic acid of claim 1, wherein the nucleic acid encodes the bispecific antibody.

7. The nucleic acid of claim 1, wherein the nucleic acid encodes the bispecific T-cell engager.

8. The nucleic acid of claim 1, wherein the nucleic acid encodes the scFv.

9. A cell comprising the nucleic acid of claim 1.

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